Demystified Series

Physics Demystified

for UTME, PUTME, SSCE, GCE and other O'level exams

Demystified Series by Dr. Timothy

Dr Aifuwa Timothy

!!! Notice – If you don't want to score high in physics, don't read this book……..

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Preface

Physics Demystified is a panacea to all pathologies in physics and a masterpiece of its kind. It is written for students enrolling physics in WAEC, UTME, PUTME and other related O'level exams. Each chapter of this book contains explicit and concise notes, comprises over 3,000 O'level exams questions with detailed/explicit explanation and JAMB past questions for self-assessment.

This book is highly recommended for all students and should be well dichotomized before the exam and also recommended for tutors in this field of study, as it will be of crucial help in teaching and inculcating the rightful knowledge and information to the students.

The comment session is open to all kind of suggestions or corrections of any erroneous statement in this book, as they will be highly accepted and appreciated.

I will strongly advice all candidates sitting for any O'level exam not to enter the exam hall, if he/she has not read Physics Demystified.

!!! The questions in this book are all exam based questions from SSCE, JAMB, PUTME and other O'level exams.

- Happy Reading!!!

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I also want express my profound gratitude to my parents : Mr and Mrs Peter AIfuwa, my sisters : Nosakhare Aifuwa and Osasere Aifuwa, Dr Ige Emmanuel, Sir O. J. Olaoye [C.E.O pedagogue house], all my friends and mentors, special thank you to you all for your support.

To God be all the glory……..

Dedication

This book is dedicated to God, my parents, friends and to all students.

Demystified Series by Dr. Timothy

Table of Contents

Chapter	Title	Pages
1	Measurement, Scalars and Vectors	5
2	Motion : Linear motion, Projectile motion, Linear momentum and Newton's laws of motion, Circular motion, Simple harmonic motion	17
3	Friction in solids, Fluids friction or viscosity, Surface Tension	38
4	Work, Energy and Power	51
5	Force field, Gravitational field	57
6	Equilibrium of forces	63
7	Elastic properties, Elasticity	72
8	Simple machines	79
9	Equilibirum in liquids : Density, Relative density and Upthrust	88
10	Pressure in solids and liquids, Kinetic theory of matter and Gas laws	97
11	Temperature and its measurement	116
12	Thermal expansivity	123
13	Thermal energy, Heat capacity, Latent heat and Vapours	131
14	Heat transfer	148
15	Wave: Wave types, Propagation and Properties	156
16	Sound waves	167
17	Light waves : Rectilinear propagation of light , Reflection of light in plane and curved or spherical surface.	182
18	Refraction of light in plane and spherical surfaces/lenses.	195
19	Optical instruments, Application of light waves	209
20	Dispersion of light, Electromagnetic spectrum	217
21	Electrostatics, Electric field, Capacitors	222
22	Electric cells, Conduction of electricity through liquids and gases	234
23	Current electricity, Electrical power and energy	240
24	Magnetic field, Magnetism and Electromagnetism	256
25	Simple A.C circuits	275
26	Elementary Atomic Physics : Atomic theories and discovery of atoms, Energy quantization and line spectra, Thermionic emission, X-ray production, Photoelectric emission, Radioactivity and nuclear reactions, Wave-particle duality.	280
27	Basic Electronics : Semi-conductors, Diodes and Transistors	299

Demystified Series by Dr. Timothy

CHAPTER 1 – MEASUREMENT, SCALARS AND VECTORS.

● **Measurement –** This is the estimation of the magnitude of a physical quantity. A measured quantity consists of its measured value or magnitude and its unit e.g the length of a rule is 1m.

● Fundamental and derived quantities and their units –

- Fundamental quantities – These are physical quantities from which others are derived. Mass, length and time are the three main fundamental quantities. The international agreed system of units for measurements are kilogram (kg) for mass, meter (m) for length and second (s) for time.

Fundamental quantity	Unit
1.Mass	Kilogram (kg)
2.Length	Meter (m)
3.Time	Seconds (s)
4.Temperature	Kelvin (K)
5.Electric current	Ampere (A)
6.Luminous intensity or brightness	Candela (cd)
7.Amount of a substance	Newton-meter(Nm)

- Derived quantities – They are derived from fundamental quantites e.g area,volume,speed density, force e.t.c. Examples of derived quantities and their units;

Derived quantity	Unit
1.Volume	cube meter(m^3)
2.Speed or velocity	meter per second (ms^{-1})
3.Force	newton($N=kgms^{-2}$)
4.Impulse or momentum	newton-second($Ns=Kgms^{-1}$)
5.Energy or workdone	joule(J=$Nm=kgm^2s^{-2}$
6.Moment or couple or torque	Newton-meter(Nm)
7.Energy density	$Jm^{-3}=Kgm^{-1}s^{-2}$

- Measurement of time – Time is measured by stop watches, clocks, ticker tape timer.

- The most accurate clocks in the world are **atomic clock and quartz crystal clocks.**

- The time scale used for astronomical purposes is called **sidereal time.**

- Ticker tape timer measures both **distance** and **time** simultaneously.

- Measurement of mass – Mass is the quantity of matter contained in a body. It is measured by beam **balance or chemical balance.** It works on the **principle of moments.**

- The mass of a body is constant and does not change with change in gravity.

- Measurement of weight – Weight is the earth's gravitational pull on a body. It is measured by a **spring balance**. It works on **Hooke's law**.

- The weight of a body varies with change in gravity.

- Measurement of length –

1. Meter rule – Measures length to **2d.p in cm** or **1d.p in mm** , (M±0.5)mm. The reading accuracy of a meter rule is **0.5mm or 0.05cm**. It is used to measure short distance like the length of a book or a piece of paper, length of a wire e.t.c.

2. Vernier caliper – Measures small lengths to **2d.p in cm** , (M±0.01)cm or **1d.p in mm**. The reading accuracy of a Vernier caliper is 0.1mm or 0.01cm. It is used to measure both the **internal and external diameter of round objects e.g a cylindrical test tube,** thickness of a meter rule or a plate, depth of a cavity and **diameter of a rod.**

- How to read a Vernier caliper –

i.Read the main scale till the mark just before the Vernier scale and take note of the reading in 1d.p in cm. **ii.**Read the Vernier scale until the point where it meets or coincides with the Vernier scale and take note of the reading in 2d.p in cm. **iii.**The reading is the sum of the main scale reading and vernier scale reading.

3.Micrometer screw guage – Measures very small length to 3d.p in cm or 2d.p in mm i.e (M±0.01)mm. The reading accuracy of a micrometer screw guage is 0.01mm or 0.001cm. It is used to measure **external or outside diameter only of a round objects e.g cylindrical test tubes, thickness of a thin converging lens or pendulum bulb, thickness of a piece of paper or rod.**

- How to read a micrometer screw guage –

i.Read the sleeve scale just before the thimble scale and take note of the reading in 1d.p in mm. **ii.**Read the thimble scale until the point where it meets or coincides with the sleeve scale and take note of the reding in 2d.p in mm. **iii.**The reading is the sum of the sleeve reading and thimble reading.

- Measurement of volume – The volume of an irregular solid is measured by a measuring cylinder i.e the volume of water displaced by the solid from the cylinder is equal to its volume.

● **Error in measurement –** The deviation of a measured value in measurement from the actual or original value. A measurement is always in the form of : $(\boldsymbol{M \pm e})$, M is the measured value, e is the error i.e the measurement within the range of $M + e$ to $M - e$.

- Absolute error=Measured value−Actual value, $\boldsymbol{e = M.V - A.V}$**.**

- Percentage error= $\frac{\boldsymbol{Absolute\ error}}{\boldsymbol{Actual\ value}} \times \mathbf{100}$

- Maximum or greatest possible error or least

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possible error or estimated uncertainty is equal to half the smallest scale division.

Examples :

1. Which of the following statements are true about the spring balance and the chemical balance? I.Both are used to measure the mass of an object II.Either of them may be used to measure the weight of an object III.the springs balance works on the principle of Hooke's law while the chemical balance works on principle of moment IV.a change in gravity changes the reading of a spring balance but not that a chemical balance A.I and IV B.II and III **C.III and IV** D.I and III E.I,II and III.

Spring balance measures the weight of an object while chemical balance measures the mass of an object. Spring balance works on the principle of Hooke's law while chemical balance works on the principle of moment. Gravity affects weight (the reading of spring balance) but does not affect mass (the reading of a chemical balance).

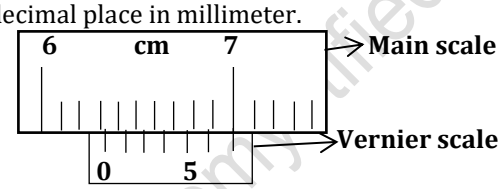
2. The basic principle of operation of a beam balance is A.Archimedes principle B.law of floatation C.Hooke's law **D.principle of moments.**

3. Which of the following instruments has a reading accuracy of 0.5mm? A.vernier calipers B.micrometer screw guage **C.meter rule** D.spring balance.

4. Which of the following represents the correct precision if the length of a piece of wire is measured with a meter rule? A.35mm **B.35.0mm** C.35.00mm D.35.01mm.

A meter rule measures the length of a material to 1 decimal place in millimeter.

5.



The diagram above represents a portion of vernier caliper. What is its reading? A.7.10cm B.7.09cm C.6.90cm D.6.34cm **E.6.27cm.**

Main scale reading=6.2cm Vernier scale reading=0.07cm. Reading=6.27cm.

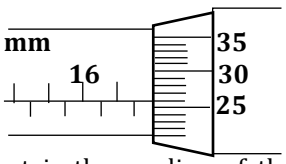
6. Which of the following readings gives the correct precision of the length of a rod using a vernier caliper? A.4.1cm **B.4.13cm** C.4.120cm D.4.125cm.

A vernier calliper measures length to the nearest 2 decimal place in centimeter.

7.The inner diameter of a small test tube can be measured accurately using? **A.pair of vernier calliper** B.micrometer screw guage C.meter rule D.pair of dividers.

A vernier caliper measures both the internal (inner) and external (outer) diamater of round or cylindrical objects.

8.



What is the reading of the micrometer screw guage? **A.17.76mm** B.16.26mm C.17.56mm D.16.25mm.

Sleeve reading=17.5mm, Thimble reading= 0.26mm. Reading=17.76mm.

9. Which of the following instruments is most suitable for measuring the outside diameter of a narrow pipe a few millimeters in diameter? A.Pair of calipers B.Metre rule **C.Micrometer screw guage** D.Tape rule.

A mircometer meter screw guage measures the external (outer) diameter of round or cylindrical objects to 2 decimal places in millimeter.

10. The thickness of the central portion of a thin converging lens can be determined very accurately by using? A.vernier caliper **B.a micrometer screw guage** C.a telescope D.a microscope.

11. I.Diameter of a small ball bearing II.Thickness of a piece of paper III.Diameter of a measuring cylinder IV.Length of a piece of wire. Which of the above can be measured using a micrometer screw guage? A.I,II and IV **B.I and II** C.II and III D.III and IV.

The diameter of a measuring cylinder can be internal or external. Since it's not specified it means both are being considered in this case.

12. The volume of a stone having an irregular shape can be determined using? A.meter rule **B.measuring cylinder** C.vernier calliper D.micrometer screw guage.

13. The volume of an irregular solid can be measured using A.burette B.pipette C.conical flask **D.displacement can.**

14. The least possible error in using a scale graduated in millimeters is A.0.1mm **B.0.5mm** C.1.0mm D.2.0mm.

The smallest scale division of a scale graduated in millimeter=1mm,

Least possible error=half the smallest scale division= $\frac{1}{2} \times 1 = 0.5$mm.

15. What is the percentage reading error in measuring a distance of 5.00cm using a meter rule marked in mm? A.10% B.5.0% **C.1.0%** D.0.5%.

Smallest scale division of a meter rule=1mm, reading error= $\frac{1}{2} \times$ smallest scale division

reading error= $\frac{1}{2} \times 1 = 0.5$mm, actual reading= 5.00cm=50mm,

$\%\text{reading error} = \frac{reading\ error}{Actual\ value} \times 100$

$\%\text{reading error} = \frac{0.5}{50} \times 100 = 1\%.$

16. A student measures the internal and external

Demystified Series by Dr Timothy

diameter of a cylindrical vessel as (100 ± 1)mm and (104 ± 1)mm repectively. Determine the percentage error in the thickness of the tube.
A.50% B.100% C.25% D.75%**.**
Percentage error $= \frac{Absolute\ error}{\text{Actual value}} \times 100$
Actual value of thickness = Ext.dia$-$Int.dia $=$ $104-100=4$mm,
Absolute error $e = e_1 + e_2 = 1+1 = 2$mm
Percentage error $= \frac{2}{4} \times 100 = 50\%$.

17. Which of the following are derived quantities? I.thrust II.temperature III.area IV.pressure A.I and IV B.II,III and IV **C.I,III and IV** D.I,II,III and IV.
Temperature is a fundamental quantity.

18. Which of the following is a derived unit? I.meter II.coulomb III.kilogram IV.ampere V.joule A.I and III **B.II and V** C.II,IV and V D.I,II III,IV and V.
The units : meter for length, kilogram for mass and ampere for electric current are all fundamental units. Coulomb for quantity of electricity or charge and joule for energy or workdone are derived units.

19. The watt is equivalent to **A.Nms^{-1}**B.Js C.Kgm^{2}s^{-2} D.Ns.
The unit for power =watt(W)= Js^{-1} = Nms^{-1}.

20. In which of the following physical quantities are the units correctly indicated? I.Weight(N) II.Energy(Nm) III.Momentum (kgms^{-1}) IV.Acceleration(Nkg^{-1}). A.I and II B.III and IV C.I,II and III **D.I,II,III and IV.**
Weight=Force=N, Energy=J=Nm, Momentum= Ns=Kgms^{-1}, Acceleration=ms^{-2}=Nkg^{-1}.

21. Which of the following quantities has the same unit as the kilowatt-hour? A.Force×time B.Force×acceleration **C.Force×distance** D.Force×velocity.
Kilowatt-hour=Power×time=Energy or work done= Force×distance. Impules = Force×time. Power = Force×velcoity.

22. Which of the following units is equivalent to the ratio of power÷acceleration? A.Js^{-1}**B.Ns** C.Nms D.Ns^{-1}.
Power/acceleration=momentum=kgms^{-1}=Ns.

● **Dimensional analysis –** Dimensional analysis shows the relations of derived quantities to the fundamental quantities from which they are obtained. In dimensional analysis, the fundamental quantities are represented by a single letter and while derived quantities are represented by a combination of these fundamental quantities. In dimension, L is used to specify length, M is used to specify mass, T is used to specify time, I is used to specify time, θ is used to specify temperature.

Steps in dimension of a physical quantity

1.Write out the formula of the quantity as related to its fundamental makeup.

2.Substitue the fundamental quantities with their individual dimension and assigning any power if present. Incase any coefficient is present e.g **2M,** ignore the number "2" and remove it from the dimension.

3.Simplify to get the overall dimension.

Examples :

1. Speed, velocity and acceleration.

a. Speed=distance/time , distance is a measure of length hence**,** Speed $= \frac{L}{T} = LT^{-1}$.

b. Velocity=displacement/time, displacement is also a measure of length hence,
Velocity $= \frac{L}{T} = LT^{-1}$.

Speed and velocity has the same dimension.

c. Acceleration = velocity/time,
$a = \frac{v}{t} = \frac{LT^{-1}}{T} = LT^{-2}$.

2.Density=mass/volume, $\rho = \frac{m}{V} = \frac{M}{L^3} = ML^{-3}$.

3. Force, Workdone, energy, power, momentum, impulse, energy density.

a. Force=mass×acceleration= $m \times a$,
$F = M \times LT^{-2} = MLT^{-2}$.

b. Workdone=Force×distance= $F \times s$,
$W = MLT^{-2} \times L = ML^2T^{-2}$.

c. Energy and workdone have the same dimension.

d. Power=workdone/time= $\frac{W}{t}$,
$P = \frac{ML^2T^{-2}}{T} = ML^2T^{-3}$.

e. Momentum=mass×velocity=$m \times v$,
$P = M \times LT^{-1} = MLT^{-1}$.

f. Impulse=Force×time= $F \times t$,
$I = MLT^{-2} \times T = MLT^{-1}$.

Impulse and momentum have the same dimension.

g. Energy density = energy/volume = workdone/volume = W/V
Energy density= $\frac{ML^2T^{-2}}{L^3} = ML^{-1}T^{-2}$.

4. Pressure, tensile stress, tensile strain, young modulus, surface tension and viscosity or coefficient of viscosity.

a. Pressure=Force/area= $\frac{F}{A}$,
$P = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$.

b. Tensile stress=Force/area= $\frac{F}{A}$,
Tensile stress= $\frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$.

c.Tensile strain=extension/original length = $\frac{e}{L}$
Tensile strain= $\frac{L}{L} = 1$. Hence, **Tensile strain is dimensionless.**

d.Young modulus=Tensile stress/Tensile strain
$E = \frac{ML^{-1}T^{-2}}{1} = ML^{-1}T^{-2}$.

Energy density, pressure, tensile stress and Young modulus all have the same dimension.

e. Surface tension=Force/Length= $\frac{F}{L}$,

Demystified Series by Dr Timothy

S.T= $\frac{MLT^{-2}}{L} = MT^{-2}$.

f. Viscosity=Force×distance/velocity×area

$\eta = \frac{F \times d}{v \times A} = \frac{MLT^{-2} \times L}{LT^{-1} \times L^2} = ML^{-1}T^{-1}$.

5.Universal gravitation constant, G

$F = \frac{Gm_1m_2}{r^2}$, $G = \frac{Fr^2}{m_1m_2} = \frac{Fr^2}{m^2}$,

$G = \frac{MLT^{-2} \times L^2}{M^2} = M^{-1}L^3T^{-2}$.

6.Specific heat capacity, c

c =Heat energy/mass×temperature change

$c = \frac{H}{m\Delta\theta} = \frac{ML^2T^{-2}}{M \times \theta} = L^2T^{-2}\theta^{-1}$.

7. Quantity of electricity or Charge, Electromotive force.

a. Quantity of electricity or charge, QQ = electric current×time= $It = IT$.

b. Electromotive force = Workdone/Charge

$E = \frac{W}{Q} = \frac{ML^2T^{-2}}{IT} = ML^2T^{-3}I^{-1}$.

- Dimensionless quantities: They are physical quantities without unit and dimension e.g mechanical advantage or force ratio, velocity ratio, coefficient of friction, tensile strain, relative density e.t.c.

- Application of dimensional analysis :

1.It is used to derive formulas from a given sets of variables.

2.Dimension is used to test the uniformity or homogeneity of an equation. An equation is said to be homogenous if it is dimensionally correct i.e the dimension of each quantity of the equation are be the same.

Rules for uniformity or homogeneity of an equation

1.If an equation is given as $P = Q$, then P and Q have the same dimensions.

2.If an equation is given as $P = Q \pm R$, then P,Q and R have the same dimension.

3.If an equation is given as $P = QR$, then the dimension of P is equal to the product of the dimensions of Q and R.

Examples :

1. The period of vibration T, of a liquid surface of a drop depends on the density (ρ), radius (r) and surface tension of the liquid (Y). Derive a formula using dimension, showing the relationship T and the other variables?.

Since period of vibration (T) depends on density (ρ), radius (r) and surface tension (T) then, $\boldsymbol{T \propto \rho^{X} r^{Y} Y^{Z}}$, where x,y and z are the powers to wich each of the quantities are raised in the formula. $T \propto \rho^{X} r^{Y} Y^{Z}$, $T = K\rho^{X} r^{Y} Y^{Z}$, K=constant of proportionality, Dimension of $T = T = M^oL^oT$, Dimension of density, $\rho = ML^{-3}$, Dimension of $r = L$ Dimension of surface tension, $Y = MT^{-2}$,

Input the dimensions into the equation,

$T = K\rho^{X} r^{Y} Y^{Z}$, to replace the variables

$M^oL^oT = K\,(ML^{-3})^{X}\,(L)^{Y}\,(MT^{-2})^{Z}$,

$M^oL^oT = K\,.\,M^{X}L^{-3X}.\,L^{Y}.\,M^{Z}T^{-2z}$,

$M^oL^oT = KM^{X+Z}.\,L^{-3X+Y}.\,T^{-2z}$,

Equate like terms on both sides of the equation;

$M^o = M^{X+Z}$, x+z=0, x= −z,

$L^o = L^{-3X+Y}$, −3x+y=0, y=3x,

$T = T^{-2z}$, 1= −2z, z= $-1/2$,

x= $-z = -(-1/2) = 1/2$, y=3x= $3 \times 1/2 = 3/2$,

Substitute "x=1/2, y=3/2 and z=−1/2" back into the original equation – $T = K\rho^{X} r^{Y} Y^{Z}$,

$T = K\,(\rho)^{1/2}.\,(r)^{3/2}.\,(Y)^{-1/2} = \frac{K\rho^{1/2}r^{3/2}}{Y^{1/2}} = K\sqrt{\frac{\rho r^3}{Y}}$.

2. If the frequency (f) of a simple pendulum is assumed to be affected by the following quantities – mass of the bulb(m), length of the bulb(L) and acceleration due to gravity (g). Derive a formula by method of dimension showing the relationship between f and the other variables?

Since frequency (f) depends on mass (m), length (L) and acceleration due to gravity (g) then, $f \propto m^{X}L^{Y}g^{Z}$, where x, y and z are the powers to which each of the quantities are raised in the formula.

$f \propto m^{X}L^{Y}g^{Z}$, $f = Km^{X}L^{Y}g^{Z}$, K=constant of proportionality. Dimension of $f = T^{-1} = M^oL^oT^{-1}$. Dimension of m=M, Dimension of L=L, Dimension of g= LT^{-2} (acceleration).

Input the dimensions into the equation,

$f = Km^{X}L^{Y}g^{Z}$, to replace the variables.

$M^oL^oT^{-1} = K(M)^{X}(L)^{Y}(LT^{-2})^{Z}$

$M^oL^oT^{-1} = K.\,M^{X}.\,L^{Y}.\,(L^{Z}\,T^{-2z})$

$M^oL^oT^{-1} = K.\,M^{X}.\,L^{Y+Z}.\,T^{-2Z}$

Equate like terms on both sides of the equation;

$M^o = M^{X}$, x=0. $L^o = L^{Y+Z}$, y+z=0, y= −z,

$T^{-1} = T^{-2z}$, $-2z = -1$, z =-1/-2=1/2,

y= −z= $-1/2$.

Substitute "x=0, y=$-\frac{1}{2}$ and z=$\frac{1}{2}$" back into the original equation – $f = Km^{X}L^{Y}g^{Z}$,

$f = K(\mathrm{m})^0.\,(L)^{-1/2}.\,(\mathrm{g})^{1/2} = \frac{Kg^{1/2}}{L^{1/2}} = \frac{K\sqrt{g}}{\sqrt{L}} = K\sqrt{\frac{g}{L}}$.

3. The velocity of a moving particle is given by : $v^2 = u^2 + 2as$. Show that the formula is dimensionally correct?.

v=final velocity, u=initial velocity, a=acceleration, s=distance

Applying rule 2, the dimension of v^2 is same as that of u^2 and as.

Dimension of $v^2 = (LT^{-1})^2 = L^2T^{-2}$

Dimension of $u^2 = (LT^{-1})^2 = L^2T^{-2}$

Dimension of $as = (LT^{-2} \times L) = L^2T^{-2}$.

2.The distance travelled by a particle is given by the equation, $s = (at^2 + bt + c)$m. What are the dimension of a, b and c.

Applying rule 2 and rule 3, dimension of s is the same as that of at^2, bt and c.

Dimension of $s = L$,

i. $s = at^2$, $L = aT^2$, $a = \frac{L}{T^2} = LT^{-2}$.

ii. $s = bt$, $L = bT$, $b = \frac{L}{T} = LT^{-1}$.

Demystified Series by Dr Timothy

iii. $s = c, \quad L = c, \quad c = L.$

Question Drill –

1. The resonant frequency f_o of a closed air column is known to depend on pressure P, the density ρ of the air and the length L of the air column. Derive a formula by method of dimension showing the relationship between f_o and the other variables? **Ans –** $\boldsymbol{f_o = \frac{K}{L}\sqrt{\frac{P}{\rho}}}$.

2. If the period of vibration T of a simple pendulum depends on the mass of the bulb m, length of the bulb L and acceleration due to gravity g. Derive a formula by method of dimension showing the relationship between T and the other variables? **Ans –** $\boldsymbol{T = K\sqrt{\frac{g}{L}}}$.

3.A wave is set up in a stretched string by plucking it. The velocity V, of the wave depends on the tension T, in the string its length L and its linear density μ. Derive a formula by method of dimension showing the relationship between T and the other variables? **NB:** Tension is a form of force, linear density is the same as mass per unit length. **Ans –** $\boldsymbol{V = \sqrt{\frac{T}{\mu}}}$.

4. The distance travelled by a particle is given by the equation;

$s = at^3 + bt^2 + ct + d$ in meters, the unit of a is A.ms^{-2} **B.ms^{-3}** C.ms^{-1} D.m.

5. Which of the following properties of steel bar can be measured in terms of dimension of length only? A.weight B.density C.pressure **D.volume.**

● Scalars and Vectors –

- Scalar quantities are quantities which have magnitude only but no specified direction. They are described by their magnitudes only.

- Vector quantities are quantities which have magnitude and specified direction. They are described by their magnitude and direction.

Vector quantities	Scalar quantities
Displacement,velocity acceleration,force, weight,momentum, impulse,moment, couple(torque), friction, all field intensities, stress, e.t.c.	Distance,speed, mass,time,length, volume,viscosity, temperature,work, energy,power,heat inertia,pressure, all potentials,velocity ratio,mechanical advantage,density, current,potential difference e.tc

- Resultant vector R – It is the single vector that gives the same effect in magnitude and direction as a set of vectors acting together at a point.

- Orientation and arrangement of vectors :

1. Parallel vectors – I. They act in the same direction in a straight line. **II.**The angle between the vectors or angular difference is 0°. **III.**Their resultant is given by the arithmetic sum of both vectors and acts in their common direction , $\mathbf{R} = \vec{A} + \vec{B}$. **IV.The resultant of two or more vectors is maximum or highest when they are parallel (θ=0°).**

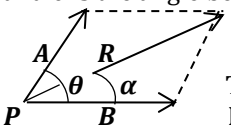
2. Anti-parallel vectors – I. They act in opposite directions in a straight line. **II.**The angle between the vectors or angular difference is 180°. **III.**Their resultant is given by the arithmetic substraction of both vectors and acts in the direction of the larger vector , $\mathbf{R} = \vec{A} - \vec{B}$. **IV.The resultant of two or more vectors is minimum or least when they are anti-parallel (θ=180°).**

- Two anti-parallel vectors of equal magnitudes will produce a zero resultant.

3. Perpendicular vectors – I. They act at right angles to each other. **II.**The angle between the vectors or angular difference is 90°. **III.**The resultant is given by pythagora's theorem i.e $\boldsymbol{R^2 = A^2 + B^2}$. The direction of the resultant vector is represented as $\boldsymbol{\theta}$.

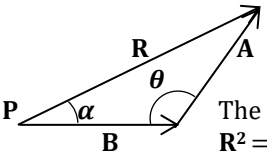
4. Diagonal vectors – I. They acts in diagonal direction and are represented by the adjacent sides of a parallelogram. **II.**The angle between the vectors is between 0° and 180° ($0° < \theta < 180°$). **III.**The diagonal of the parallelogram draw from the intersection of both vectors represents the resultant and it gotten by **cosine rule.** There are two forms of orientation of diagonal vectors :

a. The two vectors which meet at a common point P and their resultant originate the point P and θ is the angle between them;



The resultant R is given by ;
$\mathbf{R^2 = A^2 + B^2 + 2AB\ cos\ \theta}$

b. The two vectors arranged in a triangle such that the resultant from the tail of one vector to the tip of the other and θ is the angle between them.



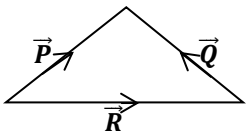
The resultant is given by :
$\mathbf{R^2 = A^2 + B^2 - 2AB\ cos\theta}$.

The direction of the resultant α is calculated using ; $\frac{sin\ \alpha}{A} = \frac{\sin\theta}{R}$ OR $\tan\alpha = \frac{Asin\theta}{B + Acos\theta}$.

- Triangle of vectors – When three concurrent vectors i.e pass through a common point, acts on body, hence they can be represented in magnitude and direction by the three sides of a triangle.

1.Two vectors and their resultant – The tip to tail rule is used in describing the resultant i.e the resultant starts from the tail of one vector and ends in the tip of the other vector.

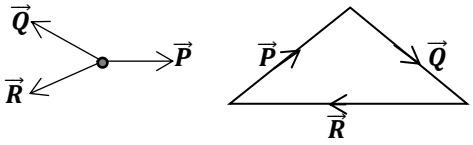
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By the tip to tail rule, the resultant of the vectors $\vec{Q}$ and $\vec{R}$ in the diagram above, is $\vec{P}$ as it starts from the tail of $\vec{R}$ and ends at the tip of $\vec{Q}$ **. The vector equation relating the vector is** :

$\vec{P} = \vec{Q} + \vec{R}$ or $\vec{R} = \vec{P} - \vec{Q}$ or $\vec{Q} = \vec{P} - \vec{R}$.

2. Three equilibrant vectors – When three vector $\vec{P}$, $\vec{Q}$ and $\vec{R}$ acts simultaneously on a body to keep it stationary i.e in equilibrium, then each vector is the equilibrant of the other two . $\vec{R}$ is the equlibrant of $\vec{P}$ and $\vec{Q}$, or $\vec{Q}$ is the equilibrant of $\vec{P}$ and $\vec{R}$.



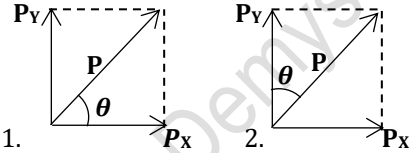
In the triangle diagram above, notice that all the vectors moves in similar direction round the triangle with the tip of one touching the tail of the other. **The vector equation of equilibrant vectors is : $\vec{P} + \vec{Q} + \vec{R} = 0$. Thus, the vector sum of equilibrant vectors is zero.**

- Equilibrant vector of two or more vectors is the single vector which will balance all other vectors keeping them in equilibrium.

- Equilibrant vector is equal and directly opposite to the resultant vector (equal in magnitude but opposite in direction).

- Resolution of vectors into components –

1. Resolution of oblique vectors – A vector P inclined at angle θ to the horizontal or vertical, has two perpendicular componets; P_Y (vertical component) and P_X (horizontal component).



In diagram 1, P is inclined at angle θ to the horizontal, $P_Y = P\sin\theta$ while $P_X = P\cos\theta$. In diagram 2, P is inclined at angle θ to the vertical, $P_Y = P\cos\theta$ while $P_X = P\sin\theta$.

- The component (axis) that makes an angle with the vector takes $\cos\theta$ while the perpendicular component takes $\sin\theta$.

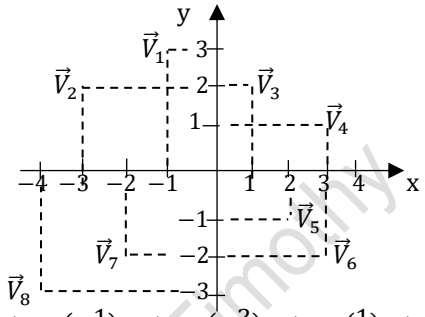
2. Resolution of parallel vectors – The component of each parallel vector along their common direction is its full positive magnitude.

3. Resolution of anti-parallel vectors – The component of each anti-parallel along the direction of the other is its full negative magnitude.

4. Resolution of perpendicular vectors – The component of each vector along the direction of the other is zero.

5. Resolution of more than two vectors or multiple vectors – Each of the vectors are resolved into their perpendicular components. The resultant is then calculated from the two perpendicular components using pythagor's theorem.

- Position vectors $\vec{V}$ – Position vector is a vector whose effect is on one point in space. Position vectors in the (x, y) coordinates is given as $\begin{pmatrix} x \\ y \end{pmatrix}$.



$\vec{V}_1 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$, $\vec{V}_2 = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$, $\vec{V}_3 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\vec{V}_4 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$,
$\vec{V}_5 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $\vec{V}_6 = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$, $\vec{V}_7 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$, $\vec{V}_8 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$.

- Relative velocity – It is the velocity of a body in respect to another body.

- One dimensional relative velocity – Relative velocity between bodies along parallel lines.

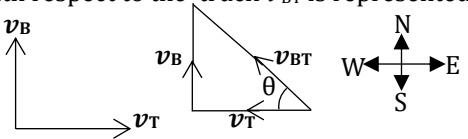
- If two bodies A and B are moving with speed v_A and v_B respectively then, the relative velocity of body A respect to body B given by :

I. $v_{AB} = v_A - v_B$, if they are moving in the same direction.

II. $v_{AB} = v_A + v_B$, if they are moving in the opposite direction.

- Two bodies are moving in opposite direction when approaching and moving in the same direction when receeding(chasing).

- Two dimensional relative velocity – Relative velocity between bodies along non-parallel lines. e.g assume a bus and truck took of from the same point with the bus travelling northwards with velocity v_B while the truck is travelling eastwards with velocity v_T.The relative velocity of the bus with respect to the truck v_{BT} is represented as –



The direction of v_T was reversed inorder for the resultant v_{BT} to obey the tip-to tial rule, i.e runs from the tail of v_T to the tip of v_B. $v_{BT}^2 = v_B^2 + v_T^2$. The direction of v_{BT} is given by – $\tan\theta = \frac{v_B}{v_T}$.

Examples :

1. Which of the following are vector quantities?
I.Work II.Displacement III.Acceleration
IV.Electric field intensity V.Magnetic induction

Demystified Series by Dr. Timothy

A.I,II and III B.II,III and IV C.III,IV and V **D.II,III,IV and V**.

Work and energy are scalar quantity. All field intensity(electrical,magnetic & gravitational) are vectors. Magnetic induction is same as magnetic flux density and it is a vector.

2. Which of the following pairs of physical quantities comprises vectors? A.Capacitance and inductance B.Force ratio and velocity ratio **C.Friction and momentum** D.Energy and power E.Electric field potential and electric field intensity**.**

Capacitance,inductance,energy,power,force ratio (mechanical advantage), velocity ratio, all potential (electric and gravitational) are all scalars. Friction, momentum and electric field intensity are vectors.

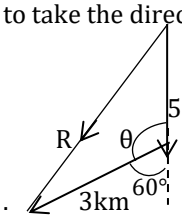
3. Which of these quantities are vectors? I.Moment II.Temperature gradient III.Density IV.Temperature **A.I and II** B.II and III C.I and IV D.II,III and IV.

Moment and temperature gradient are vectors while temperature and density are scalars.

4. A man walks 5km south and then 3km in the direction 60° west of south.His distance from the starting point is A.7.00km B.7.50km **C.8.00km** D.10.72km.

Distance is a scalar quantity and is independent of the direction. Distance=5+3=8km.

NB – If the question had stated that we should calculate for the displacement, then we will have to take the direction into consideration.



$\theta=180-60=120°$,

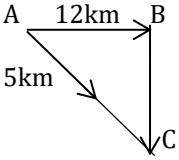
Displacement=R,

$R^2= 5^2+3^2-2\times5\times3\ cos120°$,

$R^2=25+9-30(-0.5)=34+15=49$,

$R=\sqrt{49}=7$km. **option A.**

5. A boy travels 12km eastwards to a point B and then 5km southward to another point C. Calculate the difference between the magnitude of the displacement of the boy and the distance travelled by him **A.4.0km** B.7.0km C.13.0km D.17.0km E.30.0km



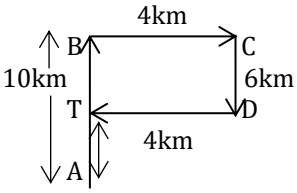
Distance=AB+BC(neglecting the direction, as it is a scalar quantity)=12+5=17km.

Displacement=AC , $AC^2=AB^2+BC^2=12^2+5^2$,

$AC^2=144+25=169$, $AC=\sqrt{169}=13$km.

Difference between the magnitude of the displacement and distance =17−13 =4km.

6. A lorry travels 10km northwards, 4km eastwards, 6km southwards and 4km westwards to arrive at a point T. What is the total displacement A.6km south **B.4km north** C.6km north D.4km east.



AB=10km, BC=4km, CD=6km, DT=4km

The lorry starts its travel from A and ends in T.

Horizontal component= BC−DT = 4−4 = 0km.

Vertical component = AB−CD = 10−6 = 4km.

Thus, the total displacement = AT = 4km due North. Total distance travelled by the lorry=10+4+6+4=24km.

7. If the angle between two vectors P and Q is 0°, the vectors are said to be **A.parallel** B.perpendicular C.intersect at angle of 45° D.intersect at angle of 60°.

Two vectors are said to be parallel when their angular difference is 0°.

8. The resultant of two forces acting on an object minimum when the angle between them is **A.180°** B.90° C.45° D.0° **.**

9. The resultant of two forces acting on an object maximum when the angle between them is A.180° B.90° C.45° **D.0° .**

10. The resultant of two forces acting on an object is maximum when the angle between them is A.180° B.90° C.45° **D.0°**.

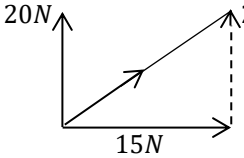
11. Two forces A and B act at a point. If their resultant is given by $(B-A)$ in the direction of B, then A.A and B are equal B.A is greater than B C.The angle between A and B is 0° D.The angle between A and B is 90° **E.The angle between *A* and *B* is 180°.**

The two vectors are anti-parallel i.e θ=180°,as their resultant is given by their arithemetic difference. Also B is greater than A, as the resultant moves in the direction of B.

12. A vector of magnitude 5 units in the north direction is combined with another vector to give a zero resultant. The other vector is A.5units in the north direction **B.5units in the south direction** C.5units in the east direction D.5units in the west direction E.−5units in the south direction.

Anti-parallel vectors(vectors acting in opposite directions), of equal magnitude produce zero resusltant.

13. What is the magnitude of the resultant two forces 20N and 15N inclined at 90° to each other? A.35N B.5N **C.25*N*** D.10N.



$R^2=20^2+15^2$,

$R^2=400+225$

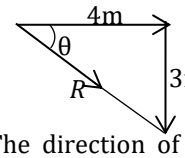
$R^2=625$,

$R=\sqrt{625}=25$N.

14. An object moves 4m eastward(E) and then

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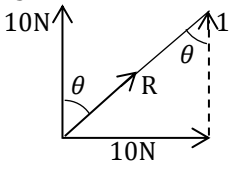
3m southward(S). Its displacement from its original position is? A.1m SE **B.5m SE** C.7m SE D.12m SE.



$R^2 = 3^2+4^2 = 9+16 = 25$,
$R^2 = \sqrt{25} = 5$m.

The direction of the resultant is in the south-eastern (SE) direction. Displacement = 5m SE.

15. Two forces each of magnitude 10N acts on a body, one towards the North and the other towards the East.What is the magnitude and direction of the resultant force? **A.$10\sqrt{2}$N, N45°E** B.$10\sqrt{2}$N, N45°W C.20N, N45°W D.20N, N45°E .

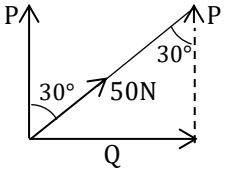


$R^2 = 10^2+10^2$,
$R^2 = 100+100 = 200$,
$R = \sqrt{200} = 10\sqrt{2}N$.

$\tan\theta = \frac{10}{10} = 1$, $\theta = \tan^{-1} 1 = 45°$,

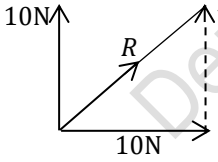
The resultant force is $10\sqrt{2}N$, $N45°E$.

16. Two forces P and Q acting perpendicularly to each other have resultant of 50N. If P makes an angle of 30° with the resultant, calculate the magnitude of Q A.55.0N B.43.3N C.40.0N **D.25.0N** E.15.8N.



$\sin 30° = \frac{Q}{50}$,
$Q = 50\sin 30°$,
$Q = 50\times 0.5 = 25$N.

17. A block of mass 2.0kg resting on a smooth horizontal plane is acted upon simultaneously by two forces, 10N due North and 10N due East. The magnitude of the acceleration produced by the forces on the block is A.0.10ms^{-2} **B.7.07ms^{-2}** C.10.00ms^{-2} D.14.10ms^{-2} E.20.00ms^{-2} .



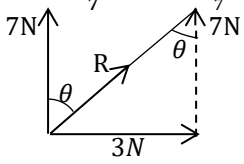
$R^2 = 10^2+10^2$,
$R^2 = 100+100 = 200$,
$R = \sqrt{200} = 10\sqrt{2}$N.
$R = F = ma$,

F=resultant force=$10\sqrt{2}$N, m=mass=2.0kg,

a=acceleration, $a = \frac{F}{m} = \frac{10\sqrt{2}}{2} = 5\sqrt{2} = 7.07$ms^{-2}.

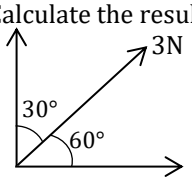
18. Two forces of magnitude 7N and 3N act at right angles to each other. The angle θ between the resultant and the 7N force is given by A.$cos\,\theta = \frac{3}{7}$ B.$sin\,\theta = \frac{3}{7}$.

C.$tan\,\theta = \frac{3}{7}$ D.$cot\theta = \frac{3}{7}$

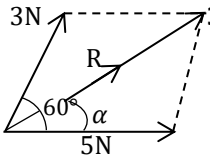


$\tan\theta = \frac{3}{7}$,
$\cot\theta = \frac{1}{\tan\theta} = \frac{7}{3}$.

19. Two forces act on a body, the one of magnitude 5N acts due east, the other of magnitude 3N acts in a direction of 030°. Calculate the resultant vector and the direction?



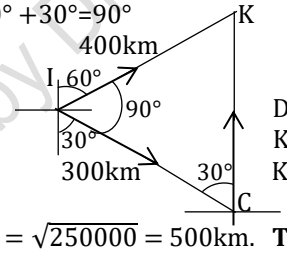
The direction of the 3N force is 030°.
Hence, the angle between the forces=90−30=60°.



$R^2 = 3^2+5^2+2\times3\times5\,cos60°$,
$R^2 = 9+25+30(0.5)$
$R^2 = 34+15 = 49$, $R = \sqrt{49}$,
$R = 7$N.

$\frac{sin\,\alpha}{3} = \frac{\sin 60°}{7}$, $\sin\alpha = \frac{3\sin 60°}{7} = \frac{2.6}{7} = 0.371$,
$\alpha = \tan^{-1} 0.371 = 21.77° \approx 21.8°$. Alternatively,
$\tan\alpha = \frac{3sin60°}{5+3cos60°} = \frac{2.6}{5+1.5} = \frac{2.6}{6.5} = 0.4$,
$\alpha = \tan^{-1} 0.4 = 21.8°$.

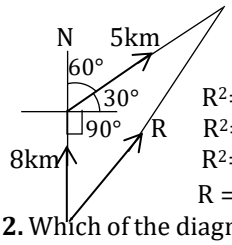
20. An aircraft travelled from Calabar to Kano as follows. It flew first to Ilorin covering a distance of 300Km, 30°West of North and then flew 400km, 60°East of North to Kano. What is the total displacement? A.567km B.594km **C.500km** D.594km E.600km.



I=Ilorion, K=Kano, C=Calabar.

Displacement=KC,
$KC^2 = 400^2+300^2$,
$KC^2 = 160000+90000$,
$KC^2 = 250000$,
$KC = \sqrt{250000} = 500$km. **Total distance of the aircraft** = 300+400= 700km (neglecting the direction as distance is a scalar quantity).

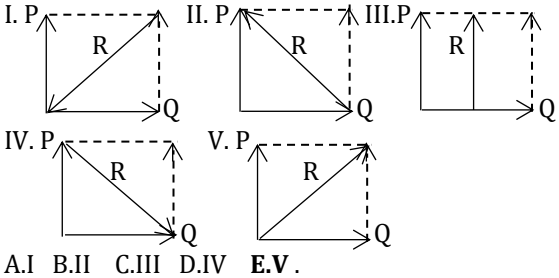
21. A man walks 8km North and then 5km in a direction 60°East of North.Calculate the total displacement of the man. **A.11.36km** B.12.36km C.13.00km D.14.36km.



R=Displacement,
θ=90° + 30° =120°,

$R^2 = 8^2+5^2-2\times8\times5\,cos120°$,
$R^2 = 64+25-80(-0.5)$,
$R^2 = 89+40 = 129$,
$R = \sqrt{129} = 11.36$km.

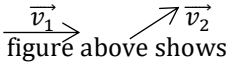
22. Which of the diagrams in the diagram vectors P and Q?

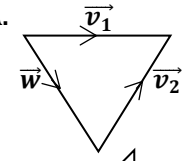


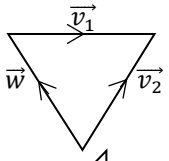
A.I B.II C.III D.IV **E.V** .

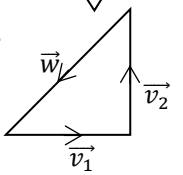
Using the tip to tail rule, in diagram V, the resultant R starts from the tail of Q and ends at the tip of P. The vector equation is : $R = P + Q$.

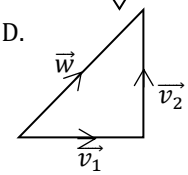
Demystified Series by Dr Timothy

23. 
The figure above shows two velocities $\vec{v_1}$ and $\vec{v_2}$. Which of the following diagrams correctly represents the vector difference: $\vec{w} = \vec{v_1} - \vec{v_2}$

A. 

B. 

C. 

D. 

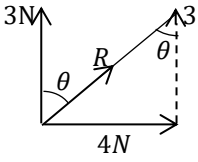
The vector difference equation given above is : $\vec{w} = \vec{v_1} - \vec{v_2}$, rearranging the vector equation we will have, $\vec{v_1} = \vec{w} + \vec{v_2}$. Hence, $\vec{v_1}$ is the resultant vector of $\vec{w}$ and $\vec{v_2}$. Using the tip to tail rule, the resultant vector $\vec{v_1}$ should start from the tail of $\vec{w}$ and end at the tip of $\vec{v_2}$ (option A).

- Option B – $\vec{v_2} = \vec{w} + \vec{v_1}$, $\vec{v_2}$ is the resultant of $\vec{w}$ and $\vec{v_1}$.

- Option C – $\vec{w} + \vec{v_1} + \vec{v_2} = 0$, they are equilibrant vectors.

- Option D – $\vec{w} = \vec{v_1} + \vec{v_2}$, $\vec{w}$ is the resultant of $\vec{v_1}$ and $\vec{v_2}$.

24. Two forces 3N and 4N act on a body in directions due to north and due east respectively. Calculate their equilibrant A.5N, 53°east of north **B.5N, 53°west of south** C.5N, 37°north of east D.7N, 37°west of north E.7N, 37°south of north



$R^2 = 3^2+4^2 = 9+16$,
$R^2 = 25$, $R = \sqrt{25}$,
$R = 5N$
$\tan\theta = 4/3 = 1.33$,
$\theta = \tan^{-1} 1.33 = 53°$

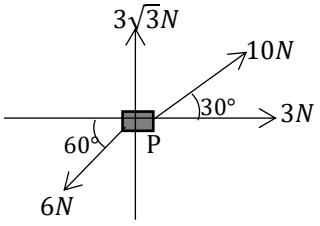
The resultant force is 5N, N53°E.

The equilibrant vector is equal in magnitude but opposite in direction to the resultant. Thus, the equilibrant force = 5N, S53°W or 53° West of South.

25. The component of vector X along the direction of vector Y is zero when A.X and Y are in opposite direction to each other B.X is at an angle of 45° to Y **C.X is perpendicular to Y** D.X and Y are parallel to each other .

The component each perpendicular vector (θ =90°) along the direction of the other is zero.

26.



The figure above shows 4 forces 3N, 10N, $3\sqrt{3}$N and 6N acting on a particle P, the resultant of the four forces is A.$10\sqrt{3}$N **B.10N** C.$5\sqrt{3}$N D.5N.

I. Resoving into the total vectors in either direction, with the vertical component R_Y taking sine and horizontal component R_X taking cosine. Also take note of the positive sign used for +Y and +X axes and negative sign used for −Y and −X axis.

$R_Y = 3\sqrt{3} + 10sin30° - 6sin60°$

$R_Y = 3\sqrt{3} + 10(\frac{1}{2}) - 6\frac{\sqrt{3}}{2} = 3\sqrt{3} + 5 - 3\sqrt{3} = 5N$

$R_X = 3 + 10cos30° - 6cos60° = 3 + 10(\frac{\sqrt{3}}{2}) - 6(\frac{1}{2})$

$R_X = 3 + 5\sqrt{3} - 3 = 5\sqrt{3}N$.

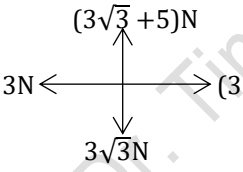
II. First resolve the oblique (inclined) forces into ther vertical and horizontal components.

Vertical component of 10N – $10sin30° = 5N$,

Horizontal component of 10N – $10cos30° = 5\sqrt{3}N$.

Vertical component of 6N – $6sin60° = 3\sqrt{3}N$,

Horizontal component of $6N$ – $6cos60° = 3N$,



The two $3\sqrt{3}$N in the vertical axis cancel out

The two $3N$ in the horizontal axis cancel out.

Net vertical force, $R_Y = 5N$. Net horizontal force, $R_X = 5\sqrt{3}N$.

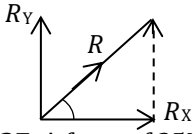
III. This is the most quick method but involves a calculator or knowledge of quadrant. The angles which meets the forces starting from the positive X-axis in an anticlockwise direction :

Vertical forces, $R_Y = 10sin30° + 3\sqrt{3}sin90° + 6sin(180+60)° + 3sin360°$,

$R_Y = 5 + 3\sqrt{3} + 6(-\frac{\sqrt{3}}{2}) + 0 = 5 + 3\sqrt{3} - 3\sqrt{3} = 5N$.

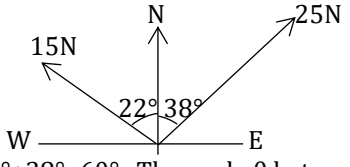
Horizontal forces, $R_X = 10cos30° + 3\sqrt{3}cos90° + 6cos(180+60)° + 3cos360°$,

$R_Y = 5\sqrt{3} + 0 + 6(-\frac{1}{2}) + 3 = 5\sqrt{3} - 3 + 3 = 5\sqrt{3}N$.

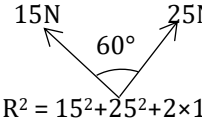


$R^2 = R_Y^2 + R_X^2 = 5^2 + (5\sqrt{3})^2$,
$R^2 = 25 + 25(3) = 25 + 75$,
$R^2 = 100$, $R = \sqrt{100} = 10N$.

27. A force of 25N inclined at N38°E and another force 15N inclined at N22°W act on an object. Calculate the magnitude of their resultant. A.20N B.25N C.30N **D.35N.**



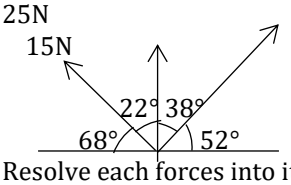
θ=22°+38°=60° . The angle θ between the 15N and 25N force is 60°.



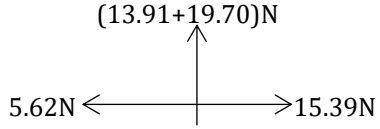
$R^2 = 15^2 + 25^2 + 2\times15\times25\cos 60°$,
$R^2 = 225 + 625 + 750(0.5) = 850 + 375$,
$R^2 = 1225$, $R = \sqrt{1225} = 35N$.

Demystified Series by Dr. Timothy

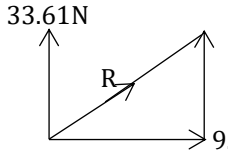
Alternatively : Resolution of multiple vectors :
25N



Resolve each forces into its vertical and horizontal component :
15N force ; $F_Y = 15\sin 68° = 13.91N$,
$F_X = 15\cos 68° = 5.62N$.
25N force ; $F_Y = 25\sin 52° = 19.70N$,
$F_X = 25\cos 52° = 15.39N$.

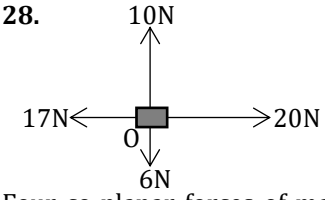


$\Sigma F_Y = 13.91 + 11.82 = 33.61N$,
$\Sigma F_X = 15.39 - 5.62 = 9.77N$,

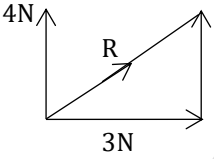


$R^2 = 33.61^2 + 9.77^2 = 1129.63 + 95.45$,
$R^2 = 1225.08$, $R = \sqrt{1225.08} = 35.00N$.

28.



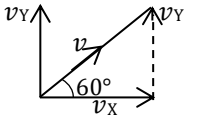
Four co-planar forces of magnitudes 10N, 17N, 6N and 20N cats at a point O as shown in the diagram above. Determine the magnitude of the resultant force A.53.0N B.21.0N C.7.0N **D.5.0N**.
$F_Y = 10 - 6 = 4N$. $F_X = 20 - 17 = 3N$.



$R^2 = 4^2 + 3^2 = 16 + 9$,
$R^2 = 25$, $R = \sqrt{25}$
$R = 5.0N$.

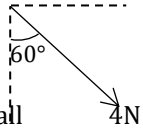
29. When a force R is resolved into two perpendicular components P and Q such that the angle between P and R is θ. Which of the following pairs represents the magnitude of P and Q respectively A.R$sin\theta$ and R$cos\theta$ **B.R$cos\theta$ and R$sin\theta$** C.R$cos\theta$ and R$tan\theta$ D.Rtanθ and R$sin\theta$ E.Rtanθ and Rcosθ.
The vector R is inclined at θ to the vertical. The vertical component P = Rcos θ while the horizontal component Q = Rsin θ.

30. A ball is moving at $18ms^{-1}$ in a direction inclined at 60° to the horizontal. The horizontal component of its velocity is A.$9\sqrt{3}ms^{-1}$ B.$6\sqrt{3}ms^{-1}$ C.$6ms^{-1}$ **D.$9ms^{-1}$**



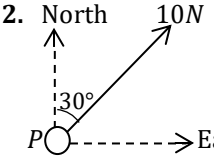
$v = 18ms^{-1}$,
$v_X = v\cos\theta = 18\cos 60°$,
$v_X = 9ms^{-1}$.
$v_Y = v\sin\theta$.

31. A boy pulls a nail from the wall with a string tied to the nail. The string is inclined at angle of 60° to the wall.If the tension in the string is 4N, what is the effective force used in pulling the nail? A.2N **B.$2\sqrt{3}$N** C.4N D.4N.



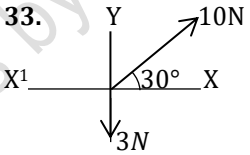
The effective force used in pulling or removing the nail, is the horizontal component of the applied force or tension in the string. $F_X = T\sin\theta$ (as the string is inclined to the wall i.e vertical). $F_X = 4\sin 60° = 2\sqrt{3}N$.

32.



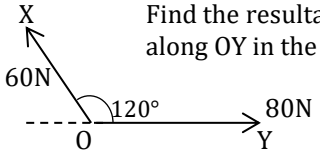
A body on the ground is acted on by a force of 10N at point P as shown in the diagram. What force is needed to stop the body from moving northward? A.5N in the direction of north B.5N in the direction south **C.$5\sqrt{3}$N in the direction of south** D.10N is the south-west direction.
The vertical(northward) component of the 10N force=$10\cos 30° = 10(\sqrt{3}/2) = 5\sqrt{3}$N. The force required to stop the body from moving northward must be acting in an opposite direction i.e southward and of equal magnitude $5\sqrt{3}$N.

33.

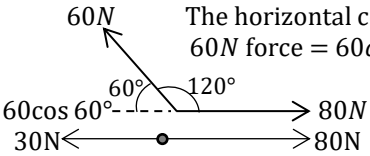


In the diagram above, the resultant force in Y-direction is **A.2N** B.5.7N C.8.0N D.11.7N .
The vertical component of the $10N = 10\sin 30° = 5N$. The resultant force in the Y-direction, $F = 5 - 3 = 2N$. (The $5N$ and $3N$ acts upward and downward respectively).

34.

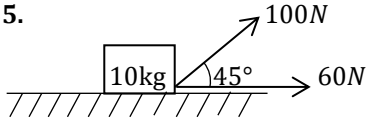


Find the resultant of the forces along OY in the diagram
A.20N B.28N C.30N **D.50N** E.110N.
In finding the resultant along the OY direction, we will have to figure out the angle which the 60N force makes with the horizontal.



The horizontal component of the $60N$ force $= 60\cos 60° = 30N$
The resultant force along OY = 80−30 = 50N.

35.



Two forces $100N$ and $60N$ act on an object of mass 10kg as illustrated in the diagram above.

Demystified Series by Dr Timothy

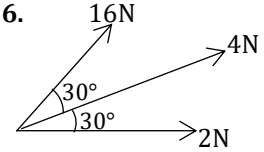
Neglecting friction, calculate the acceleration of the object A.4.0ms^{-2} **B.13.1ms^{-2}** C.16.0ms^{-2} D.22.6ms^{-2}.

The motion of the box is along the horizontal direction hence, the force causing the motion is the 60N and the horizontal component of the 100N force. Horizontal component of 100N= $100cos45° = 70.7$N.

Total force F causing motion=60+70.7=130.7N.

F = ma, a= $\frac{F}{m} = \frac{130.7}{10} = 13.07 \approx 13.1$ms^{-2}.

36.



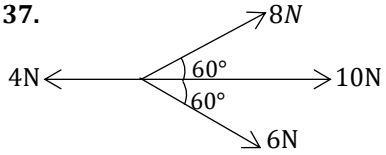
In the diagram, the resultant force along the x-axis is A. 11.46N **B.13.46N** C.19.29N D.14.02N.

Along the x-axis, the 16N force is 60° i.e (30° +30°) to the axis, the 4N force is 30° to the aixs. Resultant force along the x-axis is the total force along the horizontal.

$F_X = 16cos60° + 4cos30° + 2 = 16(0.5) + 4(\frac{\sqrt{3}}{2}) + 2$,

$F_X = 8 + 2\sqrt{3} + 2 = 10 + 2\sqrt{3} = 10 + 3.46 = 13.46$N.

37.

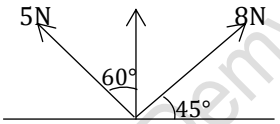


The diagram above shows forces 4N, 6N,10N and 8N which act at a point O in the direction indicated. The net horizontal force is A.7$\sqrt{3}$N B.17N C.$\sqrt{3}$N **D.13N.**

Resolving into the horizontal direction –

$F_X = 8cos60° + 6cos60° + 10 - 4 = 4 + 3 + 6 = 13N$ due east.

- Question Drill – Calculate the horizontal component of the forces in the diagram

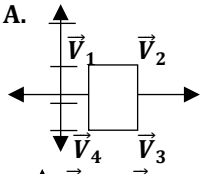


A.13.00N B.9.49N C.8.26N D.8.16N **E.1.33N**.

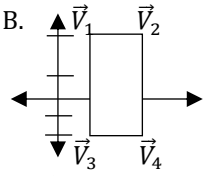
38. Which of the vector diagram below represents the following position vectors?

$\vec{V}_1 = \begin{pmatrix}1\\1\end{pmatrix}, \vec{V}_2 = \begin{pmatrix}2\\2\end{pmatrix}, \vec{V}_3 = \begin{pmatrix}2\\-2\end{pmatrix}, \vec{V}_4 = \begin{pmatrix}1\\-2\end{pmatrix}$

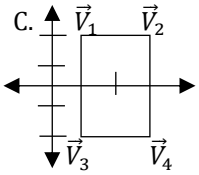
A.



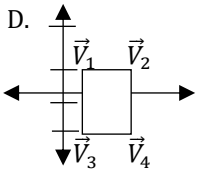
B.



C.



D.



Option A is the correct match for the position vectors.

39. The component vectors Q and R are (-3.0,5.5) and (9.2,4.4) respectively. Determine the components of Q+R A.(3.7,7.4) **B.(6.2,9.9)** C.(2.5,-4.8) D.(14.7,1.4).

Q+R = (−3.0+9.2, 5.5+4.4) = (6.2, 9.9).

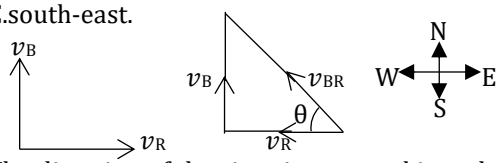
40. Two cars moving in the same direction have speeds of 100km/h and 130km/h. What is the velocity of the faster car as measured by an observer in a slower car? A.230km/h B.130km/h C.200km/h **D.30km/h.**

v_F=speed of the fast car=130km/h, v_S=speed of the slow car=100km/h, v_{FS}=speed of the faster car relative to the slower car=?, $v_{FS} = v_F - v_s$ (same direction). v_{FS}=130−100=30km/h.

41. The driver of a car moving with uniform speed of 40m/s observes a truck approaching in the opposite direction with a speed of 20m/s. Calculate the speed of the car relative to that of the truck. **A.60m/s** B.40m/s C.20m/s D.10m/s

v_C=speed of car=40m/s, v_T=speed of truck= 20m/s, v_{CT}=velocity of car relative to the truck=?, $v_{CT} = v_C + v_T$(approaching, opposite direction), v_{CT}= 40+20=60m/s.

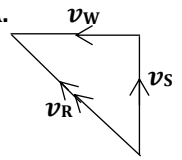
42. A boat man facing north wants to cross a flowing river to a point directly opposite his position at the other bank. If the river is flowing eastwards, in what direction should he row his boat? A.west B.east **C.north-west** D.north-east E.south-east.



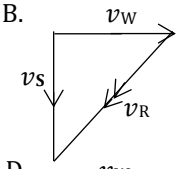
The direction of the river is reversed in order to obey the tip to tail rule, hence the direction θ of the resultant velocity of the boat v_{BR} relative to the river is north-west direction.

43. A ship moving notthward with speed v_s is acted upon by a wind blowing due west at a speed v_w. Which of the following diagrams correctly indicates the velocity v_R of the wind relative to the ship?

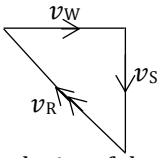
A.



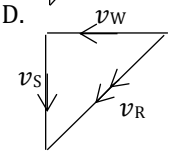
B.



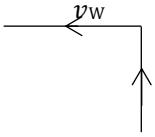
C.



D.



The velocity of the ship is v_S due north while the velocity of wind is due west.

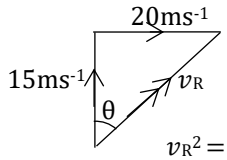


The resultant velocity of the ship relative to the wind v_{SW} or v_R is in the north-west direction. According to the tip-to-tail rule,

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v_R runs from the tail of v_S to the tip of v_W (option A).

44. A bird flying due north with a velocity of 15ms^{-1} shed a feather in an air current blowing at 20ms^{-1} due east. The feather will move with a velocity of A.25ms^{-1} east of south **B.25ms^{-1} east of north** C.25ms^{-1} west of south D.20ms^{-1} east E.15ms^{-1} north.



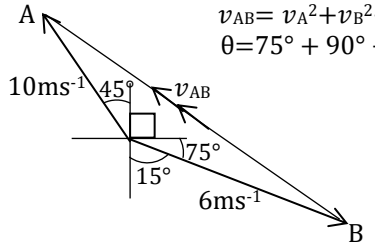
$v_R^2 = 15^2 + 20^2$,
$v_R^2 = 225 + 400$,
$v_R^2 = 625$, $v_R = \sqrt{625} = 25\text{ms}^{-1}$

$\tan\theta = \frac{20}{15} = \frac{4}{3} = 1.33$, $\theta = \tan^{-1} 1.33 = 53°$.

The velocity of the feather is 25ms^{-1} at N53°E or 53° east of north.

45. A particle A travels North west at 10ms^{-1} and another B travels 15° South of East at 6ms^{-1}. What is the velocity of A relative to B? **A.15.5ms^{-1}** B.5.7ms^{-1} C.8.7ms^{-1} D.16ms^{-1}.

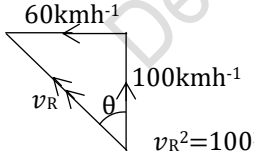
North West bearing or North East bearing or South West bearing or South East bearing is *N45°W* or *N45°E* or *S45°W* or *S45°E* respectively. Hence particle A travels 45° North of west.



$v_{AB} = v_A^2 + v_B^2 - 2v_A v_B cos\theta$,
$\theta = 75° + 90° + 45° = 210°$,

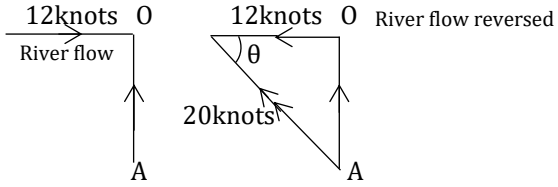
$v_{AB}^2 = 10^2 + 6^2 - 2\times10\times6cos210°$,
$v_{AB}^2 = 136 - (-103.92) = 239.92$,
$v_{AB} = \sqrt{239.92} = 15.48\text{ms}^{-1} = 15.5\text{ms}^{-1}$.

46. An aircraft attempts to fly due north at 100kmh^{-1}. If the wind blows against it from east to west at 60kmh^{-1}, its resultant velocity is A.117kmh^{-1}, N31°E B.127kmh^{-1}, N31°E
C.117kmh^{-1}, N31°W D.127kmh^{-1}, N31°W



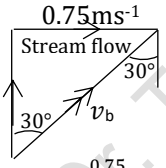
$v_R^2 = 100^2 + 60^2 = 10000 + 3600$,
$v_R^2 = 13600$, $v_R = \sqrt{13600} = 116.6 = 117\text{kmh}^{-1}$.
$\tan\theta = \frac{60}{100} = 0.6$, $\theta = \tan^{-1} 0.6 = 30.9 = 31°$.
The resultant velocity is 117kmh^{-1}, N31°W.

47. A tugboat is travelling from Asaba to Onitsha across the river Niger with a resultant speed of 20knots. If the river flows at 12knots, the direction of motion of the boat relative to the direction of water flow is A.36.87° **B.53.13°** C.90° D.136.87°.



$\cos\theta = \frac{12}{20} = 0.6$, $\theta = \cos^{-1} 0.6 = 53.13°$.

48. A stream is flowing at 0.75ms^{-1} and a boat heading perpendicularly to the stream landed at the opposite bank at an angle 30°. Calculate the velocity of the boat? **A. 1.50ms^{-1}** B.1.00ms^{-1} C.0.86ms^{-1} D.0.65ms^{-1}.



$\sin 30° = \frac{0.75}{v_b}$, $0.5 = \frac{0.75}{v_b}$, $v_b = \frac{0.75}{0.5} = 1.5\text{ms}^{-1}$.

Jamb past questions on measurements, scalars and vectors :
[1978/1,9,10,1980/11,27,1981/15,1982/3,28, 1983/2,39,1984/3,21,1985/1,7,1986/4,1987/1, 2,1988/2,4,5,1989/1,2,1990/1,2,1992/1,1993/ 1,2,4,1994/1,3,1995/1,3,4,1996/2,1997/1,1998 /1,2,3,12,2000/2,7,2001/6,9,2002/7,2003/4,12, 2004/18,2005/33,2006/20,32,2007/42,2008/1 ,2,4,7,2009/1,3,2011/5,2012/2,2013/4,2014/2, 3,4,2015/3,2016/16,23,2017/28].

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CHAPTER 2 – MOTION

● **Motion –** This is the change in position of a body with time. It is divided into two : kinematics and dynamics. **Kinematics** deals with the motion of a body neglecting the force which causes motion. **Dynamics** deals with the motion of a body and the force which causes the motion.

- Types of motion :

I. Translational motion – Motion of objects in a straight line. All points are moved in a parallel direction and with equal distance. e.g motion of a vehicle in a road.

II. Rotational motion – Motion of a body about the center of its axis e.g blade of a fan, wheels of moving cars.

- The centre of a coin as it rolls down an inclined plane undergoes translational motion only while other parts of the coin undergo rotational motion.

- A moving car is undergoing both rotational (the wheel of the tyre) and translational motion (the whole vehicle).

III. Oscillatory or vibrational motion – To and fro motion or a periodic motion of a body about a fixed point e.g simple harmonic motion, wave motion.

IV. Random motion – Haphazard or disordered movement of a body e.g motion of gass molecules, Brownian motion.

- Position x, y – It is the location of a body in a line, surface or space.

- Distance s – The separation between two points or separation between two different positions of a body in motion.

- Displacement s – The distance of a body measured in a specified direction or distance between the initial and final position of a body in motion.

- Speed v – The distance covered per unit time.

- Velocity v – The displacement of the body per unit time or speed measured in a specified direction.

- Acceleration a – The rate of change of velocity with time. The decrease in velocity with time is called **retardation** or **deceration (negative acceleration).**

● Rectilinear or Linear motion and Motion under gravity

- Linear motion – Motion of a body in a straight path in a constant direction.

- Uniform acceleration – Increase in velocity in equal time interval (constant acceleration) or constant rate of change change with time.

- Uniform velocity – velcotiy of a body when equal displacement is covered in equal time interval. It is referred to as **uniform motion.**

- Uniform speed v – speed of a body when it covers equal distances in equal time interval.

- Eqaution of linear motion :

1. $v = u \pm at$ **2.** $v^2 = u^2 \pm 2as$

3. $s = \left(\frac{v+u}{2}\right)t$ **4.** $s = ut \pm \frac{1}{2}at^2$

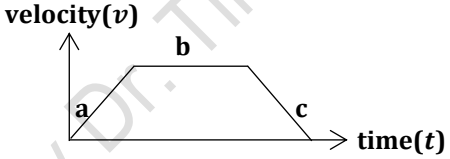
v=final velocity, u=initial velocity, s=distance, a=acceleration. **Average velocity** $= \left(\frac{v+u}{2}\right)$**.**

I. When a body moves from or at **rest or stationary,** $u = 0$**. II.** When a body is **brought to rest or stop,** $v = 0$**. III.** $a = -$**ve**, when **retarded or decelerated. IV.** If a body is moving with a **constant or uniform velocity** in a straight line, the **acceleration is zero (**$a = 0$**).**

- To convert speed or velocity from kilometer per hour(Km/hr) to meter per second(m/s) –

$v(\mathbf{m/s}) = v(\mathbf{km/hr}) \times \frac{5}{18}$**.**

- Velocity - time graph :



a.ascending line(moving upwards) represents constant acceleration, **b**.horizontal line represents constant or uniform velocity (acceleration is zero), **c.**descending line(moving downwards) represents constant deceleration.

- Acceleration of a velocity-time is given by the **slope of the ascending line**.

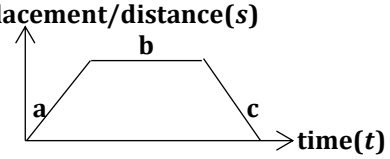
$a = \frac{change\ in\ velocity}{time\ interval} = \frac{v_2 - v_1}{t_2 - t_1}$**.**

- The slope of the horizontal line is zero i.e acceleration is zero and it shows **uniform motion(constant velocity).**

- The slope of the descending line is equal to the **deceleration or retardation (negative acceleration).**

- The total distance travelled in a velocity-time graph is the area of the figure under the graph

- Displacement(distance) - time graph :



The ascending or descending straight line **a and c** represents uniform motion i.e constant velocity, horizontal line **b** represents a **stationary body (body at rest)**.

- The slope of a displacement - time graph is **velocity** while the slope of a distance-time graph is **speed**.

- When a body is moving with a uniform acceleration in a straight line, the distance covered by a body in the nth second or last one

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second of motion is given by – $s = u + \frac{a}{2}(2t - 1)$, s=distance, u=initial velocity, a=acceleration, t=nth time.

- Instantaneous velocity v is the velocity of a body at a particular instant of time or the differential function of distance(x) and substituting the values of time(t).

- Intantaneous velocity of a moving vehicle is measured by a speedometer.

- Instantaneous acceleration a is the acceleration of a body at a particular instant of time or differential function of velocity (v) and substituting the value of time(t).

- Motion of a body under gravity – If a body is thrown upward it rises to a particular height and then falls back to the earth. All bodies moving under gravity in air, have the same acceleration i.e acceleration due to gravity, g=9.8ms^{-2} or ≈10ms^{-2}.

- Free fall – It is the vertical motion of a body downwards under the influence of gravity only.

- Two objects of different masses will fall to the ground at the **same time** from **same height neglecting air resistance** or **viscosity.**

- In the **presence of air resistance** or **viscosity** the **heavier mass experiences the greatest gravitational acceleration,** hence the **least air resistance** and **lands on the ground first before the lower mass.**

- Two bodies of diferrent masses will have the same acceleration (g=10m/s^2) at a height above the ground.

- Equation of motion under gravity :

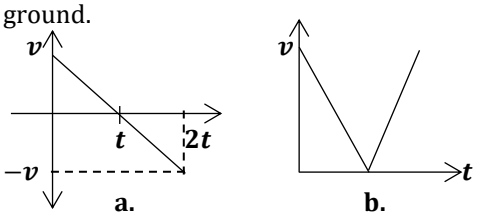
1. $v = u \pm gt$ **2.** $v^2 = u^2 \pm 2gh$

3. $h = ut \pm \frac{1}{2}gt^2$.

I. When a body falls from a height, u**=0**. **II.** At maximum height, v**=0**. **III.** g**=+ve** when a body is **falling** or **moving downwards** and g**=−ve** when a body is **moving or thrown upwards.**

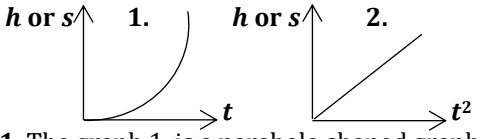
IV. Time taken to reach the maximum height = Time taken to fall from maximum height to the ground = t . **V.** Total time of flight=twice the time taken to reach maximum height=$2t$.

- Velocity-time graph of a body under gravity : The velcocity of a body decreases when thrown upward and increases when returning to the ground.



Graph a and b are both correct but **graph a** is **most appropriate** as the **direction of motion is considered(upward and downward motion)** but ignored in graph b.

- Height (displacement) - time 1, height (displacement) - square time graph 2, graph of a body falling under gravity i.e free fall is given below :



1. The graph 1, is a parabola shaped graph.

2. The slope of graph 2, equal to **half the acceleration ,** $\frac{1}{2}a$.

- Graph 1 and 2 is also valid for a body that accelerates uniformly from rest.

- Vertical projection of bodies from a height :

I. When a body is projected vertically upward from the top of a tower/cliff of height H with an initial velocity u, to the ground or foot of the tower (from a higher to a lower plane), at a time T, then – $H = \frac{1}{2}gT^2 - uT$, H=height of the tower/cliff, u=initial velocity, T=total time of flight. The landing or impact velocity v of the body at the base of the tower is given as –

$v^2 = u^2 + 2gH$.

II. When a body is projected vertically upward from the ground to the top of a tower or clif of height H with an initial velocity u, i.e lower to a higher plane, at a time T, then – $H = ut - \frac{1}{2}gT^2$. The landing or impact velocity v of the body at the top of the tower or cliff is given as –

$v^2 = u^2 - 2gH$.

Alternatively, time taken to reach maximum height H is given by – $t_{max} = \frac{u}{g}$. Time taken to return from maximum height to the ground is given by – $t_L = \sqrt{\frac{2(H_o + H)}{g}}$. The total time of flight T is given by – $T = t_{max} + t_L = \frac{u}{g} + \sqrt{\frac{2(H_o + H)}{g}}$.

Maximum height, $H_o = \frac{1}{2}g{t_{max}}^2 = \frac{u^2}{2g}$.

Examples :

1. Which of the following types of motion is oscillatory? I.a diving board when used by a diver II.the motion of the balance wheel of a wrist watch III.the motion of turn-table of a record player IV.the motion of the centre of ten kobo piece as it rolls down on an inclined plane V.the motion of a needle of a d.c ammeter into which a low a.c current is passed A.I,II and III **B.I and II** C.I,II and IV D.II,III and IV E.III,IV and V.

2. The motion of the prongs of a sounding tuning fork is A.Random B.Translational C.Rotational D.Vibrational **E.Vibrational and rotational**.

The prongs a sounding tuning fork vibrates when set in rotational motion.

A diving board, balance wheel of a wrist watch or clock, moving skin of a talking drum, swinging pendulum bob are examples of oscillatory or

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vibratory motion i.e to and fro.

3. The type of motion which a cylindrical drum rolling down an inclined plane undergoes are A.circular and translational **B.rotational and translational** C.rotational and circular D.rectilinear and circular.

4. A body can undergo the following tyoes of motion except A.random B.rotational C.translational **D.relative**.

Relative motion exist between two bodies and not a body only. Relative motion is not a type of motion.

5. A boy cycles continuously through a distance of 1.0km in 5minutes. Calculate his average speed. A.6.67ms^{-1} **B.3.33ms^{-1}** C.2.00ms^{-1} D.4.00ms^{-1}.

Distance, s=1.0km=1000m, time, t=5minutes= 5×60=300s, Average speed v=?,

$v = \frac{s}{t} = \frac{1000}{300} = 3.33\text{ms}^{-1}$.

6. The average speed of an object is determined by dividing the sum of the initial and final speeds by two, only when the acceleration is **A.uniform** B.changing C.increasing D.decreasing.

Average speed $\vec{v} = \frac{v+u}{2}$ and this holds only when the acceleration is constant or uniform.

7. A bird flies at 10ms^{-1} for 3s, 15ms^{-1} for 3s and 20ms^{-1} for 4s. Calculate the bird's average speed A.4.5ms^{-1} B.15.0ms^{-1} **C.15.5ms^{-1}** D.51.7ms^{-1}.

Distance s = velocity v × time t ,

Total distance = 10×3+15×3+20×4 ,

Total distance = 30+45+80 = 155m ,

Average speed $\vec{v} = \frac{Total\ distance}{Total\ time\ taken} = \frac{155}{3+3+4} = \frac{155}{10}$,

Average speed $\vec{v}$ = 15.5ms^{-1}.

8. A car starts travels with a constant velocity of 45km/hr for 20s. What distance does it cover in this time? **A.250m** B.500m C.750m D.650m

constant velocity=45km/hr=45× $\frac{5}{18}$=12.5ms^{-1},

t=20s, $v = \frac{s}{t}$, $s = vt = 12.5\times20 = 250$m.

9. A car accelerates uniformly from rest at 5ms^{-2}. Determine its speed after 10s **A.50.0ms^{-1}** B.25.0ms^{-1} C.2.0ms^{-1} D.0.5ms^{-1}.

u=0(from rest), a=5ms^{-2}, t=10s, v=?,

$v = u + at = 0 + 5\times10 = 50\text{ms}^{-1}$.

10. A body starts from rest and accelerates uniformly at 5ms^{-2} until it attains a velocity of 25ms^{-1}. Calculate the time taken to attain this velocity A.2.5s **B.5.0s** C.10.0s D.125.0s.

u=0(from rest), a=5ms^{-2}, v=25ms^{-1}, t=?,

$v = u + at$, $25 = 0 + 5t = 5t$, $t = \frac{25}{5} = 5.0$s.

11. A train has an initial velocity of 44ms^{-1} and an acceleration of -4ms^{-2}. Its velocity after 10sec is A.2ms^{-1} **B.4ms^{-1}** C.8ms^{-1} D.12ms^{-1} E.16ms^{-1}.

u=44ms^{-1}, a=-4ms^{-1}, t=10s, v=?,

$v = u + at = 44 + (-4\times10) = 44 - 40 = 4\text{ms}^{-1}$.

12. An object of mass 2kg moves with a uniform speed of 10ms^{-1} for 5s along a straight path. Determine the magnitude of its acceleration **A.0ms^{-2}** B.1ms^{-2} C.3ms^{-2} D.4ms^{-2}.

When a body is moving with a uniform/constant speed or velocity, the acceleration is zero as the velocity or speed is not changing with time.

13. A body accelerates from rest at the rate of 3ms^{-2} for 8s. Calculate the distance covered by the body during acceleration A.38m B.46m **C.96m** D.82m .

Initial velocity, u=0(as the body moves from rest), acceleration, a=3ms^{-2}, t=8s,

$s = ut + \frac{1}{2}at^2 = 0\times8 + \frac{1}{2}\times3\times8^2 = 1.5\times64 = 96$m.

14. A bus travelling at 15m/s accelerates uniformly at 4m/s^2. What is the distance covered in 10s? **A.350m** B.240m C.380m D.470m

u=15m/s, a=4m/s^2, t=10s, s=?,

$s = ut + \frac{1}{2}at^2 = 15\times10 + \frac{1}{2}\times4\times10^2 = 150 + 2\times100$

$s = 150 + 200 = 350$m.

15. A particle starts from rest and moves with a constant acceleration of 0.5ms^{-2}. Calculate the time taken by the particle to cover a distance of 25m. A.2.5s B.7.1s **C.10.0s** D.50.0s D.100.0s

u=0(from rest), a=0.5ms^{-2}, s=25m, t=?,

$s = ut + \frac{1}{2}at^2 = 0\times t + \frac{1}{2}at^2$, $s = \frac{1}{2}at^2$,

$t^2 = \frac{2s}{a} = \frac{2\times25}{0.5} = 100$, $t = \sqrt{100} = 10.0$s.

15. A body accelerates uniformly from rest at 2m/s^2. Calculate its velocity after travelling 9m. A.3ms^{-1} **B.6ms^{-1}** C.9ms^{-1} D.18ms^{-1}.

u=0(from rest), a=2ms^{-2}, s=9m, final velocity, v=? $v^2 = u^2 + 2as = 0^2 + 2\times2\times9 = 36$,

$v = \sqrt{36} = 6\text{ms}^{-1}$.

16. A boy moving with a velocity of 3m/s is brought to rest by a constant force, after travelling 15m. Calculate the retardation. A.0.15ms^{-2} B.0.45ms^{-2} **C.0.30ms^{-2}** D.0.75ms^{-2}.

u=3ms^{-1}, v=0(as the body is brought to rest), s=15m, a = −ve(retardation), $v^2 = u^2 - 2as$,

$0^2 = 3^2 - 2a\times15$, $30a = 9$, $a = \frac{9}{30} = 0.3\text{ms}^{-2}$.

17. A body which is uniformly retarded comes to rest in 10s after travelling a distance of 20m. Calculate its initial velocity A.0.5ms^{-1} B.2.0ms^{-1} **C.4.0ms^{-1}** D.20.0ms^{-1}

s=20m, t=10s, v=0(comes to rest), u=?,

$s = \left(\frac{v+u}{2}\right)t$, $20 = \left(\frac{0+u}{2}\right)10$, $2\times20 = 10u$,

$10u = 40$, $u = \frac{40}{10} = 4.0\text{ms}^{-1}$.

18. Two particles X and Y starting from rest cover the same distance. The acceleration of X is twice that of Y. The ratio of the time taken by X to that taken by Y is A.1/2 B.2 C.1/$\sqrt{2}$ D.$\sqrt{2}$ E.4

u_X=u_Y=0(starts from rest), s_X=s_Y(same distance), a_X=2a_Y, t_X:t_Y=?, $s = ut + \frac{1}{2}at^2$,

For X : $s_X = 0\times t_X + \frac{1}{2}a_X t_X^2 = \frac{1}{2}a_X t_X^2$,

For Y : $s_Y = 0\times t_Y + \frac{1}{2}a_Y t_Y^2 = \frac{1}{2}a_Y t_Y^2$,

$s_X = s_Y$, $\frac{1}{2}a_X t_X^2 = \frac{1}{2}a_Y t_Y^2$, $a_X t_X^2 = a_Y t_Y^2$,

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$\frac{t_X^2}{t_Y^2} = \frac{a_Y}{a_X} = \frac{a_Y}{2a_Y} = \frac{1}{2}$, $t_X{:}t_Y = \frac{t_X}{t_Y} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$ or $1:\sqrt{2}$.

19. A body starts from rest and moves with uniform acceleration of 6ms^{-2}. What distance does it cover in the third second? **A.15m** B.18m C.27m D.30m

u=0(from rest), a=6ms, t=3s(third second), s=?,

$s = u + \frac{a}{2}(2t - 1) = 0 + \frac{6}{2}(2\times3 - 1) = 3(6 - 1)$,

$s = 3(5) = 15$m.

20. If a car starts from rest and moves with a uniform acceleration of 10ms^{-1} for ten seconds, the distance it covers in the last one second of its motion is **A.95m** B.100m C.500m D.905m E.1000m.

u=0(from rest), a-10ms^{-2}, t=10s, distance covered in the last one second , s=?,

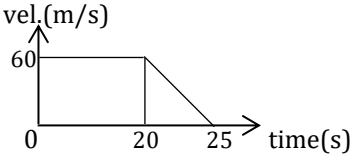
$s = u + \frac{a}{2}(2t - 1) = 0 + \frac{10}{2}(2\times10 - 1) = 5(20 - 1)$

$s = 5(19) = 95$m.

21. The area under a velocity-time graph represents A. speed B. acceleration C. moment **D. distance**.

The area under a velocity-time graph represents the total distance covered by the body during motion.

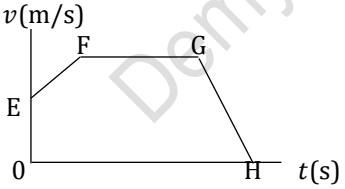
22.



From the velocity-time graph shown above, which of the following quantities cannot be determined? A.Deceleration B.Initial velocity C.Total distance travelled **D.Initial acceleration.**

The figure in the first phase of motion as seen in the graph is uniform(constant) velocity, the second phase of motion shows deceleration, the initial velocity is 60m/s i.e the body does not starts from rest. No acceleration is seen in the graph above, hence it cannot be determined.

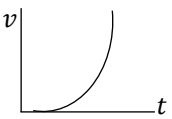
23.

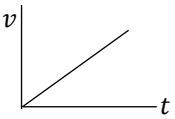


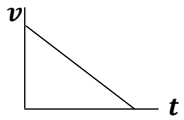
The diagram above shows a $V - T$ graph. The statement that is true about this motion is that the car A.has a constant speed between points E and F **B.has no acceleration between points F and G** C.decelerates between points F and G D.decelerates between points F and G.

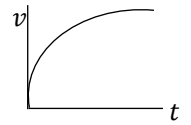
The car undergoes constant acceleration between E and F, uniform or constant velocity i.e no acceleration, between points F and G and constant deceleration between points G and H.

24. The velocity-time graph of a body moving in a straight line and decelerating uniformly to rest is represented by

A. 

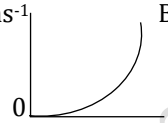
B. 

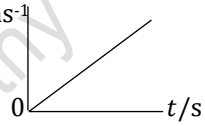
C. 

D. 

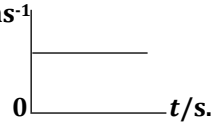
The graph in option C represents the velocity-time graph of a body decelerating or retarding uniformly in a straight line. The slope of the graph represents the deceleration or retardation.

25. Which of the velocity-time graph below correctly describes the motion of a body pulled by a force that is equal to the frictional force acting on it?

A. 

B. 

C. 

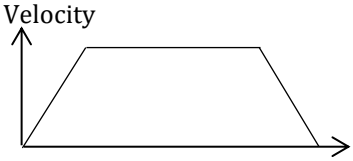
D. 

When the force pulling the body is equal to the frictional force(opposing force) acting on it, the the resultant force acting on the body will be equal to zero and the acceleration of the body will also be zero. Thus, the velocity becomes uniform or constant as acceleration is zero.

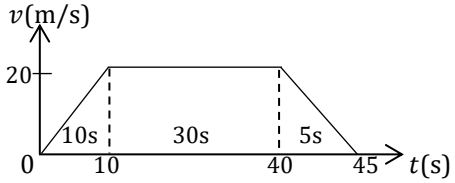
26. The slope of a straight line displacement-time graph indicates A. distance travelled **B. uniform velocity** C. uniform acceleration D. acceleration at an instant E. uniform speed.

The slope of a horizontal line displacement-time graph is zero(velocity is zero and the body is stationary).

27.



The diagram above shows the velocity-time graph of a truck which accelerates uniformly from rest at 2ms^{-2} for 10s,maintains a steady velocity for 30s and is then brought to rest in 5s under a uniform retardation.The total distance travelled is A.1500m B.1200m **C.750m** D.900m



The truck accelerated from rest at 2ms^{-2} for 10s.The final maximum velocity is calculated by:

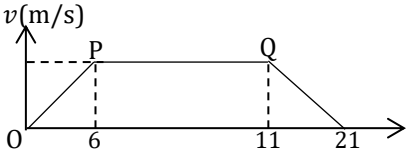
$v = u + at$ =0+2×10=20ms^{-1}

Total distance$s$ =Area of the trampezium = $\frac{1}{2}$(sum of parallel sides)height= $\frac{1}{2}(a + b)h$.

Demystified Series by Dr. Timothy

h in the graph represents maximum velocity.
$s = \frac{1}{2}(30+45)20 = 75\times10 = 750$m.

28. v(m/s)



The diagram above represents the velocity-time graph of a body in motion. The total distance travelled by the body is 195m. Calculate the acceleration of the body OP of the graph. **A.2.5ms^{-2}** B.5.0ms^{-2} C.3.5ms^{-2} D.7.5ms^{-2}.

$s = \frac{1}{2}(a+b)h$, s=195m, a=21s, b=11−6=5s, h=v=final maximum velocity, $195 = \frac{1}{2}(21+5)v$, $26v = 2\times195 = 390$, $v = \frac{390}{26} = 15$ms^{-1}.

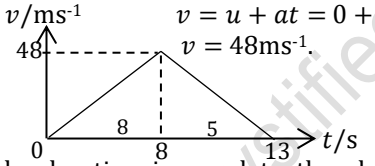
Acceleration of the body represented by the section OP = slope of OP $= \frac{15-0}{6-0} = 2.5$ms^{-2}.

29. A particle accelerates uniformly from rest at 6.0ms^{-2} for 8s and decelerates uniformly to rest in the next 5s. Determine the magnitude of the deceleration **A.9.6ms^{-2}** B.24.0ms^{-2} C.30.0ms^{-2} D.48.0ms^{-2}.

During the acceleration phase : u=0(from rest), a=6ms^{-2}, t=8s, v=?, $v = u + at = 0 + 6\times8$, $v = 48$ms^{-1}.

During the deceleration phase : final velocity of the acceleration phase=initial velocity of the decelration phase. u=48ms^{-1}, t=5s, v=0(comes to rest), a=−ve(deceleration), $v = u - at$,
$0 = 48 - 5a$, $5a = 48$, $a = \frac{48}{5} = 9.6$ms^{-2}.

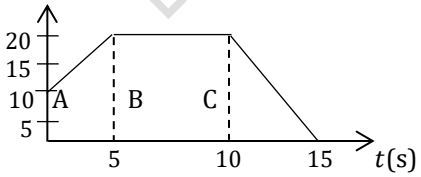
Alternatively, using graphical method :



$v = u + at = 0 + 6\times8$,
$v = 48$ms^{-1}.

The deceleration is equal to the slope of the descending line. $a = \frac{\Delta v}{\Delta t} = \frac{v_2-v_1}{t_2-t_1} = \frac{0-48}{13-8} = \frac{-48}{5}$,
$a = -9.6$ms^{-2}(negative acceleration).

30. v(m/s)

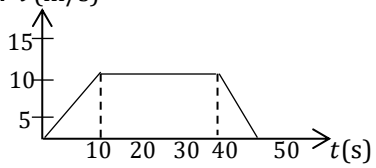


The diagram below illustrates the velocity-time graph of the motion of a graph of the motion of a body. Calculate the total distance covered by the body. A.125m B.250m **C.225m** D.500m.

Total distance covered by the body s = Area of the shape A + Area of the shape B + Area of the shape C .

$s = \frac{1}{2}(20+10)5 + 20\times(10-5) + \frac{1}{2}\times(15-10)\times20$
$s = [30\times2.5]+[20\times5]+[5\times10] = 75+100+50$
s =225m.

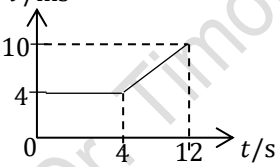
31. v(m/s)



The diagram above shows a velocity-time graph representing the motion of a car. Find the total distance covered during the acceleration and retardation periods of the motion. **A.75m** B.150m C.300m D.375m.

Distance covered during the acceleration period $= \frac{1}{2}\times(10-0)\times(10-0) = \frac{1}{2}\times10\times10 = 5\times10 = 50$m
Distance covered during the retardation period $= \frac{1}{2}\times(45-40)\times(10-0) = \frac{1}{2}\times5\times10 = 2.5\times10 = 25$m.
Distance covered in both acceleration and retardation period = 50+25 = 75m.

32. v/ms^{-1}



The diagram above illustrates the velocity-time graph of a body. Calculate the distance covered by the body during the motion. A.15m B.84m **C.72m** D.48m.

The distance covered by the body between 0 – 4s = 4×4 =16m. The distance covered by the body between 4 – 12s= $\frac{1}{2}\times(4+10)\times(12-4) = \frac{1}{2}\times14\times8$ = 7×8 = 56m. Total distance = 16+56 = 72m.

33. Two points on a velocity time graph have co-ordinates (5s,10ms^{-1}) and (20s,20ms^{-1}).Calculate the acceleration between the two points **A.0.67ms^{-2}** B.0.83ms^{-2} C.1.50ms^{-2} D.2.00ms^{-2}

v_2=20ms^{-1}, v_1=10ms^{-1}, t_2=20s, t_1=5s,
$a = \frac{change\ in\ velocity}{time\ interval} = \frac{v_2-v_1}{t_2-t_1} = \frac{20-10}{20-5} = \frac{10}{15} = 0.67$ms^{-2}.

34. A body moving with uniform acceleration has two points (5,15) and (20,60) on the velocity-time graph of its motion. Calculate the acceleration between the two points A. 0.25ms^{-2} **B. 3.00ms^{-2}** C. 4.00ms^{-2} D. 9.00ms^{-2} E. 16.00ms^{-2}.

v_2=60ms^{-1}, v_1=15ms^{-1}, t_2=20s, t_1=5s,
$a = \frac{change\ in\ velocity}{time\ interval} = \frac{v_2-v_1}{t_2-t_1} = \frac{60-15}{20-5} = \frac{45}{15} = 3.00$ms^{-2}.

35. The distance xm travelled by a particle in time, t seconds is described by the equation $x = 10+12t^2$. Find the average speed of the particle between time interval, t=2s and t=5s.
A.60ms^{-1} B.72ms^{-1} **C.84ms^{-1}** D.108ms^{-1}.

Differentiate the equation to get the derivative of the speed. Speed, $v = \frac{dx}{dt} = 0+2\times12t^{2-1} = 24t$,
Speed at t=2s, $v = 24t = 24(2) = 48$ms^{-1},
Speed at t=5s, $v = 24t = 24(5) = 120$ms^{-1}
Average speed $= \frac{48+120}{2} = \frac{168}{2} = 84$ms^{-1}.

Alternatively, first substitute the value of time to get the distance at that instant of time,

Demystified Series by Dr Timothy

Distance at t=2s, x =10+12(2²)=10+48
x =58m, Distance at t=5s, x =10+12(5²)
x =10+300=310m.
Average speed $\vec{v} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{310-58}{5-2} = \frac{252}{3}$,
Average speed=84ms^{-1}.

36. The velocity v of a particle in time t is given by the equation v =10+2t^2. Find the instantaneous acceleration after 5seconds. A.60ms^{-2} **B.20ms^{-2}** C.15ms^{-2} D.10ms^{-2}
Differentiathe the equation to get the derivative of the acceleration,
Acceleration, $a = \frac{dv}{dt} = 0+2\times 2t^{2-1} = 4t$,
Acceleration at t=5s, $a = 4t$ =4×5=20ms^{-2}.

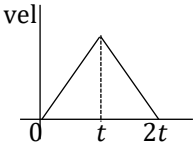
37. A body starts from rest and moves with constant acceleration. Which of the following quantities varies/vary linearly with the square of the time. I.velocity II.displacement III.momentum A.I **B.II** C.III D.I and II E.II and III.
s=displacement, $s = ut + \frac{1}{2}at^2 = 0\times t + \frac{1}{2}at^2$,
$s = \frac{1}{2}at^2$, $\frac{s}{t^2} = \frac{1}{2}a$, $s \propto t^2$, slope= $\frac{1}{2}a$ (half the acceleration).

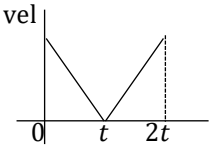
38. The distance travelled by a particle starting from rest is plotted against the square of the time elapsed from the commencement of motion. The resulting graph is linear. The slope of this graph is a measure of A.initial displacement B.initial velocity C.half the initial velocity D.acceleration **E.half the acceleration.**

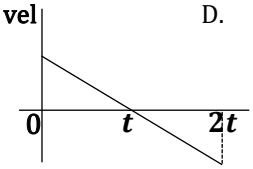
39. A body dropped from a certain height above the ground level, falls with a uniform A.speed B.velocity **C.acceleration** D.retardation.
All bodies under free fall, falls with a uniform acceleration of 9.8ms^{-2} ≈ 10ms^{-2}.

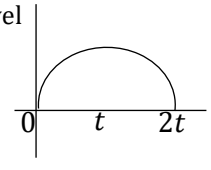
40. Two objects , one having three times the mass of the other, are dropped at the same time from a tall building.When they are above the ground, the two objects will have the same A.kinetic energy B.potential energy C.momentum **D.acceleration.**
Both bodies under free fall, will have the same acceleration(acceleration due to gravity). Their momentum, potential energy and kinetic energy are different as they all depend on the mass of the bodies.

41. A stone thrown vertically upwards returns to the ground. Which of the following figures represents the velocity-time graph?

A. 

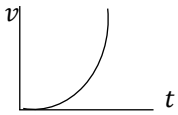
B. 

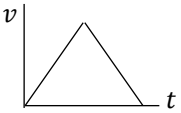
C. 

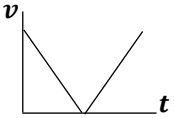
D. 

The velocity of a body thrown vertically upwards decreases as it moves upwards and increases when it is returning to the ground. Graph B and C are correct but C is most correct as it indicates the direction. Graphj A shows the velocity-time graph of a body in a lift or elevator.

42. Which of the following sketches represents the velocity-time graph of the motion of a stone projected vertically upwards and allowed to return to the point of projection?

A. 

B. 

C. 

D. 

Graph C is the correct representation of a body vertically and returning to the ground .

43. Two balls X and Y weighing 5g and 50kg respectively were thrown up vertically at the same time with a velocity of 100ms^{-1}. How will their positions be one second later A.X will be 500m ahead of Y B.X and Y will both be 500m from the point of throw C.X and Y will be 500m from each other D.Y will be 500m ahead of X .
Initial velocity of both bodies=100ms^{-1}, u_X=u_Y=100ms^{-1}, t_X=t_Y=1s, h_X=h_Y=h=?
$h = ut - \frac{1}{2}gt^2$(upward motion), g=10ms^{-2}
$h = 100\times 1 - \frac{1}{2}\times 10\times 1^2 = 100-5 = 95$m , both X and Y will be 95m from the point of throw after 1s. **None of the options are correct.**

44. A war-plane on a level flight releases a bomb X.Twenty seconds later, it releases a second bom Y. The bomb Y will reach the ground A.at the same time with X B.10s after X C.20s after X D.40s after X
The height(altitude) of the war-plane is the same for both bomb release, hence both bombs will spend the same time in air before reaching the target on the ground.Since, Y is released 20s after X, it will also reach the ground 20s after X reaches the ground.

45. A body falls from rest to the ground in 0.5s.Calculate the height from which it falls?(g=10ms^{-2}) A.0.125m B.0.5m C.1.0m **D.1.25m** E.5.0m
t=0.5s, u=0(falling from rest), g=10ms^{-2}, h=?,
$h = ut + \frac{1}{2}gt^2$(downward motion),
h =0× $t + \frac{1}{2}gt^2 = \frac{1}{2}gt^2 = \frac{1}{2}\times 10\times 0.5^2 = 5\times 0.25$,
h =1.25m.

46. A pal fruit dropped to the ground from the top of a tree 45m tall. How long does it take to reach the ground (g=10ms^{-2}) A.9s B.4.5s C.6s D.7.5s **E.3s.**
h=45m, u=0(falling from a height), g=10ms^{-2}, t=?
$h = ut + \frac{1}{2}gt^2$(downward motion),

Demystified Series by Dr. Timothy

$45 = 0\times t + \frac{1}{2}\times 10t^2$, $45 = 5t^2$, $t^2 = \frac{45}{5} = 9$,
$t = \sqrt{9} = 3$s.

47. A ball is thrown vertically upwards from the ground with an initial velocity of 20ms^{-1}. The maximum height reached by the ball is A.7.5m B.10.0m **C.20.0m** D.22.5m.
u=20ms^{-2}, v=0(at maximum height), g=10ms^{-2}, h=max.height, $v^2 = u^2 - 2gh$ (upward motion),
$0 = u^2 - 2gh$, $h = \frac{u^2}{2g} = \frac{20^2}{2\times 10} = \frac{400}{20} = 20$m.

48. A ball is dropped from a height of 45m above the ground. Calculate the velocity of the ball just before it strikes the ground (Neglect air resistance and take g as 10ms^{-2}). A.21.2ms^{-1} **B.30.0ms^{-1}** C.300.0ms^{-1} D.450.0ms^{-1} E.900.0ms^{-1}.
h=45m, u=0(falling from a height), g=10ms^{-2}, v=?
$v^2 = u^2 + 2gh$ (downward motion),
$v^2 = 0^2 + 2\times 10\times 45 = 900$, $v = \sqrt{900} = 30$ms^{-2}.

49. A body falls freely under gravity (g=9.8ms^{-2}) from a height of 40m ontop of a platform 0.8m above the ground.Its velocity on reaching the platform is A.784ms^{-1} B.80ms^{-1} C.78.4ms^{-1} D.39.2ms^{-1} **E.27.7ms^{-1}.**
u=0(falling from a height), g=9.8ms^{-2}, height of fall h=40−0.8=39.2m, v=?, $v^2 = u^2 + 2gh$,
$v^2 = 0^2 + 2gh = 2gh = 2\times 9.8\times 39.2$,
$v^2 = 768.32$, $v = \sqrt{768.32} = 27.7$ms^{-1}.

50. An object falls freely from a height of 25m onto the roof of a building 5m high. Calculate the velocity with which the object strikes the roof (g=10ms^{-2}) A.17.3ms^{-2} **B.20.0ms^{-1}** C.24,5ms^{-1} D.125.0ms^{-1} E.200.0ms^{-1}.
u=0(falling from a height), g=10ms^{-2}, height of fall h=25−5=20m, v=?, $v^2 = u^2 + 2gh$,
$v^2 = 0^2 + 2gh = 2gh = 2\times 10\times 20$,
$v^2 = 400$, $v = \sqrt{400} = 20.0$ms^{-1}.

51. A body is projected upwards with a velocity of 200ms^{-1}. How long does it take to return to the point of projection? (g=10ms^{-2}) A.2s B.4s C.10s **D.40s.**
u=200ms^{-2}, g=10ms^{-2}, time taken to return to the point of projection is the total time of flight=T.
Considering the upward motion: $v = u - gt$, at maximum height, v=0, hence $u = gt$, $t = \frac{u}{g}$, t=time taken to reach maximum height.
Total time of flight $T = 2t = \frac{2u}{g} = \frac{2\times 200}{10} = 40$s.

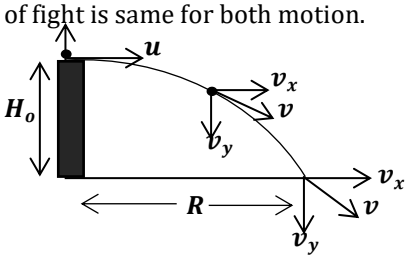
52. A ball is thrown vertically upwards from the top of a building with velocity of 10ms^{-1}. If it takes 4s for the ball to reach the ground level, the height of the building is A.5m **B.40m** C.45m D.50m.
u=10ms^{-1}, T=4s, g=10ms^{-2}, H_o=?
The ball is thrown from the top of a building to the ground i.e higher to a lower plane, the height of the tower is given by – $H_o = \frac{1}{2}gT^2 - uT$
$H_o = \frac{1}{2}\times 10\times 4^2 - 10\times 4 = 80 - 40 = 40$m.
Alternatively: $T = t_{max} + t_L = \frac{u}{g} + \sqrt{\frac{2(H_o+H)}{g}}$.
$H = \frac{u^2}{2g} = \frac{10^2}{2\times 10} = \frac{100}{20} = 5$m. $4 = \frac{10}{10} + \sqrt{\frac{2(H_o+5)}{10}}$,
$4 = 1 + \sqrt{\frac{(H_o+5)}{5}}$, $4-1 = \sqrt{\frac{(H_o+5)}{5}}$, $3 = \sqrt{\frac{(H_o+5)}{5}}$,
$3^2 = \frac{(H_o+5)}{5}$, $9\times 5 = H_o + 5$, $H_o = 45 - 5 = 40$m.

53. A ball was projected vertically from sea level into the air.It took a total of 5s to reach a 40m height cliff on its way back. Find the velocity at the top of the cliff A.13ms^{-1} B.15ms^{-1} **C.17ms^{-1}** D.19ms^{-1}.
H=40m, T=5s, g=10ms^{-2}, u=?, v=?,
The ball is thrown from the ground i.e sea level to the top of a cliff i.e lower to a higher plane, the height of the cliff is given by – $H_o = ut - \frac{1}{2}gT^2$,
$40 = u\times 5 - \frac{1}{2}\times 10\times 5^2$, $40 = 5u - 125$,
$5u = 40 + 125 = 165$, $u = \frac{165}{5} = 33$ms^{-1}.
The velocity at the top of the cliff v is the landing or impact velocity – $v^2 = u^2 - 2gH_o$
$v^2 = 33^2 - 2\times 10\times 40 = 1089 - 800$,
$v^2 = 289$, $v = \sqrt{289} = 17$ms^{-1}.

54. A body is projected vertically upwards with a speed of 10ms^{-1} from a point 2m above the ground. Calculate the total time taken for the body to reach the ground (g=10ms^{-2}) A.1.00s B.2.00s **C.2.18s** D.3.00s.
u=10ms^{-1}, H_o=2m, g=10ms^{-2}, T=?,
$H = \frac{u^2}{2g} = \frac{10^2}{2\times 10} = \frac{100}{20} = 5$m.
$T = t_{max} + t_L = \frac{u}{g} + \sqrt{\frac{2(H_o+H)}{g}} = \frac{10}{10} + \sqrt{\frac{2(2+5)}{10}}$,
$T = 1 + \sqrt{\frac{7}{5}} = 1 + \sqrt{1.4} = 1 + 1.18 = 2.18$s.
Alternatively: $H_o = \frac{1}{2}gT^2 - uT$.

● Projectile motion

- A projectile is a body which when released travels along a parabolic path called trajectory e.g a bullet fired from a gun, a ball rebouncing from a wall, an athlete doing a high jump e.t.c
- A projectile undergoes vertical and horizontal motion simultaneously.
1. Motion of an object projected horizontally from a height – Two independent motions i.e vertical motion(motion of free fall) and horizontal motion are combined here. The time of fight is same for both motion.



I.The horizontal component of the velocity is

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constant throughout the motion and its equal to the initial velocity of projection $\boldsymbol{v_x = u}$. Hence, **the horizontal motion is a uniform motion. The velocity just before reaching the ground is equal to its initial velocity $\boldsymbol{u}$.**

II. The vertical component of the velocity increases from $\boldsymbol{u_y}$**=0** at height H, to $\boldsymbol{v_y}$=$\boldsymbol{gt}$ on hitting the ground.

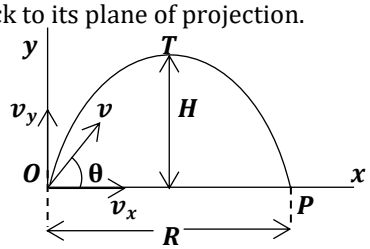
III. Horizontal acceleration is zero, a_x=0. Vertical acceleration is acceleration due to gravity, a_y=g.

IV. Height of projection , $\boldsymbol{H_o = \frac{1}{2}gT^2}$. Range or distance from the foot of platform/tower to the point of landing , $\boldsymbol{R = uT}$. T=time of flight.

- The range and height of progection are related by – $\boldsymbol{H_o = \frac{R^2 g}{2u^2}}$.

V.Instantaneous velocity $\boldsymbol{v}$ after time $t$is given by – $\boldsymbol{v = \sqrt{{v_y}^2 + {v_x}^2} = \sqrt{u^2 + (gT)^2}}$.

2. Motion of an object projected from the ground at an angle to the horizontal/vertical – The body is projected from the ground level at an angle θ, takes a parabolic path and returns back to its plane of projection.



I.Vertical acceleration is acceleration due to gravity, $a_y = \pm g$. i.e positive for downward motion(TP) and negative for upward motion (OT). Horizontal acceleration is zero, a_x =0.

II. The horizontal component of the velocity is constant throughout motion. $\boldsymbol{v_x = ucos\theta}$**.** The vertical component of the velocity changes with time. $\boldsymbol{v_y = usin\theta - gt}$.

III.Instantaneous velocity at time t is given by – $\boldsymbol{v = \sqrt{{v_y}^2 + {v_x}^2}}$. The direction of the instantaneous velocity at that point is – $\boldsymbol{\tan\beta = \frac{v_y}{v_x}}$.

At maximum height, $\boldsymbol{v_y}$ =0 and the velocity is equal horizontal component of the velocity, $\boldsymbol{v_x = vcos\theta}$.

IV. Time of flight $\boldsymbol{T}$ is the total time required for a projectile to return to its plane of projection. $\boldsymbol{T = \frac{2u\,sin\theta}{g}}$.

V.Time taken to reach maximum height $\boldsymbol{t_{max}}$. $\boldsymbol{t_{max} = \frac{u\,sin\theta}{g}}$. $\boldsymbol{T = 2t_{max}}$.

VI.Maximum height $\boldsymbol{H}$ is the highest vertical distance attained from the highest vertical point to the plane of projection. $\boldsymbol{H = \frac{u^2 sin^2\theta}{2g}}$.

- Maximum height is attained when the angle of projection $\boldsymbol{\theta = 90°}$.

- The maximum height attained by a body depends on : **i.**Angle of projection θ. **ii.**Initial velocity/speed projection u. **iii.**Aire resistance. **iv.**Acceleration due to gravity.

VII.Range $\boldsymbol{R}$ is the horizontal distance from the point of projection to the point where the projectile hits the projection plane.

$\boldsymbol{R = ucos\theta\ T = \frac{u^2 sin2\theta}{2g}}$.

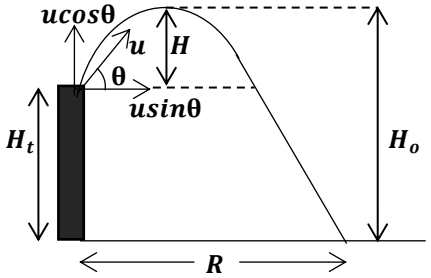
- Maximum range is attained when the angle of projection $\boldsymbol{\theta = 45°}$ or $\boldsymbol{2\theta = 90°}$..

- When two bodies A and B are projected with the same initial velocity, **if the angles are complementary i.e sum of their angle is equal to 90°** [25°& 65°,30°& 60°,75°& 15°] then, their **range is equal ($\boldsymbol{R_A = R_B}$)**.

- When the sum of their angles is greater or less than 90°, the body with the smaller angle would have the greatest range.

- The relationship between maximum height and range is – $\boldsymbol{\frac{H}{R} = \frac{tan\theta}{4}}$.

3. Motion of an object projected vertically upward from a tower or cliff of height $\boldsymbol{H_o}$ at an angle θ – The body is projected from a tower or cliff of height H_o at angle θ to the horizontal , the body initial moves upward to a maximum height H from the projection plane and then returns or falls to the ground (not the projection) plane after a period of time T i.e the total time of flight.



The body is projected from the top of a tower or cliff of height $\boldsymbol{H_o}$, the body first moved upward to a height $\boldsymbol{H}$ before returning to the ground.

I.Time taken to reach maximum height H is given by – $\boldsymbol{t_{max} = \frac{u\,sin\theta}{g}}$. Time taken to return from maximum height to the ground is given by –

$\boldsymbol{t_L = \sqrt{\frac{2(H_o + H)}{g}}}$.

II. The maximum height H above the projection plane is given by – $\boldsymbol{H = \frac{u^2 sin^2\theta}{2g}}$.

The maximum height above the ground or total height of fall of the body $\boldsymbol{H_t = H_o + H}$.

III. The total time of flight $\boldsymbol{T}$ is given by – $\boldsymbol{T = t_{max} + t_L}$ **or** $\boldsymbol{H_o = \frac{1}{2}gT^2 - usin\theta T}$.

IV. The range $\boldsymbol{R}$ is given by – $\boldsymbol{R = ucos\theta\ T}$.

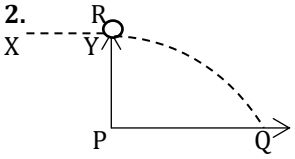
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Examples :

1. A is projected horizontally from the top of a tower. Neglecting air resistance, which of the following statements about its motion is/are correct? I.Vertical acceleration on the ball decreases as it falls II.The vertical force on the ball remains constant III.The vertical speed of ball increaseas as it falls A.I only B.II only C.I and II only **D.II and III only**.

In horizontal projection from a height : the horizontal speed is constany throughout the motion while the vertical speed increases as the body falls. The vertical acceleration(acceleration due to gravity) and the vertical force(weight of the body) remains constant.

2.



An object R leaves a platform XY with a horizontal velocity 7ms^{-1} and lands at Q. If it takes the same object 0.3s to fal freely from Y to P, calculate the distance PQ. (g=10ms^{-2}) A.7.30m B.2.55m **C.2.1m** D.1.65m E.0.45m

u=7ms^{-1}, T=0.3s, PQ=R=range, $R = uT$,

$R = 7\times0.3 = 2.1$m.

3. A ball is projected horizontally from the top of a hill with a velocity of 20ms^{-1}. If it reaches the ground 4 seconds later, what is the height of the hill? (g=10ms^{-2}) A.20m **B.80m** C.40m D.60m

u=20ms^{-1}, T=4s, H_o=?, $H_o = \frac{1}{2}gT^2$,

$H_o = \frac{1}{2}\times 10\times4^2 = 5\times16 = 80$m.

4. A body is projected horizontally from the top of a cliff 45m above the ground. If the body lands at a distance 30m from the foot of the cliff, calculate the speed of the projection (g=10ms^{-2}) **A.10ms^{-1}** B.15ms^{-1} C.20ms^{-1} D.30ms^{-1}

H_o=45m, R=30m, g=10ms^{-2}, u=? $H = \frac{R^2g}{2u^2}$,

$u^2 = \frac{R^2g}{2H_o} = \frac{30^2\times10}{2\times45} = 100$, $u = \sqrt{100}$ =10ms^{-1}

5. A stone is projected horizontally from the top of a tower with a speed of 10ms^{-1}.It lands at a horizontal distance of 20m from the foot of the tower.Calculate the height of the tower(g=10ms^{-2}) A.4.9m B.9.8m C.10.0m D.19.6m **E.20.0m**

R=20m, u=10ms^{-1}, g=10ms^{-2},H=?,

$H_o = \frac{R^2g}{2u^2} = \frac{20^2\times10}{2\times10^2} = 20$m.

6. A body is projected is projected horizontally with a velocity of 80ms^{-1} from the top of a tower. If it strikes the ground at a horizontal distance of 456.8m from the foot of the tower, find the **I.**total time of flight **II.**height of the tower and **III.**velocity 3seconds after projection.

u=80ms^{-1}, R=456.8m, t=total time of flight,

I. $R = uT$, $T = \frac{R}{u} = \frac{456.8}{80} = 5.71$ms^{-1}.

II. Height of projection, H_o=$\frac{1}{2}gt^2$=$\frac{1}{2}\times10\times5.71^2$,

H_o =163.0m

III. Instantaneous velocity after 3s, v=$\sqrt{v_y + v_x}$,

$v = \sqrt{u^2 + (gt)^2} = \sqrt{80^2 + (10\times3)^2}$,

$v = \sqrt{6400 + 900} = \sqrt{7300}$ =85.4ms^{-1}.

7. An object is projected with a velocity of 100ms^{-1} at an angle of 60° to the vertical. Calculate the taken by the object to reach the highest point(g=10ms^{-2}) **A.5.0s** B.8.7s C.10.0s D.17.3s E.20.0s

u=100ms^{-1}, α=angle of inclination to the vertical=60°, θ=angle of inclination to the horizontal=90−α =90−60=30°,t_{max}=?,

$t_{max} = \frac{u\sin\theta}{g} = \frac{100sin30°}{10} = 10\times0.5 = 5$s.

8. An object is projected with a velocity of 50ms^{-1} from the ground level at an angle α to the vertical. If the total time of flight of the projectile is 5s, what is the value of α? A.0° B.30° C.45° **D.60°** D.90° (g=10ms^{-2}).

u=50ms^{-1}, g=10ms^{-2}, α=angle of inclination to the vertical, θ=angle of inclination to the horizontal=90−α, T=5s,

$T = \frac{2u\sin\theta}{g}$, $T = \frac{2u\sin(90-\alpha)}{g}$, $5 = \frac{2\times50sin(90-\alpha)}{10}$,

$50 = 100sin(90-\alpha)$, $sin(90-\alpha) = \frac{50}{100} = 0.5$,

$(90-\alpha) = sin^{-1}0.5 = 30°$, α =90−30=60°.

OR $sin(90-\alpha) = cos\alpha$, $50 = 100cos\alpha$,

$cos\alpha = \frac{50}{100} = 0.5$, $\alpha = cos^{-1}0.5$ =60°.

9. A ball is thrown with an initial velocity u at angle 60° to the horizontal, if the time taken to reach the plane of projection is 8.66s and the horizontal range is S, determine the values of u and S (g=10ms^{-2}) **A.50.0ms^{-1} and 216.5m** B.25.0ms^{-1} and 125.0m C.20.0ms^{-1} and 108.3m D.10.0ms^{-1} and 64.3m

θ=60°, T=8.66s, u=?, R=S=?,

$T = \frac{2u\sin\theta}{g}$, $8.66 = \frac{2u\sin60°}{10}$, $86.6 = 2u\,sin60°$,

$86.6 = 2u\times0.866$, $u = \frac{86.6}{2\times0.866} = 50$ms^{-1}.

$R = \frac{u^2sin2\theta}{2g}$, $R = \frac{50^2sin2(60°)}{10}$

$R = S = \frac{2500sin120°}{10} = 250\times0.866 = 216.5$m.

OR, $R = ucos\theta\, T = 50\cos60°\times8.66 = 216.5$m.

10. An object is projected with a velocity of 80m/s at angle of 30° to the horizontal. The maximum height reached is A.20m **B.80m** C.160m D.320m.

u=80ms^{-1}, θ=30°, g=10ms^{-2}, H=?,

$H = \frac{u^2sin^2\theta}{2g} = \frac{80^2\times sin^230°}{2\times10} = \frac{6400\times(0.5)^2}{20}$,

$H = 320\times0.23.5 = 80$m.

11. A projectile attains its maximum height when the angle of projection is A.30° B.45° C.60° D.75° **E.90°**.

12. A missle is projected so as to attain its maximum range. Calculate the maximum attained if the initial velocity of projection is

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200ms^{-1} (g=10ms^{-2}) A.1500m **B.1000m** C.500m D.250m.

u=200ms^{-1}, g=10ms^{-2}, range is maximum at θ=45°, H=?, $H = \frac{u^2 sin^2\theta}{2g} = \frac{200^2 \times sin^2 45°}{2\times 10}$,

$H = \frac{40000\times(1/\sqrt{2})^2}{20} = 2000 \times \frac{1}{2} = 1000$m.

13. A shooter wants to fire a bullet in such a way that its horizontal range is three times its maximum height. At what angle should he fire the bullet to achieve this? A.64° **B.53°** C.45° D.30°.

R=3H, θ=?, $\frac{H}{R} = \frac{tan\theta}{4}$, $\frac{H}{3H} = \frac{tan\theta}{4}$, $\frac{1}{3} = \frac{\tan\theta}{4}$,

$\tan\theta = \frac{4}{3} = 1.33$, $\theta = \tan^{-1} 1.33 = 53°$.

14. At a birthday party the celebrant pops a corked fruit wine. If the cork shoots out of the bottle at an angle of 40° to the horizontal and travels a horizontal distance of 4.50m in 1.25s, calculate the initial speed of the cork A.4.2ms^{-1} **B.4.7ms^{-1}** C.5.6ms^{-1} D.7.1ms^{-1}.

θ=40°, R=4.50m, T=1.25s, u=?, R = $ucos\theta$ T ,

$4.50 = u\cos 40° \times 1.25$, $\frac{4.50}{1.25} = u\cos 40°$,

$3.6 = u\cos 40°$, $u = \frac{3.6}{\cos 40°} = \frac{3.6}{0.766} = 4.7$ms^{-1}.

15. At what angle to the horizontal must a noozle of a machine gun be kept when firing to obtain a maximum horizontal range for the bullets? A.0.0° B.22.5° C.30.0° **D.45.0°** D.90.0°.

16. Two bodies X and Y are projected on the same horizontal planes with the same initial speed but at angles 30° and 60° respectively to the horizontal. Neglecting air resistance, the ratio of the range of X to that of Y is **A.1:1** B.1:2 C.$\sqrt{3}$:1 C.1:$\sqrt{3}$.

The sum of the angles gives 90°, hence they have the same range and the ratio of their range, R_X:R_Y=1:1.

17. A stone Q is thrown with a velocity u at an angle of 75° to the horizontal. Another stone R is thrown with the same velocity u but at an angle of 15° to the horizontal. The ranges covered by the stones will be A.greater for Q B.greater for R **C.the same for Q and R** D.greater for the heavier stones.

18. A projectile is fired at an angle of 30° to the horiozontal with a velocity of 40ms^{-1}. Calculate the velocity attained after 1s. A.18.01ms^{-1} **B.36.02ms^{-1}** C.54.06ms^{-1} D.72.04ms^{-1}(g=10ms^{-2})

θ=30°, u=40ms^{-1}, g=10ms^{-2}, t=1s. The velocity attained after 1s is the instantaneous speed v. $v = \sqrt{v_y + v_x}$. v_x = $ucos\theta$ = 40cos 30°= 34.6ms^{-1}.

v_y=$usin\theta - gt$=40sin 30° −10×1=20−10=10ms^{-1}

$v = \sqrt{{v_y}^2 + {v_x}^2} = \sqrt{10^2 + 34.6^2}$,

$v = \sqrt{100 + 1197.16} = \sqrt{1297.16} = 36.02$ms^{-2}.

19. An object is projected from a height of 80m above the ground with a velocity of 60ms^{-1} at an angle of 30° to the horizontal. The time of flight is A.16s B.10s **C.8s** D.4s .

H_o=80m, u=60ms^{-1}, g=10ms^{-2}, θ=30°, T=?,

$H_o = \frac{1}{2}gT^2 - usin\theta T$,

$80 = \frac{1}{2}\times 10\times T^2 - 60sin30°T$,

$80 = 5T^2 - 30T$, $5T^2 - 30T - 80 = 0$,

$T^2 - 6T - 16 = 0$, $(T-8)(T-2) = 0$,

$T = 8$ or -2. Time of flight is 8s as time cannot be negative.

Alternatively , $T = t_{max} + t_L = \frac{u\,sin\theta}{g} + \sqrt{\frac{2(H_o+H)}{g}}$

$H = \frac{u^2 sin^2\theta}{2g} = \frac{60^2 sin^2 30°}{2\times 10} = \frac{3600\times(0.5)^2}{20} = 45$m.

$T = \frac{60\,sin30°}{10} + \sqrt{\frac{2(80+45)}{10}} = 3+5 = 8$s.

● Momentum and Newton Law's of motion

- Linear momentum – Linear momentum is the product of mass and velocity of a body. **It is the quantity of motion in a body. $P = mv$.**

- A body with a higher mass will move with a smaller speed or velocity as $m \propto \frac{1}{v}$.

- A body at rest has a linear momentum=0.

- Newton's Laws of motion :

1.Newton's First law or **Law of inertia –** It states that a body will continue in its state of rest or uniform motion in a straight line unless acted upon by a force. **Inertia** is the reluctance of a body to start moving while at rest or to stop moving while in uniform motion. **Inertia increases with mass.**

2.Newton's Second law – It states that the rate of change of momentum is directly proportional to the applied force in the direction of the force. **OR** It states that a body which is subjected to a net external force acquires an acceleration which is directly proportional to the force and inversely proportional to the mass of the body. i.e

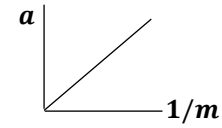
$F \propto \frac{\Delta P}{t}$, $F = k\frac{\Delta P}{t}$, k=1, $F = \frac{\Delta P}{t} = \frac{m(v-u)}{t}$.

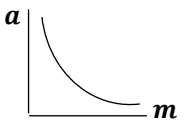
$a = \frac{(v-u)}{t}$, $F = ma$. OR $a \propto \frac{F}{m}$, $F = ma$.

mu=intial momentum, mv=final momentum, Change in momentum $\Delta P = mv - mu = m(v-u)$. On rebounds, change in momentum $\Delta P = mv + mu = m(v+u)$. Change in velocity $\Delta v = v - u$.

- On rebouncing, the direction of the velocity changes to opposite its initial direction

- Acceleration is inversely proportional to the mass of body at a constant force. $a \propto \frac{1}{m}$, (F=constant) and the graphical relationship is given below :





3.Newton's Third law – To every action there is an equal but opposite reaction i.e action and reaction are equal and opposite.

- Impulse is the product of force and its time of

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action. $\boldsymbol{I = Ft}$. $\boldsymbol{F = \frac{m(v-u)}{t}}$, $\boldsymbol{I = Ft = m(v-u)}$.

- Impulse is change in momentum (and not just momentum) while **force is rate of change of momentum.**

- The area of the the figure under a force-time graph is equal to impulse or change in momentum.

- At constant or uniform velocity, change in momentum and force is equal to zero while momentum is constant i.e initial momentum equals the final momentum.

- Coservation of linear momentum – It states that the total momentum before collision is equal to the total momentum after collision.

- Collision of objects – Elastic and inelastic;

1. Elastic collision – Momentum and kinetic energy is conserved due to conservative forces e.g motion of gas molecules and the container,two moving bodies, photon and electron,collision of billiard balls.

I. Total momentum before collision is equal to the total momentum after collision.

$\boldsymbol{m_1u_1+m_2u_2= m_1v_1+m_2v_2}$.

II. Total kinetic energy before collision is equal to the total kinetic energy after collision.

Before collision : $\boldsymbol{K.E_i=\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2}$,

After collision : $\boldsymbol{K.E_f = \frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2}$

$\boldsymbol{\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2}$.

III. Sum of the initial and final velocities of both colliding bodies are equal for bodies of equal masses or identical bodies.

$\boldsymbol{u_1+u_2= v_1+v_2}$.

2. Inelastic collision – Momentum is conserved but kinetic energy is not conserved i.e decreases due to dissipative forces e.g photoelectric effect(photon and atom), bullet and target, ball dropped onto a mud and bodies sticking together after collision.

I.Total momentum before collision=Total momentum after collision.

$\boldsymbol{m_1u_1+m_2u_2 =(m_1+m_2)V}$, V=common or composite velocity of both bodies when they stick sticked together.

II. Total kinetic energy before collision $\boldsymbol{K.E_i = \frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2}$. Total kinetic energy after collision $\boldsymbol{K.E_f = \frac{1}{2}(m_1+m_2)V^2}$.

- Kinetic energy lost $= \boldsymbol{K.E_i - K.E_f}$.

III. Ratio of the final $K.E_2$ to initial $K.E_1$ kinetic energy is – $\boldsymbol{\frac{K.E_f}{K.E_i} = \frac{(m_1+m_2)V^2}{m_1u_1^2+m_2u_2^2}}$.

- If both bodies are moving in the **same direction the velocity carry same sign** but carry **opposite signs when they are moving in opposite directions.**

- Coefficient of restitution e – It is the ratio of the final relative velocities to initial relative velocities. It is the comparison between the final and initial velocities of colliding bodies to the determine the elastic nature of the collision. $\boldsymbol{e = \frac{v_2-v_1}{u_2-u_1}}$. u_1 and $u_2$are intial velocities before collision, v_1 and v_2 are the final velocities after collision.

e =0 , $v_2 = v_1$ or $v_2 - v_1$ =0, for a perfectly inelastic collision.

e =1 , $v_2 - v_1 = u_2 - u_1$, for a perfectly elastic collision.

- Application of Newton's third law of motion:

I. Explosion Recoil of a gun – When a gun is fired, the bullets moves with a high velocity hits the target in minimum time possible due to its small mass and the gun move in opposite direction to that of the bullet with a small velocity due to its high mass.

- The recoil velocity of a gun is less than that of the bullet fired from it because of the difference in mass . Momentum of the gun=Momentum of bullet, $\boldsymbol{M_gV_g= m_bv_b}$. $\boldsymbol{m \propto 1/v}$ (a heavy body or body with higher mass a moves slowly or has a low velocity).

- The total momentum of the gun-bullet system is zero, $\boldsymbol{M_gV_g - m_bv_b = 0}$.

II. Jet and rocket propulsion – A jet engine or rocket burns and expels stream of hot gas at very high temperature.The rate of change in momentum of the exhausted or expelled gas is equal and opposite to the thrust(force) experienced by the rocket or air craft.

Thrust, $\boldsymbol{F = \frac{\Delta m}{t} \times v_{Exhaust}}$. $\frac{\Delta m}{t}$ = mass of gas ejected through the exhaust per second,Kgs^{-1}, $v_{Exhaust}$= velocity of the exhaust gas relative to the engine.

Examples :

1. Newton's first law of motion states that a body continues in its state of rest or uniform motion in a straight line unless A.The composition of the body changes B.The force of gravity changes C.There is action and reaction **D.Forces are impressed on the body** E.There is no changes in the direction of motion of the body.

Newton's first law is also called law of inertia.Inertia is the reluctance of a body to start moving while at rest or to stop moving while in uniform motion.Inertia increases with mass. Option C denotes "Newton's third law of motion".

2. A boy sits in a train moving with uniform speed in a straight track.If from his outstretched palm,he tossed a coin upwards,the coin will fall A.in front of his palm B.behind his palm C.beside his plam **D.into his palm**

A body moving with uniform or constant speed is like a body at rest(stationary body.Hence,the displaced coin returns to the position from which it was displaced.

Demystified Series by Dr Timothy

3. The quantity of motion of a body is its A.acceleration **B.momentum** C.displacement D.velocity.

4. An object of mass 5.0kg moves with a velocity of 10ms^{-1}. Calculate its momentum **A.50.0kgms^{-1}** B.15.0kgms^{-1} C.2.0kgms^{-1} D.0.5kgms^{-1}.

m=5.0kg, v-10ms^{-1}, P=?, $P = mv$,

$P = 5\times10 = 50$kgms^{-1}.

5. A constant force of 5N acts for 5s on a mass of 5kg initially at rest. Calculate the final momentum A.125kgms^{-1} **B.25kgms^{-1}** C.15kgms^{-1} D.5kgms^{-1} E.0kgms^{-1}

F=5N, t=5s, m=5kg, u=0(initially at rest),

$Ft = m(v-u) = mv - mu$, mu=0,

Final momentum, $mv = Ft = 5\times5 = 25$kgms^{-1}.

6. A resultant force of 15.00N acts for 6s on a body of mass 4kg. Calculate the change in momentum of the body within this period. A.3.75kgms^{-1} B.10.00kgms^{-1} C.22.50kgms^{-1} **D.90.00kgms^{-1}.**

m=4kg , F=15N, t=6s, ΔP=?,

Change in momentum $\Delta P = m(v-u) = Ft$

Change in momentum ΔP =15×6=90kgms^{-1}.

7. A body of mass 2kg moving vertically upwards has its initial velocity increased uniformly from 10ms^{-1} to 40ms^{-1} in 4s. Neglecting air resistance, calculate the upward vertical force **A.15N** B.20N C.35N D.45N.

m=2kg, u=10ms^{-1}, v=40ms^{-1}, t=4s, F=?,

$F = \frac{m(v-u)}{t} = \frac{2(40-10)}{4} = 15N$.

8. A force acts on a body for 0.5s changing its momentum from 16.0kgms^{-1} to 21.0kgms^{-1}, calculate the magnitude of the force A.42.0N B.37.0N C.32.0N **D.10.0N**.

t=0.5s, P_1=16kgms^{-1}, P_2=21kgms^{-1}, F=?,

$F = \frac{\Delta P}{t} = \frac{P_2-P_1}{t} = \frac{21-16}{0.5} = \frac{5}{0.5} = 10$N.

9. What change in velocity would be produced on a body of masses 4kg if a constant force of 6N acts on it for 2s? A.0.5ms^{-1} **B.3.0ms^{-1}** C.8.0ms^{-1} D.32.0ms^{-1} E.128.0ms^{-1}.

m=4kg, F=6N, t=2s, Δv=?, $Ft = m(v-u)$,

$\Delta v = (v-u) = \frac{Ft}{m} = \frac{6\times2}{4} = 3$ms^{-1}.

10. A ball of mass 0.1kg approaching a tennis player with a velocity of 10ms^{-1} is hit in the opposite direction with a velocity of 15ms^{-1}. If the time of impact between the racket and the ball is 0.01s, calculate the magnitude of the force with which the ball is hit. A.5000N B.500N **C.250N** D.50N.

m=0.1kg, u=10ms^{-1}, v=15ms^{-1}, t=0.01s, F=?,

$F = \frac{m(v-u)}{t}$, $F = \frac{m(v+u)}{t}$ (moving in opposite direction). $F = \frac{0.1(15+10)}{0.01}$ =10(25)=250N.

11. A ball of mass 5.0kg hits a smooth vertical wall normally with a speed of 2ms^{-1}. Determine the impulse experienced by the wall **A.20kgms^{-1}** B.10.0kgms^{-1} C.5.0kgms^{-1} D.2.5kgms^{-1}.

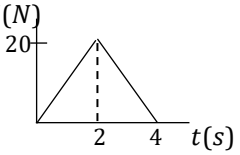
m=5kg, u=2ms^{-1}, v=2ms^{-1}(the ball rebounds on hitting the vertical wall, with the same speed), impulse, I=?, $I = Ft = m(v+u)$ i.e rebounds,

$I = 5(2+2) = 20$kgms^{-1} or 20Ns.

12. The area under a force-time graph represents A.change in kinetic energy **B.change in momentum** C.workdone D.change in internal energy.

- The area under a force-distance graph represents **workdone** which is equal to **change in kinetic energy.**

13.



The graph above illustrates a force F acting on a ball initially at rest on a smooth surface for a time t. Calculate the change in momentum of the ball in Ns after 4s. A.10Ns B.20Ns C.30Ns **D.40Ns.**

Area of a force-time(F-t) graph is equal to impulse or change in momentum. u=0(at rest), Initial momentum =mu=0.

Change in momentum = $\frac{1}{2}\times4\times20 = 40$Ns.

14. A rope is being used to pull a mass of 10kg vertically upward. Determine the tension in the rope if, starting from rest, the mass acquires a velocity of 4ms^{-1} in 8s. (g=10ms^{-2}) A.105N **B.95N** C.50N D.5N.

u=0(from rest), v=4ms^{-1}, m=10kg, g=10ms^{-2}, t=8s, $F = \frac{m(v-u)}{t} = \frac{10(4-0)}{8} = \frac{40}{8} = 5$N,

Tension $T = W - F$, weight $W = mg$,

$T = 10\times10 - 5 = 100-5 = 95$N.

15. A body of mass 4kg is accelerated from rest by a steady force of 9N. What is its speed when it has travelled a distance of 8m? A.2ms^{-1} B.4ms^{-1} **C.6ms^{-1}** D.8ms^{-1}

u=0, F=9N, s=8m, m=4kg, $v^2 = u^2+2as$,

v^2=2as (u^2=0), $a = \frac{v^2}{2s}$, $F = ma = \frac{mv^2}{2s}$

$9 = \frac{4v^2}{2\times8}$, $v^2 = \frac{9\times16}{4}$ =36, $v = \sqrt{36} = 6$ms^{-1}.

16. A car of mass 800kg moves from rest on a horizontal track and travels 60m in 20s with uniform acceleration. Assuming there were no frictional forces, calculate the accelerating force A.120N **B.240N** C.480N D.960N.

u=0, m=800kg, s=60m, t=20s, $s = ut + \frac{1}{2}at^2$

ut=0, $s = \frac{1}{2}at^2$, $a = \frac{2s}{t^2}$, $F = ma = \frac{2ms}{t^2}$,

$F = \frac{2\times800\times60}{20^2} = \frac{96000}{400} = 240$N.

17. A force of 16N applied to a 4kg block that is at rest on a smooth horizontal surface. What is the velocity of the block at t=5s? A.4ms^{-1} B.10ms^{-1} **C.20ms^{-1}** D.50ms^{-1} E.80ms^{-1}.

F=16N, m=4kg, u=0, t=5s, v=?, $v = u + at$, $v = at$ (u=0), $a = \frac{v}{t}$, $F = ma = \frac{mv}{t}$, $v = \frac{Ft}{m}$,

$v = \frac{16\times5}{4} = 20$ms^{-1}.

18. The driver in a motor car of which the total

Demystified Series by Dr Timothy

mass is 800kg and which is travelling at 20ms^{-1}, suddenly observes a stationary dog in his path 50m ahead. If the car brakes exert a force of 2000N, what will most likely happen? A.The car will be able to stop immediately the driver notices the dog **B.The car will stop 30m after hitting the dog** C.The car will stop 20m in front of the dog C.The driver will quickly reverse the car D.The car will stop 5m beyond the dog

F=2000N, u=20ms^{-1}, m=800kg, s=80m, The car comes to rest by the application of brakes, hence v=0 and a is deceleration or negative acceleration or retardation. $v^2= u^2-2as$,

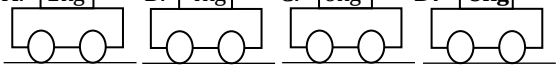
$0= u^2-2as$, $u^2=2as$, $a = \frac{u^2}{2s}$, $F = ma$,

$F = \frac{mu^2}{2s}$, $s = \frac{mu^2}{2F} = \frac{20^2\times800}{2\times2000}$ =80m.

The distance moved by the motor car when brakes is apllied is 80m, which is greaterthan the distance of the dog from the car i.e 50m. Hence, the car will stop 30m after hitting the dog.

19. Four identical trolleys are loaded with different masses and move along a straight road at the same speed. Which of the trolleys has the greatest inertia

A. 2kg B. 4kg C. 6kg **D. 8kg**



Inertia increases with mass thus, the trolley with a load of 8kg will have the greatest inertia.

20. The impact of two colliding objects is said to be perfectly elastic if **A.the total kinetic energy of the object after collision is equal to that before collision** B.the body coalesce and move together with a common speed after collision C.after collision,the kinetic energy of one object is equal to that of the other object D.it obeys law of conservation of linear momentum

Both elastic and inelaatic collisions obeys law of conservation of linear momentum i.e total momentum before and after collision are equal.Kinetic energy is conserved in elastic collision but not conserved in inelastic collision.

21. Two bodies each of mass m, moving with the same constant speed, v are approaching each other on a straight line. Their total kinetic energy E_K and momentum P are given by A.E_K=0, P =2mv B.E_K= $\frac{1}{2}mv^2$, $P = mv$ C.E_K=mv^2, $P = mv$ **D.E_K= mv^2, P =0.**

m_1=m_2=m, v_1=v_2=v , Kinetic energy is scalar quantity, hence independent of their direction. Total kinetic energy of both bodies,

E_K=2×$\frac{1}{2}mv^2$= mv^2. Momentum is a vector quantity hence, is dependent on their direction. Approaching bodies are moving in opposite direction. Total momentum $P = mv +(-mv)$=0.

22. A ball P of mass 0.25kg, loses one-third of its velocity when it makes a head on collision with an identical ball Q at rest. After the collision, Q moves off with a speed of 2ms^{-1} in the original direction of P. Calculate the initial velocity of P A.2ms^{-1} B.4ms^{-1} **C.6ms^{-1}** D.8ms^{-1}.

Elastic collision: m_1=m_2=m(as bot bodies are identical) u_1=u, loses $\frac{1}{3}u$, v_1=$u - \frac{1}{3}u$=$\frac{2}{3}u$, u_2=0(at rest), v_2=2ms^{-1}, Sum of the initial and final velocities of both colliding bodies are equal for bodies of equal masses or identical bodies. u_1+u_2= v_1+v_2 , $u + 0 = \frac{2}{3}u + 2$, $u - \frac{2}{3}u = 2$, $\frac{1}{3}u = 2$, $u = 2\times3 = 6$ms^{-1}.

23. A trolley of mass 4kg moving on a smooth horizontal platform with a speed of 1.0ms^{-1} collides perfectly with a stationary trolley of the same mass on the same platform. Calculate the total momentum of the two trolleys immediately after the collision. **A.4Ns** B.6Ns C.8Ns D.10Ns.

m_1=4kg, u_1=1.0ms^{-1}, m_2=4kg(same mass), u_2=0(stationary trolley),

Momentum before collision=Momentum after collision.

Momentum before collsion=m_1u_1+m_2u_2

Momentum after collision=m_1v_1+m_2v_2,

Total momentum after collision=Total momentum before collision.

Total momentum after collision = m_1u_1+m_2u_2 = 4×1+4×0 = 4Ns or 4kgms^{-1}.

24. A ball of mass 0.5kg moving at 10ms^{-1} collides with another ball of equal mass at rest. If the two balls move off together after impact, calculate their common velocity. A.2.5ms^{-1}**B.5.0ms^{-1}** C.7.5ms^{-1} D.10.0ms^{-1}.

m_1=0.5kg, u_1=10ms^{-1}, m_2=0.5kg, u_2=0, V=common velocity. Inelastic collision –

m_1u_1+m_2u_2=(m_1+m_2)V

0.5×10+0.5×0=(0.5+0.5)V, 5=(1)V,

V = 5ms^{-1}.

25. A ball of mass 6.00kg moving with a velocity of 10ms^{-1} collides with a 2.00kg ball moving in the opposite direction with a velocity of 5.0ms^{-1}. After the collision, the two balls coalesce and move in the same direction. Calculate the velocity of the composite body A.30.00ms^{-1} B.12.00ms^{-1} C.8.75ms^{-1} **D.6.25ms^{-1}** E.5.00ms^{-1}.

m_1=6kg, u_1=10ms^{-1}, m_2=2kg, u_2=−5ms^{-1} (opposite direction), V=?,

m_1u_1+m_2u_2=(m_1+m_2)V, 6×10+2(−5)=(6+2)V

$60-10 = 8V$, $50 = 8V$, $V = \frac{50}{8} = 6.25$ms^{-1}.

26. A particle of mass m initially at rest splits into two. If one of the particle of mass m_1 moves with velocity v_1, the second particle moves with velocity A.$\frac{m_1v_1}{m-m_1}$ **B.$\frac{m_1v_1}{m_1-m}$** C.$\frac{m_1v_1}{m}$ D.$\frac{mv}{m-m_1}$.

Initial momentum before the particle splits=0 as the body is initially at rest. mass of the first particle=m_1, velocity of first particle=v_1

mass of the second particle=m_2= $m - m_1$, velocity of second particle,v_2=?

Final momentum after the particle splits =

Demystified Series by Dr Timothy

$m_1v_1+m_2v_2= m_1v_1+(m-m_1)v_2$
Initial momentum=Final momentum,
$0= m_1v_1+(m-m_1)v_2, -(m-m_1)v_2= m_1v_1$
$(-m+m_1)v_2= m_1v_1$, $(m_1-m)v_2= m_1v_1$,
$v_2=\frac{m_1v_1}{m_1-m}$.

27. An arrow of mass 0.3kg is fired with a velocity of 100ms^{-1} into a wodden block of mass 0.7kg. Calculate the final $K.E$ after impact, given that the wooden block can freely move. **A.450J** B.500J C.600J D.700J.
m_1=0.3kg, u_1=10ms^{-1}, m_2=0.7kg, u_2=0, V=?
Momentum of arrow=Momentum of arrow and block. $m_1u_1=(m_1+m_2)V$,
$0.3\times100=(0.3+0.7)V$, $30=(1)V$, $V=30$ms^{-1}
$K.E$ after impact, $K.E_2=\frac{1}{2}(m_1+m_2)V^2$
$K.E_2=\frac{1}{2}(0.3+0.7)\times30^2=\frac{1}{2}\times1\times900=450J$.

28. Which of the following explains why the velocity of a recoiling gun is lower than that of the ejected bullet? I.The bullet must hit a target in the minimum time possible II.The mass of the bullet is much smaller than that of the gun III.The momentum of the gun-bullet system must be zero A.I B.II C.I and II D.II and III **E.I,II and III.**

29. A gun of mass 2.0kg fires a bullet of mass 1.6×10^{-2}kg due East. If the bullet leaves the nozzle of the gun with a velocity of 150ms^{-1}, what is the recoil velocity of the gun? A.150ms^{-1} due West B.1.2×10^{-4}ms^{-1} due West **C.1.2ms^{-1} due West** D.1.2ms^{-1} due East E.150ms^{-1} due East
M_g=2kg, V_g=? m_b=1.6×10^{-2}kg, v_b=150ms^{-1},
$M_gV_g= m_bv_b$, $2\times V_g=1.6\times10^{-2}\times150$,
$V_g=\frac{1.6\times10^{-2}\times150}{2}=1.2$ms^{-1}, The recoil velocity of the gun is due West as it moves in opposite direction to the propelled bullet.

30. 50g fuel is burnt each second in a rocket engine and ejected as a gas with a speed of 4,000ms^{-1}. The thrust on the rocket is A.20N B.80N **C.200N** D.500N.
Δm=50g=0.05kg, t=1s, $\frac{\Delta m}{t}=\frac{0.05}{1}$=0.05kgs^{-1},
$v_{Exhaust}$=4000ms^{-1}. Thrust, $F=\frac{\Delta m}{t}\times v_{Exhaust}$,
$F=0.05\times4000=200$N.

31. The exhaust gases ejected at a speed of 800ms^{-1} from a jet engine produces a thrust of 560N. The mass of gases ejected every second is **A.0.70kg** B.1.43kg C.7.00kg D.14.29kg.
$v_{Exhaust}$=800ms^{-1}, Thrust, F=560N, $\frac{\Delta m}{t}$=?
$F=\frac{\Delta m}{t}\times v_{Exhaust}$, $560=\frac{\Delta m}{t}\times800$,
$\frac{\Delta m}{t}=\frac{560}{800}=0.7$kg.

● Circular motion

- Motion of a body about a circular path with a constant or uniform speed.
- A body undergoing circular motion : **1.**Has a constant speed but varying direction. **2.**Have varying velocity, momentum, acceleration due to varying direction, though their magnitude is constant. **3.**Acceleration and force are directed towards the centre while velocity is tangent to the path. **4.**Acceleration and force are perpendicular to the velocity. **5.**No work is done as its movement is about a fixed point(no displacement). **6.**Kinetic energy is constant. **7.**Uniform circular motion is a conservative motion.

- **Centripetal or radial acceleration a_c**is the acceleration of a body moving a circular path and its directed towards the centre of the circular path. $\boldsymbol{a_c=\frac{v^2}{r}=\omega^2r=v\omega}$. v=uniform speed, r=radius of the circular path, ω=angular velocity or angular frequency
- **Centripetal or radial force F_c** is the inward force on a body moving in a circular path and is directed towards the centre of the path.

$\boldsymbol{F_c=ma_c=\frac{mv^2}{r}=m\omega^2r=mv\omega}$.

- **Centrifugal force** is the outwardly directed force. It is an equal and opposite force which balances the centripetal force to keep a body moving in a circular path. **The effect of centrifugal force is felt a driver in car as it turns a corner.**
- **The force holding the moon in its orbit around the earth is centripetal force.**
- **Angular velocity ω** is the angular distance covered with time. $\boldsymbol{\omega=\frac{\theta}{t}}$. Angular velocity and linear or tangential velocity are related by – $\boldsymbol{\omega=\frac{v}{r}}$, $\boldsymbol{v=\omega r}$. In one complete circle the angular distance θ= 360° or 2π radian.
- **Angular acceleration α**is the change in angular velocity with time. $\boldsymbol{\alpha=\frac{\omega_f-\omega_i}{t}}$. Angular acceleration and linear acceleration are related by – $\boldsymbol{a=\alpha r}$.
- $\boldsymbol{v=\frac{2\pi r}{T}}$, T=period or time taken for one complete circle.
- $\boldsymbol{\omega_f=\omega_i+\alpha t}$, $\boldsymbol{\omega_f{}^2=\omega_i{}^2+2\alpha\theta}$,
$\boldsymbol{\theta=\omega_it+\frac{1}{2}\alpha t^2}$. ω_f=final angular velocity (ω_f=0,when a body is brought to rest), ω_i=initial angular velocity (ω_i=0,when a body moves from rest).
- Tension in a string T on a body whirled round a horizontal circle is provided by the centripetal force and independent of the weight of the body.

Tension = Centripetal force. $\boldsymbol{T=F_c=\frac{mv^2}{r}}$.

- Tension in a string T on a body whirled round a vertical circle depends on the weight of the body and the centripetal force.

I. Tension maximum at the base or bottom of the circular path(θ =0°). $\boldsymbol{T_{max}=F_c+W}$,
$\boldsymbol{T_{max}=\frac{mv^2}{r}+mg}$.

II. Tension is minimum at the top or apex of the

Demystified Series by Dr Timothy

circular path(θ=90°). $\boldsymbol{T_{min} = F_c - W}$,
$\boldsymbol{T_{min} = \frac{mv^2}{r} - mg}$.

- A body moving undergoing circular motion will move along tangent to its path on removal of the force causing it to move.

Examples :

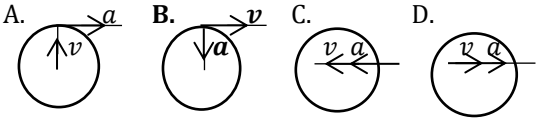
1. If a body moves with a constant speed and at the same time undergoes acceleration, its motion is said to be **A.circular** B.oscillatory C.rectilinear D.rotational

A body undergoing circular motion has a uniform or constant speed, but it direction changes i.e velocity changes. Since velocity changes it also has acceleration.

2. An object moves with uniform speed round a circle. Its acceleration has **A.constant magnitude and varying direction** B.constant magnitude and constant direction C.varying magnitude and constant direction D.varying magnitude and varying direction.

3. A body is whirled round a circular path at constant speed. Which of the following diagrams correctly indicates the direction of te body's velocity (v) and acceleration (a).

A. **B.** C. D.



The velocity and acceleration of a body undergoing circular motion are always perpendicular, with the acceleration towards the centre and the velocity tanget to the circular path as shown in option B.

4. A ball is swung in a horizontal circle of centre O with a constant speed. What is the direction of its acceleration? A.Away from O **B.Toward O** C.Along a tangent to the circle D.Along the circumference of the circle.

5. Which of the following statements is/are correct about a car moving round a circular track at a uniform speed I.its velocity is uniform II.it undergoes an acceleration III.the friction between its tyres and the ground provides the necessary centripetal force A.I B.II C.III **D.II and III** E.I,II and III.

A car moving in circle has a varying velocity and acceleration.The frictional force F_R between the road and tyres provide the necessary centripetal or **radial** force F_c , $F_R = F_c$, $\mu R = \frac{mv^2}{r}$.

6. Which of the following is true of a particle moving in a horizontal circle with constant angular velocity? **A.The energy is constant but the linear momentum varies** B.The linear momentum is constant but the energy varies C.Both energy and linear momentum are constant D.The speed and the linear momentum are both constant.

For a body undergoing circular motion : its kinetic energy and speed (do not depend on its direction) are constant while the linear momentum (depends on its direction) varies.

7. I.Its velocity is constant II.No work is done on the body III.It has constant acceleration directed away from the centre IV.The centripetal force is directed towards the centre. Which combination of the above is ture of a body moving with constant speed in a circular track? **A.II and IV** B.I and IV C.II and III D.II and III.

A body undergoing circular motion with constant speed has velocity and acceleration of constant magnitude but varying direction. The centripetal force and acceleration is directed towards the center. No work is done as the body moves about a fixed point i.e no distance or displacement.

8. A body of mass m moving round a circle of radius r with a uniform speed v experiences a centripetal force F. The workdone by the centripetal force on it is A.Fv **B.zero** C.Fmr D.Fr/m .

9. A man was moving round a circular path continuously for 4 miutes and covered the following distances in the times stated below :

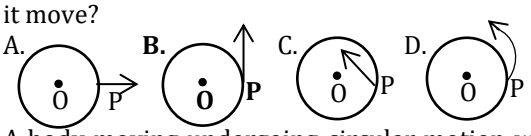
Distance(in meters) :	200	400	600	800
Tim(in miutes) :	1	2	3	4

Which of the following statements is correct about the motion of the man? **A.The man was moving with uniform speed but non-uniform velocity** B.The speed was changing constantly C.The velocity was uniform D.It was an instance of motion in which both speed and velocity are the same.

In circular motion, the speed is constant or uniform while the velocity is constantly changing (non-uniform) due to change in the direction of motion.

10. A ball attatched to one end of a string moves anti-clockwisely around a circle whose centre is O. If the string suddenly breaks when the ball is at point P, along which of the following paths will it move?

A. **B.** C. D.



A body moving undergoing circular motion will move along tangent to its path on removal of the force causing it to move.

11. A body moves along a circular path with uniform angular speed of 0.6rads^{-1} and at a constant speed of 3.0ms^{-1}.Calculate the acceleration of the body **A.0.2ms^{-2}** B.1.8ms^{-2} C.5.0ms^{-2} D.5.4ms^{-2} E.25.0ms^{-2}

ω=0.6rads^{-1}, v=3.0ms^{-1}, a_c=?,

$a_c = v\omega = 3 \times 0.6 = 1.8\text{ms}^{-2}$.

12. A body is rotating in a horizontal circle of radius 2.5m with an angular speed of 5rads^{-1}. Calculate the magnitude of the radial

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acceleration of the body. **A.62.5ms^{-2}** B.31.2ms^{-2} C.12.50ms^{-2} D.10.0ms^{-2}
ω=5rads^{-1}, r=2.5m, a_c=?, $a_c = \omega^2 r$,
$a_c = 5^2 \times 2.5 = 25 \times 2.5 = 62.5 ms^{-2}$.

13. An object of mass 2kg moves in a circular path of radius 0.5m with a speed of 1ms^{-1}. What is its angular velocity A.8rads^{-1} B.4rads^{-1} **C.2rads^{-1}** D.1rads^{-1} .
m=2kg, r=0.5, v=1ms^{-1}, ω=0.5rads^{-1},
$1 = \omega \times 0.5$, $\omega = \frac{1}{0.5} = 2 rads^{-1}$.

14. A stone is whirled round a circular path of radius 15cm, if the stone makes 30 0scillations in 10s, calculate the angular speed of the stone (π=3.14) A.9.42rads^{-1} B.12.5rads^{-1}**C.18.84rads^{-1}** D.62.80rads^{-1} E.141.30rads^{-1}.
1 oscillation = 1 cycle = 360°,
30 oscillations = 30 cycles = 30×360°
360° = 2π rad, hence 30 oscillations =30×2π =60π rad. Angular distance θ =60π rad, t=10s, ω=?, $\omega = \frac{\theta}{t} = \frac{60\pi}{10} = 6\pi = 6 \times 3.14$,
$\omega = 18.84 rads^{-1}$. Alternatively : $f = \frac{n}{t} = \frac{30}{10}$,
$f = 3Hz$. $\omega = 2\pi f = 2 \times 3.14 \times 3 = 18.84 rads^{-1}$.

15. If a wheel 1.2m in diameter rotates at one revolution per second, calculate the velocity of the wheel A.7.5ms^{-1} B.4.0ms^{-1} C.3.6ms^{-1} **D.3.8ms^{-1}.**
Diameter, d=1.2m, $r = \frac{d}{2} = \frac{1.2}{2}$ =0.6m, f=1revs^{-1},
1 revolution=1 complete cycle=360°=2π rad,
$\omega = 2\pi f = 2\pi$ rads^{-1}, $v = \omega r = 2\pi \times 0.6 = 1.2\pi$,
π=3.14, $v = 1.2 \times 3.14 = 3.8 ms^{-1}$.

16. A body of mass 12kg moves with a speed of 10ms^{-1} in a circular path of radius 5m. Calculate the magnitude of the force acting on the body A.24N B.30N C.125N **D.240N.**
m=12kg, v=10ms^{-1}, r=5m, F_c=?,
$F_c = \frac{mv^2}{r} = \frac{12 \times 10^2}{5} = 240N$.

17. A particle of mass 10^{-2}kg is fixed to the tip of a fan blade which rotates with angular velocity of 100rads^{-1}. If the radius of the blade is 0.2m, the centripetal force is A.2N **B.20N** C.200N D.400N.
m=10^{-2}kg, ω=100rads^{-1}, r=0.2m,
$F_c = m\omega^2 r = 10^{-2} \times 100^2 \times 0.2 = 20N$.

18. An object of mass 5kg moves round a circle of radius 6m. If the period of motion is πs, calculate the force towards the centre A.60N **B.120N** C.240N D.360N.
m=5kg, r=6m, Period, T= πs, F_c=?, $v = \frac{2\pi r}{T}$,
$v = \frac{2\pi \times 6}{\pi} = 12 ms^{-1}$, $F_C = \frac{mv^2}{r} = \frac{5 \times 12^2}{6} = 120N$.

19. An object of mass 6kg is whirled round a horizontal circle of radius 3m by a revolving string inclined to the vertical. Calculate the tension in the string if the uniform speed of the object is 9ms^{-1} (g=10ms^{-2}) A.27N B.54N **C.162N** D.173N.
m=6kg, r=3m, v=9ms^{-1}, Tension, T=?,
Tension in a horizontal circle is provided by the centripetal force. $T = F_c = \frac{mv^2}{r}$, $T = \frac{6 \times 9^2}{3} = 162N$.

20. An object of mass 15kg is whirled round a vertical circle of radius 3m with a constant speed of 4.5ms^{-1}. Find the maximum tension in the string (g=10ms^{-2}) A.101.3N B.179.5N C.202.5N **D.251.3N.**
Tension in a vertical circle is given by a combined effect of the centripetal force and weight. Maximum tension, $T_{max} = \frac{mv^2}{r} + mg$
$T_{max} = \frac{15 \times 4.5^2}{3} + 15 \times 10 = 101.25 + 150$,
$T_{max} = 251.25N = 251.3N$.

21. A mass attatched to a string is moving in a circular path. If the speed is doubled, the tension in the string will be A.doubles B.halved **C.four times as great** D.one-fourth as much.
$T = F_C = \frac{mv^2}{r}$, $T \propto v^2$, T_1=T, v_1=v, T_2=?, v_2=2v,
$\frac{T_1}{{v_1}^2} = \frac{T_2}{{v_2}^2}$, $\frac{T}{v^2} = \frac{T_2}{(2v)^2}$, $T_2 = \frac{T \times 4v^2}{v^2} = 4T$.
Alternatively, $T \propto v^2$, if speed is doubled then, $T \propto 2^2 = 4$, four times as great as the original value.

22. A stone is tied to one end of a light string and whirled in a vertical circle. The tension in the string is maximum when the A.stone is at the top of the circular path B.**stone is at the botton of the circular path** C.string lies in a horizontal direction D.stone is at any other point along the path .
Tension of a body moving in a vertical circle is maximum at the bottom or base of the circular path and minimum at the top or apex of the circular path.

● Simple Harmonic motion

- Simple Harmonic motion(SHM) is the periodic motion of a body or particle along a straight line such that the acceleration of the body is always directed towards a fixed point i.e centre of motion and is proportional to the displacement from that point e.g motion of a simple pendulum,vertical motion of a body suspended from a spring, vertical motion of a loaded test-tube in a liquid.
- The restoring force and acceleration is directly proportional to its displacement, $\boldsymbol{F \propto x}$, $\boldsymbol{a \propto x}$,
$\boldsymbol{F = -kx}$, $\boldsymbol{a = \omega^2 x}$.
- Amplitude $\boldsymbol{A}$ is the maximum displacement of a body from equilibrium position or rest.
- The amplitude and angular velocity/speed in SHM naturally remains constant.
- The distance between two extreme points in SHM equals twice the amplitude($2A$).
- Period $\boldsymbol{T}$ is the time taken for one complete oscillation. $\boldsymbol{T = \frac{t}{n}}$, t=time, n=number of oscillation.

Demystified Series by Dr. Timothy

- **Frequency or rate of oscillation** f is the number of oscillations in one seconds. $\boldsymbol{f = \frac{n}{t}}$.

- **Angular velocity or angular frequency** ω is the angular distance convered in time. $\boldsymbol{\omega = \frac{\theta}{t}}$, $\boldsymbol{\omega = \frac{2\pi}{T} = 2\pi f}$. T=period, f=frequency.

- The displacement of a body in SHM is represented as – $\boldsymbol{x = A cos\omega t = A cos 2\pi f t}$. x=displacement, A-amplitude, ω=angular velocity, f=frequency, t=time.

- **Relationship between SHM quantities :** **1.**Force and displacement are in phase also acceleration and displacement are in phase, [force and acceleration are maximum when displacement is maximum]. **2.**Amplitude and energy are in phase, $\boldsymbol{E \propto A^2}$ [energy is maximum when amplitude is maximum]. **3.**Velocity/energy and displacement are not in phase or antiphase i.e energy or velocity is maximum when displacement is zero or minimum. **4.**Amplitude and time are antiphase, $\boldsymbol{A \propto \frac{1}{t}}$, amplitude of an oscillating body decreases with time, due energy losses by friction.

- The velolcity of a body in SHM at a distance x from the equlibrium position : $\boldsymbol{v = \omega\sqrt{A^2 - x^2}}$

- Velocity and Kinetic energy is maximum at equilibrium position (minimum displacement), **displacement equals zero,** $\boldsymbol{x = 0}$. $\boldsymbol{v_{max} = \omega A = \omega r}$. Velocity and kinetic energy is minimum or zero at the extreme positions i.e maximum displacement.

- Acceleration and potential energy is maximum at the extreme positions or end of path, **displacement** $\boldsymbol{x}$ = **Amplitude** $\boldsymbol{A}$. $\boldsymbol{a_{max} = \omega^2 A = \omega^2 r}$. Acceleration and potential energy is minimum or zero at equilibrium position.

- **Total mechanical energy is constant.**

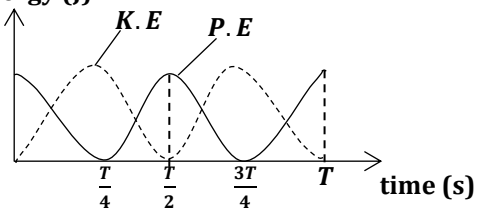
- Kinetic energy $\boldsymbol{K.E = \frac{1}{2}mv^2}$.

- Potential energy $\boldsymbol{P.E = \frac{1}{2}kx^2}$.

- The total workdone or energy stored or maximum $K.E$ or $P.E$ in SHM is : $\boldsymbol{E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2}$. k=force constant.

- **Graphical representation of the variation of potential energy and kinetic energy with time in SHM :**

Energy (J)



- **SHM in simple pendulums –**

1. Angular speed ω – $\boldsymbol{\omega = \sqrt{\frac{g}{L}} = \frac{2\pi}{T} = 2\pi f}$.

2. Period T – $\boldsymbol{T = 2\pi\sqrt{\frac{L}{g}} = \frac{2\pi}{\omega}}$.

3. Frequency or **rate of oscillation** f – $\boldsymbol{f = \frac{1}{2\pi}\sqrt{\frac{L}{g}} = \frac{\omega}{2\pi}}$.

- Period of oscillation and frequency(rate of oscillation) of a simple pendulum depends on **length and acceleration due to gravity only.**

1. $\boldsymbol{T \propto \sqrt{\frac{L}{g}}}$ thus, the period increases when the length increases and acceleration due to gravity decreases. The period of a simple pendulum in moon is greater than on earth ($g_{moon} < g_{earth}$).

- $\boldsymbol{T \propto \sqrt{L}}$, $\boldsymbol{\frac{T_1}{\sqrt{L_1}} = \frac{T_2}{\sqrt{L_2}}}$, $\boldsymbol{\frac{T_1}{T_2} = \frac{\sqrt{L_1}}{\sqrt{L_2}} = \sqrt{\frac{L_1}{L_2}}}$.

2. $\boldsymbol{f \propto \sqrt{\frac{g}{L}}}$ thus, the frequency increases when the length decreases and acceleration due to gravity increase.

- **g is greater at poles and less at the equator, hence a pendulum will have a lower period and higher frequency (swings faster) at the poles than the equator.**

- The time taken for a pendulum to pass the lowest point twice equal to the period.

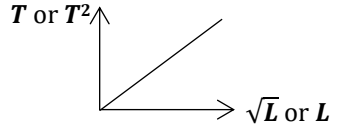
- **Simple pendula with the same length in the same location will have the same period.**

- If two pendula makes different oscillations in the same time interval, their period and number of oscillation is given by: $\mathbf{T_1 n_1 = T_2 n_2}$

- A graph of $\boldsymbol{T^2}$ **vs** $\boldsymbol{L}$ or $\boldsymbol{T}$ **vs** $\boldsymbol{\sqrt{L}}$ is a straight line graph with **slope** $= \boldsymbol{\frac{4\pi^2}{g}}$.

- A graph of $\boldsymbol{L}$ **vs** $\boldsymbol{T^2}$ or $\boldsymbol{\sqrt{L}}$ **vs** $\boldsymbol{T}$ is a straight line graph with **slope**$= \boldsymbol{\frac{g}{4\pi^2}}$.

- A graph of $\boldsymbol{log\ T}$ **vs** $\boldsymbol{log\ L}$ is a straight line graph.



NB : All these graphs originate from the origin (intercept, c =0).

- A simple pendulum as maximum $K.E$ and velocity at equilibrium position (lowest point) and maximum $P.E$ and height at extreme end (highest point).

- The maximum velocity of a simple pendulum is given by – $\boldsymbol{v_{max} = \sqrt{2gh}}$. h=vertical distance or height covered by a pendulum during oscillation.

- Potential or kinetic energy or workdone E in a simple pendulum is – $\boldsymbol{E = mgL(1 - cos\theta)}$. L=length of the pendulum, θ=angular displacement.

- **Energy loses in a simple pendulum is due to viscosity of the air.**

- **Assumptions made in simple pendulum experiment : 1.**The suspending string must be inextensible. **2.**The diameter(not mass) of the

Demystified Series by Dr Timothy

bulb must be kept small. **3.**The initial angle of oscillation should be small.

- SHM in mass-spring system –

1. Angular speed ω – $\boldsymbol{\omega = \sqrt{\frac{k}{m}} = \frac{2\pi}{T} = 2\pi f}$.

2. Period T – $\boldsymbol{T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{e}{g}} = \frac{2\pi}{\omega}}$. k=force constant or stiffness constant of spring, e=extension of the spring.

3. Frequency f – $\boldsymbol{f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{\omega}{2\pi}}$.

4.The tension in the spring T is given by the combined effect of the restoring force and weight of the body attached to the spring.

$\boldsymbol{T = F + W = ke + mg}$.

- SHM in a loaded test tube in a liquid –

1. Depth of immersion of the tube h –

$\boldsymbol{h = \frac{m}{\rho A}}$. m=mass of the test tube+mass of the liquid in the test tube, ρ=density of the liquid in which the test tube is placed, A=area of the test tube=πr^2.

2. Period T – $\boldsymbol{T = 2\pi\sqrt{\frac{m}{\rho Ag}}}$.

- Damped oscillation – This is the gradual decrease in the oscillation of an isolated body due decrease in amplitude of the body caused by energy loses due to friction.

- Forced vibration – It is the vibration of a body with a frequency of an external periodic force which is applied to keep the body oscillating rather than its own natural frequency. The amplitude and energy of forced oscillation depends on the difference between the frequency of the external periodic force and the natural frequency of the body i.e the lower the difference in their frequencies the larger the amplitude and energy , $\boldsymbol{A}$ **or** $\boldsymbol{E \propto \frac{1}{\Delta f}}$.

- Resonance – It occurs when the frequency of the external periodic force coincides or equals the natural frequency of the oscillating body. The frequency at which resonance occurs is called natural frequency. **The amplitude and energy is maximum at resonance.**

- Application of forced vibration and resonance –

I.Simple pendula with equal lengths vibrate with large amplitude while ones with different lengths vibrates with small swings.

II. Swings in play grounds for children with natural, greater and equal frequency.

III. Walking along a bridge. Walking in steps or uniformly sets the bridge to vibrate at the same frequency as that of the people walking or matching i.e resonance, hence causes the bridge to collapse.

Examples :

1. The motion of a body is simple harmonic if the A.acceleration is always directed towards a fixed point B.path of motion is a straight line **C.acceleration is directed towards a fixed point and proportional to its distance from the point** D.acceleration is constant and directed towards a fixed point.

A motion is said to be simple harmonic when the acceleration is directed towards a fixed point and is proportional to its distance from the point.

3. If a freely suspended object is pulled to one side and released, it oscillates about the point of suspension because the **A.aceleration is directly proportional to the displacement** B.motion is directed away from the equilibrium point C.acceleration is directly proportional to the square of the displacement D.velocity is minimum at equilibrium point

A body exhibiting SHM as its restoring force and displacement directly proportional to its displacement. Velocity is minimum at the extreme point and maximum and minimum at equilibrium point.

3. Which of the following statements about an object performing simple harmonic motion is correct. Its acceleration A.is minimum at the extreme ends B.is constant and directed towards a fixed point C.is zero when its is displaced from its equilibrium position **D.varies linearly with the displacement from the fixed point and is directed towards the fixed point.**

A body performing SHM has an acceleration which is proportional to its displacement and is directed towards the fixed point.The acceleration is maximum at the extreme ends and minimum at the equilibrium position. The acceleration is not constant.

4. The frequency of vibration of an oscillator is $4Hz$. If it makes 12 vibrations in t seconds, calculate the value of t A.0.33s **B.3.00s** C.3.33s C.24.00s E.48.00s

f=$4Hz$, n=12 vibrations, t=?, $f = \frac{n}{t}$,

$t = \frac{n}{f} = \frac{12}{4} = 3$s.

5. A simple harmonic oscillator has a period of 0.02s and amplitude of 0.25m. Calculate the speed in ms^{-1} at the centre of the oscillation A.1.25π**B.25.00π** C.100.00π D.400.00π

T=0.02s, A=0.25m, $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.02} = 100\pi rads^{-1}$

maximum velocity, $v_{max} = \omega A = 100\pi \times 0.25$,

$v_{max} = 25ms^{-1}$.

6. A particle executing a simple harmonic motion has a frequency of 50Hz and an amplitude of 8.0×10^{-3}m.Calculate the maximum velocity of its motion **A.2.51ms^{-1}** B.2.20ms^{-1} C.2.00ms^{-1} D.1.50ms^{-1}

f=$50HzA$=8.0×10^{-3}m, $\omega = 2\pi f$,

$v_{max} = \omega A = (2\pi f)A$,

$v_{max} = (2\pi \times 50)\times 8.0\times10^{-3} = 0.8\pi = 2.51ms^{-1}$.

7. A body moves in SHM between two points 20m

Demystified Series by Dr Timothy

on the straight line joining the points. If the angular speed of the body is 5rads^{-1}, calculate its speed when it is 6m from the centre of motion A.20ms^{-1} B.30ms^{-1} **C.40ms^{-1}** D.50ms^{-1}.
Distance between two extreme points is twice the amplitude, 20m= 2A, A=10m, ω=5rads^{-1}, x=dislplacement=6m, v=?, $v = \omega\sqrt{A^2 - x^2}$,
$v = 5\sqrt{10^2 - 6^2} = 5\sqrt{100 - 36} = 5\sqrt{64}$,
$v = 5\times8$=40ms^{-1}.

8. A body is esecuting simple harmonic motion as an angular speed of 2πrads^{-1}. Its period of oscillation is (π=3.14) A.0.05s **B.1.00s** C.2.00s D.3.14s D.6.28s
ω=2πrads^{-1}, T=?, $\omega = \frac{2\pi}{T}$, $T = \frac{2\pi}{\omega}$, $T = \frac{2\pi}{2\pi} = 1$s.

9. The angular velocity of a body undergoing simple harmonic motion with acceleration a and displacement x is A.$\frac{a}{x}$ **B.** $\sqrt{\frac{a}{x}}$ C.$\frac{x}{a}$ D.$\sqrt{\frac{x}{a}}$ E.$\frac{a^2}{x}$
$a = \omega^2 x$, $\omega^2 = \frac{a}{x}$, $\omega = \sqrt{\frac{a}{x}}$.

10. A block of mass 4.0kg causes a spiral spring to extend by 0.16m from its unstretched position. The block is removed and another body of mass 0.50kg is hung from the same spiral spring. If the spring is then stretched and released, what is the angular frequency of the subsequent motion? (g=10ms^{-2}) **A.10$\sqrt{5}$rads^{-1}** B.5$\sqrt{2}$rads^{-1} C.5rads^{-1} D.$\sqrt{5}$rads^{-1}.
First calculate for the force constant k of the spiral spring using the data given in the first sentence. m_1=4kg, e=0.16m, g=10ms^{-2},
$F = ke$, $F = W = m_1 g$, $\therefore m_1 g = ke$, $k = \frac{m_1 g}{e}$,
$k = \frac{4\times10}{0.16} = 250$Nm^{-1}. On placing the second body:
m_2=0.50kg, k=250Nm^{-1}, $\omega = \sqrt{\frac{k}{m_2}} = \sqrt{\frac{250}{0.5}}$
$\omega = \sqrt{500} = 10\sqrt{5}$rads^{-1}.

11. The bob of a simple pendulum of mass 0.025kg is displaced 0.1m from its equilibrium position.If the **angular frequency** is 4rads^{-1} and assuming SHM,calculate the energy of the system A.5×10^{-4}J **B.2×10^{-3}J** C.5×10^{-3}J D.7.9×10^{-3}J
m=0.025kg, A=0.1m, ω=4rads^{-1}, E=?,
$E = \frac{1}{2}KA^2 = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}\times0.025\times4^2\times0.1^2$,
$E = 2.0\times10^{-3}J$.

12. The amplitude of the motion of a body performing simple harmonic motion decreases with time because **A.frictional forces dissipate the energy of motion** B.the frequency of oscillation varies with time C.the period of oscillation varies with time D.energy is supplied by some external agencies
Amplitude decreases due energy loses caused by frictional forces i.e damping.

13. Which of the following with respect to a body performing simple harmonic motion are in phase? A.Displacement and velocity of the body **B.Displacement and force on the body** C.Velocity and acceleration D.Force acting on the body and acceleration.
Force and displacement are in phase. Velocity and acceleration are not in phase i.e antiphase. Velocity and displacement are not in phase i.e antiphase.

14. When the energy of the vibrating skin of a talking drum is decreased, the sound waves emanating from the drum would have a corresponding decrease in their A.wavelength **B.amplitude** C.quality D.speed.
Amplitude and energy are in phase i.e energy increases when amplitude increases.

15. Which of the following statements about the motion of a simple pendulum is true? A.It is a simple harmonic motion when the angle of displacement is large B.It passes the equilibrium position with minimum speed C.It possesses maximum kinetic energy at the extreme positions **D.It swings faster at the poles than at the equator**
Motion of a simple pendulum is simple harmonic motion when the angle of displacement is small.It passes the equilibrium position with maximum speed. Kinetic energy is minimum at the extreme positions but maximum at the equilibrium position. $f \propto \sqrt{g}$, f increases and swings faster as g increases. g is greater at poles and less at the equator, hence a pendulum will have a lower period and higher frequency (swings faster) at the poles than at the equator.

16. Which of the following processes will increase the rate of oscillation of a pendulum A.increasing the length of the pendulum **B.decreasing the length of the pendulum** C.increasing the mass of the pendulum D.decreasing the mass of the pendulum E.decreasing the amplitude of oscillation of the pendulum.
Rate of oscillation or frequency is inversely proportional to square root of the length of a pendulum i.e $f \propto \frac{1}{\sqrt{L}}$, the rate of oscillation or frequency of a pendulum increases when the length decreases.

17. Which of the following affects the period of a simple pendulum? I.mass of the pendulum II.length of the pendulum III.acceleration due to gravity A.I and II B.I and III **C.II and III** D.I,II and III.
The period T of a simple pendulum is affected by the length of pendulum and acceleration due to gravity only, $T = 2\pi\sqrt{\frac{L}{g}}$.

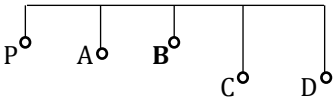
18. Two simple pendula A and B of equal lengths and masses 5g and 20g respectively are located in the same located in the same environment. The periods, T_A and T_B of their respective oscillations

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are related by the equations A.$T_A = \frac{1}{4}T_B$ B.$T_A = 4T_B$ C.$T_A = 5T_B$ **D.$T_A = T_B$.**

The period of simple pendulum is depends on the length of the pendulum and acceleration due to gravity(which depends on the location) Hence, two pendula of the same length and in the same location i.e g is constant, will have the their period will be the same, $T_A = T_B$.

19. The diagram below represents a system of simple pendula of different lengths. If *P* is set into oscillation while others are forced by it to oscillate, which of the pendula will resonate with P?



Simple pendula with equal lengths vibrate in resonance with large amplitude while ones with different lengths vibrates with small swings. Pendulum B has the same length *P* and will have the same natural frequency, hence vibrate with large swing due to resonance.

20. The length if a displaced pendulum bob which passes its lowest point twice every second is **A.0.25m** B.0.45m C.0.58m D.1.00m (g=10ms^{-2} , $\pi^2 \approx 10$)

The time taken for a pendulum to pass the lowest point twice equal to the Period. T=1s,

$T = 2\pi\sqrt{\frac{L}{g}}$, $T^2 = 4\pi^2\frac{L}{g}$, $1 = 4\pi^2\frac{L}{10}$, $L = \frac{10}{4\pi^2}$,

$L = \frac{10}{4\times 10} = 0.25$m.

21. The bob of a simple pendulum takes 0.25s to swing from its equilibrium position to one extreme end. Calculate its period. A.2.5s B.2s C.1.5s **D.1s .**

Period of a pendulum is the time taken for one complete oscillation. If it takes 0.25s to move from the equilibrium position to the extreme end, the period= $4\times 0.25 = 1$s.

22. The period of a simple pendulum X is 5s. What is the period of a simple pendulum Y which makes 50 vibrations in the same time it takes X to make 20 vibrations? A.12.5s B.2.5s **C.2.0s** D.1.2s.

T_1=5s, n_1=20 vibrations, T_2=?, n_2=50 vibrations,

Period, $T = \frac{t}{n}$. if two pendula vibrate at the same time hence, $T \propto \frac{1}{n}$, t=constant, $T_1 n_1 = T_2 n_2$,

$5\times 20 = T_2\times 50$, $T_2 = \frac{100}{50} = 2$s.

23. A boy times 30 oscillations of a certain pendulum thrown and obtained 1min 10s, 1min 12s and 1min 7s respectively. The mean peiod of oscillation of the pendulum is A.0.14s B.0.43s **C.2.32s** D.6.97s

n=30 oscillations, t_1=1min 10s = 60+10 = 80s, t_2=1min 12s=60+12=72s, t_3=1min 7s = 60+7 = 67s. Find the mean time and divide by the number of oscillations to get the mean period.

Mean time $t = \frac{t_1+t_2+t_3}{3} = \frac{80+72+67}{3} = \frac{219}{3} = 69.6$s,

Mean period $T = \frac{t}{n} = \frac{69.6}{30} = 2.32$s.

Alternatively, first calculate their respective period, then you find the mean period.

$T_1 = \frac{t_1}{n} = \frac{80}{30} = 2.33$s, $T_2 = \frac{t_2}{n} = \frac{72}{30} = 2.4$s,

$T_3 = \frac{t_3}{n} = \frac{67}{30} = 2.23$s, Mean period $T = \frac{T_1+T_2+T_3}{3}$,

$T = \frac{2.33+2.4+2.23}{3} = \frac{6.96}{3} = 2.32$s

24. A simple pendula 0.64m long has a period of 1.2s. Calculate the period of a similar pendulum 0.36m long in same location A.0.7s **B.0.9s** C.1.1s D.1.8s E.3.2s.

T_1=1.2s, L_1=0.64m, T_2=?, L_2=0.36m,

$T \propto \sqrt{L}$, $\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$ or $\frac{{T_1}^2}{{T_2}^2} = \frac{L_1}{L_2}$,

$\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$, $\frac{1.2}{T_2} = \sqrt{\frac{0.64}{0.36}}$, $\frac{1.2}{T_2} = \frac{0.8}{0.6}$,

$T_2 = \frac{0.6\times 1.2}{0.8} = 0.9$s.

25. If in a simple pendulum experiment the length of the inextensible string is increased by a factor of 4,its period is increased by a factor of A.4 B.$\pi/2$ C.1/4 D.2π **E.2.**

$T_1 = T$, $L_1 = L$, T_2=?, $L_2 = 4L$, $\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$, $\frac{T}{T_2} = \sqrt{\frac{L}{4L}}$

$\frac{T}{T_2} = \sqrt{\frac{1}{4}} = \frac{1}{2}$, $T_2 = 2T$, increased by a factor of 2.

Alternatively, $T \propto \sqrt{L}$, L is increased by a factor of 4, $T = \sqrt{4}$, T =2 i.e increased by a factor of 2.

26. A simple pendulum 0.6m long, has a period of 1.5s. What is the period of a similar pendulum 0.4m long in the same location? **A.1.5$\sqrt{(2/3)}$s** B.1.5$\sqrt{(3/2)}$s C.2.25s D.1.00s E.2.00s.

T_1=1.5s, L_1=0.6m, T_2=?, L_2=0.4m, $\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$,

$\frac{1.5}{T_2} = \sqrt{\frac{0.6}{0.4}}$, $T_2 = 1.5\sqrt{\frac{0.4}{0.6}} = 1.5\sqrt{\frac{2}{3}}$ s.

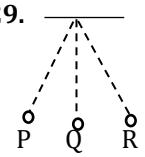
27. The period of a simple pendulum of length 80.0cm was found to have doubled when the length of the pendulum was increased by x. Calculate x A.26.7cm B.40cm C.160.0cm **D.240.0cm.**

$T_1 = T$, L_1=80cm, $T_2 = 2T$, $L_2 = (80+x)$cm,

$\frac{{T_1}^2}{{T_2}^2} = \frac{L_1}{L_2}$, $\frac{T^2}{(2T)^2} = \frac{80}{80+x}$, $\frac{1}{4} = \frac{80}{80+x}$,

$80+x = 4\times 80 = 320$, $x = 320-80 = 240$cm.

28. Which of the following assumptions is made in a simple pendulum experiment? The **A.suspending string is inextensible** B.bob has finite size C.bob has a definite mass D.initial angle of oscillation must be large.

29. 

The diagram below illustrates the equilibrium position Q and the extreme positions P and R of a swinging pendulum bob. Which of the following statements about the bob is not correct? A.the kinetic energy of

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the bob at P is equal to that at Q B.the bob possesses no kinetic energy at R **C.the bob at P has its maximum potential energy at Q** D.the mass of the bub at P is equal to its maximum at Q E.the mechanical energies at P, Q and R are equal. Potential energy or kinetic energy at P, Q and R is constant. The total mechanical energy at P, Q and R is also constant. The bob has maximum potential energy and minimum (or zero) kinetic energy at the extreme positions(P and R). The bob has maximum kinetic energy and minimum (or zero) potential energy at the equilibrium position (Q). The mass of the bob is constant i.e does not change. The bod has maximum speed at Q and minimum speed at P and R.

30. The mass of a loaded spiral spring oscillates vertically between two extreme positions P and R equidistant from the equilibrium position Q. Which of the following statements about the system is not correct? A.The momentum of the mass is maximum at Q B.The elastic potential energy of the spring is maximum at P **C.The kinetic energy is maximum at P** D.The total energy of the system is always constant.

Velocity and kinetic energy is maximum at the equilibrium position Q and minimum at the equilibrium positions P and R. Momentum is proportional to the velocity ($P \propto v$, m=constant) thus, the momemtum is maximum at Q and minimum at P and R. Potential energy is maximum at the extreme positions P and R. The total mechanical energy of the system is always constant.

31. A swinging pendulum between the rest position and its maximum displacement possesses A.kinetic energy only B.potential energy only C.gravitational energy only **D.both kinetic and potential energy.**

32. A bob of weigt $0.1N$ hangs on a massless string of length 50cm. A variable horizontal force which increases from zero is applied to pull the bob until the string makes an angle of 60° with the vertical. The workdone is **A.0.025J** B.0.050J C.0.500J D.0.250J.

$W=mg=0.1N$, L=50cm=0.5m, θ=60°, E=?,

The workdone or energy stored in a simple pendulum , $E = mgL(1 - cos\theta)$

$E = 0.1\times0.5(1-cos60°) = 0.05(1-0.5)$,

$E = 0.05\times0.5 = 0.025J$.

33. A body of mass 40g is suspended from the end of spiral spring whose force constant is 0.8Nm^{-1}. If it executes simple harmonic motion and has an amplitude of 0.3m, calculate the total energy of the system A.0.012J **B.0.036J** C.0.072J D.0.192J E.0.240J.

m=40g, k=0.8Nm^{-1}, A=0.3m, E=?,

$E = \frac{1}{2}kA^2 = \frac{1}{2}\times0.8\times0.3^2 = 0.4\times0.09 = 0.036$J.

34. The length of a simple pendulum is plotted against the square of the corresponding period. The slope of the graph is equal to A.$\frac{g^2}{4\pi}$ B.$\frac{\pi}{4g^2}$ C.$\frac{4\pi^2}{g}$ **D.$\frac{g}{4\pi^2}$.**

$T = 2\pi\sqrt{\frac{L}{g}}$, $T^2 = 4\pi^2\frac{L}{g}$, $L = \left(\frac{g}{4\pi^2}\right)T^2$, $L \propto T^2$.

The slope of the graph is $\frac{g}{4\pi^2}$.

35. An object of weight 120N vibrates with a period of 4.0s when hung from a spring . Calculate the force per unit length of the spring [g=10ms^{-2}, π=3.142] A.7.4Nm^{-1} B.9.9Nm^{-1} C.14.8Nm^{-1} **D.29.6Nm^{-1}.**

F=W=120N, T=4.0s, g=10ms^{-2}, F/L=?, $T = 2\pi\sqrt{\frac{L}{g}}$

$T^2 = 4\pi^2\frac{L}{g}$, $L = \left(\frac{g}{4\pi^2}\right)T^2 = \frac{10}{4\times3.142^2}\times4^2 = \frac{10\times16}{39.49}$,

$L = 4.05$m. Force per unit length, $\frac{F}{L} = \frac{120}{4.05}$,

$F = 29.6$Nm^{-1}.

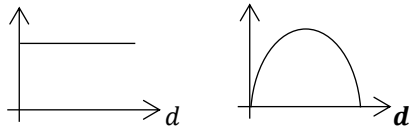
36. Marching soldiers crossing a suspension bridge are usually advised to break their step to avoid damaging the bridge owing to

A. resonance B. swinging C. vibration D. oscillation.

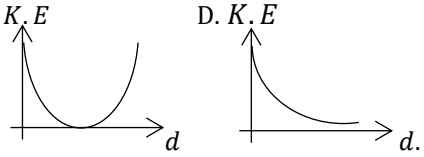
Walking in steps or uniformly sets the bridge to vibrate at the same frequency as that of the people walking or matching (resonance), hence causes the bridge to collapse.

37. Which of the following graphs correctly represents the relationship between kinetic energy ($K.E$) and the displacement (d) from the point of release of a pendulum bob?

A. $K.E$ **B. $K.E$**



C. $K.E$ D. $K.E$



The kinetic energy in SHM e,g swinging simple pendulum , is maximum at the equilibrium position and minimum at the two extreme positions as seen in the graph in option B.

Jamb past questions on Motion :

[1987/3,4,7,11,1979/49,1980/12,23,1981/25, 46,1982/11,12,13,34,1983/30,1984/1,26,36,38, 1985/2,4,5,9,12,1986/7,13,16,1987/3,4,1988/ 1,6,7,1989/6,1990/4,5,6,1991/3,4,5,1992/2,4,5, 1993/3,5,6,1994/5,6,7,1995/5,6,1997/2,4,6,5,7, 1998/5,7,9,1999/2,6,8,2000/7,8,10,11,12,2001 /9,2002/1,2003/7,10,2004/14,2005/3,28,2006 /13,19,2007/44,48,50,2008/5,6,10,2009/4,5,7, 8,9,2011/7,2012/6,7,12,2013/5,6,2014/5,7].

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CHAPTER 3 – FRICTION IN SOLIDS, FLUID FRICTION(VISCOSITY) AND SURFACE TENSION

● **Friction –** Friction is the force which acts at the surface of separation between two bodies in contact and tends to oppose the relative motion(not the force) between both bodies i.e. it acts in a direction opposite motion.

- Friction acts tangential to the surface of separation between two bodies in contact.

- Solid friction – Limiting or static friction, kinetic or dynamic or sliding friction and rolling friction.

- Limiting or static friction F_s is the frictional force that must be overcomed for a body at rest or stationary to move over another body.

$\boldsymbol{F_s = \mu_s R}$. F_s=limiting or static frictional force, R=normal reaction=weight of the body=$\boldsymbol{mg}$, μ_s=coefficient of limiting or static friction.

- Normal reaction must be perpendicular to the surface of contact while frictional force must be perpendicular to the normal reaction.

- The frictional force of a body attached to a spring balance is equal to the spring balance reading

- The force required to just move/slide a body is the static frictional force.

- If the applied force $\boldsymbol{F}$ on the rough surface exceeds the frictional force $\boldsymbol{F_s}$, it causes the body to accelerate $\boldsymbol{a}$. $\boldsymbol{F > F_s}$, $\boldsymbol{F - F_s = ma}$,

$\boldsymbol{F_s = \mu_s R = \mu_s mg}$, $\boldsymbol{F - \mu_s mg = ma}$.

- Kinetic or dynamic or sliding friction F_k is the frictional force that must be overcomed to keep a body moving in uniform or constant speed. $\boldsymbol{F_k = \mu_k R}$, F_k=kinetic or dynamic or sliding frictional force, R=normal reaction, μ_k=coefficient of kinetic friction.

- If frictional force is applied on a body inorder to bring it to rest or stop after moving a distance $\boldsymbol{s}$ with an initial speed of $\boldsymbol{u}$, then coefficient of dynamic or kinetic friction μ_k is – $\boldsymbol{\mu_k = \frac{u^2}{2gs}}$.

- Roads are banked in order to reduce the effect of centrifugal force and prevent skidding. In the absence of banking the frictional force between the tyres and the roads helps to prevent skidding. The coefficientof kinetic friction between the tyre of a vehicle moving round a circular track and the road is given by – $\boldsymbol{\mu_K = tan\theta = \frac{v^2}{gr}}$. θ= banking angle, v=speed of the vehicle round the circular track, r=radius of the circular track. $\boldsymbol{v = \sqrt{\mu_k gr}}$.

- Skidding occurs when $\boldsymbol{tan\theta > \frac{v^2}{gr}}$.

- Rolling friction is the frictional force that acts on rolling bodies. It has an extremely small value and might have a negligible effect.

- Static friction is greater than kinetic friction and kinetic friction is greater than rolling friction. Hence, it is easier to roll a body than to slide it on the ground and it is easier to keep a body in motion that to get a body at rest or stationary in motion.

- Coefficient of limiting or static friction is greater coefficient of kinetic or sliding friction.

- Properties of solid friction :

1.It depends on the nature i.e roughness or smoothness,of the surface in contact.

2.It always opposes relative motion of solids.

3.It depends on weight of the object/normal reaction.

4.It is independent of surface area of the surfaces in contact,relative velocity of the bodies and pressure.

5.Frictional force increases to the same extent as the force tending to cause motion i.e preventing the body from moving.

6.Static friction is always equal to the applied force when the object is stationary or moves with a uniform or constant speed.

7.Frictional force is proportional the normal reaction of between the surfaces(or weight of the body in contact). The slope of the graph of frictional force F_R vs normal reaction R is equal to **coefficient of friction, $\boldsymbol{\mu}$** (it has no unit).

- The coefficient of friction and friction in a perfectly smooth surface is 0 while that of perfectly rough surface is 1.

- A moving vehicle comes to rest because of the frictional force between the brakes and the wheels,frictional force between the tyres and road,external force applied opposite to the direction of motion of the vehicle and the resistance of the strong wind blowing against the vehicle.

- Advantages of friction :

I.Friction enables a nail or screw to remain in place when screwed into position.

II. Friction is utilised fan and conveyor belts used driver wheels or pulleys in machinery.

III. Friction enables the brakes to stop the car and the locomotive to pull the train.

IV. Friction enables motion e.g we walk only with the help of friction between our feet and the ground, automobile tyres are threaded(rough) to give it a good grip on road

- Disadvantages of friction :

I.Friction lead to wear and tear on moving parts of machines.

II. Friction reduces the efficiency of machines as much work is done in overcoming friction

III. Friction causes the heating of engines.

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- Reducing friction :

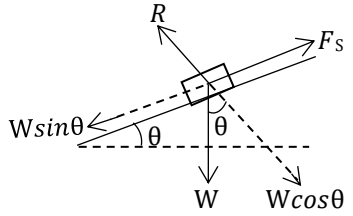
I.Lubrication : It involves the use of lubricants e.g grease,oil,air or graphite, in movingparts.

II. Balls or roller-bearings : It is based on the fact that rolling friction is much less than sliding friction i.e prevents surface from sliding.

III. Streamlining : The shape of cars, ships and aeroplanes are streamlined to lessess friction between them and their medium of motion.

IV. Polishing the surfaces.

● **Friction and Inclined planes –** An object of weight $W(mg)$ on an inclined plane is acted a set of forces, given below ;



1.Weight W(mg) acting vertically downward.

2.Normal reaction R acting upwards, perpendicular or normal to the plane.

3.Component of weight along or parallel to the plane W$sin\theta$ or $mgsin\theta$ which naturally tends to make the object slide down the plane.

4.Component of weight perpendicular or normal to the plane W$cos\theta$ or $mgcos\theta$ which counterbalances the normal reaction.

$\boldsymbol{R = W cos\theta = mgcos\theta.}$

- When the body is at rest or stationary in an inclined plane, the limiting or static frictional force F_s equals the componeent of weight along the plane. $\boldsymbol{F_s = W sin\theta = mgsin\theta.}$

- Gradual increase in the angle of inclination θ causes the object to be at the verge of sliding or just about to slide downwards. The angle at which this occurs is called the **angle of friction** and **limiting frictional force is maximum** at this point. The coefficicent of static friction is given by – $\boldsymbol{\mu_s = tan\theta}$, θ=angle of friction.

- The upward force $\boldsymbol{F_{ST}}$ required to keep the body at rest or stationary thereby preventing it from acceleration downwards is given by – $\boldsymbol{F_{ST} = W sin\theta - F_s}$, $\boldsymbol{F_{ST} = mgsin\theta - \mu_s R}$,
$\boldsymbol{F_{ST} = mgsin\theta - \mu_s mgcos\theta}$,
$\boldsymbol{F_{ST} = mg(sin\theta - \mu_s cos\theta).}$

- The acceleration $\boldsymbol{a}$ of a body down a **smooth** inclined plane(frictional force is zero), at an angle to the horizontal is – $\boldsymbol{a = gsin\theta.}$

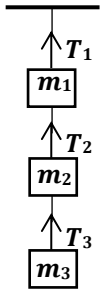
- The acceleration $\boldsymbol{a}$ of a body down a **rough** inclined plane is – $\boldsymbol{a = g(sin\theta - \mu_k cos\theta).}$

- When the body is moved up the inclined plane, the applied force $\boldsymbol{F_u}$ moving the body upward the balances the combined effect of component of weight along the plane $Wsin\theta$ and the frictional force(kinetic or sliding friction) F_k both acting downward. $\boldsymbol{F_u = W sin\theta + F_k}$.
$\boldsymbol{F_u = mgsin\theta + \mu_k mgcos\theta}$
$\boldsymbol{F_u = mg(sin\theta + \mu_k cos\theta) = 2W sin\theta.}$

- This upward force F_u prevents the body from slipping or sliding downwards.

- The acceleration $\boldsymbol{a}$ produced by this upward force is – $\boldsymbol{F_u - mg(sin\theta + \mu_k cos\theta) = ma,}$
$\boldsymbol{F_u - 2W sin\theta = ma.}$

● **Tension forces T on a string connecting bodies in equilibrium –** The tension on a string connecting a bodiesis equal to the weight of the bodies the string suspends.



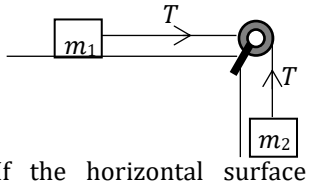
Tension $\boldsymbol{T_1}$ in the upper string suspends body m_1, m_2 and m_3.
$\boldsymbol{T_1 = W_1 + W_2 + W_3,}$
$\boldsymbol{T_1 = g(m_1 + m_2 + m_3)}$.
Tension $\boldsymbol{T_2}$ in the middle string suspends body m_2 and m_3.
$\boldsymbol{T_2 = W_2 + W_3 = g(m_2 + m_3).}$
Tension $\boldsymbol{T_3}$ in the lowest string suspends body m_3 only.
$\boldsymbol{T_3 = W_3 = m_3 g.}$

● **Systems with common acceleration :** When two bodies are connected by an inextensible string of tension $\boldsymbol{T}$, both bodies will move with a common acceleration $\boldsymbol{a}$.

- In general, common acceleration $\boldsymbol{a}$ is given by – Larger w.t − Smaller w.t = $\boldsymbol{ma}$,
$\boldsymbol{m = m_1 + m_2.}$ If the mass m_1 is on a rough surface, multiply its weight by the coefficient of friction μ of the surface. If the mass m_1 is on an inclined plane, multiply its weight by sinθ.

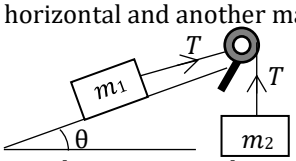
- Common acceleration is equal to 0,at the verge or point of slidding.

I. Air-track glider system – It has a mass m_1 gliding on a horizontal track via an inextensible string to another mass m_2 hanging in air.



If the horizontal surface is **rough i.e has coefficient of friction μ,** the weight of m_1 becomes $\boldsymbol{\mu m_1 g}$. The tension $\boldsymbol{T}$ can be given by : $\boldsymbol{m_2 g - T = m_2 a,}$ or $\boldsymbol{T - m_1 g = m_1 a}$ (when the surface is smooth) or $\boldsymbol{T - \mu m_1 g = m_1 a}$(when the surface is rough).

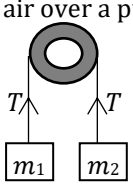
II. Inclined planes – It has a mass m_1 resting on an inclined plane inclined at an angle to the horizontal and another mass m_2 hanging in air



Weight is $\boldsymbol{m_2 g}$ and $\boldsymbol{m_1 g sin\theta}$ (if the plane is smooth) or $\boldsymbol{\mu m_1 g sin\theta}$ (if the plane is rough), $\boldsymbol{m}$ **is the sum of both masses=$\boldsymbol{m_1 + m_2}$.** The tension can be given by : $\boldsymbol{T - m_1 g sin\theta = m_1 a}$(if the plane is smooth) or $\boldsymbol{T - \mu m_1 sin\theta}$ (if the plane is rough).

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III. Pulley system – It has two masses m_1 and m_2 hanging in air over a pulley.



Weight is $\boldsymbol{m_1 g}$ **and** $\boldsymbol{m_2 g}$. If the horizontal surface is **rough i.e has coefficient of friction** $\boldsymbol{\mu}$**,** the weight of m_1 becomes $\boldsymbol{\mu m_1 g}$. The tension $\boldsymbol{T}$ can be given by : $\boldsymbol{m_2 g - T = m_2 a}$ or $\boldsymbol{T - m_1 g = m_1 a}$ (when the surface is smooth) or $\boldsymbol{T - \mu m_1 g = m_1 a}$ (when the surface is rough).

• Lifts or elevators

- Weight of a body in a lift or elevator – The body in a lift or elevator is acted upon by two forces :

I.Reaction $\boldsymbol{R}$ **or Tension** $\boldsymbol{T}$ **in the cord of the lift or apparent weight of the body** which acts upwards from the floor of the lift unto the body.

II.Weight $\boldsymbol{W(mg)}$which acts downwards.

1.If the lift is at rest or moving upwards or downwards with uniform velocity i.e acceleration is zero., the apparent weight is equal to the weight of the body.

$\boldsymbol{R = W = mg.}$

2.If the lift is accelerating upwards(ascending) the reaction force(acting upward) is greater than the weight(acting downward),then the apparent weight is more than the weight of the body.

$\boldsymbol{R - mg = ma,\ \ R = ma + mg,\ R = m(g + a).}$ **It also occurs when the lift decelerates to rest from a steady speed during downward motion.**

3.If the lift is accelerating downwards (decending), the reaction force(acting upward) is less than the weight(acting downward),then the apprarent weight is less than the weight of the body. $\boldsymbol{mg - R = ma,\ R = mg - ma,}$

$\boldsymbol{R = m(g - a)}$**. It also occurs when the lift decelerates to rest from a steady speed during upward motion.**

4.If the lift falls freely at the same rate of **g** when the cord of the lift is cut,then the apparent weight is zero i.e $\boldsymbol{a = g}$ and the body experience weightlessness**.**

- Upward deceleration or retardation gives the same reading as downward acceleration while downward deceleration or retardation gives the same reading as upward acceleration.

Examples :

1. Which of the following statements is not true of friction between two solids surfaces? **A.the frictional force between two solid surfaces is greater when one of the objects is moving than when they are both stationary** B.the frictional force between two rough surfaces is greater than that between two smooth surfaces C.the frictional force between two surfaces is independent of the area in contact D.frictional force depends on the type of material which forms the surfaces which are in contact E.frictional force is proportional to the normal reaction on one of the surfaces.

The frictional force between two solid surface is lesser when one of the objects is moving(sliding friction) than when both are stationary(static friction).

2. Which of the following statements about friction is not correct? **A.cars are less likely to skid on wet roads than on dry roads** B.nails hold boards together by friction C.limiting friction is directly proportional to the force acting between two surfaces D.sliding friction is less than limiting friction E.lubrication reduces friction

Cars are more likely to skid on wet roads than on dry roads due to friction between the tyres and roads. All other options are correct.

3. The limiting frictional force between two surfaces depends on I.the normal reaction between the surfaces II.the area of surfaces in contact III.the relative velocity between the surfaces IV.the nature of the surfaces A.I B.II C.III **D.I and IV** E.I and III.

Solid friction (limiting frictional force) depends on the normal reaction between the surfaces and the nature of the surfaces. It is independent of surface area in contact and relative velocity between the surfaces.

4. Which of the following statements are true about frictional force? I.It is always a disadvantage II.It is sometimes a disadvantage III.It always exists where there is a relative motion of bodies in contact IV.It is sometimes very useful A.I and II B.I and IV C.II and III **D.II and IV.**

Friction is not always a disadvantage it can also be an advantage. Friction does not always exists where there is a relative motion, it also exists when there is a tendency of motion.

5.The cofficient of frictionbetween two perfectly smooth surface is A.one B.half **C.zero** D.infinity.

6. Which of the following statements are true of friction? i.frictional force opposes motion ii.rolling friction between two surfaces is greater than sliding friction between the same two surfaces III.coefficient of friction is proportional to the limiting frictional force IV.it will be impossible to stop a moving car if there were no friction between the road and the tyres A.I,II,III and IV B.I,III and IV **C.I and IV** D.I,II and IV E.II,III and IV.

Rolling friction is less than sliding friction. Coefficient of friction is the ratio of the frictional

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force to the normal reaction. It is limiting frictional force that is proportional to the normal reaction.

7. Which of the following could be effectively used to reduce friction? A.water B.petrol **C.grease** D.kerosine.

8. Which of the following statements about friction is not correct? A.The force of kinetic friction is less than that of static friction B.The force of kinetic friction between two surfaces is independent of the area in contact provided the normal reaction is unchanged C.The force of rolling friction between two surface is less than the force of sliding friction **D.The angle of friction is the angle between the normal reaction and the force of friction** E.Friction may be reduced by lubrication

Angle of friction θ is the angle between the plane and the horizontal or perpendicular component of the weight and the weight at which static friction is maximum.

9. The frictional force between two bodies A.exists only when there is relative motion or tendency of motion B.acts so as to oppose the motion C.depeds on the normal reaction between the two surfaces **D.has all of these characteristics** E.is relatively indendent of the area in contact**.**

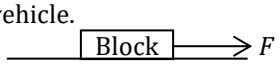
10. The surface of conveyor belts are made rough so as to **A.prevent the load from slipping** B.make the stronger C.enable them to carry more load D.protect the while carrying loads

Rough surfaces enhances friction in the belts, hence ensuring a grip on the load preventing the from slipping.

11. When the brakes in a car are apllied, the frictional force on tyres is A.disadvantage because it is in the direction of motion of the car B.a disadvantage because it is opposite in direction of motion of the car C.an advantage because it is in the direction of motion of the car **D.an advantage because it is in opposite direction of motion of the car.**

Frictional force acts opposes motion by acting in opposite direction to motion.

12. Which of the following statements is not correct? A moving vehicle comes to rest as a result of **A.the hot gases coming out of the exhaust pipe** B.the frictional force between the brakes and the wheels C.the frictional force between the tyres and the surface oon which the vehicle is moving D.the resistance of a strong nwind blowing against the vehicle E.the force applied in the opposite direction on the motion of the vehicle.

13. 

A block at rest on a horizontal surface is pulled by a horizontal force F as shown in the figure above. Which of the following statements are correct? The force of friction I.is directed to the right II.is directed to the left III.increases as F increases until the block starts to move IV.increases if a second block is placed on top of the block A.I and III B.II and IV C.II and III **D.II,III and IV.**

Frictional force opposes motion i.e acts in a direction opposite that of the force causing motion.Hence force of friction is directed to the left as F is directed right. Frictional force increases to the same extent as the force F causing motion until motion starts. If a second identical block is placed on the block, the weight W doubles,hence normal reaction R also doubles, thereby increasing the frictional force as $F_R \propto R$.

14. A body of mass 6.0kg rests on a rough horizontal surface where the coefficient of static friction is 0.25. Determine the magnitude of the limiting frictional force (g=10ms^{-2}) A.240.0N B.150.0N C.60.0N **D.15.0N.**

m=6kg, μ_s=0.25, g=10ms^{-2}, R=W=mg, F_s=?,

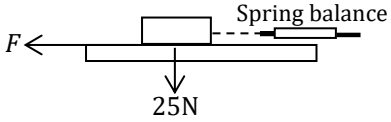
$F_s = \mu_s R = \mu_s mg = 0.25\times6\times10 = 15$N.

15. A metal block of mass 7.5kg lies on a rough horizontal platform. If a horizontal force of $12N$ is applied to the block through its centre of mass, just slides the block, calculate the coefficient of limiting friction between the block and the platform (g=10ms^{-2}) **A.0.16** B.0.63 C.1.60 D.3.66 E.6.33 .

The horizontal force just required to slide the block is equal to the frictional force. F_S=$12N$, m=7.5kg, μ_s=?, $F_s = \mu_s R$, $\mu_s = \frac{F_s}{R} = \frac{F_s}{mg}$,

$\mu_s = \frac{12}{7.5\times10} = 0.16$.

16.



In the above diagram the block just moves when the spring balance reads 15.0N. What is the value of F? A.0.6N B.10.0N **C.15.0N** D.40.0N E.375.0N.

F is the frictional force as it acts opposite to the direction of motion. The frictional force of a body attached to a spring balance is equal to the spring balance reading hence, F =15N.

- If the question had asked us to calculate the coefficient of friction of the surface in which the block moves it given as follows :

$R = W = 25N$, $\mu_s = \frac{F_s}{R} = \frac{15}{25} = 0.6$.

17. A child pulls a toy car weighing 3.00kg along a cement floor. If g=10ms^{-2} and the coefficient of friction is 0.500, the minimum force required to move the car is A.3.0N B.5.0N C.10.0N **D.15.0N** E.60.0N.

m=3kg, μ_s=0.5, g=10ms^{-1}, F_s=?,

$F_s = \mu_s R = \mu_s mg = 0.5\times3\times10 = 15$N.

18. A wooden block of weight $16N$ is placed on a rough surface. If the coefficient of friction between both surfaces is 0.25, the least

Demystified Series by Dr Timothy

horizontal force required to move the block is A.0.4N **B.4.0N** C.6.4N D.64.0N.
W=R=16N, μ_s=0.25, the least horizontal force required to move the block is equal to the frictional force. F_s=?, $F_s = \mu_s R = \mu_s W$,
$F_s = 0.25\times16 = 4$N.

19. A boy pushes a 500kg box along a floor with a force of 2000N. If the velocity of the box is uniform, the coefficient of friction between the box and the floor is A.1.0 B.0.8 C.0.65 D.0.5 **E.0.4.**
The frictional force acting on the box is kinetic or dynamic or sliding friction, as it is moving with a uniform velocity, m=500kg, g=10ms^{-2}, F_k=2000N, μ_k=?, $F_k = \mu_k R$, $\mu_k = \frac{F_k}{R} = \frac{F_k}{mg}$,
$\mu_k = \frac{2000}{500\times10} = 0.4$.

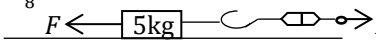
20. A body of mass 25kg moving at 3ms^{-1} on a rough horizontal floor is brought to rest after sliding through a distance of 2.50m on the floor. Calculate the coefficient of sliding friction (g=10ms^{-2}) A.0.09 **B.0.18** C.0.36 D.0.54 .
m=25kg, u=3ms^{-1}, s=2.5m, g=10ms^{-2}, μ_k=?,
$\mu_k = \frac{u^2}{2gs} = \frac{3^2}{2\times10\times2.5} = \frac{9}{50}$ =0.18.

21. A block of mass 50kg is given an initial speed of 5ms^{-1} on a rough horizontal floor. If it slides on the floor for 3m before it stops, calculate the coefficient of kinetic friction between the block and the floor (take g=10ms^{-1}) A.$\frac{3}{10}$ B.$\frac{1}{30}$ **C.$\frac{5}{12}$** D.$\frac{2}{3}$ E.$\frac{5}{6}$**.**
m=50kg, u=5ms^{-1}, s=3m, g=10ms^{-2}, μ_k=?,
$\mu_k = \frac{u^2}{2gs} = \frac{3^2}{2\times10\times2.5} = \frac{25}{60} = \frac{5}{12}$.

22. A moving car of mass 800kg experiences a frictional force of 200N. If it accelerates at 2ms^{-2}, calculate the magnitude of the force applied to the car A.600N B.1000N C.1600N **D.1800N .**
m=800kg, F_S=200N, a=2ms^{-1}, F=?,
$F - F_s = ma$, $F = ma + F_s$ =800×2+200 ,
F =1600+200 =1800N.

23. A block weighing 15N rests on a flat surface and a horizontal force of 3N is exerted on it. Determine the frictional force on the block. A.12N B.15N C.5N **D.3N.**
Frictional force opposes motion and is equal to the force tending to cause motion when the body is stationary.

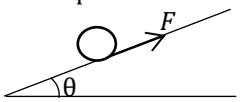
24. A force of 10N drags a mass 10kg on a horizontal table with an acceleration of 0.2ms^{-2}. If the acceleration due to gravity is 10ms^{-2}, the coefficient of friction between the moving mass and the table is? A.0.02 **B.0.08** C.0.20 D.0.80
F=10N, m=10kg, a=0.2ms^{-2}, g=10ms^{-2}, μ_s=?,
$F - F_s = ma$, $F_s = \mu_s R = \mu_s mg$.,
$F - \mu_s mg = ma$, $10-\mu_s\times10\times10=10\times0.2$,
$10-100\mu_s = 2$, $100\mu_s = 10-2 = 8$,
$\mu_s = \frac{100}{8} = 0.08$.

25. 

A block of wood of mass 5kg is pulled on a platform by force of 40N as illustrated in the diagram above. If the frictional force, F experienced by the block is 12N, calculate the magnitude of the acceleration of the block A.2.4ms^{-2} **B.5.6ms^{-2}** C.8.0ms^{-2} D.10.4ms^{-2}
F_s=12N, F=P=40N, m=5kg, a=?,
$F - F_s = ma$, $40-12 = 5a$, $28 = 5a$,
$a = \frac{28}{5} = 5.6$ms^{-2}.

26. A motorcycle of mass 100kg moves round in a circle of radius 10m with a velocity of 5ms^{-1}. Find the coefficient of friction between the road and the tyres (g=10ms^{-2}) A.25.00 B.2.50 C.0.50 **D.0.25.**
m=100kg, r=10m, v=5ms^{-1}, g=10ms^{-2}, μ_k=?,
$\mu_k = \frac{v^2}{gr} = \frac{5^2}{10\times10} = \frac{25}{100}$ =0.25.

27. A concrete block of mass 35kg is pulled along a horizontal floor with the aid of a rope inclined at an angle of 30° to the horizontal. If the coefficient of friction is 0.75, calculate the force required to move the block over the floor. A. 327.77N B.276.86N **C.211.52N** D.428.65N.
The inclined force F_p at an angle θ required to move a block of weight $W(mg)$ along a horizontal floor of coefficient of friction μ_S is given by – $\boldsymbol{F_p = \frac{\mu_s W}{\mu_s sin\theta + cos\theta}}$. μ_s=0.75, W=mg=35×10=350N, θ=30°, $F_p = \frac{0.75\times350}{0.75sin30°+cos30°} = \frac{262.5}{0.375+0.866} = \frac{262.5}{1.241}$,
F_p =211.52N.
The diagram above illustrates a ball of mass m sliding down a plank inclined at angle of θ to the horizontal. The kinetic friction between the ball and the plank F and acceleration of free fall is g.



Use the information to answer questions **28** and **29.**

28. The normal force on the ball is A.$mgsin\theta$ B.$mgtan\theta$ **C.$\boldsymbol{mgcos\theta}$** D.$mgcot\theta$
Normal force or normal reaction R is equal to the component of weight perpendicular to the plane.
$R = Wcos\theta = mgcos\theta$.

29. The net force on the ball along the plank is A.$mgsin\theta$+F B.$F - mgcos\theta$ **C.$\boldsymbol{mgsin\theta - F}$**
D.$F + mgcos\theta$.
The forces acting along the plane are – component of weight along the plane$mgsin\theta$ acting downwards and the frictional force F acting upwards. Since the ball is sliding down the plane,the component of the weight along the plane must be greater that the frictional force.
$mgsin\theta > F$, $mgsin\theta - F$ = net force ,
$mgsin\theta - F = ma$.

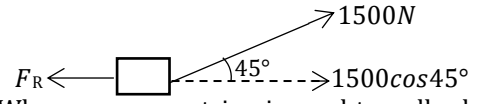
30. The limiting frictional force of a mass 4kg resting on an inclined plane is 20N. The angle of inclination θ to the plane is A.75° B.60° C.45° **D.30°.**

Demystified Series by Dr Timothy

m=4kg, g=10ms^{-1}, F_s=20N, θ=?,
When a body is at rest on an inclined plane, the limiting frictional force is completely balanced by the component of the weight of the body along the plane. $F_s = Wsin\theta = mgsin\theta$,
$20 = 4\times10sin\theta$, $sin\theta = \frac{20}{40} = 0.5$,
$\theta = sin^{-1}0.5 = 30°$.

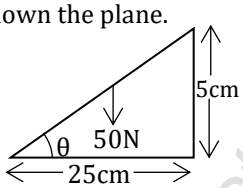
31. A body of mass 10kg rests on a rough inclined plane whose angle of tilt θ is gradually increased until the body starts to slide down the plane at 30°. The coefficient of static friction between the body and the plane is A.0.30 B.0.50 **C.0.58** D.0.87
m=10kg, θ=30°, μ_s=?, At the verge or point of sliding, θ=angle of friction.
$\mu_s = \tan\theta = tan30° = \frac{1}{\sqrt{3}} = 0.58$.

32. A load is pulled at uniform speed along a horizontal floor by a rope at 45° to the floor. If the force in the rope is 1500N, what is the frictional force on the load? A.1524N B.1350N C.1260N **D.1061N .**



When a rope or string is used to pull a body at angle θ to the floor, the frictional force F_R is equal to the horizontal component of the tension.
$F_R = Tcos\theta$. T=1500N, θ=45°, F_R=?,
$F_R = 1500cos45° = 1500(0.7071) = 1060.65$N ,
F_R =1061N.

33. The diagram below shows a block of wood resting on an inclined plane and at the point of sliding down the plane.



Calculate the coefficient of friction between the block and the plane. A.0.1 **B.0.2** C.0.3 D.0.4 E.0.5.
At the point of sliding down a plane, θ=angle of friction. $\mu_s = tan\theta = \frac{opp}{adj} = \frac{5}{25} = 0.2$.

34. A block of mass 50kg sliding down a plane is inclined at angle of 30° to the horizontal. If the coefficient of friction is 0.4, calculate the frictional force between te two surfaces (g=10ms^{-2}) A.200.00N **B.173.21N** C.100.00N D.62.50N.
m=50kg, θ=30°, μ_k=0.4(sliding friction), F_k=?,
Net force along the plane pushing the body downwards, $F_{net} = ma = mgsin\theta - F_k$,
Perpendicular/normal to the plane, R= $mgcos\theta$
$F_k = \mu_k R = \mu_k mgcos\theta = 0.4\times50\times10cos30°$
$F_k = 200\times0.866 = 173.21$N.

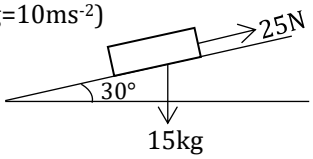
35. A block of mass 20kg slides down an incline plane of frictional force 30N. Calculate the force pushing the block down the plane, if the plane is inclined at 30° to the horizontal. (take g=9.8ms^{-2}) A.196N B.98N C.170N **D.68N**.
m=20kg, F_k=30N, θ=30°, g=9.8ms^{-2}, F_{net}=?,
$F_{net} = mgsin\theta - F_k = 20\times9.8sin30° - 30$,
F_{net} =196×0.5−30 = 98−30 = 68N.

36. A 50kg crate rests on a platform inclined at 30° to the horizontal. The coefficient of froction between the crate and the platform is 0.2. The minimum force needed to prevent the crate from sliding down the plane is A.433.0N B.250.0N **C.163.4N** D.86.6N.
m=50kg , θ=30°, μ_s=0.2. The minimum force needed to prevent the crate from sliding downwards keeps the crate stationary or at rest.
$F_{ST} = mg(sin\theta - \mu_s cos\theta)$,
$F_{ST} = 50\times10(sin30° - 0.2cos30°)$,
$F_{ST} = 500(0.5-0.2\times0.866) = 500(0.5-0.1732)$,
$F_{ST} = 500(0.3268) = 163.4$N.

37. A 5kg block released from rest on a smooth plane inclined at an angle of 30° to the horizontal. What is its acceleration down the plane **A.5.0ms^{-2}** B.5.8ms^{-2} C.8.7ms^{-2} D.25.0ms^{-2} .
m=5kg, θ=30°, g=10ms^{-2}, a=?,
The acceleration of a body down a **smooth** inclined plane is – $a = gsin\theta$(inclined to the horizontal) or $a = gcos\theta$(inclined to the vertical). $a = gsin\theta = 10sin30° = 5$ms^{-2}.

38. A block of mass 2.5kg slides down a plane inclined at 30° to the horizontal. If the coefficient of friction between the surfaces is 0.2, the acceleration of the block is **A.3.2ms^{-2}** B.4.2ms^{-2} C.5.2ms^{-2} D.12.2ms^{-2} E.21.2ms^{-2}.
m=2.5kg, θ=30°, μ_k=0.2, g=10ms^{-2}, a=?,
The acceleration of a body down a **rough** inclined plane is – $a = g(sin\theta - \mu_k cos\theta)$,
$a = 10(sin30° - 0.2cos30°) = 10\times0.326$,
a =3.26ms^{-2}=3.2ms^{-2}.

39. The diagram below shows a body resting on an inclined plane. If the body slides down the inclined plane, what will be its acceleration? (g=10ms^{-2})

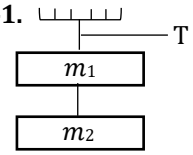


A.3.3ms^{-2} B.6.2ms^{-2} C.10.5ms^{-2} D.14.8ms^{-2} E.27.5ms^{-2}.
Since the body is sliding downwards, the 25N force acting upwards is the frictional force, as it opposes the downward motion of the body.
m=15kg, g=10ms^{-2}, θ=30°, F_K=25N, a=?,
$F_{net} = ma = mgsin\theta - F_k$,
$15a = 15\times10sin30° - 25 = 75-25 = 50$,
$a = \frac{50}{15} = 3.3$ms^{-1}.

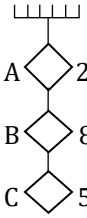
40. A body of mass 4kg is on the point of slipping down a plane which is inclined at 30° to the horizontal. What force, parallel to the plane, will just move it up the plane? (g=10ms^{-2}) A.80N B.60N **C.40N** D.20N.
m=4kg, θ=30°, g=10ms^{-2}, F_u=?,
The upward force parallel to the plane which

Demystified Series by Dr Timothy

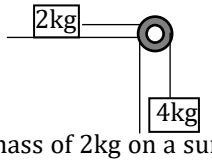
moves the body up the plane (preventing it from slipping donward) : $F_u = mg(sin\theta + \mu_k cos\theta)$
$\mu_k = tan\theta = tan30° = 0.58$.
$F_u = mg(sin\theta + \mu_k cos\theta)$,
$F_u = 4\times10(sin30° + 0.58cos30°) = 40(0.5+0.5)$
$F_u = 40$N. Alternatively, $F_u = 2Wsin\theta$,
$F_u = 2mgsin\theta = 2\times4\times10sin30° = 40$N.

41. 

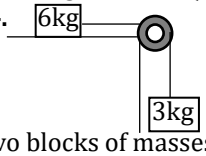
The figure below shows two connected bodies of masses m_1 and m_2 in equilibrium under gravity. The tension in the upper string is
A.$(m_1-m_2)g$ B.m_1g C.m_2g **D.$(m_1+m_2)g$** .
Tension in the string is equal to the weight suspended below by the string. $T = W_1+W_2$, $T = m_1g + m_2g = (m_1+m_2)g$.

42. 

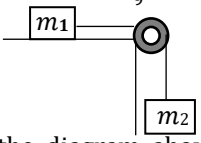
Three bodies A, B and C are connected by light inextensible strings as shown in the diagram. If the weight of the bodies are 25N, 8N and 5N respectively, what is the tension in the string connecting A and B?
A.13N B.17N C.23N D.33N E.38N.
The string connecting A and B suspends weight B and C. $T = 8+5 = 13$N.

43. 

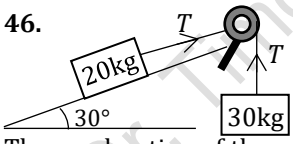
A mass of 2kg on a surface (μ=1/2) is connected to a second mass of 4kg over a a frictionless pulley asshown in the diagram. If the acceleration due to gravity is 9.8ms^{-2}, then the masses will **A.accelerate at 4.9ms^{-2}** B.remain stationary C.accelerate at 9.8ms^{-2}D.accelerate at 19.6ms^{-2} E.accelerate at 39.2ms^{-2}.
m_1=2kg, m_2=4kg, g=9.8ms^{-2}, μ=1/2, a=?,
Air-track glidder system – $m_2 > m_1$, Larger weight= m_2g, smaller weight= $\mu m_1 g$(rough surface). Larger weight−Smaller weight= ma,
$4g - \frac{1}{2}\times2g = (2+4)a$, $4g - g = 6a$, $3g = 6a$,
$a = \frac{3g}{6} = \frac{g}{2} = \frac{9.8}{2} =$ 4.9ms^{-2} , both masses will accelerate at 4.9ms^{-2}.
- The tension T in the question : $m_2g - T = m_2a$,
$T = m_2(g-a) = 4(9.8-4.9) = 4(4.9) = 19.6$N.

44. 

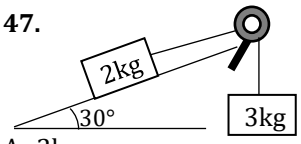
Two blocks of masses 3kg and 6kg are connected by a light inextensible string which passes over a smooth pulley as shown in the diagram. Assuming the 6kg mass is sitting on a smooth surface, the bodies will accelerate at A.10.0ms^{-2} B.6.7ms^{-2} C.5.0ms^{-2} **D.3.3ms^{-2}.**
m_1=6kg, m_2=3kg, g=10ms^{-2}, a=?,
Air-track glider system : $m_1 > m_2$, larger weight $= m_1g$(smooth surface), smaller weight= m_2g,
Larger weight−Smaller weight= ma,
$6\times10-3\times10 = (6+3)a$, $60-30=9a$,
$30 = 9a$, $a = \frac{30}{9} = 3.3$ms^{-2}.

45. 

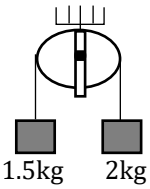
In the diagram above, the hanging mass m_2 is adjusted until m_1 is **on the verge of sliding.** The coefficient of static friction between the mass m_1 and the table is A.$\frac{m_1}{m_2}$ B.$\frac{m_1g}{m_2}$ **C.$\frac{m_2}{m_1}$** D.$\frac{m_2g}{m_1}$.
On the verge of sliding, common acceleration, a=0. m_2 is greater than m_1, since it is adjusted to make m_1 to be on the verge of sliding.
Larger weight−Smaller weight= ma,
$m_2g - \mu m_1g = (m_1+m_2)0 = 0$,
$m_2g - \mu m_1g = 0$, $\mu m_1g = m_2g$, $\mu = \frac{m_2g}{m_1g} = \frac{m_2}{m_1}$.

46. 

The acceleration of the system above is A.2ms^{-2} **B.4ms^{-2}** C.6ms^{-2} D.8ms^{-2}.
m_1=20kg, m_2=30kg, g=10ms^{-2}, θ=30°, a=?,
Inclined plane system – $m_2 > m_1$, larger weight= m_2g, smaller weight= $m_1gsin\theta$,
Larger weight−Smaller weight= ma,
$30\times10-20\times10sin30° = (20+30)a$,
$300-100 = (50)a$, $200 = 50a$, $a = \frac{200}{50} = 4$ms^{-2}.

47. 

A 2kg mass on a smooth inclined plane is connected over a smoth pulley to a 3kg mass as shown in the diagram above. The acceleration of the system is A.g B.3g/4 C.g/2 **D.2g/5.**
m_1=2kg, m_2=3kg, θ=30°, a=?,
Inclined plane system : $m_2 > m_1$, larger weight= m_2g , smaller weight= $m_1gsin\theta$,
Larger weight − Smaller weight = ma,
$3g - 2gsin30° = (2+3)a$, $3g - g = 5a$,
$2g = 5a$, $a = 2g/5$.

48. 

Two masses 2kg and 1.5kg are tied to the ends of light inextensible string which passes over a smooth light pulley which is fixed to the ceiling as shown in the diagram. The acceleration of the 2kg mass is (g=10ms^{-2}) A.10.00ms^{-2} B.5.71ms^{-2} C.4.29ms^{-2} **D.1.43ms^{-2}.**
Larger weight−Smaller weight= ma,

Demystified Series by Dr Timothy

$2\times10-1.5\times10=(2+1.5)a$, $20-15=3.5a$,
$5=3.5a$, $a=\frac{5}{3.5}=1.428=1.43\text{ms}^{-2}$.

49. A man standing on a lift that is descending does not feel any weight because **A.there is no reaction from the floor of the lift** B.there is no gravitational pull on the man in the lift C.the inside of the lift is air tight D.the lift is in vacuum
For a lift descending : $W>R$, $W-R=ma$,
$W-ma=R$, when there is no reaction on the floor , R=0, $W=ma$, the body experiences weightlessness, as the apparent weight is equal to zero or weight equals the net force.

50. A man stands on a spring balance placed in a lift. The lift descends at constant velocity. As a result, the scale reads a weight A.greater than the weight of a man **B.the same as the weight of a man** C.of zero D.less than the weight of the man E.greater than the weight of the man by 1kg
A lift descending or ascending at constant velocity i.e acceleration is zero, will read an apparent weight(or tension in the scale) equal to the weight of the body. $W-R=ma$, a=0, $W-R=0$, $W=R$.

51. A spring balance which is suspended from the roof of a lift, carries a mass of 1kg at its free end. If the lift accelerates upwards at 2.5ms^{-2}, determine the reading on the spring balance (g=10ms^{-2}) A.6.25N **B.12.5N** C.25.0N D.24.0N.
m=1kg, a=2.5ms^{-2}, g=10ms^{-2}, reading on the spring balance is the tension in the cord or apparent weight or reaction on the floor=R,
$R=m(g+a)$, accelerating upwards.
$R=1(10+2.5)=12.5$N.

52. A girl whise mass is 55kg stands on a spring weighing machine inside a lift. When the lift starts to ascend, its acceleration is 2ms^{-2}. What will be the reading on the machine? (take g=10ms^{-2}) A.22kg B.33kg C.44kg **D.66kg**
m=55kg, a=2ms^{-2}, g=10ms^{-2}, R=?,
$R=m(g+a)$, ascending in the lift,
$R=55(10+2)=55(12)=660$N.
$R=m_Rg$, $m_R=\frac{R}{g}=\frac{660}{10}=66$kg.

53. A 1000kg elevator is descending vertically with an acceleration of 1ms^{-2}. If the acceleration due to gravity is 10.0ms^{-2}, the tension in the suspending cable is A.1.0N B.10.0N **C.9000.0N** D.11000.0N.
m=1000kg, a=1ms^{-2}, g=10ms^{-2}, R=T=?
$R=T=m(g-a)=1000(10-1)=1000(9)$
$T=9000N$.

54. Calculate the apparent weight loss of a man weighing 70kg in an elevator moving downwards with an acceleration of 1.5ms^{-2} (g=10ms^{-2}) A.595N B.581N C.686N **D.105N**.
m=70kg, a=1.5ms^{-2}, g=10ms^{-2},
$W-R=ma$, a lift descending. **Apparent loss in weight for a descending body = $W-R=ma$, the resultant force.**
Apparent loss in weight $=ma=70\times1.5=105$N.

55. An elevator of mass 4800kg is supported by a cable which can safely withstand a maximum tension of 60000N. The maximum upward acceleration the elevator can have is **A.2.5ms^{-2}** B.5.0ms^{-2} C.7.5ms^{-2} D.10.0ms^{-2} (g=10ms^{-2})
m=4800kg, R=T=60000N, g=10ms^{-2}, a=?,
$R=T=m(g+a)$, a lift ascending.
$60000=4800(10+a)$, $10+a=\frac{60000}{4800}=12.5$,
$a=12.5-10=2.5\text{ms}^{-2}$.

Jamb past questions on friction in horizontal surfaces and inclined planes, lift or elevator :
[2012/3,12,13,1990/3,1981/28,1992/8,1991/2 ,1981/19,1990/11,1995/7,1987/11,1982/48, 1997/10,14,2005/4,1994/11,1985/8,1995/12, 1987/5,1988/12,1989/7,1986/11,2004/16, 2005/6,29,2008/132009/11,1978/2,1983/5, 1995/13,1983/4,1999/18,2000/3,2013/11, 2011/17,2010/9,1982/2,1986/15,1987/6,2002 /6,2005/6,2011/3,2013/7,2015/4]

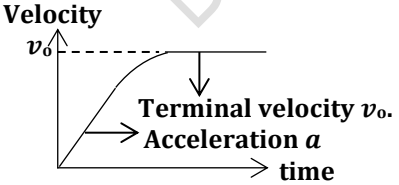
● Fluid friction and Viscosity

- Fluid friction or Viscosity – Viscosity is the friction between layers of molecules of a fluid tending to make it resist free flow.
- A more viscous liquid e.g oil,is difficult to flow or for a body to fall through it than a less viscous liquid.
- Properties of viscosity :
I.It depends on surface area and relative velocity/motion.
II. It is affected by pressure i.e it increases with pressure in the fluid.
III. It reduces with an increase in temperature.
IV. It resists the flow of liquids.
V.It is independent of weight or normal reaction.
VI. It opposes gravitational force (weight) of a falling body, as it acts upward.
- Solid friction and viscosity are similar in that they resist or oppose relative motion between surfaces and depends on the nature of the bodies in contact. **-**
Stress law – It states that shear stress $\boldsymbol{\sigma}$ is directly proportional to the velocity gradient $\boldsymbol{v}_x$.
$\boldsymbol{\sigma}\propto\boldsymbol{v}_x$, $\boldsymbol{\sigma}=\boldsymbol{\eta v}_x$, $\boldsymbol{\sigma}=\frac{F}{A}$, $\boldsymbol{v}_x=\frac{v}{L}$, $\frac{F}{A}=\boldsymbol{\eta}\frac{v}{L}$
$\boldsymbol{\eta}=\frac{FL}{Av}$, $\boldsymbol{\eta}$ is the constant called viscosity or coefficient of viscosity in **pascal-second, $\boldsymbol{Pa.s}$**, F= force, L=length or distance, A=surface area, v=velocity.
- Factors affecting viscosity : $\eta\propto\frac{tA}{vT}$.
I.Time t **II.**Surfacearea A **III.**Temperature T **IV.**Velocity v.
- The viscosity of lubricants oil is not affected temperature (viscostatic).
- The luricating effects of lubricants e.g oil and greases decreases with temperature. Water is

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not used as a lubricant because of its low viscosity.

- The bob of a simple pendulum comes to rest more quickly with a cord attached to the string than without a cord because of viscosity of air.
- A simple pendulum with a wide area comes to rest quickly due to increased viscous force oppositing the pendulums motion.
- A body falling through a viscous liquid is acted upon by three forces: **Upthrust (buoyancy force)** U and **Viscous drag(fluid friction)** V acting upward and the Weight W (mg) acting downwards. The two upward force – upthrust U and viscous force are together called **retarding force.**
- When a stone or ball bearing of mass m is falling through a viscous liquid, initially the body accelerates downwards, as the downward force (W) is greater than the upward retarding forces $(U+V)$. The resultant force experienced by the body : $W-U-V=ma$ or $W-(U+V)=ma$. a is the acceleration of the stone.
- The viscous force increases with the speed of the ball bearing, until a certain maximum speed i.e terminal velocity,where the upward forces balances the weight of the body.
- **At terminal speed(uniform or constant velocity), the upward or retarding force $(U+V)$ equals or balances the downward force (mg). The net or resultant force acting at terminal velocity is zero and the acceleration is 0.** $W=mg=U+V$ or $mg-(U+V)=0$ or $(U+V)-mg=0$.
- Raindrops whose motion is opposed by the viscosity of air attains a terminal velocity before reaching the ground. Hence, it is not harmful when it strikes a person.
- The velocity-time graph of a body falling through a viscous liquid, a parachute or a body dropped from an airplane has an initial ascending line indicating acceleration and a horizontal line indicating terminal velocity i.e a=0.



- **Stoke's law –** It states that the viscous drag force F retarding a small sphere falling through a viscous fluid at terminal velocity is proportional to the viscosity of the fluid, its radius and terminal velocity. $F=6\pi\eta Rv_o$. η=viscosity of the fluid, R=radius of the sphere, v_o=terminal velocity or steady speed of the sphere.

• Surface tension

- **Surface tension T_s –** Surface tension is the force acting tangentially per unit length at a surface. T_s or $\text{S.T}=\frac{F}{L}$.
- It occurs due to net inward cohesion of molecule on the liquid surface. It makes the surface of a liquid behave like a stretched elastic skin or membrane and contract areas of the liquid surface to the least value
- **Surface tension can also be referred to as the elasticity of a fluid.**
- **It can be explained by kinetic/molecular theory of matter :**

1.Scatterring and clinging together of hair of paint brush.
2.Water surface in a beaker is not horizontal at the glass-water surface
3.Floating of denser objects e.g needle floats on water surface
4.Dropping of nails into a glass filled with water without over flowing **.**
5.Waterproof materials e.g umbrella or tent, for keeping water out **.**
6.Molecules at the surface of a liquid have more potential energy than the molecules in the bulk
7.Washing of clothes in water is difficult without soap.
8.Water maintains a height or rises above its outside level in a capillary tube.

- **Surface tension depends on the nature of the liquid, temperature of the liquid, purity of the liquid and pressure acting on the liquid surface.**
- Surface tension is affected by temperature, $T_s \propto \frac{1}{T}$, surface tension decreases with increase in temperature.
- Surface tension is affected by the pressure acting in the liquid, $T_s \propto P$, surface tension increases with increase in pressure.
- **Reduction of surface tension –** Surface tension is reduced by the addition of : **I.**alcohol or methylated spirit **II.**soap or detergent **III.**kerosene, oil or non-polar liquids.
- **Increasing of surface tension –** Surface tension is increased by the addition of : **I.**wax **II.**Grease.
- **Adhesion or Adhesive forces –** These are the force of attraction between different molecules e.g water and glass or mercury and glass.
- **Cohesion or Cohesive forces –** These are the force of attraction between same molecules e.g between water molecules or mercury molecules.
- **Currency notes can be counted with moist finger due to adhesion of water and the currency.**
- Mercury doesn't wet glass because the cohesion between mercury molecules is greater than the adhesion between mercury and glass. Water wets glass because the adhesion between water and glasss is greater than the cohesion between

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water molecules.

● **Capillarity –** Capillarity is the rise or fall of liquid in a narrow tube.

- Water or alcohol surface forms a **concave meniscus** to air while **mercury** surface forms **convex meniscus** to air in a capiilary tube.

- A wetting liquid i.e water or alcohol, whose adhesion is greater that its cohesion, will rise above the liquid level.

- A non-wetting liquid i.e mercury, whose cohesion is greater than its adhesion, will fall below liquid level in a capillary tube.

- Application of capillarity :

I.Rise of oil through a lamp wick, rise of candle wax through a candle wick, rise or absorption of ink by a blotting paper

II. Absorption of water by a towel i.e faster rate of absorption of a drop of water on a damp cloth than a dry cloth.

III. Rise of water from the soil to the plant and flow of blood through channels.

IV. Soaking of water by sugar.

- Effects of adhesion/cohesion and surface tension on capillarity :

1.Rising of water up a capillary tube –

I.Adhesive forces pull the water up the side of the tube.

II. Surface tension holds up the water rising in the tube.

III. Water rises in the tube until surface tension balances or equals the weight of the water column. The pulling of water up the side of the tube by adhesive forces gives rise to a concave meniscus.

2.Depresion of mercury in a capillary tube –

I.Coheisve forces pulls the mercury down the side of the tube.

II. Surface tension holds down the mercury as it depresses in the tube. The pulling of the mercury down the side of the tube by cohesive forces gives rise to a convex meniscus.

- Angle of contact or contact angle – This is the angle between the wall of a capillary tuve and the tanget to the meniscus of the liquid in the tube. Angle of contact depends on the neatness of the wall of the capuillary glass tube. Angle of contact is acute i.e <90°, for water and obtuse i.e >90°, for mercury.

● **Fluids in motion –** It involves flow rates and viscosity.

- Flow rates is the time rate of flow of fluids.

- Volume rate flow $\frac{V}{t}$ in m^3s^{-1} is the time rate of flow of the volume of the fluid.

Volume rate flow= $\frac{\text{volume}}{\text{time}} = \frac{V}{t} = Av$. V=volume of the fluid, t=time of fluid flow, A=area of the vessel through which the fluid flows, v=velocity at which the fluid flows.

The volume of the liquid is given by – $V = Avt$.

- Mass rate flow $\frac{m}{t}$ in kgs^{-1} is the time rate of flow of the mass of fluid.

Mass rate flow= $\frac{mass}{time} = \frac{m}{t} = \frac{\rho V}{t} = \rho Av$. m=mass of the fluid, ρ=density of the fluid.

- Force F in a pump causing a liquid to flowis given by – $F = \rho Av^2$.

Examples :

1. The opposing force which one layer of liquid exerts on another is called A.cohesion **B.viscosity** C.limiting force D.dynamic force

Viscosity is the friction between layers of molecules of a fluid tending to make it resist free flow.

2. The property of fluid that is used to determine its suitability as a lubricant is its A.capillarity B.surface tension **C.viscosity** D.relative density.

3. Which of the following is most viscous at room temperature? **A.palm oil** B.petrol C.alcohol D.water

Palm oil at room temperature is hard to pour out of a container and an object will move slowly in it than any of the other liquids. Hence, palm oil has the highest viscosity.

4. Palm oil from a bottle flows out more earlier after it has been warmed because A.molecules are given potential energy during warming **B.friction between the oil layer is reduced** C.oil molecules force one another out D.cohesion between the oil molecules

Increase in temperature e.g by warming, decreases the viscosity of the palm oil, hence it flows easily.

5. The similarity between friction and viscosity is that both **A.oppose relative motion between surfaces** B.occur in solids C.depend on relative velocities between surfaces D.depend on normal reaction E.depend on areas of surfaces in contact.

6. As the pressure of a fluid increases, its viscosity A.decreases B.remains constant C.increases then decreases **D.increases.**

Viscosity depends on pressure, viscosity increases as pressure increases and vice-versa

7. A spherical ball falling through a viscous liquid acted upon by the I.weight of the ball II.viscous force of the liquid on the ball III.upthrust of the liquid on the ball A.I and II B.I and III C.II and III **D.I,II and III.**

The forces acting on a body falling down a viscous lioquid are ; the weight of the ball W, viscous force of the liquid V and the upthrust of the liquid U.

8. The effect of a particle in a fluid attaining its terminal velocity is that the A.acceleration is maximum **B.weight equals the retarding force** C.buoyancy force is equal to the viscous retarding

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force D.buoyancy force is more than the weight of the fluid displaced
At terminal velocity, velocity is uniform or constant,hence the acceleration is zero, also the weight equals the retarding force (upthrust or buoyancy force and viscous drag or fluid friction).

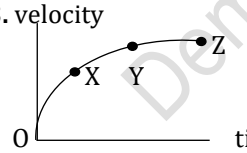
9. A parachute attains a terminal velocity when A.its density is equal to the density of air **B.the viscous force of the air and the upthrust completely counteract its weight** C.it expands as a result of reduced external pressure D.the viscous force of the air is equal to the sum of the weight and upthrust.
At terminal velocity, the upthrust and viscous force completely balances the weight of the body.

10. The terminal velocity of a ball-bearing falling through a viscous liquid is reached when the A.upthrust is equal to yhe weight of the ball B.ball accelerates uniformly C.upthrust is equal to the velocity of the ball **D.velocity is uniform.**
Terminal velocity is the maximum uniform velocity attained by a body falling through a viscous liquid. Initially it accelerates uniformly until terminal velocity is reached. The resultant or net force is zero at terminal velocity.

11. A metal ball of weight W falls through a column of glycerine of viscosity V. If the bal experiences an upthrust U and terminal velocity is attained, then A.$W > U + V$ **B.$\boldsymbol{W = U + V}$** C.$W = U - V$ D.$W < U + V$.

12. A body of mass 36kg falls through a viscous liquid which offers a drag force 260N on the body. The upthrust on the body at terminal velocity is A.50N **B.100N** C.310N D.620N (g=10ms^{-2})
Viscous drag or fluid friction, V=260N, m=36kg, g=10ms^{-2}, upthrust or buoyancy force, U=?, At terminal velocity, $W = U + V$, $U = W - V$,
$U = mg - V = 36\times10-260 = 360-260 = 100$N.

13. velocity



The motion of a falling sphere through a viscous liquid is represented by the graph. The terminal velocity is determined as the A.the value of the velocity at point X **B.value of the velocity at point along the portion YZ** C.slope of the graph at point Y D.area under the graph OXYZ.
The terminal velocity is reached when the velocity is uniform or constant over a period of time i.e portion YZ, the acceleration at this point is zero. The portion OX represent the initial acceleration phase of the falling body, an the slope of the graph at the portion OX gives the acceleration.

14. Which of the following is obeyed if a force on a sphere moving through a viscous fluid is proportional to its speed,radius and viscosity? A.Archimede's principle B.Pascal's principle **C.Stoke's law** D.Bernolli's law
Stoke's law gives the force acting on sphere falling through a viscous liquidat terminal velocity.$F = 6\pi\eta Rv_o$.

15. A hose of cross-sectional area 0.5m^2 is used to discharge water from a water tanker at a velocity of 60ms^{-1} in 20s into a container. If the container is filled completely, the volume of the container is **A.600m^3** B.6000m^3 C.240m^3 D.2400m^3.
A=0.5m^2, v=60ms^{-1}, t=20s, V=?,
Volume rate flow $= \frac{V}{t} = Av$, $V = Avt$,
$V = 0.5\times60\times20 = 600$m^3.

16. The force between molecules of the same substance is termed A.elastic force B.repulsive force **C.cohesive force** D.adhesive force.
Cohesive forces are the forces of attraction between molecules of the same substance.

17. The force between the molecules of a liquid in contact with that of a solid is **A.adhesive** B.cohesive C.magnetic D.repulsive.
Adhesive forces are the forces of attraction between molecules of different substance e.g molecules of liquids and solids.

18. Which of the following can not be explained using the concept surface tension? A.crawling of insects on the surface of a pond B.formation of spherical droplets when water drips from a tap **C.floating of balloon is air** D.formation of soap bubbles.
Floating of balloons in air is explained using the floatation principle. Other options are application surface tension

19. The following processes are explained by surface tension only except A.bristles of a paint brush cling together when removed from water B.dirts being dissolved by soap solution C.needle floating in water D.sphrical droplets of water dropping fron a tap **E.water rising up a capillary tube.**
Water rising up a capillary tube is due to greater adhesion of water molecules to the capillary tube than their cohesion and surface tension pulling the water up the inner walls of the tube until it is balanced by the weight of the water column. Hence, it cannot be explained by surface tension only.

20. Which of the following substances lowers the surface tension of water? A.metal B.sand **C.detergent** D.paper.

21. If a beaker is filled with water, it is observed that the surface of the water is not horizontal at the glass-water surface. This behavior is due to A.friction **B.surface tension** C.evaporation D.viscosity.

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Surface tension contracts area of the liquid surface to the least value, making it to behave like a streteched elastic membrane, hence the water surface is not horizontal.

22. Which of the following additions will not reduce the surface tension of water? A.Detergent **B.Grease** C.Alcohol D.Camphor.

Addition of grease to water increases the surface tension rather than decreasing it.

23. Certain insects can move about on the surface of water without sinking because A.water is denser than such insects B.the insects have tiny legs C.upthrust is greater than their weight **D.the water surface acts like an elastic membrane.**

Surface tension is the responsible for the floating of denser objects i.e objects whose dener than water e.g steel needle,razor blade, insects or a piece of wire guaze.

24. Which of the following liquids has the highest surface tension? A.Soapy water **B.Cold water** C.Hot water D.Oily water.

Surface tension decreases with temperature, hence hot water has low surface tension. Soap and Oil reduces surface tension.Hence, cold water has the highest surface tension.

25. Which of the following statememts explains why hot soapy water is more effective in cleaning oil-stained dishes? A.The oil on the dishes repels the soap **B.Soap and heat decreases the surface tension of water** C.Hot water increases the surface tension of oil D.Soap increases the surface tension of oil and water.

26. A steel needle floating on the surface of water sinks when kerosene is added to the water because A.kerosene cause an increase in the surface tension of water **B.kerosene causes a decrease in the surface tension of water** C.kerosene causes a decrease in the density of water D.kerosene causes an increases in the density of water E.a steel needle is denser than kerosene.

Steel needle floats on water surface due to surface tension. Addition of kerosene decreases the surface tension of water, hence the steel needle sinks.

27. Mercury forms little balls when it is spilled on a bench because it A.is very viscous B.adheres to the bench **C.has a high surface tension** D.has a high density E.is a liquid metal.

Surface tension contracts the area of liquid surface into small or little balls of reduced shape.

28. Water does not drip through an open umbrella of silk material unless the indide of the umbrella is touched. Which of the following phenomenon is respondible for this? **A.surface tension** B.hydrostatic upthrust C.viscosity D.diffusion E.osmosis.

When the inside of an umbrella or raincoat is touched, the surface tension between the umbrella surface and water droplets is reduced, hence the elastic skin is broken and water drips into the umbrella.

29. The rising of a liquid in an open ended glass tube of narrow bore is A.osmosis B.adhesion **C.capillarity** D.surface tension.

30. The rise or fall of a liquid in a narrow tube is because of the **A.the surface tension of the liquid** B.the friction between the walls of the tube and the liquid C.the osmotic pressure of the liquid D.the viscosity of the liquid.

Surface tension pulls up the water up the inner walls of the tube.

31. The action of a blotting paper in use is a demonstration of the principle of A.viscosity B.evaporation **C.capillarity** D.surface tension.

32. The action of a towel in drying the body after a bath is explained by A.diffusion **B.capillarity** C.osmosis D.evaporation.

33. By which of the following processes does a piece of blotting paper absorb a drop of ink from a sheet of writing paper? A.the action of the forces of cohesion between the blotting paper fibres and the ink **C.the capillarity rise of the ink into the blotting paper fibres** D.the diffusion of the ink into the blotting paper fibres E.the action of the force of viscosity between the molecules of the ink.

34. The concave meniscus of water in a glass tube is due to the fact that the **A.adhesion between water and glass molecules is greater than the cohesion between water molecules** B.adhesion between glass and water molecules is smaller than cohesion between water molecules C.molecules of water near the glass are lighter than the molecules in the central part of the water column D.weight of the water column pulls down on the central part of the water column.

The meniscus is convex as in mercury, if the cohesion between the mercury molecules is greater than the adhesion between the mercury and glass molecules.

35. In a capillary experiment, the rise of mercury in a narror capillary tube is A.higher than that of the container **B.lower than that of the container** C.equal to that of the container D.rising and falling constantly in the container.

36. Mercury which is spilled onto a glass surface forms ball-like shapes because A.mercury is a metal **B.the cohesive force between its molecules is greater than the adhesive forces between mercury and glass molecules** C.the cohesive force between its molecules is less than the adhesive forces between mercury and glass molecules D.the viscosity of mercury is high.

37. If a tube of small radius opened at both ends is placed in a liquid, the liquid will **A.fall below the liquid level if the liquid does not wet the glass** B.rise above the liquid level if the liquid

Demystified Series by Dr Timothy

does not wet the glass C.remain at the same level irrespective of whether the liquid wets the glass or not D.fall below the liquid level if the liquid wets the glass.
A non-wetting liquid e.g mercury, falls below the liquid level in a capillary tube. A wetting liquid e.g alcohol or water, rises above the liquid level in a capillary tube.

Jamb past questions on Fluid friction and surface tension :
[1987/12,1991/14,1997/17,1999/14,2003/8, 2008/12,2009/24,2011/9,2013/23,1994/2, 1998/11,2003/5,2002/4,11,2004/13,2005/49, 2006/18,2009/13]

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CHAPTER 4 -WORK , ENERGY AND POWER

● **Workdone W –** Workdone is the product of force F and **parallel distance** $\boldsymbol{s}$ moved in direction of the force. $\mathbf{W} = \boldsymbol{F} \times \boldsymbol{s}$, in **Joule(J).**

- Workdone on a body is zero when no displacement (a body at rest), even if a force is applied in a specified time e.g a boy standind with a bowl of water on his head, a body pusing a fixed or stationary wall.

- The force F can be determined by – $\boldsymbol{ma}$ (if the body is accelerated by the force) or $\mathbf{W}=\boldsymbol{mg}$ (if the body is falling freely under gravity) or $\boldsymbol{\mu R}$ (due to frictional force) or $\boldsymbol{Fcos\theta}$ (inclined to the horizontal) or $\boldsymbol{Fsin\theta}$ (inclined to the vertical).

- The distance s can be determined by – $\boldsymbol{s} = \frac{1}{2}\boldsymbol{at^2}$ or $\frac{1}{2}\boldsymbol{gt^2}$ or $\frac{v^2}{2a}$ or $\frac{v^2}{2g}$ or $\boldsymbol{h}$.

a=acceleration, t=time, v=velocity, h=height, g=acceleration due to gravity,.

- If a body rolls down a hill, slope or inclined plane, the workdone by the body depends on the vertical height of the hill. $\mathbf{W}= \boldsymbol{mgh}$, $\boldsymbol{h} = \boldsymbol{lsin\theta}$, l=length of the hill or slope or plane, θ=angle of inclination to the horizontal.

- The workdone by a body climbing a set of steps is given by – $\mathbf{W} = \boldsymbol{mgh}$, m=mass of the body, h=total in covering a flight of steps =**height of one step×number of steps.**

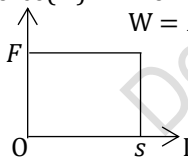
- Workdone due to friction W involves the frictional force F_R and the distance s moved by it. $\mathbf{W} = \boldsymbol{F_R} \times \boldsymbol{s} = \boldsymbol{\mu R} \times \boldsymbol{s} = \boldsymbol{\mu mg} \times \boldsymbol{s}$.

- Force-Distance(displacement) graphs – The area of the figure under a force-distance graph represents the workdone by the body.

Examples are :

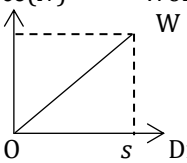
1. Constant force and varying distance moved e.g a push or a pull.

Force(N) Workdone=Area of the rectangle
W = Fs.

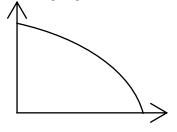


2.Variable force and varying distance moved e.g workdone in stretching an elastic material i.e spring or catapult.

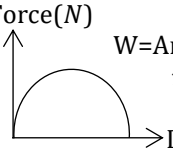
Force(N) Workdone=Area of triangle
W = ½Fe



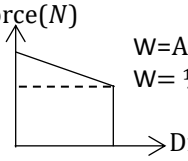
3. Force(N) W =Area of a quadrant= ¼πr^2



4. Force(N)



5. Force(N)



● **Energy –** Eenergy is the ability or capacity to do work.

- Sources of energy – Renewable and non-renewable energy source.

1.Renewable energy source – These are energy sources that are naturally and easily replenished/replaced as they are used e.g solar energy i.e energy from the sun, wind energy i.e energy in wind mills, ocean tides and waves, hydro (water) energy i.e energy generated in dams.

2.Non-renewable energy source – These are energy soures that are not easily replenished as they are depleted e.g petroleum (fossil fuels), coal, nuclear/atomic energy.

- Forms of energy – They include ;

1.Mechanical energy – It is used for mechanical work e.g potential and kinetic energy. **2.Sound energy 3.Solar energy 4.Light energy**

5.Chemical energy – Energy stored in fuels,food and cells/batteries. **6.Thermal(heat) energy –** Total internal energy of a body. **7.Atomic energy(Nuclear energy) –** Energy stored in the nucleus of an atom. **8.Electrical energy.**

- Energy conversion – Energy can be converted from one for to another. Some energy is always lost as heat energy during conversion.

From	To	Device
Mechanical	Electrical	Generator
Electrical	Mechanical	Electric motor
Heat	Mechanical	Turbine
Chemical	Heat	Bunsen burner
Electrical	Heat	Resistor
Heat	Electrical	Thermocouple
Solar	Electrical	Photocell
Chemical Atomic	Electrical Heat	Cell/battery Nuclear reactor
Electrical	Sound	Televison, electric bulb
Sound	Electrical	Microphone

- Energy conversion in electric power station – **I.**Steam engine : heat →mechanical→electrical →light and heat e.g coal or gas power station.
II. Turbine : kinetic→mechanical→electrical→

Demystified Series by Dr. Timothy

light and heat e.g hydroelectric power station.

- Mechanical energy – It is of two types;

1.Potential energy $P.E$ – Energy of a body due to is position or state of rest i.e stored up-energy. $\boldsymbol{P.E = mgh}$. m=mass of the body, h=height of the body. Examples

2.Kinetic energy $K.E$ – Energy posssed by a body due to its motion. $\boldsymbol{K.E = \frac{1}{2}mv^2}$. v=velocity of the body.

- Kinetic energy of a body is varies directly with the square of the velocity of the body $K.E \propto v^2$ (mass=constant).

- Law of conservation of energy – It states that the total energy of an isolated system is constant or energy can neither be created nor destroyed but can be converted from one form to another.

- Conservation of mechanical energy – It states that the total mechanical energy of a system is constant. Total mechanical energy $M.E$ of a system is equal to the sum of its potential energy amd kinetic energy. $\boldsymbol{M.E = P.E + K.E}$.

- Though the total mechanical energy is constant, energy can be converted from $P.E$ to $K.E$ and from $K.E$ to $P.E$.

- $P.E$ – $K.E$ conversion – I.Free fall : For a body falling under gravity, $P.E$ decreases downwards, as heights decreases while $K.E$ increases downwards, as velocity increases or $K.E$ decreases and $P.E$ increases when a body is thrown vertically upward.

- The $K.E$ of a body at a height h just before hitting the ground equals the $P.E$ at that height h or $P.E$ at the top equals K.E during falling.

- The kinetic energy of a body at half-way point $(x = H/2)$ is equal to half of its initial potential energy.

- The initial $K.E$ at the ground level of a body projected vertically upward from the ground is equal to total mechanical energy.

- Percentage of initial potential energy converted to kinetic energy $= \frac{K.E}{P.E} \times \mathbf{100}$.

- Energy dissipated as heat $= \boldsymbol{P.E - K.E}$.

- Energy lost by a falling body falling from a height H onto the floor and rebounds to a height h is given by – **Energy lost $= \boldsymbol{mg(H - h)}$**.

- Workdone-energy conversion :

I.If the workdone on a body is all utilized in setting it in motion thus, **Workdone=change in kinetic energy.** $\boldsymbol{W = \Delta K.E = K.E_2 - K.E_1}$. $W = F \times s$, initial kinetic energy, $\boldsymbol{K.E_1 = \frac{1}{2}mu^2}$, final kinetic energy, $\boldsymbol{K.E_2 = \frac{1}{2}mv^2}$. u=initial velocity, v=final velocity, $\boldsymbol{W = \frac{1}{2}m(v^2 - u^2)}$.

II. Inelastic collision of a body of mass m with another body by an apllied force after moving a distance s with a velocity v, hence the workdone by the force equals the kinetic energy of the body. $\boldsymbol{F \times s = \frac{1}{2}mv^2}$.

● **Power P** is the rate of doing work or rate at which energy is espended, in **Watt(W).**

Power $= \frac{workdone}{time} = \frac{energy}{time}$. $\boldsymbol{P = \frac{W}{t} = \frac{E}{t}}$.

$\boldsymbol{P = \frac{F \times s}{t} = Fv}$. Energy can be $K.E$ or $P.E$.

$\boldsymbol{P = \frac{P.E}{t} = \frac{mgh}{t}}$, $\boldsymbol{P = \frac{K.E}{t} = \frac{\frac{1}{2}mv^2}{t}}$.

- Horse power (Hp) is the unit of power in an engine. 1Hp=746watts=0.746kilowatts.

- The efficiency ϵ is given by –

Efficiency, $\boldsymbol{\epsilon = \frac{Power\ output}{Power\ input} \times 100 = \frac{P_o}{P_i} \times 100}$.

Power input is the power developed in the machine. Power output is the power expended or developed in the load $= \frac{mgh}{t}$.

Examples :

1. Which of the following statements correctly defines quantitatively, the workdone by a force? The product of the force and the distance moved **A.parallel ot the line of action of the force from a reference point** B.perpendicular to the line of action of the force from a reference point C.on a line through the reference point D.by the object.

Workdone involves **parallel distance** while Moment of a force involves **perpendicular distance.**

2. An object of mass 12kg is held at a height of 5 meters above the ground for 30 seconds. The work done within this period is A.600J B.200J C.60J D.**0.**

Work done within this period is zero as there is no displacement of the body.

3. A boy of weight 300N climbs to the top of a hill of height 20m.The workdone by the boy against the force of gravity **A.6000J** B.600J C.320J D.15J**.**

W=mg=300N, h=20m, W=?

$W = mgh = 300 \times 20 = 6000$J or 6kJ.

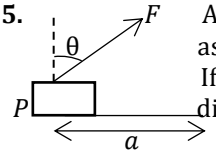
4. A body initially at rest is accelerated at the rate of $0.2ms^{-2}$ for 5s under a constant force of $50N$. The workdone on the body is A.10J B.50J **C.125**J D.250J.

u=0, a=$0.2ms^{-2}$, t=5s, F=$50N$, $s = \frac{1}{2}at^2$, W=?,

$s = \frac{1}{2} \times 0.2 \times 5^2 = 0.1 \times 25 = 2.5$m,

$W = F \times s = 50 \times 2.5 = 125$J.

5.



A force F is applied to a body P as shown in the diagram above. If the body moves through a distance a, which of the following represents the workdone

A.Fa B.$Facos\theta$ C.$Fa/cos\theta$ **D.$\boldsymbol{Fasin\theta}$** E.$Fa/sin\theta$.

The resolved component of the force F in the horizontal direction is $Fsin\theta$, distance s=a,

$W = Fsin\theta \times a = Fasin\theta$.

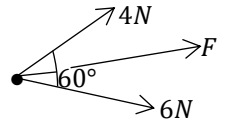
6. A load is pulled 5m along a horizontal floor by a constant force of $20N$ which acts at 30° to the

Demystified Series by Dr Timothy

floor. Cacluate the work done by the force. A.10.0J B.17.3J C.50.0J **D.86.6J** D.100.0J
s=5m, θ=30°, $F = Fcos\theta$, W=?,
$W = 20cos30° \times 5 = 100cos30° = 86.6J$.

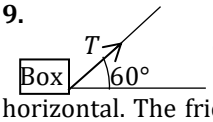
7. A 500N box rests on a horizontal floor. A constant horizontal force is exerted on it so that it moves through 8m. If the coefficient of kinetic friction is 0.22,calculate the workdone on the box **A.880J** B.440J C.400J D.110J.
W=mg=500N, s=8m, μ=0.22, W=?,
$W = F_R \times s = \mu R \times s = \mu mg \times s$,
$W = 0.22 \times 500 \times 8 = 880J$.

8. Two forces of 4N and 6N act on an object at an angle of 60° to each other. If the object is moved through a distance of 5m in the direction of the resultant force. Calculate the work done on the object by the forces. A.51.96J **B.43.59J** C.36.00J D.26.83J E.26.46J.

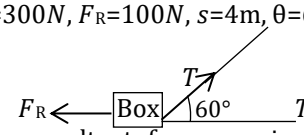


The resultant force F is gotten by cosine rule,
$F^2 = 4^2+6^2+2\times4\times6cos60°$,
$F^2 = 16+36+48(0.5)$,
$F^2 = 52+24 = 76$, $F = \sqrt{76} = 8.718$N, s=5m,
$W = F \times s = 8.718 \times 5 = 43.59$J.

9.



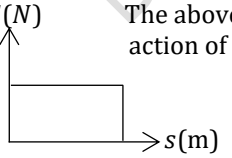
A box of mass 40kg is being dragged along the floor by a rope inclined at 60° to the horizontal. The frictional force between the box and the floor is 100N and the tension on the rope is 300N. How much work is done in dragging the box through a distance of 4m? A.680J B.400J C.200J D.100J
T=300N, F_R=100N, s=4m, θ=60°, m=40kg,



The resultant force moving the block in the horizontal direction along the floor :
$F = Tcos60° - F_R = 300cos60° - 100$,
F =150−100 = 50N, $W = F \times s$ = 50×4= 200J.

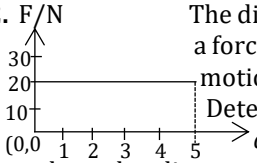
10. The area under a force-distance graph represents A.acceleration B.velocity C.power **D.work** E.momentum.

11. $F(N)$



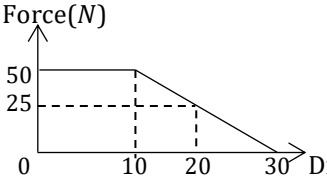
The above graph represents the action of a
A.zero force B.variable force **C.constant force** D.tangential force.

12. F/N



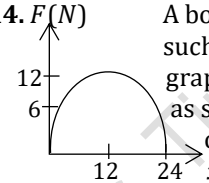
The diagram above illustrates a force-distance graph for the motion of a wooden block. Determine the workdone on the block when move through a distance of 5m. A.4J B.15J C.25J **D.100J.**
Workdone on the block = Area of the rectangle,
W = 20×5 = 100J.

13. Force(N)



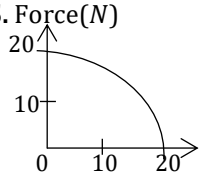
The diagram above shows the force versus displacement graph for a body. The work done on the body during the first 20m of motion is A.1000J **B.875J** C.750J D.500J .
Workdone=Area of the figure under the force-displacement graph.
Workdone during the first 10m=50×10=500J
Workdone during the next 10m i.e between 10m and 20m= $\frac{1}{2}(50+25)(20-10) = \frac{1}{2}\times75\times10 = 375J$
Total workdone during the first 20m of motion=500+375=875J.

14. $F(N)$



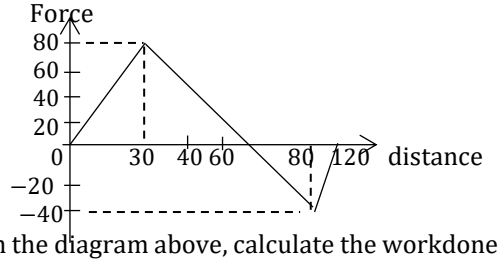
A body under the action of a force such that the force-displacement graph of the body is semicircular as shown above. The workdone on the body by the force in moving through 24m is A.36πJ **B.72πJ** C.144πJ D.288πJ.
Workdone=Area of a semi-circle= $\frac{1}{2}\pi r^2$,
diameter D=24m, $r = \frac{D}{2} = \frac{24}{2} = 12$,
$W = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi \times12^2 = \frac{1}{2}\pi \times144 = 72\pi$J.

15. Force(N)



The force displacement graph of a body is given by the quadrant of a circle,shown in the diagram abovce.The workdone by the force in moving the body through a distance of 20m is A.400πJ B.200πJ **C.100πJ** D.50πJ.
Workdone=Area of a quadrant=$\frac{1}{4}\pi r^2$, r=20m,
$W = \frac{1}{4}\pi \times20^2 = \frac{1}{4}\pi \times400 = 100\pi$J.

15. Force



From the diagram above, calculate the workdone when the particle moves from x=0m to x=80m A.1200J **B.2400J** C.6000J D.7000J.
Workdone from 0m to 70m=Area of the bigger traingle= $\frac{1}{2}\times70\times80$=2800J
Workdone from 60m to 80m=Area of smaller triangle= $\frac{1}{2}\times(80-60)\times-40 = -400$J
Total workdone=2800+(−400)=2400J.

16. Which of the following statements is not

Demystified Series by Dr Timothy

correct? A.The P.E of a body is equal to the work the body can do because of its position B.The P.E is the ability a body possesses to do work because of stored up energy **C.The P.E is the ability a body possesses to do work owing to the movement of its molecules** D.Energy cannot normally be destroyed but can be changed into another form of energy E.A body can have both kinetic and potential energy at the same time

Potential energy is the workdone by a body due to stored up energy,it involves a change in position and does not involve motion or movement of molecules. It is kinetic energy that is the workdone owing to the movement of molecules.

17. A car of mass 800kg initially at rest is accelerated at the rate of $4ms^{-2}$. The kinetic energy of the car after 5s is A.6.40×10^{3}J B.2.56×10^{4}J C.1.60×10^{5}J D.6.40×10^{5}J.

m=800kg, u=0, a=$4ms^{-2}$,t=5s, $K.E$=?,

$v = u + at = 0+at = at$,

$K.E = \frac{1}{2}mv^2 = \frac{1}{2}m(at)^2 = \frac{1}{2}\times800(4\times5)^2$

$K.E = 400\times20^2 = 1.6\times10^5$J.

18. When the speed of a car is halved, its kinetic energy is A.doubled **B.quartered** C.halved D.quadrupled.

$K.E = \frac{1}{2}mv^2$, $K.E \propto v^2$ ($\frac{1}{2}m$=constant).

If the speed v is halved ($\frac{1}{2}$), $K.E \propto v^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ hence, the kinetic energy is quartered.

19. An object of mass 0.25kg moves at a height h above the ground with a speed of $4ms^{-1}$. If its mechanical energy at this height is 12J, determine the value h (g=$10ms^{-2}$) A.0.8m **B.4.0m** C.4.8m D.5.6m.

m=0.25kg, v=$4ms^{-1}$, $M.E$=12J, g=$10ms^{-2}$, h=?,

Total mechanical energy at this height is the sum of the potential and kinetic energy of the body.

$M.E = P.E + K.E = mgh + \frac{1}{2}mv^2$,

$M.E = m(gh + \frac{1}{2}v^2)$, $12 = 0.25(10h + \frac{1}{2}\times4^2)$,

$10h + 8 = \frac{12}{0.25} = 48$, $10h = 48-8 = 40$,

$h = \frac{40}{10} = 4$m.

20. A body of mass 10.00kg falling freely attains a velocity of $9.00ms^{-1}$ at a height of 20.00m above the ground. Calculate the total energy at the height A.1800J B.2000J C.2045J **D.2405J** E.2810J

m=10kg, v=$9ms^{-1}$, h=20m, $M.E$=?,

Total energy at that height above the ground,

$M.E = P.E + K.E = m(gh + \frac{1}{2}v^2)$

$M.E = 10(10\times20+\frac{1}{2}\times9^2) = 10(200+40.5)$,

$M.E = 2405J$.

21. A diver steps off a diving platform that is 10m above water. If there is no air resistance during the fall, there will be a decrease in the diver's **A.gravitational potential energy** B.total mechanical energy C.kinetic energy D.momentum

During fall the total mechanical energy is constant, but the potential energy decreases while its kinetic energy increases.

22. A body of mass 1000kg is released from a height of 10m above the ground. Determine its kinetic energy just before it strikes the ground (g=$10ms^{-2}$) A.10J B.10^3J C.10^4J **D.10^5J.**

m=1000kg, h=10m, g=$10ms^{-2}$,

The kinetic energy just before striking the ground equals the potential energy at the height from which it was released.

$K.E = P.E = mgh = 1000\times10\times10 = 10^5$J.

23. A body of mass 0.6kg is thrown vertically upwards from the ground with a velocity of $20ms^{-1}$. Calculate its potential energy at its maximum height (g=$10ms^{-2}$) **A.120J** B.240J C.360J D.480J.

m=0.6kg, u=$20ms^{-1}$, g=$10ms^{-2}$, $P.E$=?,

$P.E$ at maximum height is equal to the initial $K.E$ from the point of through.

$P.E$ = Initial $K.E = \frac{1}{2}mu^2 = \frac{1}{2}\times0.6\times20^2$,

$P.E = 0.3\times400 = 120$J.

24. A body of mass 4kg is acted by a constant force of $12N$ for 3s. The kinetic energy gained by the body at the end of the time is **A.162J** B.144J C.72J D.81J.

m=4kg, F=$12N$, t=3s, $K.E$=?, $F = ma$, $a = \frac{v}{t}$,

$F = \frac{mv}{t}$, $v = \frac{Ft}{m}$, $K.E = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{Ft}{m}\right)^2$,

$K.E = \frac{1}{2}\times4\times\left(\frac{12\times3}{4}\right)^2 = 2\times9^2 = 162$J.

25. An object of mass 100g is projected vertically upward from the ground level has a velocity of $20ms^{-1}$ at a height of 10m. Calculate the initial kinetic energy at the ground level A.10J B.20J **C.30J** D.50J.

m=100g=0.1kg, v=$20ms^{-1}$, g=$10ms^{-1}$,

$v^2 = u^2-2gh$ (upward projection),

$u^2 = v^2+2gh$ =$20^2+2\times10\times10$=400+200,

u^2=600, Initial kinetic energy= $\frac{1}{2}mu^2$

Initial kinetic energy= $\frac{1}{2}\times0.1\times600$=30J. OR

Initial kinetic energy at the ground level equal total mechanical energy at height of 10m above the ground. Initial $K.E = M.E = P.E + K.E$,

Initial $K.E = m(gh + \frac{1}{2}v^2 = 0.1(10\times10+\frac{1}{2}\times20^2)$

Initial $K.E = 0.1(100+200) = 30J$.

26. A stone of mass 1kg is dropped from a height of 10m above the ground and falls freeky under gravity, its kinetic energy 5m above the ground is equal to A.its kinetic energy on the ground B.twice its initial potential energy C.its initial potential energy **D.half its initial potential energy.**

The kinetic energy of a body at half way point i.e 5m is equal to half its initial potential energy at 10m.

27. A ball of mass 100g falls from a height of 5m

Demystified Series by Dr Timothy

onto a floor and rebounds to a height of 3m. What energy is loss as a result of the impact on the floor? **A.2J** B.20J C.100J D.1000J E.2000J.
Energy lost by a falling body falling from a height H onto the floor and rebounds to a height h is ;
Energy lost $= mg(H-h)$, m=100g=0.1kg,
g=10ms^{-2}, H=5m, h=3m,
Energy lost $= 0.1\times10(5-3) = 1(2) = 2$J.

28. An object is moving with a velocity 5ms^{-1}. At what height must a similar body be situated to have a potential energy equal in value with the kinetic energy of the moving body? (g=10ms^{-2}) A.20.0m **B.1.3m** C.1.0m D.25.0m.
v=5ms^{-1}, m is the same for both bodies, as they are similar. $P.E = K.E, mgh = \frac{1}{2}mv^2$
$gh = \frac{1}{2}v^2$, $h = \frac{v^2}{2g} = \frac{5^2}{2\times10} = \frac{25}{20} = 1.25$m $\approx$1.3m.

29. An object of mass 2.0kg falls freely from a height of 100m onto muddy ground and comes to rest in the mud. Calculate the distance the body oenetrates if the average retarding force is 20.000N A.0.1m B.1.0m C.2.0m C.10.00m **E.100.0m.**
m=2kg, h=100m, F=20N, g=10ms^{-2}, s=?,
$P.E$ of the body equals the workdone in penetrating into the mud, $P.E = W$,
$mgh = F\times s$, $2\times10\times100 = 20\times s$,
$s = \frac{2000}{20} = 100$m.

30. A force of 10N acting continuously increases the kinetic energy of an object from $20J$ to $60J$. The distance moved by the object is then A.400m B.200m C.60m D.20m **E.4m.**
F=$10N$, initial kinetic energy, $K.E_1$=20J, final kinetic energy, $K.E_2$=60J, F=10N, s=?,
Work done-energy conversion –
Workdone=change in kinetic energy,
$W = \Delta K.E$, $F\times s = K.E_2 - K.E_1$,
$10\times s = 60-20 = 40$, $s = \frac{40}{10} = 4$m.

31. A constant force acting on a body of mass 4kg increses the velocity of the body from 20ms^{-1} to 40ms^{-1}. Calculate the workdone by the force. A.1.2$\times10^3$J **B.2.4$\times10^3$J** C.3.6$\times10^3$J D.4.8$\times10^3$J.
m=4kg, u=20ms^{-1}, v=40ms^{-1}, W=?,
$W = \Delta K.E = \frac{1}{2}m(v^2-u^2) = \frac{1}{2}\times4\times(40^2-20^2)$,
$W = 2\times1200 = 2400J = 2.4\times10^3J$.

32. A body rolls down a slope from a height of 100m. Its velocity at the foot of the slope is 20ms^{-1}. What percentage of its initial potential energy is converted into kinetic energy? A.40% B.35% **C.20%** D.15%
mass of the body=m, h=100m, v=20ms^{-1},
Initial $P.E = mgh = m\times10\times100$=$1000m$J
$K.E$ at the foot of the slope= $\frac{1}{2}mv^2$
$K.E = \frac{1}{2}\times m\times20^2$=$200m$J,
Percentange of the initial $P.E$ converted to $K.E = \frac{K.E}{P.E}\times100$=$\frac{200}{1000}\times100 = 20\%$.

33. An object slides down an inclined plane from an initial height of 30m. Its velocity at the foot of the plane is 20ms^{-1}.The percentage of its initial potential energy which is dissipated as heat is **A.33$\frac{1}{3}$%** B.50% C.66$\frac{2}{3}$% D.75%.
mass of the body=m, h=30m, v=20ms^{-1},
Initial $P.E = mgh = m\times10\times30$=$300m$J,
$K.E$ at the foot of the slope= $\frac{1}{2}mv^2$,
$K.E = \frac{1}{2}\times m\times20^2 = 200m$J. Energy dissipated as heat $= 300m - 200m = 100m$J.
Percentage of the initial $P.E$ dissipated as heat $= \frac{100m}{300m}\times100 = 33\frac{1}{3}\%$.

34. Which of the following correctly describe the energy changes in the generation of light by a hydroelectric power station? A.Electrical – mechanical – potential – light B.Potential – mechanical – electrical – light C.Mechanical – sound – electrical – light **D.Kinetic – mechanical – electrical – light.**
The kinetic energy flowing water is converted into mechanical energy which is used to rotate turbine which in turn generates a relative motion between a magnet and a coil to produce electrical and light energy.

35. Two identical boxes are to be carried up a flight of steps.A man X carries one box up in two ,but another man Y who stopped twice to rest, completed the assignment in six mimutes.What observation is correct? A.It is impossible to determine the workdone because a machine was not used **B.the power developed by X is more than the average power developed by Y** C.the energy developed by X is more three times that expended by Y D.the workdone by X is three times that done by Y E.the workdone by Y was less than that done by X.
The energy developed or workdone by both men is the same as they moved through the same flight of steps. $\boldsymbol{P \propto \frac{1}{t}}$ i.e the higher the power developed by a body the lesser the time taken to do a certain amount of work.

36. A body is pulled through a distance of 500m by a force of 20N. If the power developed is 0.4kW, calculate the time for which the force acts. A.250.0s **B.25.0s** C.2.5s D.0.5s.
s=500m, F=$20N$, P=0.4kW=400W, t=?,
$P = \frac{W}{t} = \frac{F\times s}{t}$, $t = \frac{F\times s}{P} = \frac{20\times500}{400} = 25$s.

37. A car of mass Mkg is accelerated uniformly from rest by an engine of power P. The minimum time it will take to attain a speed V is A.$\frac{MV}{P}$ B.$\frac{MV}{2P}$ **C.$\frac{MV^2}{2P}$** D.$\frac{MV^2}{P}$.
$P = \frac{K.E}{t} = \frac{\frac{1}{2}mv^2}{t} = \frac{mv^2}{2t}$, $t = \frac{mv^2}{2P} = \frac{MV^2}{2P}$.

38. Starting from rest, a car of mass 1000kg accelerates steadily to 20ms-1 in 10s. The

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average power developed in the time period is A.0.2kW B.40kW C.10kW D.15kW **E.20kW.**

m=1000kg, v=20ms^{-1}, t=10s, P=?,

$P = \frac{K.E}{t} = \frac{mv^2}{2t} = \frac{1000\times20^2}{2\times10} = 20000W$ =20kW.

39. A girl of mas 48kg runs up 25 steps, each of height 0.2m to reach the first floor of a storey building. If the power expended by the girl is 400W, calculate the time taken (g=10ms-2) A.2s B.3s C.4s D.5s **E.6s.**

m=48kg, number of steps=25, height of one step=0.2m, Total height, h=number of steps × height of one step=25×0.2=5m, P=400W,

$P = \frac{P.E}{t} = \frac{mgh}{t}$, $t = \frac{mgh}{P} = \frac{48\times10\times5}{400} = 6$s.

40. A pump lift water into an overhead tank at a height of 12m at the rate of 5kgs^{-1}. The power of the pump is A.60W **B.600W** C.720W D.1200W.

h=12m, mass per unit time = $\frac{m}{t}$ = 5kgs^{-1}, g=10ms^{-2}, P=?, $P = \frac{mgh}{t} = \frac{m}{t}gh = 5\times10\times12 = 600$W.

41. A car travelling with a uniform velocity of 30ms-1 along a horizontal road overcomes a constant frictional force of 600N. Calculate the power of the engine of the car **A.18kW** B.20kW C.180kW D.200kW.

F=frictional force overcomed=600N, v=30ms^{-1}

$P = Fv = 600\times30 = 18000W = 18kW$.

42. A pump is used to raise water from a depth of 30m to fill a reservoir of volume 1800m^3 in 5 hours. Calculate the power of the pump (density of water=1000kgm^{-3},g=10ms^{-2}) A.5kW B.10kW C.15kW **D.20kW.**

h=depth of water raised=30m, V=volume of water=1800m^3, t=5hrs=5×3600=18000s, g=10ms^{-2}, ρ=density of water=1000kgm^{-3},

$\rho = \frac{m}{V}$, $m = \rho V$, $P = \frac{mgh}{t} = \frac{\rho Vgh}{t}$,

$P = \frac{1000\times1800\times10\times30}{18000} = 20000$W = 20kW.

43. A water pump of 1.2kW rating pumps 480kg of water into an overhead tank at a height of 5m in 30 seconds. The efficiency of the pump is A.90.0% B.80.0% **C.66.7%** D.62.5%.

Power input=1.2kW=1200W, m=480kg, h=5m, t=30s, g=10ms^{-2}, Power input=$\frac{mgh}{t}$,

Power input= $\frac{480\times10\times5}{30} = 800$W,

Efficiency, $\epsilon = \frac{Power\ output}{Power\ input}\times100 = \frac{P_o}{P_i}\times100$,

Efficiency, $\epsilon = \frac{Work\ output}{Work\ input}\times100$

$\epsilon = \frac{800}{1200}\times100$= 66.7%.

44. A steam engine of efficiency 70% burns 20g of coal to produce 10kJ of energy. If it burns 200g of coal per second, calculate the output **A.70kW** B.80kW C.100kW D.200kW E.300kW.

The energy obtained by burning the coal i.e fuel, inside the steam engine is the energy input. 20g of coal give 10kJ of energy when burnt, but the steam engine actually burns 200g of coal per second.

The energy obtained when 200g of coal per second can be gotten by direct proportion.

20g of coal → 10kJ or 10000J of energy

200g of coal per second → xJ of energy

$x = \frac{200\times10000}{20} = 100000J$ per second.

100000Js^{-1} or 100000W is the power input.

$\epsilon = \frac{P_o}{P_i}\times100$, 70= $\frac{P_o}{100000}\times100$,

$P_o = \frac{70\times100000}{100} = 70000$W = 70kW.

45. The area under a force-velocity graph is equal to A.Energy B.Momentum **C.Power** D.Acceleration.

Area under force-velocity time graph = F × v ,

Power P = Force×velocity = Fv.

46. A car travelling at 30ms^{-1} overcomes a firctional force of 100N while moving. Calculate the power developed by the engine (1hp=0.75kW) A.0.23hp B.0.40hp **C.4.00hp** D.4.40hp.

F=100N, v=30ms^{-1}, P=?, $P = Fv$,

$P = 100\times30 = 3000$W = 3kW.

0.75kW → 1hp ∴ 3kW = $\frac{3kW\times1hp}{0.75kW} = 4.00$hp.

Jamb past questions on work, energy and power :

[1984/39,1983/17,1988/11,1992/6,1997/3, 2000/5,2006/29,2008/40,2009/10,2013/101984/43,1985/41,1989/6,2010/8,2014/13,1979/44,2011/2,2001/3,1994/9,1997/12,1999/3, 1991/7,2013/9,1986/5,1985/12,1987/9,2003/5,2004/12,2008/11,1984/22,1991/9,2002/5,1990/10,2003/1,2,2007/41,2013/13]

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CHAPTER 5 – FORCE FIELD AND GRAVITATIONAL FIELD

● **Force field or field forces –** These are forces which acts on a body in a region or space without contact with the body. All field forces originates from the centre and obey inverse square law e.g gravitational field forces, magnetic field forces and electric field forces.

- Gravitational field forces are always attractive only while Magnetic and electric field forces are both attractive and repulsive.

- Conservative field – is a field where workdone by a force depends on the **displacement**, **the workdone in moving a body around a closed path is zero** e.g gravitational field.

- Non-conservative field or dissipative field – is a field where the workdone by a force depends on **distance** covered by the force e.g friction and viscosity.

- Gravitational field – is a region or space around a mass in which the gravitational force of the mass can be felt. Every massive body has a gravitational field around it.

- Properties of gravitational field :

1.They act on massive bodies or planets.

2.The field lines always ends at a mass.

3.They are usually attractive.

4.The force does not cause a deflection on the massive bodies in the field.

5.The forces acts in the same direction with the field.

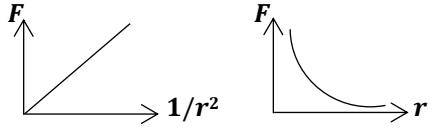
- Newton's law of universal gravitation – force exerted on two massive bodies m_1 and m_2 is directly proportional to the product of their masses and inversely proportional to the square of their distance apart r^2 . $\boldsymbol{F \propto \frac{m_1 m_2}{r^2}}$, $\boldsymbol{F = \frac{Gm_1 m_2}{r^2}}$

G=universal gravitational constant=$6.67\times10^{-11}Nm^2kg^{-2}$.

- If both masses are double and the distance between them is doubled, the gravitational force of attraction remains constant or the same.

- $\boldsymbol{F \propto \frac{1}{r^2}}$ thus, force F increases as distance r reduces. $\boldsymbol{F_1 r^2_1 = F_2 r^2_2}$.

- Graphical representation of the variation of gravitational force F between two objects and the distance r between their centres :



- A body revolving round the earth is kept on its orbit by two opposite forces: force(acting towards the centre) and centrifugal forces (acting away from the centre).

- The force with which an object is attracted to the earth is Weight or force of gravity.

- The weight of an object on earth is due to the gravitational effect of the earth on the object.

The weight of an object on the moon is due to gravitational effect of the moon on the object.

- Weightlessness – A state when a body feels no weight or in the absence of gravitational force. It occurs at infinite(great) height above the earth due to decreased force of attraction or acceleration in space , $\mathbf{W \propto \frac{1}{r^2}}$.

- During weightlessness –

I.The reaction on a body and the resultant force acting on a body is zero, the acceleration of the body equal to the gravitational acceleration , a=g.

II. The centripetal force required by the moving body is equal to or provided by the gravitational force of attraction i.e $F_C = F_g$, as the weight is equal to zero , W=0.

- Factors affecting weight W or acceleration due to gravity g (why they vary) :

I.Shape of the earth(earth is not spherical but elliptical or bulge effect) – The value of g at the poles is greater than at the eqauator $\mathbf{g_{poles} > g_{equator}}$, due to the shape of the earth as r is smaller at the poles that at the equator. Thus, a body weighs more at the poles than the equator.

II.Rotation of the earth on its axis(centripetal force effect) – The centripetal force required for the rotation of the earth is provided by part of the total gravitational force, $F_g = F_c + W$,

$F_c = F_g - W$

III. Variation (or Change) in latitude, longitude and altitude (or height).

- The density of the earth does not affect the value of g or W.

- The weight of a body depends on its mass m and acceleration due to gravity g. $\boldsymbol{W = mg}$.

- Weight of a body changes with the variation in acceleration due to gravity, $\boldsymbol{W \propto g}$ (the mass is constant).

- The weight of a body at different planets or different point in the earth surface and acceleration due to gravity at that point is related by : $\boldsymbol{W \propto g}$, $\boldsymbol{\frac{W_1}{g_1} = \frac{W_2}{g_2}}$.

- Mass of a body is constant at different locations as it is independent of acceleration due to gravity.

- The acceleration due to gravity on earth depends on the mass of the earth *M* (and not the mass of the object) and the radius of the earth *R*.

- At the earth surface – $\boldsymbol{g = \frac{GM}{R^2}}$.

- At an altitude or height h above the earth surface – $\boldsymbol{g^1 = \frac{GM}{(R+h)^2}}$. M=mass of the earth, R=radius of the earth=distance of a body from the earth's centre.

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- $\frac{g^1}{g} < 1$, $g^1 < g$ hence, acceleration due to gravity decrease with altitude/height ($g \propto 1/h$).
- The acceleration due to gravity on the earth surface g and acceleration due to gravity at height h above the earth surface g^1 are related by – $g^1 = \frac{g}{{d_R}^2}$, ${d_R}^2 = \frac{R+h}{R}$ OR $g^1 = \frac{gR^2}{(R+h)^2}$.
- The distance from the earth centre to the earth surface is equal to the radius of the earth R. If a body is at distance R from the earth surface then, its distance from the earth centre is $2R$.
- The acceleration due to gravity on moon is one-sixth the value on earth. $g_{\text{moon}} = \frac{1}{6} g_{\text{earth}}$. Hence, a body weighs more on earth than on moon.
- The acceleration due to gravity in different planets of mass M_1 and M_2 and radius R_1 and R_2 are related by – $\frac{g_2}{g_1} = \frac{M_2}{M_1}\left(\frac{R_1}{R_2}\right)^2$ [$g \propto M \propto \frac{1}{R^2}$].
- **Gravitational field intensity or force intensity E_G –** at any point in a gravitational field is defined as the force per unit mass on a body at that point. $E_G = \frac{F}{m} = \frac{GM}{r^2}$.
- $E_1 r_1^2 = E_2 r_2^2$, GM =**constant.**
- **Gravitational field intensity is the same as acceleration due to gravity (force per unit mass).**
- When a massive body moves in the same (downward) or opposite (upward) direction of a uniform gravitational force field, will experience a constant or uniform acceleration and deceleration of 9.8ms^{-2} respectively.
- **Gravitational potential V –** at a point is the workdone W in a taking a unit mass m from infinity to that same point. $V = gr = E_G r$.
- Gravitational potential at the earth surface where r =radius of the earth, R is – $V = \frac{GM}{R}$.
- The gravitational potential at any point is – $V = -\frac{Gm}{r}$. The negative sign indicvates that the potential at infinity i.e zero, is higher than that close to the mass producing the gravitational field.
- **Gravitational potential energy $P.E$ –** It is the workdone in taking a mass m from infinity to the earth surface of mass M. $P.E = -mV = -\frac{GMm}{r}$.
- **Gravitational kinetic energy $K.E$ –** This is half the gravitational potential energy but of opposite sign. $K.E = \frac{P.E}{2} = \frac{mV}{2} = \frac{GMm}{2r}$.
- **Work is done against gravity when a body is lifted or raised above the ground (energy is absorbed). Work is done by gravity when a body is brought to the ground from a height (energy is released).**
- **Escape velocity v_e -** It is the minimum velocity with which an object would just escape from the influence of the earth's gravitational field.

$v_e = \sqrt{2gR} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2P.E}{m}}$.

- **Satellite** are small bodies which move in orbits round a large astronomical body. **Natural satellites** are the earth(or other planets) moving in orbits round the sun or the moon moving in orbit round the earth. **Artificial satellites** are space crafts.
- Orbital speed or speed of objects/satellite in gravitational field is given by – $v_s = \frac{2\pi r}{T}$, r=orbital radius, T=orbital period,

$v_s = \sqrt{gR} = \sqrt{\frac{GM}{R}} = \sqrt{\frac{P.E}{m}}$.

- The escape velocity of a satellite and the speed of the satellite round its obit are related by – $v_e = v_s\sqrt{2}$.
- With a velocity of about 8kms^{-1}, a satellite moves in a circular orbit close to the earth's surface.
- With a velocity greater than 8kms^{-1} but less than 11kms^{-1}, a satellite moves in an elliptical orbit round the earth.
- With a velocity above 11kms^{-1}, the satellite completely escapes from the gravitational attraction of the earth.
- **Geosynchronous or Parking orbit – A satellite is in a parking or geosynchronous orbit if its period round its orbit exactly equals the period of rotation of the earth(as it turns about its axis within 24hrs).** The satellite will stay on the same place over the earth as the earth rotates, as they are at constant distance equal to the radius of the earth. The satellite is said to be **geostationary.**
- Parking orbits can be used to transmit television programmes continuously from one part of the world to another.
- Period T and speed of a satellite v_s is independent of size or mass of a body but depends on radius of the earth and acceleration due to gravity. $T = \sqrt{\frac{4\pi^2 r^3}{gR^2}}$, $v_s = \sqrt{\frac{gR^2}{r}}$.
- **Kepler's Laws –** Kepler's law of planetary motion describe the orbits of objects about the sun.

1st Law – The planets each travel along an ellipse with the sun at one focus.

2nd Law – The line joining the sun and the planets sweeps out equal areas in equal times interval.

3rd Law – The square of the period T of revolution of the planets are proportional to the cubes of their mean distance r from the sun i.e $T^2 \propto r^3$.

Examples :

1. Which of the following are contact forces? I.force of tension II.force of friction III.magnetic

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force IV.force of reaction A.I,II and III **B.I,II and IV** C.I,III and IV D.II,III and IV E.I and IV **.**
Magnetic force, gravitational force and electric force are field forces.

2. Which of the following statements is true of the law of universal gravitation? A.Any twobodies attract each other with a force which is proportional to the difference of their masses B.Any two bodies attract each other with a force which is inversely proportional to the square of the distance between them **C.Any two bodies attract each other with a force which is proportional to the product of their masses and inversely proportional to the square of the distance between them** D.Any two bodies attract each other with a force which is directly proportional to the square of the distance between them E.Any two bodies attract each other with a force which is proportional to the sum of their masses.

3. Which of the following statements about the universal gravitational constant, G is true? A.It is a fundamental quantity **B.It has dimension $M^{-1}L^3T^{-2}$** C.Its S.I unit is Nm^2kg^{-1} D.It is a vector quantity.
The universal gravitational constant is a derived quantity and a scalar quantity. Its unit is Nm^2kg^{-2}. Its value is constant and does not vary.

4. The mass of the earth is 6.0×10^{24}kg and that of the moon is 7.0×10^{22}kg. If the distance between them is 4.0×10^8m, calculate the force of attraction between them
[G=$6.7\times10^{-11}Nm^2kg^{-2}$] A.$1.8\times10^5$N B.$1.8\times10^{10}$N C.$1.8\times10^{15}$N **D.$1.8\times10^{20}$N .**
m_1=6.0×10^{24}kg, m_2=7.0×10^{22}kg, r=4.0×10^8m, G=$6.7\times10^{-11}Nm^2kg^{-2}$,
$F=\frac{Gm_1m_2}{r^2}=\frac{6.7\times10^{-11}\times6.0\times10^{24}\times7.0\times10^{22}}{(4\times10^8)^2}$,
$F=\frac{281.4\times10^{35}}{16\times10^{16}}=1.76\times10^{20}\approx1.8\times10^{20}$N.

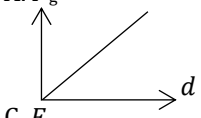
5. Two objects P and Q of masses 100kg and 200kg respectively are 1.2m apart on a horizontal surface. Calculate the net gravitational force acting on Q (G=$6.67\times10^{-11}Nm^2kg^{-2}$)
A.1.11×10^{-7}N towards P B.1.11×10^{-7}N towards Q **C.9.26×10^{-8}N towards P** D.9.26×10^{-8}N towards Q.
m_1=100kg , m_2=200kg , r=1.2m, G=6.67×10^{-11}N m^2kg^{-2}, F=?, $F=\frac{Gm_1m_2}{r^2}=\frac{6.67\times10^{-11}\times100\times200}{(1.2)^2}$,
$F=\frac{1.334\times10^7}{1.44}=9.26\times10^{-8}$N. The net gravitational force on Q is same as the gravitational force between P and Q. Gravitaional force is force of attraction, hence Q is attracted to P or move towards P.

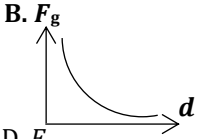
6. Calculate the force on a mass 10kg placed on the earth's surface. [Radius of the earth= 6.4×10^6m, mass of the earth=6.0×10^{24}kg, G=$6.7\times10^{-11}Nm^2kg^{-2}$) A.9.8N **B.$9.8\times10^1$N** C.$6.3\times10^8$N D.$1.0\times10^{24}$N E.$9.8\times10^{25}$N.
m_1=10kg, m_2=mass of the earth=6.0×10^{24}kg, r=radius of the earth surface=6.4×10^6m,
$F=\frac{Gm_1m_2}{r^2}=\frac{6.7\times10^{-11}\times10\times6.0\times10^{24}}{(6.4\times10^6)^2}=\frac{40.2\times10^{14}}{40.96\times10^{12}}$,
$F=0.98\times10^2=9.8\times10^1$N.

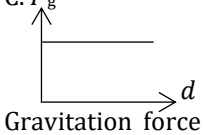
7. A planet has a mass m_1 and is at a distance r_1 from the sun. A second planet has mass m_2=$10m_1$ and is at a distance of r_2=$2r_1$ from the sun. Determine the ratio of the gravitational force experienced by the planets A.1:5 **B.2:5** C.3:5 D.4:5.
The gravitational force between m_1 and the sun of mass M is – $F_1=\frac{GMm_1}{{r_1}^2}$,
The gravitational force between m_2 and the sun of mass M is – $F_2=\frac{GMm_2}{{r_2}^2}=\frac{GM10m_1}{(2r_1)^2}=\frac{10GMm_1}{4{r_1}^2}$,
The ratio of the gravitational forces,
$F_1:F_2=\frac{F_1}{F_2}=\frac{\left(\frac{GMm_1}{{r_1}^2}\right)}{\frac{GM10m_1}{4{r_1}^2}}=\frac{GMm_1}{{r_1}^2}\times\frac{4{r_1}^2}{GM10m_1}=\frac{2}{5}$=2:5.

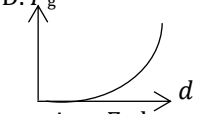
8. A rocket is fired from the earh's surface to a distant planet. By Newton's law of universal gravitation, the force F will **A.increase as r reduces** B.increases as G varies C.remains constant D.increases as r increases.
$F\propto\frac{1}{r^2}$, the force F increases as r reduces. Universal gravitational constant G does not vary, it is constant.

9. Which of the sketched graphs below illustrates the correct variation of the gravitational force F_g between two objects and the distance d between their centres?

A. 

B. 

C. 

D. 

Gravitation force of attraction F_g betweeb two objects is inversely proportional to the distance between them. Thus F_g decreases as r increases as shown in the graph in option B.

10. If the distance between two suspended masses 10kg each is tripled, the gravitational force of attraction between them is reduced by A.one-quarter **B.one-ninth** C.one half D.one third.
F_1=F, r_1=r, F_2=?, r_2=$3r$, $F_1{r_1}^2=F_2{r_2}^2$,
$F\times r^2=F_2\times(3r)^2$, $F_2=\frac{Fr^2}{9r^2}=\frac{F}{9}$, force is one-ninth its initial value. Alternatively,
$F\propto\frac{1}{r^2}$, the distance r is tripled, $F=\frac{1}{3^2}=\frac{1}{9}$, one-ninth the initial value.

11. The force of 200N acts between two objects at a certain distance apart. The value of the force when the distance is halved is A.100N B.200N

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C.400N **D.800N.**
F_1=200N, $r_1=r$, F_2=?, $r_2=r/2$, $F_1r_1^2=F_2r_2^2$,
$200\times r^2=F_2\times(r/2)^2=F_2\times r^2/4$,
$F_2=\frac{200\times r^2\times 4}{r^2}=800$N. Alternatively,
$F\propto\frac{1}{r^2}$, the distance is halved, $F=\frac{1}{(1/2)^2}=\frac{1}{1/4}$,
$F=4$, four times the original value,
$F_2=4\times200=800$N.

12. The acceleration due to gravity may be defined as the force A.of attraction of the sun on the earth B.with which the earth revolves around the sun **C.with which the earth attracts one-kilogram mass** D.of the moon on the earth.

13. A body moves in a circular orbit of radius $4R$ round the earth. Express the acceleration of free fall due to gravity of the body in terms of g [R=radius of the earth, g=acceleration of free fall due to gravity on the earth's surface] A.$\frac{16}{g}$ **B.$\frac{g}{16}$** C.$\frac{4}{g}$ D.$\frac{g}{4}$ E.$\frac{g}{8}$.
$g^1=\frac{g}{{d_R}^2}$, ${d_R}^2=\frac{R+h}{R}=\frac{4R}{R}=4$, $g^1=\frac{g}{4^2}=\frac{g}{16}$. OR
$g\propto\frac{1}{r^2}$, $g_1r_1^2=g_2r_2^2$, $g_1=g$, $r_1=R$, g_2=?, $r_2=4R$,
$g\times R^2=g_2\times(4R)^2$, $g\times R^2=g_2\times16R^2$,
$g_2=\frac{gR^2}{16R^2}$, $g_2=\frac{g}{16}$.

14. What is the acceleration due to gravity 'g' on the moon if g is 10ms^{-2} on the earth? A.0.10ms^{-2} B.0.74ms^{-2} **C.1.67ms^{-2}** D.10.0ms^{-2}.
$g_{\text{moon}}=\frac{1}{6}g_{\text{earth}}=\frac{1}{6}\times10=1.67$ms^{-2}.

15. The earth is four times the size of the moon and the acceleration due to gravity on the earth is 80 times that on the moon. The ratio of the mass of the mass of the moon to that the earth is A.1:320 **B.1:1280** C.1:80 D.1:4
R_e=radius(size) of the earth, R_m=radius(size) of the moon, g_e=acceleration due to gravity on earth, g_m=acceleration due to gravity on moon, $R_e=4R_m$, $\frac{R_e}{R_m}=4$, $g_e=80g_m$, $\frac{g_m}{g_e}=\frac{1}{80}$, $M_m:M_e$=?,
$\frac{g_2}{g_1}=\frac{M_2}{M_1}\left(\frac{R_1}{R_2}\right)^2$, $\frac{g_m}{g_e}=\frac{M_m}{M_e}\left(\frac{R_e}{R_m}\right)^2$, $\frac{1}{80}=\frac{M_m}{M_e}(4)^2$,
$\frac{1}{80}=\frac{M_m}{M_e}\times16$, $\frac{M_m}{M_e}=\frac{1}{80}\times\frac{1}{16}$, $\frac{M_m}{M_e}=\frac{1}{1280}$,
$M_m:M_e=1:1280$.

16. The mass of the moon is $\frac{1}{81}M$ and its radius is $\frac{1}{4}R$, where M and R are mass and radius of the earth respectively. Calculate the acceleration due to gravity on the surface of the moon (Take g on earth=9.8ms^{-2}) **A.1.94ms^{-2}** B.7.56×10^{-3}ms^{-2} C.1.27ms^{-2} D.9.8ms^{-2}
$M_m=\frac{1}{81}M$, $\frac{M_m}{M}=\frac{1}{81}$, $R_m=\frac{1}{4}R$, $\frac{R}{R_m}=4$, g_e=9.8ms^{-2},
g_m=?, $\frac{g_2}{g_1}=\frac{M_2}{M_1}\left(\frac{R_1}{R_2}\right)^2$, $\frac{g_m}{g_e}=\frac{M_m}{M_e}\left(\frac{R_e}{R_m}\right)^2$,
$\frac{g_m}{9.8}=\frac{1}{81}\times42$, $g_m=9.8\times\frac{1}{81}\times16=1.94$ms^{-2}.

17. What is the value of the acceleration due to gravity at a point 4000km above the earth's surface? (G=6.67×10^{-11}Nm^{2}kg^{-2}, r_E=6400km, M_E=6.0×10^{24}kg) A.6.3ms^{-2} **B.3.7ms^{-2}** D.0.4ms^{-2} D.9.8ms^{-2}
G=6.67×10^{-11}Nm^{2}kg^{-2}, radius of the earth, $r_E=R$=6400km=6400×10^{3}m, height above earth surface, h=4000km=4000×10^{3}m, mass of the earth, $M_E=M$=6.0×10^{24}kg,
g^1= acceleration due to gravity at a height h above the earth's surface,
$g^1=\frac{GM}{(R+h)^2}=\frac{6.67\times10^{-11}\times6\times10^{24}}{(6400\times10^3+4000\times10^3)^2}$,
$g^1=\frac{40.02\times10^{13}}{(10400\times10\text{^}3)^2}=\frac{40.02\times10^{13}}{1.0816\times10^{14}}=3.7$ms^{-2}.

18. If a particle of constant mass m moves in the direction of a uniform gravitational force field, its acceleration will A.increase B.decrease C.uniform **D.be constant.**
When a massive body moves in the same (downward) or opposite (upward) direction of a uniform gravitational force field, will experience a constant or uniform acceleration and deceleration of 9.8ms^{-2} respectively.

19. The acceleration of free fall due to gravity in a certain location is 9.707ms^{-2}, calculate the gravitational attraction a piece of cork of mass 50g at this location A.1.94×10^{-1}N **B.4.85×10^{-1}N** C.1.94×102N D.4.85×102N
g=9.707ms$^{\neg2}$, m=50g=0.05kg, the gravitational attraction of the piece of cork is its weight,
$W=mg=9.707\times0.05=4.85\times10^{-1}N$.

20. The mass and weight of a body on earth are 8kg and 80N respectively. Determine the mass and weight of the body on a planet where the pull of gravity is $\frac{1}{8}$ that on earth A.8kg, 8N B.1kg, 10N C.64kg, 10N **D.8kg, 10N.**
Mass of a body is constant in all location in the universe, it does not change. Pull of gravity is same as weight. Weight of the body on earth, W_e=80N, Weight of the body on planet, $W_p=\frac{1}{8}W_e=\frac{1}{8}\times80=10N$. The mass M=8kg, while its weight W_P=10N.

21. The gravitational force of the moon is one sixth that of the earth. If a body weighs 6.0N on the moon, calculate its weight on the earth A.1.0N B.2.0N C.6.0N D.12.0N **E.36.0N**
$g_e=g$, $g_m=\frac{1}{6}g_e$, W_e=?, W_m=6.0N,
$W\propto$ g, $\frac{W_1}{g_1}=\frac{W_2}{g_2}$(constant mass), $\frac{W_e}{g_e}=\frac{W_m}{g_m}$,
$\frac{W_e}{g}=\frac{6}{\frac{1}{6}g}$, $W_e=\frac{6g}{\frac{g}{6}}=\frac{6g\times6}{g}=36N$.

22. The weight of an object on the earth is 600% of its weight on the moon. If the object weighs 60N on earth, determine its weight on the moon A.0.1N B.6.0N **C.10.0N** D.36.0N
$W_e=\frac{600}{100}W_m=6W_m$, $W_m=\frac{1}{6}W_e$, W_e=60N,
$W_m=\frac{1}{6}\times60=10N$.

23. A missle weighing 400N on earth's surface is shot into the atmosphere to an altitude if 6.4×10^{6}m, taking the earth as a sphere of radius

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6.4×10^6m and assuming the inverse square law of universal gravitational , what would be the weight of the missle at that altitude? **A.100N** B.200N C.400N D.800N E.1600N.

$W=400N$, $h=6.4\times10^6$m, $R=6.4\times106$m, W^1=?,

$W^1=\frac{W}{{d_R}^2}$, ${d_R}^2=\frac{R+h}{R}$, $W^1=\frac{W}{(R+h/R)^2}$

$W^1=W\left(\frac{R}{R+h}\right)^2=400\times\left(\frac{6.4\times10^6}{6.4\times10^6+6.4\times10^6}\right)^2$,

$W^1=400\times\left(\frac{6.4\times10^6}{12.8\times10^6}\right)^2=400\times\left(\frac{1}{2}\right)^2=\frac{400}{4}$,

$W^1=100N$.

24. In the absence of gravitational force, the weight of a body is A.its mass **B.zero** C.its density D.its volume.

25. I.The earth is not spherical but elliptical in shape II.Variation in latitude and longitude III.Rotation of the earth on its axis IV.Variation in the density of the earth. On which combination of the above does the weight of an object vary on the earth's surface? A.I and II **B.I,II and III** C.II,III and IV D.I,II,III and IV.

26. An astronaut experiences weightlessness in space when he A.does not hold anything while in space B.is midway between the sun and the earth C.is free from the earth's gravitational field D.walking on the moon

27. The gravitational field intensity I.has the unit Nkg^{-1} II.is equal to the acceleration of free fall due to gravity III.increases with altitude. Which of the statements above is/are correct? A.I B.II **C.I and II** D.II and III E.I,II and III

Gravitational field intensity is equal to the acceleration of free fall due to gravity. $E_G=g=\frac{F}{m}$ i.e force per unit mass (Nkg^{-1}).

E_G and g decreases with altitude or height.

28. The magnitude of the gravitational attraction between the earth amd a particle is 40N. If the mass of the particle is 4kg, calculate the magnitude of the gravitational field intensity of the earth on that particle. A.10.0Nkg-1 B.12.6Nkg-1 C.25.0Nkg-1 D.160.0Nkg-1

$F=40N$, m=4kg, E_G=?, $E_G=\frac{F}{m}=\frac{40}{4}=10N$kg^{-1}.

29. The gravitational field intensity at a location X in space is two-fifths of its value on the earth's surface. If the weight of an object at X is 4.80N, what is its weight on the earth? A.1.92N B.2.88N C.6.72N D.8.00N **E.12.00N**.

Gravitational field intensity is same as acceleration due to gravity g. $g_X=\frac{2}{5}g_e$, W_X=4.80N, W_e=?, $\frac{W_e}{g_e}=\frac{W_X}{g_X}$, $\frac{W_e}{g}=\frac{4.8}{\frac{2}{5}g}$,

$W_e=\frac{5\times4.8\times g}{2g}=12$N.

30. The gravitational potential at a point in a gravitational field is the workdone in taking a unit mass from A.zero to infinity **B.infinity to the point** C.one point to another point D.a distance to a given point.

31. If M and R are the mass and radius of the earth respectively and G is the universal gravitational constant, the earth's gravitational potential at an altitude H above the ground level is A.$-\frac{GM}{H}$ **B.**$-\frac{GM}{R+H}$ C.$-\frac{GM}{2H}$ D.$-\frac{GM}{R-H}$.

$V=-\frac{Gm}{r}$, m=M, r=$R+H$, $V=-\frac{GM}{R+H}$.

32. In the eaerth's surface. (Radius of the earth=6.4×10^6m, mass of the earth= 6.0×10^{24}kg, G=6.67×10^{-11}Nm^{2}kg^{-2}] A.2.1×10^7Jkg^{-1} B.3.2×10^7Jkg^{-1} C.4.3×10^7Jkg^{-1} **D.6.3×107Jkg^{-1}**.

R=6.4×10^6m,M=6.0×10^{24}kg,G=6.67×10^{-11}Nm^2 kg^{-2}, V=?, Gravitational potential at the earth surface, V=$\frac{GM}{R}=\frac{6.67\times10^{-11}\times6\times10^{24}}{6.4\times10^6}=6.3\times107$Jkg^{-1}.

33. In the earth's gravitational field A.work is done when a force moves its point of application around a closed path B.bodies repel themselves when they come close C.energy is released when an object is raised above the ground **D.a body experiences no decrease in deceleration when its moves freely upwards.**

No work is done by a force around a closed path in a gravitational field(a conservative field) i.e workdone is zero round a closed path.

Gravitational force is a force of attraction, hence bodies brought close to one another attracts themselves.

When a body is raised above the ground in a gravitational field, work is done against the field and energy is absorbed. When a body is brought the ground from a height during free fall, work is done by the field and energy is released.

A body moving upward in a gravitational field experiences a constant deceleration of 9.8ms^{-2} while a body falling freely downward in a gravitational field experiences a constant acceleration of 9.8ms^{-2}.

34. A satellite of mass m moves in an orbit around the earth of mass M. If the radius of the orbit is r,the kinetic energy of the satellite can be expressed as the product of the gravitational constant and A.$-\frac{Mm}{r}$ B.$-\frac{Mm}{2r}$ C.$\frac{Mm}{r^2}$ **D.**$\frac{Mm}{2r}$ E.$\frac{2Mm}{r^2}$.

Gravitational kinetic energy is equal to half the gravitational potential but of opposite sign. $P.E=-mV=-\frac{GMm}{r}$.

$K.E=\frac{mV}{2}=\frac{GMm}{2r}=G\times\frac{Mm}{2r}$, hence the kinetic energy of the satellite can be expressed as the product of the gravitational constant G and $\frac{Mm}{2r}$.

35. The escape velocity of a rocket at location R meters above the surface of the earth of mass M and radius R, is given by A.$\sqrt{2GMR}$ B.$\sqrt{\frac{2GM}{R}}$ C.$\sqrt{\frac{GM}{2R}}$ D.$\frac{\sqrt{2GM}}{R}$ **E.**$\sqrt{\frac{GM}{R}}$.

$v_e=\sqrt{\frac{2GM}{R}}$, The distance of the rocket above the

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earth's surface is R, hence its distance from the earth's centre= $R + R$=$2R$, $v_e = \sqrt{\frac{2GM}{2R}} = \sqrt{\frac{GM}{R}}$.

36. The gravitational potential energy of a body of mass 5kg , situated at a point within the earth's gravitational field is $3.25\times10^8 J$. Calculate the magnitude of the escape velocity of the body A.5.50×104ms^{-1} **B.1.12×10^{4}ms^{-1}** C.6.25×10^{4}ms^{-1} D.3.60×10^{3}ms^{-1}.

$P.E$=$3.25\times10^8 J$, m=5kg, v_e=? $v_e = \sqrt{\frac{2P.E}{m}}$,

$v_e = \sqrt{\frac{2\times3.25\times10^8}{5}} = \sqrt{1.3\times10^8} = 1.12\times104$ms^{-1}.

37. The radius of the earth is 6.4×10^{6}m and the acceleration due to gravity is 10.0ms^{-2}. The escape velocity of a rocket launched from the earth's surface is A.4.2kms^{-1} B.4.0kms^{-1} **C.11.3kms^{-1}** D.25.3kms^{-1}.

R=6.4×10^{6}m, g=10ms^{-2}, v_e=?, $v_e = \sqrt{2gR}$

$v_e = \sqrt{2\times10\times6.4\times10^6} = \sqrt{128\times10^6}$,

$v_e = 11.3\times10^3$ms^{-1} = 11.3kms^{-1}.

38. What would happen to the earth's period of revolution if its mass was three times as great as it is? A.increase 3 times B.decrease 3 times C.decrease 6 time **D.remain the same**

The period of revolution of the earth is independent of mass or size i.e no dependent on mass, but depends on the radius of the earth and acceleration due to gravity.

39. A satellite is in a parking orbit, if its period is A.more than the period of the earth **B.equal to the period of the earth** C.the square of the period of the earth D.less than the period of the earth.

40. A geostationary satellite moves in an orbit of radius 6300km. Calculate the speed with which it moves in the orbit (π=22/7) A.115ms^{-1} B.153ms^{-1} C.229ms^{-1} **D.458ms^{-1}**.

r=6300km=6300×10^{3}m, geostationary satellite has a period is equal to the period of rotation of the earth(24hrs), T=24hrs=24×3600=86400s, v=?, $v = \frac{2\pi r}{T} = \frac{2\pi\times6300\times10^3}{86400} = 145.83\pi$,

$v = 145.\square\times\frac{22}{7} = 458$ms^{-1}.

Jamb past questions on gravitational field :

[1995/8,2001/7,2004/17,2015/1,2013/2,2010/6,2014/9,1993/7,1995/9,2003/13,2006/17, 2014/8,1997/8,2012/8,1998/6,2002/3,2005/32,2007/43,1980/1,1983/6,2015/5]

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Chapter 6 – Equilibrium of forces

● **Equilibrium of a body –** A body is said to be in equilibrium when the resultant or net force acting on the body is zero and its acceleration is zero.

- Static equilibrium – Condition of a body when the net force acting on the body is zero and the body is at rest.

- Dynamic equilibrium – Condition of a body when the net force acting on the body is zero and the body is moving with a uniform velocity in a straight line or rotating with a uniform angular velocity.

- Moment of a force – This is the turning effect produced by the force. It is the product of force and perpendicular distance in the line or point of action of the force. **Moment of a force,** $\boldsymbol{M = Fd}$.

- When the forces acting on a body are **parallel to the distance** at which they act, then the moment of the force is zero(0Nm), because **moment deals with perpendicular distance** not parallel distance.

- A force will produce zero difference in moment when it acts on a body at different points but with the same perpendicular distance.

- The resolved component of the distance perpendicular to a force will produce moment of the force e.g if a force F acts on a body at a distance d inclined to the vertical or horizontal, the component of the distance perpendicular to the force is $\boldsymbol{dsin\theta}$ **(vertical)** and $\boldsymbol{dcos\theta}$ **(horizontal)**.

- Couple – It is the combination of two parallel, equal and opposite forces, whose line of action do not coincide e.g turning of car steering wheel, turning of a tap with our fingers.

- The resultant force acting on a couple is zero. A couple produces rotational motion in a body.

- Moment of a couple also called Torque is the product of one of the forces and the perpendicular distance (called arm of a couple) from the line of actiong of the force.

Moment of a couple or Torque, $\boldsymbol{\tau = Fd}$.

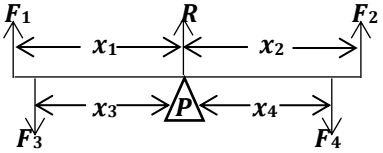
- Factors that determine the torque are : I. The magnitude of the force. **II.** The direction of the force. **III.** The perpendicular distance of the force from the pivot.

- Conditions for equilibrium of parallel coplanar forces acting on a body –

I.The algebraic sum or resultant of forces acting on a body in any given direction is zero i.e sum of upward forces equals the sum of downward forces.

II. The system must obey the **principle of moment** – Its states that if a body is in equilibrium, then the sum of clockwise moment about any point is equal to the sum of anticlockwise moment about that same point i.e the algebraic sum or resultant moment is zero.

- The moment is taken about a fixed point called the fulcrum or pivot $\boldsymbol{P}$.



I. Sum of upward forces=Sum of downward force or Sum forces acting one direction=Sum of forces acting in the other direction. $\boldsymbol{F_1+F_2 = F_3+F_4}$. The algebraic sum of forces or resultant force is zero, $\boldsymbol{F_1+F_2-F_3-F_4=0}$.

II. Sum of clockwise moment ($F_1x_1+F_4x_4$) = Sum of anticlockwise moment ($F_2x_2+F_3x_3$).

$\boldsymbol{F_1x_1+F_4x_4 = F_2x_2+F_3x_3}$.

III. The algebraic sum of moments or resultant force is zero . $\boldsymbol{F_1x_1+F_4x_4-F_2x_2-F_3x_3=0}$.

- R is the reaction of the support at P on the body. R has no moment about P, as the no perpendicular distance between R and P i.e the distance between R and P $=0$.

- Centre of gravity C.G – of a body is the point where its resultant weight(or resultant pull of the earth) is concentrated or appears to act.

- Position of the centre of gravity of regular uniform or symmetrical bodies –

I.The centre of gravity of a uniform meter rule is at the midpoint i.e 50cm mark.

II. The centre of gravity of a uniform rod, lever or beam is at the midpoint.

III. The centre of gravity of a uniform triangular body is at the point of intersection of its medians.

IV. The centre of gravity of a uniform circular or ring-shaped body is at its centre.

V.The centre of gravity of a uniform rectangular or square shaped body is at the point of intersection of its diagonals.

- The centre of gravity of a non-uniform or assymetrical body lies close to the larger side or section of the body.

● Equilibrium of non-parallel coplanar forces :

– Conditions of equilibrium of non-parallel coplanar forces –

I.They must be **coplanar** i.e they are not parallel or not collinear(not in the same line).

II. They must be **concurrent** i.e they all pass through a common point or their line of action meet at a common point.

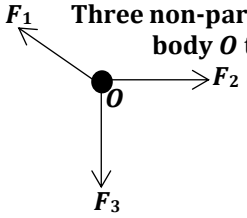
III. The sum of the resolved component along each direction is direction is equal to zero i.e the algebraic sum of components of forces in the

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vertical direction is zero and the algebraic sum of the components of forces in the horizontal direction is zero.

IV. They can be represented by the three sides of a trinagle in magnitude and direction.

V.Any one force can be the equilibrant of the other two forces and is equal in magnitude to the resultant of the other two forces.The equilibrant force is equal and opposite resultant force.



Three non-parallel forces acting on a body O to keep it equilibrium.

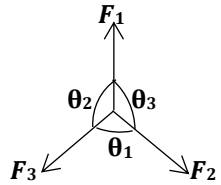
$$F_1 + F_2 + F_3 = 0.$$

F_1 can be the equilibrant of F_2 and F_3 or F_2 the equilibrant of F_1 and F_3 or F_3 the equilibrant of F_1 and F_3.

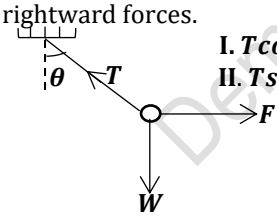
- Methods of solving questions on equilibrium of three non-parallel forces acting on a body ;

1. Lami's theorem – It states that when three concurrent forces acts on a body to keep it in equilibrium, the ratio of each force and the sine of the angle directly opposite it are equal.

Lami's theorem : $\frac{F_1}{sin\theta_1} = \frac{F_2}{sin\theta_2} = \frac{F_3}{sin\theta_3}$.



2.Resolution of forces – This involves the resolving of oblique or inclined forces into their perpendicular components.Then, equating the opposite forces in either axis i.e sum of the upward components equals sum of downward forces and sum of leftward forces equals sum of rightward forces.



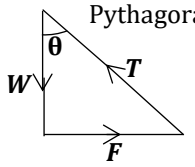
I. $Tcos\theta = W = mg$
II. $Tsin\theta = F$.

3.Triangle law of forces – It states that if three forces are in equilibrium, they can be represented by the three sides of triangle taken in order.

- Each force maintains its original position in the triangle and each angle correspondsto its original position or alternateswith its new position.

Using the same diagram used in method 2 –

Pythagora's theorem – $T^2 = W^2 + F^2$.



$sin\,\theta = \frac{F}{T}$, $cos\,\theta = \frac{W}{T}$

$tan\,\theta = \frac{F}{W}$.

● Equilibrium of rigid bodies and stability of bodies –

1.Stable equilibrium – A body is said to be in stable equilibrium if it tends to return to its original position when slightly displaced.

- A wide base and low centre of gravity(G) ensures stable equilibrium of a body e.g a cone resting on its base, a bowl or sphere in the middle of a bowl, swinging pendulum bob.

- Conditions for stable equilibrium:

I.It returns to its original position(rest) when slightly displaced i.e force tends to decrease the displacement when slightly displaced.

II. Its potential energy increases after displacement (initial P.E before displacement is minimum).

III. Wide base and low centre of gravity. Centre of gravity rises and falls inside the base of the body after displacement.

IV. It has restoring moment. Moment of weight is greater about the base i.e moment of weight is greater than moment of action about the edge of tilt.

2.Unstable equilibrium – A body is said to be in unstable equilibrium if when slightly displaced it tends to move further away from its original position.

- A narrow base and high centre of gravity(G) ensures unstable equilibrium of a body e.g a cone or egg resting on its apex or pointed end or tip, a ball or sphere resting on an inverted bowl, a tight rope walker.

- Conditions for unstable equilibrium:

I.It tip over over and does not return to its original position after displacement i.e force tends to increase displacement when slightly displaced.

II. Its potential energy decreases or lowered after displacement (Initial P.E before displacement is maximum).

III. Narrow base and high centre of gravity. Centre of gravity falls and falls outside the base of the body after displacement.

IV. No restoring moment. Moment of the weight is less about the base i.e moment of weight is less than the moment of action about the edge of tilt.

3.Neutral equilibrium – A body is said to be in neutral equilibrium if when slightly displaced, it tends to come to rest in its new position.

- A flat base and constant centre of gravity (G) ensures neutral equilibrium in a body e.g a cone or cylinder or egg resting on its side, a ball or sphere on a smoth horizontal table.

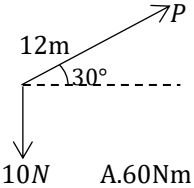
- Conditions for neutral equilibrium:

I.It remains in its new position when slieghtly displace or does not change position i.e force tends to keep the body in the same position when displaced.

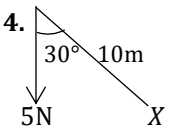
II. Its potential energy is constant or does not

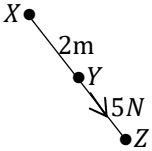
Demystified Series by Dr Timothy

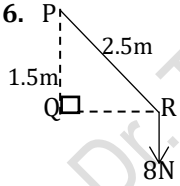
Change (Initial potential energy is zero).
III. Flat base and constant centre of gravity. Centre of gravity remains at the same position and does not change.
IV. Moment of weight is equal to the moment of action about the edge of tilt
V.Produces rolling movement on the body.
- In summary – 1.A body is most stable when its cnetre if gravity G is at the lowest point.
2.The higher the centre of gravity of a body the less stable it is and the lower the centre of gravity of a body the more stable it is.
3.A wide base favours stability as it leads to a low center of gravity.
4.If slight displacement of an object raises its centre of gravity, then the object is stable.
5.If slight displacement of an object lowers its centre of gravity then the object is unstable.
6.If slight displacement of an objecr neither raises or lowers its centre of gravity then the object is neutral equilibrium.

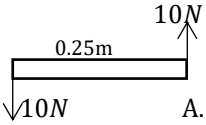
Examples :
1. An object is acted upon by a system of parallel forces.The condition(s) for static equilibrium of the object is/are that the I.algebraic sum of all the moments of the forces about a point is zero II.parallel forces must be equal in magnitude and direction III.resultant of the parallel forces is zero A.I B.II C.III **D.I and III** E.II and III.
For a body acted upon by parallel forces to be in static equilibrium: i.It must obey principle of moment i.e The algebraic sum of all moments of the forces about a point is zero or the sum of clockwise moment is equal to the sum of anticlockwise moment ii.The resultant of the parallel forces is zero or sum of in one direction(upward forces) is equal to sum of forces in the other direction(downward forces).
- The forces must not be equal in magnitude and direction to produce static equilibrium.
2. A technician applied a force of 250N at the end of a spanner of length 0.25m in order to loosen a nut.Determine the moment applied to the nut A.1000.0Nm B.625.0Nm **C.62.5Nm** D.10.0Nm.
F=250N, d=length=0.25m, M=?,
$M = F \times d = 250 \times 0.25 = 62.5$Nm.
3. 

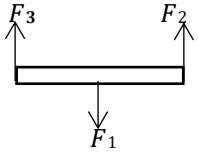
Using the diagram below, calculate the moment of the force of 10N about the point P ?
10N A.60Nm **B.60$\sqrt{3}$Nm** C.120Nm D.120$\sqrt{3}$Nm E.240Nm.
Perpendicular distance of the 10N force from P is $12sin30°$, Moment about P, $M = F \times d$,
$M = 10 \times 12cos30° = 10 \times 12 \times \frac{\sqrt{3}}{2} = 60\sqrt{3}N$m.

4. 

In the diagram shown, the moment of the 5N force about the point X is
A.25Nm B.50Nm C.25$\sqrt{3}$Nm D.50$\sqrt{3}$Nm.
Perpendicular distance of the 5N force from X is $10sin30°$, Moment about X, $M = F \times d$,
$M = 5 \times 10cos30° = 25$Nm.
5. 

A force of 5N acts at a point Y on the rod XYZ as shown in the diagram. If XY is 2m, what is the moment of the force about point X?
A.10Nm B.7Nm C.3Nm **D.0Nm.**
The moment of 5N about X is 0Nm, as the 5N force is parallel to point X or the line of action of the 5N force passes through point X. Moment deals with the product of force and perpendicular distance not parallel distance.
6. 

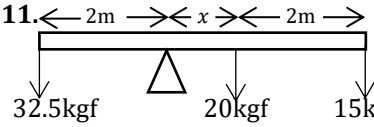
In the diagram, the difference between the moment about P of the 8N force and its moment about Q is
A.16Nm B.8Nm C.4Nm D.0Nm.
Perpendicular distance of the 8N force is QR,
$PR^2 = PQ^2 + QR^2$, $QR^2 = PR^2 - PQ^2 = 2.5^2 - 1.5^2$,
$QR^2 = 6.25 - 2.25 = 4$, $QR = \sqrt{4} = 2$m,
Moment of the 8N force about P =8×QR ,
M about P = 8×2 = 16Nm,
Moment of the 8N force about Q = 8×QR,
M about Q= 8×2 = 16Nm. The difference between the moments is zero (16−16=0).
7. The term torque means **A.the moment of a couple about an axis** B.the resultant of several forces acting on a body in equilibrium C.two equal and opposite forces whose line of action do not coincide D.two coplanar forces at right angles to each other E.the equilibrant of two concurrent forces.
8. Which of the following types of motion is produced by a couple? A.oscillatory **B.rotational** C.random D.translational.
9. 

In the figure above, calculate the magnitude of the couple
A.1.00Nm B.1.25Nm C.2.00Nm **D.2.50Nm** E.5.00Nm.
Magnitude of a couple is same as moment of a couple or torque. $\tau = Fd$, F=one of the forces=10N, d=arm of the couple or distance between the forces=0.25m, τ =10×0.25=2.5Nm.
10. 

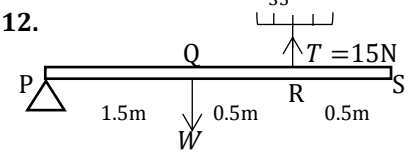
The diagram above shows three forces, F_1, F_2 and F_3 which keep the bar AB in horizontal equilibrium.
Which of the following equations is correct?

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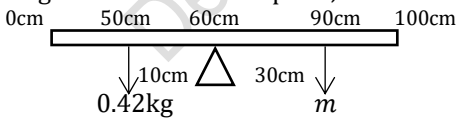
A.$F_3 = F_1 + F_2$ B.$F_2 = F_1 + F_3$ C.$F_1 = F_2 - F_3$
D.$F_1 = F_2 + F_3$
For a system acted by a set of force to be in equilibrium, the sum upwards forces equals the sum of downward forces i.e $F_2 + F_3 = F_1$ or the algebraic sum of the force equals zero i.e $F_2 + F_3 - F_1 = 0$, $F_2 = F_1 - F_3$, $F_3 = F_1 - F_2$.

11.


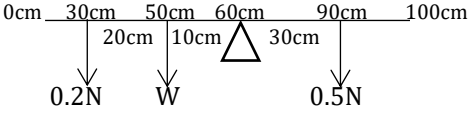
The diagram above a see-saw with three boys of weights 32.5kgf, 20kgf and 15kgf sitting on it. If the see-saw is exactly horizontal and of negligible weight, the value of x is A.0.5m **B.1.0m** C.1.5m D.2.0m.
Taking moment about the pivot,
Clockwise(C.W) moment$=15(2+x)+20x$,
Anticlockwise(AC.W) moment$=32.5(2)$,
Sum of C.W moment=Sum of AC.W moment,
$15(2+x)+20x = 32.5(2)$, $30+15x+20x = 65$,
$35x = 65-30 = 35$, $x = \frac{35}{35} = 1$m.

12.


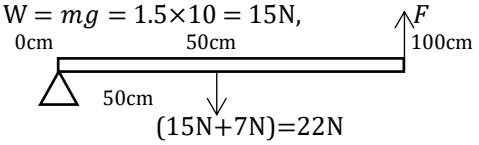
The value of W in the diagram shown above is A.7.5N B.15.0N **C.20.0N** D.30.0N
Taking moment about point P, the pivot –
Clockwise (C.W) moment$= W(1.5)$,
Anticlockwise(AC.W) moment$= T(0.5+1.5)$,
$W \times 1.5 = T(0.5+1.5) = T(2)$, $W = \frac{2T}{1.5}$,
$T = 15N$, $W = \frac{2\times15}{1.5} = 20N$.

13. A uniform meter rule of mass 0.42kg is balanced at the 60cm mark when a mass m is placed at the 90cm mark. Calculate the value of m **A.0.14kg** B.0.42kg C.0.82kg D.0.84kg E.4.20kg
The meter rule is uniform, then its mass would be concentrated at its mid-point i.e 50cm mark.
Taking moment about the pivot,


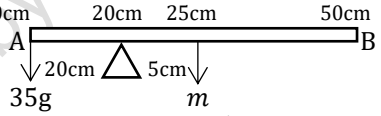
C.W moment$= m \times 30$, AC.W moment$=0.42\times10$
$m \times 30 = 0.42\times10$, $m \times 30 = 4.2$, $m = \frac{4.2}{30} = 0.14$kg.

14. Weights 0.2N and 0.5N are placed at the 30cm and 80cm marks respectively on a uniform meter rule. If the meter rule balances horizontally on a knife edge at the 60cm mark, the weight of the meter rule is A.0.1N B.0.3N C.0.4N **D.0.9N**.


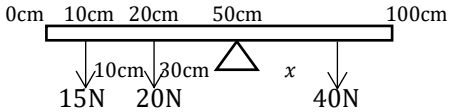
Taking moment about the pivot –
C.W moment $=0.5\times30$,
AC.W moment$=0.2(20+10)+W\times10$,
$0.5\times30 = 0.2\times30+10W$, $15 = 6+10W$,
$10W = 15-6 = 9$, $W = \frac{9}{10} = 0.9$N.

15. A uniform meter rod of mass 1.5kg is pivoted at one end. A weight of 7N is placed at the centre of the rod. The vertical force which should be applied at the other end to maintain the rod in equilibrium in equilibrium in the horizontal position is A.22.0N **B.11.0N** C.10.5N D.8.5N.
W $= mg = 1.5\times10 = 15$N,


Both the weight of the meter rod(15N) and the 7N weight are acting at the centre of the rod.
Taking moment about the pivot –
C.W moment = AC.W moment,
$22\times50 = F\times100$, $F = \frac{22\times50}{100} = 11$N.

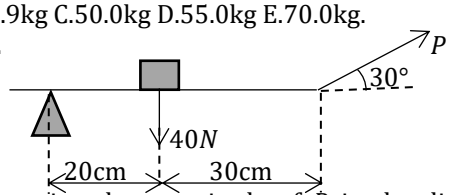
16. A uniform half meter rod AB, pivoted at a point 20cm from A, is balanced horizontally by placing a 35g mass at A. Calculate the mass of the rod. **A.140g** B.170g C.233g D.350g
A half-meter rod will have a length of 0.5m or 50cm.


$m\times5 = 35\times20$, $m = \frac{35\times20}{5} = 140$g.

17. A uniform meter rule is pivoted at its centre. Loads of 15N and 20N are placed at the 10cm and 20cm marks respectively and the meter rule is balanced horizontally by a load of 40N placed on the other side of the pivot. Calculate the distance of the 40N load from the pivot. A.15.0cm B.20.0cm **C.30.0cm** D.80.0cm.


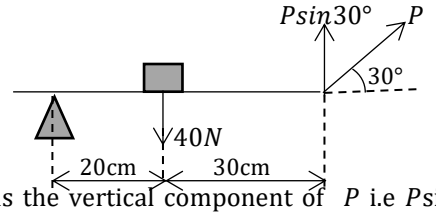
$40\times x = 15(10+30)+20(30) = 15\times40+20\times30$,
$40x = 600+600 = 1200$, $x = \frac{1200}{40} = 30$cm.

18. A uniform bar 15m long, is balanced on a pivot at its mid-point. A boy of mass 55kg sits on one arm of the bar 5m away from the pivot. What mass can be placed 2m away from the other end of the bar to keep the bar horizontal? A.5.0kg B.6.9kg C.50.0kg D.55.0kg E.70.0kg.

19.


Determine the magnitude of P in the diagram above. A.16N B.20N **C.32N** D.40N.

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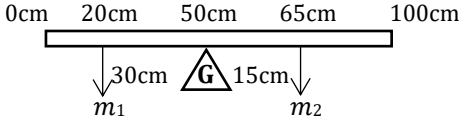
It is the vertical component of P i.e $P\sin\theta$ will produce moment about the pivot or fulcrum.

$40\times20= Psin30°(30+20)= P(0.5)(50)$,

$800=25P$, $P=\frac{800}{25}=32N$.

20. Masses m_1 and m_2 at the 20cm and 65cm marks respectively of a uniform meter rule freely suspended at its centre of gravity. If the meter rule balances horizontally, determine the ratio m_2:m_1 A.1:2 B.3:2 C.2:3 **D.2:1.**

The metre rule being suspended at its centre of gravity means it is balanced or pivoted at the mid-point of the meter rule i.e 50cm.

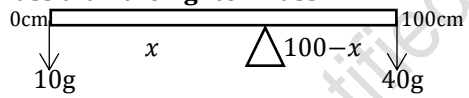


$m_2\times15= m_1\times30$, $\frac{m_2}{m_1}=\frac{30}{15}=\frac{2}{1}$, m_2:m_1=2:1.

21. A mass of 10g is attaced to the 0cm mark of a light meter rule while a mass of 40g is attached to the other end,the 100cm mark. The centre of gravity of the system is located at the A.20cm mark B.60cm mark **C.80cm mark** D.mid-point of the meter rule.

The centre of gravity of the light meter rule of negligible weight, is the point were it balances horizontally on the support or pivot.

The centre of gravity is closer to the heavier mass than the lighter mass.

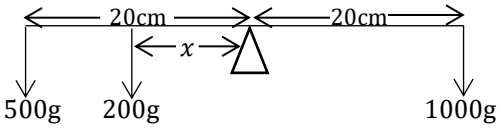


$40(100-x)=10x$, $4000-40x=10x$,

$4000=10x+40x=50x$, $x=\frac{4000}{50}=80$cm.

The center of gravity is 80cm from the 10g mass and 20cm from the 40g mass or at the 80cm mark.

22. A weight of 1000g hangs from a lever 20cm to the right of the fulcrum.At the left is 500g weight 20cm from the fulcrum.If a 200g weight is Xcm away from the fulcrum,what is the value of X that will be make the lever balance? **A.50cm** B.20cm C.10cm D.30cm E.70cm



The 200g additional weight must be placed on the lighter side inorder to balance the lever horizontally.

Taking moment about the pivot –

Clockwise moment=1000(20)

Anticlockwise moment=500(20)+200(x)

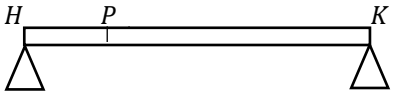
Sum of C.W moment=Sum of AC.W moment

$1000(20)=500(20)+200x$,

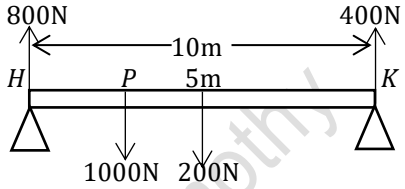
$20000=10000+200x$, $200x=20000-10000$,

$200x=10000$, $x=\frac{10000}{200}=50$cm.

23.



A uniform beam HK of length 10m and weighing 200N is supported at both ends as shows.A man weighing 1000N stands at a point P on the beam.If the reactions at H and K are respectively 800N and 400N,then the distance HP is A.4m B.3⅓m **C.3m** D.6⅔m E.7m.



Taking moments about H –

The force at H will not produce any moment, when moment is taken about H, as it don't have any perpendicular distance.Same applies to the force of K,if moment was to be taken about K.

Clockwise moment=$200\times5+1000HP$

Anticlockwise moment=400×10

Sum of CW moment=Sum of AW moment

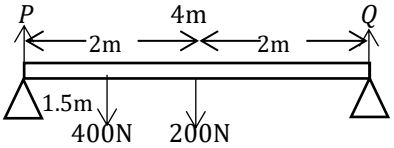
$1000+1000HP=4000$, $1000HP=4000-1000$,

$1000HP=3000$, $HP=\frac{3000}{1000}=3$m.

24. A uniform beam of length 4m and mass 20kg is supported at both ends. A girl of mass 40kg stands on the beam at a distance of 1.5m from one of the supports. The reactions at the supports are A.150N, 250N B.200N, 200N **C.250N, 350N** D.300N, 300N.

W=mg=20×10=200N, W=mg=40×10=400N,

The reactions at the supports are P and Q



Taking moment about point P –

$200\times2+400\times1.5= Q\times4$, $400+600=4Q$,

$4Q=1000$, $Q=\frac{1000}{4}=250$N,

Sum of upward force=Sum of downward forces,

$P+Q=400+200$, Q=250N,

$P+250=600$, $P=600-250=350$N.

25. Three non-parallel collinear forces, represented both in magnitude and direction by the three sides of a triangle taken in order, can A.move a body along a straight line B.rotate a body about about its axis C.make a body move randomly **D.keep a body in equilibrium** D.make a body oscillate.

26. If three coplanar non-paralle forces act on a body to keep it in equilibrium, which of the following statements is/are correct? I.the three

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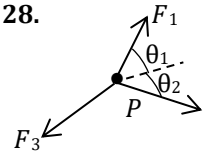
forces are collinear II.all the three forces must pass through the centre of gravity of the body III.the three forces can be represented by the three sides of a triangle A.I,II and III B.II C.II and III **D.III** E.I and II.

Three non-parallel forces acting on a body to keep is in equilibrium – are concurrent i.e pass through a common point, are not collinear i.e not on the same line and they do not pass through the centre of gravity of the body.

27. When three coplanar non-parallel forces are in equilibrium I.they can be represented in magnitude and direction by the three sides of a triangle taken in order II.their line of action meet at a point III.the magnitude of any one force equals the magnitude of the resultant of the other two IV.any of the forces is the equilibrant of the other two. Which of the statements are correct? A.I and II B.II and III C.II,III and IV **D.I,II,III and IV** E.I,II,III and IV.

When three non-parallel coplanar forces F_1, F_2 and F_3acts on a body in equilibrium,the resultant of any two forces is equal in magnitude but opposite in direction to the third force(equilibrant force).

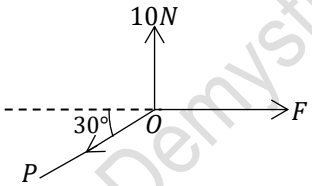
28.



A particle P is in equilibrium under the action of three forces F_1, F_2 and F_3 as shown in the diagram. Which of the following equations are correct? I.$F_1cos\theta_1 = F_2cos\theta_2$ II.$F_1sin\theta_1 = F_2sin\theta_2$ III.$F_3 - F_1cos\theta_1 - F_2cos\theta_2 = 0$ IV.$F_1+F_2+F_3=0$ A.I and II **B.II and III** C.II and IV D.II,III and IV.

Resolve the forces into their vertical and horizontal components – vertical components ; $F_1sin\theta_1 = F_2sin\theta_2$, horizontal components ; $F_3 = F_1cos\theta_1 + F_2cos\theta_2$, $F_3 - F_1cos\theta_1 - F_2cos\theta_2 = 0$.

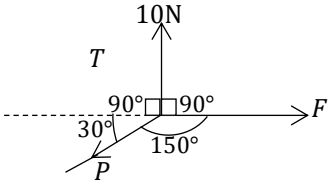
29.



The diagram above illustrates three forces acting on an object at point O. If the object is in equilibrium, determine the magnitude of the force P. A.10.5N B.11.0N C.17.3N **D.20.0N.**

I. Lami's theorem : The three forces acts concurrently on the body to keep it in θ to find the angle of of inclination of each force.

Angle opposite the $10N$ force$=180° - 30° = 150°$

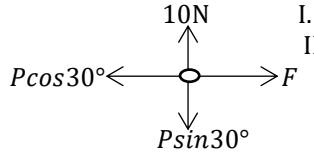


$\frac{10}{sin150°} = \frac{F}{\sin(90+30)°} = \frac{P}{sin90°}$, $\frac{10}{sin150°} = \frac{P}{sin90°}$,

$sin150° = sin(180° - 150°) = sin30°$, $sin90° = 1$.

$\frac{10}{sin30°} = \frac{P}{1}$, $P = \frac{10}{0.5} = 20\text{N}$.

II. Resolution of forces : The oblique or inclined force P is resolved into its perpendicular components and equate the forces in each axis.

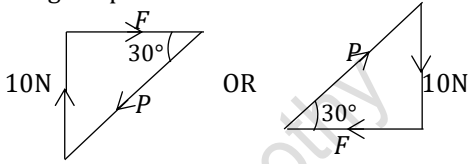


I. $10\text{N} = Psin30°$,

II. $Pcos30° = F$,

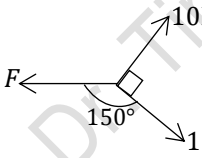
$P = \frac{10}{sin30°} = \frac{10}{0.5} = 20\text{N}$.

III. Triangle law of forces : Each force will occupy its original position, with its angle alternate or in it original position.



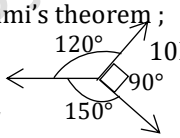
$P^2 = 10^2 + F^2$, $sin30° = \frac{10}{P}$, $P = \frac{10}{sin30°} = 20N$.

30.



The value of F in the figure above when in equilibrium is A.10N B.12N **C.20N** D.27N.

I. Lami's theorem ;



$\frac{10}{\sin 150°} = \frac{F}{\sin 90°} = \frac{17}{\sin 120°}$,

$\sin 150° = \sin 30°$,

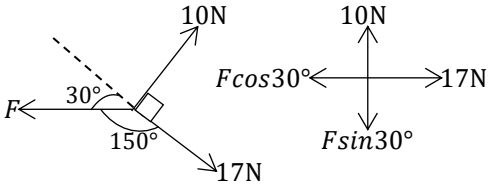
$\sin 120° = \sin 60°$,

$\sin 90° = 1$,

$F = \frac{10}{\sin 30°} = \frac{10}{0.5} = 20N$ OR $F = \frac{17}{\sin 60°} = \frac{17}{0.866}$,

$F = 19.6 \approx 20\text{N}$.

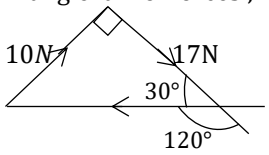
III. Resolution of forces ;



$10N = Fsin30°$, $Fcos30° = 17\text{N}$,

$F = \frac{10}{\sin 30°} = 20\text{N}$. $F = \frac{17}{\cos 30°} = 19.6 \approx 20\text{N}$.

III. Triangle law of forces ;



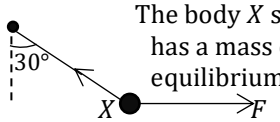
$F^2 = 17^2 + 10^2$,

$F^2 = 289 + 100 = 389$,

$F = \sqrt{389} = 19.7\text{N}$

$F \approx 20\text{N}$.

31.

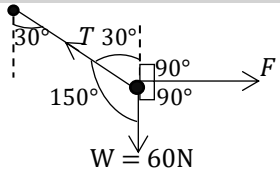


The body X shown in the diagram has a mass of 6kg and is held in equilibrium by a string and a horizontal force F.

The tension T in the string is (g=10ms^{-2}) **A.$4\sqrt{3}$N** B.12N C.$40\sqrt{3}$N D.120N.

I. Lami's theorem ; $W = mg = 6\times10 = 60\text{N}$.

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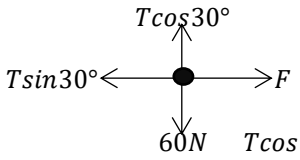


$\frac{T}{\sin 90°} = \frac{F}{\sin 150°} = \frac{60}{\sin(90+30)°}$, $\frac{T}{\sin 90°} = \frac{60}{\sin 120°}$,

$T = \frac{60}{\sin 60°} = \frac{60}{\frac{\sqrt{3}}{2}} = 60\times\frac{2}{\sqrt{3}} = \frac{120}{\sqrt{3}} = \frac{120}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}}$,

$T = \frac{120\sqrt{3}}{3} = 40\sqrt{3}$N.

II. Resolution of forces ;

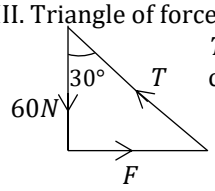


$Tcos30° = 60N$,

$T = \frac{60}{\cos 30°} = \frac{60}{\frac{\sqrt{3}}{2}} = 60\times\frac{2}{\sqrt{3}} = \frac{120}{\sqrt{3}} = \frac{120}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}}$,

$T = \frac{120\sqrt{3}}{3} = 40\sqrt{3}N$.

III. Triangle of forces :

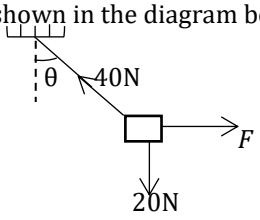


$T^2 = 60^2 + F^2$,

$\cos 30° = \frac{60}{T}$, $T = \frac{60}{\cos 30°}$,

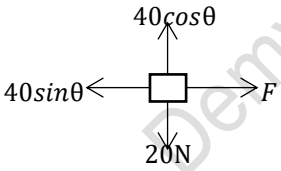
$T = 40\sqrt{3}$N.

32. For what values of F and θ will the firces shown in the diagram be in equilibrium?



A.60N and 60° B.60N and 30° **C.$20\sqrt{3}$N and 60°** D.$20\sqrt{3}$N and30°.

Resolution of the forces into their perpendicular components –



I. $40cos\theta = 20$N,

$cos\theta = \frac{20}{40} = \frac{1}{2} = 0.5$,

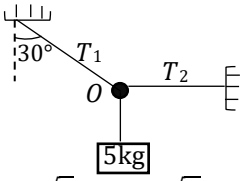
$\theta = cos^{-1}0.5 = 60°$.

II. $40sin\theta = F$, θ=60°,

$F = 40sin60°$,

$F = 40\times\frac{\sqrt{3}}{2} = 20\sqrt{3}$N.

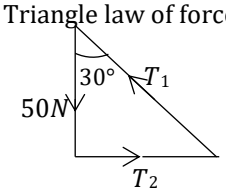
33.



If the system of forces shown in the diagram is in equilibrium, the tension T_2 is equal to (g=10ms^{-2})

A.$100/\sqrt{3}$N **B.$50/\sqrt{3}$N** C.$150/\sqrt{3}$N D.50N .

Triangle law of forces – $W = 5\times10 = 50$N



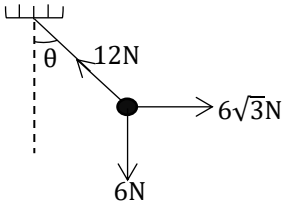
$T_2{}^2 = 50^2 + T_1{}^2$,

$tan30° = \frac{T_2}{50}$,

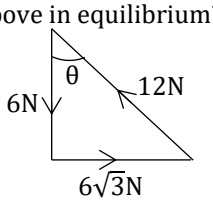
$T_2 = 50tan30° = 50\times1/\sqrt{3}$,

$T_2 = 50/\sqrt{3}$N.

34.



For what value of θ are the forces in the diagram above in equilibrium? A.15° B.30° C.45° **D.60°**.



$\sin\theta = \frac{6\sqrt{3}}{12} = \frac{\sqrt{3}}{2}$,

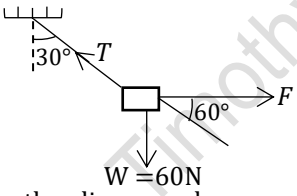
$\theta = \sin^{-1}\frac{\sqrt{3}}{2} = 60°$.

$\cos\theta = \frac{6}{12} = \frac{1}{2}$,

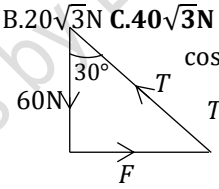
$\theta = \cos^{-1}\frac{1}{2} = 60°$.

$\tan\theta = \frac{6\sqrt{3}}{6} = \sqrt{3}$, $\theta = \sin^{-1}\sqrt{3} = 60°$.

35.



From the diagram above, a weight W=60N is suspended orizontally by a string attached to a wall. Find the tension T on the string? A.$60\sqrt{3}$N B.$20\sqrt{3}$N **C.$40\sqrt{3}$N** D.120N.



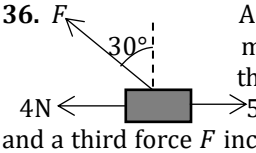
$\cos 30° = \frac{60}{T}$, $T = \frac{60}{\cos 30°} = \frac{60}{\sqrt{3}/2}$,

$T = 60\times\frac{2}{\sqrt{3}} = \frac{120}{\sqrt{3}} = \frac{120}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}}$,

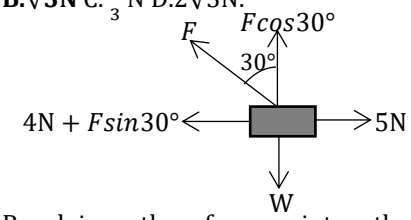
$T = \frac{120\sqrt{3}}{3} = 40\sqrt{3}$N.

• The reader should attempt question 33, 34 and 35 using the other two methods.

36.



A box X of weight W is maintained in equilibirum the action of two horizontal forces 5N and 4N and a third force F inclined at 30° to the vertical in the diagram above. The value of W is A.$\frac{\sqrt{3}}{2}$N **B.$\sqrt{3}$N** C.$\frac{\sqrt{3}}{3}$N D.$2\sqrt{3}$N.



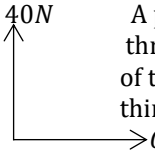
Resolving the forces into the horizontal component – $Fsin30° + 4 = 5$, $Fsin30° = 5-4$,

$Fsin30° = 1$, $F = \frac{1}{\sin 30°} = 2$N,

Resolving the forces into the vertical component – $Fcos30° = W$, $F = 2$N,

$W = 2cos30° = 2\times\frac{\sqrt{3}}{2} = \sqrt{3}$N.

37.

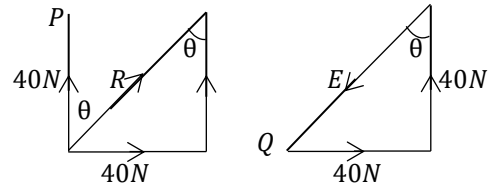


A particle is under the action of three forces in equilibrium. Two of te forces are as shown. The third force is

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A.80N along P B.160N at 60° to P **C.40$\sqrt{2}$N at 45° to either P or Q** D.0N E.40$\sqrt{2}$N at 30° to Q.

The third force required for the body to be in equilibrium is the equilibrant force. The equilibrant force E is equal in magnitude but opposite in direction to the resultant force R.

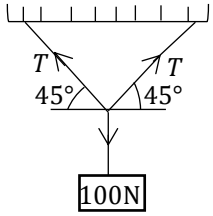


$E^2 = 40^2 + 40^2 = 1600 + 1600 = 3200$,

$E = \sqrt{3200} = 40\sqrt{2}$N, $tan\theta = \frac{40}{40} = 1$,

$\theta = tan^{-1}1 = 45°$. The third force is 40$\sqrt{2}$N at 45° to either P or Q or 40$\sqrt{2}$N at S45°W , opposite to the direction of the resultant.

38. From the diagram below, if the forces are in equilibrium, what is the value of thw tension T?



A.50.00N **B.70.71N** C.100.00N D.241.42N.

Resolving the forces vertically i.e as they are in equilibrium – $Tsin45° + Tsin45° = W$,

$W = 100N$, $2Tsin45° = 100$, $T = \frac{100}{2sin45°}$,

$T = \frac{100}{2\times 0.7071} = 70.71$N.

- The tension T increases as θ decreases. Tension T is maximum when θ is 0° and minimum when θ is 90°.

39. Which of the following actions will increase the stability of a body I.increasing the base area II.raising the centre of gravity of the body III.lowering the center of gravity of the body A.I and II **B.I and III** C.II and III D.I,II and III.

Wide base area and low centre of gravity increases the stability of a body.

40. A body was slightly displaced from its equilibrium position. Which one of the following is a condition for its stable equilibrium A.an increase in the kinetic energy of the body **B.an increase in the potential energy of the body** C.a decrease in the potential energy of the body D.an increase in the weight of the body.

The potential energy of a body increases after displacement in stable equilibrium due to increase in the height of the body, but its initial potential energy of the body before displacement is minimum.

The potential energy of a body decreases after displacement in unstable equilibrium due to decrease in the height of the body, but its initial potential energy before displacement is minimum.

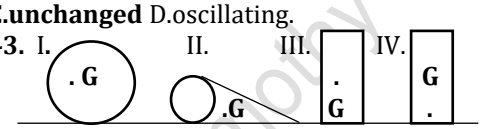
The potential energy of body in neutral equilibrium is contant and does not change after displacement, but its initial potential energy before displacement is zero.

41. Which of the following is not in stable equilibrium? A.a heavy weight suspended on a string **B.a cone resting on its slant edge** C.a heavy nased lamp table D.a beam of balance in use

A cone resting on its slant edge is neutral equilibrium as it produces rolling movement. All other options are example of stable equilibrium.

42. A cone is in neutral equilibrium has its potential energy A.decreasedC.increased **C.unchanged** D.oscillating.

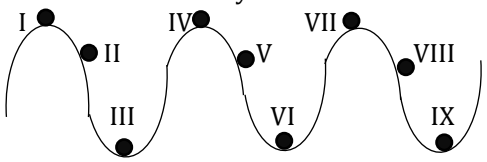
43. I. II. III. IV.



Which of the following bodies, each with centre of gravity G, lying on a horizontal table, is/are unstable equilibrium? A.II **B.IV** C.I and IV D.II and III E.III and IV.

IV represents a body in unstable equilibrium as it has a high centre of gravity G. III represents a body in stable equilibrium as it has low centre of gravity. I and II represents bodies in neutral equilibrium as they will produce rolling movements continuously on the horizontal table.

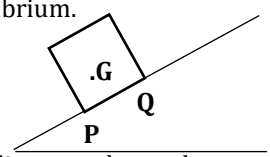
44.



The diagram above represents ball in an undulating surface. Which of the following options represents positions of stable equilibrium? A.III,IV and VIII **B.III,VI and IX** C.I,IV and VII D.II,V and VIII

Balls at I,II,IV,V,VI and VII will not return to its original position after displacement i.e unstable equilibrium. Balls at III,VI and IX will return to its original position after displacement i.e stable equilibrium.

45.



The diagram above shows a solid figure with a base PQ and centre of gravity G on an inclined plane. Which of the following statements is correct? **A.The solid will fall over if the vertical line through G lies outside the base** B.The solid will fall over of the vertical line through G lies inside the base C.The solid will not fall over if the vertical line through G lies outside the base D.The solid can never fall over

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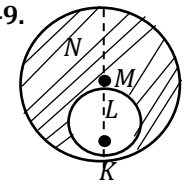
If the the vertical line through the centre of gravity G lies outside the base, the body will fall over the inclined plane i.e unstable equilibrium. If the vertical line through the centre of gravity G lies inside the base, the body will not fall over the inclined plane i.e stable equilibrium.

46. When a body is in stable equilibrium and is slightly tilted A.the vertical through the centre of gravity falls outside the base B.the vertical through the centre of gravity is not disturbed **C.the moment of the weight is greater than the moment of the action about the edge of tilt** D.the moment of the weight is less than the moment of the action about the edge of tilt E.the moment of the weight equals the moment of the action about the edge of tilt.

A body is said to be in stable equilibrium when the vertical through the centre of gravity rises and falls inside the base of the body after displacement and the moment of weight is greater about the base i.e moment of weight is greater than moment of action about the edge of tilt.

47. When a body is slightly tilted, it is found that its centre of gravity is slightly raised. What is the state of equilibrium of the body? A.unstable **B.stable** C.neutral D.cannot be determined.

48. The centre of gravity of a uniform triangular lamina is the point of intersection of A.the perpendicular bisectors of its sides B.the bisectors of its three sides **C.the medians** D.the altitudes E.all of the above.

49. 

The diagram shows a uniform circular object of centre M and diameter KN from which a circular section of L and diameter KM has been removed KL is a straight line. The centre of gravity if the resulting body is located A.at L B.at M C.between K and M **D.between *M* and *N*.**

The centre of gravity before the removal of the section L of the object is at the centre M. On removal of the section, the centre of gravity shift to the heavier side of the body, between M and N.

- When a cross-section of a circular body is cut the centre of gravity shifts to the heavier side (section not cut) of the body.

Jamb past questions on equilibrium of forces:
[1989/4,1994/10,1986/6,1999/5,2009/6,2011/8,2010/7,1981/41,1982/46,1983/3,1990/3, 1991/6,1993/11,1995/4,1997/9,2000/10,2005/7,1982/29,2008/8,2015/30,2016/34,1978/13,1979/39,1984/2,1986/9,1985/10,11,1993/12, 1994/8,1995/10,1997/11,1998/8,2003/11,2007/47,1990/7,8,1980/31,1992/7,2014/10]

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Chapter 7 – Elastic Properties, Elasticity

● **Elasticity –** Elasticity is the ability of a material to return to its original shape and size after being distorted by an external force.

- Brittle materials – These are materials that break easily i.e before getting to the elastic limit e.g glass.

- Ductile materials – These are materials that can be extended beyond the elastic limit before breaking and can be drawn into thin wires before breaking e.g copper.

- Hooke's law states that the extension of any elastic material is directly proportional to the applied force F/lod or weight, provided the elastic limit is not exceeded. $\boldsymbol{F \propto e}$, $\boldsymbol{F = ke}$, k=force or elastic constant or stiffness constant of the elastic material. $\boldsymbol{k = \frac{F}{e}}$, $\boldsymbol{\frac{F_1}{e_1} = \frac{F_2}{e_2}}$.

$\boldsymbol{e}$=extension=Final length−Original length,

$\boldsymbol{e_1 = L_1 - L_o}$, $\boldsymbol{e_2 = L_2 - L_o}$,

$\boldsymbol{\frac{F_1}{L_1 - L_o} = \frac{F_2}{L_2 - L_o}}$, $L_1$the length produced by F_1, L_2 is the length produced by F_2, L_o is the original length or natural length or unstretched length.

- Hooke's law can also be represented by – $\boldsymbol{\frac{F_2 - F_1}{F_1 - F_o} = \frac{L_2 - L_1}{L_1 - L_o}}$, **$\boldsymbol{F_o = 0}$, as no force is acting on an elastic material at rest, when $\boldsymbol{L = L_o}$.**

- If three forces are acting on the same elastic material causing a corresponding increase in the length of the material, they can be represented by $\boldsymbol{-\frac{F_3 - F_2}{F_2 - F_1} = \frac{L_3 - L_2}{L_2 - L_1}}$.

- Extension is same as change in length or increase in length by or extended by.

- Increased in length to or extended to is same as final length.

- Mass can be used in place of force in the formulas above.

● **Modulus of elasticity of elastic materials –**

- Tensile stress $\boldsymbol{S_T}$ is the ratio of the force acting on a unit cross-sectional area of an elastic material.

Tensile stress , $\boldsymbol{S_T = \frac{Force}{Area} = \frac{F}{A}}$, $\boldsymbol{A = \pi r^2}$.

- Tensile strain $\boldsymbol{S}$is the ratio of the extension to the original length. It does not have a unit.

Tensile strain $\boldsymbol{S = \frac{extension}{original\ length} = \frac{e}{L}}$.

- Young modulus of elasticity $\boldsymbol{E}$is the ratio of tensile stress to tensile strain within the elastic region.

Yound modulus, $\boldsymbol{E = \frac{Tensile\ stress}{Tensile\ strain} = \frac{S_T}{S}}$,

$\boldsymbol{E = \frac{F/A}{e/L} = \frac{FL}{Ae}}$.

- Tensile stress and Young modulus has the same unit and dimension as Pressure i.e Nm^{-2} or Pascal(Pa). Tensile Strain has no unit.

- Young modulus depends only on the type of material from which an elastic wire is made. It is independent of the dimesion of the wire, stress and strain in the wire.

- Force constant k, young molulus E, area A and length L of an elastic material is given as : $\boldsymbol{k = \frac{EA}{L}}$.

- Force per unit length=force constant×strain, $\boldsymbol{\frac{F}{L} = k \times S}$.

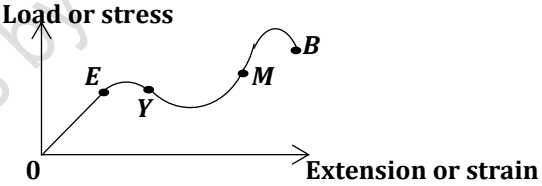
- Energy density of an elastic material or stretched wire is the ratio of its energy stored in the material to it's volume.

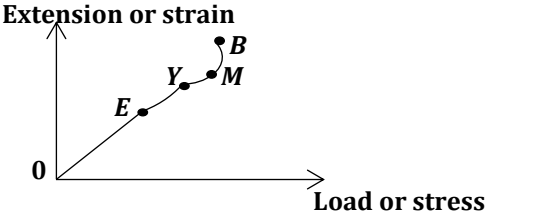
Energy density= $\left(\boldsymbol{\frac{energy}{volume}}\right) = \frac{1}{2}$(stress×strain)

- Bulk modulus is the ratio of hydrostatic stress(pressure) to volumetric strain in liquids.

- Shear modulus or Rigidity modulus is the ratio of shear/tangetial stress to shear strain.

- Force/load-extension graph or stress-strain graph of an elastic material – A plot of load(force) against extension is a straight line graph passing from the origin provided the elastic limit is not exceeded. On further increase of the load, a point is reached where the spring no longer returns to its original length and the spring snaps.





Elastic limit ($\boldsymbol{E}$) – It is the limit of force or the load beyond which the material does not return to its original length when the load is removed. **Hooke's law is obeyed up to the elastic limit i.e beyond the elastic limit the force or load is no longer proportional to the extension.**

Yield point ($\boldsymbol{Y}$) – It is the point beyodnthe elastic limit in which the elastic material has yielded all its elasticity permanently and has become plastic. **At yield point, a slight increase in load produces a large extension.**

Maximum load ($\boldsymbol{M}$) – It is the maximum load which the elastic material is able to support.

Breaking or fracture point ($\boldsymbol{B}$) – It is the force or load at which the elastic material breaks into two (fractures).

$\boldsymbol{OE}$ – Elastic region, $\boldsymbol{EB}$ – Plastic region.

- The slope of the linear portion of a force-extension graph represents the force or stiffness constant.

Demystified Series by Dr Timothy

- Tensile stress ∝ Tensile strain. The slope of the graph of a tensile stress vs tensile strain is Young modulus.

● **Workdone on an elastic material –**The workdone on an elastic material W is given by; **Workdone = Average force×extension,**

$\boldsymbol{W = \frac{1}{2}Fe = \frac{1}{2}ke^2 = \frac{F^2}{2k}}$ **.** This is the same as the elastic potetial energy stored in it.

- In a force-extension graph the area under the graph gives the workdone on an elastic material or the elastic potential energy stored in it.

Examples :

1. Hooke's law states that A.when a wire is loaded beyond the elastic limit the extension produced is inversely proportional to the tension B.the extension produced in a wire is equal to the load applied **C.the extension produced in a wire is proportional to the tension applied if the elastic limit is not exceeded** D.the enxtension produced in a wire is inverlsey proportional to the tension applied E.when a wire is loaded and unloaded it returns to its original length.

- Hooke's law can also be stated as stress is directly proportional to strain provided the elastic limit is not exceeded.

2. Which of the following is Hooke's law? A.Action and reaction are equal and opposite B.When a body is totally or partially immersed in a fluid it experiences an uothrust which is equal to the weight of fluid displaced C.For a fixed mass of gas at constant temperature, the pressure is inversely proportional to its volume **D.The extension of a material is proportional to the force applied to it if the elastic limit is not exceeded.**

3. If a load of 1kg stretches a cord by 1.2cm, what is the force constant of the cord? (g=10ms^{-2}) A.866Nm^{-1} **B.833Nm^{-1}** C.769Nm^{-1} D.667Nm^{-1}.

Load=force=$F=mg$=1×10=10N, m=1kg,

e=1.2cm=1.2×10^{-2}m, $F = ke$,

$k = \frac{F}{e} = \frac{10}{1.2\times10^{-2}} = 833\text{Nm}^{-1}$.

4. A spiral spring extends from a length of 10.00cm to 10.01cm, when a force of 20N is applied on it. Calculate the force constant of the spring. **A.2000Nm^{-1}** B.1000Nm^{-1} C.500Nm^{-1} D.250Nm^{-1}.

F=20N, L_1=10.01cm, L_o=10.00cm, k=?,

$k = \frac{F}{e} = \frac{F}{L_1-L_o} = \frac{20}{10.01-10} = \frac{20}{0.01} = 2000\text{Nm}^{-1}$.

5. A spring with a force constant of 20Nm^{-1} hanging vertically has an equilibrium length of 0.25m. A 100g mass is attatched to the spring and carefully allowed to drop so that it comes to rest. The new equilibrium length of the spring is **A.30cm** B.40cm C.50cm D.75cm (g=10ms^{-2}).

k=20Nm^{-1}, first equilibrium length=original length=L_o=0.25, m=100g=0.1kg, F=mg=0.1×10, F=1N, The new equilibrium length= L_1=?,

$F = ke = k(L_1 - L_o)$, $1=20(L_1 - 0.25)$,

$L_1 - 0.25 = \frac{1}{20} = 0.05$, $L_1 = 0.05+0.25 = 0.3$m,

$L_1 = 30$cm.

6. A spiral spring extends by 5cm under a load of 60N. When the load is replaced by a steel block, the new extension is 7cm.The weight of the steel block is A.12N B.43N**C.84N** D.96N.

e_1=5cm, F_1=60N, e_2=7cm, F_2=?,

$k = \frac{F}{e}$, $\frac{F_1}{e_1} = \frac{F_2}{e_2}$, $\frac{60}{5} = \frac{F_2}{7}$, $F_2 = \frac{60\times7}{5} = 84$N.

7. The extension in a spring when 5g wt hung from it was 0.56cm. If Hooke's law is obeyed, what is the extension caused by a load of 20g wt? A.1.12cm B.2.14cm C.2.52cm D.2.80cm **E.2.24cm**

F_1=5g wt, e_1=0.56cm, F_2=20g wt, e_2=?,

$k = \frac{F}{e}$, $\frac{F_1}{e_1} = \frac{F_2}{e_2}$, $\frac{5}{0.56} = \frac{20}{e_2}$, $e_2 = \frac{20\times0.56}{5} = 2.24$cm.

8. A 10g mass place on the pan of spring balance causes an extension of 5cm. If a 15g mass is placed on the scale pan of the same spring balance the extension is A.3.3cm B.6.5cm **C.7.5cm** D.10.8cm E.15.0cm

F_1=m_1=10g, e_1=5cm, F_2=m_2=15g, e_2=?,

$k = \frac{F}{e}$, $\frac{F_1}{e_1} = \frac{F_2}{e_2}$, $\frac{10}{5} = \frac{15}{e_2}$, $e_2 = \frac{15\times5}{10} = 7.5$cm.

9. A load of 50N attached to the free end of a spiral spring stretches it from 0.30m to 0.36m. Calculate the length of the spring when a load of 70N is suspended from the end **A.0.384cm** B.0.830cm C.0.960cm D.1.200cm.

F_1=50N,L_o=0.30cm,L_1=0.36cm,F_2=70N,L_2=?,

$k = \frac{F}{e}$, $\frac{F_1}{e_1} = \frac{F_2}{e_2}$, $\frac{F_1}{L_1-L_o} = \frac{F_2}{L_2-L_o}$,

$\frac{50}{0.36-0.3} = \frac{70}{L_2-0.3}$, $\frac{50}{0.06} = \frac{70}{L_2-0.3}$,

$L_2 - 0.3 = \frac{70\times0.06}{50} = 0.084$, $L_2 = 0.084+0.3$,

$L_2 = 0.384$cm.

10. A force of 15N stretches a spring to a total length of 30cm. An additional force of 10Nstretches the spring 5cm further. Find the natural length of the spring A.25.0cm **B.22.5cm** C.20.0cm D.15.0cm.

F_1=15N, L_1=30cm, F_2=15+10=25N,

L_2=30+5=35cm, L_o=natural length=?,

$\frac{F_1}{L_1-L_o} = \frac{F_2}{L_2-L_o}$, $\frac{15}{30-L_o} = \frac{25}{35-L_o}$, $\frac{35-L_o}{30-L_o} = \frac{25}{15}$,

$\frac{35-L_o}{30-L_o} = \frac{5}{3}$, $3(35 - L_o)=5(30 - L_o)$,

$105-3L_o = 150-5L_o$, $5L_o - 3L_o = 150-105$,

$2L_o = 45$, $L_o = \frac{45}{2} = 22.5$cm. Alternatively,

F_o=0, F_1=15N, F_2=15+10=25N, L_o=?, L_1=30cm,

L_2=30+5=35cm, $\frac{F_2-F_1}{F_1-F_o} = \frac{L_2-L_1}{L_1-L_o}$, $\frac{25-15}{15-0} = \frac{35-30}{30-L_o}$,

$\frac{10}{15} = \frac{5}{30-L_o}$, $\frac{2}{3} = \frac{5}{30-L_o}$, $2(30 - L_o)=5\times3=15$,

$60-2L_o = 15$, $2L_o = 60-15=45$, $L_o = \frac{45}{2}$,

$L_o = 22.5$cm.

11. A force of 100N stretches an elastic string to

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a total length of 200cm. If an additional force of 100N stretches the string 5cm further, find the natural length of the string **A.15cm** B.12cm C.10cm D.8cm E.5cm.

F_1=100N, L_1=20cm, F_2=100+100=200N, L_2=20+5=25cm, L_o=natural length=?,

$\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}$, $\frac{100}{20-L_o}=\frac{200}{25-L_o}$, $\frac{25-L_o}{20-L_o}=\frac{200}{100}=\frac{2}{1}$,

$25-L_o=2(20-L_o)$, $25-L_o=40-2L_o$,

$2L_o-L_o=40-25$, $L_o=15$cm. Alternatively,

F_o=0, F_1=100N, F_2=100+100=200N, L_o=?,

L_1=20cm, L_2=20+5=25cm,

$\frac{F_2-F_1}{F_1-F_o}=\frac{L_2-L_1}{L_1-L_o}$, $\frac{200-100}{100-0}=\frac{25-20}{20-L_o}$, $\frac{100}{100}=\frac{5}{20-L_o}$,

$1=\frac{5}{20-L_o}$, $20-L_o=5$, $L_o=20-5=15$cm.

12. The total length of a spring when a mass of 20g is hung from its end is 14cm, while its total length is 16cm when a mass of 30g is hung from the same end. Calculate the unstretched length of the spring assuming Hooke's law is obeyed A.9.33cm B.10.00cm C.10.66cm D.12.00cm E.15.00cm.

F_1=20g, L_1=14cm, F_2=30g, L_2=16cm, L_o=unstretched length=?, $\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}$,

$\frac{20}{14-L_o}=\frac{30}{16-L_o}$, $\frac{20}{30}=\frac{14-L_o}{16-L_o}$, $\frac{2}{3}=\frac{14-L_o}{16-L_o}$,

$2(16-L_o)=3(14-L_o)$, $32-2L_o=42-3L_o$,

$3L_o-2L_o=42-32$, $L_o=10$cm.

Alternavtively : F_o=0, F_1=20g, F_2=20g, L_o=?, L_1=14cm, L_2=16cm, $\frac{F_2-F_1}{F_1-F_o}=\frac{L_2-L_1}{L_1-L_o}$,

$\frac{30-20}{20-0}=\frac{16-14}{14-L_o}$, $\frac{10}{20}=\frac{2}{14-L_o}$, $\frac{1}{2}=\frac{2}{14-L_o}$,

$14-L_o=2\times2=4$, $L_o=14-4=10$cm.

13. The spiral spring of a spring alance is 25.0cm long when 5N hangs on it and 30.0cm long, when the weight is 10N. What is the length of the spring if the weight is 3N, assuming Hooke's law is obeyed? A.15.0cm B.17.0cm C.20.0cm **D.23.0cm.**

First calculate the original length of the spring before calculating the length of the spring when the 3N is hung from it. F_1=5N, L_1=25cm, F_2=10N, L_2=30cm, L_o=natural length=?,

$\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}$, $\frac{5}{25-L_o}=\frac{10}{30-L_o}$, $\frac{5}{10}=\frac{25-L_o}{30-L_o}$,

$\frac{1}{2}=\frac{25-L_o}{30-L_o}$, $30-L_o=2(25-L_o)=50-2L_o$,

$2L_o-L_o=50-30$, $L_o=20$cm,

F_3=3N, L_3=?, $\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}=\frac{F_3}{L_3-L_o}$,

$\frac{F_1}{L_1-L_o}=\frac{F_3}{L_3-L_o}$, $\frac{5}{25-20}=\frac{3}{L_3-20}$, $\frac{5}{5}=\frac{3}{L_3-20}$,

$1=\frac{3}{L_3-20}$, $L_3-20=3$, $L_3=3+20=23$cm.

Alternatively : F_1=5N, L_1=25cm, F_2=10N, L_2=30cm, F_3=3N, L_3=?,

$\frac{F_3-F_2}{F_2-F_1}=\frac{L_3-L_2}{L_2-L_1}$, $\frac{3-10}{10-5}=\frac{L_3-30}{30-25}$, $\frac{-7}{5}=\frac{L_3-30}{5}$,

$L_3-30=\frac{-7\times5}{5}=-7$, $L_3=30-7=23$cm.

14. Complete the table below for a body which obey's Hooke's law.

Load added	Scale reading
1g	33.7cm
3g	34.9cm
7g	?

A.35.5cm B.36.1cm **C.37.3cm** D.38.5cm E.39.1cm

First calculate the original length of the spring before calculating the length of the spring when the 3N is hung from it. F_1=1g, L_1=33.7cm, F_2=3g, L_2=34.9cm, L_o=natural length=?,

$\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}$, $\frac{1}{33.7-L_o}=\frac{3}{34.9-L_o}$,

$34.9-L_o=3(33.7-L_o)=101.1-3L_o$,

$3L_o-L_o=101.1-34.9$, $2L_o=66.2$cm,

$L_o=\frac{66.2}{2}=33.1$cm.

F_3=7g, L_3=?, $\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}=\frac{F_3}{L_3-L_o}$,

$\frac{F_1}{L_1-L_o}=\frac{F_3}{L_3-L_o}$, $\frac{1}{33.7-33.1}=\frac{7}{L_3-33.1}$, $\frac{1}{0.6}=\frac{7}{L_3-33.1}$,

$L_3-33.1=7\times0.6=4.2$, $L_3=4.2+33.1=37.3$cm.

Alternatively : F_1=1g, L_1=33.7cm, F_2=3g, L_2=34.9cm, F_3=7g, L_3=?, $\frac{F_3-F_2}{F_2-F_1}=\frac{L_3-L_2}{L_2-L_1}$,

$\frac{7-3}{3-1}=\frac{L_3-34.9}{34.9-33.7}$, $\frac{4}{2}=\frac{L_3-34.9}{1.2}$, $2=\frac{L_3-34.9}{1.2}$,

$L_3-34.9=2\times1.2=2.4$, $L_3=2.4+34.9=37.3$cm.

15. A string of length 4m is extended by 0.02m when a load of 0.4kg is suspended at its end. What will be the new length of the string when the applied force is 15N? (g=10ms^{-2}) A.5.05m B.6.08m C.4.05m **D.4.08m.**

L_o=original length=4m, e_1=0.02m, m=0.4kg, F_1=load=mg=0.4×10=4N, F_2=15N, L_2=new length of the string, $k=\frac{F}{e}$, $\frac{F_1}{e_1}=\frac{F_2}{e_2}$, $\frac{F_1}{e_1}=\frac{F_2}{L_2-L_o}$,

$\frac{4}{0.02}=\frac{15}{L_2-4}$, $L_2-4=\frac{15\times0.02}{4}=0.075$,

$L_2=0.075+4=4.075$m$=4.08$m.

16. A spiral spring of natural length 20.00cm has a scale pan hanging freely in its lower end. When an object of mass 40g is placed in the pan, its length vecomes 21.80cm. When the object is replaced with another of mass 60g, the length becomes 22.05cm. Calculate the mass of the scale pan (g=10ms^{-2}) **A.104g** B.208g C.312g D.416g.

The extension e in a spring balance is produced by a force or load which is equal the mass of the scale pan m_s and the mass of the load m placed in the scale pan.

L_o=20.00cm, $F_1=m_s+m_1=(m_s+40)$g,

L_1=21.80cm, $F_2=m_s+m_2=(m_s+60)$g,

L_2=22.05cm, $k=\frac{F}{e}$, $\frac{F_1}{e_1}=\frac{F_2}{e_2}$, $\frac{F_1}{L_1-L_o}=\frac{F_2}{L_2-L_o}$,

$\frac{m_s+40}{21.80-20.00}=\frac{m_s+60}{22.05-20.00}$, $\frac{m_s+40}{1.8}=\frac{m_s+60}{2.05}$,

$2.05(m_s+40)=1.8(m_s+60)$,

$2.05m_s+82=1.8m_s+108$,

$2.05m_s-1.8m_s=108-82$, $0.25m_s=26$,

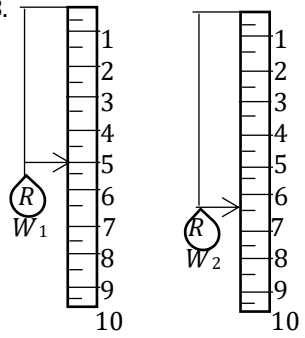
$m_s=\frac{26}{0.25}=104$g.

17. A load of mass 45g produced an extension of 6.5cm when placed on a scale pan attached to a spring. If a 60g mass produced an extension of 8cm on same spring, determine the mass of the

Demystified Series by Dr Timothy

scale pan A.35g **B.20g** C.15g D.10g .
$F_1=m_s+m_1=(m_s+45)$g, e_1=6.5cm, $F_2=m_s+m_2$ = (m_s+60)g, e_2=8cm,
$k=\frac{F}{e}$, $\frac{F_1}{e_1}=\frac{F_2}{e_2}$, $\frac{m_s+45}{6.5}=\frac{m_s+60}{8}$,
$8(m_s+45)=6.5(m_s+60)$,
$8m_s+360=6.5m_s+390$,
$8m_s-6.5m_s=390-360$, $1.5m_s=30$,
$m_s=\frac{30}{1.5}=20$g.

18.



In the figure above W_1=200g and W_2=450g. Calculate the extension of the spring per unit load A.6.0×10^{-5}mN^{-1} B.2.5×10^{-4}mN^{-1} **C.6.0×10^{-3}mN^{-1}** D.2.5×10^{-2}mN^{-1} (g=10ms^{-2}).
W_1=450g, W_1=200g, L_1=5cm, L_2=6.5cm,
Increase in load= $W_2-W_1=450-200=250$g,
F =load=250g=0.25kg=2.5N,
extension=$L_2-L_1=6.5-5=1.5$cm=1.5×10^{-2}m
Extension per unit load=$\frac{2.5}{1.5\times10^{-2}}$=6.0×10^{-3}mN^{-1}.

19. A spring has a total length of 17.5cm under a load of 250g and 20.0cm under a load of 300g. The extension per unit load is A.5×10^{-5}mN^{-1} B.8×10^{-5}mN^{-1} **C.5×10^{-2}mN^{-1}** D.8×10^{-2}mN^{-1} (g=10ms^{-2}).
W_1=300g, W_1=250g, L_1=17.5cm, L_2=20.0cm,
Increase in load= $W_2-W_1=300-200=50$g,
F =load=50g=0.05kg=0.5N,
extension=$L_2-L_1=20.0-17.5=2.5$cm=2.5×10^{-2}m
Extension per unit load=$\frac{0.5}{2.5\times10^{-2}}$ = 5×10^{-2}mN^{-1}.

- Extension per unit load is the reciprocal of force or elastic or stiffness constant.

20. The property of a material that allows it to be stretched into a wire is called A.brittleness B.malleability **C.ductility** D.fragibility.

21. Elasticity is the property which enables a material to **A.return exactly to its original shape and size on the removal of a force** B.retain its deformed shape after removal of an applied force C.be drawn out in length D.break without warning

- Plasticity is the ability of a material to remain in its new position or retain its deformed shape after distortion by an applied force.

- Ductility is the ability of a material to be drawn out in length.

- Brittleness is the ability of a material to break easily without warning.

22. The ratio of tensile stress to tensile strain is known as A.modulus of rigidity B.modulus of elasticity C.bulk modulus **D.Young's modulus** E.shear modulus.
Modulus of elasticity can be Young's modulus, shear modulus or bulk modulus.

23. The general definition of elastic modulus is **A.$\frac{stress}{strain}$** B.$\frac{strain}{stress}$ C.$stress\times strain$ D.$\sqrt{\frac{stress}{strain}}$.
Modulus of elasticity or elastic modulus generally is the ratio of stress to strain.

24. In an experiment to determine Young's modulus for a wire, several loads are attached to the wire and the corresponding extensions measured. The tensile stress in each case depends on **A.load and the radius of the wire** B.radius of the wire and extension C.load and the extension D.extension and the original length of the wire.
Tensile stress is the ratio of the tensile force or load to the area of the wire.Hence, tensile stress depends on the load or force and the cross-sectional area of the wire,which is given by its radius. Tensile strain depends on the extension and ethe original length of the wire.

25. A wire of diameter 2.8mm supports a weight W. If the tensile stress in the wire is 2×10^7Nm^{-2}, the value of W is (π=22/7) **A.123.2N** B.98.4N C.56.0N D.49.2N.
diameter, d=2.8mm=2.8×10^{-3}m, Tensile stress, S_T=2×10^7Nm^{-2}, Force=Weight=W,
Tensile stress, $S_T=\frac{Force}{Area}=\frac{F}{A}$, $A=\pi r^2$.
radius, $r=\frac{d}{2}$, $A=\frac{\pi d^2}{4}$, $S_T=\frac{F}{\frac{\pi d^2}{4}}=\frac{4F}{\pi d^2}$,
$2\times10^7=\frac{4\times W}{\frac{22}{7}\times(2.8\times10^{-3})^2}$,
$4\times W=2\times10^7\times\frac{22}{7}\times7.84\times10^{-6}=49.28\times10^1$,
$W=\frac{492.8}{4}$=123.2N.

24. The tendon in a man's leg 0.01m long, if a force of 5N stretches the tendon by 2.0×10^{-5}m, calculate the strain on the muscle A.5×10^6 B.5×10^2 C.2×10^{-3} D.2×10^{-7}.
Length,L=0.01m, extension, e=2.0×10^{-5}m, F=5N,
Tensile strain, $S=\frac{extension}{original\ length}=\frac{e}{L}$,
$S=\frac{2\times10^{-5}}{0.01}=2\times10^{-7}$.

25. An elastic spring of length 20cm stretches by 7mm under a load of 50N. The strain in the string is A.1.75×10^{-1} B.1.75×10^{-2} **C.3.5×10^{-2}** D.7.0×10^{-2}.
L=20cm, e=7mm=0.7cm, S=?,
Tensile strain, $S=\frac{e}{L}=\frac{0.7}{20}$=3.5×10^{-2}.

26. A wire of length 2.00m and radius 1.0mm is stretched by 25.0mm on application of a force of 10^3N. Calculate the Young's modulus for the wire A.5×10^{10}Nm^{-2} **B.2.5×10^{10}Nm^{-2}** C.5×10^{8}Nm^{-2} D.2.5×10^{8}Nm^{-2}.
L=2.00m, r=1.0mm=1×10^{-3}m, F=10^{3}N,

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e=25.00mm=25×10^{-3}m, Young's modulus, E=?
Yound modulus $E = \frac{Tensile\ stress}{Tensile\ strain} = \frac{S_T}{S}$,
$E = \frac{F/A}{e/L} = \frac{FL}{Ae}$, $A = \pi r^2$, $E = \frac{FL}{\pi r^2 e}$,
$E = \frac{10^3 \times 2}{\frac{22}{7}\times(1\times10^{-3})^2\times25\times10^{-3}} = \frac{2000}{\frac{22}{7}\times1\times10^{-6}\times25\times10^{-3}}$,
$E = 2.5\times10^{10}$Nm^{-2}.

27. A wire of cross-sectional area 6×10^{-5}m^2 and length of 50cm stretches by 0.2mm under a load of 3000N. The Young's modulus for the wire is A.8×10^{10}Nm^{-2} **B.1.25×10^{11}Nm^{-2}** C.2.5×10^{11}Nm^{-2} D.5×10^{11}Nm^{-2}.
A=6×10^{-5}m^2, L=50cm=0.5m, F=3000N,
e=0.2mm=0.0002m, E=?, $E = \frac{F/A}{e/L} = \frac{FL}{Ae}$
$E = \frac{3000\times0.5}{6\times10^{-5}\times0.0002} = 1.25\times10^{11}$Nm^{-2}.

28. A metallic bar 50cm long has a uniform cross-sectional area of 4.0cm^2. If a tensile force of 35kN produces an extension of 0.25mm, calculate the value of Young's modulus A. 8.75×10^{10}Nm^{-2} B.3.50×10^{11}Nm^{-2} **C.1.75×10^{11}Nm^{-2}** D.2.75×10^{11}Nm^{-2}.
L=50cm=0.5m, F=35kN=35000N, e=0.25mm=0.25×10^{-3}m, A=4.0cm^2=4×(10^{-2})^{2}m=0.0004m,
$E = \frac{F/A}{e/L} = \frac{FL}{Ae} = \frac{35000\times0.5}{0.0004\times0.25\times10^{-5}} = 1.75\times10^{11}$Nm^{-2}.

29. An elastic string of force constant 2000Nm^{-1} undergoes a strain of 0.02 under a load of 20N. The natural length of the string is **A.50cm** B.75cm C.100cm D.200cm.
k=2000Nm^{-1}, S=0.02, F=20N, L=?,
Force per unit length=force constant×strain
$\frac{F}{L} = k \times S$, $\frac{20}{L} = 2000\times0.02$, $\frac{20}{L} = 40$,
$L = \frac{20}{40} = 0.5$m=50cm.

30. A metal rod of diameter 7mm and length 100mm is made from steel of Young's modulus 2×10^{11}Nm^{-2}. The force constant for the rod is A.7.7×10^6Nm^{-1} B.3.5×10^{7}Nm^{-1} C.2.8×10^{7}Nm^{-1} **D.7.7×10^{7}Nm^{-1}**.
d=7mm=0.007m, L=100mm=0.1m,
E=2×10^{11}Nm^{-2}, k=?, $A = \pi r^2 = \frac{\pi d^2}{4}$,
$k = \frac{EA}{L} = \frac{E\frac{\pi d^2}{4}}{L} = \frac{E\pi d^2}{4L} = \frac{2\times10^{11}\times\frac{22}{7}\times0.007^2}{4\times0.1}$,
$k = 7.7\times10^7$Nm^{-1}.

- Force constant k increases as area A increases and length L decreases i.e $k \propto \frac{A}{L}$, Young's modulus E=constant.

31. A steel sample and a plastic specimen of identical dimensions are subjected to the same tensile force. The plastic specimen shows a greater extension I.plastic has a higher modulus of elasticity than steel II.steel has a higher modulus of elasticity than plastic III.steel has higher force constant than plastic IV.stell has higher demsity than plastic. Which of the above statementsis/are correct? A.I B.II C.I and IV **D.II and III.**
$F = ke$, extension, $e = \frac{F}{k}$, plastic specimen of greater extension will have a smaller force constant k than steel. $k = \frac{EA}{L}$, force constant k increases as Young's modulus E increases, hence plastic specimen has a higher Young's modulus than steel.

32. When a force of 50N is applied to the free end of an elastic cord, an extension of 4cm is produced in the cord. Calculate the workdone on the cord A.4.0J B.2.0J **C.1.0J** D.0.5J
F=50N, e=4cm=0.04cm, W=?,
W = $\frac{1}{2}Fe = \frac{1}{2}\times50\times0.04 = 1.0$J.

33. An elastic string of 20cm extends to 24cm when it supports a weight of 100N. The energy stored in the string is A.1J **B.2J** C.5J D.10J.
L_o=20cm=0.2m, L_1=24cm=0.24, F=100N,
e=L_1−L_o, W = $\frac{1}{2}Fe = \frac{1}{2}F(L_1-L_o)$,
W = $\frac{1}{2}\times100(0.24-0.2) = 50\times0.04 = 2$J.

34. An elastic string of force constant 200Nm^{-1} is stretched through 0.8m within its elastic limit. Calculate the energy stored in the string **A.64.0J** B.80.0J C.128.0J D.160.0J.
k=200Nm^{-1}, e=0.8m, W=?,
$W = \frac{1}{2}ke^2 = \frac{1}{2}\times200\times0.8^2 = 100\times0.64 = 64.0$J.

35. A spring of force constant 300Nm^{-1} is compressed such that its length shortens by 3cm. Calculate the energy stored in the spring A.0.270J B.0.405J **C.0.135J** D.0.540J
k=300Nm^{-1}, compression=e=3cm=0.03m,
W = $\frac{1}{2}ke^2 = \frac{1}{2}\times300\times0.03^2 = 0.135J$.

36. A spiral spring of natural length 30.0cm and force constant of 20Nm^{-1} is compressed to 20.0cm. Calculate the energy stored in the spring A.0.1J **B.1.0J** C.10.0J D.100.0J.
Compression involves a decrease in the length of an elastic material, L_o=30cm=0.3m, k=20Nm^{-1}, L_1=20cm=0.2m, e=L_o−L_1, W=?,
W = $\frac{1}{2}ke^2 = \frac{1}{2}k(L_o-L_1)^2 = \frac{1}{2}\times20(0.3-0.2)^2$,
W =10×0.1^2 = 1J.

37. A spring of force constant 1500Nm^{-1} is acted upon by a constant force of 75N. Calculate the potential energy stored in the spring A.5.0J B.3.8J C.3.2J **D.1.9J.**
k=1500Nm^{-1}, F=75N, potential energy stored in the spring is elastic potential energy, E=?,
$E = \frac{F^2}{2k} = \frac{75^2}{2\times1500} = 1.9$J.

38. As the tension in an elastic string increases from 100N to 180N, the string extends by 10cm. The work done in increasing the tension in the string is A.8J B.10J **C.14J** D.18J.
F_1=100N, F_2=180N, e=10cm=0.1m,
Workdone=Average force× extension,
Average force=$\frac{1}{2}(F_1+F_2) = \frac{1}{2}(100+180)$=140N
W = 140×0.1 = 14J.

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39. A force of 2.8N is applied to an elastic string of length 1.0m. If the force causes the string to increase in length by one-tenth of its original length. Calculate the elastic potential energy stored in the string **A.0.14*J*** B.0.28*J* C.0.42*J* D.0.56*J*.

F=2.8N, L=1.0m, $e=\frac{1}{10}L=\frac{1}{10}\times1$=0.1m,

$W=\frac{1}{2}Fe=\frac{1}{2}\times 2.8\times0.1$=0.14$J$.

40. Calculate the workdone to stretch an elastic string by 50cm, if a force of 12.5N produces an extension of 5cm in it A.2.250J B.3.125J C.4.450J **D.31.250J** E.44.450J.

In this question the 12.5N gives an extension of 5cm, hence we will have to calculate the force constant, before we can determine the work done in causing an extension of 50cm.

F_1=12.5N, e_1=5cm=0.05m, k=?,

$k=\frac{F_1}{e_1}=\frac{12.5}{0.05}=250\text{Nm}^{-1}$.

Work done to stretch the elastic string by 50cm,

$W=\frac{1}{2}ke_2^2$, e_2=50cm=0.5m,

$W=\frac{1}{2}\times250\times0.5^2=31.250$J.

41. A spring of length 2.0m requires a force of 6N to stretch it by 0.2cm. What work is done in stretching it by 2cm, if the elastic limit is not exceeded A.0.006J B.0.060J **C.0.600J** D.1.200J E.30.000J.

This question is solved exactly the same way question 40 was solved.

L=2.0cm, F_1=6N, e_1=0.2cm=0.002m, k=?,

$k=\frac{F_1}{e_1}=\frac{6}{0.002}=3000\text{Nm}^{-1}$,

Workdone in stretching the spring by 2cm,

$W=\frac{1}{2}ke_2^2$, e_2=2cm=0.02m,

$W=\frac{1}{2}\times3000\times0.02^2=0.6\text{J}=0.600$J.

42. A body of mass 11kg is suspended from a ceiling by an aluminium wire of length 2m and diameter 2mm. Calculate the elastic energy stored in the wirel. [Young's modulus of aluminium=7.0×10^{10}Nm^{-2}, g=10ms^{-2}, π=3.14] A.1.1×10^{-1}J **B.5.5×10^{-2}J** C.1.1×10^{-4}J D.5.5×10^{-5}J.

m=11kg, L=2m, d=2mm=0.002m, g=10ms^{-2}, π=3.14, $E=7.0\times10^{10}$Nm^{-2},

$F=\text{W}=mg=11\times10=110$N,

$k=\frac{EA}{L}$, $A=\pi r^2=\frac{\pi d^2}{4}$, $k=\frac{E\pi d^2}{4L}$,

$k=\frac{7\times10^{10}\times3.14\times0.002^2}{4\times2}=109{,}900\text{Nm}^{-1}$,

$k=1.099\times10^5$Nm^{-1}, $W=\frac{F^2}{2k}$, F=110N,

$W=\frac{110^2}{2\times1.099\times10^5}=0.055J=5.5\times10^{-2}$J.

43. The point at which the moleucles of a loaded wire begin to slide across each other resulting in a rapid increase in the extension is A.proportionality limit B.elastic point C.deformation point **D.yield point.**

Yield point is the point beyond the elastic limit where a loaded wire stretches rapidly.At yield point,a slight increase in load produces a large extension not proportional to the force.

- Beyond the yield point,the material loses it elasticity completely and is entirely plastic.
- Proportionality limit is the point up to which equal increase in load causes equal extension.
- Elastic point or limit is just beyond the proportionality limit and it is the point beyond which a stretched material does not return to its original shape and size.
- Breaking or fracture or deformation point is the point where the material breaks into two.

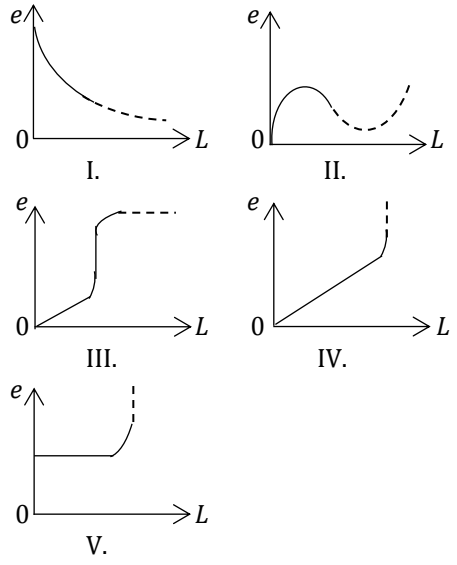
44. An additional load to an elastic wire when the elastic limit is already reached will make the wire to A.break B.recoil C.vibrate at a high amplitude **C.be strained permanently.**

An additional load to an elastic material at the elatic point or limit will extend the wire to the yield point , where it becomes plastic and would not return to its orginal shape or size when the load causing the distortion is removed.

45. A spring balance designed for a maximum load of 600N was used to weight a load of 820N. After the 820N load has been removed the pointer would not return to the zero mark on the scale because A.balance needed adjustment to return the pointer B.820N has been removed C.pointer had stopped turning D.spring balance was of a cheap type **E.elastic limit of the spring material had been exceeded.**

The elastic limitor elastic point of the spring is the maximum load it can carry (600N). If the spring balance is used to weigh a body (820N) greater than its elastic limit, the elastic limit will be exceeded and it becomes plastic, hence its pointer would not return to the zero mark.

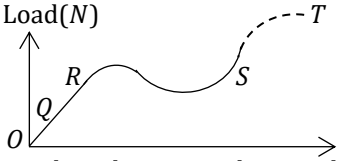
46. A piece of wire fixed to a rigid horizontal beam is loaded progressively until its breaks. Which one of the following graphs will represents the variation of extension e with load L for the wire?



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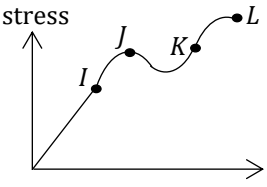
A.I B.II C.III **D.IV** E.V .
Extension-Load(or force) graph initially is a straigt line up to the elastic limit, then beyond the elastic limit it changes due to rapid increase in extension over a slight change in the load.

47. Load(N)



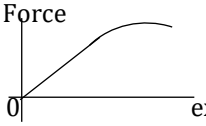
From the diagram above, the point that represents the elastic limit is A.Q B.S **C.*R*** D.T .
Point R represents the elastic limit. Hooke's law is only valid up to elastic limit.Hooke's law does not hold beyond the elastic limit. Point S represents the yield point while point T represents the breaking or fracture point.

48. The diagram below represents the stress-strain graph of a loaded wire. Which of the following is correct?



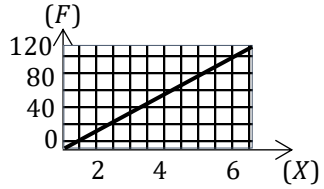
A.At I, the wire becomes plastic **B.*J* is the yield point** C.L is the elastic limit D.At K, the wire breaks.
I is the elastic limit, J is the yield point, between J and K the wire is plastic, the wire breaks at L, which is the breaking point.

49. Force



The figure shown is a graph of force against extensiom for a spiral spring. The force constant of the spring is given by the **A.slope of the linear portion of the graph** B.length of the linear portion of the graph C.area under the linear portion of the graph D.area under the entire graph.
Force constant k is given by the slope of the linear portion of the graph , $k = \frac{F}{e}$.
- The area of the entire graph gives the work done or energy stored in an elastic material.

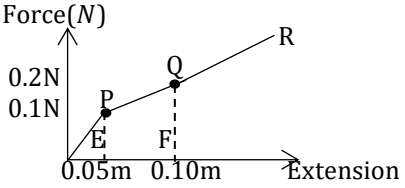
50. The diagram below represents the graph of the force(F) applied in stretching a spiral against the corresponding extension(X). The force constant of the spring is



A.40Nm^{-1} B.30Nm^{-1} **C.20Nm^{-1}** D.10Nm^{-1}.

Force constant k = Slope of the graph= $\frac{120-0}{6-0}$,
$k = \frac{120}{60} = 20\text{Nm}^{-1}$.

51. The diagram below shows the force-extension curve of a piece of wire. The energy stored when the wire is stretched from E to F?



A.1.5×10^{-2}J B.7.5×10^{-1}J **C.7.5×10^{-3}J** D.2.5×10^{-3}J.
Workdone or energy stored in an elastic material=Area of the figure under the force-extension graph. The figure between E and F is a trapezium. $A = \frac{1}{2}(a+b)h$
$W = \frac{1}{2}(0.1+0.2)(0.1-0.05) = \frac{1}{2}\times 0.3\times 0.05$,
$W = 7.5\times10^{-3}J$.

52. An object of mass 20kg is dropped on a spring which is placed 5m below. If the spring has a force constant of $5\times10^4 N\text{m}^{-1}$, the maximum compression of the spring is A.1.0m B.0.5m **C.0.2m** D.0.04m
m=20kg, h=5m, k=$5\times10^4 N\text{m}^{-1}$, compression, e=?, g=10ms^{-2} Potential energy of the object is converted into the elastic $P.E$ of the spring, $mgh = \frac{1}{2}ke^2$, $20\times10\times5 = \frac{1}{2}\times5\times10^4\times e^2$,
$e^2 = \frac{20\times10\times5\times2}{5\times10^4} = \frac{1}{25}$, $e = \sqrt{\frac{1}{25}} = \frac{1}{5} = 0.2\text{m}$.

- Question drill – 1. Use the data in the table below to determine the length X of a wire assuming Hooke's law is obeyed. **Ans=810.0mm.**

Force applied/ N	0.0	5.0	10.0	50.0
Length of wire/ mm	800.0	801.0	802.0	X

2. A spiral spring which obey's Hooke's law has an unstretched length of 20cm. If it stretches to 30cm when a mass of 500g is hung from one end, calculate the new length when a mass of 1000g replaces the 500g mass A.60cm **B.40cm** C.35cm D.20cm E.10cm.

Jamb past questions on Elastic properties :
[1978/5,1983/32,1987/7,14,1982/50,1981/43, 1987/7,1989/10,1994/16,2004/35,2005/42, 2007/30,2012/15,1992/10,1998/13,1999/14, 2006/40,2011/14,2014/16]

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Chapter 8 – Simple machines

● A simple machine is a device which enable work to be performed faster and more conveniently, where a small effort is used to overcome a large load e.g a fishing rod.

- Terms in simple machines :

1.Load L – It is the force overcomed.

2.Effort E – It is the force exerted to overcome the load.

3.Mechanical advantage $M.A$ or force ratio – It is the ratio of load to effort. $\boldsymbol{M.A = \frac{L}{E}}$.

4.Velocity ratio $V.R$ or speed or distance ratio – It is the ratio of distance moved by the effort(or effort arm) $\boldsymbol{x}$ to distance moved by the load(or load arm) $\boldsymbol{y}$. $\boldsymbol{V.R = \frac{x}{y}}$. Both $M.A$ and $V.R$ have no units.

5.Efficiency ϵ – It is the ratio of the work output to work input. $\boldsymbol{\epsilon = \frac{Work\ ouput}{Work\ input} \times 100}$.

Work output=Workdone by load or useful work done=Effort×distance moved by effort.

Work input=Workdone by effort or workdone on load or total work input=Load×distance moved by load.

- The efficiency of a machine is also given by the ratio of the mechanical advantage to velocity ratio. $\boldsymbol{\epsilon = \frac{M.A}{V.A} \times 100}$.

- Work input is always less than work input due to work done against friction i.e

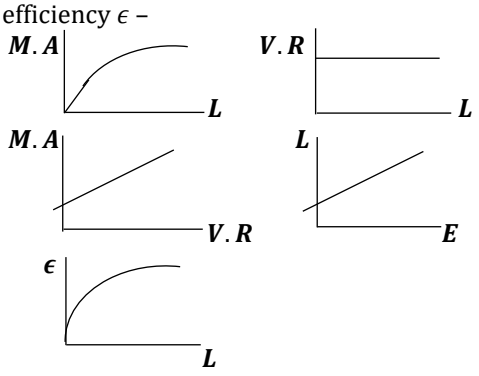
Work input=Work output+Work done against friction, $\boldsymbol{W_i = W_o + W_f}$.

- Mechanical advantage $M.A$ is always less than the velocity ratio $V.R$.

- Mechanical advantage $M.A$ and efficiency ϵ affected by friction and load , $M.A \propto L$ (not linearly), hence the efficiency can never be 100%. The highest $M.A$ is obtained in a frictionless machine.

- Velocity ratio $V.R$ depends on the size and shape of the machine parts and not affected or independent of friction and load.

- Grapical relationships of mechanical advantage $M.A$, velocity ratio $V.R$, load L, effort E and efficiency ϵ –



- $M.A$ initially increases with L and then becomes constant.

- Lubricants e.g grease and ball bearings are used in machines to reduce friction, thereby increasing $M.A$ and Effficiency but $V.R$ remains constant(as it is independent of friction).

- Classification of machines – Generally, **the basic machine of which all other machines are made of are hydraulic press, levers and inclined planes.**

● **Levers –** A small effort at a greater distance from the fulcrum is used to overcome a large load closer to the fulcrum, in order to obtain the greatest mechanical advantage. **It works on the principle of moment.**

- Classes of lever – There are three classes;

I.First class lever: The fulcrum F is between the the load L and the effort E **[LFE]** e.g crow bar, pliers,pincers,claw hammer,scissors,see saw and common balance. $M.A$ and $V.R$ is usually greater than 1 but could be less than or equal to 1.

II.Second class lever: The load L is between the effort E and the fulcrum F or pivot**[ELF]** e.g bottle opener, wheel barrow and nutcracker.$M.A$ and $V.R$ are always greater than 1.

III.Third class lever: The effort E is between the load L and the fulcrum F e.g forceps,sugar tongs,forearm of man,fishing rod and table knife.$M.A$ and $V.R$ are always less than 1.

- First and second class levers are force magnifiers while third class levers are distance magnifiers.

- In a lever system, $M.A = V.R$.

- A machine is said to be a force multiplier when the load is greater than the effort.

● **Pulley systems –** It is a machine used to raise or lower heavy loads majorly in construction sites.

- The **$V.R$ of a pulley system is equal to the number of pulleys.**

- The **$V.R$ can also be given by the number of ropes or strings supporting the pulleys, when the number of pulleys are lesser.**

- Block and tackle pulley system: It consists of fixed and movebale pulleys. **$V.R$ of a block and tackle system=number of moveable pulleys+number of fixed pulleys.**

- The greater the load to be lifted in a pulley system the greater the efficiency, as the weight of the pulleys attached to one load is less in proportion to the load.

- Efficiency increases as the mass(weight) or number of the pulleys reduces, due to decrease in friction.

● **Inclined planes –** It is a machine used to move heavy loads from a lower to a higher level up, by

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sloping a plank e.g a plank sloping from the floor of a van or truck through which heavy loads like drums of oil can be raised to the floor of the van or truck.

- The effort distance or effort arm is equal to the length of the plane L, while the load distance or load arm is equal to the height of the plane h. $\boldsymbol{V.R = \frac{L}{h}}$. $\boldsymbol{V.R = \frac{1}{sin\,\theta}}$, θ=angle of inclination of the plane to with the horizontal.

- The efficiency of an inclined plane depends on the degree of roughness of the of the inclined plane surface and not on the slope.

- The velocity ratio $\boldsymbol{V.R}$ of the inclined plane decreases as the slope or angle of inclination to the horizontal θ increases , $\boldsymbol{V.R}$ decreases as $\boldsymbol{sin\theta}$ increases.

- In a frictionless pulley, $M.A = V.R$, as efficiency is 100%. **Therefore $\boldsymbol{M.A}$ depends on the length and height of the inclined plane.**

- The effort $\boldsymbol{E}$ in a rough inclined plane i.e friction is not negligible, is given by –

$\boldsymbol{E = mgsin\theta + F_R = mgsin\theta + \mu mgcos\theta}$,

F_R=frictional force, μ=coefficient of static friction of the inclined plane.

- The load L moved up the plane $\boldsymbol{L = mg}$.

- The mechanical advantage $M.A$ is given by –

$\boldsymbol{M.A = \frac{L}{E} = \frac{mg}{mgsin\theta+\mu mgcos\theta} = \frac{1}{sin\theta+\mu cos\theta}}$.

- Efficiency ϵ of the rough inclined plane –

$\boldsymbol{\epsilon = \frac{M.A}{V.R} \times 100 = \frac{sin\theta}{sin\theta+\mu cos\theta} \times 100}$.

● **The Wedge –** It is a combination of two inclined planes used to separate bodies held together by strong and large cohesive forces e.g splitting timbers. Examples of wedges are chisels,knives,axe e.t.c.

- $\boldsymbol{V.R}$ of a wedge $= \boldsymbol{\frac{slant\ height\ of\ wedge}{thickness\ of\ wedge} = \frac{x}{th}}$.

- Neglecting frictional forces, $M.A = V.R$,

$\boldsymbol{M.A = \frac{x}{th}}$. Slant height of the edge=effort distance effort arm, thickness of wedge=load distance or load arm.

- A long(greater height), thin(lesser thickness) wedge has a higher $\boldsymbol{M.A}$ than a short(lesser height), thick(greater thickess) wedge. $\boldsymbol{M.A}$ increases i.e becomes greater as the angle θ between the slant heights decreases i.e becomes smaller.

● **ScrewJack –** It is made of an inclined plane wrapped round a cylinder, with the hypotenus of the inclined plane forming a spiral path which makes up the thread. It is used in screw jacks for lifting motor car, nuts and bolts, screw driver, spanner.

- The distance between two consecutive threads is the **pith $\boldsymbol{P}$** of the screw i.e the load distance.

- The distance through which a screw moves along its head when the head is turned through one complete revolution i.e 360° is equal to the pitch $\boldsymbol{P}$.

- Pitch $\boldsymbol{P = \frac{height\ raised\ of\ the\ load}{number\ of\ revolutions} = \frac{h}{n}}$.

- The distance moved by the effort=$\boldsymbol{2\pi r}$. r=radius of the handle or circle. or length of tommy bar.

- $\boldsymbol{V.R = \frac{2\pi r}{P}}$. $\boldsymbol{V.R \propto \frac{r}{P}}$i.e **$\boldsymbol{V.R}$ increases as length of tommy bar or radius of the circle $\boldsymbol{r}$ i.e effort arm, increases and the pitch $\boldsymbol{P}$ i.e load arm, decreases. - In a screw jack rotational motion of the effort is converted into linear motion of the load.**

● **Wheel and axle –**It is used to lift heavy loads such as anchor ships and to lift water from deep wells. The wheel and axle are connected such that they both have a common axis of rotation and rotated through one complete revolution at the same time.

- The wheel and axle have the same period and frequency i.e they make the same number of revolutions per second.

- The radius of the wheel R is the effort arm while the radius of the axle r is the load arm.

- $\boldsymbol{V.R = \frac{R}{r}}$. $\boldsymbol{M.A = \frac{R}{r}}$, i.e in the absence of friction.

- The bicycle makes use of the wheel and axle principle.

● **Gear wheels –**They are two tooted wheels which applies the wheel and axle principle between the wheels of different radii connected by a belt amd rotating on separate shafts

-They are used to change speed, direction of motion(rotation), alter $\boldsymbol{V.R}$. They also increase/magnify force and transmit motion.

- $\boldsymbol{V.R = \frac{angular\ speed\ of\ smaller\ gear}{angular\ speed\ of\ larger\ gear} = \frac{\omega_S}{\omega_L}}$,

$\boldsymbol{V.R = \frac{number\ of\ teeth\ on\ larger\ gear}{number\ of\ teeth\ on\ smaller\ gear} = \frac{N_L}{N_S}}$.

- The smaller gear(less toothed gear) is the driving gear and rotates faster than the larger gear(more toothed gear) which is the driven gear.

● **Hydraulic press/jack –** It operates on Pascal's principle of pressure in fluids.

- The pressure applied by the effort in the smaller(pump) piston of radius r and area A_1 is equal to the pressure received by the load in the larger(ram) piston of radius R and Area A_2.

- $\boldsymbol{M.A = \frac{A_2}{A_1}}$. $\boldsymbol{V.R = \frac{R^2}{r^2}}$. r=radius of the pump cylinder, R=radius of the arm cylinder. A_2=area of large piston(ram), A_1=area of small piston(pump).

Examples :

1. Which of the following definitions about simple machines is not correct? A.velocity ratio is $\frac{dist.\ moved\ by\ effort\ in\ a\ given\ time}{dist.\ moved\ by\ load\ in\ the\ same\ time}$ B.mechanical advantage is $\frac{load}{effort}$ C.efficiency is $\frac{work\ got\ off}{work\ put\ in}$

Demystified Series by Dr Timothy

D.work put in is the product of load and distance moved by the load E.efficiency is $\frac{mechanical\ advantage}{velocity\ ratio}$.

Work put in=Work input=Work done by effort =Work done on load=Effort×effort distance.

Work got off=Work output=useful work done =Work done by load=Load×load distance.

Useless work done=Work done against friction.

2. Which of the following statements about a machine is not correct? A.It is a device for doing work more conveniently **B.All machines are made of gear,wheels,levers,pulleys and screws** C.A machine often does work faster than a man D.A machine is any device by means of which a force applied at one point is made to do work at another point E.A fishing rod is a machine

All machine are made of inclined planes,lever and hydraulic press,which are the three basic machines.

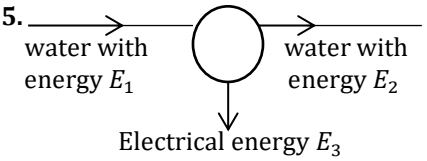
3. In practice, a machine can never be 100% efficient. This implies that; A.the mechanical advantage of any machijne is always less than 1 B.the velocity ratio of a machine is always more than 1 C.the velocity ratio of a machine is never greater than the mechanical advantage **D.the useful work obtained from a machine is always less than the work done by it** E.the ratio of the work output of a machine to the work input is always greater than 1.

Useful work obtained or work output is always less than the work input due to workdone in overcoming frictional forces. $M.A$ is always less than the $V.R$, as $M.A$ is affected by frictionwhile $V.R$ is independent of friction.

4. Which of the following statements defines correctly the efficiency of a machine? A.the load carried by the machine divided by the effort 6required in carrying the load B.the distance moved by the load divided by the distance moved by the effort **C.useful work done bymachine divided by the total work put into it** D.the movement of the load divided by the movement of the effort about the same point E.the temperature at the end of the application of the machine divided by the initial temperature.

Efficiency= $\frac{work\ output}{work\ input}$, work output is useful workdone or workdone by load, work input or total work input is work done by effort or work done on load.

5.



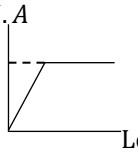
The diagram above represents the energy flow through a hydroelectric power. The efficiency of energy conversion is A.$\frac{E_2}{E_1}$ **B.**$\frac{E_3}{E_1}$ C.$\frac{E_3}{E_1+E_2}$ D.$\frac{E_2+E_3}{E_1}$.

The energy input into the machine=Initial energy of water=E_1, useful energy output from the machine=energy converted into electrical energy=E_3. E_3 is wasted energy.

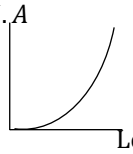
Efficency= $\frac{Energy\ output}{Energy\ input}=\frac{E_3}{E_1}$.

6. The variation of mechanical advantage with load for a simple machine with a velocity ratio of 6 is correctly illustrated as

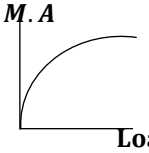
A. $M.A$



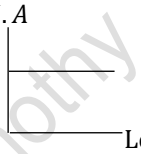
B. $M.A$



C. $\boldsymbol{M.A}$



D. $M.A$

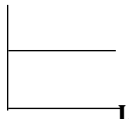


7. The variation of velocity ratio with load for a simple machine is correctly illustrated as

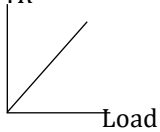
A. $V.R$



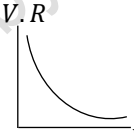
B. $\boldsymbol{V.R}$



C. $V.R$



D. $V.R$



8. A machine with a velocity ratio of 25 moves load of $3300N$ when an effort of $165N$ is applied to it. Calculate the efficiency of the machine **A.80%** B.60% C.50% D.25% E.20%

$V.R$=25, L=$3300N$, E=$165N$, ϵ=?, $M.A=\frac{L}{E}$,

Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,

$\epsilon=\frac{3300}{165}\times\frac{1}{25}\times100$=80%.

9. A machine of velocity ratio 6 requires an effort of $400N$ to raise a load of $800N$ through 1m. Find the efficiency of the machine A.55.6% B.50.0% C.22.2% C.33.3%.

$V.R$=6, E=$400N$, L=$800N$, $M.A=\frac{L}{E}$, ϵ=?,

Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,

$\epsilon=\frac{800}{400}\times\frac{1}{6}\times100$=33.3%.

10. A machine is used to lift a load of $20N$ through a height of 10m. If the efficiency of the machine is 40%, how much work is done? **A.500*J*** B.300J C.120J D.80J.

L=$20N$, load distance, y=10m, ϵ=40%, work done by the machine=work input=work done by the effort=W_i?, Work input W_o=Work done by load=load×load distance= $L\times y$,

$W_o=L\times y=20\times10=200J$, $\epsilon=\frac{W_o}{W_i}\times100$,

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$40 = \frac{200}{W_i} \times 100$, $W_i = \frac{200 \times 100}{40} = 500J$.

11. The efficiency of a simple machine is 75%. If the machine does 2400J of work, the energy spent to overcome friction is **A.600J** B.800J C.1800J D.2400J E.3200J.

ϵ=75%, work input, W_i=2400J, Work done against friction=energy spent to overcome friction, W_f=?, Work output, W_o=?,

$\epsilon = \frac{W_o}{W_i} \times 100$, $75 = \frac{W_o}{2400} \times 100$, $W_o = \frac{2400 \times 75}{100}$

$W_o = 1800J$. Work done against friction W_f,

$W_f = W_i - W_o = 2400 - 1800 = 600$J.

12. A machine has an efficiency of 80%. If the machine is required to overcome a load of 60N with a force of $40N$, calculate its velocity ratio A.0.53 B.0.83 C.1.20 D.0.50 **E.1.88.**

ϵ=80%, L=60N, E=$40N$, $V.R$=?,

Efficiency, $\epsilon = \frac{M.A}{V.R} \times 100 = \frac{L}{E} \times \frac{1}{V.R} \times 100$,

$80 = \frac{60}{40} \times \frac{1}{V.R} \times 100$, $V.R = \frac{60 \times 100}{40 \times 80} = 1.88$.

13. The efficiency of a machine is 80%. Calculate the workdone by a person using the machine to raise a load of 300kg through a height a of 4m (g=10ms^{-2}) A.7500J **B.15000*J*** C.30000J D.45000J.

ϵ=80%, L=W=300kg=3000N, load distance y=4m, Work output $W_o = L \times y$, Work done by the person using the machine=Work input W_i=?,

$\epsilon = \frac{W_o}{W_i} \times 100$, $80 = \frac{3000 \times 4}{W_i} \times 100$,

$W_i = \frac{3000 \times 4 \times 100}{80} = 15000J$.

14. A machine of velocity ratio 6 is used to lift a load of 1200N. If the efficiency of the machine is 80%, calculate the magnitude of the effort applied A.90N B.96N **C.250N** D.960N.

$V.R$=6, L=1200N, ϵ=80%, E=?,

Efficiency, $\epsilon = \frac{M.A}{V.R} \times 100 = \frac{L}{E} \times \frac{1}{V.R} \times 100$,

$80 = \frac{1200}{E} \times \frac{1}{6} \times 100$, $E = \frac{1200 \times 100}{6 \times 80} = 250$N.

15. A machine of velocity ratio 5 is used in lifting a load with an effort of 500N. If the machine is 80% efficient, determine the magnitude of the load A.2500N **B.2000N** C.900N D.625N.

$V.R$=5, E=500N, ϵ=80%, L=?,

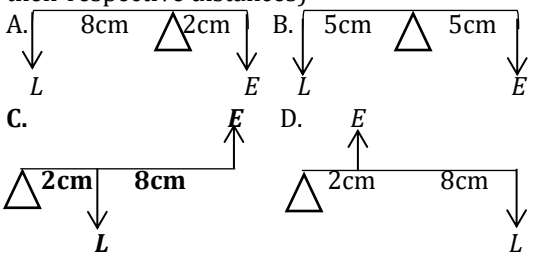
Efficiency, $\epsilon = \frac{M.A}{V.R} \times 100 = \frac{L}{E} \times \frac{1}{V.R} \times 100$,

$80 = \frac{L}{500} \times \frac{1}{5} \times 100$, $L = \frac{80 \times 500 \times 5}{100} = 2000$N.

16. It is more convenient to use a lever when the A.effort is closer to the fulcrum than the load B.load and the effort are at the same distance from the fulcrum C.effort is almost at the same distance from te fulcrum **D.load is closer to the fulcrum than the effort.**

Lever works on principle of moments. For a lever to raise a large load buy a small effort, the load must be closer to the fulcrum than the effort, $F \propto \frac{1}{d}$.

17. Which of the following levers has the greatest mechanical advantage? (The number indicate their respective distances)



In a lever system, $M.A = V.R$.Greatest$V.R$hence, greatest $M.A$ is achived in lever system in option C, where the load is closer to the fulcrum than the effort.

- Option A and B are first class lever. Option C is a second class lever. Option D is a third class lever.

18. Which of the following machines does not apply the lever principle? A.Claw hammer B.Wheel barrow **C.Single pulley** D.Sugar tong.

Claw hammer is a first class lever, wheel barrow is a second class lever while sugar tong is a third class lever. Single pulley is an example of a pulley system.

19. Which of the following instruments, in its normal use, serves as a first class lever? **A.pair of scissors** B.a pair of sugar tongs C.a bottle opener D.a knife E.a nut cracker.

Pair of sugar tongs and knife are third class levers. Bottle opener and nut craker are second class levers.

20. In wich of the following simple machines is the effort applied between the load and the fulcrum? A.claw hammer B.pliers C.wheel barrow **D.sugar tong** E.nutcracker

Effort is applied between the load and the fulcrum in a third class lever e.g sugar tong. Claw hammer and pliers are first class levers. Wheel barrow and nutcracker are second class lever.

21. Which of the following levers is not in the same class as the others in respect of the locations of the load, the effort and the fulcrum A.pincers B.see-saw C.pliers **D.sugar tongs** E.scissors.

All the other options apart from sugar tongs are first class lever.

22. At what distance from the fulcrum will a boy of mass 40kg balance another boy of mass 60kg who is sitting 5.0m from the fulcrum on the other side? A.1.5m B.3.3m C.4.0m **D.7.5m** E.20.0m.

m_1=40kg, d_1=?, m_2=60kg, d_1=5.0m,

Lever system works on principle of moment. Taking moment about the fulcrum,

$F_1d_1 = F_2d_2$ or $m_1d_1 = m_2d_2$, $40 \times d_2 = 60 \times 5$,

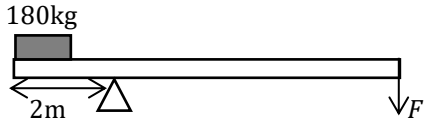
$d_2 = \frac{300}{40} = 7.5$m.

23. An effort P applied at one end of a crowbar just overcome the resistance W at the lid of a tin. The mechanical advantage of the crowbar is

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expressed as A.$W+P$ B.$P-W$ **C.**$\frac{W}{P}$ D.$\frac{P}{W}$ E.WP
Effort, E=P, Load, L=force overcomed= W,
$M.A=\frac{L}{E}=\frac{W}{P}$.

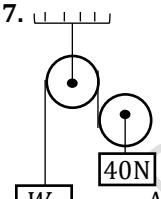
24. A lever of length 10m is used to lift a load of mass 180kg as shown in the diagram below The pivot is 2m from the load. If the efficiency of the lever is 90%, the force F required to lift the load is A.50N B.405N C.450N **D.500N.**



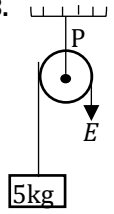
As efficnecy is not 100% hence, $M.A\neq V.R$ and principle of moment is not used to find the effort. ϵ=90%, L=180kg=1800N, load distance y=2m, effort distance x = 10−2=8m, E=F=?,
$V.R=\frac{x}{y}=\frac{8}{2}=4$. Efficiency, $\epsilon=\frac{M.A}{V.R}\times100$
$\epsilon=\frac{L}{E}\times\frac{1}{V.R}\times100$, $90=\frac{1800}{F}\times\frac{1}{4}\times100$,
$F=\frac{1800\times100}{90\times4}=500$N.

25. A block and tackle system has 6 pulleys. If the efficiency of the machine is 60%, determine its mechanical advantage A.12.0 B.10.0 **C.3.6** D.1.8
$V.R$=number of pulleys=6, ϵ=60%, $M.A$=?,
Efficiency, $\epsilon=\frac{M.A}{V.R}\times100$, $60=\frac{M.A}{6}\times100$,
$M.A=\frac{60\times6}{100}=3.6$.

26. A block and tackle system has six pulleys. A force of 50N applied to its lifts a load of weight W. If the efficiency of the system is 40%, calculate W A.300N B.200N C.140N **D.120N.**
$V.R$=number of pulleys=6, ϵ=40%, applied force E=50N, L=W=?,
Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,
$40=\frac{W}{50}\times\frac{1}{6}\times100$, $W=\frac{40\times50\times6}{100}=120$N.

27. 

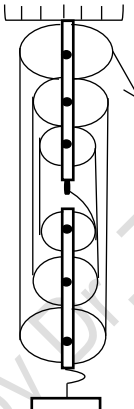
The figure shows a frictionless pulley system in which a weight in equilibrium with a weight of of 40N. Find the value of W
A.13.3N **B.20.0N** C.40.0N D.80.0N.
$V.R$=number of pulleys=2, for a frictionless pulley, $M.A$=$V.R$=2, L=40N, E=W=?,
$M.A=\frac{L}{E}$, $E=\frac{L}{M.A}=\frac{40}{2}=20$N.

28. 

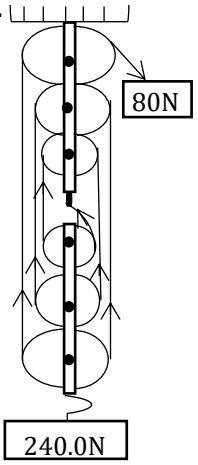
A load of mass 5kg is being lifted vertically upwards using a pulley system as shown. If the system is 80% efficient, calculate the value of the applied effort E(g=10ms^{-2})
A.80.0N **B.62.5N** C.40.0N D.6.3N E.4.0N .
L=5kg=50N, $V.R$=number of pulleys=1, ϵ=80%, E=?, Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,
$80=\frac{50}{E}\times\frac{1}{1}\times100$, $W=\frac{100\times50}{80}=62.5$N.

29. Lubricants used in machines to reduce friction A.decrease the $M.A$ **B.Increase the $M.A$** C.increase the $V.R$ D.decrease the $V.R$.
Lubricants reduce friction inorder to increase $M.A$ and efficiency of a machine. $V.R$ is not affected by friction.

30. What is the velocity ratio of a pulley system having three pulleys in the fixed block and two in the movable block? A.2 B.3 C.4 **D.5** E.6.
$V.R$ in a block and tackle system is the total number of pulleys=number of movable pulleys + number of fixed pulleys.
Number of moveable pulleys=3, number of fixed pulleys=2, $V.R$ =3+2=5.

31. 

The diagram above illustrates a block and tackle pulley system in which an effort E supports a load of 100.0N. If the efficiency of the machine is 75%, calculate the value of E
A.22.2N B.38.0N C.67.0N D.75.0N.
$V.R$=number of pulleys=6, L=100.0N, ϵ=75%, E=?, Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,
$75=\frac{100}{E}\times\frac{1}{6}\times100$, $E=\frac{100\times100}{6\times75}=22.2$N.

32. 

The diagram is a block and tackle system in which an effort of of 80N is used to lift a load of 240N. The efficiency of the machine is
A.60% **B.50%** C.40% D.33%.
L=240.0N, E=80N, $V.R$=number of pulleys=6,
Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,
$\epsilon=\frac{240}{80}\times\frac{1}{6}\times100=50\%$.

33. A machine of efficiency 80% is used to lift a box. If the effort applied by the machine is twice the weight of the box, calculate the velocity ratio of the machine A.0.21 B.0.32 **C.0.63** D.0.64
ϵ=80%, L=weight of the box=W, E =2W=2L,
Efficiency, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$

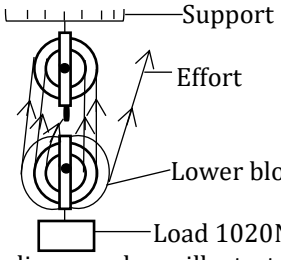
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$80 = \frac{W}{2W} \times \frac{1}{V.R} \times 100$, $V.R = \frac{W \times 100}{80 \times 2W}$,
$V.R$ =0.625=0.63.

34. Which of the following actions will improve the efficiency of a pulley system? **A.reducing the mass of the pulley** B.increasing the frictional force between the string and the pulley C.increasing the mass per unit length of the string of the pulley D.increasing the mass of the pulley
Friction increases as the mass or weight of the pulleys increases i.e $F_R \propto m \text{ or } W$. The mass of the pulleys increases as the number of the pulleys increases.Hence, reducing the mass of the pulley also reduces the frictional forces, thereby increasing the efficiency of the pulley.

35. A block and tackle system is used to raise a load of 250N through a vertical height of 30m. What is the efficiency of the system if the workdone against friction is 1500J? A.62.5% B.73.3% **C.83.3**% D.94.3% .
L=250N, load distance=30m, Work output, W_o or Work done by load=250×30=7500J, Work done against friction, W_f=1500J, Work input W_i or workdone by effort $= W_o + W_f$,
$W_i = W_o + W_f = 7500 + 1500 = 9000$J,
$\epsilon = \frac{W_o}{W_i} \times 100 = \frac{7500}{9000} \times 100 = 83.3\%$.

Use the diagram below to answer 36 – 38.



The diagram above illustrates a block and tackle system of pulleys which has a mechanical advantage of 5.

36. Determine the $V.R$ A.2 B.3 C.4 **D.6.**
$V.R$=number of strings in the pulley= 6 strings (as the number of strings (6) is greater than greater then the number of pulleys (5).

37. Calculate the magnitude of the effort A.1020N B.340N **C.204N** D.170N .
$M.A$=5, L=1020N, E=?, $M.A = \frac{L}{E}$,
$E = \frac{L}{M.A} = \frac{1020}{5} = 204$N.

38. Calculate the efficiency of machine A.100.0% **B.83.3%** C.50.0% D.33.3%.
$M.A$=5, $V.R$=6, ϵ=?,
Efficiency, $\epsilon = \frac{M.A}{V.R} \times 100 = \frac{5}{6} \times 100 = 83.3\%$.

39. A body is moved a distance Lm up an inclined plane through vertical height of hm. Which of the following equtions is correct?
A.efficiency= $\frac{h}{L} \times 100$ **B.velocity ratio** $= \frac{L}{h}$
C.mechanical advantage $= \frac{L}{h}$ D.mechanical advantage= $\frac{h}{L}$ E.$\frac{mechanical\ advantage}{velocity\ ratio} = 1$.

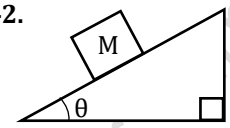
Velocity ratio of an inclined plane depends on the ratio of the length of the plane and vertical height of the plane , $V.R = \frac{L}{h}$.

- Option C and E are only true for a frictionless inclined plane ($M.A = V.R$).

40. An inclined plane used as a machine makes an angle of 30° with the horizontal. What is the velocity ratio of the machine? **A.**$\frac{1}{\sin 30°}$ B.sin 30° C.$\frac{1}{\tan 30°}$ D.$\frac{1}{\cos 30°}$ E.cos 30° .
$V.R$ of an inclined plane= $\frac{1}{\sin\theta} = \frac{1}{\sin 30°}$. θ=30°.

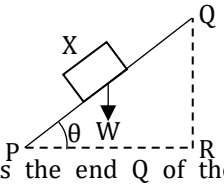
41. The mechanical advantage ($M.A$) of an inclined plane depends on A.its length B.its height C.the product of its length and height **D.the ratio of its length to its height.**
In a frictionless pulley, $M.A = V.R = \frac{L}{h}$. Hence mechanical advantage, $M.A$ of an inclined plane depends on the ratio of its length to its height.

42.



In the diagram illustrates above, a block of mass M is at rest on a plane inclined at an angle θ to the horizontal. As θ increases, the velocity ratio of the plane A.increases **B.decreases** C.remains the same D.decreases and later increases.
$V.R = \frac{1}{\sin\theta}$, $V.R \propto \frac{1}{\theta}$, as the angle of inclination θ or sin θ increases, $V.R$ decreases.

- The diagram below shows a box X of weight W resting on a plank PQ that has its lower end P hinged to a horizontal floor PR. The plank makes an angle θ with the floor. Use this information to answer questions **43** to **45**.



42. As the end Q of the plank is raised, the component of W normal to the plank will A.increase **B.decrease** C.remain the same D.be slightly greater than W E.be independent of W.
Component of the weight W normal to the plane is Wcos θ. When the end Q of the plank is raised, θ increases and cos θ deccreases thus, Wcos θ decreases.

43. If θ is equal to α at the time the box is just to slide down the plane, the coefficient of friction of static friction between the plank and the box is A.sin α **B.tan α** C.cos α D.sec α E.cot α.
When the box is about to slide down the plane, angle of fricton α equals θ and the coefficient of friction $\mu = \tan\theta = \tan\alpha$.

44. If the box is pulled from P to Q, which of the following expresses the velocity ratio of the inclined plane? A.PR/PQ B.QR/PQ C.QR/PR D.PQ/PR **E.PQ/QR.**
Velocity ratio is the ratio of the length of the

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inclined plane PQ to the height of the plane QR.

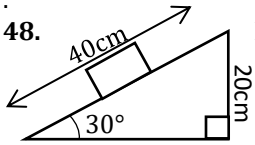
45. What is the velocity ratio of an inclined plane of length 2.6m, vertical height above the ground 1.3m, and angle of inclination to the horizontal 30°? ($\sin 30° = 0.500$, $\cos 30° = 0.866$, $\tan 30° = 0.577$) **A.2.000** B.1.300 C.1.150 D.0.577.

Length of the plane, L=2.6m, vertical height of the plane, h=1.3m, θ=30°, $V.R = \frac{L}{h}$ or $\frac{1}{\sin\theta}$,

$V.R = \frac{L}{h} = \frac{2.6}{1.3} = 2$. $V.R = \frac{1}{\sin\theta} = \frac{1}{\sin 30°} = \frac{1}{0.5} = 2$.

46. A plane inclined at angle θ has a velocity ratio of 10:1. The inclination of the plane to the horizontal is given by A.$\tan\theta = \frac{1}{10}$ B.$\cot\theta = \frac{1}{10}$ C.$\cos\theta = \frac{1}{10}$ **D.$\sin\theta = \frac{1}{10}$.**

$V.R$ =10:1= $\frac{10}{1}$, $V.R = \frac{1}{\sin\theta}$, $\frac{10}{1} = \frac{1}{\sin\theta}$, $\sin\theta = \frac{1}{10}$.

48. 

From the diagram, find the the mechanical advantage of the system, if the plane is smooth A.0.50 B.0.86 C.1.15 **D.2.00.**

A smooth plane is a frictionless plane i.e a plane with no friction. Hence, $M.A = V.R$,

$M.A = \frac{L}{h}$ or $\frac{1}{\sin\theta}$, L=40cm, h=20cm, θ=30°,

$M.A = \frac{L}{h} = \frac{40}{20} = 2$. $M.A = \frac{1}{\sin\theta} = \frac{1}{\sin 30°} = \frac{1}{0.5} = 2$.

49. A plane inclined at angke of 30° to the horizontal has an effieicny of 60%. A force of magnitude F is used to push a load of 15kg up the plane. Calculate F. A.45N B.75N **C.125N** D.360N (g=10ms^{-2}).

L=15kg=150N, E=F, ϵ=60%, θ=30°,

$V.R = \frac{1}{\sin\theta} = \frac{1}{\sin 30°} = \frac{1}{0.5} = 2$,

Efficiency, $\epsilon = \frac{M.A}{V.R} \times 100 = \frac{L}{E} \times \frac{1}{V.R} \times 100$

$60 = \frac{150}{F} \times \frac{1}{2} \times 100$, $F = \frac{150\times100}{2\times60} = 125$N.

50. The distance through which a screw moves along its head when the screw head is turned through 180° is A.the pitch **B.half the pitch** C.one and half the pitch D.twice the pitch E.thrice the pitch.

The distance through which a screw moves along its head when the head is turned through one complete revolution i.e 360° is equal to the pitch P. Hence, the distance moved by the head of the screw when it rotates 180° i.e half of 360° is half the pitch.

51. Which of these statements is correct abput a screw jack used in raising a car? A.velocityratio= $\frac{pith\ of\ screw}{2\pi\times length\ of\ effort\ handle}$ B.mechanicaladvantage= $\frac{effort\ applied}{weight\ of\ car}$ **C.velocity ratio can be increased by using a jack with a smaller pitch** D.velocity ratio can be decreased by increasing the length of the effort arm E.mechanical advantage can be increased by increasing the effort.

Velocity ratio of a screw jack is given by – $V.R = \frac{2\pi r}{P}$, r=length or radius of effort handle or tommy bar, P=pitch. $V.R \propto \frac{r}{P}$ i.e velocity ratio increases as the radius or length of effort arm or tommy bar increases and decreases as the pitch increases. Hence, using a jack with a smaller pitch and increased length of the effort arm will increase the $V.R$. $M.A = \frac{Weight\ of\ the\ car}{Effort\ applied}$.

$M.A$ increases as the weight of the car or load increases and the effort decreases.

52. A spanner of length 20cm is used to turn a nut. If the spanner were replaced by another of length 15cm and held at its end with the same effort as in the first case, the A.velocity ratio would increase **B.the velocity ratio would decrease** C.mechanical advantage would increase but the velocity ratio would remain constant D.mechanical advantage would decrease but the velocity ratio would remain constant E.mechanical advantage and velocity ratio would remain constant.

A spanner works with the same principle as a screw jack i.e screw system. $V.R = \frac{2\pi r}{P}$,

Using a spanner of length 20cm, r=20cm,

$V.R = \frac{2\pi\times20}{P} = \frac{40\pi}{P}$.

Using a spanner of length 15cm, r=15cm,

$V.R = \frac{2\pi\times15}{P} = \frac{30\pi}{P}$.

Hence, $V.R$ decreases as $\frac{40\pi}{P} > \frac{30\pi}{P}$ OR $V.R \propto r$, velocity ratio increases when the radius or length of the screw increases and decreases when radius or length of the screw decreases. Hence $V.R$(15cm)< $V.R$(20cm) i.e $V.R$ decreases. The mechanical advantage $M.A$ is constant, as the effort is the same in both case and they act on the same nut i.e the same load.

53. A screw jack with a Tommy bar of length 12cm is used to raise a car through a vertical height of 25cm turning the Tommy bar through 50 revolutions. Calculate the approximate velocity ratio of the jack [π=3.14] A.453 B.302 **C.151** D.75.

Length of tommy bar, r=12cm, vertical height of load, h=25cm, number of revolution, n=50, pitch, $P = \frac{h}{n} = \frac{25}{50} = 0.5$cm, π=3.14, $V.R$=?,

$V.R = \frac{2\pi r}{P} = \frac{2\times3.14\times12}{0.5} = 150.72 = 151$.

54. Two spanners X and Y of lengths 15cm and 20cm respectively are used in turn to give a screw of pitch 2mm one complete rotation. If R_X and R_Y are the respectively velocity ratios of the spanners, what is the ratio R_X:R_Y? A.2:3 B.3:2 **C.3:4** D.4:3

r_X=15cm, r_Y=20cm, P_X=P_Y=2mm i.e they have the same pitch, $V.R_X$=R_X, $V.R_Y$=R_Y,

$V.R = \frac{2\pi r}{P}$, $V.R \propto r$ i.e $\frac{2\pi}{P}$ = constant,

$\frac{V.R_X}{r_X} = \frac{V.R_Y}{r_Y}$, $\frac{R_X}{15} = \frac{R_Y}{20}$, $\frac{R_X}{R_Y} = \frac{15}{20} = \frac{3}{4}$, R_X:R_Y =3:4.

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55. The handle of a screw jack of pitch 4mm turns through a circle of radius 21cm when the jack is used to raise a load. If $\pi=\frac{22}{7}$, the velocity ratio of the jack is A.66 B.165 C.264 **D.330**

P=4mm=0.4cm, r=21cm, $\pi=\frac{22}{7}$, $V.R$=?,

$V.R=\frac{2\pi r}{P}=\frac{2\times\frac{22}{7}\times21}{0.4}=\frac{132}{0.4}$ =330.

56. The pitch of a screw jack is 0.45cm and the arm is 60cm long. If the efficiency of the jack is $\frac{75}{\pi}$%. Calculate the mechanical advantage A.400 B.300 **C.200** D.150 .

P=0.45cm, r=60cm, $\epsilon=\frac{75}{\pi}$% , $M.A$=?,

$V.R=\frac{2\pi r}{P}=\frac{2\pi\times60}{0.45}=\frac{120\pi}{0.45}=\frac{800\pi}{3}$,

Efficiency, $\epsilon=\frac{M.A}{V.R}\times100$, $\frac{75}{\pi}=\frac{M.A}{\frac{800\pi}{3}}\times100$,

$M.A=\frac{75}{\pi}\times\frac{800\pi}{3}\times\frac{1}{100}$ =200.

57. In an ideal wheel and axle system, R stands for the radius of the wheel and r is the radius of the axle. The mechanical advantage is **A.**$\frac{R}{r}$ B.$\left(\frac{r}{R}\right)^2$ C.$\frac{r}{R}$ D.$\left(\frac{R}{r}\right)^2$.

An ideal machine is a mchine with 100% efficiency. Hence, in an ideal wheel and axle system, $M.A=V.R=\frac{R}{r}$.

58. The radius of a wheel is 30.0cm and that of its axle is 6.0cm. Calculate the effort required to lift a load of 120.0N using this machine, assuming 100% efficiency A.600.0N B.40.0N **C.24.0N** D.20.0N.

Radius of wheel, R=30.0cm, radius of axle, r=6.0cm, L=120.0N, E=?,

$M.A=V.R$,as the effieicny of the machine is 100%. $M.A=\frac{L}{E}, V.R=\frac{R}{r}\therefore\ \frac{L}{E}=\frac{R}{r}$,

$\frac{120}{E}=\frac{30}{6}$, $E=\frac{120\times6}{30}=24.0$N.

59. In a wheel and axle mechanism, the diameter of the wheel and axle are 40cm and 8cm respectively. Given that the machine is 80% efficient, what effort is required to lift a load of 100N? A.20N **B.25N** C.50N D.80N.

R=diameter of the wheel=40cm, r=diameter of the axle=8cm, ϵ=80%, L=100N, E=?,

$V.R=\frac{R}{r}=\frac{40}{8}=5$, Efficiency, $\epsilon=\frac{M.A}{V.R}\times100$,

$\epsilon=\frac{L}{E}\times\frac{1}{V.R}\times100$, $80=\frac{100}{E}\times\frac{1}{5}\times100$,

$E=\frac{100\times100}{80\times5}=25$N.

60. The wheel and the axle of of a wheel-and- axle system have radii 50cm and 10cm respectively. If one-fifth of the work done by the effort is used in overcoming friction, an effort of 500N applied to the wheel will raise a load of A.2500N **B.2000N** C.1000N D.500N.

R=50cm, r=10cm, E=500N, L=?,

Workdone by effort=Work input W_i, Work done against friction $W_f=\frac{1}{5}W_i$, Work output or work done by load , $W_o=W_i-W_f=W_i-\frac{1}{5}W_i=\frac{4}{5}W_i$,

Efficiency, $\epsilon=\frac{W_o}{W_i}\times100=\frac{\frac{4}{5}W_i}{W_i}\times100=80\%$ OR If one-fifth(or $\frac{1}{5}$ of 100% i.e 20%) of the workdone by the effort is expended in overcoming friction, efficiency=80%

$V.R=\frac{R}{r}=\frac{50}{10}=5$, $\epsilon=\frac{M.A}{V.R}\times100=\frac{L}{E}\times\frac{1}{V.R}\times100$,

$80=\frac{L}{500}\times\frac{1}{5}\times100$, $L=\frac{80\times500\times5}{100}=2000$N.

61. Gear wheels are used in machines for varying I.mechanical advantage II.speed III.velocity ratio A.I and II **B.II and III** C.I and III D.I,II and III E.none of the above.

Gear wheels are used to change or vary speed, direction of motion(rotation) and velocity ratio. They also increase or magnify force and transmit motion.

62. A 20 toothed gear wheel drives a 50 toothed one. If the angular speed of the smaller wheel is 120revs^{-1}, the angular speed of the larger wheel is A.3revs^{-1} **B.40revs^{-1}** C.360revs^{-1} D.2400revs^{-1}.

Number of teeths in smaller(driving) gear N_S=20, number of teeths in larger(driven) gear N_L=50, angular speed of smaller gear ω_S=120revs^{-1}, angular speed of larger gear ω_L=?, $V.R=\frac{\omega_S}{\omega_L}=\frac{N_L}{N_S}$

$\frac{120}{\omega_L}=\frac{50}{20}$, $\omega_L=\frac{120\times20}{50}=40$revs^{-1}.

63. In a hydraulic press A.a small downward force on the smaller piston produces a large downward force on the larger piston B.a large upward force on the smaller piston produces a large upward force on the larger piston C.a small upward force on the smaller piston produces a small upward force on the larger piston**D.a small downward force on the smaller piston produces a large upward force on the larger piston** E.a large downward force on the larger piston produces a small upward force on the larger piston.

In a hydraulic press, a small downward force called the effort, on the smaller piston produces a large upward force called the load, on the larger piston.

64. In a hydraulic press, a force of 40N is applied to the smaller piston of area 10cm^2. If the area of the larger piston is 200cm^2, calculate the force obtained **A.800N** B.500N C.80N D.50N.

E=force applied in the smaller piston=40N, A_1=area of the smaller piston=10cm^2, L=force obtained in the larger piston=?, A_2=area of the larger piston=200cm^2. $M.A=\frac{L}{E}=\frac{A_2}{A_1}$,

$\frac{L}{E}=\frac{A_2}{A_1}$, $\frac{L}{40}=\frac{200}{10}$, $L=\frac{40\times200}{10}=800$N.

65. A hydraulic press has a large circular piston of radius 0.8m and a circular plunger of 0.2m. A force of 500N is exterted by the plunger. Find the force exerted by the piston **A.8000N** B.4000N C.2000N D.31N.

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Radius of piston=radius of larger section R=0.8m, radius of plunger=radius of smaller section r=0.2m, force exerted by plunger= effort E=500N, force exerted by the piston =load L =?, $M.A = \frac{L}{E} = \frac{A_2}{A_1}$, $A_2 = \pi R^2$, $A_2 = \pi r^2$, $\frac{L}{E} = \frac{\pi R^2}{\pi r^2}$, $\frac{L}{500} = \frac{\pi \times 0.8^2}{\pi \times 0.2^2}$, $L = \frac{500 \times 0.64}{0.04} = 8000$N.

Jamb past questions on simple machines :
[1979/2,1981/10,1982/14,1987/13,1988/8,13, 1994/13,1995/15,2001/8,2005/27,2006/15, 2015/36,2017/24,1983/36,1986/12,1989/9, 1992/9,1999/2,2000/10,2004/11,2007/46,200 8/14,2011/12,2015/8,1984/20,1991/8,1994/ 12,1998/12,1999/4,2002/6,2014/15,2015/36, 1990/12,1993/9,2001/4,2009/12,2010/10, 2013/12,2014/14]

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Chapter 9 – Equilibrium in liquids [Density, Relative density and Upthrust]

● **Density** $\boldsymbol{\rho}$ is the mass per unit volume of a substance. $\boldsymbol{\rho = \frac{m}{V}}$**.** Its unit is Kgm^{-3} or gcm^{-3}.

- Relative density $\boldsymbol{R.D}$ or **specific gravity** is the ratio of the mass or weight of a given volume of a substance to the mass or weight of an equal volume of water. It is also the ratio of the density of a substance to the density of water.

$\boldsymbol{R.D = \frac{mass\ of\ a\ given\ volume\ of\ a\ substance}{mass\ of\ an\ equal\ volume\ of\ water} = \frac{m_s}{m_w}}$**,**

$\boldsymbol{R.D = \frac{weight\ of\ a\ given\ volume\ of\ a\ substance}{weight\ of\ an\ equal\ volume\ of\ water} = \frac{W_s}{W_w}}$**,**

$\boldsymbol{R.D = \frac{density\ of\ a\ substance}{density\ of\ water} = \frac{\rho_s}{\rho_w}}$**.**

- $R.D$is dimensionless or has no unit.

- The density of water is 1gcm^{-3} or 1000kgm^{-3}.

- The density of a substance in gcm^{-3} is numerically equal to its relative density e.g if the $R.D$ of mercury is 13.6, its density is equal to 13.6gcm^{-3}. $\boldsymbol{\rho(gcm^{-3}) = R.D}$.

- The density of a substance in kgm^{-3} is equal to its relative density times 1000 e.g $R.D$ of mercury is 13.6, its density is 13600kgm^{-3} or 13.6×10^3kgm^{-3}. $\boldsymbol{\rho(Kgm^{-3}) = R.D \times 1000.}$

- Density of a regular solid is determined by a beam balance(mass) and measurement (volume).

- Density of an irregular solid is determined by a beam balance(mass) and volume of liquid displaced (volume of the body).

- Density bottle is used for accurate measurement of the density/relative density of a substance in the powder or granular form.

- If a liquid is heated i.e increase in temperature its volume increase while the density increases, while its mass remaining constant, then its density and volume before and after heating are related by – $\boldsymbol{\rho_1V_1 = \rho_2V_2}$ **-** If two substance have the same mass, their densities and volumes are related by – $\boldsymbol{\rho_1V_1 = \rho_2V_2}$ **.**

- The volume of a solid substance is determined from its thickness or length and its area. **Volume=thickness or length×Area.**

- If two substances of different densities are mixed together, the density of the resulting mixture is calculated as: $\boldsymbol{\rho_{mixture} = \frac{m_{mixture}}{V_{mixture}}}$**,**

$\boldsymbol{\rho_{mixture} = \frac{m_1+m_2}{V_1+V_2}}$**,** $\boldsymbol{m_1 = \rho_1V_1}$ **,** $\boldsymbol{m_2 = \rho_2V_2}$

OR $\boldsymbol{\rho_1V_1 + \rho_2V_2 = \rho_{mixture}V_{mixture}}$ **,** $\boldsymbol{V_{mixture} = V_1 + V_2}$OR $\boldsymbol{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2} = \frac{m_1+m_2}{\rho_{mixture}}}$**.**

● Upthrust and Archimede's principle :

- Upthrust or buoyancy force or apparent loss in weight $\boldsymbol{U}$ is the upward force exerted by a fluid on a body immersed completely or partially in the fluid. **Upthrust** $\boldsymbol{U = \rho_LV_og}$**.**

- Upthrust on a body depends on the density of the fluid(liquid) $\boldsymbol{\rho_L}$ **and the volume of the object or body immersed in the fluid or volume of the liquid displaced** $\boldsymbol{V_o}$**.**

- Upthrust is due to the difference in pressure between the upper and lower faces of the object, with the pressure being greater at the lower face. Hence, upthrust increases with pressure, $U \propto P$.

- Upthrust is independent of temperature.

- Archimede's principle – It states that a body immersed(completely or partialy) in a fluid experiences an upthrust(upward force) which is equal to the weight of the fluid displaced.

- Real and Apparent weight – Real weight of a body is its weight in air. Apparent weight of a body is its weight when submerged or immersed in a fluid(liquid).

- An object tends to weighs less when in a fluid(liquid) than in air, hence on immersion of a body attached to a string on water and attempting to lift the body might break the string as the weight in air (real weight) is greater than its weight in water (apparent weight).

- A body appears to weigh lighter when completely or partially immersed in water due to upthrust.

- Upthrust in a liquid equal to the apparent loss in weight of a body immersed in a fluid or the weight of the fluid displaced.

Upthrust=Real weight – Apparent weight,

real weight or weight in air = $\boldsymbol{W_A = m_Ag}$,

apparent weight or weight of a body in fluid= $\boldsymbol{W_{Ap} = m_{Ap}g}$ **,** $\boldsymbol{U = W_A - W_{Ap}}$**.**

- Real weight or weight of the body in air, $\boldsymbol{W_A = m_Ag = \rho_oV_og}$.

- Apparent weight or weight of the body in a fluid, $\boldsymbol{W_{Ap} = m_{Ap}g = (\rho_o - \rho_L)V_og}$**.**

- Apparent mass , $\boldsymbol{m_{Ap} = (\rho_o - \rho_L)V_o}$**.**

- Bodies immersed or submerged in a fluid – The forces acting on a body immersed in a fluid (liquid) are : **weight** $\boldsymbol{W}$ of the body acting downwards, **upthrust or buoyancy force** $\boldsymbol{U}$ acting upwards. The magnitude of the two forces determines how the body will behave in the fluid (liquid) :

1. If the weight is greater than the upthrust $\boldsymbol{(W > U)}$ or if density of object is greater than density of liquid $\boldsymbol{(\rho_o > \rho_L)}$, the body has a positive apparent weight and sinks in the liquid.

2. If a body is **completely immersed** and the weight is less than the upthrust $\boldsymbol{(W < U)}$ or $\boldsymbol{(\rho_o < \rho_L)}$**,** the body has a negative apparent weight, hence rises and begins to float.

3. If a body is **partially immersed** and the

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weight of the body is equal to the upthrust **($W = U$)**, the apparent weight of the body is zero and the body floats i.e principle of floatation of floatation of a body.

- An addition force called **tension T** also acts upward on a body immersed in a fluid if the body is suspended by a string or spring balance.

$T + U = W$, $T = W - U$.

- For objects immersed completely in a liquidthe weight of object equals weight of fluid displaced and the volume of object equals the volume of fluid displaced.

- The volume of the body immersed or volume of liquid displaced by the body decreases as the density of the liquid increases **[$V \propto \frac{1}{\rho}$].**

- Volume of the body above liquid surface increases as the volume of body immersed or volume of liquid displaced decreases.

- The weight of liquids displaced by a body decreases as the relative density(specific gravity) or density of the liquid increases.

- Rise in volume of a liquid on complete or total immersion of an object in a fluid equals the volume of the object immersed.

- The volume of fluid displaced by a body on complete immersion is also equal to the volume of the vessel.

- In complete immersion of bodies, upthrust is the same at all positions and depths within the liquid as the volume of the liquid displaced at all depths are the same.

- In partial immersion of bodies, upthrust varieswith the fraction of the body submerged i.e upthrust increasing with increase in the fraction submerged.

- Principle of floatation – It states that a floating body **partially submerged** will displace a weight of the fluid equal to its own weight i.e a body floats in a liquid when its weight is equal to the upthrust acting on it by the liquid, **$W = U$.**

- Factors affecting floatation of a body :

1.Density of the body – A body will float in a fluid(liquid or gas) if the density of the body is less than that of the liquid.

2.Shape of the body – A body denser than a liquid can still float if the body is shaped in a way that its volume can displace its own weight of the liquid.The volume of the body will be so large compared to its weight.

- A ship floats in water because its larger volume displaces a large volume of water whose weight balances the weight of the ship. Also a balloon filled with a gas lighter than air will float if the weight of the balloon and the and the gas contentequals the upthrust of air on the ballon.

- When a body floats in a liquid, an increase in temperature (by heating), will decrease the density of the liquid, thereby increasing the fraction of the volume of the body immersed and decreasing the fraction of the volume above water.

- Fraction of volume of an object submerged in a liquid is the ratio of the volume of the liquid displaced to the volume of the object or the ratio of the density of the object to the density of the liquid.

Fraction of volume submerged $= \frac{volume\ of\ liquid\ displaced}{volume\ of\ object} = \frac{density\ of\ object}{density\ of\ liquid}$,

Fraction submerged$= \frac{V_{Ld}}{V_o} = \frac{\rho_o}{\rho_L}$.

- Relative density of a substance based on Archimede's principle –

1.Relative density of solids: A spring balance is used to measure the weight of the solid in air or true weight W_A and its weight in water W_w. The relative density of the solid is given by –

$R.D_s = \frac{weight\ in\ air}{upthrust\ in\ water} = \frac{W_A}{U_w} = \frac{W_A}{W_A - W_w}$.

- The relative density of the solid can also be calculated from the extension produced by the weight in the spring balance, when in air e_A and water e_w. $R.D_s = \frac{e_A}{e_A - e_w}$.

2.Relative density of liquids – The weight of a solid as recorded by a spring balance in air W_A, in water W_w and in the liquid W_L, whose reltive density is to be determined. The relative density of the liquid is given by –

$R.D_L = \frac{upthrust\ in\ liquid}{upthrust\ in\ water} = \frac{U_L}{U_w} = \frac{W_A - W_L}{W_A - W_w}$.

- The relative density of the liquid can also be determined by the extension produced in air e_A, water $e_w$and in the liquid e_L. $R.D_s = \frac{e_A - e_L}{e_A - e_w}$.

● **Hydrometers –** A hydrometer is used to measure the relative density of liquids e.g acids in battery.

- It works on floatation principle.

- The weight of the hydrometer is equal to the weight of the fluid displace by it i.e upthrust on it when immersed in liquid or in water.

- The length of the test-tube submerged in a liquid in a hydrometer is inversely proportional to the density if the liquid i.e $l \propto \frac{1}{\rho}$. Hence, the higher the density of the liquid, the shorter the length of the tube submerged or denser the liquid, lesser the hydrometer sinks and the lesser the volume displaced.

- The wide bulb of a hydrometer ensures that the upthrust of any liquid is large enough to supports its weight.

- The narrow stem of a hydrometer ensures that the instrument is very sensitive to small changes in density.

- The weighted end (lead shot) ensures that the hydrometer floats upright and stable in the liquids.

- The density scale has unequal divisions and

Demystified Series by Dr Timothy

decreases upwards.
- Hydrometers are used to test the purity of liquid whose density are known and the quality of milk.
- When a hydrometer is placed upright in a liquid, it sinks freely to a certain depth h_L in which it floats and the depth it sinks freely into water h_w, the relative density of the liquid is given by –

$R.D_L = \frac{depth\ immersed\ in\ water}{depth\ immersed\ in\ liquid} = \frac{h_w}{h_L}$.

- The relative density of the liquid can also be calculated as the ratio of the fraction of the volume submerged in water to the fraction submerged in liquid.

$R.D_L = \frac{fraction\ of\ volume\ submerged\ in\ water}{fraction\ of\ volume\ submerged\ in\ liquid} = \frac{V_w}{V_L}$.

- Fraction of height or volume immersed in water+Fraction of height or volume above water=1.
- A hydrometer reading is independent of temperature.Hence, a hydrometer submerged in water and cooling the water will not result in any rise or fall below the initial level as the volume change of the liquid compensate the expansivity change of the tube.

Examples :

1. The density of a solid is defined as the A.mass of the solid compared to the mass of an equal volume of water B.amount of water displaced when a unit mass of solid is immersed in it C.weight per unit volume of the solid D.volume per unit mass of solid **E.mass per unit volume of the solid .**
Density of substance is the mass per unit volume of the substance. Option A defines relative density.

2. Density is a measure of the A.quantity of matter an object contains B.space an object occupies **C.compactness of matter with respect to its size** D.surface area of an object.

3. What is the mass of 1.025 litres of water? A.1025kg B.102.5kg C.10.25kg **D.1.025kg** E.0.1025kg.
Density of water ρ =1gcm^{-3}, volume of water V=1.025litre=1.025(1000)cm^3=1025cm^3,
$\rho = \frac{m}{V}$, $m = \rho V$ =1×1025=1025g=1.025kg.

4. A rectangle metal block of density 2.5×10^{3}kgm^{-3} measures 10cm by 8cm by 5cm. Calculate its mass A.4kg **B.1kg** C.0.4kg D.0.16kg E.0.004kg.
Density of the block ρ=2.5×10^{3}kgm^{-3}, dimension of 10cm by 8cm by 5cm means, length L=10cm=0.1m, breadth b=8cm=0.08m, height h=5cm=0.05m, volume of the metal block= Lbh , volume of the metal block= 0.1×0.08×0.05=0.0004m^3=4×10^{-4}m^3,
$m = \rho V$ =2.5×10^3×4×10^{-4}=1kg.

5. What volume of alcohol with a density of 8.4×10^{2}kgm^{-3} will have the same mass as 4.2m^3 of petrol whose density is 7.2×10^{2}kgm^{-3} A.1.4m^3 **B.3.6m^3** C.4.9m^3 D.5.0m^3 E.5.8m^3.
Volume of alcohol V_a=?, density of alcohol, ρ_a=8.4×10^{2}kgm^{-3}, volume of petrol V_p=4.2m^3,density of petrol ρ_p=7.2×10^{2}kgm^{-3}. As they both have the same mass, $\rho_1 V_1 = \rho_2 V_2$, mass is constant, $\rho_a V_a = \rho_a V_a$,
8.4×10^2× V_a = 7.2×10^2×4.2,
$V_a = \frac{7.2\times10^2\times4.2}{8.4\times10^2}$ = 3.6m^3.

4. Colad oil of volume 600cm^3 has a density of 0.9gcm^{-3}. When heated, the density of oil decreases to 0.6gcm^{-3}. The volume of the heated oil is A.324cm^3 B.400cm^3 C.720cm^3 **D.900cm^3**
Increase in temperature i.e by heating, decreases the density of a liquid and increases the volume with the mass being constant.
$\rho_1 V_1 = \rho_2 V_2$, ρ_1=0.9gcm^{-3}, V_1=600cm^3, ρ_2=0.6gcm^{-3}, V_2=?, 0.9×600 = 0.6× V_2 ,
$V_2 = \frac{0.9\times600}{0.6}$ = 900cm^3.

5. The relative density of gold,silver,copper and zinc are given as 19.3, 10.5, 8.9 and 7.1 respectively. A piece of ornamental metal weighs 0.445kg and displaces 5×10^{-5}m^3 when completely immersed in water. The metal is A.zinc **B.copper** C.silver D.gold .
Weight or mass of the ornament m=0.445kg , volume of water displaced when totally immersed equals the volume of the ornament, V=5×10^{-5}m^3, $\rho = \frac{m}{V} = \frac{0.445}{5\times10^{-5}}$ = 8900kgm^{-3},
$R.D$ of metal = $\frac{density\ of\ metal}{density\ of\ water} = \frac{\rho_m}{\rho_w}$,
Density of water=1000kgm^{-3} or 1gcm^{-3} ,
$R.D = \frac{8900}{1000}$ =8.9 i.e the metal is copper.

6. A tank of capacity 27m^3 initially filled with water is drained at the rate of 5kgs^{-1}. If the density of water is 10^{3}kgm^{-3}, the time it takes to empty the tank is A.13.5min B.54min **C.90min** D.135min .
V=27m^3, mass rate flow $\frac{m}{t}$ =5kgs^{-1}, ρ=10^{3}kgm^{-3},
mass rate flow = $\frac{m}{t} = \frac{\rho\times V}{t}$, $5 = \frac{10^3\times27}{t}$,
$t = \frac{10^3\times27}{5}$ = 5400s = $\frac{5400}{60}$ = 90min

7. 1000 identical drops of oil of density 500kgm^{-3} have a total mass of 5×10^{-4}kg. One of the drop form a thin film of area 0.5m^2 on water. The thickness of the film is A.2×10^{-8}m **B.2×10^{-9}m** C.2×10^{-7}m D.3×10^{-9}m
Density of oil ρ=500kgm^{-3}, mass of 1000 drops of oil=5×10^{-4}kg, mass of one drop of oil m = $\frac{5\times10^{-4}}{1000}$ =5×10^{-7}kg, Area of the film A=0.5m^2, thickness of the film th=?, volume=Area×thickness, $V = A \times th$,
$\rho = \frac{m}{V} = \frac{m}{A\times th}$, $500 = \frac{5\times10^{-7}}{0.5\times th}$, $th = \frac{5\times10^{-7}}{0.5\times500}$,
th = 2×10^{-9}m^3.

8. Alcohol of mass 21g and density 0.7gcm^{-3} is mixed with 10g of water. Determine the density

Demystified Series by Dr Timothy

of the mixture [Density of water= 1.0gcm^{-3}] **A.0.775gcm^{-3}** B.0.780gcm^{-3} C.0.875gcm^{-3} D.0.880gcm^{-3}.

m_1=21g, ρ_1=0.7gcm^{-3}, m_2=10g, ρ_2=1gcm^{-3},

$\rho_{mixture} = \frac{m_{mixture}}{V_{mixture}}$, $\rho_{mixture} = \frac{m_1+m_2}{V_1+V_2}$, $m_1=\rho_1 V_1$, $m_2=\rho_2 V_2$, $V_1 = \frac{m_1}{\rho_1} = \frac{21}{0.7} = 30\text{cm}^3$, $V_2 = \frac{m_2}{\rho_2} = \frac{10}{1}$, $V_2 = 10\text{cm}^3$, $\rho_{mixture} = \frac{21+10}{30+10} = \frac{31}{40} = 0.775\text{gcm}^{-3}$

OR $\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2} = \frac{m_1+m_2}{\rho_{mixture}}$, $\frac{21}{0.7} + \frac{10}{1} = \frac{21+10}{\rho_{mixture}}$, $30+10 = \frac{31}{\rho_{mixture}}$, $40 = \frac{31}{\rho_{mixture}}$, $\rho_{mixture} = \frac{31}{40} = 0.775\text{gcm}^{-3}$.

9. A liquid of volume 2.00m^3 and density 1.00×10^3kgm^{-3} is mixed with 3.00m^3 of another liquid of density 8.00×10^3kgm^{-3}. Calculate the density of the mixture (Assuming there is no chemical reaction) A.5.20×10^2kgm^{-3} B.8.80×10^2kgm^{-3} C.1.13×10^3kgm^{-3} **D.5.20×10^3kgm^{-3}.**

V_1=2.00m^3, ρ_1=1.00×10^3kgm^{-3}, V_2=3.00m^3, ρ_2=8.00×10^3kgm^{-3}, $\rho_{mixture} = \frac{m_{mixture}}{V_{mixture}}$, $\rho_{mixture} = \frac{m_1+m_2}{V_1+V_2}$, $m_1=\rho_1 V_1$, $m_2=\rho_2 V_2$,

$m_1 = \rho_1 V_1 = 1.00\times10^3\times2.00 = 2.00\times10^3$kg,

$m_2 = \rho_2 V_2 = 8.00\times10^3\times3.00 = 24.00\times10^3$kg,

$\rho_{mixture} = \frac{2\times10^3+24\times10^3}{2+3} = \frac{26\times10^3}{5} = 5.2\times10^3\text{kgm}^{-3}$

OR $\rho_1 V_1 + \rho_2 V_2 = \rho_{mixture}(V_1 + V_2)$, $\rho_{mixture} = \frac{\rho_1 V_1+\rho_2 V_2}{V_1+V_2} = \frac{1\times10^3\times2+8\times10^3\times3}{2+3}$, $\rho_{mixture} = \frac{2\times10^3+24\times10^3}{5} = 5.2\times10^3\text{kgm}^{-3}$.

10. In an experiment to determine the relative density of cork, the following recordings were made I.weight of sunker in water=x, II.weight of sinker in water and cork in air=y, III.weight of both sinker and cork in water=x. Which of the following fractions gives the relative density of the cork? A.$\frac{y}{z-x}$ B.$\frac{x}{y-z}$ **C.$\frac{y-x}{y-z}$** D.$\frac{z}{x-y}$ E.$\frac{z-y}{x-z}$

Weight of sinker in water= x, Weight of sinker in water and cork in air= y,

Weight of the cork in air $W_A = y - x$, Weight of both sinker and cork in water= z, Weight of an equal volume of water= $y - z$.

$R.D_{cork} = \frac{weight\ of\ a\ given\ volume\ of\ cork}{weight\ of\ an\ equal\ volume\ of\ water} = \frac{y-x}{y-z}$.

11. In an experiment using the relative density bottle, the following results were recorded:

Mass of empty bottle+stopper=50g

Mass of bottle full of water+stopper=80g

Mass of bottle full of liquid Q +stopper=74g

Use these results to calculate the relative density of liquid Q A.0.60 **B.0.80** C.1.25 D.0.93 E.1.60

Mass of empty bottle m_b + stopper m_s =50g, Mass of bottle full of water $(m_b + m_w)$ + stopper m_s = 80g, Mass of water m_w = 80−50=30g , Mass of bottle full of liquid Q $(m_b + m_Q)$ + stopper m_s = 74g, Mass of liquid Q m_Q = 74−50=24g.

$R.D_Q = \frac{mass\ of\ a\ given\ volume\ of\ liquid\ Q}{mass\ of\ an\ equal\ volume\ of\ water} = \frac{m_Q}{m_w}$,

$R.D_Q = \frac{24}{30} = 0.8$. $\rho_Q = 0.8\text{gcm}^{-3}$ or 800kgm^{-3}.

12. The mass of a specific gravity bottle is 15.2g when empty. It is 24.8g when filled with kerosene and 27.2g when filled wit distilled water. Calculate the relative density of kerosene A.1.25 B.1.10 C.0.90 **D.0.80.**

Specific gravity bottle is same as a relative density bottle.

Mass of empty bottle=15.2g, Mass of bottle when filled with kerosene=24.8g, Mass of kerosene only m_k =24.8−15.2=9.6g, Mass of bottle when filled with water=27.2g, Mass of water only m_w = 27.2−15.2=12.0g.

$R.D_{kerosene} = \frac{mass\ of\ kerosene}{mass\ of\ water} = \frac{m_k}{m_w} = \frac{9.6}{12.0} = 0.8$.

13. An empty relative density bottle has a mass of 30g. When filled with paraffin, its mass is 70g. Calculate the mass of the bottle when it is filled with water. [Relative density of paraffin=0.8] A.20g B.40g C.60g **D.80g**.

Mass of empty bottle=30g, Mass of bottle when filled with paraffin=70g, Mass of paraffin only=70−30=40g, relative density of paraffin=0.8,

$R.D_{paraffin} = \frac{mass\ of\ paraffin}{mass\ of\ water} = \frac{m_p}{m_w}$, $0.8 = \frac{40}{m_w}$, $m_w = \frac{40}{0.8} = 50$g , Mass of water only=50g, Mass of bottle when filled with water=50+30=80g.

14. When a body is completely immersed in a liquid, it displaces an amount of the liquid that is equal to the I.volume of the body II.weight of the body III.mass of the body. Which of the above is/are correct? A.I B.II C.III **D.I and II** E.I and III.

When a body is completely immersed in a liquid, its displaces volume and weight of the liquid displaced are equal to its volume and weight respectively.

15. An empty 60 litre petrol tank has a mass of 10kg. Its mass when full of relative density 0.72 is A.7.2kg B.33.2kg C.43.2kg **D.53.2kg**.

mass of empty petrol tank=10kg, volume of petrol when full=volume of the tank=60litre= 60(1000)cm^3=60000cm^3, relative density of the petrol $R.D$=0.72, density of the petrol ρ=0.72gcm^{-3}, mass of the petrol only, $m = \rho V$, m =0.72×60000 = 43200g=43.2kg. Mass of the petrol tank when full=mass of the empty tank+mass of the petrol=10+43.2=53.2kg.

16. Why does a a piece of cork float on water? A.there is no upthrust on it when completely immersed in water **B.the upthrust on it when completely immersed in water is greater than its weight in air** C.upthrust on it when completely immersed water is less than its weight D.its specific gravity is greater than that of water E.its specific gravity is same as that of water.

Demystified Series by Dr. Timothy

A body completely immersed in a liquid will float when the upthrust on it is greater than its weight in air or when its density or specific gravity (relative density) is greater than that of water. Option C is the condition necessary for a body to sink.

17. A piece of wood is floating on water. The forces acting on the wood are A.upthrust and reaction B.weight and reaction **C.weight and upthrust** D.upthrust and viscosity E.weight and viscosity.

The forces acting on a body floating on a liquid (water) are weight and upthrust. A body partially immersed in a liquid will float when it weight balances or equals the upthrust acting on it.

18. A solid suspended by a piece of string is completely immersed in water. On attempt to lift the solid out of the water, the string breaks when the solid is partly out of the water, this is because A.the tension in string decreases as the solid if lifted B.the mass of the solid is increased **C.the solid apparently weighs less when completely immersed in water than when partially immersed** D.part of the solid still in water is exerting more force on the string E.the solid is no longer in equilibrium with water.

A solid appears to weigh less when immersed completely in liquids than when partially immersed or in air due to the upthrust exerted by the liquid on the body.Hence, when the body is partly out of water its weight exceeds the tension in the string and the string breaks.

19. A balloon filled with hydrogen rises when released into the air because A.the upthrust of the balloon is less than the weight of the balloon B.the upthrust on the balloon is greater than the weight of the balloon **C.the upthrust on the balloon equal to the weight of the balloon** D.the density of the balloon and hydrogen is greater than the density in air E.the pressure below the balloon is greater than that above the balloon.

The ballon rises and floats in air because the upthrust of air on the balloon equal its weight , $U = W$.

20. A 300g rubber balloon containing 200g of hydrogen moves vertically upwards with a constant velocity. Neglecting air resistance, calculate the magnitude of the force pulling the balloon upwards (g=10ms^{-2}) A.0.5N B.1.0N C.2.0N D.3.0N **E.5.0N.**

Mass of the balloon=300g, mass of the hydrogen gas in the balloon=200g, Total mass of the balloon=mass of the balloon + mass of the gas content= 300+200=500g=0.5kg

The force pulling the ballon upward is the upthrust or fluid thrust and its equal to the weight of the ballon, as the balloon floats in air. $U = W = mg = 0.5\times10 = 5.0$N.

21. The forces acting on a body which is suspended freely in still air from a support are A.gravitational and tension only B.frictional and gravitational force only **C.gravitational, tension and fluid thrust only** D.fluid thrust and tension only E.frictional, gravitational and tension only.

The forces acting on a body suspended i.e by a string, in still air are gravitational forces (weight) tension(in the string) and fluid thrust (upthrust in air).

22. The upthrust experienced by an object in a liquid depends on the I.density of the liquid II.volume of the object III.volume of the liquid displaced A.I B.I and II C.I and III D.II and III **E.I,II and III.**

Upthrust of a liquid on an object, $U = \rho_L V_o g$. Hence, upthrust depends on the density of the liquid ρ_L, the volume of the object V_o or volume of the liquid displaced by the body.

23. A block of ice floats with nine-tenth of its volume submerged in water. In order to find its relative density, one also need to know A.total volume of the ice block B.the actual volume of the ice under water C.the weight of the ice block **D.the density of the ice block** E.nothing else.

$R.D_{ice} = \frac{\rho_{ice}}{\rho_w}$, the density of water is already known, hence inorder to determine the relative density of ice we also need to know the density of the ice block.

24. A uniform solid cube material 10cm on each side of mass 700g is submerged in water. Which of the following best described the behavior of the cube in water? A.the cube will rest at an equilibrium position in water B.The cube will melt after a period of time C.The cube will sink in water **D.The cube will float.**

Length of the cube L=10cm, Volume of the cube $V = L^3$=10^3=1000cm^3, mass of the cube m=700g, density of the cube $\rho_{cube} = \frac{m}{V} = \frac{700}{1000}$, density of the cube $\rho_{cube} = 0.7$gcm^{-3}, density of the liquid(water) ρ_{water} =1gcm^{-3}. Since the density of the solid cube submerged in water is less than the density of water ($\rho_{cube} < \rho_{water}$), the solid cube will have a negative apparent weight and will rise and begin to float in the water.

- The solid cube will sink if its density is higher that of water.

25. A body weighing 14N in air is partially immersed in water. If the mass of the water displaced in the process is 200g, calculate the upthrust on the body (g=10ms^{-2}) **A.2.0*N*** B.3.0N C.3.5N D.7.0N

In partial or complete immersion of a body in a fluid(liquid), the weight of the fluid(liquid) displaced W_{fd} by the body is equal to the upthrust U. Mass of water displaced m_{fd}=200g=0.2kg, g=10ms^{-2}, upthrust is equal to

Demystified Series by Dr Timothy

the weight of the water displaced.
$U = W_{fd} = m_{fd}g = 0.2\times10 = 2.0$N.

26. A body of volume 0.046m^3 is immersed in a liquid of density 980kgm^{-3} with $\frac{3}{4}$ of its volume submerged. Calculate the upthrust on the body (g-10ms^{-2}) A.11.27N B.33.81N C.112.70N **D.338.0N.**
Volume of body V_o=0.046m^3, density of liquid ρ_L=980kgm^{-3}, volume submerged=volume of liquid displaced $V_{Ld}=\frac{3}{4}V_o$, upthrust U=?,
$U = \rho_L V_{Ld} g = \rho_L \frac{3}{4}V_o g = 980\times\frac{3}{4}\times0.046\times10$
$U = 338.10$N.

27. A solid has a weight of 8.0N in air and 2.0N in a certain liquid. Calculate the mass of the liquid displaced by the solid (g=10ms^{-2}) A.10g B.40g **C.600g** D.1000g.
Weight in air W_A=8.0N, Weight in liquid or apparent weight W_{Ap}=2.0N,
Upthrust=Real weight−Apparent weight,
$U = W_A - W_{Ap}$. Upthrust is equal to the weight of the fluid displaced by the body, $U = W_{fd} = m_{fd}g$,
$m_{fd}g = W_A - W_{Ap}, m_{fd}\times10=8-2,\ m_{fd}\times10=6,$
$m_{fd} = \frac{6}{10} = 0.6$kg=600g.

28. An object of volume 1m^3 and mass 2kg is totally immersed in a fluid of density 1kgm^{-3}. Calculate its apparent weight A.20N **B.10N** C.2N D.1N.
V_o=1m^3, m_o=2kg, $\rho_o = \frac{m_o}{V_o} = \frac{2}{1} = 2$kgm^{-3},
ρ_L=1kgm^{-3}, $W_{Ap} = (\rho_o - \rho_L)V_o g$,
$W_{Ap} = (2-1)\times1\times10 = 10$N.

29. A metal object of relative density 9.0 and volume 2×10^{-4}m^3 is completely immersed in water. Its apparent weight is A.18N **B.16N** C.4.5N D.2.0N.
$R.D_o$=9.0, ρ_o=9000kgm^{-3}, $V_o=2\times10^{-4}$m^3,
ρ_L=density of water=1000kgm^{-3},
$W_{Ap} = (\rho_o - \rho_L)V_o g$,
$W_{Ap} = (9000-1000)\times2\times10^{-4}\times10 = 16$N.

30. If the relative density of a metal is 19. What will be the mass of 20cm^3 of the metal when immersed in water? A.400g B.380g **C.360g** D.180g E.39g.
$R.D_o$=19, ρ_o=19gcm^{-3}, V_o=20cm^3, ρ_L=1gcm^{-3}, mass when immersed in water=apparent mass,
$m_{Ap} = (\rho_o - \rho_L)V_o = (19-1)\times20 = 360$g.

31. A body whose mass is 2kg and has a volume of 500cm^3 just floats when completely immersed in a liquid. Calculate the density of the liquid A.4.0×10^2kgm^{-3} **B.4.0×10^3kgm^{-3}** C.1.0×10^3kgm^{-3} D.1.0×10^6kgm^{-3}.
m_o=2kg, V_o=500cm^3=500×10^{-6}m^3, ρ_L=?,
Upthrust=weight of the fluid displaced by it. The body is completely immersed hence, the weight of the fluid equal to the weight of the body.
$U = W,\ U = \rho_L V_o g,\ W = m_o g,\ \rho_L V_o g = m_o g,$
$\rho_L V_o = m_o,\ \rho_L\times500\times10^{-6} = 2,$
$\rho_L = \frac{2}{500\times10^{-6}} = 4000$kgm^{-3}=$4.0\times10^3$kgm^{-3}.

32. A piece of iron weighs 250N in air and 200N in a liquid of density 1000kgm^{-3}. The volume of the iron is A.2.0×10^{-3}m^3 B.2.5×10^{-3}m^3 C.4.5×10^{-3}m^3 **D.5.0×10^{-3}m^3.**
W_A=250N, W_{Ap}=200N, ρ_L=1000kgm^{-3}, g=10ms^{-2}, V_o=?, $U = \rho_L V_o g = W_A - W_{Ap}$,
$1000\times V_o\times10 = 250-200 = 50,\ V_o = \frac{50}{1000\times10},$
$V_o = 5.0\times10^{-3}$m^3.

33. An object weighs 22kg in water and 30kg in air. What is the upthrust exerted by the liquid on the object? A.520N B.220N **C.80N** D.20N (g=10ms^{-2}).
m_{Ap}=22kg, m_A=30kg, g=10ms^{-2}, U=?,
$W_A = m_A g,\ W_{Ap} = m_{Ap}g,\ U = W_A - W_{Ap},$
$U = m_A g - m_{Ap}g = g(m_A - m_{Ap}),$
$U = 10(30-22) = 80N.$

34. A test tube of radius 1.0cm is loaded to 8.8g, if it is placed upright in water, find the depth to which it would sink **A.2.8cm** B.5.2cm C.25.5cm D.28.0cm (g=10ms^{-2}, density of water=1000kgm^{-3} or 1gcm^{-3}).
m=8.8g, r=1.0cm, ρ=1gcm^{-3}, g=10ms^{-2}, h=?,
Upthrust on the test tube=Weight of water displaced. $U = W, U = \rho Vg = \rho(Ah)g,\ W = mg,$
$\rho(Ah)g = mg,\ \rho(Ah) = m, A = \pi r^2,\ \pi = \frac{22}{7},$
$1(\frac{22}{7}\times1^2\times h)=8.8,\ \frac{22}{7}h = 8.8,\ h = \frac{7\times8.8}{22} = 2.8$cm.

35. If a spherical metal bob of radius 3cm is fully immersed in a cylinder containing water and the water level rises by 1cm, what is the radius of the cylinder? A.12cm **B.6cm** C.3cm D.1cm.
Radius of the spherical bob r=3cm, height of water rise h=1cm, radius of the cylinder R=?,
Rise in volume of a liquid on complete or total immersion of an object in a fluid equals the volume of the object immersed.The volume of fluid displaced by a body on complete immersion is also equal to the volume of the vessel. Thus, volume of the spherical bob displaced=volume of the liquid displaced.
$\frac{4}{3}\pi r^3 = \pi R^2 h,\ \frac{4}{3}r^3 = R^2 h,\ \frac{4}{3}\times3^3 = R^2\times1,$
$R^2 = \frac{4}{3}\times27=36,\ R = \sqrt{36} = 6$cm.

36. A piece of wood of mass 40g and uniform cross-sectional area of 2cm^2 floats upright in water of density 1gcm^{-3}. The length of the wood immersed is A.80cm B.40cm **C.20cm** D.2cm
Weight of the wood displaced=upthrust of water.
$W = U,\ U = \rho Vg = \rho(Ah)g,\ W = mg,\ m = \rho(Ah),$ m=40g, A=2cm^2, ρ=1gcm^{-3}, length of the wood immersed h=?,
$40=1(2\times h),\ h = \frac{40}{2} = 20$cm.

37. A metal cube of volume 10^{3}mm^3 is lowered into a measuring cylinder containing water. If the internal cross-sectional area of the cylinder is 1.5×10^2mm^2, by how much does the water level

Demystified Series by Dr Timothy

rise in the cylinder? **A.6.67×10⁰mm** B.8.50×10³mm C.1.15×10³mm D.2.50×10³mm E.1.50×10⁵mm.

Volume of the metal cube V=10³mm³, internal cross-sectional area of the cylinder A=1.5×10²mm², height of water rise h=?, volume of the metal cube= volume of water displaced(i.e the volume of the cylinder),

$V = Ah$, $10^3 = 1.5\times10^2\times h$, $h = \frac{10^3}{1.5\times10^2}$,

$h = 6.67\text{mm} = 6.67\times10^0\text{mm}$.

38. When a ship sails from salt water into fresh water, the fraction of its water above the water surface will A.increase **B.decrease** C.remain the same D.increase then decreases.

Sea or salt water has a higher density i.e denser, than fresh water. Volume of a body submerged (or immersed) or volume of the liquid displaced increases as the density of the liquid decreases $V \propto \frac{1}{\rho}$. The volume of the body above the liquid increases as the volume of the body submerged or volume of liquid displaced decreases. Hence, when a ship sails from salt water to fresh water i.e decrease in density, the volume of the ship submerged increases and the volume above the liquid decreases.

39. A solid object floats in liquid X of relative density 0.9 but sinks in liquid Y of relative density 0.7. It can be inferred that the A.volume of X displaced is less than that of Y B.volume of X displaced is greater than that of Y C.weight of X displaced is less than that of Y D.weight of X displaced is greater than that of Y.

The upthrust in liquid X or the weight of liquid X displaced by the solid is equal to its weight while the upthrust in liquid Y or the weight of liquid Y displaced by the solid is less than its weight. Hence, upthrust or weight of liquid X displaced is greater than the upthrust or weight of liquid Y displaced.

40. A piece of wood floats inside water at room temperature with a fraction of it above the liquid surface. As the temperature of the water is raised, the temperature of the water is raised, the part if the wood above the liquid will **A.Decrease because the density of water decreases with temperature** B.Increases because the density of water decreases with temperature C.Decreases because the density of water increases with temperature D.Increases because the density of water increases with temperature .

The density of water decreases with increase in temperature, hence the body sinks further and the fraction of the body above the water surface decreases.

41. A block of volume 3×10⁻⁵m³ and density 2.5×10³kgm⁻³ is suspended from a spring balance with $\frac{2}{3}$ of its volume immersed in a liquid of density 900kgm⁻³. Determine the reading of the spring balance (g=10ms⁻²) A.0.18N **B.0.57N** C.0.75N D.0.93N.

V_o=3×10⁻⁵m³, ρ_o=2.5×10³kgm⁻³=2500kgm⁻³, volume of liquid displaced V_{Ld}= $\frac{2}{3}V_o$, g=10ms⁻², ρ_L=900kgm⁻³, reading of the spring balance is the apparent weight or weight of the body in the liquid. $W_{Ap} = W_A - U = \rho_o V_o g - \rho_L V_{Ld} g$,

$W_{Ap} = \rho_o V_o g - \rho_L \frac{2}{3} V_o g = \left(\rho_o - \frac{2}{3}\rho_L\right) V_o g$,

$W_{Ap} = (2500 - \frac{2}{3}\times900)\times3\times10^{-5}\times10$,

$W_{Ap} = (2500-600)\times3\times10^{-5}\times10 = 0.57\text{N}$.

42. An object of mass 400g and density 600kgm⁻³ is suspended with a string so that half of it is immersed in paraffin of density 900kgm⁻³.The tension in the string is **A.1.0N** B.3.0N C.4.0N D.5.0N.

$U + T = W$, $T = W - U$, T=tension

Mass of the object m_o=400g=0.4kg,

Weight of the object $W = mg$ =0.4×10=4N

Upthrust U =weight of the fluid displaced, $U = \rho_L V_{Ld} g$, ρ_o=600kgm⁻³, Volume of object $V_o = \frac{m_o}{\rho_o} = \frac{0.4}{600}\text{m}^3$, Volume of liquid displaced $V_{Ld} = \frac{V_o}{2}$, $U = \rho_L V_{Ld} g = \rho_L \frac{V_o}{2} g$,

$U = 900\times\frac{1}{2}\times\frac{0.4}{600}\times10 = 3\text{N}$.

$T = W - U = 4-3 = 1\text{N}$. Alternatively,

$\boldsymbol{T = (m_o - \rho_L V_{Ld})g} = (0.4-900\times\frac{1}{2}\times\frac{0.4}{600})10$

$T = (0.4-0.3)10 = 1\text{N}$.

43. A solid weighs 45N and 15N respectively in air and water. Determine the relative density of the solid A.0.33 B.0.50 **C.1.50** D.3.00.

Weight in air W_A=45N, Weight in water W_w=15N, relative density of the solid, $R.D_s$=?

$R.D_s = \frac{weight\ in\ air}{upthrust\ in\ water} = \frac{W_A}{U_w} = \frac{W_A}{W_A - W_w}$,

$R.D_s = \frac{45}{45-15} = \frac{45}{30} = 1.50$.

44. A piece of metal of relative density 5.0 weighs 60N in air. Calculate its weight when fully immersed in water A.4N B.5N **C.48N** D.60N.

$R.D_s$=5.0, W_A=45N, W_w=?,

$R.D_s = \frac{weight\ in\ air}{upthrust\ in\ water} = \frac{W_A}{U_w} = \frac{W_A}{W_A - W_w}$,

$5.0 = \frac{60}{60-W_w}$, $60 - W_w = \frac{60}{5} = 12$,

$W_w = 60-12 = 48\text{N}$.

45. A spiral spring loaded with a piece of metal, extends by 10.5cm in air. When the metal is fully submerged in water, the metal extends by 6.8cm. Calculate the relative density of the metal (assume Hooke's law is obeyed) A.8.52 B.5.68 **C.2.84** D.1.42.

Extension in air e_A=10.5cm, Extension in water e_w=6.8cm, $R.D_s$=?, Based on Hooke's law – F(or W) $\propto e$, hence extension can be used in place of force(or weight).

$R.D_s = \frac{extension\ in\ air}{upthrust\ in\ water} = \frac{e_A}{U_w} = \frac{e_A}{e_A - e_w}$,

Demystified Series by Dr Timothy

$R.D_s = \frac{10.5}{10.5-6.8} = \frac{10.5}{3.7} = 2.84.$

46. An object weighs 60.0N in air, 48.2N in a certain liquid X and 44.9N in water. Calculate the relative density of X A.3.300 B.1.279 C.0.932 **D.0.781.**

Weight in air W_A=60.0N, Weight in liquid XW_X=48.2N, Weight in water W_w=44.9N, relative density of the liquid $XR.D_X$=?,

$R.D_X = \frac{upthrust\ in\ liquid\ X}{upthrust\ in\ water} = \frac{U_X}{U_W} = \frac{W_A - W_X}{W_A - W_w}$,

$R.D_X = \frac{60.0-48.2}{60.0-44.9} = \frac{11.8}{15.1} = 0.781.$

47. A solid of weight 0.600N is totally immersed in oil and water respectively. If the upthrust in oil is 0.210N and the relative density of oil is 0.875. Find the upthrust in water A.0.600N B.0.360N **C.0.240N** D.0.180N .

W_A=0.600N, Upthrust in oil U_o=0.210N, relative density oil $R.D_o$=0.875, Upthrust in water U_w=?,

$R.D_o = \frac{upthrust\ in\ oil}{upthrust\ in\ water} = \frac{U_o}{U_w}$, $0.875 = \frac{0.210}{U_w}$,

$U_w = \frac{0.210}{0.875} = 0.240.$

48. A copper cube weighs 0.25N in air, 0.17N when completely immersed in paraffin oil and 0.15N when completely immersed in water. The ratio of upthrust in oil to upthrust in water is **A.4:5** B.3:5 C.13:5 D.7:10

Weight in air W_A=0.25N, Weight in paraffin oil W_o=0.17N, Weight in water W_w=0.15N, the ratio of the upthrust in paraffin oil to upthrust in water is equal to the relative density of paraffin oil.

$R.D_o = \frac{upthrust\ in\ oil}{upthrust\ in\ water} = \frac{U_o}{U_w} = \frac{W_A - W_o}{W_A - W_w}$,

$R.D_o = \frac{0.25-0.17}{0.25-0.15} = \frac{0.08}{0.1} = \frac{4}{5} = 4{:}5.$

49. A solid weighs 4.8g in air, 2.8g in water and 3.2g in kerosene. The ratio of density of the solid to that of kerosene is A.12 **B.3** C.2 D.$\frac{3}{2}$ E.$\frac{5}{4}$

Weight in air W_A=4.8g, Weight in water W_w=2.8g, Weight in kerosene W_k=3.2g, relative density of the solid $R.D_s$,

$R.D_s = \frac{weight\ in\ air}{upthrust\ in\ water} = \frac{W_A}{U_w} = \frac{W_A}{W_A - W_w}$,

$R.D_s = \frac{4.8}{4.8-2.8} = \frac{4.8}{2} =$ 2.4, density of the solid ρ_s=2.4gcm^{-3} or 2400kgm^{-3}

relative density of kerosene $R.D_k$,

$R.D_k = \frac{upthrust\ in\ kerosene}{upthrust\ in\ water} = \frac{U_k}{U_W} = \frac{W_A - W_k}{W_A - W_w}$,

$R.D_k = \frac{4.8-3.2}{4.8-2.8} = \frac{1.6}{2} =$ 0.8, density of kerosene ρ_k=0.8gcm^{-3} or 800kgm^{-3}.

Ratio of the density of the solid to that of kerosene= $\frac{\rho_s}{\rho_k} = \frac{2.4}{0.8} = 3$ or 3:1.

50. A piece of cork floats in a liquid. What fraction of its volume will be immersed in the liquid (density of the cork=0.25×10^{3}kgm^{-3}, density of the liquid=1.25×10^{-3}kgm^{-3}) A.0.1 **B.0.2** C.0.5 D.0.8.

Density of the cork ρ_o=0.25×10^{3}kgm^{-3}, density of the liquid ρ_L=1.25×10^{-3}kgm^{-3}, volume of the cork V_o , volume of the liquid displaced V_{Ld} . When the cork floats, the weight of the cork equals the weight of the liquid displaced, $W_o = W_{Ld}$, $\rho_o V_o = \rho_L V_{Ld}$, Fraction of volume submerged = $\frac{V_{Ld}}{V_o} = \frac{\rho_o}{\rho_L} = \frac{0.25\times10^3}{1.25\times10^3} = \frac{1}{5} = 0.2.$

51. A wooden object whose density is 0.06gcm^{-3} was dropped into water contained in a tall beaker. How much of the wooden object is in the water? A.One-quarter B.One-sixth C.Two-thirds **D.A small fraction .**

Density of the wooden object ρ_o=0.06gcm^{-3}, density of the liquid(water) ρ_L=1gcm^{-3}, volume of the wooden object V_o, volume of the liquid(water) displaced V_{Ld}. When the wooden block floats in water, the weight of the block equals the weight of the water displaced. $W_o = W_{Ld}$, $\rho_o V_o = \rho_L V_{Ld}$,

Fraction of volume submerged= $\frac{V_{Ld}}{V_o} = \frac{\rho_o}{\rho_L}$,

Fraction of wooden block submerged= $\frac{0.06}{1}$,

Fraction of wooden block submerged= $\frac{3}{50}$, a small fraction of the volume.

52. A hydrometer is used for measuring the A.depth of water in a vessel **B.relative density of a liquid by method of floatation** C.relative density of a liquid by finding the apparent weight D.relative humididty in air.

Hydrometers are used for measuring relative density. Its operation is based on principle of floatation.

53. The relative density of the acid in a car battery can be measured with a A.hygrometer B.manometer **C.hydrometer** D.barometer.

54. Which of the following statements is/are correct about the common hydrometer? I.It has a long narrow stem II.The relative density values marked on the stem increases down the the stem III.It has a wide bulb IV.The bulb is loaded to make the hydrometer float vertically A.II B.II and IV C.I,II and III D.II and IV **E.I,II,III and IV.**

55. A loaded tube of uniform cross-sectional area 4.0cm^2 floats upright in water with 15.5cm of its length immersed and then in liquid L with 18.0cm of its length immersed, if the density of water is 1.0gcm^{-3} is A.25.0g B.47.2g C.34.8g **D.62.0g.**

Cross-sectional area of the loaded tube A=4.0cm^2, depth of immersion in water h_w=15.5cm, depth of immersion in liquid Lh_L=18.0cm, ρ_w=1.0gcm^{-3}, The loaded test-tube as stated in the question is a hydrometer. In a hydrometer, by principle of loatation the weight of the tube is equal to the weight of the water displaced by it. Weight of the tube=Weight of water displaced, $W = W_{Ld}$, $mg = \rho_w V_{Ld} g$, $m = \rho_w V_{Ld}$, volume of liquid displaced $V_{Ld} = Ah_w$, mass of the tube m=?,

$m = \rho_w A h_w$ =1×4×15.5 = 62.0g.

Demystified Series by Dr Timothy

- The relative density of the liquid L can be gotten by – $R.D_L = \frac{h_w}{h_L} = \frac{15.5}{18.0} = 0.86$. Hence, the density of L is 0.86gcm^{-3} or 860kgm^{-3}.

56. A hydrometer of mass 3.60kg and volume 6.00×10^{-5}m^3 floats in a liquid with $\frac{1}{5}$ of its volume above the liquid. Calculate the density of the liquid A.4.50×10^5kgm^{-3} B.3.00×10^5kgm^{-3} C.1.50×10^5kgm^{-3} **D.7.50×10^4kgm^{-3}.**

Based on the principle of floatation,

Weight of hydrometer=Upthrust of the liquid (or Weight of the liquid displaced by it). $m = \rho_L V_{Ld}$, mass of hydrometer m=3.60kg, Volume of the hydrometer=6.00×10^{-5}m^3, if $\frac{1}{5}$ of the volume of the hydrometer is above the liquid, then $\frac{4}{5}$ of its volume is immersed. Volume of the liquid displaced $V_{Ld} = \frac{4}{5}$ of the volume of the hydrometer. $3.60 = \rho_L \times \frac{4}{5} \times 6.00\times10^{-5}$,

$\rho_L = \frac{3.60\times5}{4\times6.00\times10^{-5}} = 7.50\times10^4$kgm^{-3}.

57. A hydrometer is placed in water and 6.4cm of its stem is immersed. What is the depth of immersion when it is placed in a liquid of relative density 0.8 A.60cm B.70cm **C.8cm** D.9cm (density of water=1gcm^{-3}).

Depth of immersion in water h_w=6.4cm, relative density of the liquid $R.D_L$=0.8, depth of immersion in liquid h_L=?,

$R.D_L = \frac{h_w}{h_L}$, $0.8 = \frac{6.4}{h_L}$, $h_L = \frac{6.4}{0.8} = 8$cm.

58. A plastic sphere floats in water with 50% of its volume submerged. If it floats in glycerine with 40% of its volume submerged, the density of the glycerine is A.1400kgm^{-3} **B.1250kgm^{-3}** D.1000kgm^{-3} C.500kgm^{-3} (density of water =1000kgm^{-3}).

Fraction of volume submerged in water V_w= 50% = $\frac{50}{100}$=0.5, Fraction of volume submerged in glycerine V_g=40%= $\frac{40}{100}$=0.4, ρ_g=?,

$R.D_L = \frac{fraction\ of\ volume\ submerged\ in\ water}{fraction\ of\ volume\ submerged\ in\ liquid}$,

$R.D_L = \frac{\rho_g}{\rho_w} = \frac{V_w}{V_L}$, $\frac{\rho_g}{1000} = \frac{0.5}{0.4}$, $\rho_g = \frac{0.5\times1000}{0.4}$,

ρ_g =1250kgm^{-3} .

59. A uniform cylindrical block of wood floats in water with one third of its height above the water level. In a liquid of relative density 0.8, what fraction of its height will be above the liquid level? **A.$\frac{1}{6}$** B.$\frac{1}{5}$ C.$\frac{1}{3}$ D.$\frac{4}{5}$ E.$\frac{5}{6}$.

Height above water=$\frac{1}{3}h$, Height submerged in water h_w=$\frac{2}{3}h$, Height above the liquid =x, Height submerged in liquid h_L=$(1-x)h$, relative density of the liquid $R.D_L$=0.8,

$R.D_L = \frac{h_w}{h_L}$, $0.8 = \frac{\frac{2}{3}h}{(1-x)h} = \frac{\frac{2}{3}}{1-x}$,

$1-x = \frac{2}{3}\times\frac{1}{0.8} = \frac{2}{3}\times\frac{10}{8} = \frac{5}{6}$, $x = 1 - \frac{5}{6} = \frac{1}{6}$.

60. A body of mass 25g floats in water of density 1.0×10^3kgm^{-3} with $\frac{1}{4}$ of its volume submerged. Calculate the volume of the body above the water surface **A.75cm^3** B.60cm^3 C.50cm^3 D.25cm^3 E.6.25cm^3

m_o=25g, ρ_w=1.0×10^3kgm^{-3}=1gcm^{-3}, fraction of volume submerged in water or volume of the liquid displaced $V_{Ld} = \frac{1}{4}V_o$,

Floatation principle – Weight of the body = Weight of the water displaced by it,

$m = \rho_w V_{Ld} = \rho_w \frac{1}{4}V_o$, $25 = 1\times\frac{1}{4}V_o$,

V_o =4×25=100cm^3. Volume of body above water surface= $\frac{3}{4}V_o = \frac{3}{4}\times100 = 75$cm^3.

61. A reactangular block of wood floats in water with two third of its volume immersed. When placed in another liquid it floats with half its volume immersed. Calculate the relative density of the liquid A.1.50 **B.1.33** C.1.00 D.0.83 E.0.66.

Fraction of volume submered in water V_w=$\frac{2}{3}$, Fraction of volume submerged in liquid V_L=$\frac{1}{2}$, relative density of the liquid $R.D_L$=?,

$R.D_L = \frac{fraction\ of\ volume\ submerged\ in\ water}{fraction\ of\ volume\ submerged\ in\ liquid}$,

$R.D_L = \frac{V_w}{V_L} = \frac{2/3}{1/2} = \frac{2}{3}\times2 = 1.33$.

Jamb past questions on Eqilibrium in fluids (density, relative density and upthrust) :

[1978/14,1979/1,1980/29,30,1981/50,1982/15,20,45,1983/8,28,31,1984/4,1985/16,17,1986/3,1987/15,1988/14,1989/12,1990/14,1991/12,1992/12,1994/18,19,1995/20,1997/16,19,1998/16,1999/15,2000/4,2001/5,2002/10,2003/3,2004/34,2005/46,2006/41,46,2007/34,2008/18,2009/17,2010/17,2011/15,2012/16,18,2013/8,15,2015/13,27].

Demystified Series by Dr. Timothy

Chapter 10 – Pressure in solids and liquids, Gas laws and Kinetic theory of matter

● Pressure in solids –

- **Pressure** $\boldsymbol{P}$ is the normal or perpendicular force F per unit area A acting on a surface.

$\boldsymbol{P = \frac{F}{A}}$, $\boldsymbol{F = W = mg}$, $\boldsymbol{A = \pi r^2}$.

- Pressure in solids depends on the force exerted and area of the body.

- Pressure on a body increases with an increase in force acting on it i.e $\boldsymbol{P \propto F}$.

- Pressure decreases with an increase in area, $\boldsymbol{P \propto \frac{1}{A}}$**. Hence, a man standing on one toe of his foot (small area) exertes a larger pressure, cutting objects with sharp knifes (small area) exerts a great pressure.**

- Solid pressure is exerted downwars only.

- The units for pressure and their relationship:

i.1Pascal(Pa)=1Nm^{-2} **ii.**1Bar=10^5Nm^{-2} =10^5Pa. **iii.**1mmHg = 1Torr **iv.**760mmHg = 1Atmospher(Atm).

● Pressure in liquids or Hydrostatic pressure –

The pressure exerted in a liquid is given by : $\boldsymbol{P = \rho gh}$**.** ρ=density of the liquid, g=acceleration due to gravity, h=depth below liquid surface or height of liquid column.

- Pressure in liquids depends on density (relative density or specific gravity) of the liquid, depth or height of liquid column and acceleration due to gravity. It is independent of the shape of the container or total volume of liquid and cross-sectional area or diameter of the vessel.

- Characteristics of liquid pressure –

1.Pressure at all points at the same level within a liquid is the same.

2.Pressure in a liquid is exerted equally in all directions at any point in a liquid.

3.Pressure in liquids icreases with depth of the liquid.

4.Pressure at the same level in different liquids increases with their density.

● Atmospheric pressure – The atmosphere exerts some pressure on the earth surface and objects on the earth surface. This is also the pressure exerted by air on objects.

- Atmospheric pressure decreases with increasing altitude (height) above sea level and increases with increasing depth below sea level.

- Evidence of atmospheric pressure i.e experiments to show that the atmosphere (air) exerts pressure –

1.Crushing of a can when the air in it is evacuated.

2.Cardboard paper maintaining its position to an inverted beaker filled with water.

3.Bleeding of the nose at high altitude(height)

4.Maintaining the height or altitudes of aeroplanes high above sea level.

- Pressure at the surface of any liquid is only atmospheric pressure i.e $\boldsymbol{P_{atm}}$ **or** $\boldsymbol{H}$**mmHg or** $\boldsymbol{H}$**mmH$_2$O.**

- Pressure at a point h above the atmospheric end or above sea level –

$\boldsymbol{P = (P_{atm} - \rho gh)N}$**m^{-2}** or $\boldsymbol{(H - h)}$**mmHg or mmH$_2$O.**

- Pressure at a point below atmospheric end or below sea level or at the base of a liquid –

$\boldsymbol{P = (P_{atm} + \rho gh)N}$**m^{-2} or** $\boldsymbol{(H + h)}$**mmHg or mmH$_2$O.**

- The pressure exerted at the base of a liquid surface exposed to an enclosed air column is –

$\boldsymbol{P =}$ **Pressure due to air column +** $\boldsymbol{\rho gh}$**.**

- The pressure change in air of density ρ_{air} and in a liquid of density ρ_L or between different liquids is given by – $\boldsymbol{P_{air} = P_L}$**,** $\boldsymbol{\rho_{air} gH = \rho_L gh}$**.**

- Measurment of atmospheric pressure – Atmospheric pressure is measured by a **barometer.** Types of barometer –

1.Simple or Torricellian barometer – Thick walled glass tube filled with mercury and inverted over a dish of mercury. A vacuum exist above the mercury level in the tube.The atmospheric pressure supports a mercury column of 760mmHg or 76cmHg.

2.Fortin barometer– It is a simple barometer where a leather bag is used as a reservoir of mercury instead of a mercury dish. It has an ivory pointer or index at the zero mark to indicate the level of mercury and locate the zero mark of the barometer scale.

3.Aneroid barometer – It has no barometric fluid and measures slight variations of atmospheric pressure through a partially evacuated and sealed metal can. It can be used as an **altimeter** to measure height or altitude) and in **weather forcasts** i.e high pressure indicates dry weather while low pressure indicates rainy weather.

- In a fautless barometer, the space above the barometric liquid is a complete vacuum and when tilted the vertical height of the mercury column remains constant i.e 76cm 0r 760mm.

- In a faulty barometer, the space above the barometric liquid is filled with air bubble and when tilted the vertical height of the mercury column falls below 76cm 0r 760mm.

- When a barometer is used as as an altimeter in determining the height(altitude) of mountains or aeroplanes H_P above sea level, the relation used

Demystified Series by Dr Timothy

is – $\boldsymbol{H_P = \frac{\rho_{Hg}}{\rho_{air}}(H_o - H_1)}$. H_P=height of mountain or aeroplane, ρ_{Hg}=density of mercury, ρ_{air}=density of air, H_o=barometric height at sea level (larger height) or pressure at sea level, H_1= barometric height above sea level at H_P (smaller height) or pressure above sea level.

- Mercury is a better barometric liquid than water because its density is several times greater than that of water, it needs no colouring, has a low vapour pressure i.e does not vaporize easily, and does not wet glass.

- The reading of pressure on a mercury barometer depends on atmospheric pressure, density of mercury,acceleration due to gravity and temperature of mercury.It is independent of the cross-sectional area of the tube.

- Applications of atmospheric pressure –

1.Hare's apparatus and a U-tube – For comparing the densities of two immiscible liquids or determining the relative density of a liquid. If a liquid L of density ρ_L and water of density ρ_w are introduced into a U-tube or Hare's apparatus, the relative density of the liquid L is given by – $\boldsymbol{R.D_L = \frac{\rho_L}{\rho_w} = \frac{h_w}{h_L}}$. h_L=height of liquid column or height of liquid atmospheric pressure supports, h_w=height of water column or height of water atmospheric pressure supports.

2.Siphon – This is a deviced used to remove liquids from receptacles or vessels which cannot be easily lifted or reached or emptied converniently. For a siphon to function efficiently, the tube must be filled by the liquid, the recipient vessel must be lower than the original vessel and the height of liquid column must be less than the barometric height of the liquid.

3.Lift pumpor common pump – It is used to lift water from wells or lifting kerosene from a tall tank. It raises water to a height less than 10m because atmospheric pressure cannot support water column above 10m high.

4.Force pump – It raises water to a height greater than 10m.It has an air chamber which ensures a steady flow of water.

- In both a lift and force pump, upward stroke opens the lower valve and closes the upper valve while downward stroke closes the lower valve and open the upper valve

5.Compression pump – Increases pressure by forcing air into an enclosure e.g bicycle pump and syringe.

6.Vacuum or evacuation pump – Decreasing or lowering pressure by evacuating air from an enclosure e.g crushing can experiment.

– Lift pump, force pump, compression pump and vacuum pump are all mechanical pumps.

8.Pressure cooker – It is an enclosed air-tight pot used to cook food at high temperature as pressure inside the cooker is increased above atmospheric pressure. **Water in a pressure cooker boils at a high temperature greater than its normal boiling point at sea level i.e 100**°C**.** Pressure cooker cooks food faster, thereby saving time and fuel.

- Measurement of gas pressure – Gas pressure is measured by a **manometer.**

- The pressure of a gas in a manometer –

1.If the arm exposed to the atmosphere is higher than or at the same level the arm exposed to the gas, the gas pressure is given by – $\boldsymbol{P = P_{atm} + \rho gh}$ or $\boldsymbol{H + h}$.

2.If the arm exposed to the atmosphere is lower than the arm exposed to the gas, the gas pressure is given by – $\boldsymbol{P = P_{atm} - \rho gh}$ or $\boldsymbol{H - h}$. H=Height of liquid column due to atmospheric pressure, h=the difference in the liquid level between the two arms of the manometer.

- If pressure is required in cm or mm of H_2O and the atmospeheric pressure is in cm or mm of Hg. The atmospheric pressure should be converted to cm or mm of H_2O using the equation – $\boldsymbol{\rho_{Hg}h_{Hg} = \rho_{H_2O}h_{H_2O}}$.

- Mercury is used for measuring high pressure while water is used measuring low pressure due to their density.

- Transmission of pressure – Pascal's principle of transmission of pressure states that the pressure applied to a fluid which is confined in a container will be transmitted equally to all parts of the container.

- Pascals principle is applied in hydraulic press for lifting heavy loads, motor car-foot brakes or hydraulic brakes is slowing down or stopping moving automobiles or cars and hydraulic jacks for lifting cars.

- Based on Pascal's principle, the pressure exerted at a smaller piston of area A_1 is equal to that exerted in the larger piston A_1. Hence, $\boldsymbol{P_1 = P_2 = P}$, $\boldsymbol{\frac{F_1}{A_1} = \frac{F_2}{A_2}}$.

Examples :

1. The stylus of a photograph record exerts a force of 7.7×10^{-2}N on a groove of radius 10^{-5}m. Compute the pressure exerted by the stylus on the grove A.2.24×10^{9}Nm^{-2} B.4.90×10^{8}Nm^{-2} **C.2.45×10^{8}Nm^{-2}** D.3.45×10^{8}Nm^{-2}

F=$7.7\times10^{-2}N$, r=10^{-5}m, $A = \pi r^2$, $P = \frac{F}{A}$,

$P = \frac{7.7\times10^{-2}}{\pi(10^{-5})^2} = \frac{7.7\times10^{-2}}{\pi10^{-10}} = \frac{7.7\times10^{8}}{\pi} = \frac{7.7\times10^{8}}{3.142}$,

P =2.45×10^{8}Nm^{-2}

2. A rectangular water tank of weight $4.5\times10^{3}N$ measures 2.0m by 1.5m by 1.2m. Calculate the minimum pressure it can exert when resting on a

Demystified Series by Dr Timothy

horizontal surface A.$1.9\times10^3 Nm^{-2}$ B.$2.5\times10^3 Nm^{-2}$ C.$3.0\times10^3 Nm^{-2}$ **D.$1.5\times10^3 Nm^{-2}$.**
F=W=4.5×10^{3N}, minimum pressure is exerted when the area is maximum. The maximum area is given by the dimension 2.0m by 1.5m, A =2.0×1.5=3.0m^2, $P=\frac{F}{A}=\frac{W}{A}$,
$P=\frac{4.5\times10^3}{3.0}$ =1.5×10^3Nm^{-2}.

3. A man exerts a pressure of 2.8×10^3Nm^{-2} on the ground and has 4×10^{-2}m^2 of his feet in contact with the ground. The weight of the man is A.140N **B.112N** C.102N D.70N.
P=$2.8\times10^3 Nm^{-2}$, A=4×10^{-2}m^2, F=W=?,
$P=\frac{F}{A}=\frac{W}{A}$, W = PA = $2.8\times10^3\times4\times10^{-2}$= 112N.

4. A rectangular block of dimensions 2.0m×1.0m×0.5m weighs 200N. Calculate the maximum pressure exerted by the block on the horizontal floor A.100Nm^{-2} B.200Nm^{-2} C.300Nm^{-2} **D.400Nm^{-2}.**
F=W=200N, maximum pressure is exerted when the area is minimum. The minimum area is given by the dimension 1.0m×0.5m, A =1.0×0.5 ,
A =0.5m^2, P=?, $P=\frac{F}{A}=\frac{W}{A}=\frac{200}{0.5}=400$Nm^{-2}.

5. A man will exert the greatest pressure on a bench when he **A.stands on the toes of one foot** B.stands on one foot only C.stands on both feet D.lies flat on his back E.lies flat on his belly.
The greatest pressure will be exerted when the area is minimum i.e standing on the toes of one foot.

6. Which of the following statements is correct? **A.pressure exerted is higher with a narrow heel shoe than a flat heel shoe** B.no pressure is exerted in both narrow and flats heel shoes C.pressure exerted is lower with a narror heel shoes than a flat heel shoe D.pressure exerted is the same in both narrow and flat heel shoes.
A narrow heel shoe has a small area than a flat heel shoes, hence will exert the greatest pressure.

7. The hydrostatic blood pressure diffenrence between the head and feet of a boy standing straight is 1.65×10^4Nm^{-2}. Find the height of the boy (Density of the blood=1.1×10^3kgm^{-3}, g=10ms^{-2}) A.0.6m B.0.5m **C.1.5m** D.2.0m
P=1.65×10^4Nm^{-2}, ρ=1.1×10^3kgm^{-3}, g=10ms^{-2}, height of the boy h=?, $P=\rho gh$,
$1.65\times10^4=1.1\times10^3\times10\times h$,
$h=\frac{1.65\times10^4}{1.1\times10^3\times10}=1.5$m.

8. A reservoir is filled with a liquid of density 2000kgm^{-3}. Calculate the depth at which the pressure in the liquid will be equal to 9100Nm^{-2} (g=10ms^{-2}) A.0.262m **B.0.455m** C.0.664m D.0.819m.
ρ=2000kgm^{-3}, P=9100Nm^{-2}, g=10ms^{-2}, h=?,
$P=\rho gh$, $h=\frac{P}{\rho g}=\frac{9100}{2000\times10}=0.455$m.

9. A weightless vessel of dimensions 4m×3m ×2m is filled with a liquid of density 1000kgm^{-3} and sealed. What is the maximum pressure thus container can exert on a flat surface? (g=10ms^{-2}) A.9×10^4Nm^{-2} **B.4×10^4Nm^{-2}** C.3×10^4Nm^{-2} D.2×10^4Nm^{-2}.
Maximum pressure is exerted when the vessel rest on its least area i.e 3m×2m, hence the height h=4m, g=10ms^{-2}, ρ=1000kgm^{-3}, P=?,
$P=\rho gh=1000\times10\times4=40000=4\times10^4 Nm^{-2}$.

10. A liquid of mass 1.0×10^3kg fills a rectangular tank of length 2.5m and width 2.0m. If the tank is 4m high what is the pressure at the middle of the tank? (g=10ms^{-2}) A.1.0×10^4Nm^{-2} B.2.0×10^3Nm^{-2} C.1.5×10^3Nm^{-2} **D.1×10^3Nm^{-2}.**
Mass of the liquid m=1.0×10^3kg, length of the tank l=2.5m, width or breadth of the tank b=2.0m, height of the tank h=4m, Volume of the liquid=Volume of the tank, $V=l\times b\times h$
V =2.5×2×4=20m^3, density of the liquid,
$\rho=\frac{m}{V}=\frac{1\times10^3}{20}=$ 50kgm^{-3}, height at the middle of the tank $h=\frac{1}{2}(4)$=2m, Pressure at the middle of the tank, $P=\rho gh=50\times10\times2=1\times10^3$Nm^{-2}.

11. Determine the pressure, due to water, at the bottom of a tank which 25m deep and is four-fifths full of water (Density of water= 1000kgm^{-3}, g=10ms^{-2}) A.5×10^2Nm^{-2} B.4×10^2Nm^{-2} C.2.5×10^4Nm^{-2} **D.2×10^5Nm^{-2}** E.2.5×10^5Nm^{-2}.
h=25m, depth of liquid in the tank= $\frac{4}{5}h$,
ρ=1000kgm^{-3}, g=10ms^{-2}, $P=\rho g\frac{4}{5}h$,
$P=1000\times10\times\frac{4}{5}\times25=2\times10^5$Nm^{-2}.

12. The pressure exerted by a liquid in a container is dependent on the A.mass of the liquid B.cross-sectional area of the container C.volume of the liquid **D.density of the liquid**
The pressure exerted by a liquid depends on the density of the liquid ρ, acceleration due to gravity g and depth of the liquid h.

13. Which of the following about liquid pressure is not correct? The pressure A.At a point is proportional to the depth B.At any point in a liquid is the same at the same level C.Is exerted equally in all directions at any point D.Of a liquid at any point on the wall of its container acts in a direction perpendicular to the wall **E.At a particular depth depends on the shape of the vessel.**
Pressure in liquids at a particular depth is independent of the shape of the vessel. All other options are true for liquid pressure.

14. Which of these statements are correct for the pressures in liquids? I.Pressure in a liquid at a point acts equally in all directions II.Pressure increases with depth III.Pressure at a depth depends on the shape of the container IV.Pressure at the same depth in different liquids are proportional to the densities of any liquids A.I,II and III **B.I,II and IV** C.I,III and IV D.II,III and IV.

Demystified Series by Dr Timothy

Pressure in liquids at a depth is independent of the shape of the container.

15. I.Density of the liquid II.Acceleration due to gravity III.Type of container of the liquid IV.The constituents of the liquid. Which of the above conditions will not affect the pressure of liquid? A.I and III B.II and III **C.III and IV** E.I and II.

Pressure in liquids is affected by the density of the liquid, acceleration due to gravity and depth below the surface of the liquid.

16. Air was pumped out of sealed can and the can collapsed. Which of these statements is the best explanation for this? A.Pumping out of the air weakened the can B.The material used in making the canw as very weak B.The remaining air inside the can condensed to form water **D.The air pressure inside the can became less than the pressure outside the can.**

This is the crushing can experiment. Pumping air out of the sealed can reduces the pressure inside it, hence the atmospheric pressure exceeds the pressure inside the can and causes it to collapse.

17. When cold water is poured on a can containing hot water, the can collapse because the A.steam condenses and occupies the partial vacuum in the can B.external air pressure counterbalances the pressure within the can C.steam expands to occupy the vacuum remaining in the can **D.external pressure becomes greater than the pressure within the can.**

18. The instrument used for measuring atmospheric pressure is A.hydrometer **B.barometer** C.manometer D.photometer E.ratemeter

Barometer is used to measure atmospheric pressure while manometer is used to measure gas pressure.

19. Which of the following is suitable to use as an altimeter? A.a mercury barometer B.a Fortin barometer C.a mercury manometer **D.an Aneroid barometer**

Aneroid barometer is used as an altimeter for measuring the altitude (or height) of bodies like mountains or air planes above the ground or sea level.

20. Which of the following devices are used to measure pressure? I.Aneroid barometer II.Hydrometer III.Hygrometer IV.Manometer A.I and III **B.I and IV** C.II and III D.III and IV.

21. Mercury is a suitable barometric fluid than water because I.It needs no colouring II.Its density is many times greater than that of water III.It does not wet glass IV.It has a high vapour pressure V.It has a high boiling point. Which of the above statements are true? A.I and II **B.I,II and III** C.III and IV D.IV and V E.III,IV and V

Mercury is a better barometric fluid than water due to its high density, does not wet glass, low vapour pressure and need no colouring.

22. Mercury is suitable as a barometric fluid because it A.expands uniformly **B.is opaque C.is several times denser than water** D.is a good conductor of heat

The high density of mercury is the major reason why it is used a barometric fluid.

23. Which of the following statements is correct? The reading of pressure on a mercury barometer is independent on **A.the cross-sectional area of the tube** B.the atmospheric pressure C.the density of mercury D.the temperature of mercury D.acceleration due to gravity.

Pressure in a mercury barometer depends on density of mercury, acceleration due to gravity and height of mercury. The density of mercury is affected by temperature hence, temperature also has an effect on the reading

24. A simple barometer reads 74.6cm when a standard Fortin barometer reads 76.0cm of mercury. This may be because the A.tube of the simple barometer is not vertical **B.simple barometer contains some air in the space above the mercury** C.cross-sectional area of the tube of the simple barometer is greater than that of the Fortin barometer D.cross-sectional rea of the tube of the simple barometer is smaller than that of the Fortin barometer E.atmospheric air exerts less pressure on the simple barometer than on the fortin barometer

The height of mercury colu.mn supported by atmospheric pressure is 76.0cm. A faulty simple barometer with air in the space above the mercury will record a height of mercury column less than 76.0cm i.e 74.6cm, due to the air bubbles which decreases the height of mercury column.

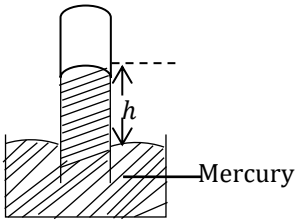
25. The ivory pointer of a Fortin barometer is used to A.adjust the level of mercury in the leather bag B.read the scale of the barometer **C.locate the zero mark of the scale** D.adjust the Vernier scale E.open and close the brass casing.

26. The height of the mercury column in a simple barometer is *h*. If the tube is slightly raised, which of the following statement is tru? A.*h* will change, but the space above the mercury will remain constant B.*h* will remain constant, but the space above the mercury is will change C.Both *h* and the space above the mercury will change **D.Both *h* and the space above the mercury will remain constant** E.None of the above is true.

27. Which of the following devices used to measure pressure contain(s) no liquid? I.Simple barometer II.Fortin barometer III.Aneroid barometer IV.Manometer. A.I B.IV C.II and III D.I and II **E.III**.

28. The diagram below is of a faultless simple barometer.

Demystified Series by Dr Timothy



Which of the following is not correct? A.A vacuum exists above the mercury in the tube B.The atmospheric pressure supports a column of mercury h C.The height of the column will not be the same if another loquid is used instead of mercury **D.If the tube is tilted, the vertical height of the mercury column will change** E.As the barometer is taken up a mountain, h decreases.

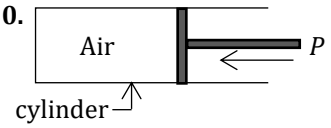
A fautless barometer has a vacuum above the mercury in th tube while a faulty barometer has air above the mercury in the tube.

The atmospheric pressure will support a column of mercury equal to h and the height h of the column will not change if the tube is tilted. The height of the column is different for different liquids due to their density difference i.e h is 76cm for mercury and 1033.6cm for water. Pressure decreases with increase in altitude(h), hence barometer reading indicated by h decreases.

29. Mechanical pumps cannot A.move fluids in a horizontal direction B.move fluids in a vertical direction C.compress gases in a container **D.increase the atmospheric pressure** E.decrease the pressure in an enclosure

Atmospheric pressure cannot be increased or decreased by a mechanical pump. Mechanical pumps can be used to move fluids in horizontal direction when flow rate is low. Mechanical pumps , lift and force pumps, can be used to move fluids in a vertical direction. Mechanical pumps e.g vacuum or evacuation pump, can be used to decrease the pressure in an enclosure by evacuating air. Mechanical pressure e.g compression pump(bicycle pump) can be used to increase the pressure in a vessel by compressing air or gases into a container.

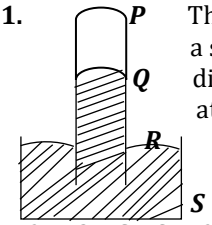
30.



In the diagram above, the piston P experiences a resistance to motion as it is pushed into the cylinder because **A.as the air is compressed its pressure increases** B.as the volume of the air is reduced its kinetic energy increases C.the increase in pressure increases the friction at the walls of the cylinder D.the decrease in the volume of the air increases its viscosity E.pressure is transmitted equally in all directions.

The diagram above is a compression pump. Compression of the air increases the pressure in the cylinder which offers resistance to the motion of the piston P as it is pushed into the cylinder.

31.



The diagram above illustrates a simple barometer. Which distance measures the atmospheric pressure?

A.PQ **B.QR** C.RS D.QS .

The height of mercury column QR in the tube of a simple barometer is the atmospheric pressure.

32. A faulty barometer reads 72.6cmHg when the atmospheric pressure is 75.0cmHg. Calculate the atmospheric pressure when this barometer reads 72.0cmHg A.79.4cmHg **B.74.4cmHg** C.74.2cmHg D.69.4cmHg.

First barometer reading h_1=72.6cm, Atmospheric pressure at first reading H_1=75.0cmHg, Second barometer reading h_2=72.0cmHg, Atmospheric pressure at second barometer reading H_2=?,

Atmospheric pressure H is a measure of the barometer reading or barometric height h, hence $H \propto h$, $\frac{H_1}{h_1} = \frac{H_2}{h_2}$. $\frac{75.0}{72.6} = \frac{H_2}{72.0}$,

$H_2 = \frac{75.0 \times 72.0}{72.6} = 74.38\text{cmHg} = 74.4\text{cmHg}$.

33. A simple barometer is constructed using a liquid Q of density 16.2gcm^{-3}. Calculate the least length of the tube to be used to measure pressure of 1.5 atmosphere [Density of mercury=13.6gcm^{-3}, 1 atmosphere=76.0cm of mercury] A.1358mm B.957mm C.760mm D.638mm

ρ_Q=16.2gcm^{-3}, ρ_{Hg}=13.6gcm^{-3}, P=1.5atm, 1atm=76.0cmHg, $\therefore P$ =1.5×76.0=114cmHg, height of mercury in barometer h_{Hg}=pressure in height of mercury=114cmHg, height of liquid Q inin barometer h_Q=?,

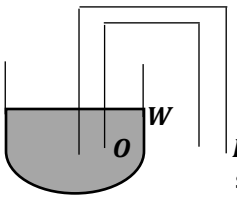
Pressure in barometer when mercury or liquid Q is used as the barometric liquid is the same.

$P_{Hg} = P_Q$, $\rho_{Hg} g h_{Hg} = \rho_Q g h_Q$,

$\rho_{Hg} h_{Hg} = \rho_Q h_Q$, 13.6×114=16.2× h_Q ,

$h_Q = \frac{13.6 \times 114}{16.2} = 95.7\text{cm} = 957\text{mm}$.

34.



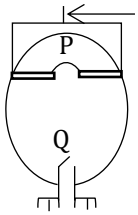
The figure above shows an inverted U-tube with the open end, O of one limb below the level W, of water in a tank. In order that water should begin to flow from the tank it is necessary that A.The U-tube is completely filled with water and P should be higher than W **B.P should be lower than O and W** C.P should be lower than W and O should reach to the bottom of the veseel D.The U-tube is completely filled with water and P should be

Demystified Series by Dr Timothy

lower than W E.The U-tube is completely filled with water and O should reach the bottom of the vessel.

The diagram is an example of a siphon. For water to flow out of P – the U-tube must be completely filled with water due to suction pressure, P should be lower than O and W as liquid flows under pressure ρgh, h is the difference between levels of W and P , O should be lower than W.

- A siphon cannot function in space as atmospheric pressure at the surface of the liquid is zero.

35. 

The diagram shows a lift pump with valves P and Q. During a downward stroke of the piston,

A.both valves are opened **B.P is opened while Q is closed** C.P is closed while Q is opened D.both valves are closed.

The two valves in lift and force pumps opens and clioses alternarively during the upward and downward stroke. In upward stroke, the upper valve P closes while the lower valve opens. In downward stroke, the upper valve P opens while the lower valve Q closes.

36. Normal atmospheric pressure at sea level is 10^5Nm^{-2} and the acceleration due to gravity is approximately 10ms^{-2}. If the atmosphere has a uniform density of 1kgm^{-3}. What is its height?

A.100m B.1000m **C.10000m** D.100000m E.1000000m

Atmospheric pressure P=10^5Nm^{-2}, g=10ms^{-2}, ρ=1kgm^{-3}, h=?, $P_{atm} = \rho gh$,

10^5=1×10× h, $h = \frac{10^5}{10} = 10^4$m=10000m.

37. Calculate the length of the liquid in a barometer tube that would support an atmospheric pressure of 204000Nm^{-2}, if the density if the liquid is 5200kgm^{-3} (g=10ms^{-2})

A.0.75m B.0.76m **C.3.92m** D.39.23m

Atmospheric pressure P=204000^{N}m^{-2}, g=10ms^{-2}, ρ=5200kgm^{-3}, h=?, $P_{atm} = \rho gh$,

$h = \frac{P}{\rho g} = \frac{204000}{5200\times10} = 3.92$m.

38. A diver is 5.2m below the surface of water of density 1000kgm^{-3}. If the atmospheric pressure is 1.02×10^5Pa, calculate the pressure on the diver (g=10ms^{-2})

A.1.02×10^5Pa **B.1.54×10^5Pa** C.5.20×10^5Pa D.6.20×10^5Pa

P_{atm}=1.02×10^5Pa, g=10ms^{-2}, ρ=1000kgm^{-3}, h=5.2m, Pressure at a depth 5.2m below the water surface, $P = P_{atm} + \rho gh$,

P =1.02×10^5+1000×10×5.2,

P =1.02×10^5+5.2×10^4=10^4(1.02×10+5.2),

P =10^4(10.2+5.2)=10^4(15.4)=1.52×10^5Pa or 1.52×10^5Nm^{-2}.

39. A circular membrane lies 24cm below the open surface of a pool of mercury of relative density 13.6.If the barometric height is 76cm of mercury,the pressure on the membrane is

A.3.26×10^4Nm^{-2} B.1.03×10^4Nm^{-2} **C.1.36×10^5Nm^{-2}** D.1.36×10^7Nm^{-2}

Barometric height or height supported by atmospheric pressure H =76cm=0.76m

depth below water surface h=24cm=0.24m

$\rho = R.D$ ×1000=13.6×1000=1.36×10^4kgm^{-3}

Pressure on the membrane = Atmospheric pressure +$\rho gh = P_{atm} + \rho gh$=$\rho gH + \rho gh$,

$P = \rho g(H + h)$ =1.36×10^4×10(0.76+0.24) ,

P =1.36×10^5×1 = 1.36×10^5Nm^{-2}.

40. The barometric reading at a place is 73.5cmHg. Calculate the pressure at a point 30m below the surface of water contained in a reservoir at the place (g=10ms^{-2}, density of mercury=1.3×10^4kgm^{-3}, density of water= 1.0×10^3kgm^{-3}) **A.4.0×10^5Nm^{-2}** B.3.0×10^5Nm^{-2} C.2.0×10^5Nm^{-2} D.1.0×10^5Nm^{-2}

Barometric reading at the place=atmospheric pressure at that place=73.5cmHg=0.73mHg, height of mercury at the place corresponding to atmospheric pressure h_{Hg}=0.73m, density of mercury ρ_{Hg}=1.3×10^4kgm^{-3}, g=10ms^{-2}, depth of waterh_{H_2O}=30m, density of water ρ_{H_2O}=1.0×10^3kgm^{-3} ,

Atmospheric pressure in Nm^{-2}= $\rho_{Hg} g h_{Hg}$,

Pressure due to depth of water = $\rho_{H_2O} g h_{H_2O}$,

∴ Pressure at the point 30m below the surface of water $P = P_{atm} + \rho_{H_2O} g h_{H_2O}$,

$P = \rho_{Hg} g h_{Hg} + \rho_{H_2O} g h_{H_2O}$,

P =1.3×10^4×10×0.735+1.0×10^3×10×30,

P = 0.9555×10^5+3.0×10^5=(0.955+3.0)10^5,

P =3.9555×10^5 = 4.0×10^5Nm^{-2}.

41. A rectangular tank contains water to the depth of 2m. If the base is 4m×3m, calculate the force on the base (density of water= 10^3kgm^{-3}, g=10ms^{-2}) **A.2.4×10^5N** B.2.4×10^4N C.2.0×10^4N D.1.7×10^3N.

h=2m, base dimension 4m×3m, base area A=4×3=12m^2, ρ=10^3kgm^{-3}, g=10ms^{-2}, F=?,

$P = \frac{F}{A} = \rho gh$, $F = A(\rho gh)$,

F =12(10^3×10×2) = 24×10^4 = 2.4×10^5N.

42. The horizontal door of a submarine at a depth 500m has an area of 0.4m^2.Calculate the force exerted by the sea water on the door at this depth. (Relative density of sea water=1.03, density of pure water=1000kgm^{-3} Atmospheric pressure=1.0×$10^5$$N$m^{-2}, g=10ms^{-2})

A.6.3×$10^6$$N$ B.3.2×$10^6$$N$ **C.2.1×$10^6$$N$** D.1.2×$10^6$$N$

h_{sea}=500m, A=0..4m^2, $R.D$ of sea=1.03, ρ_{sea}=1.03×10^3kgm^{-3}, P_{atm}=1.0×$10^5$$N$m^{-2},

∴ Pressure at a depth of 500m in sea water ,

$P = P_{atm} + P_{sea} = P_{atm} + \rho_{sea} g h_{sea}$,

$P = \frac{F}{A} = P_{atm} + \rho_{sea} g h_{sea}$,

$F = A(P_{atm} + \rho_{sea} g h_{sea})$,

Demystified Series by Dr Timothy

$F = 0.4(1.01\times10^5+1.03\times10^3\times10\times500)$,
$F = 0.4(1.01\times10^5+51.50\times10^5)$,
$F = 0.4\times52.51\times10^5 = 21.004\times10^5 = 2.1\times10^6$N.

43. The hatch door of a submarine has an area of 0.5m^2. The specific gravity of sea water is 1.03, assume that g=10ms^{-2} and neglect the atmospheric pressure, the force exerted by the sea water on the hatch door at depth of 200m is **A.1.0×10^{6}Nm^{-2}** B.1.03×10^{4}Nm^{-2} C.2.06×10^{5}Nm^{-2} D.2.06×10^{6}Nm^{-2} E.1.03×10^{3}Nm^{-2}.

A=0.5m^2, specific gravity or relative density of sea $R.D$=1.03, ρ_{sea}=1.03×10^{3}kgm^{-3}, h=200m, g=10ms^{-2}, Neglecting atmospheric pressure P_{atm}, $P = P_{sea} = \rho_{sea}gh_{sea}$, $P = \frac{F}{A} = \rho_{sea}gh_{sea}$,
$F = A(\rho_{sea}gh_{sea}) = 0.5(1.03\times10^3\times10\times200)$
$F = 0.5(206\times10^4) = 103\times10^4 = 1.03\times10^6$Nm^{-2}.

44. Two divers G and H are at depths 20m and 40m repectively belowe the water surface in a lake. The pressure on G is P_1 while the pressure on H is P_2. If the atmospheric pressure is equivalent to 10m of water, the value of P_2/P_1 is A.0.50 B.0.60 **C.1.67** D.2.00 E.3.00.

Depth of diver G h_1=20m, depth of diver H h_2=40m, Atmospheric pressure H=10m of water. $P = H + h$, P_1=10+20=30m, P_2=10+40=50m, $P_2/P_1 = 50/30 = 1.67$.

45. At a fixed point below liquid surface, the pressure downward is P_1 and the pressure upward is P_2. It can be deduced that **A.$P_1 = P_2$** B.$P_1 > P_2$ C.$P_1 < P_2$ D.$P_2 > P_1$.

Pressure at all points at the same level or a fixed points within a liquid is the same. Hence, pressure downward P_1 is equal to pressure upward P_2 as both are at a fixed point or same level(or depth) below the liquid surface.

46. A pilot records the atmospheric pressure outside the plane as 63cm of Hg while a ground observer records a reading of 75cmHg for the atmospheric pressure on the ground. Assuming that the density of the atmosphere is constant, calculate the height of the plane above the ground (Relative density of Hg=13.0 and of the air=0.00013) **A.12000m** B.6300m C.7500m D.13800m

Pressure change for mercury is equal to pressure change for air. $P_{Hg} = P_{air}$,
$R.D_{Hg}$=13.0, ρ_{Hg}=13.0gcm^{-3}, atmospheric pressure at the ground or sea level=75cmHg, pressure above sea level= 63cmHg, height above the ground H=?, $R.D_{air}$=0.00013, ρ_{air}=0.00013gcm^{-3}, $\rho_{Hg}gh_{Hg} = \rho_{air}gH$,
$\rho_{Hg}h_{Hg} = \rho_{air}H$, $13.0\times(75-63) = 0.00013\times H$,
$13.0\times12 = 0.00013\times H$, $H = \frac{13.0\times12}{0.00013}$,
$H = 1200000$cm=12000m. Alternatively :
$H_P = \frac{\rho_{Hg}}{\rho_{air}}(H_o - H_1)$, height of plane above the ground H_P=?, ρ_{Hg}=13.0gcm^{-3}, ρ_{air}=0.00013gcm^{-3}, barometric height at sea level (larger height) or pressure at sea level H_o=75cmHg , barometric height above sea level at H_P (smaller height) or pressure above sea level H_1=65cmHg,
$H_P = \frac{\rho_{Hg}}{\rho_{air}}(H_o - H_1) = \frac{13.0}{0.00013}(75-63)$,
$H_P = 100000(12) = 1200000$cm = 12000m.

47. The height at which the atmosphere ceases to exist is about 80km. If the atmospheric pressure on the ground level is 760mmHg, the pressure at a height of 20km above the ground level is A.380mmHg **B.570mmHg** C.190mmHg D.480mmHg (ρ_m=13.68gcm^{-3}, ρ_{air}=0.00013 gcm^{-3}).

Pressure change for mmercury is equal to pressure change for air. $P_{Hg} = P_{air}$,
ρ_{Hg}=13.68gcm^{-3}, atmospheric pressure at ground level=760mmHg=76cmHg, pressure above the ground level H_1=?, height above the ground H=20km=2×10^{6}cm, ρ_{Hg}=13.68gcm^{-3}, ρ_{air}=0.00013gcm^{-3},
$\rho_{Hg}gh_{Hg} = \rho_{air}gH$, $\rho_{Hg}h_{Hg} = \rho_{air}H$,
$13.68(76-H_1)$=0.00013×2×10^6
$76-H_1 = \frac{0.00013\times2\times10^6}{13.68} = 19.0$, $H_1 = 76-19$,
$H_1 = 57$cmHg = 570mmHg. Alternatively,
$H_P = \frac{\rho_{Hg}}{\rho_{air}}(H_o - H_1)$, height of plane above the ground H_P=12km=2×10^{6}cm, ρ_{Hg}=13.68gcm^{-3}, ρ_{air}=0.00013gcm^{-3}, barometric height at sea level (larger height) or pressure at sea level H_o=75cmHg , barometric height above sea level at H_P (smaller height) or pressure above sea level H_1=?, $2\times10^6 = \frac{13.68}{0.00013}(75-H_1)$,
$(76-H_1) = \frac{0.00013\times2\times10^6}{13.68} = 19$,
$H_1 = 76-19 = 57$cmHg = 570mmHg.

48. In the Hare's apparatus, water rises to a height of 26.5cm in one limb. If a liquid rises to a height of 20.4cm in the other limb, what is the relative density of the liquid? A.0.8 B.1.1 C.1.2 **D.1.3.**

height of water column h_w=26.5cm, height of liquid column h_L=20.4cm, $R.D_L = \frac{\rho_L}{\rho_w} = \frac{h_w}{h_L}$,
$R.D_L = \frac{h_w}{h_L} = \frac{26.5}{20.4} = 1.3$.

49. Water and kerosene are drawn respectively into two limbs of a Hare's apparatus. The density of water is 1.0gcm^{-3} and the density of kerosene is 0.80gcm^{-3}. If the height of the water column is 20.0cm, calculate the height of the kerosene column A.1.6cm B.16.0cm C.20.0cm D.20.8cm **E.25.0cm.**

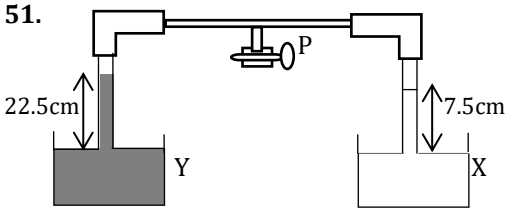
ρ_w=1.0gcm^{-3}, ρ_k=0.80gcm^{-3}, h_w=20.0cm, h_k=?
$R.D_k = \frac{\rho_k}{\rho_w} = \frac{h_w}{h_k}$, $\frac{0.80}{1.0} = \frac{20.0}{h_k}$, $h_k = \frac{20.0}{0.80}$,
$h_k = 25.0$cm.

50. Atmospheric pressure supports height y and $2y$ of water and liquid Q respectively. The relative density liquid Q is A.4.0 B.2.0 C.1.0 **D.0.5.** Height of water atmospheric pressure supports

Demystified Series by Dr Timothy

h_w=y, height of liquid Q atmospheric pressure supports h_Q=2y, $R.D_Q = \frac{h_w}{h_Q} = \frac{y}{2y} = 0.5$.

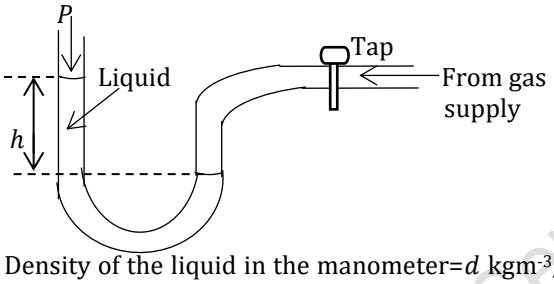
51.



The figure above shows the height of two liquids X and Y when some air is sucked out of the apparatus through the pump P. The diameter of the tube in X is twice that of tube in Y. What is the relative density of liquid X with respect to liquid Y A.1/3 B.2/3 **C.3** D.6.

The pressure is independent of the diameter or area of the tube. The pressure at the surface of both liquids are equal as they both exposed to the atmosphere. $P_Y = P_X$, $\rho_Y g h_Y = \rho_X g h_X$, $\rho_Y h_Y = \rho_X h_X$, $R.D_X = \frac{\rho_X}{\rho_Y} = \frac{h_Y}{h_X} = \frac{22.5}{7.5} = 3$.

52. What is the pressure of the gas in the diagram below?

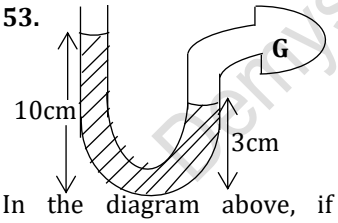


Density of the liquid in the manometer=d kgm^{-3}, atmospheric pressure=P Nm^{-2}, acceleration due to gravity=g ms^{-2}. A.$(P + h)$Nm^{-2} B.$(P - h)$Nm^{-2} **C.$(P + hdg)$Nm^{-2}** D.$(P + hd)$Nm^{-2} E.hdgNm^{-2}.

The total gas pressure = Atmospheric pressure+pressure due to liquid column

Total gass pressure= $(P + hdg)$Nm^{-2}.

53.



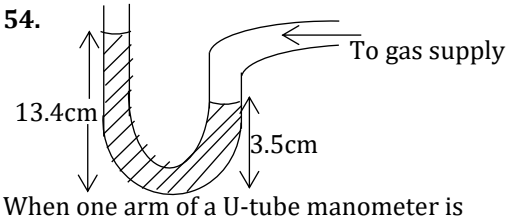
In the diagram above, if the atmospheric pressure is 760mm, the pressure in the chamber G is A.660mm B.690mm **C.830mm** D.860mm.

h = difference in the level of liquids in both arms=10cm−3cm=7cm=70mm, Atmospheric pressure H=760mm,

Gas pressure in chamber G, $P = H + h$,

P =760+70=830mm.

54.

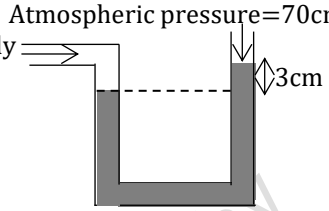


When one arm of a U-tube manometer is connected to a gas supply, the levels of mercury in the two arms of the U-tube are as shown in the diagram above. If the atmospheric pressure is 76.0cmHg, what is the gas pressure?

A.62.6cmHg B.72.5cmHg C.79.5cmHg **D.85.9cmHg.**

h = difference in the level of liquids in both arms=13.4cm−3.5cm=9.9cm, Atmospheric pressure H=76.0cm, Gas pressure in chamber G, $P = H + h$ =76.0+9.9=85.9cmHg.

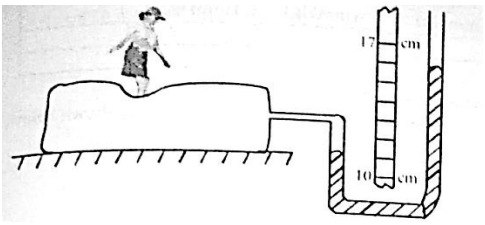
55. Atmospheric pressure=70cm mercury



The water manometer in the figure above is measuring the pressure of the gas supply. If the specific gravity of mercury is 13.6 and the atmospheric prerssure is 70cm of mercury, what is the total pressure of the gas supply in cm of water? A.76cm B.73cm C.949cm D.952cm **E.955cm.**

Pressure of the gas supply is the pressure at the liquid level in the left arm of the manometer. Pressure at the left arm and the right arm of the manometer are equal at the point where the liquid level in both are the same. Atmospheric pressure=70cmHg, the equivalent of atmospheric pressure in cmH_2O is required as stated in the question. $\rho_{Hg} g h_{Hg} = \rho_{H_2O} g h_{H_2O}$, $\rho_{Hg} h_{Hg} = \rho_{H_2O} h_{H_2O}$,ρ_{Hg}=13.6gcm^{-3}, h_{Hg}=70cm, ρ_{H_2O}=1gcm^{-3}, h_{H_2O}=?, 13.6×70=1× h_{H_2O} , h_{H_2O} =952cm, Atmospheric pressure H =952cmH_2O, h=3cm, Total gas pressure $P = H + h$ =952+3=955cmH_2O.

56.



A girl stands on an air bag connected to a manometer as illustrated in the diagram above. Determine the pressure exerted on the bag by the girl [Atmospheric pressure = 75cmHg] A.10cmHg B.17cmHg C.70cmHg **D.80cmHg.**

Pressure exerted by the girl is same as the total gas pressure P=?, atmospheric pressure H=75cmHg, difference in the level of mercury of the arm exposed to the atosphere and that connected to the air bag h =16−11=5cmHg

$P = H + h$ =75+5 = 80cmHg.

57. When pressure is applied by means of a piston to the surface of a liquid contained in a cylinder, the pressure transmitted to the liquid is A.greater at the surface thanm the bottom of the

Demystified Series by Dr. Timothy

liquid B.all at the bottom of the liquid **C.the same at every point in the liquid** D.only at the surface of the liquid E.smaller at the surface than at the bottom of the liquid.
Pressure applied at a point in a liquid is transmitted equally in all directions in the liquid, (Pascal's principle).
58. The areas of cross-section of two pistons in a hydraulic press are $2cm^2$ and $150cm^2$. If a downward force of $20N$ is exerted on the smaller piston, calculate the force om the larger one A.112500N **B.1500N** C.150N D.15N E.3.75N.
F_1=$20N$, F_2=?, A_1=$2cm^2$, A_2=$150cm^2$,
$\frac{F_1}{A_1} = \frac{F_2}{A_2}$, $\frac{20}{2} = \frac{F_2}{150}$, $F_2 = \frac{150\times20}{2} = 1500$N.

● Gas laws and kinetic theory of matter

- Kinetic theory of gases :
1.The temperature of a gas is a measure of the average kinetic energy of the gas molecules.
2.The volume of gas molecules is negligible compared to the volume of the gas container.
3.The attractive force between gas molecules are negligible as they are far apart relative to their sizes (attraction between gas molecules is negligible).
4.The molecules of a gas move in straight lines colliding with one another and with the walls of the container.
5.The collisions of gas molecules are perfectly elastic i.e kinetic energy and momentum of the system are conserved.
6.The duration between the gas molecules is negligible rather than the time between collision.
- The collision of gas molecules with the walls of the container constitute the pressure.
- Expansion of gases – Gases expands more than liquids and solids for every rise in temperature. Gas expansion is accompanied by a change in temperature,pressure,volume and density.
- Collision of gas molecules :
1.The molecules of gases are in constant motion but the number of collisions is not constant i.e Number of collision increases as the volume decreases.
2.Number of collision increases with increase in temperature.
3.Number of molecules of gas does not increase with increase in pressure.
4.Number of collision per second i.e rate of collision, increases with pressure.
- Mean free path is the mean or average distance which a molecule travels between successive collisions.
- Kinetic theory of matter :
1.Matter is made up of tiny paryicles called atoms and molecules.
- Atoms are not capable of free existence while molecules are capable of free existence
2.Energy is required to overcome and break the attractive forces between molecules of matter.
- Evidence of molecular or kinetic theory of matter – Diffusion,change of state(melting, boiling,evaporation),osmosis,heat transfer (conduction and convection), Brownian motion,expansion.
- Evidence of intermolecular forces – Friction,surface tension,cohesion and adhesion,capillarity, elasticity and viscosity.
- Radiation cannot be explained by molecular or kinetic theory of matter as it involves electromagnetic radiations in space and not particles of matter.
- Temperature is a physical property and is a measure of the average kinetic energy.
- Heat or Thermal energy is a form of energy and is a measure of the total internal energy.
- The density of a gas depends on the separation between its molecules i.e intermolecular distance.
- Matter exists in three states namely solid, liquid and gas. The three states of matter differ in their degree of movement,shape, compressibility,intermolecular bonds and energy of molecules.
- The major difference in solids, liquids and gases is accounted for by the spacing of their molecules and intermolecular forces acting between the molecules.
Solids – I. Molecules of solids are in fixed positions and closely packed i.e they have definite shapes. **II.**Solid molecules moves vibrates or oscillates about a their mean position and has the least kinetic energy. **III.**Molecules of a solid are bounded by the strongest intermolecular forces. **IV.**Solids are not compressile.
Liquids – I. Molecules of liquids are free to move within the liquid i.e assume the shape of its container and loosely packed. **II.**Liquid molecules move randomly or vibrate about different mean positions within the liquid and has more kinetic energy than solids. **III.**Molecules of liquids are bounded by stronger intermolecular forces but less than that of solids. **IV.**Liquids are not compressible.
Gases – I. Molecules of gases are farther apart due to the large intermolecular distance between its molecules. **II.**Gas molecules move randomly or disorderly colliding with one another and with the walls of the container and has the highest kinetic energy. **III.**The intermolecular forces between gas molecules are relatively small or negligible. **IV.**Gases are compressible.
- Diffusion – Diffusion is movement of molecules(solids,liquids or gases), from a more crowded region to a less crowded region, till

Demystified Series by Dr Timothy

equilibrium is attained. Molecules diffuse from a refion of higher concentration to a region of lower concentration.

- The difference in the concentration of both mediums is called **diffusion of concentration gradient.** Rate of diffusion increases as the diffusion or concentration gradient increases.

- Factors affecting the rate of diffusion R :

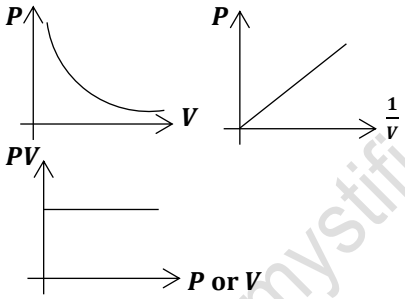
1.size,mass or density, $R \propto \frac{1}{m}$ or $R \propto \frac{1}{\rho}$, rate of diffusion increases as the size, mass or density of decreases. **2.**Temperature, $R \propto T$, the rate of diffusion increases as the temperature increases. **3.**Hydrostatic or osmotic pressure , pressure that increases the rate of diffusion or balances osmosis.

- The rate of diffusion of solids, liquids and gases increases in the order – solids<liquids< gases. Hence, solids has the least rate of diffusion while gases has the highest rate of diffusion.

- Brownian motion is a random motion motion of molecules(solid,liquid or gas). **It gives evidence of tiny particles of matter and they are in constant state of random motion.**

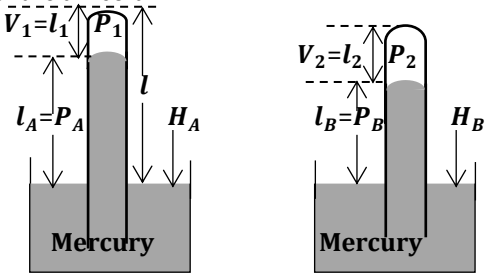
- Boyle's law gives the relation between pressure and volume of a fixed mass at constant temperature. It states that the pressure of a fixed mass of a gas, is inversely proportional to its volume provide the temperature is kept constant. $P \propto \frac{1}{V}$, PV =**constant,** $P_1V_1 = P_2V_2$.

- Graphs of Boyle's law :



- Applications of Boyle's law :

1.Simple barometer – The pressure in a simple baromter with air trapped above can be varied with the volume of air, represented as the length of the air column.



First day (A) Second day (B)

l = length of the tube above mercury in the container(mercury reservoir), H_A and H_B are the atmospheric reading of the 1st and 2nd days respectively, P_A(or l_A) and P_A(or l_A) are the barometric readings of the faulty barometer for the 1st and 2nd days respectively. $P_1 = H_A - P_A$, $P_2 = H_B - P_B$, $V_1 = l_1 = l - l_A$, $V_2 = l_2 = l - l_B$.

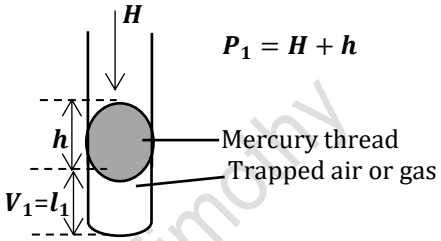
Applying Boyle's law – $P_1V_1 = P_2V_2$,

P_1 and P_2 are the pressures of the trapped air for the first and second days respectively.

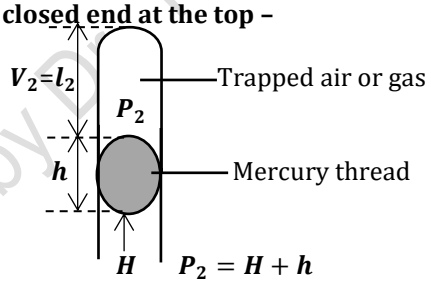
V_1 and V_2 are the volume of trapped air for the first and second day respectively.

2. Capillary tubes(tubes with one end closed) – The pressure exerted depends on the plane of inclination of the tube.

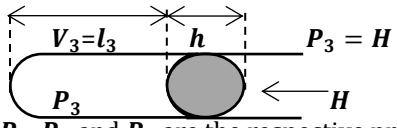
a.Vertically, with open end facing upward or closed end at the bottom –



b.Vertically, with open end facing downward or closed end at the top –



c. Horizontally lying flat –



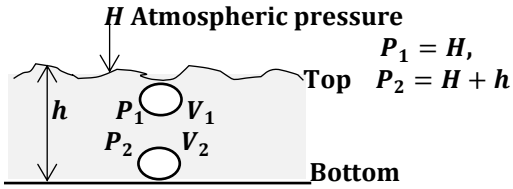
P_1, P_2 and P_3 are the respective pressures of the trapped air. V_1, V_2 and V_3 are the respective volumes of the trapped air and is equal to their respective length of air trapped by the capillary tube i.e l_1, $l_2$and l_3 , as the tube is of uniform cross-sectional area.

H =atmospheric pressure, h=height of mercury thread in the capillary tube.

Applying Boyle's law – $P_1V_1 = P_2V_2 = P_3V_3$.

NB – The length of entrapped air decreases as the pressure increases.

3.Air bubbles in water bodies – The air bubbles just below the surface of water is under the influence of atmospheric pressure only while air bubbles at the bottom is under the influence of atmospheric pressure and pressure due to depth of water.

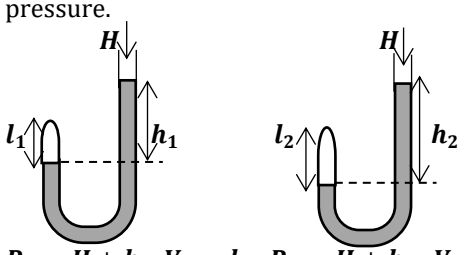


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P_1=pressure at the top=atmospheric pressure H, P_2=pressure at the bottom=atmospheric pressure + pressure due to depth of water = $H + h$, V_1=volume of air bubbles at the top, V_2=volume of air bubble at the bottom.

Applying Boyle's law – $P_1V_1 = P_2V_2$.

4.Manometer – The volume of air trapped at one-arm of a manometer can be varied with pressure.



$P_1 = H + h_1, V_1 = l_1, P_2 = H + h_2, V_2 = l_2$.

- In a manometer, when the liquid in both arms of the tube are at the same level, the pressure is atmospheric pressure only.

5. Evacuation pump – Boyle's law applies to an evacuation pump. $P_1{V_1}^n = P_2{V_2}^n$, $P_1{V_1}^n = P_2(V_1 + V_P)^n$. P_1=Initial pressure of air in the vessel, V_1=volume of vessel to be exhausted,P_2=final pressure of air on the vessel,V_2=volume of the vessel+volume of evacuation pump=$V_1 + V_P$,n=number of strokes of the pump.

- The slope of a Boyle's law graph i.e a graph of pressure plotted agains the reciprocal of volume or the product of pressure and volume is equal to energy or workdone by an expanding gas at constant pressure.

$W = P\Delta V = P(V_2 - V_1)$.

- Pressure increases with depth from the top to the bottom of a water body,hence the volume decreases simultaneously.

● **Charles' law** gives the relation between volume and temperature of a fixed mass of gas,at constant pressure. It states that the volume of a fixed mass of gas is directly proportional to its absolute temperature at constant pressure. Charles' law can also be stated as, the volume of a fixed mass of gas increases by $\frac{1}{273}$ of its volume at 0℃ per degree celcius rise in temperature, provided its pressure remains constant.

$V \propto T$, $\frac{V}{T} =$ **constant,** $\frac{V_1}{T_1} = \frac{V_2}{T_2}$.

- Volume or cubic expansivity or coefficient of cubic expansion γ of a gas at constant pressure is the fractional increase in the volume a gas from 0℃ per kelvin or degree celcius rise in temperature.

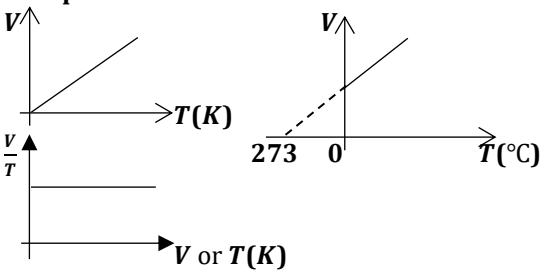
$\gamma = \frac{increase\ in\ volume\ from\ 0℃}{volume\ at\ 0℃ \times temp.\ rise} = \frac{V_\theta - V_o}{V_o\theta}$.

- Cubic expansivity of a gas in a capillary tube can also be given in terms of the length of air column.

$\gamma = \frac{l_\theta - l_o}{l_o\theta}$.

V_θ or l_θ=volume of gas or length of gas column at θ, $V_o$or l_o=volume of gas or length of ga column at 0℃, θ=temperature rise, γ=cubic expansivity in $℃^{-1}$ or K^{-1}.

- Graphs on Charles' law –



● **Pressure law** gives the relation between pressure and temperature of a fixed mass of gas, at constant volume. It states that the pressure of a fixed mass of gas is directly proportional to its absolute temperature provided its volume remains constant.

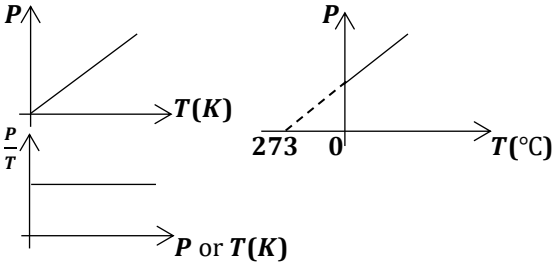
$P \propto T$, $\frac{P}{T} =$ **constant,** $\frac{P_1}{T_1} = \frac{P_2}{T_2}$.

- Pressure expansivity or coefficient of pressure expansion β of a gas at constant volume is the fractional increase in the pressure of a gas from 0℃ per kelvin or degree celcius rise in temperature.

$\beta = \frac{increase\ in\ pressure\ from\ 0℃}{pressure\ at\ 0℃ \times temp.\ rise} = \frac{P_\theta - P_o}{P_o\theta}$.

P_θ=pressure of gas at θ, P_o=pressure of gas at 0℃, θ=temperature rise, β=pressure expansivity in $℃^{-1}$ or K^{-1}.

- Graphs on pressure law –



- An inexpnasible vessel has a constant volume.

- When solving questions in gas laws, always convert from ℃ to K before solving.

$T(K) = t℃ + 273$.

- General gas law is a combination of Boyle's law, Charles' law and pressure law. General gas equation is given as: $\frac{PV}{T} =$ **constant,** $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$.

- The ideal gas equation is given as – $PV = nRT$, P=pressure, V=volume, n=number or amount of moles, T=temperature, R=molar gas constant= $8.313 J mol^{-1}K^{-1}$.

- In summary, the equation of gas laws is represented by – $P^xV^yT^z$ =constant K.

1.In Boyle's law $[PV = K]$, $x = 1, y = 1, z = 0$.

2.In Charles' law $[\frac{V}{T} = K]$, $x = 0, y = 1, z = -1$.

3.In Pressure law $[\frac{P}{T} = K]$, $x = 1, y = 0, z = -1$.

4.General gas law $[\frac{PV}{T} = K]$, $x = 1, y = 1, z = -1$.

Demystified Series by Dr. Timothy

- When an ideal gas expands at constant temperature, the volume, average separation between molecules increases while pressure, average number of collisions per unittime, average number per unit volume and momentum decreases. Average kinetic energy and average speed is constant as temperature is constant.
- The pressure of gas decreases when cooled (decrease in temperature) at constant volume, because thegas molecules collide less frequently with the walls of the container, as they are less energetic.
- At constant pressure, density of a fixed mass of gas is inversely proportional to its temperature, as volume varies inversely with density at constant mass.

- Standard temperature and pressure(S.T.P) – Standard temperature s.t is **0**℃ **or 273*K*** while standard pressure is **1atmosphere(1atm)** or **760mmHg(76cmHg)** or **1.013×10^5Pa or Nm^{-2}**.

Examples :

1. The process whereby molecules of different substances move randomly is called A.surface tension **B.diffusion** C.capillarity D.osmosis.

2. A few grains of table salt were put in a cup of cold water, kept at constant temperature and left undisturbed.Eventually all the water tasted salty. This action is due to A.convention B.osmosis C.capillarity **D.diffusion** E.conductivity.

Salt molecules diffuse from an area of higher concentration to that of lower concentration, until it is evenly distributed in the solution.

3. I.Mass II.Density III.Temperature IV.Nature of substance. Which of the above affect diffusion? A.I and II B.I,II and IV C.II,III and IV **D.I,II,III and IV.**

4. When the surface of a piece of chalk is scraped, the tiny particle that flakes off are known as A.atoms **B.molecules** C.elements D.matter.

Matter is made up of tiny particles called atoms and molecules. Molecules can exist alone e.g tiny particles of the chalk that flakes off, but atoms cannot exist alone.

5. Which of the following phenomena cannot be explained by the molecular theory of matter? A.Expansion B.Conduction C.Convention D.Evaporation **E.Radiation.**

6. I.Change of state II.Diffusion III.Radiation IV.Osmosis. Which of the processes above can be explained using the kinetic theory? A.I,III and IV **B.I,II and IV** C.I,II,III and IV D.I,II,III.

7. Which of the following statements is not correct about matter? A.molecules of solid move to and fro about a fixed point B.energy is required to break the forces of attraction between molecules of matter **C.molecules of solid move more freely than molecules of liquids** D.molecules of liquids are more closely packed than that of gases E.molecues of gases move more randomly in their containers.

Molecules of liquids move more freely than molecules of solids, as liquid molecules are far apart that those pof solids.

8. Which of the following statements is correct about the movement of molecules in a liquid? I.all the molecules have the sae kinetic energy II.the number of molecules leaving a liquid surface is always the same as the number entering the liquid III.near the surface molecules with kinetic energy above average break away from the liquid and remain as molecules of vapour. A.I B.II **C.III** D.I and II E.II and III.

Molecules of a liquid have different kinetic energy i.e some are more energetic that others. Molecules of liquid near the surface with kinetic energy above the average, evaporates i.e break away from the liquid and remain as molecules of vapour. Statement II is only valid when the liquid is in dynamic equilibrium with its own vapour i.e saturated vapour.

9. The differences observed in solids, liquids and gases may be accounted for by A.their relative sizes **B.the spacing and force acting between the molecules** C.their melting point D.the different molecules in each of them.

10. Which of the following is/are significantly different in solids, liquids and gases?

I.the arrangement of the molecules in the entire bulk of the material III.the motion of the molecules III.the distances between the molecules A.I B.II C.I and II D.I and III **E.I,II and III.**

Solids are more tightly or close packed than liquids and liquids are more closely packed than gases i.e difference in the arrangement of molecules.

Solids oscillate or vibrate about a fixed position. Liquids moves randomly or vibrate at different mean positions i.e translational motion, within the container. Gases move randomly colliding with one another and with the walls of its container i.e difference in motion of the molecules.

The intermolecule distance between molecule increases from solids-liquids-gases i.e difference in ther distances between the molecules.

11. Which of the following statements about the molecular theory of matter is not correct? A.gases exerts negligible force on one another B.solids are closer together than those of liquids C.gases are in random motion **D.liquids undergo only vibratory motion** E.solids vibrates about their mean position.

Liquids vibrates and oscillates about different mean points within the liquid.

12. Which of the following statements about kinetic theory of matter is not correct? **A.when a**

Demystified Series by Dr Timothy

body is heated, the average kinetic energy of its molecules decreases B.the molecules of gas move more freely than tose of a liquid C.molecules of a liquid move more freely than those of a solid D.the molecules of matter are always in motion E.matter is made up of very tiny particles called molecules.

When a body is heated, the temperature increases, hence the average kinetic energy of its molecules also increases.

13. In which of the following media would the average speed of the molecules of water be highest? A.ice B.water **C.steam** D.ice-water mixture E.water-steam mixture.

The average speed of molecules increases from solids-liquids-gases, due to decrease in the intermolecular forces binding the molecules. Hence, the molecues of steam (gaseous water) has the highestaverage speed.

14. I.It is a rapid, constant and irregular motion of tiny particles II.It gives evidence than any tiny particles of matter called molecules exists III.It takes place only in gases IV.It gives evidence that molecules are in a constant state of random motion. Which of the combinations above is correct about Brownian motion? A.I,II and III **B.I,II and IV** C.I,III and IV D.II,III and IV.

Brownian motion takes place in solids, liquids and gases.

15. From the kinetic theory of gases, temperature is a A.Form of energy and is proportional to the total kinetic energy of the molecules B.Form of energy and is proportional to the average kinetic energy of the molecules C.Physical property and is proportional to the total kinetic energy of the molecules **D.Physical property and is proportional to the average kinetic energy of the molecules.**

Heat is a form of energy and is proportional to the total kinetic energy of the molecules.

16. Which of the following is not correct about the assumptions of the kinetic theory of gases? A.the attraction between the molecules is negligible B.the volume of molecules is negligible compared with the volume occupied by the gas **C.the molecules of the gas behave like perfectly elastic spheres** D.the duration of a collision is negligible compared with the time between collisions.

It is the collision of gas molecules that is perfectly elastic.

17. One valid assumption of the kinetic theory of gases is that A.the molecules of a gas are constantly in a state of motion and the number of collisions remain constant B.as the temperature increases, the number of collisions made by the gas remains constant **C.the molecules of a gas are all identical and very small in size** D.the number of molecules of a gas increases with increasing pressure.

Molecules of a gas are in a state of constant random motion but the number of collisions is not constant. Increase in pressure and decrease in volume increases the number of collisions. Number of collisions per seconds increases with pressure, not number of molecules. Number of collisions also increases with temperature.

18. According to the kinetic theory of gases, the collision of the molecules of a gas with the walls of the container is mainly responsible for the I.temperature of the gas II.viscosity of the gas III.pressure of the gas A.I B.II **C.III** D.II and III E.I,II and III.

19. Which of the following statements about the gaseous state of matter is not correct? The molecules of a substance in gaseous state A.are in a constant state of motion B.have different speeds C.have a temperature which is a measure of the average kinetic energy **D.have a pressure which is determined by the number if molecules present** E.have no intermolecular attractive force.

The pressure in gases is determined by the collisions of the gas molecules with the walls of the container and not the number of molecules present.

20. Which of the following quantities will increase if a tight-fitting piston is pushed into a cylinder closed at one end? I.the number of collisions per second between the air molecules II.the average distance distance between the air molecules III.the number of collisions per second of the air molecules against the walls of the cylinder IV.the volume of the air A.I and II **B.I and III** C.II and III D.I,II and III E.II,III and IV.

Pushing the piston into the cylinder compresses the air in the cylinder.Hence the pressure increases while the volume decreases. The number of collisions per second between the molecules and the number of collisions per second between the molecules and the walls of the container increases as pressure increases. The average distance between the molecules decreases as the volume decreases.

21. What hapens when a gas expands at a constant temperature **A.It pressure decreases** B.The total momentum of its molecules remains constant C.Its pressure decreases and the total momentum of its molecules remains constant D.Its pressure decreases and the total kinetic energy of its molecules decreases.

Expansion of gas at constant temperature increases the volume, hence the pressure decreases by Boyle's law. The momentum change per second decreases as pressure decreases. The average kinetic energy is constant as temperature is constant.

22. When a fixed of an ideal gas expands at

Demystified Series by Dr Timothy

constant temperature, which property of the gas molecules increases? A.Average number per unit volume B.Average kinetic energy C.Average speed **D.Average separation.**

Expansion of gas at constant temperature increases the volume and decreases the pressure by Boyle's law. The average separation between the gas molecules increases as the volume increases. The average number per unit volume and number of collision per second increases with pressure The average speed and average kinetic energy are constant as temperature is constant.

23. As the average molecular speed of a fixed volume of a gas at a very low pressure increases, its **A.pressure increases** B.pressure decreases C.temperature decreases D.temperature remains constant E.pressure remains constant.

Increase in the average molecular speed causes the average kinetic energy to increase. Hence, the temperature also increases. Increase in temperature at constant or fixed volume brings about increase in pressure by pressure law.

24. If a given mass of gas at constant pressure obeys the relation, volume is proportional to the absolute temperature, the gas is said to obey A.Hooke's law B.Boyle's law **C.Charles' law** E.Newton's law.

Charles' law gives the relation betweenthe volume and temperature of a fixed mass of gas , $V \propto T$.

25. A ballon inflated with helium gas at ground level is released. As it rises through a constant temperature atmosphere A.its pressure reduces but volume remains constant **B.the product of pressure and volume remains constant** C.both pressure and volume increase D.its pressure remains constant but volume decreases E.both pressure and volume remains constant.

The pressure of the inflated ballon decreases as it rises i.e pressure decreases with altitude. The volume of the inflated balloon increases as pressure decreases, at constant temperature by Boyle's law. Also the product of the pressure and volume is constant according to Boyle's law i.e $P \propto \frac{1}{V}$, PV =K.

26. If the volume of a fixed mass of gas is kept constant, the pressure of the gas A.is inversely proportional to its centigrade temperature B.is inversely proportional to its absolute temperature C.is directly proportional to its centigrade temperature **D.is directly proportional to its absolute temperature** E.remaibs constant also.

The pressure of a fixed mass of gas is directly proportional to its absolute kelvin temperature at constant volume by pressure law , $P \propto T$.

27. In a gas experiment, the pressure of the gas is plotted against the reciprocal of the volume of the gas at constant temperature. The unit of the slope is A.force B.force/m **C.work** D.force/m^2 E.energy/m^2.

The slope is equal to work or energy and it unit is Joule.

28. Which of the following gas laws is equivalent to the workdone? A.Pressure law B.Van der waal's law C.Charles' law **D.Boyle's law .**

Boyle's law constant is equivalent to workdone or energy , $PV = K =$ W.

29. A product of PV where P is pressure and V is the volume has the same unit as **A.energy** B.power C.acceleration D.force.

30. A gas with initial volume 2×10^{-6}m^3 is allowed to expand to six times its initial volume at constant pressure of 2×10^5Nm^{-2}, what's the workdone? A.12.0J B.4.0J **C.2.0J** D.1.2J

P=2×10^5Nm^{-2}, V_1=2×10^{-6}m^3, V_2=$6V_1$,

$W = P\Delta V = P(V_2 - V_1) = P(6V_1 - V_1)$,

$W = P(5V_1) = 2\times10^5(5\times2\times10^{-6}) = 2.0$J.

31. A gas at a volume V_o in a container at pressure P_o, is compressed to one fifth of its volume. What will be its pressure if it maintains its original temperature T? A.$\frac{P_o}{5}$ B.$\frac{4}{5}P_o$ C.P_o **D.5P_o.**

P_1=P_o, V_1=V_o, P_2=?, $V_2 = \frac{1}{5}V_o$, At constant temperature, Boyle's law applies i.e $PV = K$.

$P_1V_1 = P_2V_2$, $P_o \times V_o = P_2 \times \frac{1}{5}V_o$,

$P_2 = \frac{5P_oV_o}{V_o} = 5P_o$, five times the initial value.

32. The volume of an air bubble increases from the bottom to the top of a lake at constant temperature because A.atmospheric pressure acts on the surface of the lake **B.pressure increases with depth** C.density increases with depth C.density remains constant with pressure D.the bubble experiences an upthrust.

Increase in the volume of air bubble from the bottom of the lake to the top proportionately causes a decrease in the pressure at constant temperature. Pressure at the bottom of the lake is greater than the pressure at the top, as pressure increases with depth of lake.

33. An air bubble of volume 4cm^3 if formed 20cm under water. What will be its volume when it rises to just below the surface of the water, if the atmospheric pressure is equivalent to height of 10cm of water? A.2cm^3 B.4cm^3 C.6cm^3 D.8cm^3 **E.12cm^3.**

Atmospheric pressure H=10cmH_2O, depth of water h=20cm, pressure just below the surface of water P_1=H, volume of air bubble just below the surface of water V_1=?, pressure at the bottom of water P_2=$H + h$, volume of air bubble at the bottom of water V_2=4cm^3,

Boyle's law – $P_1V_1 = P_2V_2$, $(H)V_1 = (H + h)V_2$,

$10\times V_1 = (10+20)\times4$, $10\times V_1 = 30\times4=120$,

$V_1 = \frac{120}{10} = 12$cm^3.

34. An air bubble rises from the bottom to the top

Demystified Series by Dr Timothy

of a water dam which is 40m deep. The volume of the bubble just below the surface is 2.5cm^3. Find its volume at the bottom of the dam, if atmospheric pressure is equivalent to 10m of water A.10.0cm^3 B.2.0cm^3 C.1.6cm^3 D.0.625cm^3 **E.0.5cm^3.**

H=10m, h=40m, P_1=H, V_1=2.5cm^3, P_2=$H+h$, V_2=?, Boyle's law – $P_1V_1 = P_2V_2$,

$(H)V_1 = (H+h)V_2$, $10\times2.5 = (10+40)V_2$,

$25 = 50V_2$, $V_2 = \frac{25}{50} = 0.5\text{cm}^3$.

35. An air bubble of radius 3cm rises from a depth of 7.0m in a swimming pool. What will be the radius of the bubble just before it reaches the water surface? (Assume that the water temperature is uniform, atmospheric pressure=10.34 metres of water) **A3.6cm** B.7.1cm C.5.0cm D.5.3cm.

Air bubble is spherical. The volume of a sphere is given by – $\frac{4}{3}\pi r^3$, r=radius. P_1=10.34mH_2O, r_1=?, V_1=$\frac{4}{3}\pi {r_1}^3$, P_2=10.34+7 =17.34mH_2O, r_2=3cm, V_2= $\frac{4}{3}\pi {r_2}^3$=$\frac{4}{3}\pi\times3^3$, V_2=36πcm^3, $P_1V_1 = P_2V_2$,

$10.34\times\frac{4}{3}\pi {r_1}^3 = 17.34\times36\pi = 624.24\pi$,

${r_1}^3 = \frac{3\times624.24}{10.34\times4} = 45.28$, $r_1 = \sqrt[3]{45.28}$,

$r_1 = 3.56\text{cm} = 3.6\text{cm}^3$.

36. A narrow, uniform glass tube, closed at one end, contains air which is sealed by a thread of mercury. The tube is placed first horizontally and then vertically with the open end at the top. Which of the following statements is correct? A.The length of the entrapped air in the horizontal position is less than that in the vertical position **B.The length of the entrapped air in the vertical position is less than that in the horizontal position** C.The pressure of the entrapped air is atmospheric pressure in both cases D.The length of the mercury thread becomes shorter in the horizontal position than in the vertical position E.The product of the pressure and the volume of the entrapped air is not the same in both cases.

The pressure of the entrapped air in the horizontal position is atmospheric pressure only while the pressure of the entrapped air in the vertical position is the atmospheric pressure and the pressure due to the column of mercury thread. Hence, pressure in the vertical position is greater than the pressure in the horizontal position. The length of the entrapped air i.e the volume of the entrapped air as $V \propto l$, in the vertical position is less than in the horizontal position by Boyle's law. The length of the mercury thread is always constant. The product of the pressure and the volume of the entrapped air is the same or constant in both cases by Boyle's law.

37. Some quatity of a gas is trapped in a cylinder of volume 1.0m^3 at pressure of 2.0 atmospheres. The piston of the cylinder is moved inwards, reducing the volume of the gas to 0.50m^3. What is the pressure of the gas in this case if the temperature is constant? A.1.0 atmosphere B.1.5 atmospheres C.2.0 atmospheres D.3.0 atmospheres **E.4.0 atmospheres.**

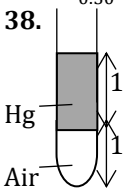
P_1=2.0atm, V_1=1.0m^3, P_2=?, V_2=0.50m^3,

Boyle's law applies at constant temperature,

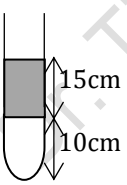
$P_1V_1 = P_2V_2$, $2.0\times1.0 = P_2\times0.50$,

$P_2 = \frac{2.0\times1.0}{0.50} = 4.0\text{atm}$.

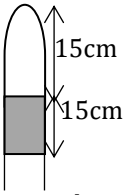
38.



An air column 10cm in length is trapped into the sealed end of a capillary tube by a 15cm column of mercury with the tube held vertically as shown above. On inverting the tube, the air column becomes 15cm long. What is the atmospheric pressure during this experiment? A.90cm B.76cm **C.75cm** D.60cm.



I.Vertical



II. Inverted

Atmospheric pressure H=?, length of mercury thread h=15cm, length of air column l,

1st diagram – P_1=$H+h$=H +15, V_1=l_1=10cm,

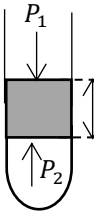
2nd diagram – P_2=$H-h$=H −15, V_2=l_2=15cm,

$P_1V_1 = P_2V_2$, $(H+15)10 = (H-15)15$,

$10H+150 = 15H-225$, $15H-10H = 150+225$,

$5H = 375$, $H = \frac{375}{5} = 75\text{cm}$.

39.

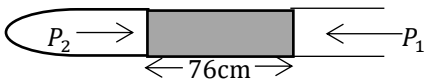


In the figure above, a thread of mercury 76cm long is used to trap some dry air in a capillary tube of uniform cross-section closed at one end. Calculate the ratio of the pressure P_2/P_1 [Atmospheric pressure is 76cm of mercury] **A.2** B.1 C.0.5 D.0.2.

Atmospheric pressure H=76cmHg, length of mercury thread h=76cmHg, P_1 is the atmospheric pressure, P_1=H, P_2 is the pressure exerted by the trapped air on the mercury thread, $P_2 = H+h$,

$\frac{P_2}{P_1} = \frac{H+h}{H} = \frac{76+76}{76} = \frac{152}{76} = 2$.

40. The diagram below shows a horizomtal tube containing a column of mercury 76cm long. Calculate the ratio of P_2/P_1 .



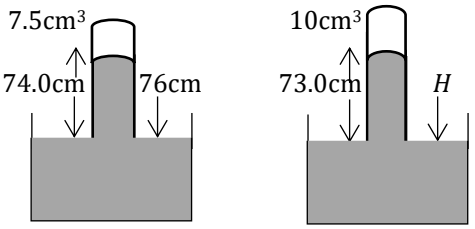
(Atmospheric pressure is 76cm of mercury) A.4 B.3 C.2 **D.1.**

Atmospheric pressure H=76cmHg, length of mercury column or thread h=76cmHg, P_1 is the atmospheric pressure, P_1=H, P_2 is the pressure

Demystified Series by Dr. Timothy

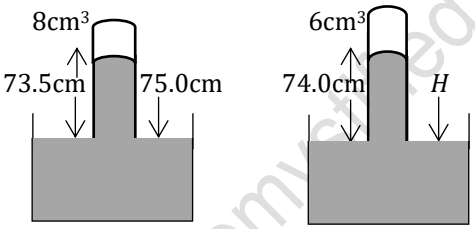
exerted by the trapped air on the mercury thread. In the horizontal position pressure exerted by the trapped air is equal to the atmospheric pressure. Hence. $P_2=H$, $\frac{P_2}{P_1}=\frac{H}{H}$ =1:1 or 1.

41. A little quantity of air is trapped on top of the mercury column in a barometer. When the volume of the trapped air is 7.5cm^3 and the atmospheric pressure is 76cm, the barometer reads 74.0cm. When the volume of the trapped air is 10cm^3 and the barometer reads 73.0cm, the correct atmospheric pressure is A.75.7cm B.75.0cm **C.74.5cm** D.74.0cm.



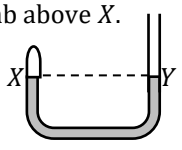
H=76cm, h=74.0cm, H=?, h=73.0cm,
$P_1=H-h$=76−74.0, $P_2=H-h=H-73.0$,
P_1=2cm, V_1=7.5cm^3, V_2=10cm^3,
$P_1V_1=P_2V_2$, 2×7.5= $(H-73)10$,
$15=10H-730$, $10H=15+730=745$,
$H=\frac{745}{10}=74.5$cm.

42. As a result of air at the top of a barometer, the height of the mercury column is 73.5cm when it should be 75.0cm, the volume of the space above the mercury is 8cm^3. Calculate the correct barometric height when the barometer reads 74.0cm and the volume of the space above mercury is 6.0cm^3 A.72.0cm B.74.5cm C.75.1cm **D.76.0cm**



H=75.0cm, h=73.5cm, H=?, h=74.0cm,
$P_1=H-h$=75.0−73.5, $P_2=H-h=H-74.0$,
P_1=1.5cm, V_1=8cm^3, V_2=6cm^3,
$P_1V_1=P_2V_2$, 1.5×8= $(H-74.0)6$,
$12=6H-444$, $6H=12+444=456$,
$H=\frac{456}{6}=76.0$cm.

43. In the J-tube below, Y and X are on the same horizontal level and 30cm^3 of air is trapped above Y when the atmospheric pressure is 75cmHg. Calculate the volume of air trapped above Y when 15cmHg is now poured into the limb above X.



A.15cm^3 **B.25cm^3** C.35cm^3 D.45cm^3.

The pressure initially acting on both arms are the same, as the liquid level in both arms are the same. The arm above Y is exposed to the atmosphere, hence the pressure acting on that arm is atmospheric pressure.

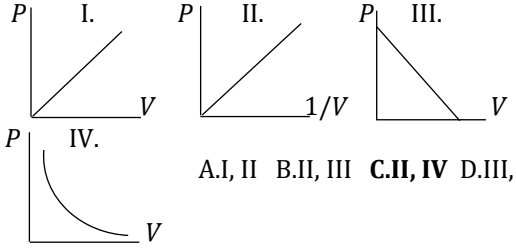
Atmospheric pressure H=75cmHg, $P_1=H$= 75cmHg, V_1=30cm^3, on pouring mercury of 15cm into the limb above X, the pressure increases. P_2=75+15=90cmHg, V_2=?, $P_1V_1=P_2V_2$,
75×30=90× V_2, $V_2=\frac{75\times30}{90}=25$cm^3.

44. A certain pump has a barrel whose volume is 300cm^3 and it is being used to exhaust a vessel of volume 3dm^3 which contains air initially at a pressure of 71.50cm of mercury. Assuming that the operation takes place at a constant temperature,what will be the pressure in the vessel after **one exhaust stroke** of the pump A.7.15cmHg **B.65.0cmHg** C.79.44cmHg D.715.0cmHg E.786.5cmHgInitial pressure of air in thevessel P_1=71.50cm of mercury=7150cmHg, volume of vessel to be exhausted V_1=3dm^3= 3000cm^3, volume of pump V_P=300cm^3, $V_2=V_1+V_P$=3000+300=3300cm^3, number of strokes n=1,
$P_1{V_1}^n={P_2}{V_2}^n$, $P_1{V_1}^n=P_2(V_1+V_p)^n$,
$71.5\times(3000)^1=P_2\times(3000+300)^1$,
$214500=P_2\times3300$, $P_2=\frac{214500}{3300}=65$cmHg.

45. A fixed mass of gas occupying a certain volume has its pressure reduced to 25 percent of its original value while the temperature is maintained constant. The ratio of the new volume to the original volume is A.2:1 B.1:2 C.1:4 **D.4:1.**

$P_1=P$, $P_2=25\%P=\frac{25}{100}P=\frac{1}{4}P$, $V_2:V_1$=?, $P_1V_1=P_2V_2$
$P\times V_1=\frac{1}{4}P\times V_2$, $\frac{V_2}{V_1}=\frac{4\times P}{P}=4$ or 4:1.

46. A fixed quantity of gas is subjected to various pressure P and the corresponding volume V is measured at constant temperature. Which of the following graphs best represents the results?



A.I, II B.II, III **C.II, IV** D.III, IV.

47. At constant temperature the density of a fixed mass of gas is A.consatbt with temperature B.proportional to its volume **C.inversely proportional to its temperature** D.independent of its volume.

According to Charles' law, volume is directly proportional to absolute temperature at constant pressure i.e $V\propto T$.

Density $\rho=\frac{mass}{volume}=\frac{m}{V}$, at a fixed or constant mass, $\rho\propto\frac{1}{V}$. Hence $\rho\propto\frac{1}{T}$ i.e density is inverlsey

Demystified Series by Dr. Timothy

proportional to its absolute temperature.

48. The equation $P^xV^yT^z =$ constant is Charles' law, when A.x=1, y=−1, z=1 **B.x=0, y=1, z=−1** C.x=1, y=0, z=1 D.x=0, y=1, z=1.

Charles' law – $V \propto T$, $\frac{V}{T}$ = constant. It can also be written as $\frac{P^0V^1}{T^1}$ or $P^0V^1T^{-1}$ = constant.

$P^xV^yT^z = P^0V^1T^{-1}$, x=0, y=1, z=−1.

49. A fixed mass of gas of volume 546cm³ at 0°C is heated at constant pressure. What is the volume of the gas at 2°C. A.1092cm³**B.550cm³** C.548cm³ D.542cm³ E.273cm³

V_1=546cm³, T_1=0°C=0+273=273K, V_2=?, T_2=2°C=2+273=275K, Charles' law holds at constant pressure – $\frac{V_1}{T_1} = \frac{V_2}{T_2}$, $\frac{546}{273} = \frac{V_2}{275}$,

$2 = \frac{V_2}{275}$, $V_2 = 2\times275 = 550\text{cm}^3$.

50. At what temperature will the volume of a given mass of air be 3 times its volume at 0°C when the pressure is kept constant? **A.546.0**°C B.273.0°C C.136.0°C D.3°C E.−273.0°C.

V_1=V, V_2=3V, T_1=0°C=273K, T_2=?, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$,

$\frac{V}{273} = \frac{3V}{T_2}$, $T_2 = \frac{3V\times273}{V} = 819K$,

$T_2 = 819-273$ =546.0°C.

51. A gas occupies a certain volume at 27°C. At what temperature will its volume be doubled assuming that its pressure remains constant? A.54°C B.60°C C.108°C D.327°C E.600°C.

V_1=V, V_2=2V, T_1=27°C=27+273=300K, T_2=?,

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$, $\frac{V}{300} = \frac{2V}{T_2}$, $T_2 = \frac{2V\times300}{V}$ =600K,

$T_2 = 600-273 = 327$°C.

52. Dry hydrogen is trapped by pellet of mercury in a uniform capillary tube closed at one end. If the column of hydrogen at 27°C is 1.0m, at what temperature will the length be 1.20m? A.22.5°C B.32.4°C C.77.0°C **D.87.0**°C E.360.0°C.

$V \propto l$, as the capillary tube is of uniform cross-sectional area. V_1=l_1=1.0m, T_1=27°C, T_1=27+273=300K, V_2=l_2=1.2m, T_2=?,

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$, $\frac{1}{300} = \frac{1.2}{T_2}$, $T_2 = 1.2\times300 = 360K$,

$T_2 = 360-273 = 87$°C.

53. A column of air 10.0cm long is trapped in a tube at 27°C. What is the length of the column at 100°C? **A.12.4cm** B.13.7cm C.18.5cm D.37.0cm.

$V \propto l$, as the capillary tube is of uniform cross-sectional area. V_1=l_1=10.0m, T_1=27°C, T_1=27+273=300K, V_2=?, T_2=100°C=100+273, T_2=373K, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$, $\frac{10.0}{300} = \frac{V_2}{373}$, $V_2 = \frac{373}{30} = 12.4$cm.

54. The cubic expansivity of a certain gas at constant pressure is $\frac{1}{273}K^{-1}$. If a given mass of the gas is held at constant pressure and its volume at 0°C is 273m³, determine the volume of the gas at 273°C A.273m³**B.546m³** C.819m³ D.1092m³.

Cubic expansivity γ=$\frac{1}{273}K^{-1}$, temperature rise θ=273−0=273°C, volume at 0°C V_o=273cm³, volume at θ V_θ=?,

$\gamma = \frac{increase\ in\ volume\ from\ 0°C}{volume\ at\ 0°C\times temp.\ rise} = \frac{V_\theta - V_o}{V_o\theta}$,

$\frac{1}{273} = \frac{V_\theta-273}{273\times273}$, $V_\theta - 273 = \frac{273\times273}{273} = 273$,

$V_\theta = 273+273 = 546\text{cm}^3$.

55. A uniform capillary tube of negligible expansion is sealed at one end, contains air trapped by a pellet of mercury. The trapped air column is 13.7cm long at 0°C and 18.7cm long at 100°C.Calculate the cubical expansivity of the air at constant pressure A.14.56×10$^{-3}K^{-1}$ B.10.95×10$^{-3}K^{-1}$ C.7.30×10$^{-3}K^{-1}$ **D.3.65×10$^{-3}K^{-1}$.**

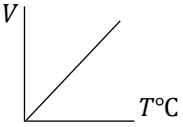
Length at 0°C l_o=13.7cm, length at θ l_θ=18.7cm, temperature rise θ=100-0=100°C,

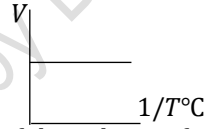
Cubic expansivity $\gamma = \frac{l_\theta - l_o}{l_o\theta} = \frac{18.7-13.7}{13.7\times100}$,

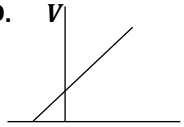
$\gamma = \frac{5}{1370} = 3.65\times10^{-3}K^{-1}$.

56. Which of the following graps is consistent with Charles' law?

A. 

B. 

C. 

D. 

57. If the volume of a fixed mass of a gas is kept constant, the pressure of the gas A.is inversely proportional to its centigrade B.is inversely proportional to its absolute temperature C.is directly proportional to its centigrade temperature **D.is directly proportional to its absolute temperature** E.remains constant also.

58. A fixed mass of air at s.t.p is heated in such a way that its volume remains unchanged. Calculate the temperature if its pressure is increased to 114cm of mercury A.0°C B.1.5°C C.109.2°C **D.136.5**°C E.273°C.

P_1=s.p=76cmHg, T_1=s.t=273K, P_2=114cmHg,

T_2=?, $\frac{P_1}{T_1} = \frac{P_2}{T_2}$, $\frac{76}{273} = \frac{114}{T_2}$, $T_2 = \frac{114\times273}{76}$,

$T_2 = 409.5K$, $T_2 = 409.5-273 = 136.5$°C.

59. The tyre of a car was found to be 150cmHg in the morning at a temperature of 27°C. Hard driving in the afternoon raised the temperature of the tyres to 57°C. The tyre pressure had by afternoon **A.increased by 15.0cmHg** B.decreased by 15cmHg C.increased by 22.50cmHg D.decreased by 22.50cmHg E.remains constant.

P_1=150cmHg, T_1=27°C=27+273=300K, P_2=?, T_2=57°C=57+273=330K, $\frac{P_1}{T_1} = \frac{P_2}{T_2}$,

$\frac{150}{300} = \frac{P_2}{330}$, $P_2 = \frac{150\times330}{300} =$ 165cmHg, The tyre pressure is increased by 15cmHg i.e $P_2 - P_1 = 165-150 = 15$cmHg.

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60. The pressure of a fixed mass of an ideal gas at 27°C is 3Pa. Tha gas is heated at a constant volume until its pressure is 5Pa. Determine the mew temperature of the gas A.100°C **B.227**°C C.273°C D.500°C

P_1=3Pa, V_1=27°C=27+273=300°C, P_2=5Pa, T_2=?, $\frac{P_1}{T_1}=\frac{P_2}{T_2}$, $\frac{3}{300}=\frac{5}{T_2}$, $T_2=\frac{5\times300}{3}=500K$,
$T_2=500-273=227$°C.

61. A motor tyre is inflated to a pressure of 2.0×10^5Nm^{-2} when the temperature of air is 27°C. What will be the pressure in it at 87°C assuming that the volume of the tyre does not change? A.2.6×10^5Nm^{-2} **B.2.4×10^5Nm^{-2}** C.2.2×10^5Nm^{-2} D.1.3×10^5Nm^{-2}.

P_1=2.0×10^5Nm^{-2}, T_1=27°C=27+273=300K, P_2=?, T_2=87°C=87+273=360K, $\frac{P_1}{T_1}=\frac{P_2}{T_2}$,
$\frac{2.0\times10^5}{300}=\frac{P_2}{360}$, $P_2=\frac{2.0\times10^5\times360}{300}$=$2.4\times10^5$Nm^{-2}.

62. The pressure of a certain mass of gas at constant volume increases from 250mmHg at 20°C to 400mmHg at 50°C. Calculate the pressure expansivity of the gas. A.0.05°C^{-1} B.0.04°C^{-1} C.0.03°C^{-1} **D.0.02°C^{-1}.**

Pressure at 20°C P_o=250mmHg, pressure at 50°C P_θ=400mmHg, temperature rise θ=50−20=30°C, pressure expansivity β=?,
$\beta=\frac{P_\theta-P_o}{P_o\theta}=\frac{400-250}{250\times30}=\frac{150}{7500}=0.02$°C^{-1}.

63. The pressure P, the volume V, and the temperature T, of a given mass of air change simultaneously. Which of the following equations is correct? A. $PV=\frac{constant}{T}$ B. $\frac{P}{T}=constant\times V$ C. $\frac{P}{V}=\frac{constant}{T}$ D. $\frac{1}{T}=\frac{1}{PV}$ **E.** $\frac{V}{T}=\frac{constant}{P}$.

General gas law – $\frac{PV}{T}$ = constant, Rearranging the equation, $\frac{V}{T}=\frac{constant}{P}$.

64. When the volume of a given gas is halved and its temperature doubled, the pressure A.remains constant **B.increased by a factor of 4** C.increases by a factor of 3 D.decreased by a factor is 3.

P_1=P, V_1=V, T_1=T, P_2=?, V_2=$\frac{1}{2}V$, T_2=$2T$,
$\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$, $\frac{P\times V}{T}=\frac{P_2\times\frac{1}{2}V}{2T}$, $P_2=\frac{P\times V\times2T}{T\times\frac{1}{2}V}$,
$P_2=P\times2\times2=4P$, four times its initial value or increased by a factor of 4. Alternatively,
$\frac{PV}{T}$ = constant. V is halved i.e $\frac{1}{2}$, T is douled i.e 2, hence $P=\frac{T}{V}=\frac{2}{1/2}=2\times2=4$ i.e four times its initial value or increased by a factor of 4.

64. The volume of a given mass of gas 27°Cand 800mmHg is 76cm^3. Calculate its volume at s.t.p A.100.0cm^3 **B.72.8cm^3** C.60.0cm^3 D.36.4cm^3.

P_1=800mmHg, V_1=76cm^3, T_1=27°C=27+273, T_1=300K, P_2=s.p=760mmHg, V_2=?, T_2=s.t=273K,
$\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$, $\frac{800\times76}{300}=\frac{760\times V_2}{273}$, $V_2=\frac{800\times76\times273}{300\times760}$,
V_2 = 72.8cm^3.

65. A quantity of gas occupies a certain volume when the temperature is −73°Cand the pressure is 1.5 atmospheres. If the pressure is increased to 4.5 atmospheres and the volume is halved at the same time, what will e the new temperature of the gas? A.573°CB.327°C C.300°C D.110°C**E.27**°C

P_1=1.5atm, V_1=V, T_1=−73°C=−73+273=200K,
P_2=4.5atm, V_1=$\frac{1}{2}V$, T_2=?, $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$,
$\frac{1.5\times V}{200}=\frac{4.5\times\frac{1}{2}V}{T_2}$, $T_2=\frac{200\times4.5\times\frac{1}{2}V}{1.5\times V}=300K$,
$T_2=300-273=27$°C.

66. A gas volume 700cm^3 at a temperature of 30°C has a pressure of 606mmHg. If the volume of the gas is reduced by 300cm^3 when the temperature increases to 47°C. Calculate the pressure of the gas. A.280mmHg B.560mmHg **C.1120mmHg** D.2240mmHg.

V_1=700cm^3, T_1=30°C=30+273=303K, P_1=606mmHg, the volume is reduced by 300cm^3, $V_2=V_1-300$=700−300=400cm^3, temperature is reduced to 47°C , T_2=47°C, T_2=47+273=320K, P_2=?, $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$, $\frac{606\times700}{303}=\frac{P_2\times400}{320}$,
$P_2=\frac{606\times700\times320}{303\times400}=1120$mmHg.

67. The pressure of a given mass of a gas changes from 300Nm^{-2} to 120Nm^{-2} while the temperature drops from 127°C to −73°C. What is the ratio of the final volume to the initial volume? A.2:5 **B.5:4** C.5:2 D.4:5.

P_1=300Nm^{-2}, T_1=127°C=127+273=400K, P_2=120Nm^{-2}, T_2=−73°C=−73+273=200K, the ratio of the final volume to initial volume V_2:V_1=?, $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$, $\frac{300\times V_1}{400}=\frac{120\times V_2}{200}$, $\frac{V_2}{V_1}=\frac{300\times200}{400\times120}$,
$\frac{V_2}{V_1}=\frac{5}{4}$ or 5:4.

68. A given mass of ideal gas occupies a volume V at a temperature T and under a pressure P. If the pressure is increased to $2P$ and the temperature reduced to $\frac{1}{2}T$, then what is the percentage change in the volume of the gas? A.0% B.25% **C.75%** D.300% E.400%.

P_1=P, V_1=V, T_1=T, P_2=$2P$, V_2=?, T_2=$\frac{1}{2}T$,
$\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$, $\frac{P\times V}{T}=\frac{2P\times V_2}{\frac{1}{2}T}$, $V_2=\frac{P\times V\times\frac{1}{2}T}{T\times2P}$,
$V_2=\frac{1}{4}V=0.25V$. OR $\frac{PV}{T}$ = constant. P is doubled i.e 2, T is halved i.e $\frac{1}{2}$ hence, $V=\frac{T}{P}=\frac{1/2}{2}=\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}=0.5$ i.e $V_2=0.5V$.
Percentage change in volume $\%\Delta V=\frac{V_1-V_2}{V_1}\times100$
$\%\Delta V=\frac{V-0.5V}{V}\times100=\frac{0.75V}{V}\times100=75\%$.

69. At 27°C and 2 atmospheres pressure, the density of a gas is 0.40g per litres. What is its density at 4 atmospheric pressure and 227°C. A.0.12g per litre B.0.24g per litre C.0.36g per litre **D.0.48g per litre**

P_1=2atm, T_1=27°C=27+273=300K,ρ_1=0.40g per

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litre, P_2=4atm, T_2=227℃=227+273, T_2=500K, ρ_2=?, $\frac{PV}{T}$ = constant, density= $\frac{mass}{volume}$, $\rho = \frac{m}{V}$, $\rho \propto \frac{1}{V}$ or $V = \frac{1}{\rho}$. Hence, $\frac{P}{T\rho}$ = **constant,** $\frac{P_1}{T_1\rho_1} = \frac{P_2}{T_2\rho_2}$, $\frac{2}{300\times0.40} = \frac{4}{500\times\rho_2}$, $\rho_2 = \frac{300\times0.40\times4}{2\times500}$, ρ_2 = 0.48g per litre.

Jamb past questions on Pressure in solids and fluids :
[1978/8,1979/22,1981/5,1982/1,21,1983/9,37,50,1985/14,15,1986/22,1988/15,1989/11,1990/13,1991/1,13,1992/11,1993/10,1994/15,17,1995/16,1997/18,1998/14,15,1999/12,2000/6,2001/2,2002/8,2004/15,32,2005/40,2006/50,2007/38,2008/16,17,2009/14,2010/12,2011/10,2012/17,2013/14]

Jamb past questions in gas laws and kinetic theory :
[1978/20,1979/17,38,1980/24,33,1981/6,36,1982/5,41,1983/14,1984/28,1985/20,21,26,1986/24,1987/19,20,1988/16,1989/14,15,17,1990/17,1991/18,21,1992/16,1993/16,1994/21,26,1995/17,21,1997/22,1998/19,23,1999/10,13,2001/16,19,2002/18,2005/44,47,2006/47,2007/29,33,37,2008/21,2010/22,2011/19,24,2012/21,2013/18,2014/21,2015/12,2016/11]

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Chapter 11 – Temperature and its measurement

● **Heat** is a measure of the total internal energy of a body. It is a form of energy transferred from one body to another due to difference in temperature between bodies.

● **Temperature** is the degree of hotness or coldness of a body. It is a measure of the average kinetic energy of the molecules of a body. Temperature determines the direction of heal flow between two bodies in contact.

- Temperature is accurately measured by a **thermometer.**

- The substance whose physical property is used in determining temperature is called a thermometric substance.

- **Fixed temperature –** All thermometers have two fixed points or reference points called upper fixed temperature point and lower fixed temperature point.

- **Upper fixed temperature point or Steam point U.F.P or *U*** is the temperature of steam from pure boiling water at standard atmospheric pressure of 760mmHg. Upper fixed point is determined using a **hypsometer**.

- **Lower fixed temperature point or Ice point L.F.P or *L*** is the temperature of a pure melting ice at standard temperature of 760mmHg. Lower fixed point is determined using a glass funnel and ice shavings or pure melting ice.

- **Temperature Scales –** Temperature scales are used for temperature measurement. Types of temperature scales are namely ; Fahrenheit scale, Absolute or Thermodynamic or Kelvin scale, Centigrade or Celcius scale, Rhemur scale and Rankin scale.

- **Fundamental or Thermometric interval $F.I$** is the difference or interval between the upper fixed point (U.F.P) and lower fixed point (L.F.P) of a scale.

1.Fahrenheit scale – U.F.P is 212°F, L.F.P is 32°C, fundamental interval $F.I$ is 180°F.

2.Absolute or Themodynamic or Kelvin scale – U.F.P is $373K$, L.F.P is $273K$, fundamental interval $F.I$ is $100K$

3.Centigrade or Celcius scale– U.F.P is 100°C, L.F.P is 0°C, fundamental interval $F.I$ is 100°C.

4.Rhemur scale – U.F.P is $80°R$, L.F.P is $0°R$, fundamental interval $F.I$ $80°R$.

5.Rankin scale – U.F.P is $672°R$, L.F.P is $492°R$, fundamental interval $F.I$ is $180°R$.

- **A temperature change of 1°C is the same as 1K change in temperature, as they have the same fundamental interval of 100.**

- **A temperature change of 1°C is the same as $9/5$°F.**

- In conversion of temperature scale P to another temperature scale Q and vice-versa the expression below is used – $\frac{t_{\theta_P}-t_{L_P}}{t_{U_P}-t_{L_P}}=\frac{t_{\theta_Q}-t_{L_Q}}{t_{U_Q}-t_{L_Q}}$ OR $\frac{t_{\theta_P}-t_{L_P}}{F.I_P}=\frac{t_{\theta_Q}-t_{L_Q}}{F.I_Q}$, t_{θ_P}=temperature in scale P, t_{L_P}=lower fixed point or ice point of scale P, t_{U_P}=upper fixed point or ice point of scale P, t_{θ_Q}=temperature in scale Q, t_{L_Q}=lower fixed point or ice point in scale Q, t_{U_Q}=upper fixed point or steam point in scale Q, $F.I_P$=fundamental interval of scale P, $F.I_Q$=fundamental interval of scale Q.

- **Qualities of a good thermometer –** A good thermometer must have: low thermal capacity or high heat conductivity to measure wide temperature range, high sensitivity or short response time, suitability i.e purpose of usage, high accuracy, reproducibility i.e equivalent fixed points on different thermometer and consistency i.e agreement between different readings on the same thermometer.

- **Types of thermometer, thermometric property and thermometric substance used :**

1.Optical or infrared thermometer – Variation in light intensity, colour or radiant heat with temperature. It is used for high temperature measurement

2.Bimetallic thermometer – Change in expansion(length) of two metals which forms a bimetallic strip, with temperature. For bimetallic thermometer ; $\frac{L_\theta-L_L}{L_U-L_L}=\frac{\theta-\theta_L}{\theta_U-\theta_L}$.

3.Resistance thermometer – Change in electrical resistance of a platinum wire with temperature. For resistance thermoemeter ; $\frac{R_\theta-R_L}{R_U-R_L}=\frac{\theta-\theta_L}{\theta_U-\theta_L}$.

4.Thermoelectric thermometer – Change in electromotive force E.M.F between two joined dissimilar metals e.g copper and silver, which forms a thermocouple, with temperature. The variation of E.M.F with temperature is referred to as **See-beck effect. Thermoelectric thermometer is the most convenient thermometer for measuring high and varying(or rapidly changing) temperature e.g temperature of the wings of an airplane.** For Thermoelectric thermometer ; $\frac{E_\theta-E_L}{E_U-E_L}=\frac{\theta-\theta_L}{\theta_U-\theta_L}$.

5.Gas or Absolute thermometer – Change or variation in pressure at constant volume i.e constant volume gas thermometer, or change or varaition in volume at constant pressure i.e constant pressure gas thermometer.It is very sensitive and accurate, hence used for more accurate temperature measurement. **All thermometer are calibrated against gas thermometer or used as standard for**

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calibrating other thermometer. Thrmometric substance is helium (widely used) and hydrogen. For gas thermometer ; $\frac{P_\theta - P_L}{P_U - P_L} = \frac{\theta - \theta_L}{\theta_U - \theta_L}$(constant volume gas thermometer), $\frac{V_\theta - V_L}{V_U - V_L} = \frac{\theta - \theta_L}{\theta_U - \theta_L}$(constant pressure gas thermometer.

6.Liquid-in glass thermometer – Change in volume or length i.e expansion of liquid e.g mercury or alcohol, with temperature.

- Suitability of a liquid as a thermometric liquid – I. The liquid must be easily seen in glass. **II.**The liquid should expand or contract uniformly with temperature. **III.**The liquid should have a high boiling point and low melting point to ensure a wide range of temperature measurement.

- Mercury is preferred to alcohol as the thermometric liquid because – I. Mercury has small heat capacity hence, high thermal conductivity than alcohol. **II.**Mercury does not vapourize easily (it has a high boiling point than alcohol). **III.**Mercury is opaque i.e can be easily seen, making it eeasy to read temperature while alcohol is transparent, hence has to be coloured. **IV.**Mercury does not wet glass. **V.**Mercury measures a wide range and high temperature due to its high boiling point(319℃).

- Alcohol is preferred to mercury because – I. Alcohol can measure low temperature due to its low freezing point(-115℃) than mercury(-39℃). **II.**It expands more than mercury (alcohol expands six times as more than mercury but not uniformly as mercury).

- Water is not a good thermometric liquid because it exhibits anomalous expansion, colourless and wets glass, small range of expansion due to small temperature interval ($0℃ - 100℃$).

- Factors increasing the sensitivity of a liquid in glass thermometer : I. Large and thin bulb for volume of liquid and conductivity i.e the bulb is made of thin glass so it can conduct heat quickly and assume the heat of the surrounding quickly. **II.**Thin walled stem bore and narrow tube for visibility and capillary rise i.e the capillary tube is very narrow and of uniform bore so that small changes in temperature can lead to large changes in the length of the liquid. **III.**High expansivity of liquid i.e the thermometric liquid should have high expansivity.

- Clinical thermometer –It is a form of liquid in glass thermometer which contains mercury as the thermometric liquid (a mercury in glass thermometer). It is used in hospitals for measuring the human body temperature. **It has a short temperature range($35℃ - 45℃$), a kink or constriction (bent region) which prevents the back flow of the liquid,therby altering the temperature reading.**

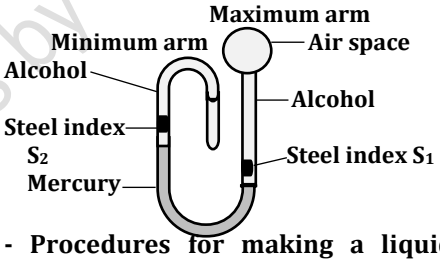
- A clinical theromometer is not sterilized in pure boiling water at normal atmospheric pressure of 760mmHg because at 760mmHg water boils at 100℃ and the range of temperature of a clinical thermometer is below 100℃ and will break the glass due to excessive expansion of mercury.

- Six's maximum and minimum thermometer – It is a liquid-in glass thermometer which is used to measures the temperature of the weather or the maximum and minimum temperature for a day. It consists of two long arms region which contains alcohol and a U-shaped arm region which contains mercury.

- The steel index in the minimum arm(left arm) moves upward while that in the maximum arm (left arm) remains stationary, when the temperature falls or at a low temperature.

- The steel index in the maximum arm (right arm) moves upward while that in the minimum arm (left arm) remains stationary, when the temperature rises or at a high temperature.

- The steel indices are returned to their original positions by a magnet.



- Procedures for making a liquid-in-glass thermometer –

1.Bulb making – One end is sealed and blowed for storage of the thermometric liquid

2.Liquid filling – The liquid is filled in steps by eliminating the air inside and allowing a large quantity of liquid to move in. The filling process is done by: **i.**Inverting the tube in a glass jar containing the liquid with the bubl facing upwards or at the uppermost and heating gently. **ii.**Removing the thermometer and boil the liquid in the tube by heating the bulb. **iii.**Inverting the tube again in the glass jar form complete filling of the tube.

3.End sealing – The thermometer is heated to a temperature above the maximum temperature required and sealing of the open end.

- Absolute zero temperature is the temperature in the kelvin scale where thermal motion ceases corresponding to the complete absence of translational motion of molecular motion. It is the temperature at which the average kinetic energy of particles making up a body is zero. **Absolute zero temperature is equal to $0K$ or $-273℃$.**

Demystified Series by Dr. Timothy

Examples :

1. In the study of physics, temperature and heat are often confused with each other. Which of the following statements correctly defines these two elements? A.Temperature is a measure of the average kinetic energy of molecules B.Heat is a measure of the total kinetic energy of molecules in a system **C.Different materials require different amounts of heat to cause a change in temperature** D.all of the above E.A and B only.

2. Temperature is a measure of the A.quantity of heat transferred into the molecules of an object **B.mean kinetic energy of any individual molecules of the object** C.kinetic energy of any individual molecules of the substance D.amount of workdone by the molecule of the object.

3. Which of the following properties cannot be utilized for temperature measurement? The A.Length of a liquid column B.Volume of a gas at constant pressure C.Pressure of a gas at constant volume D.Resistance of a metallic wire **E.Current produced in a photoelectric effect.**

Length of a liquid column is utilized in liquid-in -glass thermometer. Volume of a gas at constant pressure and pressure of a gas at constant volume are utilized in gas/absolute thermometer. Resistance of a metallic wire is utilized in resistance thermometer. Photoelectricity is used for light intensity.

4.Which of the following physical properties is not normally used to measure temperature? Change in the A.length of a column of liquid B.volume of a fluid **C.mass of a solid** D.pressure of a gas E.electrical resistance of a metal.

5. The thermometric property of the thermocouple is that its A.pressure changes with resistance **B.e.m.f changes with temperature** C.volume changes with temperature D.resistance changes with temperature.

6. The variation in the thermometric property of an optical pyrometer depends on the **A.radiant heat** B.volume of the trapped gas C.pressure of the trapped gas D.volume of the thermometric liquid.

Optical pyrometer uses changes in light intensity, radiant heat and colour for temperature measurement. Volume of trapped gas is used in constant pressure gas thermometer, pressure of trapped gas is used in constant volume gas thermometer. Volume of thermometric liquid is used in a liquid-in-glass thermometer.

7. The thermometer whose reading is indicated by a change in colour of its thermometric property is the A.bimetallic strip thermometer B.platinum resistance thermometer **C.optical pyrometer** D.thermocouple.

8. The thermometric substance of an absolute thermometer is A.alcoholB.mercury **C.helium** D.platinum

Helium and hydrogen are used as the thermometric substance of an absolute or gas thermometer. Alcohol and mercury are thermometric substance used in liquid-in-glass thermometer. Platinum is used in a resistance thermometer.

9. Which of the following statements about a gas thermometer and a mercury in glass thermometer is correct? A.mercury thermometer can not be calibrated against a constant volume gas thermometer B.mercury thermometers are much more accurate than gas thermometers **C.gas thermometers has a wide temperarure range than mercury thermometer** D.gas thermometers are less cumbersome than mercury in glass thermometer.

Mercury thermometer and all other thermometers are calibrated against a gas thermometer. Gas thermometer are of higher accuracy and wide temperature range of measurement than mercury in glass thermometer and all other thermometers. Gas thermometer are cumbersome hence they are limited to industrial use only.

10. Which of the following about the constant volume gas thermometer are correct? The thermometer is I.used as standard for calibrating other types of thermometer II.less convenient to manipulate when compared with the use of the liquid in glass thermometer III.less sensitive than liquid in glass thermometer **A.I and II** B.I and III C.II and III D.I,II and III.

Gas thermometer are more sensitive than liquid in glass thermometer and are used as standard for calibrating other themometers.

11. I.reproducibility II.sensitivity III.high thermal capacity IV.high accuracy. The qualities of a good thermometer includes A.I,II and III **B.I,II and IV** C.I,III and IV D.II,III and IV.

A good thermometer must be reproducible i.e gives equivalent fixed points in different thermometers, sensitive i.e responds to temperature changes quickly, low thermal capacity or high thermal conductivity, high accuracy i.e comparing the measured value to the standard.

12. I.High thermal capacity II.High sensitivity III.Easy readability IV.Accuracy over a wide range of temperatures. From the statements above, the qualities of a good thermometer is A.I and II B.I,III and IV **C.II,III and IV** D.I,II,III and IV

A good thermometer must have low thermal capacity or high thermal conductivity to measure a wide range of temperature, high sensitivity, easy readability and high accuracy over a wide temperature range.

13. The air in the stem of a mercury in glass thermometer should be expelled before

Demystified Series by Dr Timothy

introducing mercury so that A.the mercury may not be oxidized B.the mercury may not evaporate **C.enough mercury may enter the bulb** D.the mercury rise in the tue may not be subjected to varying air pressure E.the glass of the bulb may not crack if the mercury expands to compress the air.

Air is excluded from the bulb used for storing thermometric liquid, inorder to reduce the pressure and hence for more volume of the bulb to accommodate mercury.

14. Which of the following is not a reason for using mercury in preference to alcohol as a thermometric liquid? A.Mecury has a higher boiling point than alcohol **B.Mercury has a higher expansivity than alcohol** C.Mercury is a better conductor than alcohol D.Mercury does not wet glass E.Mercury is opaque.

Alcohol has a higher expansivity than mercury but the expansivity of mercury is uniform while that of alcohol is irregular.

15. Which of the following properties of mercury is not a reason for using it as a thermometric liquid? A.It is visible B.It does not wet the glass C.It allows for a wide temperature range to be measured **D.It has a high density** E.It has a high conductivity.

The density of mercury does not affects its usage as a thermometric liquid. The high density of mercury makes is suitable as a barometric liquid.

16. Which of the following is not a reason why mercury is preferred to alcohol when used as a thermometric liquid? Mercury A.has a small thermal capacity B.does not vaporize C.is opaque and so easily seen D.is a good conductor of heat **D.freezes at -39℃.**

Mercury is preferred to alcohol as a thermometric liquid because it has ; a low or small thermal capacity i.e high temperature range, does not vaporize easily i.e high boiling point, opaque i.e easily seen, good thermal conductivity. The disadvantages of mercury as a thermometric liquid are ; low expansivity i.e alcohol expands more than mercury, and low freezing point -39℃.

17. One special advantage of alcohol over mercury as a thermometric liquid is A.high specific heat capacity B.low boiling point **C.low freezing point** D.low density.

The low freezing point of alcohol i.e -115℃, hence makes it suitable in a thermometer for measuring very low temperature.

18. Water is a poor thermometric liquid because it A.has low vapour pressure B.is opaque C.is a poor conductor D.wets glass.

Water is not a good thermometric liquid because it exhibits anomalous expansion (between 0℃ − 4℃), colourless and wets glass, small range of expansion due to small temperature interval (0℃ − 100℃).

19. Which of the following features does not increase the sensitivity of a liquid-in-glass thermometer? A.A large bulb **B.A thick-walled tube** C.A capillary tube with a narrow bore D.A thin-walled bulb E.A liquid with a high expansivity.

Sensitivity of a liquid-in-glass thermometer increases when a large and thin-walled bulb, thin capillary tube or stem with a narrow bore and a liquid of high expansivity.

20. In the construction of a liquid-in-glass thermometer, the sensitivity increases if the A.thermal conductivity of the liquid is low B.the boiling point of the liquid is high **C.volume expansivity of the liquid is high** D.diameter of the capillary tube is large.

Sensitivity of a liquid-in-glass thermometer increases if the diameter of the capillary tube is small (thin-capillary tube) and volume expansivity of the liquid is high.

21. A short response time is obtained in a liquid-in-glass thermometer when the A.bulb is large and thick walled B.stem is long and thin C.bulb is small and thick-walled D.liquid is of high density and the bore is large **E.bulb is thin-walled and the liquid is a good conductor of heat.**

A liquid-in-glass thermometer with a short response time indicates a very high sensitive thermometer. Sensitivity increases when the bulb is large and thin-walled, the stem or tube is thin-walled and norrow bore,liquid of high volume expansivity and good conductivity.

22. I.Use a liquid with a high melting point II.Use a liquid of high volume expansivity III.Use a capillary tube of large diameter. Which of the above best describes how the sensitivity of a liquid-in-glass thermometer can be enhanced? A.I **B.II** C.I and III D.II and III.

23. Clinical thermometer are examples of A.constant pressure gas thermometer B.resistance thermometer C.alcohol thermometer **D.mercury-in-glass thermometer.**

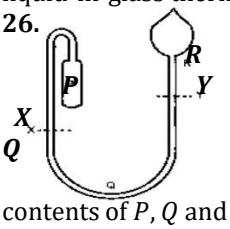
24. A clinical thermometer is different from other mercury in glass thermometer owing to A.its long stem **B.the constriction on its stem** C.its wide range of temperatures D.the grade of mercury use in it .

Clinical thermometer has a constriction in its stem which prevents the mercury from falling back into the scale.

25. The clinical thermometer differs from other mercury in glass thermometer because it has I.a constriction II.a narrow bore III.a wide range IV.a short range. Which of the above are correct A.I and II **B.I and IV** C.II and III D.I,II and IV E.I,II,III and IV.

Demystified Series by Dr Timothy

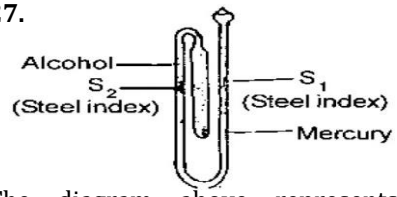
Clinical thermometer has a constriction or kink or bent region which prevents the back flow of mercury into the scale thereby preventing the temperature read by it from falling back. It also has a short temperature range (35℃ − 45℃). All liquid-in-glass thermometers have narrow bore.

26. 

The diagram above shows a maximum and minimum thermometer divided into three portions *P*, *Q* and *R*. Which of the following is true about the respective of contents of *P*, *Q* and *R*? **A.Alcohol, mercury and alcohol** B.Air, alcohol and mercury C.Mercury, alcohol and mercury D.Air, mercury and alcohol

NB – The bulb above *R* contains air.

27.



The diagram above represents the Six's maximum and minimum thermometer. If the temperature of the surrounding falls, which of the following correctly states how the steel indices in the thermometer would respond to the change in temperature? A.S_1 would be moved up, S_2 would not move **B.S_1 would not move but S_2 would move upward** C.both S_1 and S_2 would move upwards D.S_1 would move upwards and S_2 would move downwards E.S_1 would move downwards and S_2 would move upwards

When the temperature of the surrounding falls, the steel index S_2 in the minimum(or left) arm moves upwards while the steel index S_1 in the maximum(or right) arm would not move.

When the temperature of the surrounding rises, the steel index S_1 in the maximum(or right) arm moves upwards while the steel index S_2 in the minimum(or left) arm would not move.

28. The absolute zero temperature is defined as the temperature at which A.the pressure of a real gas is maximum B.the temperature of a gas is 273°C **C.thermal motion ceases** D.the volume of real gases is maximum E.ice melts.

At absolute zero i.e $0K$ or -273℃, the volume of a gas is theoretically zero.

29. Absolute zero temperature can be defined as the temperature A.at which pure water and steam co-exist **B.at which the average kinetic energy of particles making up a body is zero** C.of zero degree Celsius scale D.at which pure water changes to ice at standard atmospheric pressure.

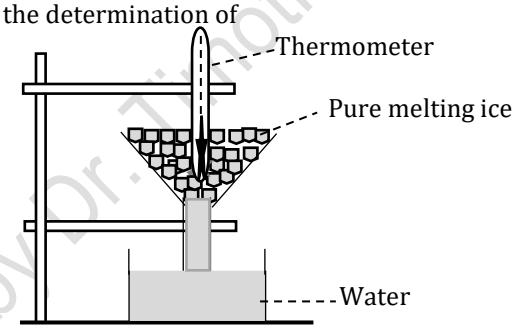
30. Which of the following statements about the lower fixed point of a thermometer using the Celcius scale is not correct? It is A.different from absolute zero of the temperature B.the zero mark of this scale **C.the temperature of a mixture of ice and salt** D.the temperature of pure melting ice at a pressure of 760mm of mercury D.the freezing point of pure water at a pressure of 760mm of mercury

Lower fixed point or ice point is the temperature of a pure melting ice at a pressure of 760mmHg or the freezing point of pure water at a pressure of 760mmHg. It is the zero mark of the Celcius scale i.e 0°C. The absolute zero in Celcius scale is -273°C, hence it is different from the lower fixed poin which is 0°C.

31. A hypsometer is used for determining **A.steam point** B.relative humidity C.ice point D.relative density.

Hypsometer is used for determining the upper fixed point or steam point.

32. The diagram below represents the process for the determination of



A.upper fixed point **B.lower fixed point** C.melting point D.freezing point.

The diagram shows the use of a glass funnel and pure melting ice in determining the lower fixed point or ice point.

33. A temperature of 20°C is the same as A.36°F **B.68**°F C.11.1°F D.43.1°F

$\frac{t_{\theta_C}-t_{L_C}}{t_{U_C}-t_{L_C}}=\frac{t_{\theta_F}-t_{L_F}}{t_{U_F}-t_{L_F}}$, temperature in Celcius scale t_{θ_C}=20°C, L.F.P in Celcius scale t_{L_C}=0°C, U.F.P in Celcius scale t_{U_C}=100°C, temperature in Fahrenheit scale t_{θ_F}=?, L.F.P in Fahrenheit scale t_{L_F}=32°F, U.F.P in Fahrenheit scale t_{U_F}=212°F,

$\frac{20-0}{100-0}=\frac{t_{\theta_F}-32}{212-32}$, $\frac{20}{100}=\frac{t_{\theta_F}-32}{180}$,

$t_{\theta_F}-32=\frac{2}{10}\times 180=36$, $t_{\theta_F}=36+32=68$°F.

OR $\theta=\frac{F_\theta-F_L}{F_U-F_L}\times 100$, temperature in celcius scale θ=20°C, L.F.P in Fahrenheit scale F_L=32°F, U.F.P in Fahrenheit scale F_U=212°F, temperature in Fahrenheit scale F_θ=?,

$20=\frac{F_\theta-32}{212-32}\times 100$, $20=\frac{F_\theta-32}{180}\times 100$,

$F_\theta-32=\frac{20\times 180}{100}=36$, $F_\theta=36+32=68$°F.

34. A faulty thermometer registers 102.5°C at 100°C. If the thermometer has zero error, what temperature will it register at 55°C? A.54.6°C B.55.0°C C.56.0°C **D.56.4**°C.

The thermometer has U.F.P or steam point error but no L.F.P or ice point or zero error. Hence, for

Demystified Series by Dr. Timothy

the faulty thermometer, U.F.P or U=102.5℃, L.F.P or L=0℃, θ_f=?, for the fautless thermometer, U.F.P or U=100℃, L.F.P or L=0℃, θ=55℃,
$\frac{\theta_f-0}{102.5-0}=\frac{55-0}{100-0}$, $\frac{\theta_f}{102.5}=\frac{55}{100}$,
$\theta_f=\frac{55}{100}\times 102.5=56.375℃=56.4℃$.

35. The lower fixed point of a faulty thermometer reads 2℃ while the upper fixed point is 100℃. What is the true temperature when the thermometer reads 51℃? A.52℃**B.50**℃ C.49℃ D.46℃.
Hence, for the faulty thermometer: U.F.P or U=100℃, L.F.P or L=2℃, θ_f=51℃, for the fautless thermometer: U.F.P or U=100℃, L.F.P or L=0℃, θ=?, $\frac{51-2}{100-2}=\frac{\theta-0}{100-0}$, $\frac{49}{98}=\frac{\theta}{100}$, $\theta=\frac{49}{98}\times 100$
$\theta=50℃$. OR $\theta=\frac{t_\theta-t_L}{t_U-t_L}\times 100$, $\theta=\frac{51-2}{100-2}\times 100$
$\theta=\frac{48}{98}\times 100=50℃$.

36. A thermometer with an arbitrary scale of equal divisions registers $-30°S$ at the ice point and $+90°S$ at steam point. Calculate the Celsius temperature corresponding $60°S$ A.25.0℃ B.50.0℃ C.66.7℃ **D.75.0**℃.
$t_L=-30°S$, $t_U=+90°S$, $t_\theta=60°S$, θ_U=100℃, θ_L=0℃, θ=?, $\frac{t_\theta-t_L}{t_U-t_L}=\frac{\theta-\theta_L}{\theta_U-\theta_L}$,
$\frac{60-(-30)}{90-(-30)}=\frac{\theta-0}{100-0}$, $\frac{60+30}{90+30}=\frac{\theta}{100}$, $\frac{90}{120}=\frac{\theta}{100}$,
$\theta=\frac{90}{120}\times 100=75℃$. OR
$\theta=\frac{S_\theta-S_L}{S_U-S_L}\times 100=\frac{60-(-30)}{90-(-30)}\times 100$,
$\theta=\frac{60+30}{90+30}\times 100=\frac{90}{120}\times 100=75℃$.

37. The ice and steam points on a mercury-in-glass thermometer are 10cm and 30cm repectively. Calculate the temperature in degree celcius, when the mercury meniscus is at the 14cm mark **A.20**℃ B.30℃ C.34℃ D.70℃.
L_L=10cm, L_U=30cm, L_θ=14cm, θ_L=0℃, θ_U=100℃, θ=?, $\frac{L_\theta-L_L}{L_U-L_L}=\frac{\theta-\theta_L}{\theta_U-\theta_L}$,
$\frac{14-10}{30-10}=\frac{\theta-0}{100-0}$, $\frac{4}{20}=\frac{\theta}{100}$, $\theta=\frac{4}{20}\times 100$,
$\theta=20℃$. OR $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100=\frac{14-10}{30-10}\times 100$
$\theta=\frac{4}{20}\times 100=20℃$.

38. The length of mercury thread when it is at 0℃, 100℃ and at an unknown temperature θ is 25mm, 225mm, 175mm respectively. The value of θ is A.85.0℃ B.80.0℃**C.75.0**℃ D.70.0℃
L_L=25mm, L_U=225mm, L_θ=175mm, θ_L=0℃, θ_U=100℃, θ=?, $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100$,
$\theta=\frac{175-25}{225-25}\times 100=\frac{150}{200}\times 100=75.0℃$.

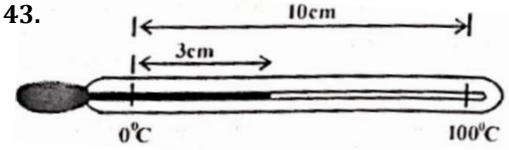
39. The ice and steam points of a certain thermometer are 25cm apart. If the length of mercury thread is 10cm above the ice point mark, calculate the temperature recorded by the thermometer in kelvin A.252K **B.313K** C.382K D.422K E.523K.
L_U-L_L=25cm, $L_\theta-L_L$=10cm, θ_L=0℃, θ_U=100℃, θ=?, $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100$,
$\theta=\frac{10}{25}\times 100=40℃$, $\theta=40+273=313K$.

40. Calculate the length which corresponds to a temperature of 20℃. If the ice and steam points of an ungraduated thermometer are 400mm apart A.20mm B.30mm C.60mm **D.80mm**
L_U-L_L=400mm, $L_\theta-L_L$=L=?, θ_L=0℃, θ_U=100℃, θ=20℃, $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100$,
$20=\frac{L}{400}\times 100=\frac{L}{4}$, $L=20\times 4=80$mm.

41. The lengths of the mercury thread in a thermometer when its bulb is placed in melting ice and steam are 4.0cm and 29.4cm respectively. If the length of the tread, when the bulb is place in a liquid is 19.0cm, calculate the temperature of the liquid A.74.8℃ B.65.5℃ C.59.1℃ D.40.9℃
L_L=4.0cm, L_U=29.4cm, L_θ=19.0cm, θ_L=0℃, θ_U=100℃, θ=?, $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100$,
$\theta=\frac{19.0-4.0}{29.4-4.0}\times 100=\frac{15.0}{25.4}\times 100=59.055℃$,
$\theta=59.1℃$.

42. The heights of the mercury thread in a mercury in glass thermometer when in melting ice and then in steam are 3cm and 18cm repectively. At a temperature 60℃ the height would be A.7.5cm B.9.0cm C.10.8cm **D.12.0cm** E.12.6cm.
L_L=3cm, L_U=18cm, L_θ=?, θ_L=0℃, θ_U=100℃, θ=60℃, $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100$, $60=\frac{L_\theta-3}{18-3}\times 100$,
$60=\frac{L_\theta-3}{15}\times 100$, $L_\theta-3=\frac{60\times 15}{100}=9$,
$L_\theta=9+3=12.0$cm.

43.



The diagram above illustrates the mercury-in-glass thermometer used for determining the temperature of a room. Use the data on it to determine the room temperature A.3.3℃ B.7.0℃ C.13.0℃ **D.30.0**℃.
L_U-L_L=10cm, $L_\theta-L_L$=3cm, θ_L=0℃, θ_U=100℃, θ=?, $\theta=\frac{L_\theta-L_L}{L_U-L_L}\times 100$, $\theta=\frac{3}{10}\times 100=30℃$.

44. A well lagged bar of 100cm has its ends maintained at 100℃ and 40℃ respectively, what is the temperature at a point 60cm from the hotter end? A.58℃ B.62℃**C.64**℃ D.76℃
A point 60cm from the hotter end 100℃ is same as 40cm from the colder end 0℃.
L_U-L_L=100cm, $L_\theta-L_L$=40cm, θ_L=40℃, θ_U=100℃, θ=?, $\frac{L_\theta-L_L}{L_U-L_L}=\frac{\theta-\theta_L}{\theta_U-\theta_L}$, $\frac{40}{100}=\frac{\theta-40}{100-40}$,
$\frac{40}{100}=\frac{\theta-40}{60}$, $\theta-40=\frac{40}{100}\times 60=24$,
$\theta=24+40=64℃$.

45. The readings on the pressure scale at the steam and ice points are 800mmHg and

Demystified Series by Dr Timothy

300mmHg respectively. Determine the corresponding temperature in degree Celsius when it reads 450mmHg A.15°C B.30°C C.45°C D.60°C E.80°C

P_U=800mmHg, P_L=300mmHg, P_θ=450mmHg, θ_U=100°C, θ_L=0°C, θ=?, $\frac{P_\theta - P_L}{P_U - P_L} = \frac{\theta - \theta_L}{\theta_U - \theta_L}$,

$\frac{450-300}{800-300} = \frac{\theta - 0}{100-0}$, $\frac{150}{500} = \frac{\theta}{100}$,

$\theta = \frac{150}{500} \times 100 = 30°C$. **OR** $\boldsymbol{\theta = \frac{P_\theta - P_L}{P_U - P_L} \times 100.}$

46. The pressure on the gas of a constant volume gas thermometer at ice point is 325mm of mercury and at the steam point is 875mm of mercury. Find the temperature when the pressure of the gas is 490mm of mercury A.30K B.243K C.300K **D.303K.**

P_U=875mmHg, P_L=325mmHg, P_θ=490mmHg, θ_U=100°C, θ_L=0°C, θ=?, $\theta = \frac{P_\theta - P_L}{P_U - P_L} \times 100$

$\theta = \frac{490-325}{875-325} \times 100 = \frac{165}{550} \times 100 = 30°C$, $\theta = 30+273 = 303K$.**OR**

P_U=875mmHg, P_L=325mmHg, P_θ=490mmHg, θ_U=373K, θ_L=273K, θ=?, $\frac{P_\theta - P_L}{P_U - P_L} = \frac{\theta - \theta_L}{\theta_U - \theta_L}$,

$\frac{490-325}{875-325} = \frac{\theta - 273}{373-273}$, $\frac{165}{550} = \frac{\theta - 273}{100}$,

$\theta - 273 = \frac{165}{550} \times 100 = 30$, $\theta = 30+273 = 303K$.

47. A thermometer record 680mmHg at steam point and 440mmHg at ice point. The temperature it records at 380mmHg is **A.-25**°C B.-20°C C.20°C D.25°C.

P_U=680mmHg, P_L=440mmHg, P_θ=380mmHg, θ_U=100°C, θ_L=0°C, θ=?, $\theta = \frac{P_\theta - P_L}{P_U - P_L} \times 100$,

$\theta = \frac{380-440}{680-440} \times 100 = \frac{-60}{240} \times 100 = -25°C$.

48. The gas in a gas thermometer occupies 54cm^3 at 0°C and 74cm^3 at 100°C. What is the temperature when it occupies 46cm^3 if its pressure is constant A.-45°C**B.-40**°CC.40°C D.45°C.

V_U=74cm^3, V_L=54cm^3, V_θ=46cm^3, θ_U=100°C, θ_L=0°C, θ=?, $\theta = \frac{V_\theta - V_L}{V_U - V_L} \times 100$,

$\theta = \frac{46-54}{74-46} \times 100 = \frac{-8}{20} \times 100 = -40°C$.

49. A plantinum resistance thermometer has a resistance of 4Ω at 0°C and 10Ω at 100°C. Assuming the resistance changes uniformly with temperature, calculate the resistance of the thermometer when the temperature is 45°C A.14.0Ω **B.6.7Ω** C.3.2Ω D.2.7Ω.

R_U=10Ω, R_L=4Ω, R_θ=?, θ_U=100°C, θ_L=0°C, θ=45°C, $\theta = \frac{R_\theta - R_L}{R_U - R_L} \times 100$, $45 = \frac{R_\theta - 4}{10-4} \times 100$

$45 = \frac{R_\theta - 4}{6} \times 100$, $R_\theta - 4 = \frac{45 \times 6}{100} = 2.7$,

$R_\theta = 2.7+4 = 6.7\Omega$.

50. A plantinum resistance thermometer wire has a resistance of 5ohms at 0°C and 5.5ohms at 100°C. Calculate the temperature of the wire when the resistance is 5.2ohms A.80°C B.60°C C.40°C D.10°C.

R_U=5.5Ω, R_L=5Ω, R_θ=5.2Ω, θ_U=100°C, θ_L=0°C, θ=?, $\theta = \frac{R_\theta - R_L}{R_U - R_L} \times 100 = \frac{5.2-5}{5.5-5} \times 100$,

$\theta = \frac{0.2}{0.5} \times 100 = 40°C$.

51. Which of the following types of thermometers can be used to measure a range of temperatures from -50°C to 50°C? I.Clinical II.Mercury-in-glass III.Alcohol-in-glass A.I B.II **C.III** D.I and II

The range of a clinical thermometer is from 35°C to 43°C. Mercury freezes at -39°C, hence cannot measure a temperature below -39°C i.e 50°C, but boils at 3357°C. Alcohol freezes at -115°C and boils at 78°C, hence it is the most appropriate for measure a range of temperature from -50°C to 50°C.

52. A change in temperature of 45°C in equivalent to a change in temperature on the kelvin scale of A.25K **B.45K** C.81K D.318K.

53. The temperature of an object is raused by 120°C. The resulting increase in its absolute temperature is A.50K **B.120K** C.200K D.393K.

A temperature change of 45°C is the same as 45K change in temperature, as they have the same fundamental interval of 100.

53. A change in temperature of 90°C is equivalent to a change in temperature on the Fahrenheit scale of A.50°F B.90°F**C.162**°FD.194°F

A division on the Celcius scale is $\frac{9}{5}$ times the size of a division on the Fahrenheit scale.

A temperature change of 90°C is the same as $\frac{9}{5} \times 90°F = 162°F$.

Jamb past questions on temperature and its measurement :

[1978/17,1979/14,1980/7,1981/1,42,1983/35, 1984/18,29,1987/16,1989/13,1990/15,1991/ 15,1992/13,1993/14,1994/20,1995/18,23,26, 1997/20,1998/17,2000/21,2005/43,2006/48, 2007/31,39,2008/1,2009/18,2010/15,2012/19 ,2013/16,2014/19]

Demystified Series by Dr Timothy

Chapter 12 – Thermal Expansivity

● **Thermal expansivity –** Thermal expansivity is the increase in size or dimension of an object when heat is applied to it.

- Thermal expansion is due to an increase in the amplitude of vibration and motion of molecules of objects when heated.

- Expansion in solids – Expansion in solids is regular i.e expands when heated and contracts when cooled. Expansion is solids can be linear or length, area or superficial and volume or cubic expansion.

1.Linear or length expansivity α – This is the increase in length per unit length per degree rise in temperature.

$\boldsymbol{\alpha = \frac{increase\ in\ length}{original\ length \times temp.\ rise} = \frac{\Delta L}{L_1 \Delta\theta} = \frac{L_2 - L_1}{L_1(\theta_2 - \theta_1)}}$.

ΔL=change in length or linear expansion of solid $= L_2 - L_1$, L_2=final length at θ_2, L_1=initial length at θ_1. $\boldsymbol{\Delta L = \alpha L_1 \Delta\theta}$, $\boldsymbol{L_2 = L_1(1 + \alpha\Delta\theta)}$.

- When the temperature decreases and the length also decreases , $\theta_2 < \theta_1$ or $L_2 < L_1$, the formula becomes – $\boldsymbol{\alpha = \frac{L_1 - L_2}{L_1(\theta_1 - \theta_2)}}$.

- Safety gap or gap between metals is same as expansion $= \Delta L = L_2 - L_1$.

- Change in d iameter is a linear expansion not area or superficial expanision.

- The relationship between the linear expansivity α, Young modulus E, area A, force or tension F acting on a solid is given by – $\boldsymbol{F = EA\alpha\Delta\theta}$.

2.Area or superficial expansivity β – This is the increase in area per unit area per degree rise in temperature.

$\boldsymbol{\beta = \frac{increase\ in\ area}{original\ area \times temp.\ rise} = \frac{\Delta A}{A_1 \Delta\theta} = \frac{A_2 - A_1}{A_1(\theta_2 - \theta_1)}}$.

ΔA=increase in area=$A_2 - A_1$, A_2=final area at θ_2 , A_1=initial area at θ_1. $\boldsymbol{\Delta A = \beta A_1 \Delta\theta}$, $\boldsymbol{A_2 = A_1(1 + \beta\Delta\theta)}$.

- When the temperature decreases and the area also decreases , $\theta_2 < \theta_1$ or $A_2 < A_1$, the formula becomes – $\boldsymbol{\beta = \frac{A_1 - A_2}{A_1(\theta_1 - \theta_2)}}$.

- Area expansivity of a material is twice its linear expansivity. $\beta = 2\alpha$.

3.Volume or cubic expansivity γ – This is the increase in volume per unit volume per degree rise in temperature.

$\boldsymbol{\gamma = \frac{increase\ in\ volume}{original\ volume \times temp.\ rise} = \frac{\Delta V}{V_1 \Delta\theta} = \frac{V_2 - V_1}{V_1(\theta_2 - \theta_1)}}$.

ΔV=increase in volume=$V_2 - V_1$, V_2=final volume at θ_2 , V_1=initial volume at θ_1.

$\boldsymbol{\Delta V = \gamma V_1 \Delta\theta}$, $\boldsymbol{V_2 = V_1(1 + \gamma\Delta\theta)}$.

- When the temperature decreases and the volume also decreases , $\theta_2 < \theta_1$ or $V_2 < V_1$, the formula becomes – $\boldsymbol{\gamma = \frac{V_1 - V_2}{V_1(\theta_1 - \theta_2)}}$.

- Volume expansivity of a material is thrice its linear expansivity. $\gamma = 3\alpha$.

- Volume expansivity and area expansivity of a material are related by – $2\gamma = 3\beta$.

- Percentage in the volume of a solid after expansion $\%\Delta V = \frac{\Delta V}{V_1} \times 100 = \frac{V_2 - V_1}{V_1} \times 100$,

$\%\Delta V = \gamma\Delta\theta \times 100$.

- In calculating for percentage change in length or area, replace volume V with length L or area A and volume expansivity γ with linear expansivity α or area expansivity β.

- Night time temperature is same as the lowest temperature while day time temperature is same as the highest temperature.

- Useful application of expansion in solids – I. Fitting of wheels on rims**. II.**Riveting of steel plates. **III.**Bimetallic strips used in thermometers and thermostart(i.e fire alarms,compensation of expansion in watches and clocks,laundry iron e.t.c). **IV.**Removal of tight glass stoppers. **V.**Roller bearings in metal bridges

- Unuseful application of expansion in solids – I. Bursting of water pipes in cold weather. **II.**Collapsing of metal bridges. **III.**Cracking of thick glasses due to uneven expansion i.e inner wall expanding more than the outer wall. **IV.**Sagging of telegraph wire. **V.**Buckling of rail way tracks.

- Thermostats – They are used to control temperature of laundry iron,aquaria for tropical fish,heated apartments and hot water storage tanks. **It is not used in a pressure cooker.**

- A **bimetallic strip** consists of two metals with different expansivities i.e copper(or brass) and steel(or iron). **The metal with high expansivity (brass or copper) occupies the upper part while the other metal occupies the lower part(steel or iron).**

- When the bimetallic strip is heated, the metal with high expansivity occupies the convex part and also occupies the concave part when the strip is cooled.

- The order of increasing expansivity in some common solids is – **iron→copper→silver→ aluminium.**

- Expansion in liquids – Expansion in liquids unlike solids, is irregular. **Liquids expands more than solids.**

- When a vessel containing a liquid is heated, the liquid level first drops,due to the expansion of the vessel,the liquid level later rises as it also expands as more heat is supplied.

- The expansion in liquids involves apparent cubic expansion (expansion of the liquid only) and real cubic expansivity (expansion of liquid and container).

- Apparent cubic expansivity γ_a of a liquid is the apparent increase in volume per unit volume per degree rise in temperature when the liquid is

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heated in an inexpansible vessel. It is amount (mass or volume or weight) of liquid expelled to the amount(mass or volume or weight) remaining per degree rise in temperature.

$\gamma_a = \frac{apparent\ increase\ in\ volume}{original\ volume \times temp.\ rise}$,

$\gamma_a = \frac{amount\ of\ liquid\ expelled}{amount\ of\ liquid\ remaining \times temp.\ rise}$,

Amount of liquid expelled=Amount of liquid at initial temp − Amount of liquid at the final temp. Amount of liquid remaining is the amount of the liquid at final temp. The amount of liquid can be in mass or volume.

- Real or Absolute cubic expansivity γ_r of a liquid is the increase in volume per unit volume per degree rise in temperature in an expansible vessel.

- Real cubic expansivity of a liquid is always greater than its apparent cubic expansivity ($\gamma_r > \gamma_a$).

- Real cubic expansivity of a liquid is equal to the sum of the apparent cubic expansivity of the liquid and the cubic expansivity of the of the vessel or container. $\gamma_r = \gamma_a + \gamma_c$. The cubic expansivity of the vessel can be gotten from its linear expansivity , $\gamma_c = 3\alpha_c$.

- The density ρ of substance can be used in place of volume in volume or cubic expansivity as they are inversely related – Hence, $\rho_1 = \rho_2(1 + \gamma\Delta\theta)$.

- Anomalous or Abnormal expansion of water : Most liquids expand with temperature rise (heated) and contract with temperature fall (cooled) over all ranges of temperatures.

- Water contracts from 0℃ to 4℃ (temperature rise) and expands from 4℃ to 0℃ (temperature fall) (anomalous or abnormal expansion and contraction).The density increases while the volume decreases between 0℃ to 4℃. Hence, **anomalous or abnormal behavior of water occurs between 0℃and 4℃.**

- Below 0℃ and above 4℃, water behaves like all other liquids i.e expands with temperature rise and contracts with temperature fall.

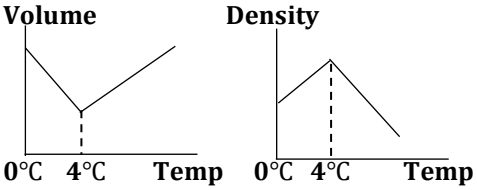
- Water has its maximum or highest volume and minimum or least density at 0℃while it has its minimum or least volume and maximum or highest density at 4℃.

- Hope's experiement using Hope's apparatus is used for the illustration and verification of the anomalous behavior of water.

- The anomalous behavior or expansion of water makes ponds, lakes or rivers to freeze from the top surface rather than from the bottom.Hence, marine lives can survive during winter since ice forms at the surface of the water while the bottom of the water remains at 4℃, a temperature warm enough for aquatic life.

- The cubic expansivity of water is negative between 0℃ and 4℃ (the change volume is negative when water is heated from 0℃ to 4℃).

- Graphical representation of anomalous expansion of water –



Examples :

1. Which of the following is true about the expansion of matter when heated? A.The distances between the molecules decrease B.The force of attraction between the molecules increases C.The kinetic energy of the molecules increases without an appreciable increase in their size D.The molecules decrease **E.The molecules increase in size.**

When a substance is heated , the kinetic energy of the molecules increase and they overcome the force of attraction between the molecules i.e the force of attraction decreases, and then move further apart i.e distance between molecules increases, resulting in an increase in size of the substance.

2. The magnitude of the expansion or contraction of a substance depends on the I.temperature change II.nature of the substance III.size of the substance. Which of the statements above are correct? A.I and II B.I and III C.II and III **D.I,II and III.**

$\alpha = \frac{increase\ in\ length}{original\ length \times temp.\ rise} = \frac{\Delta L}{L_1 \Delta\theta}$,

Expansion/contraction or increase in length, $e = \Delta L = L_1\alpha\Delta\theta$, hence expansion or contraction depends on temperature change or rise in temperature $\Delta\theta$, nature of the substance or linear expansivity of the substance α and original length or size of the substance L_1.

3. A hollow steel piston fits exactly into a steel cylinder when both are at 60℃. Whe the temperature just falls to 19℃, A.The piston contracts and the cylinder expands B.Both contract but the cylinder contracts more than the piston C.Both contract but the piston contract more than the cylinder **D.The piston still fits the cylinder exactly** E.Only the cylinder contracts, the piston remains the same.

The cylinder and the piston are both made of steel hence, they have the same linear expansivity. Therefore, when the temperature of the cylinder and the piston falls from 60℃ to 19℃, they both contract to the same extent and the piston fits the cylinder exactly.

4. In which of the following devices is the expansion of solid a disadvantage? I.Fire alarms II.Thermostat III.Simple pendulum IV.Bimetallic thermometer A.I B.II **C.III** D.I and IV.

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Expansion of solids is a disadvantage in simple pendulum,balance wheel of a watch i.e makes it to lose time,buildings,bridges,railway lines, electric transmission and telegraph wires.
Expansion of solids is an advantage in fitting steel tyres on steel wheels, rivets or riveted joint of steel plates,thermostat and fire alarm system.

5. The expansion of solids can be considered a disadvantage in the A.fire alarm system **B.balance wheel of a watch** C.fitting wheels on rims D.riveting of steel plates.

6. Large metal bridges are often mounted in roller bearings in order to A.prevent the bridges from sinking into the ground B.prevent the bridges from changing theur lengths **C.allow for the expansion and contraction of the bridges** D.allow movement of bridges from one side to another.
Roller bearings and contraction of bridges to prevent the bending of bridges.

7. Which of the following is not an application ethermometer B.Temperature control in an electric iron **C.Sagging of telegraph wires** D.Compensated balance wheel of a watch.
Sagging of telegraph wires is an undesirable event which leads to snapping of telegraph wires or electric transmission cables.

8. During summer, the balance wheel of a clock expands. What effect does this have on the accuracy of the clock? A.The clock gains time B.The accuracy of the clock is not affected **C.The clock loses time** D.The clock stops working.
Increase in temperature increases the diameter of the balance wheel and weakens the elasticity of the hairspring, thus causing it to lose time.

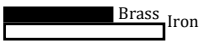
9. One of the most important applications of bimetallic strip is found in the construction of A.An altimeter B.A thermocouple C.A hygrometer **D.A thermostat.**

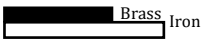
10. Which of the following statements is not true? Thermostats are used to control the temperature of **A.pressure cookers** B.laundry irons C.hot water storage tanks D.aquaria for tropical fishes E.heated apartments
Thermomstat is used when it is desirable to maintain a system at a certain steady temperature. Thermostats are not used in pressure cookers.

11. The flasher unit of a motor vehicle works on the principle that A.metals are good conductors of heat **B.a bimetallic strip expands and bends when heated** C.the linear expansivity of one metal of a bimetallic strip changes while that of the other remains constant when heated D.the linear expansivity of a bimetallic strip changes as the temperature changes E.the linear expansivity of two metals always add up to unity.
The flasher unit of a motor vehicle is made of a bimetallic strip and works on the principle that expands and bends inorder to complete the circuit hence, switches the flash light on.

12. The design of the thermostat of an electric iron is based on the A.apparent expansion of a body B.change in the resistance of a body on heating C.thermo-electric effect D.change in the density of a body when heated **E.expansion of metals when heated.**

13. A bimetallic strip of iron and brass was heated. Which of the following diagrams accurately after heating?

A. 

B 

C. 

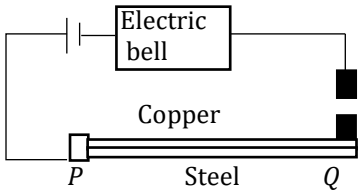
D. 

When a bimetallic strip is heated, the metal at the convex or upper part i.e brass, of the curved bimetallic strip is of higher expansivity strip is of higher linear expansivity that the other metal i.e iron.**Option C best describes it**
Option D illuustrates a bimetallic strip when cooled.

14. 

The figure shows a composite bar of two metal strips *X* and *Y* which has been heated. The curvature observed is as shown because A.the two strips of metal expand in the same direction B.*X* expands more than *Y* for the same temperature C.the two strips become softer when heated and the weight causes bending D.Y expands more than *X* at the same temperature E.*Y* expands more than *X* at different temperatures.
X and *Y* are two different metals, hence their coefficient of linear expansivity is also different i.e expands differently.In a bimetallic strip, the metal in the upper or convex part has a greater linear expansivity or expands more than the other metal at the lower part.

15. Which of the following statements cearly describes the behaviour of thw fire alarm as shown below, given that the linear expansivities of copper and steel are $2.0\times10^{-5}°C^{-1}$ and $1.2\times10^{-5}°C^{-1}$?



I.The bimetallic strip will not be able not close the circuit when there is fire II.The bimetallic strip will close the circuit when there is fire III.If the copper and steel are interchanged, the circuit will close when there is fire A.I B.II C.III **D.I and III** E.II and III.
The bimetallic strip upon heating when there is fire, will bend away from the key of the electric

Demystified Series by Dr Timothy

bell, as copper has higher linear expansivity than steel (copper expands more than steel). If the copper and steel are interchanged (copper is outside while the steel inside), the bimetallic strip will bend towards the key and close the circuit upon heating.

16. The difference in length between two metal rods of different materials is constant at all tempetures. Which of the following conclucisons is correct? A.The linear expansivities of the two metal rods are the same B.The increase in length of both metal rods is the same at all temperatures C.The difference in length is equal to the length of the shorter rod D.The sum of their lengths is always constant E.The increase in length at all temperature is equal to the new length of the shorter rod.

Diffrence in length is same as increase or change in length hence, the increase in length of both metal rods is the same at all temperatures.

17. A rod of initial length 2m at a temperature of 25°C is heated to 80°C. Calculate the increase in length of the rod if its linear expansivity is $4.0\times10^{-3}K^{-1}$ A.0.26m **B.0.44m** C.0.53m D.0.84m.

L_1=2m, θ_1=25°C, θ_2=80°C, $\alpha=4.0\times10^{-3}K^{-1}$, increase in length ΔL=?, $\alpha=\frac{\Delta L}{L_1\Delta\theta}$, $\Delta L=\alpha L_1\Delta\theta$,

$\Delta L=\alpha L_1(\theta_2-\theta_1)$ $=4.0\times10^{-3}\times2(80-25)$

$\Delta L=8.0\times10^{-3}\times55=0.44$m.

18. A metal rod 800mm I heated from 10°C to 95°C expands by 1.36mm, the linear expansivity of the metal is A.$2.0\times10^{2}K^{-1}$ B.$2.0\times10^{-2}K^{-1}$ C.$5.0\times10^{-3}K^{-1}$ **D.$2.0\times10^{-5}K^{-1}$.**

L_1=800mm, θ_1=10°C, θ_2=95°C, increase in length or expansion ΔL=1.36mm, α=?,

$\alpha=\frac{\Delta L}{L_1\Delta\theta}=\frac{\Delta L}{L_1(\theta_2-\theta_1)}=\frac{1.36}{800(95-10)}=2.0\times10^{-5}K^{-1}$.

19. A metal rod 80cm long lengthens by 0.09cm when its temperature rises by 93.6°C. What is the linearv expansivity of the metal? A.11×10^{-6}°C^{-1} **B.12×10^{-6}°C^{-1}** C.15×10^{-6}°C^{-1} D.9.6×10^{-6}°C^{-1} E.10.4×10^{-6}°C^{-1}.

L_1=80cm, $\Delta\theta$ =93.6°C, ΔL=0.09cm, α=?,

$\alpha=\frac{\Delta L}{L_1\Delta\theta}=\frac{0.09}{80\times93.6}=\frac{0.09}{7488}=12\times10^{-6}$°C^{-1}.

20. A brass rod is 2m long at a certain temperature. What is the length for a temperature rise of $100K$, if the linear expansivity of brass is $18\times10^{-6}K^{-1}$**A.2.0036m** B.2.0018m C.2.1800m D.2.0360m

L_1=2m, $\Delta\theta$=100K, $\alpha=18\times10^{-6}K^{-1}$, L_2=?,

$L_2=L_1(1+\alpha\Delta\theta)=2(1+18\times10^{-6}\times100)$,

$L_2=2(1+0.0018)=2\times1.0018=2.0036$m.

21. What is the linear expansivity of a solid whose temperature rises from 10°C to 100°C and whose length increases from 110.0cm to 110.11cm? A.0.0011×10^{-6}°C^{-1} B.0.011×10^{-6}°C^{-1} C.0.11×10^{-6}°C^{-1} D.1.1×10^{-6}°C^{-1}E.11×10^{-6}°C^{-1}

θ_1=10°C, θ_2=100°C, L_1=110.00cm, L_2=110.11cm, α=?, $\alpha=\frac{\Delta L}{L_1\Delta\theta}=\frac{L_2-L_1}{L_1(\theta_2-\theta_1)}=\frac{110.11-110.0}{110.0(100-10)}$,

$\alpha=\frac{0.11}{110.0\times90}=1.1\times10^{-5}$°C^{-1} $=0.11\times10^{-6}$°C^{-1}.

22. A wire of length 20m is cooled from 45°C to −5°C. Find the change in its length if its linear expansivity is $0.000020K^{-1}$. A.0.16m **B.0.020m** C.0.160m D.0.200m E.1.600m.

L_1=20m, θ_1=45°C, θ_2=−5°C, $\alpha=0.000020K^{-1}$, α $=2.0\times10^{-5}K^{-1}$, change in length ΔL=?,

The temperature decreases i.e cooling, hence the body contracts. $\alpha=\frac{L_1-L_2}{L_1(\theta_1-\theta_2)}=\frac{\Delta L}{L_1(\theta_1-\theta_2)}$,

$\Delta L=\alpha L_1(\theta_1-\theta_2)$ $=2.0\times10^{-5}\times20(45-(-5)$,

$\Delta L=40.0\times10^{-5}(50)=2.0\times10^{-2}$m $=0.020$m.

23. A wire of length 100.0m at 30°C has a linear expansivity of $2\times10^{-5}K^{-1}$. Calculate the length of the wire at a temperature of −10°C A.100.08m B.100.04m C.99.96m **D.99.92m**.

L_1=100.0m, $\alpha=2.0\times10^{-5}K^{-1}$, θ_1=30°C, θ_2=−10°C, L_2=?, The temperature decreases i.e cooling, hence the body contracts. $\alpha=\frac{L_1-L_2}{L_1(\theta_1-\theta_2)}$,

$L_1-L_2=\alpha L_1(\theta_1-\theta_2)$, $L_2=L_1-\alpha L_1(\theta_1-\theta_2)$,

$L_2=100-2.0\times10^{-5}\times100(30-(-10)$,

$L_2=100-2.0\times10^{-5}\times100\times40=100-0.08$,

$L_2=99.92$m.

24. When a brass of length 100cm at 50°C is heated, its length changes to 100.054cm at [linear expansivity of brass = $0.000018K^{-1}$] A.51°C B.54°C C.72°C**D.80**°C E.83°C.

L_1=100.0m, θ_1=50°C, L_2=1000.54cm, $\alpha=0.000018K^{-1}=1.8\times10^{-5}K^{-1}$, θ_2=?,

$\alpha=\frac{\Delta L}{L_1\Delta\theta}=\frac{L_2-L_1}{L_1(\theta_2-\theta_1)}$, $(\theta_2-\theta_1)=\frac{L_2-L_1}{L_1\alpha}$,

$\theta_2-50=\frac{100.054-100}{100\times1.8\times10^{-5}}=\frac{0.054}{0.0018}=30$,

$\theta_2=30+50=80$°C.

25. An iron rod is 2.58m long at 0°C. Calculate the length of a brass rod at 0°C, if the difference between the lengths of the two rods must remain the same at all temperature [linear expansivity of iron=$1.2\times10^{-5}K^{-1}$, linear expansivity of brass=$1.9\times10^{-5}K^{-1}$] A.1.61m B.1.62m **C.1.63m** D.1.64m.

For iron – original length of iron L_I=2.58m, linear expansivity of iron $\alpha_I=1.2\times10^{-5}K^{-1}$, increase or difference in length of iron ΔL_I , temperature rise $\Delta\theta$, $\therefore\alpha_I=\frac{\Delta L_I}{L_I\Delta\theta}$,

For brass – original length of brass L_B=?, linear expansivity of brass $\alpha_B=1.9\times10^{-5}K^{-1}$, increase or difference in length of brass ΔL_B , temperature rise $\Delta\theta$, $\therefore\alpha_B=\frac{\Delta L_B}{L_B\Delta\theta}$,

Temperature rise $\Delta\theta$ for both iron and brass is the same, also the increase or difference in lengths of iron and brass are equal , $\Delta L_I=\Delta L_B$, $\Delta L_I=\alpha_I L_I\Delta\theta$, $\Delta L_B=\alpha_B L_B\Delta\theta$,

$\therefore\alpha_I L_I\Delta\theta=\alpha_B L_B\Delta\theta$, $\alpha_I L_I=\alpha_B L_B$,

$L_B=\frac{\alpha_I L_I}{\alpha_B}=\frac{1.2\times10^{-5}\times2.58}{1.9\times10^{-5}}=1.63$m.

26. The length of an iron bar is 100cm at 20°C. At what temperature will its length increase by 0.01%? [linear expansivity of iron=1.2×10^{-5}°C^{-1}]

Demystified Series by Dr. Timothy

A.48.0°C B.38.0°C **C.28.3**°C D.23.0°C.
L_1=100cm, θ_1=20°C, L_2 is increased by 0.01% of L_1, $L_2 = L_1 + 0.01\%L_1 = L_1 + \frac{0.01}{100}L_1$, change in length $\Delta L = L_2 - L_1 = \frac{0.01}{100}L_1$,
α=2.0×10^{-5}°C^{-1}, $\alpha = \frac{\Delta L}{L_1\Delta\theta} = \frac{\Delta L}{L_1(\theta_2-\theta_1)}$,
$(\theta_2 - \theta_1) = \frac{\Delta L}{\alpha L_1}$, $\theta_2 - 20 = \frac{\frac{0.01}{100}L_1}{1.2\times10^{-5}\times100}$,
$\theta_2 - 20 = \frac{0.01}{100}\times 100 \times \frac{1}{1.2\times10^{-5}\times100} = 8.3$,
$\theta_2 = 8.3+20 = 28.3$°C.

27. Stee; bars, each of length 3m at 28°C are to be used for constructing a rail line. If the linear expansivity of the steel is $1.0\times10^{-5}K^{-1}$, what is the safety gap that must be left between successive bars if the highest temperature expected is 40°C? A.1.2×10^{-1}cm B.7.2×10^{-2}cm C.6.0×10^{-2}cm **D.3.6×10^{-2}cm** E.1.8×10^{-2}cm.
L_1=3m=300cm, θ_1=28°C, α=1.0×10$^{-5}K^{-1}$,
θ_2=40°C, safety gap = expansion or change in length ΔL. $\alpha = \frac{\Delta L}{L_1\Delta\theta}$, $\Delta L = \alpha L_1\Delta\theta$,
$\Delta L = \alpha L_1(\theta_2 - \theta_1)$ =1.0×10^{-5}×300(40−28) ,
ΔL = 3.0×10^{-3}×12= 36×10^{-3}cm = 3.6×10^{-2}m.

28. A bridge made of steel is 600m long. What is the daily variation in its length if the night time and day time temperatures are 10°C and 35°C repetively (linear expansivity of steel is 0.000012°C^{-1}) A.0.18cm B.1.80cm **C.18cm** D.1800cm.
L_1=600m, α=0.000012°C^{-1}=1.2.0×10^{-5}°C^{-1},
θ_1=10°C, θ_2=35°C, daily variation in length = increase in length ΔL=?,
$\alpha = \frac{\Delta L}{L_1\Delta\theta}$, $\Delta L = \alpha L_1\Delta\theta = \alpha L_1(\theta_2 - \theta_1)$,
ΔL =1.2×10^{-5}×600(35−10) = 720×10^{-5}×25,
ΔL = 0.18m = 18cm.

29. A 10km railwayline is laid when the temperature is 2°C. If the maximum temperature in the region is 32°C, what should be the minimum total length of all gaps between the iron rails if the line is laid in strips? (Linear expansivity of iron= 0.000012°C^{-1}) A.7.68m B.4.08m C.3.84m **D.3.6m** E.0.24m
L_1=10km=10000m, θ_1=2°C, θ_2=32°C, α=0.000012°C^{-1}=1.2×10^{-5}°C^{-1}, gap between rails = expansion or increase in length ΔL=?,
$\alpha = \frac{\Delta L}{L_1\Delta\theta}$, $\Delta L = \alpha L_1\Delta\theta = \alpha L_1(\theta_2 - \theta_1)$,
ΔL =1.2×10^{-5}×10000(32−2)=1.2×30=3.6m.

30. The ratio of the linear expansivity of copper to that of iron iss approximately 1.5. A specimen of iron and a specimen of copper expands by the same amount per unit rise in temperature. The ratio of their length is A.3 B.1.5 C.1.32 D.1 **E.0.67**
For copper – $\Delta L_C = \alpha_C L_C\Delta\theta$, For iron – $\Delta L_I = \alpha_I L_I\Delta\theta$, the expansion or increase in length of both copper amd iron are equal.
$\Delta L_C = \Delta L_I$, $\alpha_C L_C\Delta\theta = \alpha_I L_I\Delta\theta$,
$\frac{L_C}{L_I} = \frac{\alpha_I\Delta\theta}{\alpha_C\Delta\theta} = \frac{\alpha_I}{\alpha_C}$, α_C:$\alpha_I = \frac{\alpha_C}{\alpha_I} = 1.5$, $\frac{\alpha_I}{\alpha_C} = \frac{1}{1.5}$
$\frac{L_C}{L_I} = \frac{1}{1.5} = \frac{2}{3} = 0.67$, L_C:L_I = 0.67:1 or 0.67.

31. The linear expansivity of a metal P is twice that of another metal Q.When these materials are heated through the same temperature change, their increase in length is the same. Calculate the ratio of the original length of P to that of Q A.1:4 **B.1:2** C.2:1 D.4:1 E.6:1.
For metal P – $\Delta L_P = \alpha_P L_P\Delta\theta$, For metal Q – $\Delta L_Q = \alpha_Q L_Q\Delta\theta$, the increase in length of both metal P and Q are equal. $\Delta L_P = \Delta L_Q$,
$\alpha_P L_P\Delta\theta = \alpha_Q L_Q\Delta\theta$, $\frac{L_P}{L_Q} = \frac{\alpha_Q\Delta\theta}{\alpha_P\Delta\theta} = \frac{\alpha_Q}{\alpha_P}$, $\alpha_P = 2\alpha_Q$,
$\therefore \frac{L_P}{L_Q} = \frac{\alpha_Q}{2\alpha_Q}$, $\frac{L_P}{L_Q} = \frac{1}{2}$, L_P:L_Q = 1:2.

32. The ratio of the coefficient of linear expansion of two metals $\frac{\alpha_1}{\alpha_2}$ is 3:4. If when heated through the same temperature change, the ratio of the increase in lengths of two metals $\frac{e_1}{e_2}$ is 1:2, the ratio of the original lengths $\frac{L_1}{L_2}$ is A.$\frac{3}{2}$ B.$\frac{3}{8}$ C.$\frac{8}{3}$ **D.$\frac{2}{3}$.**
$\frac{\alpha_1}{\alpha_2} = 3{:}4 = \frac{3}{4}$, $\frac{\Delta L_1}{\Delta L_2} = \frac{e_1}{e_2} = 1{:}2 = \frac{1}{2}$, $\frac{L_1}{L_2}$=? ,
$\frac{\Delta L_1}{\Delta L_2} = \frac{e_1}{e_2} = \frac{\alpha_1 L_1\Delta\theta}{\alpha_2 L_2\Delta\theta} = \frac{\alpha_1 L_1}{\alpha_2 L_2} = \frac{\alpha_1}{\alpha_2}\times\frac{L_1}{L_2}$, $\frac{1}{2} = \frac{3}{4}\times\frac{L_1}{L_2}$,
$\frac{L_1}{L_2} = \frac{1\times4}{2\times3}$, $\frac{L_1}{L_2} = \frac{2}{3}$ or 2:3.

33. A metal sheet of area 100cm^2 was heated through 70°C. Calculate its new area if the linear expansivity is $0.000017K^{-1}$ A.100.06cm^2 B.100.12cm^2**C.100.24cm^2** D.100.36cm^2.
A_1=100cm^2, $\Delta\theta$=70°C, α=0.000017K^{-1}, β=2α,
β=2×0.000017=0.000034K^{-1}=3.4×10$^{-5}K^{-1}$,
A_2=?, $\beta = \frac{\Delta A}{A_1\Delta\theta} = \frac{A_2-A_1}{A_1(\theta_2-\theta_1)}$,
$A_2 = A_1(1 + \beta\Delta\theta) = 100(11+3.4\times10^{-5}\times70)$,
$A_2 = 100(1+0.00238) = 100\times1.00238$,
$A_2 = 100.238$cm^2= 100.24cm^2.

34. A thin aluminium plate of surface area 1.500m^2 at 20°C. What will be its surface area when it is cooled to −20°C? (Take the linear expansivity of aluminium to be $2.5\times10^{-5}K^{-1}$) A.1.503m^2 B.1.500m^2 C.1.498m^2 **D.1.497m^2** E.1.490m^2.
A_1=1.500m^2, θ_1=20°C, θ_2=−20°C, α=2.0×10$^{-5}K^{-1}$, β=2α, β=2×2.5×10^{-5}=5.0×10$^{-5}K^{-1}$, temperature decreases and area also decreases.
$\beta = \frac{\Delta A}{A_1\Delta\theta} = \frac{A_1-A_2}{A_1(\theta_1-\theta_2)}$, $A_1 - A_2 = \beta A_1(\theta_1 - \theta_2)$,
$A_2 = A_1 - \beta A_1(\theta_1 - \theta_2)$
$A_2 = 1.5 - 5.0\times10^{-5}\times1.5(20-(-20)$,
$A_2 = 1.5 - 7.5\times10^{-5}\times40 = 1.5 - 0.003 = 1.497$m^2.

35. A thin square sheet of metal of sides 0.20m is heated from 15°C and 75°C. If the linear expansivity of the metal is $1.2\times10^{-5}K^{-1}$ find the increase in its area A.6.05×10^{-5}m^2 **B.5.76×10^{-5}m^2** C.5.05×10^{-5}m^2 D.3.75×10^{-5}m^2.
L_1=0.20m, A_1=L^2=0.20^2= 0.04m^2, θ_1=15°C, θ_2=75°C, α=1.2×10$^{-5}K^{-1}$, β=2α, β=2×1.2×10^{-5} = 2.4×10$^{-5}K^{-1}$, change or increase in area ΔA=?,

Demystified Series by Dr Timothy

$\Delta A = \beta A_1 \Delta\theta = \beta A_1(\theta_2 - \theta_1)$,
ΔA =2.4×10^{-5}×0.04(75−15)= 9.6×10^{-8}×60
ΔA =5.76×10^{-5}m^2.

36. A cube made of a metal of linear expansivity α is heated through a temperature θ. If the initial volume of the cube is V_o , the expressiom for the increase in volume of the cube A.$\frac{1}{3}\alpha V_o\theta$ B.$\frac{1}{2}\alpha V_o\theta$ C.$2\alpha V_o\theta$ **D.$3\alpha V_o\theta$.**
Cubic expansivity $\gamma = \frac{\Delta V}{V_1\Delta\theta}$, γ=3α , V_1=V_o , $\Delta\theta$=θ ,
$3\alpha = \frac{\Delta V}{V_o\theta}$, $\Delta V = 3\alpha V_o\theta$.

37. A blacksmith heated a metal whose cubic expansivity 6.3×10$^{-6}K^{-1}$. The area expansivity is A.6.3×10$^{-6}K^{-1}$ **B.4.2×10$^{-6}K^{-1}$** C.2.1×10$^{-6}K^{-1}$ D.2.0×10$^{-6}K^{-1}$.
γ=6.3×10$^{-6}K^{-1}$, β=?, $2\gamma = 3\beta$, 2×6.3×10^{-6} =3β,
$\beta = \frac{2\times6.3\times10^{-6}}{3} = 4.2\times10^{-6}K^{-1}$.

38. If L, S and V are the linear, area and volume expansivities of a given metal respectively, which of the following equation is correct? A.$L - S = 0$ B.$V - 2S = 0$ **C.$S - 2L = 0$** D.$2S - L = 0$ E.$3V - L = 0$.
Linear expansivity $\alpha = L$, area expansivity $\beta = S$, volume expansivity $\gamma = V$,
β=2α, $\therefore S = 2L$, $S - 2L = 0$.
γ=3α, $\therefore V = 3L$, $V - 3L = 0$.
$2\gamma = 3\beta$, $\therefore 2V = 3S$, $2V - 3S = 0$.

39. If the cubic expansivity of a brass between 27°C and 100°C is 5.7×10$^{-5}K^{-1}$, what is the linear expansivity? A.2.85×10$^{-5}K^{-1}$**B.1.90×10$^{-5}K^{-1}$** C.1.86×10$^{-5}K^{-1}$ D.1.70×10$^{-5}K^{-1}$ E.1.62×10$^{-5}K^{-1}$
γ=5.7×10$^{-5}K^{-1}$, α=?, γ=3α, $\alpha = \frac{\gamma}{3}$,
$\alpha = \frac{5.7\times10^{-5}}{3} = 1.90\times10^{-5}K^{-1}$.

40. Calculate the increase in volume when 1500cm^3 of steel is heated from 0°C to 40°C[linear expansivity of steel=1.2×10^{-5}°C^{-1}] A.0.12cm^3 B.0.72cm^3 C.1.44cm^3 **D.2.16cm^3** E.1500.72cm^3.
V_1=1500cm^3, θ_1=0°C, θ_2=40°C, α=1.2×10^{-5}°C^{-1},
γ=3α=3×1.2×10^{-5}=3.6×10^{-5}°C^{-1}, ΔV=?,
$\Delta V = \gamma V_1\Delta\theta = \gamma V_1(\theta_2 - \theta_1)$,
ΔV =3.6×10^{-5}×1500(40−0) ,
ΔV =3.6×10^{-5}×1500×40 = 2.16cm^3.

41. A solid metal cube of side 10cm is heated from 10°C to 60°C. If the linear expansivity of the metal is 1.2×10$^{-5}K^{-1}$, calculate the increase in its volume A.0.6cm^3 B.1.2cm^3 **C.1.8cm^3** D.3.6cm^3 E.6.0cm^3
L_1=10cm, V_1=$L_1{}^3$=10^3=1000cm^3, θ_1=10°C,
θ_2=60°C, α=1.2×10$^{-5}K^{-1}$, γ=3α=3×1.2×10^{-5},
γ=3.6×10$^{-5}K^{-1}$, ΔV=?,
$\Delta V = \gamma V_1\Delta\theta = \gamma V_1(\theta_2 - \theta_1)$,
ΔV =3.6×10^{-5}×1000(60−10) ,
ΔV =3.6×10^{-5}×1000×50 = 1.8cm^3.

42. A piece of brass of mass 170kg has its temperature raised from 0°C to 30°C. Calculate its increase in volume, given density of brass at 0°C is 8.5×10^{3}kgm^{-3} and its cubic expansivity as 5.7×10$^{-5}K^{-1}$ **A.3.4×10^{-5}m^3** B.4.3×10^{-5}m^3 C.3.4×10^{-4}m^3 D.3.4×10^{-3}m^3 E.3.4×10^{-2}m^3.
Mass m=170kg, density ρ=8.5×10^{3}kgm^{-3}, original volume $V_1 = \frac{m}{\rho} = \frac{170}{8.5\times10^3} =$ 0.02m^3,
θ_1=0°C, θ_2=30°C, γ=5.7×10$^{-5}K^{-1}$, ΔV=?,
$\Delta V = \gamma V_1\Delta\theta = \gamma V_1(\theta_2 - \theta_1)$,
ΔV =5.7×10^{-5}×0.02(30−0) ,
ΔV =5.7×10^{-5}×0.02 ×30 = 3.4×10^{-5}m^3.

43. When a metal balls is heated through 30°C, its volume becomes 1.0018cm^3. If the linear expansivity of the ball is 2.0×10$^{-5}K^{-1}$, calculate the original volume **A.1.0000cm^3** B.1.0020cm^3 C.1.0036cm^3 D.1.0180cm^3.
V_2=1.0018cm^3, $\Delta\theta$=30°C, α=2.0×10$^{-5}K^{-1}$,
γ=3α=3×2.0×10^{-5}=6.0×10$^{-5}K^{-1}$, V_1=?,
$V_2 = V_1(1 + \gamma\Delta\theta)$, $V_1 = \frac{V_2}{(1+\gamma\Delta\theta)}$,
V_1=$\frac{1.0018}{1+6.0\times10^{-5}\times30} = \frac{1.0018}{1+0.0018} = \frac{1.0018}{1.0018}$= 1.0000cm^3.

44. The linear expansivity of brass is 2.0×10^{-5}°C^{-1}. If the volume of a piece of brass is 15.00cm^3 at 0°C, what is the volume at 100°C? A.16.03cm^3 B.16.00cm^3 **C.15.09cm^3** D.15.03cm^3
α=2.0×10^{-5}°C^{-1}, γ=3α=3×2.0×10^{-5}, γ =6.0×10^{-5}°C^{-1}, V_1=15.00cm^3, θ_1=0°C, θ_2=100°C,
V_2=?, $\gamma = \frac{\Delta V}{V_1\Delta\theta} = \frac{V_2 - V_1}{V_1(\theta_2 - \theta_1)}$,
$V_2 = V_1\big(1 + \gamma(\theta_2 - \theta_1)\big)$,
V_2 = 15(1+6.0×10^{-5}(100−0)) ,
V_2 = 15(1+0.006) = 15×1.006 = 15.09cm^3.

45. A rectangular metal block of volume 10^{-6}m^3 at 273K is heated to 573K. If its coefficient of linear expansion is 1.2×10$^{-5}K^{-1}$, the percentage change of its volume is A..1.5% **B.1.1%** C.0.4% D.0.1%.
V_1=10^{-6}m^3, θ_1=273K, θ_2=573K, α=1.2×10$^{-5}K^{-1}$,
γ=3α=3×1.2×10^{-5}=3.6×10$^{-5}K^{-1}$,
$\gamma = \frac{\Delta V}{V_1\Delta\theta} = \frac{\Delta V}{V_1(\theta_2 - \theta_1)}$, $\frac{\Delta V}{V_1} = \gamma(\theta_2 - \theta_1)$,
Percentage change in volume $\Delta V\% = \frac{\Delta V}{V_1} \times 100$,
$\Delta V\% = \gamma(\theta_2 - \theta_1) \times 100$,
$\Delta V\%$ =3.6×10^{-5}(573−273) ×100 ,
$\Delta V\%$ =3.6×10^{-5}×300×100 = 1.08% = 1.1%.

46. When very hot water is poured into a thin-walled glass container, it is less likely to break because A.thin glass does not expand fast B.glass can withstand a very high temperature **C.all parts of the container get heated uniformly** D.the outer part of the container is cooler than the inner part E.the molecules of glass are too far apart to cause breakage of the container.
The inner and outer walls of a thin-walled glass container expands uniformly or equally as heat is supplied uniformly to all parts.

47. Which of the following explains why a thick glass cup cracks when water is poured into it? A.large increase in the heat capacity of the cup B.high density of water **C.unequal expansion of the interior walls of the cup D.anomalous**

Demystified Series by Dr Timothy

expansion of water E.greater specific heat capacity of water compared with that of glass.
A thick-walled glass cup or container expandsunequally or non-uniformly when heat is supplied i.e the interior wall expands more than the exterior wall.

48. Consider a thick glass thumbler and a thin glass thumbler which are made of the same kind of glass. Some hot liquid is poured into them. Which of the following statements is correct? A.The thick thumbler is less likely to crack because it is stronger B.The thick thumbler is less likely to cracks because it insulatea the heat better, glass being a poor conductor of heat C.Both thumblers are equally likely to crack because they are made of the same kind of glass **D.The thick thumble is more likely to crack because of uneven expansion persists for a longer time the thicker glass** E.The thick thumbler is more likely to crack because it is heavier.

49. What is likely to happen if the glass of a thermometer expands more upon heating than the liquid inside? A.The glass will shrink **B.The liquid will go down in the stem** C.The liquid will rise up in the stem D.The will break
Expansion of the glass makes its volume to increase, hence the liquid level falls down the stem to fill the increased volume of the glass.

50. Which of the following statement is /are correct? I.When a liquid expands its density falls II.The apparent expansion of a liquid is greater than its real cubic expansion III.When a liquid is heated to a sufficiently high temperature it changes into a solid **A.I** B.II C.III D.I and II E.II and III.
When a liquid expands, its volume increases and its density decreases or falls.The apparent cubic expansion of is less than its real cubic expansivity or the real cubic expansivion of a liquid is greater than its apparent cubic expansion. When a liquid is heated to a sufficiently high temperature it changes into a gas.

51. The cubic expansivity of mercury is $1.8\times10^{-4}K^{-1}$ and the linear expansivity of glass is $8.0\times10^{-6}K^{-1}$, calculate the apparent cubic expansivity of mercury in a glass container A.$1.00\times10^{-4}K^{-1}$ **B.$1.56\times10^{-4}K^{-1}$** C.$1.72\times10^{-4}K^{-1}$ D.$2.04\times10^{-4}K^{-1}$.
Real cubic expansivity of mercury γ_r=$1.8\times10^{-4}K^{-1}$, linear expansivity of glass α_c=$8.0\times10^{-6}K^{-1}$, cubic expansivity of glass container $\gamma_c = 3\alpha_c = 3\times8.0\times10^{-6}=24\times10^{-6}=0.24\times10^{-4}K^{-1}$, apparent cubic expansivity γ_a=?, $\gamma_r = \gamma_a + \gamma_c$,
$\gamma_a = \gamma_r - \gamma_c = 1.8\times10^{-4}-0.24\times10^{-4}$,
$\gamma_a = 1.56\times10^{-4}K^{-1}$.

52. The mass of an empty density bottle is 30g. When filled with a liquid, the mass of the bottle and the liquid is 40g. On heating the filled bottle through 40°C,the mass reduces to 38g. Calculate the apprarent cubic expansivity of the liquid A.$6.25\times10K^{-1}$ **B.$6.25\times10^{-3}K^{-1}$** C.$2.50\times10^{-3}K^{-1}$ D.$2.50\times10^{-4}K^{-1}$.
Mass of density bottle=30g, mass of density bottle+liquid=40g, mass of density bottle+ liquid after heating=38g, mass of liquid expelled $M_e = 40-38 = 2$g, mass of liquid remaining(or left) $M_r = 38-30 = 8$g, temperature rise $\Delta\theta$=40°C, γ_a=?,
$\gamma_a = \frac{mass\ of\ liquid\ expelled}{mass\ of\ liquid\ remaining\times temp.\ rise} = \frac{M_e}{M_r\times\Delta\theta}$,
$\gamma_a = \frac{2}{8\times40} = 0.00625K^{-1} = 6.25\times10^{-3}K^{-1}$.

53. A relative density bottle of volume 50cm^3 is completely filled with a liquid at 30°C. It is then heated to 80°C such that 0.75cm^3 of the liquid is expelled. Calculate the apparent cubic expansivity of the liquid **A.$0.00030K^{-1}$** B.$0.00032K^{-1}$ C.$0.01970K^{-1}$ D.$0.02030K^{-1}$.
Initial volume=50cm^3, volume expelled V_e=0.75cm^3, volume remaining V_r= $50-0.75$, V_r=49.25cm^3, θ_1=30°C, θ_2=80°C, γ_a=?,
$\gamma_a = \frac{volume\ of\ liquid\ expelled}{volume\ of\ liquid\ remaining\times temp.\ rise}$,
$\gamma_a = \frac{V_e}{V_r\times\Delta\theta} = \frac{V_e}{V_r\times(\theta_2-\theta_1)} = \frac{0.75}{49.25\times(80-30)}$,
$\gamma_a = \frac{0.75}{49.25\times50} = 0.00030K^{-1}$.

54. A specific gravity bottle contains 0.055kg of a liquid at 20°C and 0.050kg at 70°C. If the cubic expansivity of the glass is $0.00003K^{-1}$, determine the real cubic expansivity of the liquid A.$0.00183K^{-1}$ B.$0.00192K^{-1}$ **C.$0.00203K^{-1}$** D.$0.00205K^{-1}$.
Mass of liquid at 20°C=0.055kg, mass of liquid at 70°C=mass of liquid remaining M_r=0.050kg, mass of liquid expelled M_e =$0.055-0.050$, M_e=0.005kg, θ_1=20°C, θ_2=70°C, γ_a=?,
$\gamma_a = \frac{mass\ of\ liquid\ expelled}{mass\ of\ liquid\ remaining\times temp.\ rise}$,
$\gamma_a = \frac{M_e}{M_r\times\Delta\theta} = \frac{M_e}{M_r\times(\theta_2-\theta_1)} = \frac{0.005}{0.050\times(70-20)}$,
$\gamma_a = \frac{0.005}{0.050\times50} = 0.002K^{-1}$, $\gamma_r = \gamma_a + \gamma_c$,
γ_c= $0.00003K^{-1}$, γ_r =0.00003+0.002,
$\gamma_r = 0.00203K^{-1}$.

55. The temperature of a glass vessel containing 100cm^3 of mercury is raised from 10°C to 100°C. Calculate the apparent cubic expansion of the mercury. (Real cubic expansivity=$1.82\times10^{-4}K^{-1}$, cubic expansivity of glass=$2.4\times10^{-5}K^{-1}$) A.0.52cm^3 **B.1.42cm^3** C.1.87cm^3 D.5.22cm^3 E.1.22cm^3.
Original volume V_1=100cm^3 , θ_1=10°C, θ_2=100°C, γ_r=$1.82\times10^{-4}K^{-1}$, γ_c=$2.4\times10^{-5}K^{-1}$,apparent cubic expansion=apparent increace in volume ΔV=?, $\gamma_a = \frac{apparent\ increase\ in\ volume}{original\ volume\times temp.\ rise} = \frac{\Delta V}{V_1(\theta_2-\theta_1)}$,
$\Delta V = \gamma_a V_1(\theta_2-\theta_1)$, $\gamma_a = \gamma_r - \gamma_c$,
$\therefore \Delta V = (\gamma_r - \gamma_c)V_1(\theta_2-\theta_1)$,
$\Delta V = (1.82\times10^{-4}-2.4\times10^{-5})\times100(100-10)$,
$\Delta V = 1.58\times10^{-4}\times100\times90 = 1.42$cm^3.

Demystified Series by Dr Timothy

56. Water shows anomalous behaviour A.below 0°C **B.between 0°C and 4°C** C.at exactly 4°C D.between 4°C and 100°C E.above 100°C.

57. In what range of temperature is the expansion of water anomalous? A.+208°C to +212°C B.−80°C to −76°C C.−40°C to 0°C D.+96°C to +100°C **E.0°C to +4°C.**

58. Fishes can survive in the sea when the temperature of the atmosphere is below 0°C because A.their body temperature can melt the ice B.only a thin layer of the water at the bottom freezes **C.only a thin layer of the water at the top freezes** D.the water never freezes at all E.the salts in sea water keep it warm.

Anomlous behaviour of water makes water to freeze from its surface downwards. As the water cools from 4°C to 0°C, it expands and its density decreases, hence it rises to the water surface where it freezes at 0°C while the water below remains as liquid. Therefore aquatic life e.g fishes, can survive in deep layers of water in sea.

59. When left in a freezer, a bottle full of water cracks on freezing into ice because of the A.decrease in the volume of water B.contraction of the bottle C.expansion of the bottle **D.increase in the volume of water.**

Water expands on freezing to ice hence, the volume of the ice now becomes greater than the volume of the bottle making it to crack.

60. At 4°C, the volume of a fixed mass of water is A.constant B.maximum **C.minimum** D.zero

Water has a maximum density at and minimum volume at 4°C and has a minimum density and maximum volume at 0°C.

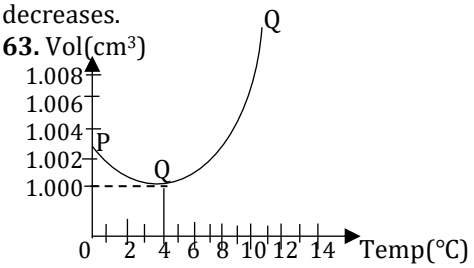
61. If the temperature of water is gradually increased from 0°C to 4°C, the density water within this range A.increases for a while and then decreases B.decreases for a while and then increases **C.increase gradually** D.decreases gradually E.remains the same

The density of water increases from 0°C to 4°C and is maximum at 4°C. Above 4°C, water behaves like other liquids when heated i.e its volume increases whileits density decreases.

62. As water is cooled from 10°C to 2°C, its density A.increases B.decreases C.first decreases then increases **D.first increases and then decreases.**

The density of water is maximum at 4°C, hence on cooling from 10°C its density increases till it reaches 4°C. When cooled below 4°C, its density decreases.

63.

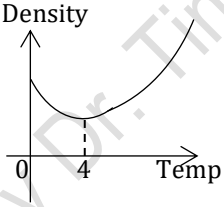


The graph above show the expansion of water as the temperature increases from 0°C. Which of the following deductions from the graph are true? I.water has its maximum density at Q II.the volume of water is greater at 0°C than at 4°C III.the volume of water decreases uniformly when cooled from 100°C to 0°C IV.when water solidifies, its volume increases A.I and II B.I and III C.II and III D.I,II and III **E.I,II and IV**

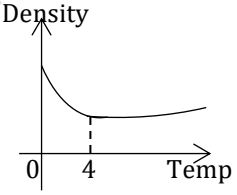
Water has its minimum volume i.e 1.000cm³ at 4°C hence, its density will be maximum at 4°C. The volume of water at 0°C is 1.002cm³ while its volume at 4°C is 1.000cm³ hence, its volume at 0°C is greater than its volume at 4°C. The volume of water decreases when cooled from 100°C to 4°C and then increases from 4°C to 0°C. When water solidifies i.e changes to ice, its volume increases.

64. The graph of density against temperature for water is correctly illustrated by

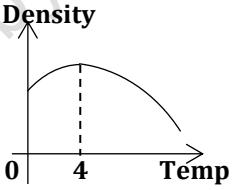
A. Density



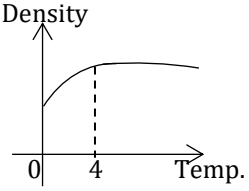
B. Density



C. Density



E. Density



The density of water increases from 0°C to 4°C and decreases above 4°C, due to anomalous expansion as shown in option C.

Jamb past questions on thermal expansivity :

[1978/22,1979/9,34,1980/44,1982/22,1983/22,1982/16,1983/42,1985/18,22,1986/18,1987/18,1990/16,1991/16,1992/14,15,1993/15,1994/23,1995/19,1999/18,2000/21,25,2004/33,38,2005/43,48,2006/44,2007/6,28,36,2008/19,2009/19,2010/16,2011/23,2013/17,2014/20,2015/46]

Demystified Series by Dr. Timothy

Chapter 13 – Thermal energy, Heat Capacity, Latent heat and Vapours

● **Thermal or heat energy H –** This is the amount or quantity of heat required to raise or reduce the temperature of a body. It is the energy required to change the average position or kinetic energy of molecules of a substance.

- Thermal or Heat capacity C – is the quantity of heat required to raise the temperature of a body by $1K$ or 1℃. Thermal or heat capacity is also the quantity heat lost by a body when its temperature falls by $1K$. $\boldsymbol{H = C\Delta\theta}$, $\boldsymbol{C = \frac{H}{\Delta\theta}}$. H=quantity of heat, C=heat or thermal capacity (JK^{-1}), $\Delta\theta$=temperature rise= $(\theta_2 - \theta_1)$.

- Specific heat capacity c – is the quantity of heat required to raise the temperature of a unit (1kg) mass of a body by $1K$ or 1℃. Specific heat capacity is also the quantity of heat lost by a unit (1kg) mass of a body when its temperature falls by $1K$ or 1℃. $\boldsymbol{H = mc\Delta\theta}$, $\boldsymbol{c = \frac{H}{m\Delta\theta}}$, c=specific heat capacity ($Jkg^{-1}K^{-1}$ or $Jg^{-1}K^{-1}$), m=mass of the body.

- The specific heat capacity is constant for substances of the same material but differ for substance of different materials.

- Thermal capacity C and specific heat capacity are related by – $\boldsymbol{C = mc}$**.**

- When equal masses or weights of bodies are supplied with the same amount of heat, the body with a higher thermal capacity or specific heat capacity have lower temperature difference or temperature rise, as they are inversely proportional , $\boldsymbol{C \propto \frac{1}{\Delta\theta}}$ **or** $\boldsymbol{c \propto \frac{1}{\Delta\theta}}$**.**

- Thermal equilibrium is a condition where two bodies in thermal contact are at the same temperature and there is no net flow of heat between both bodies.

- Measurement of specific heat capacity is done by – Electrical method, Mixture method.

I. Electrical method or Electrical calorimetry : When an electrical device (heater or kettle) is used to supply heat to a liquid or object, the heat supplied by the heater equals the heat gained by the liquid or object. The heat supplied by the electrical device can be calculated using : $\boldsymbol{Pt}$ **or** $\boldsymbol{IVt}$ **or** $\boldsymbol{\frac{V^2t}{R}}$ **or** $\boldsymbol{I^2Rt}$**.**

Heat supplied by electrical heater = Heat gained by substance. $\boldsymbol{Pt = IVt = \frac{V^2t}{R} = I^2Rt = mc\Delta\theta}$. P=electrical power or rate of heat supplied or rating of the heater, t=time, I=current. V=potential difference or voltage, R=resistance.

II. Mixture method – When a hotter body (body at high temperature) is added to or mixed with a colder body (body at low temperature), heat is transferred from the hot to cold body i.e heat lost by hot body is equal to the heat gained by cold body, until both attain an equilibrium or same temperature or thermal equilibrium.

Heat lost by hotter body H_H = Heat gained by colder body H_C, $\boldsymbol{m_Hc_H\Delta\theta_H = m_Cc_C\Delta\theta_C}$ **,** m_H=mass of hotter body, c_H=specific heat capacity of hotter body, $\Delta\theta_H = \theta_H - \theta$, $\Delta\theta_H$=temperature change of hotter body, θ_H=temperature of hotter body, m_C=mass of colder body, c_C=specific heat capacity of colder body, $\Delta\theta_C = \theta - \theta_C$, θ_C=temperature of colder body, θ=equilibrium temperature or temperature of mixture or final temperature.

Hence, $\boldsymbol{m_Hc_H(\theta_H - \theta) = m_Cc_C(\theta - \theta_C)}$ **.**

- Waterfall from a height – When water falls from a height in a dam, the potential energy of at the top and the kinetic energy of falling water is converted to heat energy(internal energy) at the bottom of the dam,hence the water becomes slightly warm due to slight increase in temperature. $\boldsymbol{P.E = H, K.E = H,}$ $\boldsymbol{mgh = mc\Delta\theta}$ or $\boldsymbol{\frac{1}{2}mv^2 = mc\Delta\theta}$**.** m=mass of water, g=acceleration due to gravity ($10ms^{-2}$), h=height of waterfall, v=velocity or speed of waterfall.

- A **calorimeter** is an apparatus used in experiments that involve exchange of heat between two bodies at different temperature.

- A good calorimeter should be made of a material e.g copper or aluminium, with high thermal or heat conductivity and low specific heat capacity.

- Methods of reducing heat loss in a calorimeter – I. Lagging the calorimeter i.e surrounding it with an insulator e.g cotton wool or felt,to reduce heat loss by conduction. **II.**Air insulating lid which prevents heat loss by convention and evaporation. **III.**Polishing the outer surface to reduce heat loss by radiation. **IV.**Shielding the calorimeter from draught**.**

- Calorific value of a substance is the therma; energy required per mass or volume of the source. Calorific value is in Jg^{-1}or J/m^3.

Calorific value $= \boldsymbol{\frac{Thermal\ energy\ required}{mass\ or\ volume\ of\ the\ source}}$,

Calorific value $= \boldsymbol{\frac{H}{m\ or\ V} = \frac{mc\Delta\theta}{m\ or\ V}}$**.**

- 1 calorie (1cal) is equal to 4.2J.

● Latent heat and change of state

- Latent heat also called hidden heat of a substance is the heat gained or lost to change the state of that substance at the same temperature (without a change in temperature). It is the heat

Demystified Series by Dr Timothy

supplied or removed from a substance with no temperature difference. Latent heat is the energy required to keep the molecules of a substance in their mean position or their potential energy.

- Latent heat causes a change in the density and volume of a substance without any change in mass.

- **Specific latent heat L** is the heat supplied or removed to change the state of a unit (1kg) mass of a substance at a constant temperature. $\boldsymbol{H = mL}$, $\boldsymbol{L = \frac{H}{m}}$ $\boldsymbol{(Jkg^{-1})}$.

- **Latent heat of fusion** is the quantity of heat required to convert a solid to its liquid state at melting point at constant temperature.

- **Specfic latent heat of fusion L_f** is the heat required to change the state of 1kg mass of a solid substance to its liquid state at its melting point at constant temperature.

- **Specific latent heat of vaporization L_v**is the heat required to change the state of 1kg mass of a liquid substance to its gaseous state at its boiling point at constant temperature.

- **Specific latent heat of fusion of a substance is less than its specific latent heat of vaporization and vice-versa,** because the intermolecular bonds in a solid is reduced when transformed from solid to liquid but is completely broken when transformed from liquid to gas.

- **The apparatus involved in determining the specific latent heat of vaporization includes**: **I.**Boiling flask:Where water is heated to produce wet steam. **II.**Connecting tube:Wet steam passes through it to the water or steam trap. **Steam or water trap dries the steam and collects the water that condenses.**

- When heat is supplied from an electrical heater to cause a change of state, then –

$\boldsymbol{Pt = IVt = \frac{V^2t}{R} = I^2Rt = mL}$.

- **Change of state –** Matter exists as solid, liquid and gas. Change of state occurs at constant temperature and latent heat is either gained or lost.

- **Change of state is accompanied by change in density,volume and heat content without any change in the mass and temperature.**

- During change of state from solid→liquid→ gas – latent heat is absorbed or supplied.

- During change of state from gas→liquid→ solid – latent heat is removed or given off.

- **Melting or fusion point** of a substance is the temperatureat which the solid substance changes into the liquid state.

- When a solid substance is heated to its melting point, the temperature increases before and after melting while temperature remains constant during melting.

- Substance contracts(decrease in volume) on melting and expands(increase in volume) on freezing or solidification.

- Density of substance increase on melting and decreases on freezing or solidification.

- Increase in pressure and presence of impurities decreases the melting point or freezing point of a substance.

- The effect of pressure on the melting point of a solid is shown using **Regelation or Refreezing experiment.** Regelation is the refreezing of asubstance after melting by the application of pressure.

- When a ice cubes are introduced into a container filled to the brim and the ice floats,when the ice cube melts the water level remain the same as the ice cube displaces its volume of water equal to its volume before floating which is now filled when it melts

- When a container is filled with ice cubes to the brim, when the ice melts, the water level drops because ice occupies more volume than water (anomalous expansion) and due to the fact that the air spaces between them is completely filled with water when it melts.

- **Evaporation** is a process where a liquid turns spontaneously into vapour or gaseous state below its boiling point. It is the escape of molecules at the liquid surface with energy above the average kinetic energy from the liquid.

- **Evaporation occurs at all temperatures.**

Evaporation is the change from liquid to unsaturated vapour and brings about cooling.

- Rate of evaporation $\boldsymbol{E}$ is affected by nature of the liquid, wind or force convention of air over a liquid W, surface area of liquid exposed A, humidity or dryness of air H, temperature T, pressure P and density of the liquid ρ, $\boldsymbol{E \propto \frac{WAT}{P\rho H}}$.

Rate of evaporation of liquid increases as wind, area and temperature increases while pressure, density and humidity decreases.

- Blowing air over a liquid increases the surface area.

- Rate of evaporation is high in an unsaturated environment than a saturated environment, as the air can still accommodate more vapour.

- Rate of evaporation is not affected by osmotic or hydrostatic pressure.

- When a volatile liquid e.g ether, methylated spirit or alcohol, is poured on the skin or hands, latent heat of vaporization is drawn from the skin making the person feel cool.

- Volatile liquids have low boiling points and low latent heat.

- Steam causes a more severe burning or damage on the skin because : I.the heat content of steam or internal energy (thermal energy and latent heat) is greater that of boiling water(thermal energy only). **II.**Steam is at a

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higher temperature than boiling water. **Condition (i) is most acceptable.**

- Refrigerators works on the cooling effect of evaporation. The latent heat generated in a refrigerator is removed by conduction, convention and radiation to the surrounding. A thermostat inside a refrigerator controls the rate of vaporization and degree of cooling.

- **Boiling point or Ebullition point** of a liquid is the temperature at which the vapour pressure of the liquid equals the atmospheric or external pressure.

- Boiling at higher temperature than the boiling point means the atmospheric pressure is increased, it has to beheated rapidly for the vapour pressure to equalize it.

- At normal atmospheric pressure(760mmHg or 1atm),water boils or evaporates at 100℃.

- Boiling point increases with pressure and presence of impurities. Pressure increases with liquid below sea level and decreases with liquid above sea level.

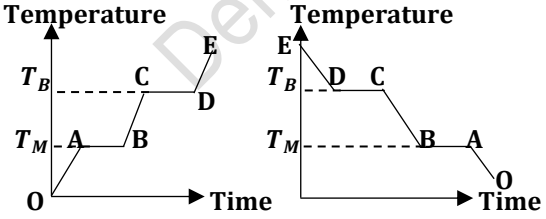
- **Pressure cooker** saves both time and fuel in cooking because inside the cooker the boiling point of water is raised due to increased pressure.

- Evaporation and Boiling both involves a change of state from liquid to gaseous state.

- Evaporation occurs at any temperature while Boiling occurs at a definite temperature. Evaporation occurs at the surface of a liquid (surface phenomenon) while Boiling involes the whole volume or throughout the body of the liquid.

- Evaporation depends on wind and surface area of liquid while Boiling is independent of wind and surface area of liquid.

- **Heating and cooling curves –** Curves which shows the variation of temperature with time of a substance when heated or cooled to a change in state.



Section OA – It represents the solid phase. Temperature changes when heated or cooled.

Section AB – It represents the melting or freezing region i.e a mixture of solid and liquid. Temperature is constant here and is equal to the melting point T_M.

Section BC – It represents the liquid phase. Temperature changes when heated or cooled.

Section CD – It is the boiling or liquefaction region i.e a mixture of liquid and gas(vapour). Temperature is constant and is equal to the boiling point T_B of the liquid.

Section DE – It represents the gaseous or vapour phase. Temperature changes when heated or cooled.

The heat supplied or removed in section OA, BC and DE is thermal energy. The heat supplied or re supplied or removed in section AB and CD are latent heat of fusion and vaporization repectively.

●Vapours (Saturated and Unsturated)

- A liquid heated in a closed vessel, evaporates and vapour is formed in the space above the liquid. The vapour molecules exerts pressure on the liquid surface and the walls of the vessel called **vapour pressure.**

- When the space above the liquid is crowded with vapour as the container is closed, initially the rate of vaporization exceeds the rate of condensation, the vapour is said to be **unsaturated** at this point. A point is attained where the rate of vaporizaation equals the rate of condensation, the vapour is said to be **saturated.**

- **Unsaturated vapour** is a vapour that can still hold more vapour molecules within a cofined space. The number of molecules leaving a liquid per second is more than the number of molecules rerturning to the liquid per second i.e rate of vaporization exceeds the rate of condensation. The vapour is not in dynamic equilibrium with its own liquid in a cofined space **(vapour formed during evaporation).**

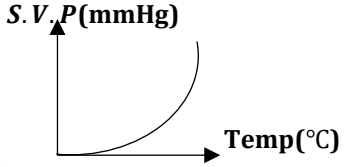
- **Saturated vapour** is a vapour which can no longer hold any more vapour molecules within a cofined space. The number of molecules escaping per second equals the number of molecules returning per second i.e rate of vaporization equals rate of condensation.The saturated vapour is in in dynamic equilibrium with its own liquid.

- **Unsaturated vapours obey gas law (Boyle's law and Charles' law) while Saturated vapours do not obey gas laws (Boyle's law and Charles' law).**

- **Saturated vapour pressure $S.V.P$ or equilibrium vapour pressure** is the pressure exerted by molecules of a saturated vapour at at a definite temperature.

- **Saturated vapour pressure is affected by temperature only. Saturated vapour pressure increases with temperature i.e $S.V.P \propto T$.**

- **Saturated vapour pressure at any given temperature is independent of the volume of the vapour or external pressure.**



Demystified Series by Dr Timothy

- A liquid boils when its saturated vapour pressure $S.V.P$ equals the external atmospheric pressure $A.P$.

- Boiling point or ebullition point of a liquid is the temperature at which the saturated vapour pressure of the liquid equals the external atmospheric pressure.

- A liquid with high $S.V.P$ evaporates faster and reaches its boiling point quickly that a liquid with low $S.V.P$.

- The pressure of water vapour in air can never exceeds the $S.V.P$.

- Sublimation is the change of solids substance directly to the gaseous state.

- Condensation is the change from the gas (vapour) to the liquid state.

- Condensation of water vapour in the atmosphere – The amount of water vapour air can hold decreases with a decrease in temperature. Hence, cooling air below saturation or lowering of temperature decreases its ability to hold water, which results the formation of water droplets in form of dew, mist, fog, frost and cloud.

1.Dew is moisture or water droplets formed on when air is cooled below a critical temperature due to variation in temperature between the day and night. Dew occurs due to land breezes that occur at night.

2.Mist is thin layer tiny water droplets close to the earth surface formed by cooling air above but near the earth.

3.Fog is a thick and dense mist which reduces visibility. Car drivers put on their head lamps and drives slowly due to impaired vision caused by fog.

4.Cloud is water droplets or ice crystals high up in the sky or atmosphere i.e floats in air due to the small size, formed by condensation of water vapour in the atmosphere.

5.Rain is water droplets which falls from large clouds.

6.Frost is tiny crystals of ice formed on surfaces, when water vapour changes directly to ice below the freezing point due to fall in air temperature.

7.Hail is cloud of large water droplets which solidifies to ice pellets on falling to the ground.

- Dew point is the temperature at which the water vapour present in air is just sufficient to saturate it.

- Dew point is dependent on the prevailing atmospheric conditions like temperature, wind and the amount of water vapour in the atmosphere.

- When air is cooled below dew point, water vapour condenses. Dew point is always less than air temperature.

- Humidity is the amount of water vapour in the atmosphere or the mass of water vapour in a unit volume of the air in the atmosphere.

- A wet or moist air is referred to as a humid air.

- Relative humidity $R.H$ is the ratio of the mass of water vapour present in a certain volume of air to the mass of water vapour required to saturate the same volume of air ar at the same temperature. Relative humidity is the degree of humidity of air or the nearness of the water vapour present in atmosphere to saturation.

- Relative humidity is expressed in percentage

$$R.H = \frac{mass\ of\ water\ vapour\ in\ a\ given\ vol.\ of\ air}{mass\ of\ water\ vapour\ required\ to\ saturate},$$

$$R.H = \frac{S.V.P\ of\ water\ at\ dew\ point}{S.V.P\ of\ water\ at\ original\ air\ temperature},$$

$$R.H = \frac{vapour\ pressure(\ partial\ pressure)\ of\ air}{S.V.P\ at\ the\ same\ temperature\ of\ the\ air},$$

- Relative humidity $R.H$ increases with $S.V.P$ at dew point and decreases with $S.V.P$ at air temperature.

- High humidity – Wet/moist atmosphere during rainy or winter season.

- Low humidity – Dry aymosphere during harmattan or summer season.

- Relative humidity is measured with a **hygrometer** e.g wet and dry bulb hygrometer.

- Relative humidity is inversely proportional to temperature difference between the wet and dry bulb hygrometer, $R.H \propto \frac{1}{\Delta T}$ i.e relative humidity $R.H$ is greatest when the difference in temperature between the dry and wet bulb hygrometer ΔT is least.

Examples :

1. The amount of heat required to raise the temperature of a body is A.thermal capacity B.specific heat capacity **C.thermal energy** D.heat loss E.thermal conduction.

2. The amount of heat needed to raise the temperature of 10kg of copper by1K is its A.internal energy **B.heat capacity** C.specific heat capacity D.latent heat.

Heat capacity is amount of heat required to raise the whole mass of a body by 1K or 1°C.

3. The quantity of heat required to raise the temperature of unit mass of a substance thrpugh unit temperature is **A.specific heat capacity** B.specific heat of sublimation C.specific heat of fusion D.specific heat of fission.

4. Equal masses of copper and rubber are raised to the same temperature. After sometimes, the copper was observed to be at a lower temperature because A.the specific heat capacity of copper is lower than that of rubber B.copper expand more than rubber **C.the specific heat capacity of rubber is lower than that of copper** D.rubber expands more than copper.

Bodies with higher specific capacity has a lower temperature difference, $c \propto \frac{1}{\Delta\theta}$. Hence, copper is

Demystified Series by Dr Timothy

at a lower temperature than rubber, because the specific capacity of copper is higher than that of rubber or the specific heat capacity of rubber is lower than that of copper.

5. When equal weights of iron and water are subjected to an equal supply of heat, it is found that the piece of iron becomes much hotter than water after a shorter time because A.the specific heat of iron is higher than that of water **B.the specific heat of water is higher than that of iron** C.the specific heat of iron is infinite D.iron is in solid form E.water is in liquid form

Bodies with higher specific capacity has a lower temperature difference, $c \propto \frac{1}{\Delta\theta}$. Hence, iron is much hotter i.e at higher temperature, than water, because the specific heat capacity of iron is lower than that of water or specific heat capacity of water is higher than that of iron.

6. How much heat is given out when a piece of iron of mass 0.0500kg and specific heat capacity 460Jkg^{-1}℃$^{-1}$ cools from 80.0℃ to 20.0℃? **A.1380J** B.2760J C.13800J D.23000J E.27600J.

m=0.0500kg, c=460Jkg^{-1}℃$^{-1}$, θ_1=20.0℃, θ_2=80.0℃, H=?, $H = mc\Delta\theta = mc(\theta_2 - \theta_1)$,

$H = 0.05\times460(80-20) = 0.05\times460\times60$=1380J.

7. A copper ball of heat capacity $400JK^{-1}$ is heated fro, 20℃ to 100℃. Calculat ethe quantity heat absorbed A.40.0kJ **B.32.0kJ** C.2.0kJ D.8.0kJ E.3.2kJ.

C=400JK^{-1}, θ_1=20℃, θ_2=100℃, H=?,

$H = C\Delta\theta = C(\theta_2 - \theta_1) = 400(100-20)$,

$H = 400\times80 = 32000$J= 32.0kJ.

7. A body of mass 200g and specific heat capacity 0.4$Jg^{-1}K^{-1}$ cools from 37℃ to 31℃. Calculate the quatiity of heat released by the body A.4800J B.1200J **C.480J** D.202J.

m=200g, c=0.4Jg^{-1}℃$^{-1}$, θ_1=31K, θ_2=37K, H=?,

$H = mc\Delta\theta = mc(\theta_2 - \theta_1) = 200\times0.4(37-31)$,

$H = 200\times0.4\times6 = 480$J.

8. A liquid of mass 1000g requires 3.36×10^5Jof heat to raise its temperature by 80K. Calculate its specific heat capacity A.5.25×10^{-2}J$g^{-1}K^{-1}$ **B.4.20×10^{0}J$g^{-1}K^{-1}$** C.2.41×10^1J$g^{-1}K^{-1}$ D.3.25×10^1J$g^{-1}K^{-1}$ E.$2.69\times10^5Jg^{-1}K^{-1}$.

m=1000g, H=3.36×10^5J, $\Delta\theta$=80K, c=?,

$c = \frac{H}{m\Delta\theta} = \frac{3.36\times10^5}{1000\times80} = 4.2$ or 4.20×10^0J$g^{-1}K^{-1}$,

9. A body of mass m, has a specific heat capacity s and a heat capacity C. If the temperature of the body changes by θ℃ , which of the following equations is correct? A.$ms\theta = mC^{-1}$ **B.$ms = C$** C.$ms = C\theta$ D.$ms = s\theta$.

Mass m, specific heat capacity c=s , thermal capacity C, temperature change $\Delta\theta$=θ℃ ,

$C = mc$, $\therefore C = ms$.

10. Water of mass 1.5kg is heated from 30℃ to 80℃ using electric kettle which is rated 5A, 230V, calculate the time taken to reach the final temperature (specific heat capacity of water=4200J$kg^{-1}K^{-1}$) **A.4.57minutes** B.5.02minutes C.7.30minutes D.10.04minutes.

m=1.5kg, θ_1=30℃, θ_2=80℃,I=5A, V=230V, c=4200J$kg^{-1}K^{-1}$, t=?,

Heat supplied by electric kettle = Heat gained by water. $IVt = mc\Delta\theta = mc(\theta_2 - \theta_1)$,

$5\times230\times t = 1.5\times4200(80-30)$,

$1150t = 6300\times50$, $t = \frac{6300\times50}{1150} = 273.91$s

$t = \frac{273.91}{60} = 4.57$min.

11. A electrical heater 220V, 1000W is immersed into a bucket full of water.Calculate the mass of the water if the temperature changes from 30℃ to 100℃ and current flows for 300seconds. (specific heat capacity of water=4200J$kg^{-1}K^{-1}$) A.428kg B.42.86kg **C.1.02kg** D.7.14kg.

V=220V, P=1000W, θ_1=30℃, θ_2=100℃,t=300s, c=4200J$kg^{-1}K^{-1}$, m=?,

Heat supplied by heater = Heat gained by water.

$Pt = mc\Delta\theta = mc(\theta_2 - \theta_1)$,

$1000\times300 = m\times4200(100-30)$,

$300000 = m\times4200\times70 = 294000m$,

$m = \frac{300000}{294000} = 1.02$kg.

12. A block of aluminium of mass m is heated electrically by 25W heater. If the temperature rises by 10℃ in 5minutes. Calculate the heat capacity of aluminium A.1250JK^{-1} **B.750JK^{-1}** C.125JK^{-1} D.50JK^{-1}.

P=25W, $\Delta\theta$=10℃, t=5min=300s, C=?,

Heat supplied by heater = Heat gained by water.

$Pt = C\Delta\theta$, $25\times300 = C\times10$,

$C = \frac{25\times300}{10} = 750JK^{-1}$.

13. An immersion heater rated 400W, 220V, is used to heart a liquid of mass 0.5kg. If the temperature of the liquid increases uniformly at the rate of 2.5℃ per second, calculate the specific heat capacity of the liquid (Assume no heat is lost) A.1100J$kg^{-1}K^{-1}$ **B.320J$kg^{-1}K^{-1}$** C.200J$kg^{-1}K^{-1}$ D.176J$kg^{-1}K^{-1}$.

P=400W, V=220V, m=0.5kg, rate of temperature increase $\frac{\Delta\theta}{t}$=2.5℃s^{-1}, c=?, $Pt = mc\Delta\theta$, $P = mc\frac{\Delta\theta}{t}$

$400 = 0.5\times c\times2.5$, $c = \frac{400}{0.5\times2.5} = 320Jkg^{-1}K^{-1}$.

14. An electrical heater dipped into 500g of water in a plastic beaker raised the temperature from 20℃ to 50℃ in 3minutes. Determine the rate at which heat is given out by the heater **A.350Js^{-1}** B.700Js^{-1} C.3500Js^{-1} D.7000Js^{-1} (specific heat capacity of water =4.2 J$g^{-1}K^{-1}$).

m=500g, θ_1=20℃, θ_2=50℃, t=3min=180s,

Heat supplied by heater = Heat gained by water,

$H = mc\Delta\theta = mc(\theta_2 - \theta_1) = 500\times4.2(50-20)$

$H = 2100\times30 = 63000$J.

Rate of heat given out = Power = $\frac{H}{t} = \frac{63000}{180}$,

Rate of heat given out =350Js^{-1} or 350W.

15. Study the data below and use it to answer question 15.

Demystified Series by Dr Timothy

Mass of metal: 1.5kg, Rating of heating coil: 75W, Time of heating: 8minutes, Initial temperature: 27℃, Final temperature: 47℃. Calculate the specific heat capacity of the metal A.2520.0J$kg^{-1}K^{-1}$ **B.1200.0J$kg^{-1}K^{-1}$** C.250.0J$kg^{-1}K^{-1}$ D.142.9J$kg^{-1}K^{-1}$

m=1.5kg, P=75W, t=8min=480s, θ_1=27℃, θ_2=47℃, c=?, $Pt = mc(\theta_2 - \theta_1)$,

$75\times480 = 1.5\times c(47-27)$, $36000 = 30c$,

$c = \frac{36000}{30} = 1200.0\text{J}kg^{-1}K^{-1}$.

16. A 500W heater is used to heat 0.6kg of water from 25℃ to 100℃ in t_1 seconds. If another 1000W heater is used to heat 0.2kg of water from 10℃ to 100℃ in t_2 seconds, find t_1/t_2 A.50 **B.5** C.5/4 D.1/5.

Heat supplied by heater = Heat gained by water,

$Pt = mc\Delta\theta = mc(\theta_2 - \theta_1)$,

$500\times t_1 = 0.6\times c(100-25) = 0.6c\times75$

$t_1 = \frac{0.6c\times75}{500} = 0.09c$,

$1000\times t_2 = 0.2\times c(100-10) = 0.2c\times90$

$t_2 = \frac{0.2c\times90}{1000} = 0.018c$. $\frac{t_1}{t_2} = \frac{0.09c}{0.018c} = 5$.

17. A 1.2kW immersion heater is used to supply energy for 75s. The energy supplied is used to heat 800g of parrafin in a calorimeter of heat capacity 42.5JK^{-1} from 20℃ to 66℃. Calculate the specific heat capacity of paraffin A.1651J$kg^{-1}K^{-1}$ B.2369J$kg^{-1}K^{-1}$ **C.2393J$kg^{-1}K^{-1}$** D.2446J$kg^{-1}K^{-1}$ E.2508J$kg^{-1}K^{-1}$.

Heat supplied by heater = Heat gained by calorimeter + Heat gained by paraffin,

$Pt = mc_p(\theta_2 - \theta_1) + C_C(\theta_2 - \theta_1)$,

P=1.2kW=1200W, t=75s, mass of paraffin m=800g=0.8kg, heat capacity of calorimeter C_C=42.5JK^{-1},θ_1=20℃, θ_2=66℃, specific heat capacity of paraffin c_p=?,

$1200\times75 = 0.8\times c_p(66-20) + 42.5(66-20)$,

$90000 = 36.8c_p + 1955$,

$36.8c_p = 90000-1955 = 88045$,

$c_p = \frac{88045}{36.8} = 2392.5 = 2393Jkg^{-1}K^{-1}$.

18. Two thermos flasks of volume V_X and V_Y are filled with liquid water at an initial temperature of 0℃. After some time, the temperature were found to be θ_X and θ_Y repectively. Given that $\frac{V_X}{V_Y} = 2$ and $\frac{\theta_X}{\theta_Y} = \frac{1}{2}$. The rate of the rate of heat flow into the flasks is A.1/4 B.1/2 C.1 D.2 **E.1.**

Heat flow in flask X, $H_X = m_X c\Delta\theta_X$, Heat flow in flow in flast Y, $H_Y = m_Y c\Delta\theta_Y$, density $= \frac{mass}{volume}$, as the liquid in both flask is water and its density is 1gcm^{-3}, volume of the flask can be used in place of mass. Heat flow in flask X, $H_X = V_X c\Delta\theta_X$, Heat flow in flask Y, $H_Y = V_Y c\Delta\theta_Y$,

Ratio of heat flow, $\frac{H_X}{H_Y} = \frac{V_X c\Delta\theta_X}{V_Y c\Delta\theta_Y} = \frac{V_X c(\theta_X-0)}{V_Y c(\theta_Y-0)} = \frac{V_X\theta_X}{V_Y\theta_Y} = \frac{V_X}{V_Y}\times\frac{\theta_X}{\theta_Y} = 2\times\frac{1}{2} = 1$.

19. When two objects P and Q supplied with the same quantity of heat, the temperature change in P is observed to be twivce that in Q. If the masses of P and Q are the same, calculate the ratio of the specific heat capacities of P to Q A.1:1 **B.1:2** C.1:3 D.1:4.

For P – $H_P = m_P c_P\Delta\theta_P$, H_P=quantity of heat supplied to P, m_P=mass of P, c_P=specific heat capacity of P, $\Delta\theta_P$=temperature change in P,

For Q – $H_Q = m_Q c_Q\Delta\theta_Q$, H_Q=quantity of heat supplied to Q, m_Q=mass of Q, c_Q=specific heat capacity of Q, $\Delta\theta_Q$=temperature change in Q.

$H_P = H_Q$, $\Delta\theta_P = 2\Delta\theta_Q$, $m_P = m_Q$,

$m_P c_P\Delta\theta_P = m_Q c_Q\Delta\theta_Q$, $c_P(2\Delta\theta_Q) = c_Q\Delta\theta_Q$,

$\frac{c_P}{c_Q} = \frac{\Delta\theta_Q}{2\Delta\theta_Q} = \frac{1}{2}$, $\therefore c_P{:}c_Q = 1{:}2$.

20. Two metals X and Y lose the same quantity of hear when their temperature dropped from 25℃ to 20℃. If the specific heat capacity of X is hyalf that of Y, calculate the ratio of the mass of X to that of Y A.1:2 **B.2:1** C.1:3 D.3:1.

$H_X = m_X c_X\Delta\theta_X$, $H_Y = m_Y c_Y\Delta\theta_Y$, the two metals lose the same quantity of heat with the same temperature change or drop. $\therefore H_X = H_Y$,

$\Delta\theta_X = \Delta\theta_Y$. $m_X c_X\Delta\theta_X = m_Y c_Y\Delta\theta_Y$,

$m_X c_X = m_Y c_Y$, $c_X = \frac{1}{2}c_Y$, $m_X\times\frac{1}{2}c_Y = m_Y c_Y$,

$\frac{1}{2}m_X = m_Y$, $\frac{m_X}{m_Y} = \frac{2}{1}$, $m_X{:}m_Y = 2{:}1$.

21. A piece of copper of mass 0.1kg is heated to 100℃ and then trasnsferred to a well lagged copper can of mass 0.05kg containing 0.2kg of water at 10℃. Calculate the final steady temperature of the mixture [Specific heat capacity of water=4.2×10^{3}J$kg^{-1}K^{-1}$, specific heat capacity of copper= 4.0×10^{2}J$kg^{-1}K^{-1}$] A.7℃ **B.14**℃ C.28℃ D.42℃.

Mass of copper m_C=0.1kg, initial temperature of copper θ_1=100℃, mass of copper can M_C=0.5kg, mass of water m_W=0.2kg, initial temperature of copper can and water θ_o=10℃, specific heat capacity of copper c_C=4.0×10$^2Jkg^{-1}K^{-1}$, specific heat capacity of water c_W=4.2×10$^3Jkg^{-1}K^{-1}$, final steady temperature of the mixture θ=?,

Heat lost by copper = Heat gained by copper can + Heat gained by water,

$m_C c_C\Delta\theta_1 = M_C c_C\Delta\theta_o + m_W c_W\Delta\theta_o$,

$\Delta\theta_1 = \theta_1 - \theta$, $\Delta\theta_o = \theta - \theta_o$,

$0.1\times4.0\times10^2(100-\theta) = 0.05\times4.0\times10^2(\theta-10) + 0.2\times4.2\times10^3(\theta-10)$,

$40(100-\theta) = (\theta-10)(20+840)$,

$40(100-\theta) = (\theta-10)860$,

$(100-\theta) = (\theta-10)\frac{860}{40} = (\theta-10)21.5$,

$100-\theta = 21.5\theta - 215$,

$100+215 = 21.5\theta+\theta$, $315 = 22.5\theta$,

$\theta = \frac{315}{22.5} = 14$℃.

22. Two liquids P at a temperature of 20℃ and Q at a temperature of 80℃ have specific heat capacities of 1.0Jkg^{-1}℃$^{-1}$ and 1.5Jkg^{-1}℃$^{-1}$

Demystified Series by Dr Timothy

repectively. If equal masses of P and Q are mixede in a lagged calorimeter, then the equilibrium temperature is A.44℃ B.50℃**C.56**℃ D.60℃ E.70℃.

m_P=m_Q=m, θ_P=20℃, θ_Q=80℃, θ=?,
c_P=1.0Jkg^{-1}℃$^{-1}$, c_Q=1.5Jkg^{-1}℃$^{-1}$,
Heat lost by Q = Heat gainhed by P ,
$m_Q c_Q(\theta_Q - \theta) = m_P c_P(\theta - \theta_P)$,
$m \times 1.5(80 - \theta) = m \times 1(\theta - 20)$,
$120 - 1.5\theta = \theta - 20$, $120+20 = \theta + 1.5\theta$,
$140 = 2.5\theta$, $\theta = \frac{140}{2.5} = 56$℃.

23. A metal of mass 200g at a temperature of 100℃ is placed in 100g of water at 25℃ in a container of negligible heat capacity. If the final steady temperature is 30℃, calculate the specific heat cdapacity of the ,metal (specific heat capacity of water is 4200J$kg^{-1}K^{-1}$) **A.150J$kg^{-1}K^{-1}$** B.300J$kg^{-1}K^{-1}$ C.320J$kg^{-1}K^{-1}$ D.1960J$kg^{-1}K^{-1}$.

m_m=200g=0.2kg, θ_m=100℃, m_W=100g=0.1kg
θ_W=25℃, c_W=4200$Jkg^{-1}K^{-1}$, θ=30℃,c_m=?,
Heat lost by metal = Heat gained by water,
$m_m c_m(\theta_m - \theta) = m_W c_W(\theta - \theta_W)$,
$0.2 \times c_m(100-30) = 0.1 \times 4200(30-25)$,
$14c_m = 2100$, $c_m = \frac{2100}{14} = 150Jkg^{-1}K^{-1}$.

24. When 100g of liquid L_1 at 78℃ was mixed with Xg of liquid L_2 at 50℃, the final temperature was 66℃.Given that the specific heat capacity of L_2 is half that of L_1, find X A.50g B.100g **C.150g** D.200g .

m_1=100g, θ_1=78℃, θ_2=50℃, c_2=$\frac{1}{2}c_1$, θ=66℃,
m_2=X, Heat lost by L_1 = Heat gained by L_2,
$m_1 c_1(\theta_1 - \theta) = m_2 c_2(\theta - \theta_2)$,
$100c_1(78-66) = X \times \frac{1}{2}c_1(66-50)$,
$1200 = 8X$, $X = \frac{1200}{8} = 150$g.

25. 100g of water at 70℃ is added to 200g of water at 10℃, and the mixture is well stirred. Neglecting the heat absorbed by the container, calculate the final temperature of the mixture A.25℃ B.30℃ C.40℃ D.50℃ E.55℃

m_H=100g, θ_H=70℃, m_C=200g, θ_C=10℃, θ=?,
c_H=c_C, as both liquids are water.
Heat supplied by water at 70℃ = Heat gained by water at 10℃.
$m_H c_H(\theta_H - \theta) = m_C c_C(\theta - \theta_C)$,
Hence, $m_H(\theta_H - \theta) = m_C(\theta - \theta_C)$,
$100(70 - \theta) = 200(\theta - 10)$, $70 - \theta = 2(\theta - 10)$,
$70 - \theta = 2\theta - 20$, $70+20 = 2\theta + \theta$, $90 = 3\theta$,
$\theta = \frac{90}{3} = 30$℃.

26. A tap supplies water at 30℃ while another supplies water at 86℃. If a man wishes to bath with water at 44℃, calculate the ratio of the mass of hot water to that of cold water required **A.1:3** B.3:1 C.3:7 D.7:3

θ_H=86℃, θ_C=30℃, θ=44℃, m_H:m_C=?,
Heat supplied by water at 86℃ = Heat gained by water at 30℃. $m_H(\theta_H - \theta) = m_C(\theta - \theta_C)$
$m_H(86-44) = m_C(44-30)$, $42m_H = 14m_C$,
$\frac{m_H}{m_C} = \frac{14}{42} = \frac{1}{3}$, m_H:m_C = 1:3.

27. A tap supplies water at 25℃ while another supplies water at 75℃. If a man wishes to bath with water at 40℃, the ratio of the mass of cold water to the mass of hot water required is A.1:3 B.3:1 **C.7:3** D.15:18

θ_H=75℃, θ_C=25℃, θ=40℃, m_C:m_H=?,
Heat supplied by water at 75℃ = Heat gained by water at 25℃. $m_H(\theta_H - \theta) = m_C(\theta - \theta_C)$
$m_H(75-40) = m_C(40-25)$, $35m_H = 15m_C$,
$\frac{m_C}{m_H} = \frac{35}{15} = \frac{7}{3}$, m_C:m_H = 7:3.

28. Hot water is added to three times its mass at 10℃ and the resulting temperatur is 20℃ What is the initial temperature of the hot water? A.100℃ B.80℃**C.50**℃ D.40℃

Mass of water at 10℃=three times its mass of hot water, m_C=3m_H , θ_C=10℃, θ=20℃, θ_H=?,
Heat supplied by hot water = Heat gained by water at 10℃, $m_H(\theta_H - \theta) = m_C(\theta - \theta_C)$
$m_H(\theta_H - 20) = 3m_H(20-10)$, $\theta_H - 20 = 30$,
$\theta_H = 30+20 = 50$℃.

29. Water at 10℃ is added to 3 times its mass at 30℃, calculate the temperature of the mixture A.5℃ B.10℃ C.15℃ D.25℃

Mass of at water 30℃=3 times its mass at 10℃,
m_H=3m_C , , θ_C=10℃, θ_H=30℃, θ=?,
$m_H(\theta_H - \theta) = m_C(\theta - \theta_C)$,
$3m_C(30 - \theta) = m_C(\theta - 10)$, $90 - 3\theta = \theta - 10$,
$90+10 = \theta + 3\theta$, $100 = 4\theta$, $\theta = \frac{100}{4} = 25$℃.

30. What mass of water at 100℃ should be added to 15g of water at 40℃ to make thetemperature of the mixture 50℃? (Negelect heat losses to the surroundings) **A.3.0g** B.4.0g C.4.5g D.6.0g E.7.0g.

m_H=?, m_C=15g , θ_H=100℃, θ_H=40℃, θ=50℃,
$m_H(\theta_H - \theta) = m_C(\theta - \theta_C)$,
$m_H(100-50) = 15(50-40)$, $50m_H = 150$,
$m_H = \frac{150}{50} = 3.0$g.

31. The water at the bottom of a waterfall is slightly warmer than that at the top because part of the A.potential energy raises the water temperature B.watr absorbs latet heat from the atmosphere C.kinetic energy is changed into internal energy D.dropping mass of water increases the heat capacity **E.A and C.**

When water falls through a height, the intial potential energy at that height and the kinetic energy of falling water is converted to internal or heat energy at the bottom, hence water becomes slightly warmed at the bottom.

32. A waterfall is 420m high. Calculate the difference in temperature of the water between the top and bottom of the waterfall. Neglect heat losses (g=10ms^{-2}, specific heat capacity of

Demystified Series by Dr Timothy

water=4.20×10³J$kg^{-1}K^{-1}$) A.0.1°C **B.1.0**°C C.4.2°C D.42.0°C E.100.0°C.
h=420m, g=10ms^{-2}, c=4.20×10³$Jkg^{-1}K^{-1}$, $\Delta\theta$=?,
$P.E = H$, $mgh = mc\Delta\theta$, $gh = c\Delta\theta$,
$10\times420 = 4.20\times10^3\times\Delta\theta$, $\Delta\theta = \frac{10\times420}{4.2\times10^3}$,
$\Delta\theta = 1.0$°C.

33. A body of specific heat capacity $450Jkg^{-1}K^{-1}$ falls to the ground from rest through a vertical height of 20m. Assuming conservation of energy, calculate the change in temperature of the body on striking the ground level (g=10ms^{-2}) A.0°C B.$\frac{2}{9}$°C **C.$\frac{4}{9}$**°C D.$\frac{9}{4}$°C E.$\frac{9}{2}$°C .
c=$450Jkg^{-1}K^{-1}$, h=20m, g=10ms^{-2}, $\Delta\theta$=?,
$P.E = H$, $mgh = mc\Delta\theta$, $gh = c\Delta\theta$,
$10\times20 = 450\times\Delta\theta$, $\Delta\theta = \frac{200}{450} = \frac{4}{9}$°C .

34. A good calorimeter should be of A.low specific heat capacity and low heat conductivity **B.low specifc heat capacity and high conductivity** C.high specific heat capacity and low conductivity D.high specific heat capacity and high conductivity D,dull surface and low heat conductivity.
A good calorimeter should respon quickly to temperature changes i.e high thermal conductivity, and should be able to raise the temperature of a body to a very high value hence, must be made of low specific heat capacity.

35. Which of the following processes does not reduce heat loss from a liquid in a calorimeter? A.Lagging the calorimeter B.Using an insulator lid C.Shielding the calorimeter from draughts **D.Constantly stirring the liquid** E.Polishing the inner and outer surfaces of the calorimeter.

36. During the transformation of matter from the solid to the liquid state,the heat supplied does not produce a temperature increase because A.the heat energy is quickly conducted away B.the heat capacity has become very large as the substance melts C.the heat gained is equal to the heat lost by the substance **D.all the heat is used to break the bonds holding the molecules of the solid together**
During change of state, the heat supplied is used to break the bond holding molecules together without any change in temperature of the body.

37. Thelatent heat of fusion of a substance is the heat required to A.raise the temperature of unit mass of the substance by 1°C B.maintain unit mass of the substance a constant temperature **C.change unit mass of substance from solid to liquid at constant temperature** D.raise the temperature of unit mass by 100°C.

38. In the process of changing a piece of ice to water at 0°C A.some quantity of heat is given out B.there is usually a temperatuyre change C.some quantity of heat is taken in D.the quantity of heat given out is definite E.the quantity of heat taken in varies with the temperature of the ice.
Latent heat is absorbed or taken in when ice (solid) changes to water (liquid) at constant temperature.

39. Which of the following occurs bwhen a substance solidifies? A.The latent heat of fusion is absorbed B.There is a rise in temperature **C.The latent heat of fusion is given off** D.There is a fall in temperature E.The specific heat is given off.
When a substance solidifies of freezes it gives off latent heat of fusion but when a substance melts is absorbs latent heat of fusion.

40. What happens when a certain quantity of pure ice is completely changed to water at 0°C? **A.Latent heat is absorbed, the mass remains constant and the volume decreases** B.Latent heat is given out, the mass remains constant and the volume decreases C.Latent heat is given out, the mass increases and the volume remains constant D.Latent heat is absorbed, the mass decreases and the volume increases.
When ice melts to water at a constant temperature of 0°C, latent heat is absorbed, the mass remains constant, the volume decreases and its density increases.

41. The specific latent hear of suion of lead is the amount of heat required to A.Melt lead at its melting point B.Heat a unit mass of lead through 1°C **C.Change the state of a unit mass of lead at its melting point** D.Change the state of a unit mass of lead at its boiling point.

42. Which of the following statements is true of the latent heat of fusion of a body? A.It is the heat required to change the temperature of the body **B.It is the energy required to breakdown the forces which hold the atoms in a regular pattern without a change in temperature** C.It is the energy required to bring the body to its melting point D.It is the energy required to change the temperature of the body after melting E.It is the energy required to break down the forces which hold the atoms of a unit mass of the body in a regular pattern at a definite temperature.
Option E gives the definition of specific latent heat of fusion.

43. An ice cube floats in a glass of water filled to the brim. What happens when the ice cube melts? **A.The water level remain the same** B.There is a drop in the level of water in the glass due to condensation outside C.The water level in the glass overflows D.The level of water drops because melted ice occupies less volume E.None of the above is correct.
The ice cube displaces volume of water equal to its own volume before floating. Hence, after melting, it occupies the volume of water originally displaced.

44. If a container is filled with ice to the brim,

Demystified Series by Dr Timothy

what happens to the level of water when the ice melts completely? A.The water in the glass overflows **B.The level of waterdrops** C.The level of water remains unchanged D.The level of water goes up

The water level drops because ice occupies more volume than water (anomalous expansion) and due to the fact that the air spaces between them is completely filled with water when it melts.

45. Changes of state is accompanied by a change of A.temperature B.volume C.heat content D.temperature and volume **E.volume and heat content.**

Change of state is accompanied by change in density,volume and heat content without any change in the mass and temperature.

46. The specific latent heat of vaporization of a substance is the heat needed to A.change a unit mass of the substance at 100°C to vapour without any change at temperature B.raise the temperature of a unit mass of the substance by 1°C C.change the subtstance from liquid to vapour at its boiling point without any change in temperature D.change the temperature of the substance from 0°C to 100°C**E.change a unit mass of the substance at its boiling point from liquid to vapour without aby change in temperature**

Option C the definition of latent heat of vaporization.

47. A steam trap is a component of a steam generator that can be used to determine the specific latent heat of vapourization of water. In the steady state, steam trap A.stores the steam for future use B.prevents the steam from escaping **C.collects the water that condenses** D.warms the steam up to 100°C E.determine the quality of steam used

Steam or water trap dries steam so that only dry steam rather than condensed(wet) steam or liquid water, passes through the delivery tube into the calorimeter.

48. Which of the following observations is not correct? A.When water freezes its volume increases **B.When ice melts the mass of the water produced is less** C.When ice melts, its volume decreases D.An increase in pressure lowers the melting point of ice E.An increase in pressure increases the boiling point

When water freezes into ice, its density decreases while its volume increases, the opposite hold when ice melts. When ice melts the mass of the water produced is same as that of the ice. An increase in pressure increases the boiling point but decreases the melting point.

49. Which of these statements is not correct? A.When the temperature of a solid is increased, each molecule in the solid increases its amplitude of oscillation **B.When a liquid is heated to a high enough temperature, the vibrations become so violent that the molecules are forced to leave the surface** C.The temperature of a solid is constant during melting because the heat supplied is used up in keeping the molecules in their mean positions D.The temperature of liquid falls during evaporation because the more energetic molecules leave the liquid while the less energetic one left behind E.The average kinetic energy of a liquid increases with temperature.

Heating a solid to increase its temperature cuases its molecules to vibrate at a high amplitude. When a liquid is heated to high enough temperature, the molecules changes to the gaseous state. Evaporatiion which is the escape of liquid molecules from a surface does not depend on heating, it takes place at all temperature and bring about a cooling effect i.e fall in temperature.

50. If a weight is supported by a loop of copper wire which passes over a block of ice, the wire gradually passes through the ice but the ice still remains in one piece. This phenomenon is known as A.freezing B.conduction C.melting **D.regelation** E.convention.

The experiment shows regelation which is the effect of pressure on the melting point of a solid.

51. The following are some units I.Ns II.Nm III.Nm^{-2} IV.JK^{-1} V.Jkg^{-1}. What are the units of latent heat? A.I B.II C.III **D.IV** E.V.

52. How much heat is required to convert 20g of ice at 0°C to water at the same temperature? [Specific latent heat of fusion of ice=$336Jg^{-1}$] A.1.35×10^3J B.5.38×10^3J **C.6.72×10^3J** D.7.06×10^3J E.8.06×10^3J.

m=20g, L=$336Jg^{-1}$, H=?, $H = mL$,

$H = 20\times336 = 6720\text{J} = 6.72\times10^3\text{J}$.

53. The latent of fusion of ice is $80\text{cal}g^{-1}$. How much heat is required to change 10g of ice at 0°C into water at the same temperature A.8cal B.10cal C.80cal D.400cal **E.800cal.**

L=$80\text{cal}g^{-1}$, m=10g, H=?, $H = mL$,

$H = 10\times80 = 800\text{cal}$.

54. Ice of mass 50g at 0°C was completely melted by 70% of the total heat energy supplied. Calculate the total energy supplied. [Specific latent heat of fusion of ice=$336\text{J}g^{-1}$] A.2400J B.6800J C.7251J **D.24000J.**

m=50g, 70% of H supplied to melt the ice,

$70\%H = mL$, $\frac{70}{100}H = mL$, $H = \frac{100mL}{70}$,

$H = \frac{100\times50\times336}{70} = 24000\text{J}$.

54. The melting point of a solid is given as 80°C. If 10^5J of heat energy is required at this temperature to melt 10g of the solid, what is the specific latent heat of fusion of the solid? A.$1.00\times10^3\text{J}kg^{-1}$ B.$1.25\times10^5\text{J}kg^{-1}$ **C.$1.00\times10^7\text{J}kg^{-1}$** D.$8.00\times10^8\text{J}kg^{-1}$.

Demystified Series by Dr Timothy

H=10^5J, m=10g=0.01kg, L=?, $H = mL$,
$L = \frac{H}{m} = \frac{10^5}{0.01}$ =$1.00\times10^7 Jkg^{-1}$.

55. A 90W immersion heater is used to supply energy for 5 minutes. The energt supplied is used to completely melt 180g of a solid at its melting point. Neglecting energy losses to the surrounding, calculate the specific latent heat of fusion of the solid A.0.5Jg^{-1} B.2.5Jg^{-1} C.15.0Jg^{-1} **D.150.0Jg^{-1}** E.450.0Jg^{-1}.
P=90W, t=5minutes=300s, m=180g, L=?,
$Pt = mL$, $90\times300 = 180\times L$, $L = \frac{90\times300}{180}$,
$L = 150.0$Jg^{-1}.

56. What amount of current would pass through a 10Ω coil if it takes 21s for the coil to just melt a lump of ice of mass 10g at 0°C if there are no heat losses? [latent heat of fusion of ice=$336Jg^{-1}$] A.2A **B.4A** C.8A D.16A.
R=10Ω, t=21s, m=10g, L=$336Jg^{-1}$, I=?,
$I^2Rt = mL$, $I^2\times10\times21 = 10\times336$,
$I^2 = \frac{10\times336}{10\times21} = 16$, $I = \sqrt{16} = 4A$.

57. An electric heater is used to melt a block of ice of mass 1.5kg. If the heater is powered by a 12V battery and a current of 20A flows through the coil, calculate the time taken to melt the block of ice at 0°C [Specific latent heat of fusion of ice=$336\times10^3 Jkg^{-1}$] A.76.0min **B.35.0min** C.21.0min D.2.9min.
m=1.5kg, V=12V, I=20A, L=$336\times10^3 Jkg^{-1}$, t=?,
$IVt = mL$, $20\times12\times t = 1.5\times336\times10^3$,
$t = \frac{1.5\times336\times10^3}{20\times12} = 2100$s = 35min.

58. Calculate the quantity of heat released when 100g of steam 100°C condenses to water (Take the specific latent heat of vapourization of water as $2.3\times10^6 Jkg^{-1}$) A.2.3×10^1J B.2.3×10^2J C.2.3×10^4J **D.2.3×10^5J** E.3.3×10^7J.
m=100g=0.1kg, L=2.3×10^6Jkg^{-1}, H=?,
$H = mL$, $H = 0.1\times2.3\times10^6 = 2.3\times10^5$J.

59. A quantity of steam at 100°C condenses to water at the same temperature by releasing 6.9×10^4J of energy. Calculate the mass of the condensed steam (Specific latent heat of vaporization of water=2.3×10^6Jkg^{-1}) A.3.0×10^{-1}kg **B.3.0×10^{-2}kg** C.3.0×10^{-3}kg D.3.0×10^{-4}kg.
H=$6.9\times10^4 J$, L=2.3×10^6Jkg^{-1}, m=?,
$H = mL$, $m = \frac{H}{L} = \frac{6.9\times10^4}{2.3\times10^6} = 3.0\times10^{-2}$kg.

60. A heater marked 50W will evaporate 0.005kg of boiling water in 50 seconds.What is the specific latent heat of vaporization of water in J/kg A.1.0×10^6B.2.5×10^6 C.2.5×10^5**D.5.0×10^5** E.5.0×10^6.
P=50W, t=50s, m=0.005kg, L=?, $Pt = mL$,
$50\times50 = 0.005\times L$, $L = \frac{50\times50}{0.005}$ =5.0×10^5 J/kg

61. All the heat generated in a 5Ω resistor by a 2A flowing for 30 seconds is used to evaporate 5g of a liquid at its boiling point. What is the correct value of specifc latent heat of the liquid? A.120J B.60Jg^{-1} **C.120Jg^{-1}** D.1500J E.1500Jg^{-1}.
R=5Ω, I=2A, t=30s, m=5g, L=?, $I^2Rt = mL$,
$L = \frac{I^2Rt}{m} = \frac{2^2\times5\times30}{5} = 120Jg^{-1}$.

62. A quantity of ice at −10°C is heated until the temperature of the heating vessel is 90°C. Which of the following constants is not required to determine the quantity of heat supplied to the vessel? A.specific heat capacity of ice B.specific heat capacity of water C.specific latent heat of fusion **D.specific latent heat of vaporization.**
Heating of ice from −10°C to 90°C involves –
I.Conversion of ice at −10°C to 0°C(m.p of ice) i.e $H_1 = mc_i\Delta\theta_i$(specific heat capacity of ice).
II.Conversion of ice at 0°C to water at 0°C i.e $H_2 = mL_i$ (latent heat of fusion of ice),
III.Conversion of water at 0°C to water at 90°C i.e $H_3 = mc_W\Delta\theta_W$ (specific heat capacity of water).
Hence, latent heat of vaporizatiojn is inot required, as the temperature of liquid water is not up to its boiling point(100°C).

63. Ice of mass 10g at −5°C was completely converted to water at 0°C. Calculate the quantity of heat used (Specific heat capacity of ice=2.1J$g^{-1}K^{-1}$, specific latent heat of fusion of ice=336Jg^{-1}) A.105J B.3255J **C.3465J** D.16821J.
m=10g, c_i=2.1J$g^{-1}K^{-1}$, L_i=336Jg^{-1}, H=?,
Total heat energy H required to convert ice at −5°C to water at 0°C includes –
Conversion of ice at −5°C to ice at ice at 0°C : $H_1 = mc_i\Delta\theta_i$ (specific heat capacity),
Conversion of ice at 0°C to water at 0°C :
$H_2 = mL_i$ (specific latent heat of fusion).
$H = H_1 + H_2 = mc_i\Delta\theta_i + mL_i = m(c_i\Delta\theta_i + L_i)$,
$H = 10(2.1(0-(-5)+336) = 10(2.1(5) + 336)$,
$H = 10(10.5+336) = 3465$J.

64. How much heat energy is needed to change 3g of ice at 0°C to steam? (Specific latent heat of fusion=336Jg^{-1}, specific latent heat of vaporization of water=2260Jg^{-1}, specific heat capacity of water=4.2J$g^{-1}K^{-1}$) A.2268J B.3016J C.7788J D.7800.6J **E.9048J.**
m=3g, L_i=336Jg^{-1}, L_s=2260Jg^{-1},
c_W=4.2J$g^{-1}K^{-1}$, H=?,
Total heat energy H required to convert ice at 0°C to steam at 100°C includes –
Conversion of ice at 0°C to water at 0°C : $H_1 = mL_i$ (specific latent heat of fusion),
Conversion of water at 0°C to water at 100°C : $H_2 = mc_W\Delta\theta_W$ (specific heat capacity of water),
Conversion of water at 100°C to steam at 100°C : $H_3 = mL_s$ (specific latent heat of vapourization).
$H = H_1 + H_2 + H_3 = mL_i + mc_W\Delta\theta_W + mL_s$,
$H = m(L_i + c_W\Delta\theta_W + L_s)$,
$H = 3(336+4.2(100-0)+2260)$,
$H = 3(336+420+2260) = 9048J$.

65. Calculate the amount of heat required to convert 2kg of ice at −2°C to water at 0°C (specific heat capacity of ice=2090Jkg^{-1}°C^{-1},

Demystified Series by Dr Timothy

specific latent heat of fusion of ice= 333kJkg^{-1}) A.666J B.8360J C.666000J **D.674360J.**
m=2kg, L_i=333$KJkg^{-1}$=333000Jkg^{-1},
c_i=2090Jkg^{-1}°C^{-1}, H=?,
Total heat energy H required to convert ice at −2°C to water at 0°C includes –
Conversion of ice at −2°C to ice at ice at 0°C : $H_1 = mc_i\Delta\theta_i$ (specific heat capacity),
Conversion of ice at 0°C to water at 0°C :
$H_2 = mL_i$ (specific latent heat of fusion).
H=$H_1 + H_2 = mc_i\Delta\theta_i + mL_i$, $H = m(c_i\Delta\theta_i + L_i)$
$H = 2(2090(0-(-2)+333000)$,
$H = 2(2090(2) + 333000) = 2(4180+333000)$
H =674360J.

66. How much heat energy is needed to convert 20g of ice at −15°C to steam at 100°C? (specific heat of ice=2.2$Jg^{-1}K^{-1}$, specific heat of water=4.18$Jg^{-1}K^{-1}$, specific latent heat of ice=0.334KJg^{-1}, specific latent heat of steam=2.260KJg^{-1}) A.77.28kJ **B.60.90kJ** C.60.32kJ D.52.55kJ E.51.92kJ.
m=20g, c_i=2.2$Jg^{-1}K^{-1}$, c_W=4.18$Jg^{-1}K^{-1}$,
L_i=0.334KJg^{-1}=334Jg^{-1}, L_s=2.260KJg^{-1}= 2260 Jg^{-1},
Total heat energy H required to convert ice at −15°C to steam at 100°C includes –
Conversion of ice at −15°C to ice at 0°C : $H_1 = mc_i\Delta\theta_i$ (specific heat capacity of ice),
Conversion of ice at 0°C to water at 0°C : $H_2 = mL_i$ (specific latent heat of fusion),
Conversion of water at 0°C to water at 100°C :
$H_3 = mc_W\Delta\theta_W$ (specific heat capacity of water),
Conversion of water at 100°C to steam at 100°C :
$H_4 = mL_s$ (specific latent heat of vapourization).
$H = H_1 + H_2 + H_3 + H_4$,
$H = mc_i\Delta\theta_i + mL_i + mc_W\Delta\theta_W + mL_s$,
$H = m(c_i\Delta\theta_i + L_i + c_W\Delta\theta_W + L_s)$,
H=20(2.2(0−(−15)+334+4.18(100−0)+2260
$H = 20(2.2(15)+334+4.18(100)+2260$,
H= 20(33+334+418+2260) = 60900J= 60.90KJ.

67. A solid of mass 0.5kg initially at 50°C melt at 60°C. Calculate the quantity of the heat required to melt it completely (Specific heat capacity of the solid=2.0$KJkg^{-1}K^{-1}$, Specific latent heat of fusion of the solid=1.0$KJkg^{-1}$)
A.1500J B.10000J **C.10500J** D.15000J
m=0.5kg, c=2.0$KJkg^{-1}K^{-1}$=2000 $Jkg^{-1}K^{-1}$,
L=1.0$KJkg^{-1}$=1000Jkg^{-1}, H=?,
Total heat energy H required to melt the solid at 60°C –
Conversion of the solid at 50°C to 60°C : $H_1 = mc\Delta\theta$ (specific heat capacity of solid),
Conversion of the solid at 60°C to the liquid at 60°C (melting) : $H_2 = mL$ (specific latent heat of fusion). $H = H_1 + H_2 = mc\Delta\theta + mL$,
$H = m(c\Delta\theta + L) = 0.5(2000(60-50)+1000)$,
H =0.5(2000(10)+1000) =10500J.

68. Calculate the heat energy lost when 10g of boiling water changes to ice at 0°C (specific latent heat of ice=336Jg^{-1}, specific heat capacity of water=4.2J$g^{-1}K^{-1}$) A.840J B.3360J C.3420J D.4200J **E.7560J.**
m=10g, L_i=4.2$Jg^{-1}K^{-1}$, c_i=336Jg^{-1}, H=?,
Total heat energy H lost by boiling water –
Conversion of boiling water at 100°C to water at 0°C : $H_1 = mc_W\Delta\theta_W$,
Conversion of water at 0°C to ice at 0°C (freezing) : $H_2 = mL_i$. $H = H_1 + H_2$,
$H = mc_W\Delta\theta_W + mL_i = m(c_W\Delta\theta_W + L_i)$,
H =10(4.2(100−0)+336) = 10(4.2(100)+336)
$H = 10(420+336) = 7560J$.

69. What is the difference in the amount of heat given out by 4kg of steam and 4kg of water when both are cooled from 100°C to 80°C? (specific latent heat of steam= 2260000Jkg^{-1}, specific heat capacity of water=4200$Jkg^{-1}K^{-1}$) A.4200J B.2260000J **C.9040000J** D.9380000J.
L_s=2260000Jkg^{-1}, c_W=4200$Jkg^{-1}K^{-1}$, m=4kg ,
Cooling steam from 100°C to 80°C involves –
Conversion of steam at 100°C to water at 100°C (condensation) : $H = mL_s$, Conversion of water at 100°C to water at 80°C : $H = mc_W\Delta\theta_W$.
Total heat given out by cooling steam = $mL_s + mc_W\Delta\theta_W$,
While cooling water from 100°C to 80°C, heat given out = $mc_W\Delta\theta_W$. Hence difference in the amount of heat energy given when steam and water are cooled from 100°C to 80°C = $(mL_s + mc_W\Delta\theta_W) - mc_W\Delta\theta_W = mL_s$,
Difference in heat energy = 4×2260000 ,
Difference in heat energy = 9040000J.

70. 1kg of copper is transferred quickly from boiling water to a block of ice. Calculate the mass of ice melted, neglecting heat loss (specific heat capacity of copper = 400J$kg^{-1}K^{-1}$, latent heat of fusion of ice= 333×10^3Jkg^{-1}] A.60g B.67g **C.120g** D.133g.
c_c=400J$kg^{-1}K^{-1}$, L_i=333×10^3Jkg^{-1}, m_c=1kg, m_i=?, $\Delta\theta_c$=100−0=100°C,
Heat lost by copper at 100°C = Heat required to melt ice at 0°C. $m_c c_c\Delta\theta_c = m_i L_i$,
$1\times400\times100 = m_i \times333\times10^3$,
$m_i = \frac{400\times100}{333\times10^3} = 0.120\text{kg} = 120\text{g}$.

71. When comparing the specific latent heats of vapourization L_v of a substance to its specific latent heat of fusion L_f , which of the following statements is true A.$L_v < L_f$ B.$L_v = L_f$ **C.$L_v > L_f$** D.L_v and L_f can't be measured.
Specific latent heat of vaporization L_v of a substance is always greater than its specific latent heat of fusion L_f.

72. A person suffers a more severe burn from steam than from boiling water because A.steam is at a higher temperature than boiling water B.steam spreads more easily over the skin than boiling water C.steam penetrates more deeply into the skin than boiling water **D.steam posses**

Demystified Series by Dr Timothy

greater heat content per unit mass than boiling water.
The internal energy or heat content per unit mass of steam is greater than that of boiling waterb as steam posseses both thermal energy of boiling water and latent heat of vaporization. Hence, it causes more severe burn than boiling water. Steam is also at a higher temperature than boiling water.

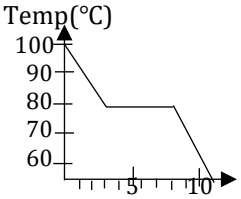
73. Steam from from boiling water causes more damage on the skin than deos boiling water because A.steam has latent heat of fusion **B.the steam is at a higher temperature than water** C.steam brings heat more easily by convention D.water has a high specific heat .
Steam causes more damage on the skin than boiling water because it has latent heat of vaporization and is at a higher temperature than boiling water. Temperature of steam can exceed 100°C but but boiling water can not go above 100°C.

74. The melting point of a substance is equivalent to its A.liquidification temperature **B.solidification temperature** C.solidification pressure D.vapour pressure.
Melting point of a substance is the same as its freezing point or solidification temperature.

75. A glass of soft drink can be cooled more effectively by adding melting ice than by adding the same mass of water at 0°C because A.ice makes better thermal contacts with water B.melting ice is at lower temperature than the water C.water has higher specific heat capacity than ice **D.ice absorbs latent heat as it melts.**
The melting ice asbsorbs the latent heat of fusion needed to melt it from the soft drink, hence decreasing the temperature of the drink i.e cooling it. The water and the melting ice are at the same temperature i.e 0°C.

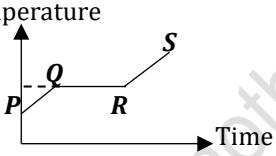
76. The statements below were made by some students describing what happened during the determination of the melting points of solids 1.The temperature of the solid was constanbt until melting 2.The temperature pf the solid rose until melting started 3.During melting, the temperature was rising 4.The temperature was constant 5.The temperature continued to rise after all the solid had melted 6.The temperature stopped rising after all the solid had melted. Which of the following gives the correct statements in the right order? **A.2,4 and 5** B.2,3 and 6 C.1,3 and 5 D.1,3 and 6 E.1,4 and 5.
Temperature of a solid rises before and after melting but remains constant during melting.

77. In an experiment in which molten naphthalene is allowed to cool. The cooling curve in the figure below was obtained. The temperature 80°C is known as the



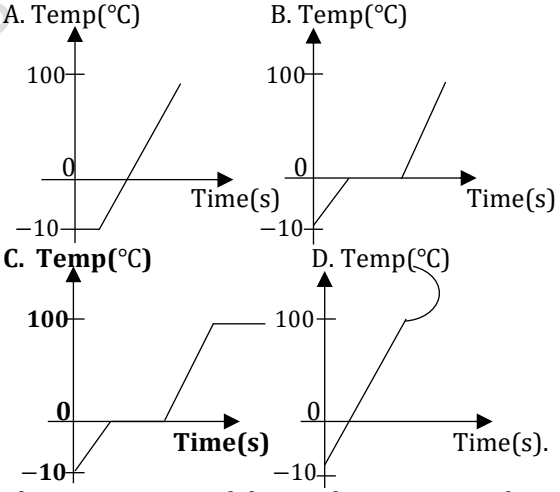
A.Cooling temperature B.Boiling point **C.Melting point** D.Vaporization point .
The temperature 80°C indicates the temperature at which the molten naphthalene solidifies(freezes). Freezing point of a substance is equal to its melting point.

78. The diagram below is the heating curve for a pure solid , ***QR*** is the



A.boiling region B.evaporation region **C.melting region** D.sublimation region.
Temperature during melting is constant and is represented by a horizontal line in the heating curve.

79. A mass of ice at −10°C is heated at a constant rate unitl it all melts and then boils away. Which of the following temp[erature-time graphs correctly represents the process?



The temperature of the ice first increases from −10°C to 0°C. The ice at 0°C then melts to water at 0°C. The temperature of water increases from 0°C to 100°C, wher it finally boils off as steam at 100°C.

80. Evaporation can be explained on the basis of the kinetic theory by the following statements I.The fast-moving molecules of the liquid escape II.Some molecules have greater energy than the average III.The temperature of the remaining liquid falls IV.The average kinetic energy of the remaining molecules falls. What is the correct order in which the statements should be placed? A.I,II,III,IV B.II,III,IV,I **C.II,I,IV,III** D,I,II,IV,III E.I,III,IV,II.

Demystified Series by Dr. Timothy

Evaporation is the escape of vapour molecules from the surface of a liquid below the boiling point of the liquid. According to the kinetic theory, some molecules having kinetic energy greater than average kinetic energy move to the liquid surface and escape from the surface as vapour. The average kinetic energy of the remaining liquid molecules then drops hence decrease in the temperature.

81. Evaporation of a liquid produces cooling of the liquid because the A.density of the liquid decreases B.volume of the liquid decreases C.vapour pressure above the liquid rises D.attraction between the vapour molecules and the liquid molecules is great **E.average kinetic energy of the molecules of the liquid is reduced.**

Evaporation is the escape of molecules with energy greater than the average kinetic energy from the liquid surface, this results in a decrease in the average kinetic energy of the remaining molecules in the liquid and gives a cooling effect due to reduction in temperature.

82. Which of the following about evaporation is not correct? A.It is the escape of high energy molecules from the liquid surface B.It lowers the average internal energy C.It causes the temperature of the liquid to fall **D.It occurs only at the boiling point of the liquid** E.It occurs less where the attractive forces of the molecules are very great.

Evaporation occurs at any temperature below the boiling point of a liquid. It occurs less where the attractive forces of the molecules are very great, as the liquid molecules are strongly attracted to one another.

83. Which of the following is not a factor that can increase the rate of evaporation of water in a lake? **A.Increase in the pressure of the atmosphere** B.Rise in temperature C.Increase in the average speed of the molecules of water D.Increase in the kinetic energy of the energy

The rate of evaporation increases with increase in temperature. Temperature increases as the average speed and kinetic energy increases. Rate of evaporation decreases as the pressure of the atmosphere increases, $E \propto \frac{WAT}{HP\rho}$.

84. The process whereby a liquid turns spontaneously into vapour is called A.boiling **B.evaporation** C.regelation D.sublimation.

85. What does not affect the rate of evaporation of a liquid? A.exposed surface area of the liquid B.humidity of the liquid C.nature of the liquid **D.volume of the liquid** E.temperature of the liquid.

Rate of evaporation depends on wind, exposed surface area,temperature,humidity, pressure and the density of the liquid. It also depends on the nature of the liquid.

86. Which of the bfollowing does not affect the rate of evaporation of a liquid? A.the surface area of the liquid B.force-convention of the air around the liquid C.the temperature of the surroundings D.density of the liquid **E.the osmotic pressure of the liquid.**

Evaporation depends on the surface area of the liquid, wind or force-convention on the liquid, density of the liquid and temperature of the surroundings. It does not depend on the osmotic pressure of the liquid.

87. Blowing air over a liquid aids evaporation by A.decreasing its vapour pressure B,decreasing its density **C.increasing its surface area** D.increasing its temperature

Blowing air or wind over a liquid surface sweeps away vapour from the liquids surface, hence increasing the exposed surface area for more molecules to evaporate.

88. I.Increase in melting point of the liquid II.Increase the boiling point of the liquid III.Decrease the melting of the liquid IV.Decrease the boiling point of the liquid, Which of the statements above about the effects of increase in pressure in a liquid are correct? **A.II and III** B.I and II C.III and IV D.I and III.

Increase in pressure increases the boiling point of a liquid and decreases the melting or freezing point of a solid.

89. Which of the following conditions will make water boil at a temperature of 100°C when the atmospheric pressure is 750mmHg? **A.increasing the external pressure** B.reducing the external pressure C.heat more readily at the same pressure D.reduce the pressure by a quarter.

Water boils at 100°C at normal atmospheric pressure of 760mmHg. When the pressure is lowered to 750mmHg, water tends to boil at a temperature lower 100°C. Hence, the external pressure has to be increased for water to boil at 100°C.

90. The boiling point of water exceeds 100°C when A.It is slowly heated **B.It is rapidly heated** C.Its vapour pressure is greater than atmospheric pressure D.Its vapour pressure is less than the atmospheric pressure.

When water is rapidly heated it means the pressure is increased hence, the boiling point will exceeds 100°C.

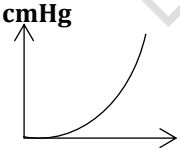
91. A quantity of water was found to boil at 103°C when heated in a beaker. This was because the A.specifc heat capacity of the beaker was high **B.water contained a quanity of dissolved salt** C.beaker contained distilled water D.water was being heate on top of a mountain E.specific heat

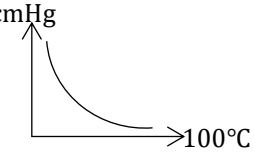
Demystified Series by Dr Timothy

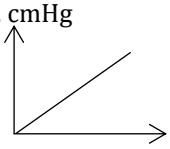
capacity of the mercury in the thermometer was high.
The presence of impurity e.g salt, increases the boiling points of liquids.
92. Food cook quicker in salt water than in pure water because of the effect of A.Food nutrients on the thermal energy B.Salt on the thermal conductivity of water C.Atmospheric pressure on the boiling point **D.Dissolved substances on the boiling point.**
Dissolved substance or impurities increases the boiling point of liquids,
93. What will happen to the boiling point of pure water when it is heated in a place 30m below the sea level? A.It will be less than 100°C B.It will still be 100°C C.It will still be fluctuating **D.It will be more than 100**°C**.**
Presssure increases with depth below sea level and decreases above sea level. Hence, at a place 30m below sea level pressure increases and the boiling point increases.
93. A red hot heated metal is quickly transferred into a beaker containing boiling water. What effect will the hot metal have on the boiling water? **A.The rate of vaporization of the boiling water increases** B.The boiling point rises sharply C.The boiling rises and then fall D.The boiling point rises slightly.
The water is already boiling at a fixed temperature(100°C) and the water is undergoing vaporization. Thus, transferring the red heated metal into the boiling water will increase the rate of vaporization of the boiling water, as the heat supplied to the boiling water becomes more intense.
94. Which of the following is common to evaporation and boiling? They A.take place at any temperature B.are surface phenomena **C.involve change of state** D.take place at a definite temperature.
Both boiling and evaporation involves the change of state from liquid phase into the gaseous phase. Evaporation takes place at any temperature while boiling takes place at a definite temperature. Evaporation is a surface phenomena i.e occurs at the liquid surface while boiling occurs throughout the whole volume of liquid.
95. Which of the following statements gives the true differences between evaporation and boiling? I.evaporation occurs at all temperatures while boiling occurs at fixed temperature for a given pressure II.evaporation is a surface phenomenon while boiling is an interior phenomenon III.evaporation is affected by surface area while boiling is not A.I and II C.I and III C.II and III **D.I,II and III.**
Evaporation occurs at all temperature whule boiling or vaporization occurs at a definite temperature. Evaporation occurs ar the surface only while boiling occurs throughout the whole volume of the liquid. Evaporation is affecred by surface area while boiling is independent of surface area.
96. If a strong flow of air is passed through alcohol contained in a small beaker, water droplets soon form and freezes on the outside of the beaker. This is because alcohol has a A.high latent heat of vaporization B.low specific heat capacity C.high specific heat capacity **D.low latent heat of vaporization** E.very high relative humidity.
Passing air or wind over alcohol increases its rate of evaporation. The latent heat of vaporization needed to evaporate it is drawn from the air around the beaker hence the formation of water droplets around the beaker.
97. Why does a body feel cold when splashed with petrol? A.is naturally cold B.removes heat from the body C.has a very low latent heat of vaporization **D.obtains its latent heat of vaporization from the body** E.has a low freezing point
The splashed petrol draws the latent heat of vaporization from the body, hence decreasing the temperature of the body surface making feel cool.
98. Which of the following substances is the most volatile at room temperature? A.water B.diesel **C.petrol** D.kerosene
A volatile liquid is a liquid which evaporates spontaneously without heat. Volatile liquids have low latent heat of vaporization and low boiling point. Petrol is the most volatile than the other liquids given in the option.
99. A vapour is said to be saturated when A.more molecules return to the liquid than leave it **B.the number molecules leaving the liquid for the vapour are equal in number to those leaving the vapour for the liquid** C.higher number of moleculesreturn to the vapour than the molecules leaving it for the liquid D.the vapour pressure equal the atmospheric pressure.
Vapour is said to be saturated when the number of molecules leaving the liquid equals the number of molecules returning to the liquid.
100. Vapour is said to be saturated on top of an enclosed liquid if **A.the rate of condensation is equal to that of vaporization** B.there is neither condensation nor vaporization of the liquid C.the rate of condensation is less than that of vaporization D.the rate of condensation is greater than that of vaporization.
Vapour is said to be saturated when the rate of vaporization of liquid is equal to the rate of condensation. Vapour is said to be unsaturated when the rate of vaporization is greater than the rate of condensation or the rate of condensation is less than rate of vaporization.

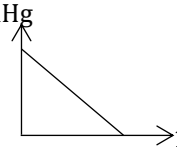
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101. A vapour is said to be saturated when A.more molecules return to the liquid than leave it **B.a dynamic equilibrium exists between the molecules and the vapour molecules at a given temperature** C.the vapour pressure is atmospheric D.the vapour pressure is atmospheric E.all molecules are moving with the same speed.
Saturated vapour is in dynamic equilibrium with its own liquid in a confined space.
102. A vapour whose molecules are in dynamic equilibrium with those of its own liquid is said to be A.unsaturated B.gaseous **C.saturated** D.diffused.
103. Which of the following statements is not correct? **A.An unsaturated vapour is one that is in contact with its own liquid** B.A liquid boils only when its saturation vapour pressure is equal to the external pressure C.The dew point is not fixed but merely describes the condition of the atmosphere at a particular time D.Still air favours the formation of dew E.Relative humidity can be measured by using a hygrometer
An unsaturated vapour is not in contact with its own liquid as it does not tends to escape from the vessel as in evaporation.
104. The saturation vapour pressure of a liquid depends on its A.volume **B.temperature** C.mass D.density E.pressure.
Saturated vapour pressure of a liquid is affected by temperature of the liquid only. It is independent of the volume of the vapour or the external pressure.
105. The sarurated vapour pressure of a liquid increases as the A.volume of the increases B.volume of the liquid decreases **C.temperature of the liquid increases** D.temperature of the liquid decreases.
Saturated vapour pressure increases with the temperature of the liquid.
106. Which of the following correctly represents the variation of the vapour pressure of water with temperature?

A. cmHg


B. cmHg


C. cmHg


D. cmHg


107. Which of the following statements is/are not true? I.A saturated vapour is one that is in equilibrium with its liquid in a closed space II.A vapour contains molecues with varying kinetic energy and their distances apart are smaller than than those of the molecules of a liquid III.Relative humidity is $\frac{saturated\ vapour\ pressure\ at\ dew\ point}{saturated\ vapour\ pressure\ at\ air\ temperature} \times 100\%$ IV.The saturated vapour pressure of water increases with temperature A.I **B.II** C.II and III D.III and IV E.II,III and IV.
The molecules of vapour(or gases) have varying kinetic energy but their average kinetic energy is the same, also their molecules are at a greater distance apart than a liquid.
108. Which of the following is not correct? A.boiling occurs when the saturated vapour pressure of the liquid involved equals the external pressure **B.both the boiling point and the saturated vapour pressure depends on the external pressure** C.saturated vapour pressure rises with increase in temperature D.the saturated vapour pressure is independent of volume availiable for the vapour E.it is possible to boil water at a lower temperature than 100℃ in altitude.
Boiling point depends on external pressure while saturated vapour pressure is independent of external pressure and volume of the vapour available. Pressure decreases with altitude or height, hence at high altitude pressure decreases and the boiling point of water decreases below 100℃.
109. Water boils at a lower temperature when heated at the top of a mountain than at sea level because at the top of a mountain A.saturation vapour pressure of water is higher than at sea level B.sun adds more heat to the water C.temperature is lower than at sea level D.relative humidity of the atmosphere is lower **E.pressure of the atmosphere is lower.**
Pressure decreases with height above sea level. At the top of a mountain i.e high altitude, the pressure is lowered which inturn lowers the boiling point of water.
110. The following is a table of the $S.V.P$ of water from 10℃ to 50℃.

Temp/℃	10	20	30	40	50
$S.V.P$/mmHg	9.2	17.5	31.8	55.0	92.5

On a particular day, when the atmospheric pressure is 78.85cm of Hg, some water vapour is present in a barometer tube which reads 72.62cm of mercury. What is temperature of the surrounding? A.10℃ **B.30**℃ C.40℃ C.45℃ D.55℃.
Atmspheric pressure P_{atm}=78.85cmHg, barometric reading = height of mercury column h=72.67cmHg. **The total pressure is equal to the atmospheric pressure which is given by the sum of the saturated vapour pressure due to the water vapour and the pressure due to the height of the mercury column.** $\boldsymbol{P_T = P_{atm} = S.V.P + h}$, $\boldsymbol{S.V.P = P_{atm} - h}$,
$S.V.P = 78.85 - 72.67 = 3.18\text{cmHg} = 31.8\text{mmHg}$.

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The temperature which corresponds to the saturated vapour pressure of water from the table is 30°C.

111. A simple barometer contains a thin layer of water on top of the mercury column at 20°C. If the atmospheric pressure is 75.8cm of mercury and the saturation vapour pressure of water vapour at 20°C s 1.7cm of mercury, calculate the height of the mercury column A.77.5cm B.77.1cm C.76.1cm D.74.4cm **E.74.1cm.**

P_{atm}=75.8cmHg, $S.V.P$=1.7cmHg, h=?,

$P_{atm} = S.V.P + h$, $h = P_{atm} - S.V.P$,

$h = 75.8 - 1.7 = 74.1$cmHg.

112. A pressure cooker saves both time and fuel in cooking because inside the cooker the **A.boiling point of water is raised** B.pressure is constant C.heat is completely trapped D.volume of steam varies E.temperature is evenly distributed.

Pressure cooker is used to cook foods at high altitudes where pressure lowered. The pressure in the pressure cooker makes water to boil above its boiling point i.e 100°C, hence food is cooked at a higher temperature saving both time and fuel. Mountain dwellers cook with pressure cooker as atmospheric pressure is low at high altitudes e.g mountain tops, making to water boil below 100°C.

113. The main reason why rice cooks faster in a pressure cooker than in cooking pot is that A.less heat escapes from the cooker B.the vapour pressure in the cooker is constant C.the rice is not affected by draught over the cooker D.the vapour pressure in the cooker decreases **E.the boiling point of water in the cooker is raised.**

114. Clouds formation is the direct result of A.precipitation B.vaporization C.fusion D.sublimation **E.condensation.**

115. The phenomenon whereby water droplets in the atmosphere combine with dust particles to reduce visibility is **A.fog** B.hail C.mist D.cloud.

116. The small droplets of water that form on the grass in the early hours of the morning is A.hail **B.dew** C.fog D.mist.

Dew is condensed water vapour formed on object in contact with the ground while mist is suspended water droplets in air.

117. The temperature at which the water vapour present in the air saturates the air and begins to condense is known as A.boiling point B.melting point C.triple point **D.dew point** E.critical temperature.

Dew point is the temperature at which the water vapour present in air is just sufficient to saturate it.

118. Dew point is the temperature at which water vapour in the atmosphere A.Turns into steam B.Soliddifes into ice pellets **C.First condenses into liquid form** D.Is just sufficient to cause cooling E.Has a relative of fifty percent.

Dew point is the temperature or point at which water vapour or moisture in the atmospehere first condenses into the liquid form.

119. If a room is saturated with water vapour, the temperature of the room must be A.at 0°C B.at 100°C C.above the dew point **D.below or at the dew point.**

Air is unsaturated with water vapour above dew point. Air is be saturated with water vapour at dew point. Air is supersaturated with water vapour below dew point and at this temperature dew is formed as the air cdan no longer hold water vapour causing it to condense into dew.

120. At low humidity in an environment, the human skin is usually A.dry and smooth B.damp and smooth C.damp and rough **D.dry and rough.**

At low humidity, the rate of evaporation is high and the skin is dry and rough while at high humidity, the rate of evaporation is low and the skin is damp and smooth.

121. On a day when humidity of the air is very high, evaporation from a lake will A.be rapid **B.be slow** C.be nil D.increase with temperature D.decrease with temperature.

122. When the relative humidity of the atmosphere is described as being 100%, it implies that **A.the air in the atmosphere is saturated with water vapour** B.water can easily evaporate into the atmosphere C.the atmosphere contains very little water vapour D.the air in the atmosphere is very dry.

Relative humidity is the degree of humidity or the degree of water vapour present in the atmosphere. A relative humidity of 100% menas theair in the atmosphere is 100% humid or wet and the air in the atmosphere is completely saturated with water vapour.

123. The mass of water vapour in a given volume of air is 0.05g at 20°C, while the mass of water vapour required to satiurate it at the same temperature is 0.15g. Calculate the relative humidity of the air A.3.33% B.5.55% **C.33.33%** D.55.55%.

Mass of water vapour in air m=0.05g, mass of water vapour required to saturate it M=0.15g, relative humidity $R.H$=?,

$$R.H = \frac{mass\ of\ water\ vapour\ in\ a\ given\ vol.\ of\ air}{mass\ of\ water\ vapour\ required\ to\ saturate}$$

$$R.H = \frac{m}{M} \times 100 = \frac{0.05}{0.15} \times 100 = 33.33\%.$$

124. On a certain day, the dew point is 10°C when the temperature of the air is 32°C. Calculate the relative humidity of the air [$S.V.P$ of water at 32°C and 10°C are 16.2mm and 9.8mm of mercury respectively] A.10.4% B.12.8% C.25.2% **D.60.4%** E.74.1%.

$S.V.P$ at dew point of 10°C=9.8mmHg, $S.V.P$ at air temperature of 32°C=16.2mmHg, relative humidity $R.H$=?

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$R.H = \frac{S.V.P\ of\ water\ at\ dew\ point}{S.V.P\ of\ water\ at\ air\ temperature} \times 100$,

$R.H = \frac{9.8}{16.2} \times 100 = 60.4\%$.

125. If the partial pressure of water vapour at 27°C is 18mmHg and the saturated vapour pressure at the same temperature is 24mmHg the relative humidity at this temperature is A.25% B.33% **C.75%** D.82%.

Partial pressure of water vapour at 27°C= 18mmHg, $S.V.P$ of water at 27°C=24mmHg, relative humidity $R.H$=?,

$R.H = \frac{Partial\ pressure\ of\ air}{S.V.P\ of\ the\ air} \times 100$,

$R.H = \frac{18}{24} \times 100 = 75\%$.

126. The table above shows the saturation vapour pressure against temperatuire in a certain town.

T°C	0	5	10	15	20	40	60
S.V.P.(mmHg)	4.58	6.51	8.94	12.67	17.50	55.10	149.00

If the vapour pressure in this town at 20°C is 10mmHg, what is the relative humidity? A.170.0% **B.57.0%** C.17.5% D.10.0%.

Vapour pressure $V.P$ of air at 20°C=10mmHg, $S.V.P$ of air at 20°C=17.50mmHg,

$R.H = \frac{vapour\ presssure\ of\ air}{S.V.P\ of\ the\ air} \times 100$,

$R.H = \frac{V.P}{S.V.P} \times 100 = \frac{10}{17.50} \times 100 = 57.0\%$.

127. At which of the following temperatures will the relative humidity of the air in a room be least if the dew point in the room is 20°C A.10°C B.15°C C.20°C D.25°C **E.30°C.**

Air temperature is always greater than the dew point or dew point is always less than the air temperature. Hence, the air temperature can either be 25°C or 30°C which are above the dew point 20°C. Saturated vapour pressure increases with temperature $S.V.P \propto T$, therefore $S.V.P$ at 30°C is greater than $S.V.P$ at 25°C.

$\boldsymbol{R.H \propto \frac{S.V.P\ of\ water\ at\ dew\ point}{S.V.P\ of\ water\ at\ air\ temperature}}$, relative humidity increases with $S.V.P$ at dew point and decreases with $S.V.P$ at air temperature. Relative humidity is least when $S.V.P$ at air temperature is greatest or maximum i.e at 30°C.

128. Which of the following may be used to determine the relative humidity in a physics laboratory? I.Manometer II.Wet and dry bulb hygrometer III.Hair hygrometer IV.Hydrometer A.I **B.II** C.II and III D.III E.II,III and IV.

129. Which of the following instruments is used to determine the relative amount of water vapour present in the atmosphere? A.hypsometer B.hydrometer C.barometer **D.hygrometer** E.manometer.

Relative humidity or relative amount of water vapour present in the atmosphere is measured by a hygrometer.

130. When the temperature difference between the wet and dry bulbs of a hygrometer is high, this indicates that A.the relative humidity is high **B.the relative humidity is low** C.it is about to rain D.there is plenty of sunshine

Relative humidity is inversely proportional to the temperature difference between the wet and dry bulbs of a hygrometer $R.H \propto \frac{1}{\Delta T}$. When the temperature difference is high it indicates a low relative humidity and when the temperature difference is low is indicates a high relative humidity.

- Relative humidity is 100% when the temperature difference is zero i.e the wet and dry bulb are at the same temperature.

131.

	Day 1	Day 2	Day 3
Dry	30°C	29°C	25°C
Wet	22°C	22°C	21°C

The readings above are for three consecutive days from wet and dry bulb hygrometer. It can be concluded that the relative humidity for the three days is A.decreased steadily B.remained unchanged C.increased steadily **D.was least on day 1.**

Relative humidity is higher when the temperature difference between the wet and dry bulb is lower. $R.H \propto \frac{1}{\Delta T}$.

Day 1 – $\Delta T = 30 - 22 = 8$°C.

Day 2 – $\Delta T = 29 - 22 = 7$°C.

Day 3 – $\Delta T = 25 - 21 = 4$°C.

Relative humidity is least at day 1 as the temperature is greatest at day 1 (8°C).

Jamb past questions on thermal energy, latent heat and vapours :

1978/15,18,19,21,24,1979/5,10,42,1980/15,32, 1981/2,3,16,37,1982/4,17,27,1983/38,44,1984 /13,17,19,27,1985/19,23,24,25,1986/18,19,20, 21,25,1987/21,22,23,1988/17,18,19,1989/16, 18,1990/18,19,1991/17,18,20,1992/17,18, 1993/17,18,19,1994/22,24,25,1995/22,24,25, 1997/21,23,24,25,26,1998/21,22,1999/11,16, 24,2000/15,24,26,2002/20,2003/14,17,22,25, 2004/39,2005/41,45,50,2006/26,43,45,49, 2007/35,40,2008/22,23,24,2009/21,23,2010/ 19,20,21,2011/16,18,20,2012/22,23,24,2013/ 19,22,2014/22,23,2015/14,47,2016/45].

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Chapter 14 – Heat transfer

● **Heat transfer or flow –** Heat flows naturally from a hot region (high temperature region) to a cold region (low temperature region).

- Heat flow between two bodies in thermal contact when there is a **temperature gradient** (difference in temperature per unit distance between the two regions).

- Heat transfer process continues until both bodies in thermal contact attain the same temperature (thermal equilibrium), where there is no net flow of heat.

- **Thermal equilibrium** is a condition where two bodies in thermal contact are at the same temperature and there is no net flow of heat between both bodies.

- **Modes of heat transfer –** Conduction, convection, radiation. Heat is also transferred partly by evaporation and condensation.

1.Conduction – Conduction involves the transfer of heat through a material medium by the vibratory motion of the molecules about their mean position with higher frequency without movement of the medium.

- It occurs in solids, liquids and gases but majorly in solids.

- Heat transfer by conduction is by the free electrons.

- Heat cannot be conducted through a vacuum.

- **Thermal conductivity** is the extent to which a material can conduct heat.It is dependent on the type of material.The decreasing order of conductivity for common substance is; **Silver→Copper→Aluminium→Brass→Iron (or steel)→Wood→Glass. Glass has the least thermal conductivity.**

- Rate of heat flow $\boldsymbol{R}$ or heat flux by conduction varies directly with temperature gradient $\frac{\Delta\theta}{L}$, cross sectional area $\boldsymbol{A}$ and the nature of the material or thermal conductivity $\boldsymbol{K}$. $\boldsymbol{R = \frac{dQ}{dt} \propto A\frac{\Delta\theta}{L}}$, $\boldsymbol{R = \frac{dQ}{dt} = KA\frac{\Delta\theta}{L}}$.

$\boldsymbol{K}$ is the constant of proportionality called the **thermal conductivity of the conductor.** The unit is **Joule per second per meter per kelvin [$Js^{-1}m^{-1}K^{-1}$]** or **Watt per meter per kelvin [$Wm^{-1}°C^{-1}$]**.The unit for rate of heat flow is Js^{-1} or W.

- **Temperature gradient $\boldsymbol{T_G}$** is the difference in temperature between the hot and cold end per the distance between both ends. $\boldsymbol{T_G = \frac{\Delta\theta}{L} = \frac{\theta_H - \theta_C}{L}}$ Its unit is **Kelvin per meter [Km^{-1}].** .

- **Application of conduction – I.**Insulators or poor conductors of heat(e.g plastic,wood, rubber) are used in handle of cooking utensils or kitchen wares while the body is made of a metal of good thermal conductivity as it responds to temperature change quickly rather than increase in size. **II.**Better conductors tends to conduct heat away from a body rapidly compared to insulators, hence the body feels cold when in contact with such conductors. **III.**Ignition point of gas e.g wire gaze,fire fighting suits,miner's safety lamp. **IV.**Calorimeter and refrigerator lagging, plastic covering of hot water storage tank.

- **Heat transfer by conduction is similar to wave motion because no permanent transfer of materials is involved and energy is transferred.**

- **Newton's law of cooling –** It states that the rate of heat loss of a body in air is directly proportional to difference in temperature between the body and its surrounding or excess temperature over the surrounding.

- Rate of heat loss $\frac{dQ}{dt}$ depends on difference in temperature between the body and its surrounding or excess temperature over the surrpunding, nature of surface i.e dull or polished, and exposed area of the surface.

Rate of heat loss $\boldsymbol{\frac{dQ}{dt} = KA(\theta - \theta_S)}$**.** K=constant depending on the nature of the surface, A=exposed area of the surface, $\theta - \theta_S$= difference in temperature between the body and its surrounding or excess temperature over the surrpunding, θ=temperature of the body, θ_S=temperature of the surrounding.

- Rate of heat loss can also be expressed in terms of of thermal capacity and specific heat– **Rate of heat loss** $\boldsymbol{\frac{dQ}{dt} = C\frac{\Delta\theta}{t} = mc\frac{\Delta\theta}{t}}$ **.** $\frac{\Delta\theta}{t}$ = rate of temperature fall.

2.Convection – Convection is the heat transfer through a material medium in fluids, due to translational movement of the medium or migration molecules of the medium from hot to colder region.

- It takes place in fluids i.e liquids and gases, but majorly/most rapidly in liquids than gases. It does not occur in solids.

- **Convection is heat transfer due to density and temperature gradient.**

- **Convection is depends or affected by : I.**Nature of the surface. **II.**Nature or type of fluid. **III.**Fluid properties(volume of the fluid, density, fluid viscosity, specific heat capacity and thermal conductivity). **IV.**Fluid velocity/velocity of molecules. It is **indepdent** of **nature and shape of the vessel**.

- **Archimedes' principle applies in convection as it involves fluids.**

- Convection cannot occur in a vacuum.

- **Natural convection –** When water is heated on a heater, the waterin contact with the hot base or

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bottom of the vessel rises to the top because it is less dense than cold water, while the cold dense water moves down and a **convectional current is set up.** Natural convection current are also set up as land and sea breeze due to difference in the densities between warm and cold air.

- Sea breeze – Sea breeze occurs during the day. It is the cool air/breeze which blows from the sea to land during the day. Land is warmer than the sea during day time due to the lower thermal capacity of land, hence warm less dense air from land rises and is replaced by cool dense air from sea setting up convectional current i.e sea breeze.

- Land breeze – Land breeze occurs during the night. It is the cool air/breeze which blows from land to sea during the night. Sea retains much of its heat due to its high thermal capacity and is warmer than land at night, hence warm less dense air from the sea rises and is replaced by cool dense air from land setting up convectional current i.e land breeze.

- Forced convection – A hot fluid could be forced to circulate to a cold region and a cold fluid also circulating to a hot region continuously through the action of a fan.

- Application of convection in liquids – I. Domestic water boilers. **II**.Cooling of motor car engine by a radiator. **III.**Land and sea breeze.

- Application of convection in gases – I. Ventilation of industrial buildings. **II.**Chimneys in kitchens. **III.**Domestic hot water supply system.

3.Radiation – Radiation is the heat transfer from hotter to colder place without heating of the intervening medium. It is the heat transfer through a body without a material medium or heat transfer through a vacuum.

- It is the heat transfer by electromagnetic waves in form of **infrared waves**.

- Radiation cannot be explained by molecular or kinetic theory of matter, as it doesn't require a material medium.

- Heat energy given out by radiation is radiant heat or radiant energy.

- All surfaces with temperature greater than abolute zero temperature ($0K$ or -273℃), emits radiant heat.

- The intensity of radiant heat energy emmited by a surface depends on – I. Temperature of surface T **II.**Surface area A **III.**Nature/colour of surface [black(dull) or bright(polished) surface].

- Black or dull surfaces e.g a surface covered with soot, are good radiators, absorbers and emmiters of heat and are poor reflectors.

- Silvered or polished surface are good reflectors of heat but poor absorbers, emmiters and radiators of heat.

- Painting a hot metal ball black will increase the heat lost instead of preventing it, as black surfaces are good radiators of heat.

- Temperature and surface area effect on radiatiom is combined by **Josef Stefan-Boltzmann's law**. It states that the radiant flux $\boldsymbol{H}$ (rate of emission of radiant heat energy per unit time), emitted from a body per unit area $\boldsymbol{A}$ is directly proportional to emmisivity $\boldsymbol{\varepsilon}$, and fourth power of the absolute Kelvin temperature $\boldsymbol{T}$.

$\frac{\boldsymbol{H}}{\boldsymbol{A}} \propto \boldsymbol{\varepsilon T^4}$, $\boldsymbol{H} \propto \boldsymbol{A\varepsilon T^4}$, $\boldsymbol{H = \sigma A\varepsilon T^4}$. H in W or Js^{-1}, T in K, A in m^2, σ=stefans-Boltzmann's constant $=5.67\times10^{-8}Wm^2K^{-4}$.

- Emmisivity ε of a body ranges from 1 for a perfectly black or dull surface to 0 for a perfectly white or polished body.

- The emission and absorption of radiant energy can be investigated with a **Leslie's cube**. Radiant heat can be detected by a **thermopile**.

- Application of radiation : I. Silvered surface at the base of iron and tea pot. **II.**Use of aluminium paints(silvered) in the roofs of buildings. **III.**Direct transfer of heat from the sun to the earth or from a charcoal fire to an occupant in the room. **IV.**Greenhouse effect. **V.**Radiator of a motor cooling itself as it is coloured black.

- Thermos flask or **vacuum flask –** is used to keep its content at the suitable temperature (keep hot liquids from cooling down and cold liquids form getting warm), by preventing heat loss and gain mainly by conduction, convention and radiation and partly by evaporation.

- Features for preventing heat loss/gain in a thermos flask – I. Double wall or vacuum space by evacuating the space between the glass walls prevents heat loss/gain by conduction and convection. **II.**Polished surface/silvered double wall prevents heat loss/gain by radiation. **III.**Insulaing cork at the top or cork stopper prevents heat loss/gain by conduction. It also prevents heat loss by evaporation and convection. **IV.**Insulating cork at the bottom or cork suppport of low thermal conductivity, prevents heat loss/gain by conduction.

- The main sources of heat loss in a thermos flask are the thin glass walls at the neck and the poorly conducting cork at the bottom.

- The working principle of refrigerator utilizes the cooling effect of evaporation. **Heat is lost in a refrigerator by conduction, convection and radiation. A thermostat in the refrigerator is used to control the rate of vaporization, hence degree of cooling.**

- A hot cup of tea placed on a table loses heat by **conduction** through the walls of the cup, by **convection** to air currents in contact with the outer surface of the cup and **evaporation** from the exposed surface of the tea, and by **radiation** from all exposed surfaces of the cup and its

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content.
- A hot metal ball suspended in the open air loses heat as it cools by **conduction** and **radiation**.
- Heat reaches the food which is being cooked in a pot or saucepan chiefly by **conduction** and **convection**.
- When some ice cubes are added to a glass of warm water, the glass is cooled by conduction and convection.
- The human body loses heat by **conduction** through the skin, by **convection and radiation** from the surface of the skin. Heat is also lost via **evaporation of sweat** from the skin.

Examples :

1. The mechanism of heat transfer from one point to another through vibration of the molecules of the medium is A.convection **B.conduction** C.radiation D.diffusion.
Conduction involves the molecular vibration of nolecules of a substance.

2. When heat is applied to one end of a metal rod, the molecules at the other end soon begin to vibrate with greater amplitude than before because heat has been transferred by A.radiation B.convection **C.conduction** D.evaporation.
The statement is the molecular explanation of conduction using kinetic theory.

3. In which of the following are the substance arranged in descending order of their thermal conductivities? **A.copper, steel, glass** B.steel, copper, glass C.steel, glass, copper D.copper, glass, steel.
Metals generally conduct heat than non-metals. The best heat conductors in descending order are : silver, copper, gold, aluminium and steel.

4. Which of the following makes metal ideal for cooking utensils? A.high coefficient of expansion **B.good conduction of heat** C.poor radiator of heat.
Metals being good conductors of heat transfer heat and respond to temperature changes quickly .

5. Cooking pots are usually made of metals because metals A.have high coefficient of expansion B.have low specific heat capacity **C.are good conductors of heat** D.are poor conductors of heat.

6. Which of the following cannot be explained by using the concept of conduction of heat? A.calorimeter lagging **B.land and sea breezes** C.the working of a miner's safety lamp D.refrigeration E.the use of fighting suits.
Land and sea breezes are natural conventional currents and can be explained by convection.

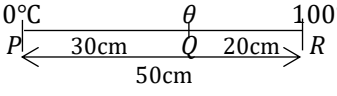
7. An iron bar at 0°C feels colder in the hand than a piece of wood at 0°C because A.iron is heavier than wood B.the molecules of iron vibrate more violently than the molecules of wood C.molecules of iron move more slowly than those of wood **D.iron is a better conductor of heat than wood** E.iron has a higher expansivity than wood.
The iron bar conducts heat away from the hand than the piece of wood, being a better conductor of heat, hence the hands feels cool when holding the iron bar and warm when holding the piece of wood.

8. The blade of a hoe feels colder to touch in the morning than the wooden handle because the A.handle contains stored energy in form of heat **B.blade is a better conductor of heat that the handle** C.blade is placed at a lower temperature than the handle D.handle is a better conductor of heat than the handle.
The blade of a hoe being a metal conducts heat away from hence making the hands feel cool. The hande of a hoe is made of wood a poor conductor of heat.

9. A concerte floor feels colder than a rugged floor on a cold morning because A.rug loses heat to the bare feet at a faster rate than the concrete B.rug loses heat to the bare feet at a slower rate than concrete C.rug is a better conductor of heat than concrete **D.concrete is a better conductor of heat than rug.**
Poor thermal conductors have high thermal capacities while good thermal conductors have low thermal capacities.

10. The temperature gradient across a copper rod of thickness 0.02m, maintained at two temperature junctions of 20°C and 80°C respectively is A.3.0×10^{2}Km^{-1} **B.3.0×10^{3}Km^{-1}** C.3.0×10^{4}Km^{-1} D.5.0×10^{3}Km^{-1}.
Thickness of the rod L=0.02m, θ_H=80°C , θ_C=20°C , temperature gradient=?,
Temperature gradient $T_G = \frac{\Delta\theta}{L} = \frac{\theta_H-\theta_C}{L}$,
$T_G = \frac{80-20}{0.02} = \frac{60}{0.02}$ =3000°Cm^{-1} ,
T_G =3.0×10^{3}°Cm^{-1} or 3.0×10^{3}Km^{-1}.

11. A lagged steel bar of length 50cm has its ends immersed in melting ice and boiling water.The temperature at a point 20cm from the hot end is A.40°C B.50°C **C.60**°C D.80°C.
Temperature of the end immersed in boiling water θ_H=100°C , temperature of the end placed in melting ice θ_C=0°C, length of the steel bar L=50cm, Temperature gradient $T_G = \frac{\Delta\theta}{L} = \frac{\theta_H-\theta_C}{L}$
$T_G = \frac{100-0}{50}$ = 2°Ccm^{-1}. The point 20cm from the hot end is same 30cm from the cold end. Hence, the temperature at that point, θ = 2°Ccm$^{-1}\times$30cm = 60°C or 40°C from the cold end.
Alternatively – Given that the length of the lagged (well insulated), steel bar is 50cm and it ends are immersed in melting ice at 0°C and boiling water at 100°C.



Demystified Series by Dr Timothy

Temperature gradient $= \frac{\Delta\theta}{L} = \frac{\theta_H - \theta_C}{L}$,

Temperature gradient T_G is constant across any section along the length of a bar. Hence, temperature gradient of section PQ is equal to that of section PR.

T_G across section $PQ = \frac{\theta-0}{30} = \frac{\theta}{30}$,

T_G across section $PR = \frac{100-0}{50} = \frac{100}{50} = 2$ Km^{-1},

T_G across section $PQ = T_G$ across section PR,

$\frac{\theta}{30} = 2$, $\theta = 2\times30 = 60°$.

12. What are the units of thermal conductivity A.kg m sec^2 **B.Joule sec^{-1} m^{-1} K^{-1}** C.kg m D.Newton sec^{-1} m^{-1} K^{-1} E.m^2 .

13. An iron plate 2cm thick has a cross-sectional area of 0.5m^2 with temperature of 15°C and 150°C on both sides. Calculate the amount of heat that passes through the plate each second (Thermal conductivity of iron= $59W$m$^{-1}K^{-1}$) A.18.4W B.36.9W **C.184.4W** D.368.8W.

Thickness L=2cm=0.02m, A=0.5m^2, θ_C=15°C, θ_H=150°C, K=$59W$m^{-1}K^{-1}, amount of heat passing through the plate each second $R = \frac{dQ}{dt}$=?,

$R = \frac{dQ}{dt} = KA\frac{\Delta\theta}{L} = KA\frac{\theta_H - \theta_C}{L}$,

$R = \frac{dQ}{dt} = 59\times0.5\times\frac{150-15}{0.02} = 29.5\times\frac{125}{0.02}$,

$R = \frac{dQ}{dt} = 184.375\text{W} = 184.4\text{W}$.

14. If 1.2×10^6J of heat energy is given off in 1sec from a vessel maintained at a temperature gradient of 30Km^{-1}, the surface area of the vessel is [thermal conductivity of the vessel=$400W$m$^{-1}K^{-1}$] **A.1.0×10^{2}m^2** B.9.0×10^{2}m^2 C.1.0×10^{3}m^2 D.9.0×10^{4}m^2.

Q=1.2×10^6J, t=1s, T_G=30Km^{-1}, K=$400W$m^{-1}K^{-1}, A=?, $R = \frac{dQ}{dt} = KA\frac{\Delta\theta}{L} = KAT_G$,

$\frac{1.2\times10^6}{1} = 400\times A\times30$, $A = \frac{1.2\times10^6}{400\times30} = 1.0\times10^2$m^2.

15.

100°C	iron	copper	0°C
	θ		

The diagram above illustrates a composite bar of iron and copper. The bar is insulated along its sides and it has a diameter of 10mm. The length and thermal conductivity of the iron are 0.15m and 40Wm^{-1}K^{-1} respectively and those of copper are 0.05m and 360Wm^{-1}K^{-1}, respectively. If the free ends of the iron and copper are kept at 100°C and 0°C respectively, calculate the

I. Temperature θ at the interface between the bars ?

L of iron=0.15m, K of iron=40Wm^{-1}K^{-1}, θ_H of iron=100°C, θ_C of iron=θ, L of copper=0.05m, K of copper=360Wm^{-1}K^{-1}, θ_H of copper=θ, θ_C of copper=0°C, diameter=10mm, radius r=$\frac{10}{2}$=5mm =5×10^{-3}m, θ=equilibrium temperature. As the bar is insulated, the heat lost by the iron=heat gained by copper. $R = \frac{dQ}{dt} = KA\frac{\Delta\theta}{L} = KA\frac{\theta_H - \theta_C}{L}$,

$40\times\pi(5\times10^{-3})^2\times\frac{100-\theta}{0.15} = 360\times\pi(5\times10^{-3})^2\times\frac{\theta-0}{0.05}$,

$40\times25\pi\times10^{-6}\times\frac{100-\theta}{0.15} = 360\times25\pi\times10^{-6}\times\frac{\theta}{0.05}$,

$\frac{40(100-\theta)}{0.15} = \frac{360\theta}{0.05}$, $\frac{40(100-\theta)}{360\theta} = \frac{0.15}{0.05}$, $\frac{100-\theta}{9\theta} = 3$,

$100-\theta = 3\times9\theta = 27\theta$, $100 = 27\theta+\theta = 28\theta$,

$\theta = \frac{100}{28} = 3.57°\text{C} \approx 3.6°\text{C}$.

II. Rate of heat flow along the bar ?

$R = \frac{dQ}{dt} = KA\frac{\Delta\theta}{L} = KA\frac{\theta_H - \theta_C}{L}$. It can be calculated using the parameters of copper or iron.

$R = \frac{dQ}{dt} = 360\times\pi(5\times10^{-3})^2\times\frac{3.6-0}{0.05}$,

$R = \frac{dQ}{dt} = 360\times25\pi\times10^{-6}\times\frac{3.6}{0.05} = 0.648\pi$,

$R = \frac{dQ}{dt} = 0.648\times\frac{22}{7} = 2.04$Js^{-1} or 2.04W.

16. The time rate of loss of heat by a body is proportional to the **A.difference in temperature between the body and its surroundings** B.temperature of its surrounding C.ratio of the temperature of the body to that of its surroundings D.temperature of the body

According to Newton's law of cooling : the rate of heat loss $\frac{dQ}{dt}$ of a body in air is directly proportional to difference in temperature between the body and its surrounding or the excess temperature over the surrounding $\Delta\theta_S$. $R = \frac{dQ}{dt} \propto \Delta\theta_S$,

$R = \frac{dQ}{dt} \propto (\theta - \theta_s)$.

17. On which of the following will the rate of heat loss of a body to the surrounding? I.exposed surface area of the body II.the shape of the body III.the density of the body IV.the temperature difference between the body and the surrounding A.I and II **B.I and IV** C.II and III D.III and IV.

Rate of heat loss of a body depends on the exposed surface area of the body, difference in temperature between the body and its surrounding or excess temperature over the surrpunding and nature of surface(dull or shiining/polished surface). It does not depend on the shape or density of the body.

18. According to Newton's law of cooling, the heat energy lost per unit area of a body depends on the I.Surface area II.Nature of the surface III.Temperature of the surface IV.Excess temperature over the surroundings A.I and II B.II and IV C.I,II and III **D.I, II and IV.**

Based on Newton's law of cooling : the rate of heat loss depends in the excess temperature over the surrounding, exposed surface area and the nature of the surface. It does not depend on the temperature of the surface.

19. Thermal equilibrium between two objects exists when **A.the temperature of both objects are equal** B.the quantity of heat in both bodies are the same C.the heat capacities of both bodies are the same D.one object loses heat continuously to the other.

20. Two bodies P and Q are in thermal equilibri-

Demystified Series by Dr. Timothy

um. Which of the following statements about the bodies is correct? A.The temperature of Q is higher than that of P B.P and Q ave the same heat capacity C.P and Q have the same mass **D.P and Q are at the same temperature.**

21. Two bodies P and Q are in thermal contact. Heat can flow from one body to another if A.the temperature of the bodies are in equilibrium B.the bodies have the same thermal conductivity **C.a temperature gradient exists between the bodies** D.the bodies have the same thermal capacity.

Heat can only flow between two bodies at different temperatures i.e temperature gradient exists between them. Heat flows from a body at higher temperature to a body at a lower temperature. No net flow of heat occurs when the temperature of bodies in thermal contact are in equilibrium or same.

22. Heat transfer conduction is similar to wave motion A.transfer of materials is involved B.there is increases in temperature **C.no transfer of materials is involved** D.charged particles vibrate at right angles.

Heat transfer by conduction and wave motion are similar because no transfer of materials is involved and energy is transferred in both. Heat transfer by conduction involves increase in temperature and chanrged particles vibrate but not at right angles to the direction of heat flow. Wave motion involves no increase in temperature and charged particles vibnrate at right angles to the direction of wave motion i.e transverse wave.

23. Heat transfer by convection in a liquid is due to the **A.translatory motion of the molecules of the liquid** B.increased vibratory of the molecules of the liquid about their mean position C.expansion of the liquid as it is heated D.latent heat of vaporization E.ability of the liquid to evaporate at all temperature

Convection involves the translational motion of molecules of a the liquid. Convection is heat transfer by molecular motion.

24. Heat transfer by convection in a liquid is due to the A.increased vibration of the molecules of the liquid about their mean position **B.variation of the density of the liquid** C.expansion of the liquid as it is heated D.latent heat of vaporization of the liquid.

25. Heat transfer by convection will occur most rapidly in one of the following subatnces A.oxygen **B.water** C.wood D.mercury

Convection occurs in fluids i.e liquids and gases but majorly in liquids. Heat transfer in mercury is by conduction, as mercury is a liquid metal and a conductor.

26. Which of the following is/are not correct? I.A hot water circulation system transfers heat by conduction II.Convection occurs without any change in the average position of the particles of the medium concerned II.The principle of the miner's safety lamp is based on the fact that a metal guaze is a good conductor of heat A.I B.II C.III **D.I and II** E.I and III.

A hot water circulating system trasfers hear by convection due to difference in densities between warm and cold water. Convection occurs with a change in the average position of the particles of the medium involved. Miner's safety lamp is based on heat transfer by conduction.

27. In domestic water boilers, heat transfer is mainly by A.conduction **B.convection** C.transmission D.radiation.

Domestic water boilers are used to supply hot and cold water in homes. It is done by the rise of hot less dense water and fall of cold dense waterb in the boiler.

28. Ice wrapped in guaze placed at the bottom of a tube filled with water remains unmelted when heat is applied to the top of the tube. This experiment shows that A.ice cannot sink in water **B.water is a bad conductor of heat** C.water is a good medium to show convention D.ice is a solid that conducts heat.

Heat transfer in water and other liquids is by convection and not conduction. When the water molecules at the top of the test tube is heated, remains afloat at the top as it is less dense than the cold liquid below. Hence, heat is not transferred to the bottom of the liquid and the ice will not melt. If the test tube was heated from the top rather than the top, the ice will melt as it is heated directly and the warm less dense water molecules at the bottom rises to the top to replace the cold dense water molecules

29. Land breeze occurs A.during the day B.as a result of heat transfer by radidation **C.as a result of heat transfer by convection** D.as a result of heat transfer by conduction.

Land breeze occurs during the night while sea breeze occurs during the night. Both land and sea breeze are as a result of heat transfer by convection.

30. Cold breeze across a sandy beach in a sunny afternoon because A.the heat of the sun is absorbed by the sea B.the sea is warmer at that time of the day C.cool air blows from the beach of the sea **D.warm air on the beach is replaced by cooler air from the sea.**

Cool breeze across a sandy beach in a sunny afternoon is sea breeze which occur during the day. It is the conventional current set up when warm less dense from the land /beach is replaced by cool dense air from the sea.

31. In the formation of sea breeze, wind blows from A.sky to land B.sea to sky C.land to sea

Demystified Series by Dr Timothy

D.sea to land.

32. The mode of heat transfer which does not require a material medium is **A.radiation** B.propagation C.conduction D.convection.

33. Heat may be transferred by conduction, convection and radiation. By which of these methods does heat travel through vacuum? A.conduction only B.convection only **C.radiation only** D.all three methods E.None of the methods.

34. Which of the following cannot be explained by the molecular theory of matter? A.conduction B.convection C.evaporation D.expansion **E.radiation.**

35. The heat from the sun reaches the earth by the process of A.insulation B.precipitation C.conduction **D.radiation** E.convection

Heat from the sun reaches the earth by radiation as no material medium is needed.

36. A man standing 2m from a charcoal fire is warmed mainly by A.convection **B.radiation** C.refraction D.conduction.

37. Heat is radiated by all hot objects in the form of A.light energy B.solar energy **C.infrared rays** D.x-rays.

38. Which of the following statements about heat transfer is true? A.conduction occurs due to translational motion of the molecules of the medium B.fresh air can be made to circulate in a room by the use of radiation process **C.a hot water circulating system trasnfers heat by convection** D.heat from the sun reaches the earth by conduction process.

Conduction occurs due to vibrational motion of the molecules of the medium about its mean position. Fresh air can be made to circulate in a room by forced convention e.g using a fan. Heat from the sun reaches the earth by radiation.

39. The ground is always cold at night because A.sun no longer shines at night B.atmosphere reflects sun's energy at night C.atmosphere absorbs the sun's energy **D.earth radiates heat to the atmosphere at night.**

The earth absorbs radiant heat from the sun during the day and radiates the heat to the to back to the air at night, hence making the earth cool at night.

40. The radiator of a motor car is cooled by A.radiation and conduction B.conduction **C.radiation** D.convection

The radiator of a motor car being blackened is cooled by radiation. The radiator cools the engine of an automobile or motor car by convection.

41. A room is heated by means of a charcoal fire. A man standing away from the fire is warmed by A.conduction only B.conduction and convection C.convection only **D.convection and radiation** E.radiation only

The man is heated mainly by radiation as the fire emits radiant heat and it reached the man directly. The man is also heated partly by convection as the charcoal fire warms air above it, which rises and circulates round the room by convectional currents.

42. Which of the following statements is not correct? A.Good emitters of radiant energy are also good absorbers B.Radiant heat can be detected by a thermopile C.Infrared rays are heat rays D.Heat is transmitted through a vacuum by radiation **E.Shiny surfaces are better absorbers than dull surfaces.**

Good emitters of radiant energy are also good absorbers and good radiators. Radiant heat can be detected by a thermopile. Infra-red rays are also called heat rays. Polished or shiny surfaces are poor absorbers than dull or darkened surfaces.

43. A perfect emitter or absorber of radiant energy is a A.white bodyB.red body **C.black body** D.conductor.

Black body are perfect emitter or absorber or radiator. White body are perfect reflector.

44. Highly polished silvery surface are **A.poor absorbers and poor emitters of radiation** B.good absorbers and good emitters of radiation C.good absorbers but poor emitters of radiation D.poor absorbers but good emitters of radiation

Highly polished silvery surface are poor emitters and poor absorbers of radiation but good reflectors.

45. It is advisable to wear white dresses in tropics because white can A.absorb sweat readily B.absorb all radiant heat C.conduct heat away more readily from the body **D.reflect radiant heat** E.absorb the rays from the sun.

The temperature in tropics is very high, hence it is advisable to wear white dresses which are good reflectors and poor absorbers of radiant heat.

46. Which of the following phenomena explains the fact that a house whose roof is coated with white paint will be cooler than one painted with black paint? A.conduction B.convection **C.reflection** D.refraction.

White paints are good reflector and poor absorber or radiator or emitterof radiant energy, hence it reflects most the heat incident on the roof, preventing the temperature of the house from being raised. Black paints are poor reflector and good absorber or radiator or emitter of radiant energy, hence it absorbs most of the heat incident on the roof , raising the temperature of the house.

47. Which of the following statements about radiant heat is/are correct? I.Radiant heat cannot travel through a vacuum II.Rough surfaces emits radiant heat more than polished surfaces III.Dark surfaces absorbs radiant heat better than bright surfaces A.I B.II C.III D.I and II

Demystified Series by Dr Timothy

E.II and III.

48. Two similar kettles containing equal masses of boiling water are placed on a table. If the surface of one is highly polished and the surface of the other is covered with soot, which of the following observations is correct A.the two kettles will cool down at the same rate B.the polished kettle cools down more quickly by conduction **C.the kettle covered with soot cools down more quickly because it is a good radiator of heat** D.the kettle covered with soot cools down more quickly by the process of heat convection.

The kettle covered with soot(black surface) cools i.e its tempeeratrure falls down quickly, faster than the highly polished kettle, as the kettle covered with soot radiates or loses and absorbs heat faster.

49. Two closed vessels contain equal volumes of boiling water and are identical except that one is blackened on the outside while the other is polished. Which of the following is correct? A.The polished vessel will cool faster because a polished surface is a good radiator B.The polished vessel will cool faster because because it will absorb less heat from the room C.The black vessel will cool faster because it is a better conductor of heat D.The polished vessel will cool faster because it reflects heat better **E.The black vessel will cool faster because a black surface radiates heat faster than a polished one.**

50. Heat radiation occurs most rapidly from a surface which is A.shinning black B.white and black C.white and black **D.dull and black.**

51. Which of the following is not a suitable method for reducing heat loss from a hot metal ball? A.placing it in a vacuum **B.painting it black** C.placing it on a rubber support D.wrapping it with cotton wool.

Placing the hot metal ball in a vacuum prevents heat loss by conduction, as not material medium is present. Painting the metal block black will aid heat loss by radiation and not prevent heat loss, as black bodies are good radiators. Placing the hot metal ball on a rubber support or wrapping it with cotton wool both of which are insulators, prevents heat loss by conduction.

52. Which of the following methods is not suitable for reducing loss of heat from a piece of hot iron A.wrapping it in cotton wool B.placing it on rubber supports **C.painting it black** D.keeping it in a closed wooden box E.placing it in a vacuum.

53. Which of the following statements is not correct? A.The circulation of air which gives rise to sea breeze during the day is due to convection B.An air space can be used between a heated material and the lagging to improve insulation C.Cotton materials are better than woolen materials for hot weather D.Convection currents play a large part in the cooing process of a motor car engine **E.The vacuum space in a thermos flask helps to reduce heat losses by radation**

Air is a poor conductor or insulator, hence can be used between a heated material and the lagging. Cotton materials tends to absorbs heat less than woolen materials making them suitable for hot weather. The vacuum space in a thermos flast helps to reduce heat loss by conduction and convection, as they require material medium.

54. Which of the following statements are correct? I.Land and sea breezes are natural convection currents II.Convection may occur in liquids or gases but not in solids III.The vacuum in a thermos flask prevents heat loss due to convection only **A.I and II** B.I and IIII C.II and III D.I,II and III.

55. Which of the following statements is correct? Silvered walls of a vacuum flask are used to prevent **A.heat loss due to radiation** B.heat loss due to conduction C.vacuum loss D.heat entry into the inner flask due to convection E.heat loss due to convection.

The silvered wall surface prevents heat loss/ gain by radiation, as silvered surface are poor radiators or emitters or absorbers of heat.

56. Heat loss by radiation from a hot liquid in a Thermos flask is reduced mainly by the A.vacuum between the glass walls **B.silvering of the glass walls** C.outer container D.insulating cork cover E.glass which is transparent.

57. A vacuum is needed between the two glass walls of a vacuum flask A.to prevent heat loss by radiation B.to prevent heat gained by radiation C.becasue air is a conductor of heat and will conduct heat to the outside D.because vacuum is a good absorber of heat **E.to reduce heat losses by conduction and convection**.

58. The thermos flask is designed to A.store hot tea or water **B.prevent heat loss or gain by conduction, convection and radiation** C.prevent heat gain by conduction, convection and radiation D.prevent heat loss by conduction, convection and radiation.

59. The space between the double glass walls of a thermos flask is evacuated and the two surfaces facing the evacuated space are silvered. The residual source of heat loss takes place by A.convention B.radiation from the surfaces **C.conduction through the stopper and the glass** D.conduction across the walls.

The double glass walls with an evacuated space (vacuum) prevents heat loss by convection and conduction. The silvered walls prevents heat loss by radiation. Hence, heat can only be lost by conduction through the cork stopper or insulating cork at the top and the cork support or insulating cork at the bottom through the glass to

Demystified Series by Dr Timothy

the surrounding.

60. The main reason for making the cover of a vacuum flask air tight is to prevent heat loss by A.conduction **B.evaporation** C.radiation D.convection.

A Thermos or vacuum flask not poorly closed or not air tight will lead to evaporation of the liquid molecules of its content, leading to the cooling or the content.

61. In a good Thermos flask, the main cause of heat loss is **A.conduction through the cork** B.the plastic base of the Thermos flask C.the silvered walls and shiny metals D.the outer cover or jacket.

The main causes of heat loss in a Thermos flask are the thin glass walls at the neck and the poorly conducting cork at the bottom or cork stopper.

62. The coils and fins at the back of a refrigerator are painted black so that the refrigerator A.can produc e ice blocks B.may not lose its cool temperature C.can absorbs air for effective cooling **D.can loose heat more rapidly by radiation.**

63. The intensity of heat radiated by a surface depends on the following except A.surface area B.absolute temperature **C.thermal conductivity** D.nature of the surface

Radiation depends on nature of the surface, surface area and absolute temperature. It is independent of thermal conductivity.

64. Which of the following has no effect on radiation A.Temperature B.Surface area C.Nature of the substance **D.Density.**

Jamb past questions on heat transfer :

[1981/20,1982/43,1986/1,1987/24,1987/24, 25,26,1988/20,1990/20,1991/22,1992/19, 1994/27,1995/26,1999/26,2000/22,2002/25, 2003/26,2004/36,42,2007/32,2008/25,2009/ 25,2010/18,23,2011/17,22,2013/24,2014/25]

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Chapter 15 – Wave : Types, Propagation and Properties

● **Wave** is a disturbance which travels through a medium or without a medium, transferring or propagating energy from one point to a other without any permanent displacement of the medium.

- A wave is a combination of both **translational and oscillatory/vibratory motion.** The particles of the medium vibrate or oscillate about their equilibrium position while the wave itself spreads forward over a long distance i.e transaltional motion.

- Classification of waves – Waves are classified based on the requirement of a medium for propagation i.e mechanical and electromagnetic waves, direction of vibration of the particles of the medium relative to the wave motion i.e transverse and longitudinal wave.

- Mechanical waves – They require a material medium for propagation e.g water waves, sound waves,wave in strings.

- Electromagnetic waves – They don't require a material medium for propagation or they travel through a vacuum e.g gamma rays, x-rays,ultraviolet rays, light rays,infra red rays, radio waves and mircro-waves (**GXULIRM**)**.** The order of the electromagnetic spectrum GXULIRM, shows an **increasing wavelength and decreasing frequency or penetration power** hence, Gamma rays has the highest frequency or penetration power and lowest wavelength. **They all moves with the same speed in a vacuum equal to the speed of light ($3\times10^8 ms^{-1}$).**

- Transverse waves are waves with direction of motion perpendicular to the direction of propagation e.g water waves, electromagnetic waves.

- Longitudinal waves are waves with direction of motion is parallel to the direction of propagation e.g sound wave, wave in a spring.

- Wave in strings is both mechanical and transverse, wave in springs is both mechanical and longitudinal, sound wave is both mechanical and longitudinal, and light wave is both electromagnetic and transverse.

- Water waves is both mechanical and transverse wave. It can be studied in the laboratory by a ripple tank. A stroboscope is used to make the water waves appear stationary.

- Water waves travels faster in deep water than in shallow water, $v \propto \lambda \propto$ depth.

- Terms used to describe wave –

- Pulse is the sudden increase in the magnitude of a physical quantity shortly followed by a rapid decrease.Pulse is seen mostly in water waves. **A pulse consists of a single crest.**

- Amplitude A – is the maximum displacement of wave from rest or equilibrium or mean position.

- Phase – Particles or points are said to be in phase when they occupy similar positions or at the same vertical distance from their equilibrium position or are moving in the same direction.

- Cycle is a complete oscillation of a vibrating particle.

- Wavelength λ - is the distance between two consecutive crests or troughs or compressions or rarefractions. It is the distance between two successive points in phase or distance advanced when a wave completes one cycle.

- The distance between adjacent crests or troughs or compressions or rarefractions is equal to one wavelength.The distance between adjacent compressions and rarefractions or adjacent crests and troughs is equal to half wavelength.

- Crest and trough are respectively maximum upward and downward displacement in transverse waves while compressions and rarefractions are respectively maximum upward and downward displacement in longitudinal waves.

-Frequency f of a wave is the number of waves completed in one second. **Frequency is the fundamental property of a wave which depends only on the source of wave and not the medium of propagation hence, frequency of wave equals the frequency of source producing wave**.

- The frequency of devices with regularly ruled lines or holes e.g **stroboscope, gramophone records, disc siren, and spoke of bicycles or cart** is given by – $\boldsymbol{f = nv}$, f=frequency in Hertz, n=number of holes or regularly ruled lines, v=speed in revolution per second(rev/s)

- Period T is the time taken for the wave for one complete wave or time taken to move one complete wavelength. $\boldsymbol{T = \frac{1}{f}}$.

-Wavefront is any line or section taken through an advancing wave in which all the particles are vibrating in the same phase or in the same state of disturbance. **Wavefront shows how energy is being transmitted from one point to another. The direction of propagation of a wave or the direction of energy flow in wave motion is drawn perpendicular or normal to the wavefront.**

- Wave speed v is the distance s travelled by the wave in one second $t = 1$. $\boldsymbol{v = \frac{s}{t} = \lambda f = \lambda/T}$.

- Particle velocity is the displacement of the particles about their mean position in one second. $\boldsymbol{V = \omega A}$, ω=angular speed or angular frequency of the particle, A=amplitude.

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- Waves can be described based on their degree of movement or restriction i.e progressive and stationary wave.
- **Progressive or travelling waves** are waves that spreads out from a source transferring energy without any restriction**.**
- **Energy is the property of a wave that is propagated in a progressive or travelling wave.**
- **Standing or stationary waves** are produced by the overlapping or interference two equal progressive of equal amplitude and frequency travelling in opposite direction. It is alsoproduced when progressive wave is reflected from a surface.
- **Stationary waves do not transfer energy.**
- The amplitude of a progressive wave can be constant while that of stationary wave is always constantly changing or varying.
- The amplitude of the resultant wave is the algebraic sum of the amplitude of both waves.
- The superimposition of both waves creates a region of zero resultant disturbance as both waves are of the same amplitude and frequency and are out of phase.
- **Nodes (N)** are the points with minimum displacement or zero amplitude or at rest while **Antinodes (A)** are points with maximum displacement or amplitude in a stationary wave.
- **The distance between two consecutive nodes NN or antinodes AA is half wavelength, $\frac{\lambda}{2}$. The distance between a node and antinode NA is quarter wavelength, $\frac{\lambda}{4}$.**
- A stationary wave can be produced by both transverse and longitudinal waves.
- **Equation of progressive wave –** A wave can either travel in the positive x-direction or negative x-direction. The motion of a progressive wave is generally represented by the equation : $\boldsymbol{y = A\,sin(\theta \pm \phi)}$, y=vertical displacement of particles, A=amplitude of wave, θ=angular displacement= ωt, ϕ=phase angle or difference= kx.
- Other equations derived from the general equation above are given below –

1. $\boldsymbol{y = A\,sin(\omega t \pm kx)}$, ω=angular frequency or angular speed or velocity, $\boldsymbol{\omega = 2\pi f = \frac{2\pi}{T}}$, t=time, k=wave constant or wave number, $\boldsymbol{k = \frac{2\pi}{\lambda}}$, x=displacement of the wave from origin.

2. $\boldsymbol{y = A\,sin(2\pi ft \pm \frac{2\pi x}{\lambda})}$, $\boldsymbol{y = A\,sin(\frac{2\pi}{T}t \pm \frac{2\pi x}{\lambda})}$

- **Phase difference or angle** $\boldsymbol{\phi = kx = \frac{2\pi x}{\lambda}}$.

3. $\boldsymbol{y = A\,sin\frac{2\pi}{\lambda}(x \pm vt)}$ or $\boldsymbol{A\,sin\frac{2\pi}{\lambda}(vt - x)}$,

4. $\boldsymbol{y = A\,sin(\frac{\omega t}{v} \pm \omega x) = A\,sin\,\omega(\frac{t}{v} \pm x)}$,

$\boldsymbol{y = A\,sin\,2\pi f(\frac{t}{v} \pm x)}$.

- The sign is negative, when the wave moves from left to right i.e positive-x direction and positive, when the wave moves from right to left i.e negative-x direction.
- **Phase velocity,** $\boldsymbol{v = \frac{angular\ frequency}{wave\ number} = \frac{\omega}{k}}$ **.**
- **Properties of wave –** All waves exhibit the the properties: reflection, refraction, diffraction and interference.

1. Reflection is the bouncing back of a wave from a surface in a different direction, that can't be penetrated or with minimum absorption. The velocity, frequency and wavelength of the wave is unchanged.
- Reflection principle is applied in determining the depth of a sea.

2. Refraction is the change in direction,velocity and wavelength of a wave as it travels from one medium to the other with different densities. **Frequency of wave is the only property of wave which is constant during refraction**.

3. Diffraction is the spreading out or bending of waves around corners or obstacles.
- **Diffraction is pronounced and noticeable when:** the width of the opening or aperture is about the same order as the wavelength of the waves and the opening is narrow i.e small opening.
- Diffraction does not takes place when the opening between two barriers or slit is large or has a wide aperture.
- Sound waves requires a larger opening than water waves and water waves larger than light waves due to their wavelengths i.e the wavelength of sound waves is greater than that of water waves, which in turn is greater than that of light waves, $\lambda_S > \lambda_W > \lambda_L$.
- **Examples of diffraction: I.**Formation of blurred and brighter image in a pin hole camera with a large hole. **II.**Distributing of sound round the corners in a room through large openings. **III.**Bending of water waves through a narrow opening.

4. Interference is the combination/overlapping of identical waves of same amplitude,frequency and wavelength which are travelling in the same or opposite direction.
- **Conditions for interference – I.Coherent sources** : sources emmiting waves of equal frequency, wavelength and amplitude in same phase or with a constant phase difference. **II.**More than one vibrating source at a small distance apart i.e close to each other.
- **Constructive interference or reinforcements** occurs when two waves meets in phase or step i.e crests meets with crests and troughs meets with troughs, and reinforce to produce a larger wave with large amplitude. **Phase difference between the waves is equal to one cycle or**

Demystified Series by Dr Timothy

360° or 1 wavelength, 1λ.

- Destructive interference or cancellations occurs when two waves meets out of phase or step i.e crests meets with troughs to produce a wave of zero disturbance. **Phase difference between the waves is equal to half cycle or 180° or half wavelength $\frac{1}{2}\lambda$.**

- Nodal and antinodal lines are occurrence in superimposition of two identical waves travelling in the same direction i.e they occur in interference of waves. Antinodal lines are the lines joining all points of constructive interference while Nodal lines are the lines joining all points of distructive interference.

- Examples of interference – I.Beats: regular rise and fall representing constructive and destructive interference. It occurs when two sounds of nearly equal frequency are combined. **II.**The colours seen in the thin film of oil on the road and in soap bubbles. **III.Blooming of lenses:** production of different reflected beams from a lens coated with fluoride salt. **IV.**Circular colour rings formed when white light falls on a converging lens of large radius of curvature placed on a plane glass.

- Superposition is the combination of waves of different frequency.

5. Polarization or **sidedness effect of a wave** is the restriction of an incident wave into vibrating in one plane or straight lines parallel to direction of vibration.It is the production of a wave whose vibration occur in only in one plane. It is also a procedure for changing incident wave of constant intensity to an emergent wave of varying intensity.

- Polarization is exhibited by transverse waves only.

- Malu's law of intensity of polarization states that the transmission of light through a polarizer placed at an angle to the analyzer varies as the square of the cosine of the angle. $\boldsymbol{I = I_o cos^2\theta}$, I=intensity of incident unpolarised light in candela (cd), I_o=intensity of emergent polarised light in candela (cd), θ=angle between the incident light and axis of polaroid.

- Polarization of light – Polarized light can be produced using polarizers [e.g tourmaline crystal,quartz or a polaroid], selective absorption by crystals, double refraction, scattering through fine particles and by reflection.

- Polarizers allows light vibrating in only one particular direction to pass through them and absorbs light vibrating in other directions.

- Polaroids are used : **I.**sunglases,camera lens,side windows of a vehicle or plane and head lamps to reduce the intensity of incident light and reduces reflected light or glare. **II.**To produce 3-dimensional films. **III.**Polarimeter in determining the concentration of crystals e.g sugar solution.

- Reflection and refraction of light occur at the sides of a glass not the front. Reflection and refraction of light at water or glass surface is always accompanied by partial polarization of light.

- Polarization by reflection– Polarization of light can also be produced when light is incidented on a polished surface of a glass plate at angle of incidence of about 57°. The reflected light is said to be polarized. **The reflected and refracted rays are at right angles to each other** i.e angle between them is 90°, and the **angle of incidence equals the polarizing angle.**

- The refractive index of the glass surface producing the polarized light is calculated using **Brewster's law.** It states that the refractive index of the glass material $\boldsymbol{n}$ is equal to the tanget of the angle of polarization $\boldsymbol{\theta_p}$. **Brewster's law:** $\boldsymbol{n = tan\,\theta_p}$. n=refractive index of the glass medium, θ_p=incident or polarizing angle.

Examples :

1. A stone is dropped into the middle of a pool of water. Which of the following statements is/are correct? I.Spherical waves are set up in water II.The water moves outwards to the side of the pool III.Energy is transmitted outwards from the centre of disturbance A.I B.II C.III **D.I and III** E.II and III.

A stone dropped into a pool of water produces circular or spherical wave fronts.The water waves moves forward and outwards transmitting energy over a long distance from the centre disturbance.Hence,water waves does not necessarily move to the side of the pool.

2. Which of the following is not true about waves? A.waves transfer energy from one place to another B.electromagnetic waves have the same velocity in the same medium C.particles of a medium vibrate if there is a wave motion in the medium D.the velocity of a sound wave depends on the temperature of the air **E.all waves require a material medium for their propagation.**

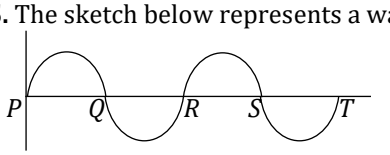
Energy is propagated in a travelling wave froom one point to another,while the particles vibrate about a mean point without being displaced in the medium. Only mechanical waves require material medium for their propagation. Electromagnetic waves do not require material medium.

The velocity of sound wave is directly proportion to the square root of its absolute temperature , $V \propto \sqrt{T}$.

3. Which of the following is not a mechanical wave? A.wave produced in a stretched string

Demystified Series by Dr. Timothy

B.waves in closed pipes C.water waves **D.radio waves** E.sound waves.
Radio waves are electromagnetic waves.
4. Which of the following statements are correct of the production and propagation of waves? I.vibration produces waves II.waves transmit energy III.the medium through which the wave travels does not travel with the wave IV.waves do not require any medium for transmission A.I and IV B.II and IV C.III and IV **D.I,II and III** E.II,III and IV.
Vibration of a medium produces waves which transmit energy with the particles of the medium vibrating and oscillating about a fixed point and does not travel with the wave. Some waves require material medium i.e mechanical waves while some do not i.e electromagnetic waves.
5. A transverse wave and a longitudinal wave travelling in the same direction in a medium differ essentially in their A.amplitude **B.direction of vibration of the particles of the medium** C.period of vibration of the particles of the medium D.frequency.
A transverse wave travels in a direction perpendicular to the direction of vibration of the particles of the medium. A longitudinal wave travels in a direction parallel to the direction of vibration of the particles of the medium.
6. Which of the following statements is/are correct? I.In a transverse wave, the particles of the medium vibrate perpendicularly to the direction of propagation of the wave II.In a longitudinal wave , the particles of the medium vibrate parallel to the direction of propagation of the wave III.In a stationary wave, the particles of the medium do not vibrate but are stationary **A.I and II** B.I and III C.II and III D.III E.I,II and III.
Stationary or standing waves as the name implies, are waves which do not move or are stationary though the particles of the of the medium vibrate about ther mean position.
7. In which of the following cases is a transverse wave formed? I.A stone is thrown into a pool of water II.A tuning fork is struck at the end III.A sonometer wire is plucked at the centre A.I B.II C.III D.I and II **E.I and III.**
A stone thrown into a pool of water produces water waves which are transverse wave. A tuning fork struck at the end produces sound waves which are longitudinal waves. A sonometer wire iss made of strings and when plucked at the centre produces transverse waves.
8. Which of the following waves are longitudinal waves? I.ripples on the surface of water II.wave produced by a tuning fork vibrating in air III.light waves IV.wave produced by a flute A.I and II B.I and III C.II and III **D.II and IV** E.III and IV.
Ripples on the surface of water are water waves i.e transverse waves. Wave produced by a flute due to the vibration of a air column in the medium are sound waves i.e longitudinal waves.
9. Which of the following is correct about light and sound waves? **A.Both of them are associated with energy transfer** B.Both of them need material media for propagation C.They are both longitudinal waves D.Their velocities in air are equal E.Both of them can be polarized.
All waves transfer energy as they travel. Light waves are electromagnetic wave and transverse wave while sound waves are mechanical and longitudinal wave. Their velocity of liqght in air is about $3\times10^{8}ms^{-1}$ while that of sound is about $333ms^{-1}$. Only light wave which is travserse wave can be polarized. Sound wav, a longitudinal wave cannot be polarized.
10. Which of the following is both mechanical and trasverse? A.Infra-red rays B.Gamma rays C.Sound waves **D.Water waves** E.Microwaves.
11. In a wave, the maximum displacement of particles from their equilibrium position is called A.frequency **B.amplitude** C.period D.wavelength E.wave velocity.
12. The fundamental property of wave which depends only on the source and not the medium of propagation is the A.wavelength B.harmonics **C.frequency** D.velocity.
13. Any line or section taken through an advancing wave in which all the particles are in the same phase is called the A.wavelength **B.wavefront** C.wavetrough D.wave amplitude E.wavecrest.
Wave front is a line or surface in the path of an advancing wave in which all the particles are in the same phase or state of disturbance. Wave length is the distance between two points in phase. Wavecrest is the maximum upward displacement while wave trough is the maximum downwards displacement of transverse wave.
14. Which of the following statements about a wave is/are correct? I.A wavefront is a line which contains all particles whose vibrations are in phase II.The direction of propagation of a wave is the line drawn parallel to the wavefront III.A wavefront is a circle which is common to all points that are in the same state of disturbance A.I B.II C.III D.I and II **E.I and III.**
The direction of propagation of a wave is the line drawn normal or perpendicular to the wavefront front. Statement 'III' describes a circular or spherical wave front.
15. The sketch below represents a wave profile

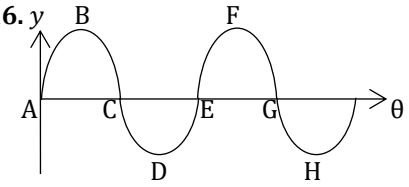


Which points are vibrating in phase? A.*P* and *Q* **B.*P* and *R*** C.*P* and *S* D.*R* and *S* E.*S* and *T*.
Two or more points or particloes are said to be in

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phase when they occupy similar positions and travelling in the same direction. Points *P*, *R* and *T* are in phase and their phase difference of phase angle is 0°, as they occupy similar positions and are travelling in the same direction. Points *Q* and *T* are also in phase.

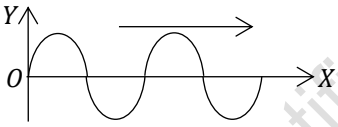
16.



The diagram above represents a trasverse travelling wave. Which two points are 180° out of phase? A.A and E B.B and D C.C and F D.D and H.
Two points are out of phase when they are not occupying similar positions and not travelling in the same direction. Two points which are 180° out of phase have a phase difference of 180° or π rad, half wavelength $\frac{1}{2}\lambda$ or half cycle apart and their vertical displacement from equilibrium position are equal in magnitude but opposite in direction.
Points B and D and points C and F are 180° out of phase. Points A and E and points D and H are in phase.

17. I.Wavelength II.Medium of propagation III.Wave velocity IV.Frequency V.Energy. Which of the above are used for characterizing waves? A.I,II and IV B.I,III and V C.I and IV **D.I,III and IV**
Wavelength, wave velocity, frequency, period, amplitude and wavefront are used for chnaracterizing waves.

18.



The above figure shows the motion of a progressive wave along a string. The particle motion of the medium is in the direction A.parallel to *OX* **B.parallel to *OY*** C.60° to *OX* D.60° to *OY*.
The figure above represents a transverse wave i.e wave in a string. Transverse waves travel perpendicular to the direction of vibration of particles of the medium. Hence, the particle motion is parallel to *OY* and perpendicular or at 90° to *OX*.

19. A string is sustaining a stationary wave when A.the frequency of the wave is not proportional to the wavelength of the wave B.the amplitude of vibration is always zero **C.two equal waves are travelling in opposite directions** D.the waves are longitudinal E.the frequency is directly proportional to the length of the string.
A stationary wave is produced when two identical waves of the same wavelength, amplitude, frequency travelling in opposite direction are superimposed.

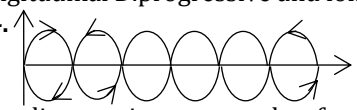
20. Two wave trains will produce a standing wave in a medium if they interfere with A.the same frequency, different amplitude and travelling in the same direction B.the same frequency, the same amplitude and travelling in the same direction **C.the same frequency, the same amplitude and travelling in opposite direction** D.different frequencies, the same amplitude and travfelling in the same direction E.different frequencies, different amplitudes abd travelling in the same direction.

21. Which of the following is a characteristic of stationary waves? A.The antinode is a point of minimum displacement B.The distance between two successive nodes is one wavelength **C.They are formed by two identical waves travelling in opposite directions** D.They can be trasverse or longitudinal.
Antinode in a stationary wave is a point of maximum displacement. The distance between two consecutive nodes or antinodes is half wavelength $\frac{1}{2}\lambda$. Sationary wave can be produced by a transverse or longitudinal wave

22. Which of the following statements is/are not correct about a standing wave? I.It is formed by the superimposition of two progressive waves travelling in the same direction II.all points on the wave are vibrating with maximum displacement III.it is formed by the superposition of two progressive waves of different amplitudes travelling in opposite direction A.I B.II C.I and II D.II and III **E.I,II and III.**
Sationary wave is formed by superposition or interference of two identical progressive wave i.e with the same amplitude, frequency, wavelenbth, travelling in opposite direction. Some points are with minimum disolacement or at rest i.e nodes while some points are with maximum displacement i.e antinodes.

23. One end of a long wire is fixed while vibrator is attached to the other end. When the vibrator is energized, the types of waves generated in the wire are **A.stationary and transverse** B.progressive and transverse C.stationary and longitudinal D.progressive and longitudinal.

24.



The diagram is an example of a A.beat pattern **B.standing wave** C.diffraction pattern D.progressive wave.

25. Which of the properties of waves is/are common to all waves? I.diffraction II.refraction III.interference A.I B.III C.I and II **D.I,II and III**
All waves exhibit reflection,refraction, diffraction and interference.

26. Which of the following statements about electromagnetic waves is not correct? A.They include X-rays **B.They cannot be diffracted**

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C.Infra-red rays are electromagnetic D.They can undergo interference E.Radio waves are electromagnetic waves
All waves can be diffracted or can undergo diffraction.

27. Ripples on water are similar to light waves in that they both A.have the same wavelength B.longitudinal C.cannot be reflected D.travel at the same speed **E.can be refracted and diffracted.**
Ripples on water are water waves. Water waves and light waves are both transverse, they can both be reflected,refracted amd diffracted. The wavelength of water wave is greater than that of light wave.

28. Which of the following characteristics of wave is used in the measurement of the depth of the sea? A.Diffraction B.Interference C.Refraction **D.Reflection.**

29. Transverse waves can be distinguished from longitudinal waves using the characteristics of A.diffraction **B.polarization** C.reflection D.refraction.
Only trasverse waves udergoes polarization.

30. Vibration in a stretched spring cannot be polarized because they are A.transverse waves B.mechanical waves C.stationary **D.longitudinal waves**
Longitudinal waves cannot be polarized, only transverse waves can polarized.

31. Which of the following is the exclusive property of a transverse wave? A.diffraction B.reflection C.compression **D.polarization.**

32. Which of the following is not a property of longitudinal waves? A.compression B.reflection C.refraction **D.polarization** E.diffraction.

33. The difference between sound and light waves is that sound waves A.are transverse while light wave are longitudinal B**.require a medium to travel while light waves do not** C.can be diffracted but light cannot D.cannot be reflected but light waves can.
Sound waves are longitudinal while light waves are transverse. Sound waves are mechanical i.e require medium while light wave are electromagnetic i.e do not require medium. Both sound and light waves can be diffracted and reflected.

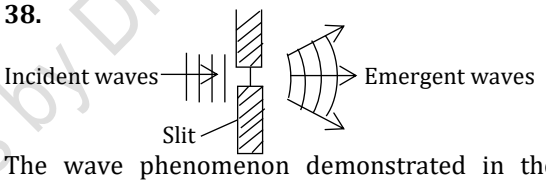
34. The change in the direction of a wave front because of a change in the velocity of the wave in another medium is called A.reflection **B.refraction** C.diffraction D.interference E.dispersion.

35. A wave disturbance travelling in air enters a medum in which its velocity is less than that in air. Which of the following statements is true about the wave in the medium? A.The frequency of the wave is decreased while the wavelength is increased B.The frequency of the wave is decreased while the wavelength is unaltered C.Both the frequency of the wave and the wavelength are decrease **D.The frequency of the wave is unaltered while the wavelength is decreased.**
When a wave passes the boundary between to media it undergoes refraction. Refraction is accompanied by a change in the direction of wavefront, velocity and wavelength while the frequency is constant. Velocity is directly proportional to wavelength at constant frequency i.e $v = \lambda f$, $v \propto \lambda$ (f=constant). Hence, the wavelength of the wave decreases in the medium as the velocity decreases.

36. Which of the following remain unchanged as light travels from one medium to the other? I.Speed II.Wavelength III.Frequency A.I B.II **C.III** D.I and II E.II and III.

37. The phenomenon of light bending round an obstacle is A.reflection B.polarization C.refraction **D.diffraction.**
Diffraction grating is used to measure wavelengths of light waves and to study the structutr and intensity of spectra lines.

38.



The wave phenomenon demonstrated in the diagram is A.reflection B.deflection **C.diffraction** D.refrraction.
The incident waves on passing through the slit undergoes diffraction and spreads out as an emergent waves.

39. Which of the following is/are necessary for the production of interference with two wave trains? I.The sources must be close to each other II.The waves must have the same frequency III.The waves must have the same amplitude IV.The waves must travel with the speed of light A.I and II **B.I,II and III** C.I,II an d IV D.I,III and IV E.II,III amd IV.
Intereference occurs between two waves when their source are close to each other and are coherent i.e source that are always in phase or constant phase difference,have the same amplitude, frequency and wavelength.

40. I.Coherence II.Same frequency III.Same wavelength IV.Same intensity. Which of the conditions above are necessary to produce interference fringes? A.I and II B.II and III **C.I,II and III** D.I,II and IV.

41. I.They should be identical B.They should originate from the same source III.They should be coherent IV.The should be monochromatic. From the statements above, the conditions for two waves ot interfere are **A.I,II and III** B.I,III and IV C.II,III and IV D.I,II,III and IV.

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Interference occurs with identical waves i.e same wavelength,amplitude and frequency, same source or different source and the source must be coherent. Monochromatic is when a light is of one colour or wavelength, hence it improves interference in light wave only.

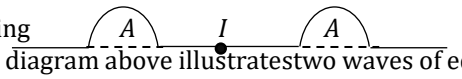
42. When two waves are superimposed on each other, the occurences are possible, except A.stationary waves B.antinodal lines C.nodal lines **D.dispersion.**

Superimposition of identical waves travelling in the same direction is interference of waves. Nodal and antinodal lines represents destructive and constructive interference respectively. Dispersion is the separation of white light when passed through a prism, it does not involve superimposition of waves.

43. Two identical waves travelling in the same direction are superimposed. What should be the phase difference between the waves for maximum destructive interference to occur A.20° B.45° **C.180°** D.225° E.270°.

Destructive interference occurs for two waves out pf phase, with a phase difference or phase angle of 180° or π rad or distance of $\frac{1}{2}\lambda$ or half cycle. Constructive intefrence occurs for two waves in phase angle of 360° or 2π or distance of 1λ or 1 cycle.

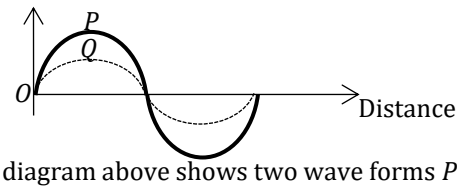
44.



The diagram above illustratestwo waves of equal amplitude A and frequencies aproching each other. When the two waves meet at a point I midways between them, the vertical displacement of the resulting wave will be A.0 B.$\frac{A}{2}$ C.$\frac{A}{4}$ **D.2*A*.**

The two waves as shown in the diagram above will interfere constructively, as both wave are of the same frequency and amplitude and starts at the same time. The waves will interfere with the crest and trough on each wave metting each other. The vertical displacement of the resulting wave is equal to the sum of their amplitude : $A + A = 2A$.

45. Particle displacement



The diagram above shows two wave forms P and Q at a particular instant of time. The two waves will interfere A.destructively to produce a wave of a larger amplitude B.destructively to produce a wave of smaller amplitude **C.constructively to produce a wave of a larger amplitude** D.constructively to produce a wave of a small amplitude.

Constructive interference occurs when two waves meets in phase i.e crest meets crest and troughs meets troughs, and it results to a higher resultant disturbance or larger amplitude.

46. The angle between two successive crests of a wave is A.90° B.180° C.270° **D.360°.**

The distance between two successive crest is equal to one wavelength (1λ) or one cycle which is equivalent to an angular difference of 360° or 2π rad.

47. The colours seen in the film of oil on the road and in soap bubbles are due to A.reflection **B.interference** C.diffraction D.polarization.

Colours seen in the film of oil on the road and in soap bubbles are due to interference from the sky or cloud.

48. Whenever light waves wave are restricted to a specific plane, they are said to be A.diffracted B.refracted C.regularly reflected **D.plane-polarized.**

A wave is plane-polarized if its vibration occurs only in one plane.

49. Which of the following are correct of plane-polarized light? I.Plane polarization of light is the formation of hydrogen bubbles on the particles of light II.Plane polarization of light is the splitting of light into its spectra components III.Plane-polarized light vibrates in one plane only IV.Polarization is characteristic of transverse waves A.I and II B.I and III C.I and IV D.II and III **E.III and IV**.

50. A narrow beam of unpolarized light of intensity 10.2cd is incident at an angle 30° to the axis of the polaroid. What is the intensity of the emergent polarized ray? A.3.83cd **B.7.65cd** C.11.48cd D30.6cd.

Intensity of unpolarized light I_o=10.2cd, angle between the incident light and polaroid axis θ=30°, Intensity of the emergent polarized light ray I=?, Malu's law : $I = I_o cos^2\theta$,

I=10.2× $cos^2$30°=10.2× $\left(\frac{\sqrt{3}}{2}\right)^2$ =10.2× $\frac{3}{4}$ = 7.65cd.

51. A parallel beam of unpolarized light is incident on a plane glass surface at an angle of 58° to the normal. If the reflected beam is completely polarized, calculate the refractive index of the glass A.0.5 B.0.8 **C.1.6** D.3.2.

Angle of incidence or polarizing angle θ_p=58°, refractive index of plane glass surface n=?,

Brewster's law: $n = tan\ \theta_p = tan\ 58° = 1.6$.

52. If the polarizing angle for a given optical medium is 49°, calculate the refractive index of the material A.1.75 **B.1.15** C.0.75 D.0.66.

θ_p=49°, n=?, Brewster's law: $n = tan\ \theta_p$,

$n = tan\ 49° = 1.15$.

53. Which of the following statements is not correct about sound and light waves? A.The wavelength of light waves is generally shorter than that of sound waves B.The frequency of

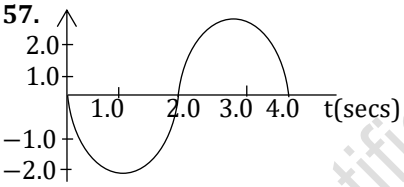
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sound waves is generally lower than that of light waves C.Light is visible sound is not D.Sound waves causes particles of the medium to vibrate in the direction of propoagation of the waves **E.Light waves have the same speed as sound waves.**

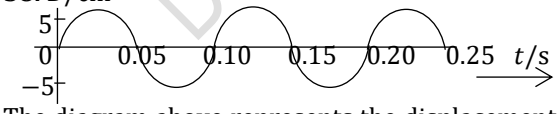
54. The distance between successive crest and trough of a wave is A.quarter wavelength B.twice wavelength C.full wavelength **D.half wavelength**
The distance between two suceesive crests or troughs is equal to one full wavelength. The distance between successive crest and trough is half wavelength. The distance between three successive crests or troughs is equal to twice wavelength.

55. What is the frequency of the sound made by a siren having a discs with 32 holes and making 25 revolutions per second? A.80Hz B.600Hz **C.800Hz** D.1600Hz.
Number of holes n=32, speed in revolution per second v=25rev/s, frequency f=?,
$f = nv = 32\times25 = 800$Hz.

56. A revolving gear wheel has 22 evenly spaced teeth. A piece of card is held against the teeth when the wheel is rotating at 360 revolutions per minute. Calculate the frequency of the note emitted A.22Hz **B.132Hz** C.360Hz C.360Hz D.1320Hz E.7920Hz.
Number of teeths n=22, speed in revolutions per second v=360 $\frac{rev}{1min}\times\frac{1min}{60s}$ = 6rev/s,
$f = nv = 22\times6 = 132$Hz.

57.


The frequency of the wave in the diagram above is **A.0.25Hz** B.0.17Hz C.1.05Hz D.1.57Hz
Time take to complete 1 cycle=period T=4.0s ,
Frequency $f = \frac{1}{T} = \frac{1}{4.0} = 0.25$s^{-1} or 0.25Hz.

58. D/cm


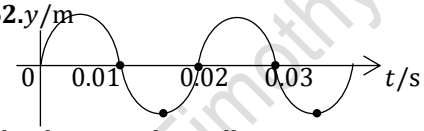
The diagram above represents the displacement D versus time t, graph of a progressive wave. Deduce the frequency of the wave A.20Hz **B.10Hz** C.5Hz D.4Hz.
Time take to complete 1 cycle=period T=0.10s
Frequency $f = \frac{1}{T} = \frac{1}{0.10} = 10$s^{-1} or 10Hz.

59. A wave has a frequency of 2Hz and a wavelength of 30cm. The velocity of the wave is A.60.0ms^{-1} B.6.9ms^{-1} C.1.5ms^{-1} **D.0.6ms^{-1}**
f=2Hz, λ=30cm=0.3m, v=?, $v = \lambda f$
$v = 0.3\times2 = 0.6$ms^{-1}.

60. The velocity of wave having a frequency of 1000Hz and a wavelength of 5.000cm is A.0.05ms^{-1} B.0.20ms^{-1} C.5.00ms^{-1} **D.50.00ms^{-1}** E.200.00ms^{-1}.
f=1000Hz, λ=5.000cm=0.050m, v=?,
$v = \lambda f = 0.050\times1000 = 50.00$ms^{-1}.

61. If the wavelength of a wave with a velocity of 360ms^{-1} is 60m. The period of the wave is A.6s B.3.6s **C.0.17s** D.0.6s E.3s
v=360ms^{-1}, λ=60m, T=?, $v = \frac{\lambda}{T}$, $T = \frac{\lambda}{v}$,
$T = \frac{60}{360} = 0.166$s = 0.17s.

62. A wave travcelling with a speed of 360ms^{-1} has a wavelength of 60cm. The period of the wave in seconds is A.1.5×10^{-3} B.1.7×10^{-3} C.6.1×10^{-3} D.1.7×10^{-2} E.3.6×10^{-2}.
v=360ms^{-1}, λ=60cm=0.6m, T=?, $v = \frac{\lambda}{T}$,
$T = \frac{\lambda}{v} = \frac{0.6}{360} = 1.66\times10^{-3} = 1.7\times10^{-3}$s.

62.


The diagram above illustrates a variation of the displacement y of a wave particle with time t. If the velocity of the wave is 250ms^{-1}, calculate the distance between two successive particles in phase A.2.0m B.2.5m C.3.0m **D.5.0m** E.50.0m
v=250ms^{-1}, time taken to comlete 1 cycle is equal to the period T=0.02s, distance between two points in phase is equal to the wavelength λ=?,
$v = \frac{\lambda}{T}$, $\lambda = vT$, $\lambda = 250\times0.02 = 5.0$m.

64. The distance between two successive troughs of a wave is 0.4m. if the frequency of the source is 825Hz, calculate the speed of the waves A.165ms^{-1} **B.330ms^{-1}** C.412.5ms^{-1} D.825ms^{-1}.
Distance betweeb two successive troughs = wavelength λ=0.4m, frequency of source = frequency of wave$f$=825Hz, v=?, $v = \lambda f$,
$v = 0.4\times825 = 330$ms^{-1}.

65. A boat is roked by waves of speed 30ms^{-1} whose successive crests are 10m apart. Calculate the rate at which the boat receives the waves A.30s^{-1} B.10s^{-1} **C.3s^{-1}** D.1s^{-1}.
v=30ms^{-1}, distance between successive crest = wavelength λ=10m, rate at which the boat receives wave is the frequency f of the wave=?,
$v = \lambda f$, $f = \frac{v}{\lambda} = \frac{30}{10} = 3$s^{-1} or 3Hz.

66. A boat at anchor is rocked by waves whose crests are 100m apart and whose velocity is 25ms^{-1}. At what interval does the wave crest reach the boat? A.0.24s **B.4.00s** C.75.00s D.2500.00s.
Distance between crests λ=100m, v=25ms^{-1}, interval at which the creast reaches the boat is period T=?, $v = \frac{\lambda}{T}$, $T = \frac{\lambda}{v} = \frac{100}{25} = 4.00$s.

67. A radio wave has a wavelength of 150m. If the velocity of radio waves in free space 3×10^{8}ms^{-1},

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calculate the frequency of the radio wave A.4.5×10^{10}Hz B.5.0×10^{9}Hz C.4.5×10^{9}Hz **D.2.0×10^{6}Hz.**

λ=150m, v=c=3×10^{8}ms^{-1}, f=?, $v = c = \lambda f$,

$f = \frac{c}{\lambda} = \frac{3\times10^{8}}{150} = 2.0\times10^{6}$Hz.

68. Sixty complete waves pass a particular point in 4s. If the distance between three successive troughs of the waves is 15m, calculate the speed of the waves A.300.0ms^{-1} B.225.0ms^{-1} **C.112.5ms^{-1}** D.75.0ms^{-1} E.16.0ms^{-1}.

Number of waves n=60, time taken t=4s, frequency $f = \frac{number\ of\ waves}{time\ taken} = \frac{n}{t} = \frac{60}{4}$,.

f=15Hz, distance between three successive troughs is equal to two wavelength. 2λ=15m, $\lambda = \frac{15}{2} = 7.5$m, $v = \lambda f = 7.5\times15 = 112.5$ms^{-1}.

69. Water waves are generated by dropping stones at regular intervals at a point in a pool of water.The first crest reaches another point 8m away in 4s.If the distance between two successive crests is 0.5m,the frequency of the wave is A.2Hz **B.4Hz** C.8Hz D.16Hz.

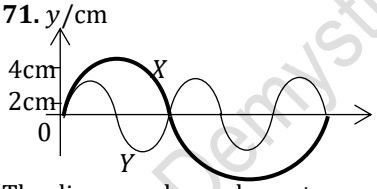
Distance s=8m, time t=4s, distance between two successive crest=wave length λ=0.5m, f=?, $v = \lambda f = \frac{s}{t}$, $0.5\times f = \frac{8}{4} = 2$, $f = \frac{2}{0.5} = 4$Hz.

70. In a ripple tank experiment, a vibrating plate is used to generate ripples in the water. If the distance between two successive troughs is 3.5cm and the wave travels a distance of 31.5cm in 1.5s, calculate the frequency of the vibrator A.3.0Hz **B.6.0Hz** C.12.0Hz D.27.0Hz E.73.5Hz.

λ=3.5cm, s=31.5cm, t=1.5s, f=?,

$v = \lambda f = \frac{s}{t}$, $3.5\times f = \frac{31.5}{1.5} = 21$, $f = \frac{21}{3.5} = 6.0$Hz.

71. y/cm



The diagram above shows two wave forms X and Y. If the frequency of X is 30Hz, what is the frequency of Y? A.12Hz B.15Hz C.60Hz D.75Hz

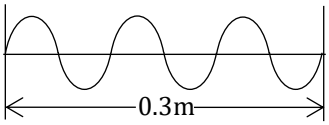
The digram shows that wave form X and Y starts from the same point and completes their oscillation at the same time t.

Frequency $f = \frac{number\ of\ cycles/waves}{time\ taken} = \frac{n}{t}$,

$f \propto n$ (t=constant), $\frac{f_X}{n_X} = \frac{f_Y}{n_Y}$, f_X=30Hz, n_X=1 cycle, n_X=2½ cycle, f_Y=?,

$\frac{30}{1} = \frac{f_Y}{2\frac{1}{2}}$, $f_Y = 30\times2\frac{1}{2} = 30\times\frac{5}{2} = 75$Hz.

72.



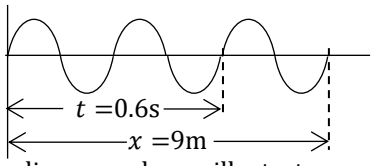
The diagram above represents a transverse electromagnetic wave travelling with a speed 3.0×10^{8}ms^{-1}.What is the frequency of the wave A.3.0×10^{7}Hz B.9.0×10^{7}Hz C.1.0×10^{9}Hz **D.3.0×10^{9}Hz .**

3 complete cyles = 3 wavelengths, $3\lambda = 0.3$m,

$\lambda = \frac{0.3}{3} = 0.1$m, v=3.0×10^{8}ms^{-1}, f=?,

$v = \lambda f$, $f = \frac{v}{\lambda} = \frac{3\times10^{8}}{0.1} = 3.0\times10^{9}$Hz.

73.



The diagram above illustrates a wave form. Determine the speed of the wave A.5.4ms^{-1} B.8.4ms^{-1} C.9.6ms^{-1} **D.10.0ms^{-1}.**

From the diagram above, 0.6s is the time taken for 2 complete cycles. Time taken to complete 1 cycle= period $T = \frac{0.6}{2} = 0.3$s. 3 complete cycles = 3 wavelengths, $3\lambda = 9$m,

$\lambda = \frac{9}{3} = 3$m, $v = \frac{\lambda}{T} = \frac{3}{0.3} = 10$ms^{-1}.

74. Determine the distance between the consecutive antinodes XX. If the wavelength is 60cm A.15cm **B.30cm** C.60cm D.120cm

λ=60cm, distance between two consecutive antinodes AA is half the wavelength,

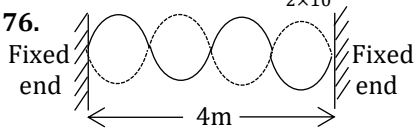
$AA = XX = \frac{\lambda}{2} = \frac{60}{2} = 30$cm.

75. A wave of frequency 10Hz forms a stationary wave pattern in a medium where the velocity is 20cms^{-1}. The distance between adjacent nodes is **A.1.0cm** B.1.5cm C.2.0cm D.5.0cm

f=10Hz, v=20cms^{-1}, distance between adjacent nodes NN is half the wavelength, $NN = \frac{\lambda}{2}$, $\therefore \lambda = 2(NN)$, $v = \lambda f = 2(NN)\times f$

$20 = 2(NN)\times10$, $NN = \frac{20}{2\times10} = 1.0$cm.

76.



The diagram above illustrates a wave set up between two fixed ends 4.0m apart. If the speed of the wave is 1.0ms^{-1}, its wavelength and frequency respectively are A.1m; 0.25Hz B.2m; 0.25Hz **C.2m; 0.50Hz** D.4m; 0.50Hz.

The diagram above represents a stationary wave. The 4m is the distance between 4 nodes i.e $4NN$, distance between 2 consecutive nodes $NN = \frac{\lambda}{2}$,

$\therefore 4NN = 4$m, $4\frac{\lambda}{2} = 4$m, $\lambda = \frac{4\times2}{4} = 2$m,

v=1.0ms^{-1}, $v = \lambda f$, $f = \frac{v}{\lambda} = \frac{1}{2} = 0.5$Hz.

77. The equation of trasverse wave travelling along a string is given by y=0.3sin $(0.5x - 50t)$ where y and x are in cm and t is in seconds, find the maxium displacement of the particles from the equilibrium position A.50.0cm B.2.5cm

Demystified Series by Dr Timothy

C.0.5cm **D.0.3cm.**
The maxium displacement of the particles from equilibrium position is the amplitude A of the wave. Comparing y=0.3sin $(0.5x - 50t)$ with the equation $y = A\,sin(\omega t - kx)$, it shows that the amplitude $A = 0.3$cm.

78. A wave is represented by the equation $y = A\sin\frac{2\pi x}{\lambda}$. Which of the following statements about the wave is not correct? A.A represents the maximum displacement B.λ= represents the distance between two consecutive particles which are in phase C.x is the distance of the particles from origin **D.$\frac{2\pi x}{\lambda}$ represents the wave number** E.The unit of y is the same as the unit of x.
$\frac{2\pi x}{\lambda}$ represents the phase difference or phase angle. Wave number or wave constant $k = \frac{2\pi}{\lambda}$.

79. A travelling wave moving from left to right has an amplitude of 0.15m, a frequency of 550Hz and a wavelength of 0.01m. The equation describing the wave is
A.$y = 0.15sin200\pi(x - 5.5t)$
B.$y = 0.15sin200\pi(0.01x - 5.5t)$
C.$y = 0.15sin5.5\pi(x - 200t)$
D.$y = 0.15sin\pi(550x - 0.01t)$.
Amplitude A=0.15m, frequency f=550Hz, wavelength λ=0.01m, the wave is moving from left to right i.e positive-x direction, hence the sign is negative. Substituting into the equation:
$y = A\,sin\frac{2\pi}{\lambda}(x - vt)$, $v = \lambda f$ =0.01×550,
$v = 5.5\text{ms}^{-1}$, $y = 0.15\sin\frac{2\pi}{0.01}(x - 5.5t)$,
$y = 0.15\sin 200\pi(x - 5.5t)$.

80. A wave has an amplitude of 4.0m, angular speed $\frac{1}{3}\pi$ rads^{-1} and phase angle $\frac{2}{3}\pi$ rad. The displacement y of the wave particle is given as **A.$y = 4\sin\frac{\pi}{3}(t + 2)$** B. $y = 4\sin\frac{\pi}{3}(t + \frac{2}{\pi})$ C.$y = 4\sin\frac{\pi}{3}(2t + 1)$ D.$y = 4\sin\frac{\pi}{3}(t - 2)$
Amplitude A=4.0m,angular speed ω=$\frac{1}{3}\pi$ rads^{-1} phase angle ϕ=$\frac{2}{3}\pi$ rad, Substituting into the equation: $y = A\sin(\theta \pm \phi) = A\sin(\omega t \pm \phi)$
$y = 4\sin\left(\frac{1}{3}\pi t \pm \frac{2}{3}\pi\right) = 4\sin\frac{\pi}{3}(t \pm 2)$.
$y = 4\sin\frac{\pi}{3}(t + 2)$.

81. The equation of a wave travelling in a horizontal direction is expressed as
$y = 15\sin\frac{2\pi}{5}(60t - x)$. What is the wavelength? A.60m B.15m **C.5m** D.2m
Comparing $y = 15\sin\frac{2\pi}{5}(60t - x)$ with the equation $y = A\,sin\frac{2\pi}{\lambda}(vt - x)$, $\frac{2\pi}{\lambda} = \frac{2\pi}{5}$, $\lambda = 5$m.

82. The equation of a wave travelling along the positive x-direction is given by
$y = 0.25\times10^{-3}\sin(500t - 0.025x)$. Determine the angular frequency of the wave motion
A.0.25×10^{-3}rads^{-1} B.0.25×10^{-1}rads^{-1}
C.5.00×10^{2}rads^{-1} D.2.5×10^{2}rads^{-1}.
Comparing $y = 0.25\times10^{-3}\sin(500t - 0.025x)$ with the equation $y = A\sin(\omega t - kx)$,
Angular frequency or angular velocity or speed $\omega = 500\text{rads}^{-1} = 5.00\times10^{2}\text{rads}^{-1}$.

83. A plane progressive wave is represented by the equation $y = 0.2\sin(200\pi t - 0.5x)$, where x and y are measured in cm and t in s. What is the frequency of the wave? A.0.5Hz B.4.0Hz C.10.0Hz **D.100.0Hz** E.200.0Hz.
Comparing $y = 0.2\sin(200\pi t - 0.5x)$ with the equation $y = A\,sin(2\pi ft - \frac{2\pi x}{\lambda})$,
$2\pi ft = 200\pi t$, $f = \frac{200}{2} = 100.0$Hz.

84. The equation of a certain progressive wave $y = 2\sin 2\pi\left(\frac{t}{0.01} - \frac{x}{30}\right)$, where x and y are in cm and t in seconds. Calculate the period of the wave A.0.001s **B.0.010s** C.10.000s D.100.000s.
Comparing $y = 2\sin 2\pi\left(\frac{t}{0.01} - \frac{x}{30}\right)$ with the equation $y = A\,sin(\frac{2\pi}{T}t - \frac{2\pi x}{\lambda})$,
$y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$, $\frac{t}{T} = \frac{t}{0.01}$, T =0.010s.

85. The equation of a wave is
$y = 0.005\sin[\pi(0.5x - 200t)]$, where x and y are in meters and t is in seconds. What is the velocity of the wave? A.4000ms^{-1} B.400ms^{-1} C.250ms^{-1} D.40ms^{-1}.
Comparing $y = 0.005\sin[\pi(0.5x - 200t)]$ with the equation $y = A\,sin\frac{2\pi}{\lambda}(x - vt)$,
$\frac{2\pi}{\lambda}vt = 200\pi t$, $v = \frac{200\lambda}{2} = 100\lambda$, $\frac{2\pi x}{\lambda} = 0.5\pi x$,
$\lambda = \frac{2\pi x}{0.5\pi x} = 4$m, $v = 100\lambda = 100\times4 = 400\text{ms}^{-1}$.

86. The distance between two points P and Q, along a wave is 0.05m. If the wavelength is 0.10m, determin the phase angle between P and Q in radians A.0.1π B.π C.2π D.10π.
x=0.05m, λ=0.10m, Phase angle ϕ=?,
Phase angle ϕ= $\frac{2\pi x}{\lambda} = \frac{2\pi\times0.05}{0.10} = \pi$rad or 180°.

87. A progressive wave is represented by
$y = 10\sin(100\pi t - \frac{\pi x}{34})$, Two layers of the wave separated by 153cm have a phase difference of A.270° B.180° C.90° D.45°
Compare $y = 10\sin(100\pi t - \frac{\pi x}{34})$ with the equation $y = A\sin(2\pi ft \pm \phi)$, $\phi = \frac{\pi x}{34}$,
x=153cm, $\phi = \frac{\pi\times153}{34}$ =4.5π, π=180°,
$\phi = 4.5\times180 = 810°$, ϕ=810° −(2×360°)=90°.

88. A progressive wave equation is represented by $y = A\sin(150\pi t - \frac{\pi x}{4})$. If the phase difference is 45°, the value of x in the equation is A.1cm B.2cm C.3cm D.4cm
Comparing $y = A\sin(150\pi t - \frac{\pi x}{4})$ with the equation $y = A\sin(2\pi ft \pm \phi)$, $\phi = \frac{\pi x}{4}$,
ϕ=45°, x=?, $45° = \frac{\pi x}{4}$, $x = \frac{45°\times4}{\pi}$, π=180°,

Demystified Series by Dr Timothy

$x = \frac{45° \times 4}{180°} = 1\text{cm}.$

Jamb past questions on Waves : types, propagation and properties :
[1978/49,1979/6,36,40,48,1981/4,22,1982/19, 1983/13,49,1984/45,1986/29,34,49,1987/27 ,28,1988/35,1989/19,22,31,1991/24,1992/20, 23,1993/22,23,2000/36,2001/17,2002/17,26, 2004/40,2006/34,2007/4,8,2008/26,27,28, 2009/26,2730,2010/24,2011/21,2012/26,27, 2013/25,2013/26,2014/26,2015/37]

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Chapter 16 – Sound waves

● **Sound waves** – Sound waves is produced by vibrating bodies e.g gutitar strings or struked tuning fork.

- Sound wave is a mechanical wave, it requires a material medium for its propagation. Hence, sound waves cannot travel through a vacuum.
- Themedium through which sound waves travels are solid, liquid and gases.
- Sound waves in air comprises of regions with air pressure or densities above normal called compressions and regions with air pressure or densities below normal called rarefractions.
- Sound waves are longitudinal waves.
- **Sound waves have a longer wavelength than light waves.**
- Sound waves and also light waves exhibits **Doppler effect.**
- **Doppler effect** is the apparent change in the frequency/pitch of a source of wave due to the relative motion between the source of wave and the observer.
- The sudden drop in frequency/pitch of a train when sounding its whistle and just passes an observer, is a common example of Doppler effect.
- **Application of Doppler effect – I.**Radar used in large airports for recording the height and distance of incoming planes about to land. **II.**Police speed traps.

The apparent frequency f^1 in Doppler effect is given by :

1.When both the source and observer are moving or heading towards each other i.e approaching each other in opposite direction, then : $\boldsymbol{f^1 = \left(\frac{v+v_o}{v-v_s}\right) f}$.

2.When both the source and observer are moving away from each other i.e chasing or receding in the same direction, then : $\boldsymbol{f^1 = \left(\frac{v-v_o}{v-v_s}\right) f}$.

f^1=new or apparent frequency of source, f=initial frequency of source, v=speed of sound in air, v_s=speed of source, v_o=speed of observer.

- **When the source is stationary, $\boldsymbol{v_s = 0}$.**
- **When the observer is stationary, $\boldsymbol{v_o = 0}$.**
- According to Doppler effect, the frequency increases when the source moves towards the observer and decreases when the source moves away from the observer.
- **Speed or velocity of sound –** Speed of sound v is the ratio of the distance traveled x by sound wave to the time taken t. $\boldsymbol{v = \frac{x}{t} = \lambda f}$ **.**
- The speed of sound at 0°C is about 33ms^{-1} in air, 1500ms^{-1} in water and 5000ms^{-1} in steel.
- **Factors affecting the speed or velocity of sound :**

1.Wind : speed of sound increases with wind coming from the source to the observer or wind moving in the same direction with the sound and decreases when the wind is in opposite direction as the sound.

2.Density $\boldsymbol{\rho}$: speed sound increases as the density of the medium decreases, $\boldsymbol{v \propto \frac{1}{\sqrt{\rho}}}$.

- The speed of sound in solid,liquid and gas decreases from solid-liquid-gas. Hence, sound travels faster through solid than in liquid and faster in liquid than in air(gas) i.e $\boldsymbol{v_S > v_L > v_g}$.

3.Elasticity of the medium i.e Young modulus E for solids : speed of sound increases as the elasticity of the medium increases, $\boldsymbol{v \propto \sqrt{E}}$.

4.Humidity H : speed of sound increases with humidity $\boldsymbol{v \propto H}$.

5.Temperature T : speed of sound increases as the temperature of the medium increases. **The speed of sound is directly proportional to the square root of its absolute Kelvin temperature.** $\boldsymbol{v \propto \sqrt{T}}$, $\boldsymbol{\frac{v_1}{\sqrt{T_1}} = \frac{v_2}{\sqrt{T_2}}}$, $\boldsymbol{\frac{{v_1}^2}{T_1} = \frac{{v_2}^2}{T_2}}$.

- **Speed of sound is independent of pitch or frequency,air pressure and loudness or intensity.**
- The speed of sound v in solids is related to the Young modulus E and density ρ of the solid medium is given by – $\boldsymbol{v = \sqrt{\frac{E}{\rho}}}$.
- During thunderstorm, we see the flash of lightening before the sound of thunder because the speed of light(about 3×10^8ms^{-1}) is far greater than the speed of sound(about 340ms^{-1}).
- The distance $\boldsymbol{s}$ between us and the point of generation of the lightning and thunder or the storm-centre is given by : $\boldsymbol{s = vt}$, v=speed of sound in air , t=time interval between the flash of lightning and sound of thunder**.**
- Sound waves undergo reflection, refraction, diffraction and interference. Sound waves do not exhibit polarization.
- **Reflection of sound waves –** The reflection of sound waves from a plane surface gives rise to echo. Irregular or rough surfaces will cause regular reflection of sound waves.
- **Echo** is the sound heard after reflection of sound wave from a plane hard surface. It is also called reflected sound.
- **Reverberation** is the perserverance of sound due to multiple reflection from a surface i.e multiple echo with diminishing intensity,after the source ceases. It is indistinct echo or sound that cannot be differentiated from the original sound.
- **A hall where reverberation occurs is said to have poor acoustical properties.**
- Reverberation is reduced in halls by hanging curtains round hall or lined with thick felt, having

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more openings in the walls or covering boards which absorbs the sound quickly, hence minimize reflection of sound waves.

- Application of Echoes –

1.Determination of the speed of sound in air – $\boldsymbol{v = \frac{2x}{t}}$, v=speed of sound in air, x=distance between the source and the reflecting surface $2x$=total distance moved by the sound before echo is heard due to reflection, t=time interval taken for the echo to be heard.

2.Determination of the depth of a sea-bed.

3.Detection of submarine.

4.Exploration of earth for mineral deposits, gas and oil.

- Refraction of sound waves – Sound waves are refracted when they travel from to medium to another of different densities. Sound coming from a distance is heard easily at night than in the day due to refraction of sound waves towards the ground by cooler air at night than warmer air at day.

- Diffraction of sound waves – Sound waves are diffracted when they travel across an obstacle ot bend e.g distribution of sound round corners in a room through a large opening.

- During refraction of sound waves, the frequency is constant hence, speed of sound is directly propotional to its wavelength. Thus, when sound moves from a cold air region to hot air region i.e increase in temperature, speed and wavelength increase but frequency remains constant.

-When sound waves moves from less dense (air) to a more dense medium(water), the velocity and wavelength increases and vice-versa, but the frequency remains constant.

- Inteference of sound waves – Superimposition of sound waves gives rise to interference effects. Constructive and destructive interference of sound waves produces loud and reduced or zero sound respectively.

- Beat is the alterations in loudness of sound produced when two notes or sound waves of nearly equal frequencies are superimposed. Hence, beats are due to interference of two sound waves of close frequency.

- Beat frequency $\boldsymbol{f_B}$is the number of intense beats(sounds) heard per second. It is equal to the difference in the frequency between the two wave sources. $\boldsymbol{f_B = \frac{n}{t}}$.

$\boldsymbol{f_B = f_1 - f_2 = \frac{v}{\lambda_1} - \frac{v}{\lambda_2} = v\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)}$.

- Beat period $\boldsymbol{T_B}$ is equal to the reciprocal of the beat frequency. $\boldsymbol{T_B = \frac{1}{f_B}}$.

- Musical note and Noise – Sound can either be grouped as either musical note or noise.

- Musical note is produced by vibrations or sound of regular frequency or pitch, which is pleasant to the ear.

- Noise is produced by vibrations or sound of irregular frequency or pitch, which is unpleasant to the ear.

- Characteristics of musical note or sound –

1.Pitch – Pitch of a note is its position in a musical scale. **Pitch depends on frequency of sound** i.e a low pitched note has a low frequency while a high pitched note has high frequency.The human ear hear sound wave of frequency ranging from **20Hz to 20kHz**only. Sound of frequency above hearing rang e i.e above 20Hz are called **ultrasonic sound.**

2.Intensity – Intensity of sound is the rate of flow of energy per second per unit area perpendicular to the direction of propagation of sound wave. $\boldsymbol{I = \frac{E}{t \times A} = \frac{P}{A}}$. It is measured in **watt per square meter** ($\boldsymbol{W m^{-2}}$).

- Intensity depends on the square of the amplitude of the vibrating body $\boldsymbol{I \propto A^2}$.;

3.Loudness – Loudness is the magnitude of sensation resulting from a sound reaching the ear.

- Loudness depends on intensity$\boldsymbol{L \propto I}$, mass of air which can be set into vibration$\boldsymbol{L \propto M_{air}}$, and square of the amplitude of the vibrating body $\boldsymbol{L \propto A^2}$.

- Intensity and loudness varies inversely with the square of distance $\boldsymbol{r}$ from the source $\boldsymbol{L}$ or $\boldsymbol{I \propto \frac{1}{r^2}}$.

- Amplitude of sound varies inversely with the distance from the source of sound $\boldsymbol{A \propto \frac{1}{r}}$.

4.Quality or Tone or Timbre – This is the characteristic note of a musical instrument which differentiates it from another note of the same pitch and loudness produced by another instrument.

- A well made tuning fork has a pure tone i.e single or fundamental frequency or with free vibration and is not affected by quality.

- A musical note is a mixture of several frequencies namely: fundamental frequency, harmonics and overtones.

- Fundamental frequency $\boldsymbol{f_o}$ is the lowest, strongest single audible frequency of a vibrating body.

- Harmonics H are frequencies which are multiple of the fundamental frequency, which is the first harmonic. 1st harmonic $= f_o$, 2nd harmonic$=2f_o$, 3rd harmonic $=3f_o$ e.t.c.

- Overtones are the next higher frequencies or mode of vibration above and multiple of fundamental quantity. If, f_o =5Hz, overtones are 20Hz, 30Hz, 40Hz i.e multiple of f_o. Also, 1st overtone=2nd harmonic, 2nd overtone = 3rd harmonic, 3rd overtone=4th harmonic.

- Overtones are quitter than the fundamental frequency.

- Quality or timbre or tone depends on

Demystified Series by Dr Timothy

number of harmonics and overtones.

- Musical intervals is the ratio of the frequencies of two notes.

- Octave is a sound of frequency twice the fundamental frequency. If two musical notes have frequency in the ratio of 2:1, they are said to differ by an octave.

● **Forced vibrations –** Forced vibrations are those vibrations resulting from an external periodic force acting on a system and setting the system vibrating at the same frequency as the external periodic force. A louder sound is heard due to the large area of surface in contact and the large mass of air molecules set into vibration e.g vibration of a table when a vibrating tuning fork is placed in contact with the table top.

- Resonance - Reesonance is a special form of forced vibration at which the frequency of the forced vibration is the same as or coincides with the natural frequency of the body set into vibration. **It is characterised by maximum amplitude of vibration and loudness**.

● **Vibrations in strings –** When a string fixed at both ends is plucked at the middle, **stationary transverse wave** is produced in the string which in turn produces **progressive longitudinal** sound waves in air.

- The strings can be made to vibrate in differents modes/loop which gives different wave forms with nodes *N* and antinodes *A* –



1st position or loop **2nd position or loop**

- The first and simplest mode of vibration is the fundamental frequency or first harmonic. Other higher mode of vibration are all multiples of the fundamental frequency or first harmonic. 1st harmonic=1 loop=f_o, 2nd harmonic=2 loops=1st overtone=$2f_o$, 3rd harmonic=3loops=2nd overtone=$3f_o$ e.t.c.

- Wavelenght λ_n of a n harmonics or loops of a vibrating string is given by : $\boldsymbol{\lambda_n = \frac{2L}{n}}$, n=number of harmonics, L=length of string.

- Factors affecting the frequency of vibrating strings/a sonometer –

1.Length L : Frequency varies inversely with the length of the string hence, frequency decreases as length increases $\boldsymbol{f \propto \frac{1}{L}}$.

2.Tension or force on string T : Frequency varies directly with the square root of tension hence, frequency increases when tension increases $\boldsymbol{f \propto \sqrt{T}}$ or $\boldsymbol{f^2 \propto T}$.

3.Linear density or mass per unit length or thickness μ : Frequency varies inversely with the square root of linear density or mass per unit length $\boldsymbol{f \propto \frac{1}{\sqrt{\mu}}}$, $\boldsymbol{\mu = \frac{m}{L}}$, $\boldsymbol{f \propto \frac{1}{\sqrt{m/L}}}$.

- Combining these relationships it gives –

$\boldsymbol{f \propto \frac{1}{L}\sqrt{\frac{T}{\mu}}}$, $\boldsymbol{f = \frac{k}{L}\sqrt{\frac{T}{\mu}}}$. k=constant which depends on the mode of vibration or number of loops.

- When two strings are of the same length, frequency will be proportional to the square root of the mass , $\boldsymbol{f \propto \frac{1}{\sqrt{m}}}$.

- The frequency f_n of n loops or number of harmonics of a vibrating string is given by – $\boldsymbol{f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$.

- Fundamental frequency f_0 of a string is given by : $\boldsymbol{f_0 = \frac{1}{2L}\sqrt{\frac{T}{\mu}} = \frac{1}{\lambda}\sqrt{\frac{T}{\mu}} = \frac{v}{2L}}$.

- The velocity or speed of the transverse wave set up along the stretched string is – $\boldsymbol{v = \sqrt{\frac{T}{\mu}}}$.

- Audible sound are sound that can be hearable or sound that can be distinctly separated. **Audible sound can be produced can be produced with lower frequencies**. Any factor that lowers frequency, can be used to produce audible sound.

● **Vibrations of pipes –**Vibration of air column in a tube or pipe produces a **stationary and longitudinal waves** along the length of the tube or pipe.It can be open or closed pipes.

- The open end of the tube is an antinode (point of maximum displacement) while the closed end or water surface is a node(point of zero or minimum displacement).

- Open pipes: Pipes with both ends opened. Both open ends of an open pipe is an antinode. It has **all multiple** of frequencies or harmonics just like a vibrating string hence, 1st harmonic=0 overtone=f_o, 2nd harmonic =1st overtone=$2f_o$, 3rd harmonic=2nd overtone=$3f_o$, 4th harmonic=3rd overtone= $4f_o$ e.t.c.

- Frequency $f_n$of nth overtone of open pipeis given by: $\boldsymbol{f_n = (n_o + 1)\frac{v}{2L}}$, n_o=number of overtones, L=length of air column or length of pipe.

-Wavelength λ_n of n harmonics of an open pipe is given by : $\boldsymbol{\lambda_n = \frac{2L}{n}}$, n=number of harmonics.

- Close pipes: Pipes with one end closed and the other end opened.The closed end is a node while the open end is an antinode. It has **odd multiple** of frequencies or harmonics only, 1st harmonic=0 overtone=f_o, 3rd harmonic=1st overtone=$3f_o$, 5th harmonic= 2nd overtone=$5f_o$, 7th harmonic=3rd overtone=$7f_o$ e.t.c.

- The frequency of the longitudinal wave in the air column increases with the length of the air column.

- Frequency f_n of nth overtone of close pipe, is given by: $\boldsymbol{f_n = (2n_o + 1)\frac{v}{4L}}$, n_o=number of overtones, L=length of air column or length of pipe.

Demystified Series by Dr Timothy

- Wavelength λ_n of n harmonics of closed pipe is given by : $\boldsymbol{\lambda_n = \frac{4L}{n}}$, n=number of **odd** harmonics.
- The strength of resonance in both open and close, decreases as the length of the air column increases, $\boldsymbol{R \propto \frac{1}{L_{air}}}$. Thus , **the first position of resonance is stronger and louder than the second position.**
- **Resonance tube experiment:** A resonance tube (a closed pipe) is filled with water and the volume or length of air in the tube can be varied lowering or raising the water level in the tube, $\boldsymbol{v_{air}}$ **or** $\boldsymbol{L_{air} \propto \frac{1}{L_{water}}}$. A tuning fork of frequency f, which vibrates at the top of the tube until a loud sound is heard,indicating the first position of resonance (fundamental note) $\boldsymbol{L_1 = \frac{\lambda}{4}}$. Further increase in the length of air column by lowering the water level produces another loud sound,indicating the second position of resonance (first overtone),

$\boldsymbol{L_2 = \frac{3\lambda}{4}}$. L_1 and L_2 are the length of air column at the first and second resonanace position respectively.
- **The shortest air column which produces resonance is equal to the first resonance length ,** $\boldsymbol{L = \frac{\lambda}{4}}$**.**
- The difference of both position is used to determine the wavelength λ - $\boldsymbol{\lambda = 2(L_2 - L_1)}$**.**
- The speed of sound in air from resonance tube experiment is given by – $\boldsymbol{v = 2f(L_2 - L_1)}$**.**
- **End correction or end effect** $\boldsymbol{e}$ is the small distance at which the antinode extends beyond the edge of the tube. End correction is significat only in closed pipes(resonance tube) as it cancels out at both ends of an open pipe. It is determined by – $\boldsymbol{2e = L_2 - 3L_1}$**.**

NB – The fundamental note of an open pipe is an octave higher than that of a closed pipe of the same length.

- **Musical instruments –** They are classified as:

i.Wind instrument – Air is vibrated in wind instrument e.g flute,organ pipe,trumpet, saxophone,whistle,clarinet trombone and bugle.

ii.Precussion instrument – A hard body is vibrated against another body in percussion instrument e.g bell,cymbal,drum, tuning fork and xylophone.

iii.String instrument – String is vibrated in string instrument e.g piano,harp,guitar,violin and sonometer.

Examples :

1. Which of the following statements about sound is not true? A.Sound is a longitudinal wave B.Sound can be reflected and refracted like light **C.Sound travels through a vacuum** D.Sound moves faster in a steel wire than in air E.Sound can only be produced by vibrating objects.

Sound wave is a mechanical and longitudinal wave. It cannot travel through a vacuum. Sound waves are produced by vibrating body. Sound waves undergo reflection and refraction. Speed of sound in solids e.g steel is greater that its speed in gases or air.

2. The sound of an electric bell dies down slowly when air is slowly pumped out from a bottle becasuse A.sound cannot pass through the centre **B.sound needs a material medium** C.sound can pass through a vacuum D.the wavelength of sound becomes greater in the bottle.

Pumping air out of thr bottle creates a vacuum in it and sound waves cannot travel through a vacuum hence, the sound dies down.

3. The sound from a bell in an enclosed jar gradually faints away while the jar is being evacuated. Which of the following explains this observation? A.the pressure is reduced B.the sound waves are pumped out C.the waves are absorbed in the inner walls **D.there is no more material medium** E.the air is disturbed.

4. Thunder is usually heard some seconds after lightning is seen because A.the human eye is more sensitive to light than the ear to sound B.sound and light travel in different media C.thunder occurs after lightning **D.sound travels more slowly than light** E.sound travels in the form of waves while light dies not.

The speed of light in air is greater than that of sound in air, hence lightning(a form of light) is seen first before thunder(a form of sound) is heard.

5. Which of the following media allow the transmission of sound waves through them? I.air II.liquid III.solids A.I and II B.I and III C.II and III **D.I,II and III.**

Sound waves can travel through solids, liquids and gases(air). Sound waves travels faster in solids, than in liquids and faster in liquids than in gases(air).

6. The speed of sound travelling in various media increases in the order : A.air,mercury, water,iron bar B.air, water, iron, mercury **C.air, water, mercury, iron bar** D.water, mercury, air, iron bar E.water, mercury, iron bar, air.

The increasing order of the speed of sound is : gases–liquids–solid, $v_g < v_L < v_S$. Hence, speed of sound is greatest in solids.

7. Two boys are communicating with each other by steetching a string passing through a hole punched in the bottom of each of two time cans. The physical principle employed is that sound travels A.mainly through air B.fainter in stretched string C.faster through gases than in solids and liquids D.with greater ease through a string than in air E.none of the above is correct.

Demystified Series by Dr Timothy

Sound travels faster with greater ease through a string (a solid) than in air (gas).

8. If a source of sound is moving, a stationary listener will hear a sound of different frequency. This is called A.ultra-sound B.resonance **C.Doppler effect** D.rarefraction E.diffraction.

Doppler effect is the apparent change in the frequency of sound form a source when there is relative motion between the source and an observer.

9. ***X*** ⟵ ***Y*** ***Z***

Suppose a car, travelling at a uniform speed, sounds its horn at point *Y* while towards *X* in the above figure. Assuming there is no wind, which of the following observationas is correct? A.The man at point *X* will receive a lower pitch B.The driver at point *Y* will receive a lower pitch **C.The man at point *Z* will receive a lower pitch** D.The pitch will be the same at points *X* and *Z* E.None of the above.

According to Doppler effect, as relative motion exists between the car i.e the source of sound, and the man, there will be a change in frequency. The frequency decreases as the car is moving away from the man. Hence, the frequency at point *Z* will be lower. The pitch at point *Z* is also lowered as pitch depends on frequency.

10. I.Temperature II.Density of air molecules III.Pressure IV.Pitch. Which of the above will affect the velocity of sound in air? **A.I and II** B.II and IV C.I,II and IV D.I,II,III and IV.

Speed or velocity of sound in air is affected by temperature, density of air molecules, wind and elasticity. Pressure and pitch will not affect speed of sound.

11. Which of the following factors affects speed of sound in air? I.temperature II.pressure III.frequency **A.I** B.II C.I and II D.II and III

Pressure and frequency will not affect the speed of sound.

12. Which of the following statements is not correct? **A.The velocity of sound in air deceases with increase in temperature** B.Pressure has no effect on the velocity of sound in air C.Sound waves require a material medium for transmission D.The velocity of sound in water is greater than that in air E.Sound waves have longer wavelengths than light.

The velocity of aouns in air increase with increase in temperature, $v \propto \sqrt{T}$.

13. Which of the following factors will affect the velocity of sound? A.an increase in the pitch of the sound B.an increase in the loudness of the sound **C.wind travelling in the same direction of the sound** D.change in the atmospheric pressure at constant temperature

Velocity of sound increases with wind travelling in the same direction of the sound.

14. The velocity of sound in air will be doubled if its absolute temperature is A.doubled B.halved C.constant **D.quadrupled.**

Velocity of sound is directly proportional to the square of root of its absolute temperature $v \propto \sqrt{T}$. When v is doubled then, $2 = \sqrt{T}$, $\sqrt{T} = 2$, $T = 2^2 = 4$ i.e four times the original value or quadrupled.

15. The velocity of sound in air at 15°C is 340ms^{-1}. Calculate the velocity at 47°C A.790ms^{-1} B.602ms^{-1} **C.358ms^{-1}** D.322ms^{-1}.

T_1=15°C=15+273=288K, v_1=340ms^{-1},
T_2=47°C=47+273=320K, v_2=?,
$v \propto \sqrt{T}$, $\frac{v_1}{\sqrt{T_1}} = \frac{v_2}{\sqrt{T_2}}$, $\frac{{v_1}^2}{T_1} = \frac{{v_2}^2}{T_2}$,
$\frac{340^2}{288} = \frac{{v_2}^2}{320}$, ${v_2}^2 = \frac{340\times340\times320}{288} = 128444.44$,
$v_2 = \sqrt{128444.44} = 358.39 = 358$ms^{-1}.

16. The velocity of sound in air of 16°C is 340ms^{-1}. What will it be when the pressure is doubled and its temperature rises to 127°C. A.4.000ms^{-1} **B.400ms^{-1}** C.160,000ms^{-1} D.8,000ms^{-1}.

Velocity of sound is independent of pressure hence, doubling the pressure will not affect the velocity.

T_1=16°C=16+273=289K, v_1=340ms^{-1},
T_2=127°C=127+273=400K, v_2=?,
$v \propto \sqrt{T}$, $\frac{v_1}{\sqrt{T_1}} = \frac{v_2}{\sqrt{T_2}}$, $\frac{{v_1}^2}{T_1} = \frac{{v_2}^2}{T_2}$, $\frac{340^2}{289} = \frac{{v_2}^2}{400}$,
${v_2}^2 = \frac{340\times340\times400}{289} = 160000$,
$v_2 = \sqrt{160000} = 400$ms^{-1}.

17. The velocity of sound wave at 27°C is 360ms^{-1}. Its velocity at 127°C is A.120$\sqrt{3}$ms^{-1} B.240ms^{-1} **C.240$\sqrt{3}$ms^{-1}** D.720$\sqrt{3}$ms^{-1}.

T_1=27°C=27+273=300K, v_1=360ms^{-1},
T_2=127°C=127+273=400K, v_2=?,
$v \propto \sqrt{T}$, $\frac{v_1}{\sqrt{T_1}} = \frac{v_2}{\sqrt{T_2}}$, $\frac{{v_1}^2}{T_1} = \frac{{v_2}^2}{T_2}$, $\frac{360^2}{300} = \frac{{v_2}^2}{400}$,
${v_2}^2 = \frac{360\times360\times400}{300} = 172800$,
$v_2 = \sqrt{172800} = 240\sqrt{3}$ms^{-1}.

18. Given that Young's modulus for aluminium is $7.0\times10^{10}N$m^{-2} and density is 2.7×10^3kgm^{-3}, find the speed of the sound produced if a solid bar is struck at one end with a hammer **A.5.1×10^3ms^{-1}** B.4.2×10^3ms^{-1} C.3.6×10^3ms^{-1} D.2.8×10^3ms^{-1}.

Young's modulus E=$7.0\times10^{10}N$m^{-2}, density ρ=2.7×10^3kgm^{-3}, speed of sound v=?, $v = \sqrt{\frac{E}{\rho}}$,
$v = \sqrt{\frac{7.0\times10^{10}}{2.7\times10^3}} = \sqrt{2.59\times10^7} = 5.1\times10^3$ms^{-1}.

19. In a sound wave in air the adjacent rarefractions and compressions are separated by a distance of 17cm. If the velocity of sound wave is 340ms^{-1} determine the frequency? A.10Hz B.20Hz **C.1000Hz** D.5780Hz.

The distance between adjacent compressions and rarefractions or adjacent crests and troughs is equal to half wavelength.

$\frac{\lambda}{2}$=17cm, λ = 17×2=34cm=0.34cm, v=340ms^{-1},

Demystified Series by Dr Timothy

$v = \lambda f$, $f = \frac{v}{\lambda} = \frac{340}{0.34} = 1000$Hz.

20. A source of sound produces waves in air of wavelength 1.65m. If the speed of sound in air is 330ms^{-1}, the period of vibration in second is A.200s B.2.0s C.0.5s D.0.02s **E.0.005s**

λ=1.65m, v=330ms^{-1}, T=?, $v = \frac{\lambda}{T}$, $T = \frac{\lambda}{v}$,

$T = \frac{1.65}{330} = 0.005$s.

21. Calculate the wavelength of a note which is **one octave lower** than a note of 512.0Hz in a certain medium in which the speed of sound is 384.0ms^{-1} A.0.375m B.0.670m **C.1.500m** D.2.670m E.6.000m.

Frequency of the note f= $\frac{512}{2}$ = 256Hz i.e one octave lower than 512.0Hz, v=384.0ms^{-1}, λ=?, $v = \lambda f$, $\lambda = \frac{v}{f} = \frac{384}{256} = 1.500$Hz.

22. The speed of sound in air is 330ms^{-1}. How far from the centre of a storm is an observer who hears a thunderclap 2s after lightning flash? (Neglect the time taken by light to travel to the observer) A.1320m **B.660m** C.600m D.330m E.165m.

v=330ms^{-1}, time interval between lightning flash and thunderclap t=2s, distance of the observer from centre of stor s=?, $s = vt = 330\times2 = 660$m.

23. The speed of sound in air at sea-level is 340ms^{-1} while that of light is 300,000kms^{-1}. How far(to the nearest metre) from the centre of a thunder storm is an observer who hears a thunder 2s after a lighning flash? A.170m B.340m C.600m **D.680m.**

v=340ms^{-1}, t=2s, s=?, $s = vt = 340\times2 = 680$m.

24. The physical property of sound waves can best be described by **A.Reflection and diffraction** B.Polarization and reflection C.Polarization and diffraction D.Polarization and refraction.

Sound waves are longitudinal waves, hence cannot be polarized.

25. Which of the following sets of characteristics is/are associated with sound waves? A.Reflection,refraction and resonance B.Refraction,interference and resonance C.Interference,diffraction and resonance D.Diffraction,reflection and resosnace **E.All of the above.**

26. Which of these statements is/are correct? I.Echoes are produced by the reflection of sound on a hard surface II.Refraction of a wave is caused by a change in the velocity of the wave III.Diffraction can be observed when a wave passed through a large slit A.I B.II C.III **D.I and II** E.II and III .

Echoes are produced when sound waves are reflected by a plane surface. Refraction of wave is caused by a change in direction of wavefront, velocity and wavelength. Diffraction will not occur when a wave is passed thorugh a large slit.

27. Which of the following characteristics of a wave is used in the measurement of the depth of the sea? A.diffraction B.intefernce C.refraction **D.reflection.**

The depth of a sea can be determined by echo, which is reflected sound waves.

28. Which of the following statements about echoes is not correct? They can be used A.in underwater exploration for oil B.to measure the speed of sound in air **C.to differentiate between transverse and longitudinal waves** D.to locate shoals in fish E.to determine the depth of the sea bed.

Polarization is used to differentiate between transverse and longitudinal waves not echoes.

29. Which onr of the following is not a characteristic of a transverse wave? **A.Pitch** B.Reflection C.Refraction D.Interference E.Wavelength.

Pitch is perculiar to sound waves which are longitudinal waves.

30. The pheneomenon that makes sound persist when its source has been removed is known as A.rarefraction B.acoustic vibration **C.reverberation** D.echo.

Reverberation is the persistence of sound waves due to multiple echoes or reflection of sound.

31. During a thunderstorm, the sound is heard over a long time. This phenomenon is referred to as A.refraction of sound **B.reverberation** C.superposition D.diffraction of sound.

32. In large telecommunications auditorium, perforated absorbent materials are used to line the ceiling so as to **A.reduce the reverberation of sound in the hall** B.reduce the height of the ceiling from the floor C.increase the reverberation of sound in the hall D.increase the amount of echo in the hall.

The auditorium is perforated to absorbs sound and reduce reflection of sound, thus reducing reverberation.

33. The walls of a sound-recording studio are usually lined with thick felt so that the sound waves can be A.evenly distributed B.easily reflected **C.absorbed to prevent reflection** D.reflected to prevent absorption E.rasily refracted in the whole room.

34. A gun is fired in a ship some distance from a cliff. The echo is heard 0.500s later in the ship..What is the distance of the ship from the cliff? (Velocity of sound in air=330ms^{-1}) A.66.0m **B.82.5m** C.165.0m D.329.5m E.330.5m.

t=0.500s, v=330ms^{-1}, x=?, for echo or reflection of wave, $v = \frac{2x}{t}$, $x = \frac{vt}{2}$, $x = \frac{330\times0.5}{2} = 82.5$m.

35. A herdsman yelling out to fellow herdsman heard his voice reflected by a cliff 4s later. What is the velocity of sound in air if the cliff is 680m away? A.170ms^{-1} B.136ms^{-1} **C.340ms^{-1}** C.680ms^{-1} E.510ms^{-1}.

Demystified Series by Dr. Timothy

t=4s, x=680m, v=?, $v=\frac{2x}{t}$, $v=\frac{2\times680}{4}=340\text{ms}^{-1}$.

36. In order to find depth of the sea. A ship sends out a sound wave and receives an ehco after one second. If the velocity of sound in water is 1500ms^{-1}, what is the depth of the sea? **A.0.75km** B.1.50km C.2.20km D.3.00km E.3.75km.

t=1s, v=1500ms^{-1}, x=?, $v=\frac{2x}{t}$, $x=\frac{vt}{2}$,

$x=\frac{1500\times1}{2}=750\text{m}=0.75\text{km}$.

37. The human ear can only distinguish between sound if they occur at least 0.10s apart. If the velocity of sound in air is 340ms^{-1}, calculate the least distance from a reflecting surface at which a listener can detect an echo A.3400m B.170m C.68m D.34m **E.17m.**

t=0.10s, v=340ms^{-1}, x=?, $v=\frac{2x}{t}$, $x=\frac{vt}{2}$,

$x=\frac{340\times0.10}{2}=17\text{m}$.

38. A fathometer is used to send a wave down to the sea bed. The reflected wave is received 0.5seconds. Calculate the depth of the sea (Speed of sound in water=1500ms^{-1}) **A.375m** B.390m C.400m D.450m E.500m.

t=0.5s, v=15000ms^{-1}, $v=\frac{2x}{t}$, $x=\frac{vt}{2}$,

$x=\frac{1500\times0.5}{2}=375\text{m}$.

39. A sound wave is produced from a source and an echo is heard t seconds after wards. If d ia the distance of the reflecting surface from the source, v the speed, λ the wavelength and T the period of the wave then A.$d=\frac{2t}{\lambda}$ **B.$d=\frac{\lambda t}{2T}$** C.$d=\frac{2T}{\lambda t}$ D.$d=\frac{\lambda T}{2t}$.

$v=\frac{2x}{t}=\frac{\lambda}{T}$, x=d, $\frac{2d}{t}=\frac{\lambda}{T}$, $d=\frac{\lambda t}{2T}$.

40. A man clapping his hands at regular intervals observes that the echo of a clap coincides with the next clap. If the reflecting cliff is 160m away and the speed of sound is 320ms^{-1}, what is the frequency of the clapping? A.1Hz B.2Hz C.4Hz D.8Hz

As the echo of a clap coincides with the next clap, the time taken is equal to the period.

x=160m, v=320ms^{-1}, $v=\frac{2x}{t}=\frac{2x}{T}$, $t=T$,

$T=\frac{2x}{v}$, $T=\frac{1}{f}$, $\frac{1}{f}=\frac{2x}{v}$, $f=\frac{v}{2x}=\frac{320}{2\times160}=1\text{Hz}$.

41. A man hears his echo from a nearby hill 2s after he shouted. If the frequency of his voice is 260Hz and the wavelength of his voice is 1.29m, how far is the hill? A.330.0m **B.335.4m** C.660.0m D.670.8m.

t=2s, f=260Hz, λ=1.29m, x=?, $v=\frac{2x}{t}=\lambda f$,

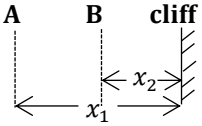
$x=\frac{\lambda ft}{2}=\frac{1.29\times260\times2}{2}=335.4\text{m}$.

42. A car B moves towards a stationary car A. If B produces an ultrasonic sound at a point and it takes 5.6×10^{-3}s for a beep to be heard in B, calculate the distance between the two cars at that instant (speed of sound in air= 340ms^{-1}) A.0.119m B.0.238m C.0.476m **D.0.952m**

The beep heard in car B is the echo of the ultrasonic sound of car B after reflection by car A. t=5.6×10^{-3}s, v=340ms^{-1}, distance between the two cars x=?, $v=\frac{2x}{t}$, $x=\frac{vt}{2}$,

$x=\frac{340\times5.6\times10^{-3}}{2}=0.952\text{m}$.

43. A ship travelling towards a cliff receives the echo of its whistle after 3.5s. A short while later, it receives the echo after 2.5s. if the speed of sound in air under the prevailing conditions is 250ms^{-1}, how much closer is the ship to the cliff? A.10m **B.125m** C.175m D.350m E.1000m.

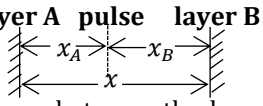


x_1=initial distance of the ship from the cliff, x_2=final distance of the ship from the cliff ,

t_1=3.5s, t_2=2.5s, v=250ms^{-1}, "how much closer is the ship to the cliff" is the difference between the initial and final distance of the ship from the cliff, $x_1=\frac{vt_1}{2}$, $x_2=\frac{vt_2}{2}$, The ship is closer to the cliff by - $x=x_1-x_2$, $x=\frac{vt_1}{2}-\frac{vt_2}{2}=\frac{v}{2}(t_1-t_2)$,

$x=\frac{250}{2}(3.5-2.5)=125\times1=125\text{m}$.

44. A sound pulse sent vertically downwards into the earth is reflected from two different layers of the earth such that echoes are heard after 1.2s and 1.4s. Assuming the speed of the pulse is 2000ms^{-1}, calculate the distance between the layers A.200m B.400m C.2400m **D.2600m** E.2800m.



distance between the layer A and the pulse x_A, time taken to hear echo after reflection at layer A t_A=1.2s, distance between the layer B and the pulse x_B, time taken to hear echo after relfection at layer B t_B=1.4s, distance between layer A and B x=x_A+x_B=?, v=2000ms^{-1},

$x_A=\frac{vt_A}{2}=\frac{2000\times1.2}{2}=1200\text{m}$,

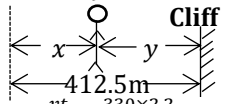
$x_B=\frac{vt_B}{2}=\frac{2000\times1.4}{2}=1400\text{m}$, $x=x_A+x_B$,

$x=1200+1400=2600\text{m}$. Alternatively ,

Total time taken for pulse to travel between both layers t=t_A+t_B=1.2+1.4=2.6s, $x=\frac{vt}{2}$,

$x=\frac{2000\times2.6}{2}=2600\text{m}$.

45. A hunter 412.5m away from a cliff, moves a distance x towards the cliff and fires a gun. He hears the echo from the cliff after 2.2seconds. Calculate the value of x (speed of sound in air 330ms^{-1}) **A.49.5m** B.99.0m C.363.0m D.412.5m



y=new distance btw the hunter and cliff.

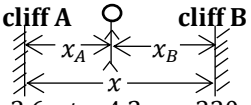
t=2.2s, v=330ms^{-1},

$y=\frac{vt}{2}=\frac{330\times2.2}{2}=363\text{m}$, $x+y=412.5\text{m}$,

$x=412.5-y=412.5-363=49.5\text{m}$.

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46. A man standing between two cliffs, claps his hands and hears two echoes after 3.6s and 4.2s respectively. If the speed of sound in air is 330ms^{-1} calculate the distance between the cliffs A.594m B.693m **C.1287m** D.2574m.



t_A=3.6s, t_B=4.2s, v=330ms^{-1}, x=?,

$x_A = \frac{vt_A}{2} = \frac{330\times3.6}{2} = 594$m,

$x_B = \frac{vt_B}{2} = \frac{330\times4.2}{2} = 693$m, $x = x_A + x_B$,

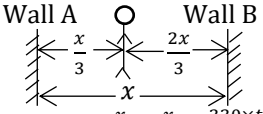
$x = 594+693 = 1287$m. Alternatively ,

$t=t_A + t_B$=3.6+4.2=7.8s , $x = \frac{vt}{2}$,

$x = \frac{330\times7.8}{2} = 1287$m.

47. A man stands one-third of the way between two walls and claps his hands.He heard two distinct echoes 1s apart.If the speed of sound is 330ms^{-1},the distance between the walls is A.33m B.110m C.220m D.330m **E.495m.**

The two distinct echoes heard are due to a distance of $\frac{x}{3}$ from one wall and $\frac{2x}{3}$ from the other wall.



For distance $\frac{x}{3}$; $\frac{x}{3} = \frac{330\times t_1}{2}$, $t_1 = \frac{2x}{990}$,

For distance $\frac{2x}{3}$; $\frac{2x}{3} = \frac{330\times t_2}{2}$, $t_2 = \frac{4x}{990}$,

Time interval between echoes= $t_2 - t_1$ =1s ,

$\frac{4x}{990} - \frac{2x}{990} = 1$, $\frac{4x-2x}{990} = 1$, $\frac{2x}{990} = 1$, $x = \frac{990}{2}$,

$x = 495$m.

48. When sound wave passes from air into water its A.speed and frequency increase but its wavelength remains constant **B.speed and wavelength increase but its frequency remains the same** C.speed decreases D.speed remains the same but its frequency and wavelength change E.speed increases but its frequency and wavelength decrease

Speed of sound in water is greater than its speed in air, hence when sound waves passes air into water, its speed increases. During refraction frequency is constant and speed is directly proportional to wavelength, $v \propto \lambda$. Hence, the wavelength also increases.

49. Which of the following is true of sound? A.sound travel faster in air at 20°C than at 30°C B.the frequency of a given sound wave changes when it crosses the boundary separating two media C.the wavelength of a given sound wave in air decreases as the temperature increased D.sound waves can not be reflected.

None of the options are correct.

Speed of sound increases with absolute temperature $v \propto \sqrt{T}$, hence sound travels faster at higher temperature i.e 30°C than at low temperature i.e 20°C. Wavelength increases with the velocity $v \propto \lambda$, hence wavelength will also increase with temperature $\lambda \propto \sqrt{T}$. Frequency is constant during refraction, as it crosses the boundary speparating two media. All waves including sound waves can be reflected.

50. If a sound wave goes from a cold-air region to a hot air region, its wavelength **A.increase** B.decrease C.decrease then increase D.remain constant .

Wavelength increases with temperature $\lambda \propto \sqrt{T}$, hence when sound waves goes from cold-air region to hot-air region i.e increase in temperature, its wavelength increases.

51. The main difference between a musical note and a noise is that A.the amplitude of the noise is geater than that of the musical note **B.the frequency of musical note is regular while that of noise is irregular** C.both the amplitude and the frequency of noise is greater than those of a musical noise D.the frequency of noise is higher than that of a musical note E.the wavelength of a musical note is irregular while that of a noise is regular.

Musical note is a combination of sound of regular frequency while noise is a combination of sound of irregular frequency.

52. Which of the following is/are characteristic of sound? I.pitch II.loudness III.quality IV.noise A.I B.II C.I and II **D.I,II and III** E.I,II,III and IV.

Pitch,intensity,loudness and quality are all characteristics of sound.

53. The pitch of a sound note depends on A.timbre B.harmonics C.quality **D.frequency**

Pitch of a note depends on frequency.

54. The pitch of an acoustic device can be increased by **A.increasing the frequency** B.increasing the amplitude C.decreasing the loudness D.decreasing the intensity

Pitch of a note increases with frequency.

55. What changes(s) is/are observed in the sound produced by a siren when the speed of revolution is increased? I.The pitch becomes higher II.The wavelength increases III.The velocity remains constant **A.I** B.II C.III D.I and II E.I and III .

The frequency of a disc siren $f = nv$, v is the speed in revolutions per second, hence the frequency is directly proportional to the speed of revolution $f \propto v$. Increasing the speed of revolution increases the frequency and thus, the pitch becomes higher, as pitch depends on frequency.

56. The characteristic of a vibration that determines its intensity is the A.frequency B.overtone C.wavelength **D.amplitude**

Intensity depends on amplitude i.e intensity is proportional to the square of the amplitude of a vibrating body $I \propto A^2$.

57. Loudness of a musical note depends on its

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A.overtones B.frequency **C.amplitude** D.fundamental notes E.harmonics.
Loudness of a musical note depends on its amplitude i.e loudness is proportional to the square of the amplitude $L \propto A^2$.

58. Which of the following is true of the loudness of sound? It **A.depends on the square of the amplitude of the vibrating body** B.is proportional to the distance of the observer from the source of the sound C.is greater in vacuum D.is independent of frequency.
Loudness depends on intensity and square of the amplitude of the vibrating body. Loudness and intensity is inversely proportional to the square of the distance from a source $L \propto \frac{1}{r^2}$. Loudness of sound is lowest in a vacuum as there is no material medium e.g air, for sound to travel, as sound is a mechanical wave.

59. If the distance from a point source of sound is doubled, by what factor does the intensity decreases? A.4.00 B.2.00 C.0.50 **D.0.25.**
Intensity of sound I varies inversely with the square distance r from the source $I \propto \frac{1}{r^2}$.
$I_1 {r_1}^2 = I_2 {r_2}^2$, I_1=I, r_1=r, I_2=?, r_2=$2r$,
$I \times r^2 = I_2 \times (2r)^2$, $Ir^2 = I_2 \times 4r^2$,
$I_2 = \frac{Ir^2}{4r^2} = \frac{I}{4}$ or $0.25I$ i.e the intensity decreases by one-fourth of its original value or by a factor of 0.25. Alternatively :
$I \propto \frac{1}{r^2}$, distance r is doubled , $I = \frac{1}{2^2} = \frac{1}{4} = 0.25$.

60. If the same note is sounded on a piano, an organ or any other musical instrument,the source of sound can be recognized because of the characteristic of sound called A.pitch B.frequency **C.quality** D.intensity E.wavelenght
Quality/timbre/tone is a characteristic note of a musical instrument which differentiates it from another note of the same pitch or frequency and loudness produced by another instrument.

61. Notes of the same frequency sounded on a piano and a flute may differ in A.loudness II.quality III.pitch A.I **B.II** C.I and II D.II and III E.I,II and III.

62. The quality(timbre) of sound depends on A,amplitude B.frequency **C.harmonics** D.velocity E.wavelength.
The quality(timbre) of sound depends on overtones or harmonics.

63. What determines the quality of a note produced by a vibrating string? A.The pitch of the note B.The tension of the string C.The material of the string **D.The presence of overtones** E.The thickness of the string.

64. Which of the following is not correct? A.The pitch of a note depends on the frequency of a vibrating source **B.The speed of the wind has no effect on the speed of sound in air** C.The overtones present in a note determines its quality E.The velocity of sound in air does not depend on the intensity produing it.
The speed of the wind affects the speed of sound in air i.e speed of sound increases when wind travels in its direction.

65. Which of the following is not correct? I.The pitch of a sound note depends on the frequency of vibrations II.The intensity of a sound note is proportional to the amplitude of vibrations III.Beats are produced by two sources sound because one wave is travelling faster than the other IV.When two sources of sound of frequencies 500Hz and 502Hz are sounded together, a beat frequency of 2Hz is observed V.The first harmonic of a note has double the frequency of the fundamental note A.I and II **B.II and III** C.I and III D.III and IV E.IV and V.
Pitch depends on frequency of vibration. Intensity is proportional to the square of the amplitude. Beats are produced when two sources of nearly equal frequency are sounded together. Beat frequency is the difference between the frequency of the two sources, $f_B = 502 - 500 =$ 2Hz. The first harmonic of a note is same as the fundamental frequency of the note. Hence, statement "II,III and V" are not correct.

66. All of the following frequencies are overtones of 320Hz except **A.520Hz** B.640Hz C.960Hz D.1280Hz E.1600Hz.
Overtones are multiples of frequency above the fundamental frequency. Hence overtones of 320Hz are multiples of frequency above it.
1st overtone $f_1 = 2 \times 320 = 640$Hz,
2nd overtone $f_2 = 3 \times 320 = 960$Hz,
3rd overtone $f_3 = 4 \times 320 = 1280$Hz,
4th overtone $f_4 = 5 \times 320 = 1600$Hz.
520Hz is not a multiple of 320Hz, hence is not an overtone of 320Hz.

67. A note is the octave of another note if it has **A.a frequency twice that of the first note** B.a frequency half of the first note C.the same fundamental frequency D.a frequency eight times that of the first note.
An octave is a sound of frequency twice its fundamental frequency of frequency of its first note.

68. A slight loading of a tuning fork has the effect of A.decreasing its amplitude B.increasing its amplitude **C.decreasing its frequency** D.increasing its frequency.
Loading e.g by placing a piece of plasticine on it or adding gums to the prong, so as to increase the mass, decreases the frequency while the amplitude remains constant. The frequency of the unloaded tuning fork is greater than the frequency of loaded tuning fork.

69. Two sound waves have frequencies of 12Hz and 10Hz. Calculate their beat period **A.0.5s** B.1.0s C.1.2s D.2.0s.

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f_1=12Hz, f_2=10Hz, beat period T_B=?,
Beat frequency $f_B = f_1 - f_2 = 12-10 = 2$Hz or 4 beats per second. Beat period $T_B = \frac{1}{f_B} = \frac{1}{2} = 0.5$s.

70. Of two identical tuning forks with natural frequency 256Hz,one is loaded so that 4 beats per second are heard when they are sounded together.What is the frequency of the loaded tuning fork? A.260Hz **B.252Hz** C.248Hz D.264Hz E.258Hz.
Loading a tuning decreases its frequency. The loaded tuning fork has a lower frequency than the unloaded tuning fork. Frequency of unloaded tuning fork f_1=256Hz, beat frequency f_B=4 beats per second or 4Hz, frequency of loaded tuning fork f_2=?, $f_B = f_1 - f_2$, $4 = 256 - f_2$,
$f_2 = 256-4 = 252$Hz.

71. A sounding tuning fork is brought near the end of a pipe containing air column and the loudness of the sound increases. This effect is due to A.diffraction B.echo **C.resonance** D.interference E.reverberation.
Resonance occurs when the natural frequency of a vibrating body coincides or equals the frequency of the external periodic force causing it to vibrate.It is characterized by a increase in loudness and amplitude.

72. When the bottom tip of vibrating tuning fork is held in contact with a wooden box, a louder sound is heard. This phenomenon is known as A.beats B.echoing **C.resonance** D.reverberation.

73. Marching solidiers crossing a suspension bridge are usually advised to break their steps to avoid damaging the bridge owing to A.swinging B.vibration C.oscillation **D.resonance.**
When the soldiers match in step, the bridge is forced to vibrate in resonance at a frequency same as the natural frequency of the soldiers, which thereby causes the bridge to collapse.

74. Which of the following statements is correct about resonance? Resonance occurs when a body A.vibrates with its natural frequency **B.is forced to vibrates at its own natural frequency by another body vibrating at the same frequency** C.vibrates at the frequency of the body causing vibration D.vibrate with an irregular frequency.
Option A and C describes free and forced vibration respectively.

75. Which of the diagrams below represents the second overtone of a vibrating string fixed at both ends? A. *N A N* B. *N A N A N*
C. *N A N A N A N* D. *N A N A N A N A N*
Second overtone is same as third harmonic or third position or three loops,as indicated by the digram in option C. Option A indicates the fundamental or first harmonic i.e one loop, option B indicates the first overtone or second harmonic i.e 2 loops, option D indicates third overtone or fourth harmonic i.e four loops.

76. A guitar string is 75cm long. The wavelength of its fundamental note is A.75cm **B.150cm** C.37.5cm D.112.5cm E.50cm .
Length of the string L=75cm, number of harmonics n=1 i.e fundamental note, wavelength of a fundamental note λ_1=?, $\lambda_n = \frac{2L}{n}$,
$\lambda_1 = \frac{2\times75}{1} = 150$cm.

77. A string is fastened tightly between two walls 24cm apart. The wavelength of the second overtone is A.24cm **B.16cm** C.12cm D.8cm
Distance between the two walls=Length of the string L=24cm, number of harmonics n=3 i.e 2nd overtone or 3rd harmonic, wavelength of the 2nd overtone λ_3=?, $\lambda_n = \frac{2L}{n}$, $\lambda_3 = \frac{2\times24}{3} = 16$cm.

78. If a sonometer has fundamental frequency of 450Hz, what is the frequency of the fifth overtone? **A.2700Hz** B.456Hz C.444Hz D.75Hz.
A sonometer is an instrument used for the experimentation of factors affecting the frequency of a vibrating string. Fundamental frequency f_o=450Hz, fifth overtone f_5 =sixth harmonics= $6f_o = 6\times450 = 2700$Hz.

79. The lowest note emitted by a stretched string has a frequency of 40Hz. How many overtones are there between 40Hz and 150Hz? A.4 B.3 **C.2** D.1.
Frequency of lowest note=fundamental frequency f_o=40Hz, 1st overtone $f_1 = 2f_o = 2\times40 = 80$Hz, 2nd overtone $f_2 = 3f_o = 3\times40 = 120$Hz, 3rd overtone $f_3 = 4f_o = 4\times40$,
$= 160$Hz. Hence, the number of overtones between 40Hz and 150Hz are 2 i.e 80Hz and 120Hz.

80. A steel wire of length 0.5m is stretched between two fixed points and its fundamental frequency is 200Hz. The speed of the wave in the wire is A.100ms^{-1} B.120ms^{-1} **C.200ms^{-1}** D.250ms^{-1}.
L=0.5m, f_o=200Hz, $\lambda_n = \frac{2L}{n}$, n=1, $\lambda_1 = \frac{2L}{1}$,
λ_1=2L, $v = \lambda f = 2Lf_o = 2\times0.5\times200 = 200$ms^{-1}.

81. Find the frequencies of the first three harmonics of a piano string of length 1.5m, if the velocity of the waves on the string is 120ms^{-1} **A.40Hz,80Hz,120Hz** B.80Hz,160Hz, 240Hz C.180Hz,360Hz,540Hz D.360Hz,180Hz, 90Hz.
L=1.5m, v=120ms^{-1}, $\lambda_n = \frac{2L}{n}$, n=1, $\lambda_1 = \frac{2L}{1}$,
λ_1=2L, $v = \lambda f$, $f_o = \frac{v}{\lambda} = \frac{v}{2L} = \frac{120}{2\times1.5} = 40$Hz,
1st harmonic f_o = 40Hz, 2nd harmonic $2f_o$ = 2×40 = 80Hz, 3rd harmonic $3f_o$ = 3×40 = 120Hz.

82. What factors determined the frequency of a note emitted by a vibrating string? A.Amplitude of vibration,force constant of string and length of string B.Amplitude of vibration,force constant of string and tension in string **C.Mass per unit length of string, tension in string and length**

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of string D.Force constant of string,tension in string and length of string.
The frequency of a vibrating string is determined by its length, linear density or mass per unit length and tension or force in the string.

83. Which of the following statements is true? The frequency of a vibration is A.proportional to its length B.inversely propotional to the tension within it C.proportional to the amplitude of vibration D.inversely propotional to its cross section **E.proportional to the square root of the tension within it.**
Frequency of a vibrating string is proportional to the reciprocal of its length,square of the tension within it and square root of the reciprocal of its linear density or mass per unit length , $f \propto \frac{1}{L}\sqrt{\frac{T}{\mu}}$.

84. Under which of the following conditions will a stringed instrument produce a high frequency note. When the string is A.long,thick and loose B.long,thin and loose C.long,thin and taut D.short,thick and taut E.short,thin and taut
$f \propto \frac{1}{L}\sqrt{\frac{T}{\mu}}$ hence, decrease in length i.e short string, decrease in thickness or linear density i.e thin string and increase in tension i.e tout, will give a note with high frequency or pitch.

85. If the load at the end of a sonometer wire is immersed in a bucket of water,the original fundamental frequency of the wire could be restored by **A.decreasing the length of the wire** B.increasing the length of the wire C.increasing the mass per unit length of the wire D.changing the temperature of water
Immersion of the load in water, decreases the weight of the load due to upthrust. Hence, weight or tension T is reduced.
$f \propto \frac{1}{L} \propto \sqrt{\frac{T}{\mu}}$, $f^2 \propto \frac{1}{L^2} \propto \frac{T}{\mu}$, $f^2 \propto \frac{T}{L^2\mu}$,
$T \propto f^2L^2\mu$, $T \propto L^2\mu$, where f=constant, as original fundamental frequency is to be restored.
$T \propto L^2\mu$or $T \propto L^2 \propto \mu$ i.e Tension T is directly proportional to square of the length L and directly proportional to the linear density or mass per unit length μ.
Decreasing the weight or tension of the load , will cause a corresponding decrease in the length of the wire and decrease in linear density or mass per unit length.

86. When the length of a vibrating string is doubled, its frequency of vibration A.remains the same **B.becomes half the former value** D.becomes double former value E.becomes four times the former value
Frequency is inversely proportional to the length of a string, $f \propto \frac{1}{L}$. Hence, when the length L is doubled, $f = \frac{1}{2}$ i.e frequency is reduced by half or becomes half the former value.

87. Under constant tention and constant mass per unit length, the note produced by a plucked string is 500Hz when the length of the string is 0.90m. At what length is the frequency 150Hz? **A.3m** B.4m C.5m D.6m.
f_1=500Hz, L_1=0.90m, f_2=150Hz, L_2=?,
$f \propto \frac{1}{L}$ at constant T and μ , $f_1L_1 = f_2L_2$, $500\times0.9 = 150\times L_2$, $L_2 = \frac{500\times0.9}{150} = 3$m.

88. When the tension in a vibrating string is increased, its frequency of vibration **A.increases as the square root of the tension** B.decreases as the square root of the tension C.increases as the square of the tension D.decreases as the square of the tension E.is directly proportional to the tension.
$f \propto \sqrt{T}$, hence frequency is directly proportional to the square root of tension.

89. When the tension in a sonometer wire is doubled the ratio the new frequency to its initial frequency is A.$1/\sqrt{2}$ B.$1/2$ C.$\sqrt{2}$ D.2
T_1=T, T_2=$2T$, f_1:f_2=?, $f \propto \sqrt{T}$, $\frac{f_1}{\sqrt{T_1}} = \frac{f_2}{\sqrt{T_2}}$,
$\frac{f_2}{f_1} = \frac{\sqrt{T_2}}{\sqrt{T_1}} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{2T}{T}} = \sqrt{\frac{2}{1}} = \sqrt{2}$:1 or $\sqrt{2}$.
Alternatively, $f \propto \sqrt{T}$, when tension T is doubled, then $f = \sqrt{2}$ i.e frequency is increases by a factor of $\sqrt{2}$ or the ratio of its final to initial frequency is $\sqrt{2}$ or $\sqrt{2}$:1.

90. The fundamental frequency of a plucked wire under a tension of 400N is 250Hz. When the frequency is changed to 500Hz at constant length, the tension is A.40N B.160N C.400N **D.1600N.**
f_1=250Hz, T_1=400N, f_2=500Hz, T_2=?,
$f \propto \sqrt{T}$, $\frac{f_1}{\sqrt{T_1}} = \frac{f_2}{\sqrt{T_2}}$ or $\frac{{f_1}^2}{T_1} = \frac{{f_2}^2}{T_2}$,
$\frac{250^2}{400} = \frac{500^2}{T_2}$, $T_2 = \frac{500\times500\times400}{250\times250} = 1600N$.

91. The note produced by a stretched string has a fundamental frequency of 400Hz. If the length of the string is doubled while the tension in the string is increased by a factor of 4,the frequency is A.200Hz **B.400Hz** C.800Hz D.1600Hz
$f \propto \frac{1}{L}\sqrt{\frac{T}{\mu}}$, $f \propto \frac{1}{L}\sqrt{T}$ i.e when mass per unit length or linear density μ is constant. Thus, when the length L is doubled while the tension T increased by a factor of 4 , the frequency $f = \frac{1}{2}\sqrt{4} = \frac{1}{2}\times 2 = 1$.The ratio of the final to initial frequency f_2:f_1 = 1:1 i.e the frequency is constant or unaltered. The new frequency f_2 =400Hz.

92. Which of the following actions is necessary in order to tune a string to produce a note of an octave higher than its fundamental? A.Doubling the length of the string B.Reducing the tension by half C.Reducing the mass per unit length **D.increasing the tension four times**

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Doubling the length of the string decreases the frequency by half i.e $f \propto \frac{1}{L}$, $f = \frac{1}{2}$.
Reducing the tension by half decreases the frequency by a factor of $\sqrt{\frac{1}{2}}$, $f \propto \sqrt{T}$, $f = \sqrt{\frac{1}{2}}$
Reducing the mass per unit length increases the frequency $f \propto \frac{1}{\mu}$.
Increasing the tension four times increases the frequency by a factor of 2 or the frequency is twice the fundamental i.e an octave higher than its fundamental, $f \propto \sqrt{T}$,
$f = \sqrt{4} = 2$ or $2f_o$.

93. Two strings of the same length and under the same tension give notes of frequency in the ratio 4:1. The masses of the strings are in the corresponding ratio of A.2:1 B.1:2 C.1:4 **D.1:16**

$f \propto \frac{1}{L}\sqrt{\frac{T}{\mu}}$ or $f \propto \frac{1}{L}\sqrt{\frac{T}{m/L}}$, thus, $f \propto \sqrt{\frac{1}{m}}$ or $f \propto \frac{1}{\sqrt{m}}$, when length L and tension T are constant, f_1:f_2=4:1, m_1:m_2=?,

$f_1\sqrt{m_1} = f_2\sqrt{m_2}$, $\frac{f_1}{f_2} = \frac{\sqrt{m_2}}{\sqrt{m_1}} = \sqrt{\frac{m_2}{m_1}}$, $\frac{4}{1} = \sqrt{\frac{m_2}{m_1}}$,

$\frac{4^2}{1^2} = \frac{m_2}{m_1}$, $\frac{m_2}{m_1} = \frac{16}{1}$, $\frac{m_1}{m_2} = \frac{1}{16}$,

m_1:$m_2 = 1{:}16$. Alternatively,

$f \propto \frac{1}{\sqrt{m}}$, $\sqrt{m} \propto \frac{1}{f}$ or $m \propto \frac{1}{f^2}$, as frequency is in the ratio of 4:1 or $\frac{4}{1}$ = 4, hence, $m = \frac{1}{4^2} = \frac{1}{16}$

or m_1:$m_2 = 1{:}16$.

94. A piano wire 0.5m long has a total mass of 0.01kg and is stretchyed with a tension of $800N$. Calculate the frequency of the wire when it sounds its fundamental note **A.200Hz** B.100Hz C.4Hz D.2Hz

L=0.5m, m=0.01kg, T=$800N$, f_o=?, $f_o = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

$f_o = \frac{1}{2L}\sqrt{\frac{T}{m/L}} = \frac{1}{2L}\sqrt{\frac{TL}{m}} = \frac{1}{2\times0.5}\sqrt{\frac{800\times0.5}{0.01}}$,

$f_o = \sqrt{\frac{400}{0.01}} = \sqrt{40000} = 200$Hz.

95. A string of length 1.0m vibrate in 10 loops. If the total mass of the string is 1.0×10^{-3}kg and the tension in it is 10N,calculate the frequency of the vibration A.50HzB.100Hz C.250Hz D.400Hz **E.500Hz.**

number of loops n=10 loops, L=1m, T=$10N$, frequency of vibration f_n= f_{10},

$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}} = \frac{n}{2L}\sqrt{\frac{T}{m/L}} = \frac{n}{2L}\sqrt{\frac{TL}{m}}$,

$f_{10} = \frac{10}{2L}\sqrt{\frac{TL}{m}} = \frac{10}{2\times1}\sqrt{\frac{10\times1}{1\times10^{-3}}} = 5\sqrt{\frac{10}{10^{-3}}}$,

$f_{10} = 5\sqrt{10\times10^3} = 5\sqrt{10^4} = 5\times10^2 = 500$Hz.

Alternatively, $f_o = \frac{1}{2L}\sqrt{\frac{T}{\mu}} = \frac{1}{2L}\sqrt{\frac{T}{m/L}} = \frac{1}{2L}\sqrt{\frac{TL}{m}}$

$f_o = \frac{1}{2\times1}\sqrt{\frac{10\times1}{1\times10^{-3}}} = 0.5\sqrt{\frac{10}{10^{-3}}} = 0.5\sqrt{10\times10^3}$

$f_o = 0.5\sqrt{10^4} = 0.5\times10^2 = 50$Hz,

$f_{10} = 10f_o = 10\times50 = 500$Hz.

96. A sonometer wire of linear density 0.08kgm^{-1} subjected to a tension of $800N$ is plucked. Calculate the speed of a pulase which moves from one end of the wire to the other A.179.0ms^{-1} B.100.0ms^{-1} C.15.8ms^{-1} D.8.0ms^{-1}

μ=0.08kgms^{-1}, T=$800N$, v=?, $v = \sqrt{\frac{T}{\mu}}$,

$v = \sqrt{\frac{800}{0.08}} = \sqrt{10000} = 100$ms^{-1}.

97. A transverse wave is applied to a string whose mass per unit lemgth is 3×10^{-2}kgm^{-3}. If the string is under a tension of $12N$,the speed of propagation of the wave is A.40ms^{-1} B.30ms^{-1} C.20ms^{-1} D.5ms^{-1}

μ=3×10^{-2}kgm^{-3}, T=$12N$, v=?, $v = \sqrt{\frac{T}{\mu}}$,

$v = \sqrt{\frac{12}{3\times10^{-2}}} = \sqrt{4\times10^2} = 40$ms^{-1}.

98. Which of the following statements about the 3rd overtone of a vibrating air column of an open pipe is correct? It has **A.4 nodes** B.5 nodes C.3 nodes D.4 antinodes.

Sketch the diagram of the 3rd overtone or 4th harmonic, using the fact that the wavelength increase by half , $\frac{1}{2}$ which is equivalent to two consecutive nodes inevery overtone. An open pipe always have an antinodes at both open ends. Therefore, the third overtone or 4th harmonic has 4 nodes and 5 antinodes.

99. An organ pipe of length 40cm opened at both ends sounds its fundamental note. Neglecting end corrections, calculate the frequency of the note (Speed of sound in air=340ms^{-1}) A.213Hz B.272Hz C.380Hz **D.425Hz** E.544Hz

L=40cm=0.4m, v=340ms^{-1}, wavelength of an open pipe $\lambda_n = \frac{2L}{n}$, n=1 for fundamental note,

$\lambda_1 = \frac{2L}{1} = 2L$, $v = \lambda f_o$, $f_o = \frac{v}{\lambda} = \frac{v}{2L}$,

$f_o = \frac{340}{2\times0.4} = 425$Hz. Alternatively, frequency of n_o overtones of an open pipe is given by,

$f_n = (n_o + 1)\frac{v}{2L}$, f_n=f_o , number of overtones n_o=0 overtones or first harmonic or fundamental, $f_o = (0{+}1)\frac{340}{2\times0.4} = 425$Hz.

100. An organ pipe of length 40cm opened at both ends sounds its second overtone.Neglecting end correttions,calculate the frequency of the note. [Speed of sound in air=340ms^{-1}] A.213Hz B.425Hz **C.1275Hz** D.850Hz

L=40cm=0.4m, n_o=2 i.e 2nd overtone, v=340ms^{-1}, second overtone f_2=?,

$f_n = (n_o + 1)\frac{v}{2L}$, $f_2 = (2{+}1)\frac{340}{2\times0.4}$,

$f_2 = 3\times425 = 1275$Hz.

101. An organpipe open at both ends has a fundamental frequency of 400Hz. The frequencythe third overtone is A.400Hz **B.1600Hz** C.800Hz D.1200Hz .

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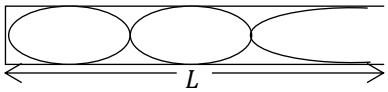
Open pipes has all multiples(odd and even) of harmonics. 3rd overtone f_3=4th harmonic
$f_3 = 4f_o = 4\times400 = 1600$Hz.

102. In a closed organ pipe producing a musical note, an antinode is always at A.the closed end **B.the open end** C.middle D.All the parts of the pipe.
At the open end of a closed pipe is an antinode while at the closed end is a node.

103. An organ pipe closed at one end is 8cm long. Neglecting end corrections, calculate the wavelength of its fundamental note A.2cm B.4cm C.8cm D.16cm **E.32cm.**
L=8cm, wavelength of n harmonics of a closed pipe $\lambda_n = \frac{4L}{n}$, n=1 for fundamental note, $\lambda_1 = \frac{4L}{1} = 4L = 4\times8 = 32$cm.

104. The wavelength of the first overtone of a note in a closed pipe of length 33cm is **A.44cm** B.33cm C.22cm D.17cm.
L=33cm, $\lambda_n = \frac{4L}{n}$, n=3 for 1st overtone or 3rd harmonic(odd harmonics), $\lambda_2 = \frac{4\times33}{3} = 44$cm.

105. From the digram of a closed pipe below, the frequency produced by the wavelength of the length L is known as



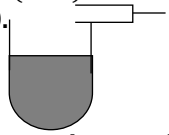
A. fundamental B.second harmonic C.third harmonic **D.fifth harmonic.**
The diagram of the closed pipe indicates the 3rd overtone which is the 5th harmonic as closed pipe form odd harmonic only.

106. A pipe of length 45cm is closed at one end. Calculate the fundamental frequency of the sound generated in the pipe if the velocity of sound in air is 360ms^{-1} (Neglect end corrections) A.5.5Hz B.148.5Hz **C.200.0Hz** D.550.0Hz.
L=45cm=0.45m, number of overtones n_o=0 for fundamental frequency, v=360ms^{-1}, f_o=?,
$f_n = (2n_o + 1)\frac{v}{4L}$, $f_o = (2\times0+1)\frac{360}{4\times0.45}$,
$f_o = (0+1)200 = 200$Hz.

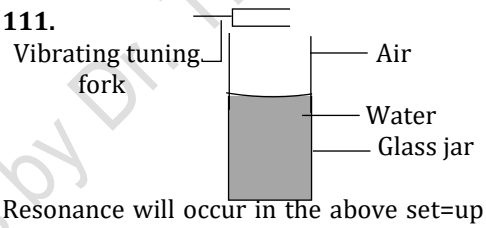
107. If the fundamental frequency of a closed pipe organ on a day when the speed of sound is 340ms^{-1} is 170Hz, then the length of the pipe is **A.50cm** B.70cm C.100cm D.150cm E.200cm.
v=340ms^{-1}, f_o=170Hz, n_o=0, L=?,
$f_n = (2n_o + 1)\frac{v}{4L}$, $f_o = (2\times0+1)\frac{v}{4L} = \frac{v}{4L}$,
$170 = \frac{340}{4L}$, $L = \frac{340}{4\times170} = 0.5\text{m} = 50\text{cm}$.

108. When vibration occur in an air column, the distance between a node and an antinode is equal to **A.One-quarter of the wavelength** B.One-half the wavelength C.The wavelength D.Twice the wavelength E.Four times the wavelength
The distance between a node and an antinode is one-quarter the wavelength, $NA = \frac{1}{4}\lambda$.

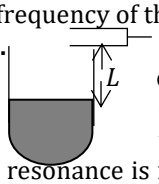
109. A closed pipe 60cm long is ¾ full of water. If the velocity of sound in air is 340ms^{-1} what is the frequency of the fundamental note produced by the pipe ? (Neglect end correction) A.188.89Hz B.204.00Hz **C.566.67Hz** D.755.56Hz E.2266.67Hz.
Length of closed pipe=60cm, length of water in the tube = $\frac{3}{4}\times60$ =45cm, length of air column L=60−45=15cm=0.15m, v=340ms^{-1}, f_o=?, n_o=0,
$f_n = (2n_o + 1)\frac{v}{4L}$, $f_o = (2\times0+1)\frac{340}{4\times0.15}$,
$f_o = (0+1)566.67 = 566.67$Hz.

110. 
The air column as shown in the figure vibrates by the tuning fork. In this state of resonance the waves in the air column will be A.stationary and transverse **B.stationary and longitudinal** C.progressive and transverse D.progressive and longitudinal.
The vibration of the tuning fork sets up stationary and transverse wave in the air column in the resonance tube.

111.



Resonance will occur in the above set=up when the natural frequency of A.the water column is the same as the frequency of the tuning fork B.the water column and the air column is the same as the frequency of the tuning fork C.the air column is higher that the frequency of the tuning fork **D.the air column is the same as the frequency of the tuning fork** E.the air column is less than the frequency of the tuning fork.

112. 

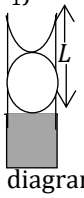
In the figure shows, a resonance tube experiment is performed using one tuning fork. As the water level is lowered the first resonance is is obtained when the length of the column $L = \lambda/4$. The second resonance is obtained when L equals A.$\lambda/2$ **B.$3\lambda/4$** C.λ D.$3\lambda/2$.
The first resonance length is equivalent with the fundamental note of a closed pipe, $L_1 = \frac{\lambda}{4}$
The second resonance length is equivalent with the 1st overtone or 3rd harmonic of a closed pipe, $L_2 = \frac{3\lambda}{4}$.

113. If the first resonance position of resonance in a resoanace tube is 16.50cm from the open end, the distance from the open end to the next position where resonance occurs will be A.8.25cm B.22.00cm C.24.75cm D.33.00cm **E.49.50cm.**
First position of resonance or first resonance length L_1=16.50cm, second position of resonance

Demystified Series by Dr Timothy

or second resonance length L_2=?,
$L_1 = \frac{\lambda}{4}$, $\lambda = 4L_1 = 4\times16.50 = 66.0$cm
$L_2 = \frac{3\lambda}{4} = \frac{3\times66.0}{4} = 49.50$cm. Alternatively,
$L_1 = \frac{\lambda}{4}$, $L_2 = 3\left(\frac{\lambda}{4}\right) = 3L_1 = 3\times16.5 = 49.50$cm.

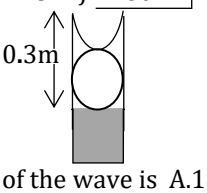
114. In a resonance tube experiment, the effective length of the air column for the first resonance is 20cm when set into vibration by a tuning fork of frequency 480Hz. Neglecting end effect, the velocity of sound in air is A.96ms^{-1} B.225ms^{-1} C.340ms^{-1} **D.384ms^{-1}.**
First resonance length L_1=20cm=0.2m, frequency of vibrating air column f equals thefrequency of tuning fork at resonance, f=480Hz, v=?, $v = \lambda f$, $L_1 = \frac{\lambda}{4}$, $\lambda = 4L_1$, $v = 4L_1 f = 4\times0.2\times480 = 384$ms^{-1}.

115. 

The diagram above shows a stationary wave of wavelength 40cm in a closed tube.The length of the resonating air column is A.40cm B.10cm C.20cm D.30cm.
The diagram shows the second position of resonance $L = \frac{3\lambda}{4}$, λ=40cm, $L = \frac{3\times40}{4} = 30$cm. Alternatively, the closed tube has 1 antinode at the top and 2 nodes i.e 1 at the middle and 1 at the bottom. Thus, $L = AN + NN = \frac{\lambda}{4} + \frac{\lambda}{2}$,
$L = \frac{\lambda+2\lambda}{4} = \frac{3\lambda}{4} = \frac{3\times40}{4} = 30$cm.

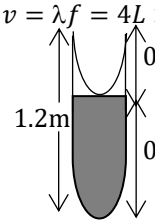
116. In a resonance tube experiment, if the fundamental frequency of a vibrating air is f, the possible overtones are A.$\frac{1}{2}f, \frac{1}{4}f, \frac{1}{8}f$ e.t.c B.$2f, 3f, 4f, 5f, 6f$ e.t.c C.$2f, 4f, 6f, 8f$ e.t.c **D.$3f, 5f, 7f, 9f$ e.t.c.**
Resonating tube is a close pipe and will produce only odd harmonics of overtones.
the tuning fork.

117. In a resosnace tube experiment, a tube of fixed length is closed at one end and several tuning forks of increasing frequency used to obtain resonance at the open end. If the tuning fork with the lowest frequency which gave resonance had a frequency of f_1 and the next tuning fork which gave resonance had a frequency of f_2, find the ratio of f_2:f_1 A.8 **B.3** C.2 D.1/2 E.1/3
A resonance tube is a closed pipe.
First resonance position f_1 = fundamental frequency f_o , Second resonance position f_2 = first overtone = third harmonic $3f_o$,
$\frac{f_2}{f_1} = \frac{3f_o}{f_o} = 3{:}1 = 3$.

118. 

In a resonance tube experiment in the figure the frequency of the tuning fork is 750Hz and the resonating length of the air column is 0.3m as shown. The velocity of the wave is A.1850ms^{-1} B.225ms^{-1} **C.300ms^{-1}** D.53×10^{-5}ms^{-1} E.2500ms^{-1}.
f=750Hz, L=0.3m, the diagram indicates the second position of resonance $L = \frac{3\lambda}{4}$, $\lambda = \frac{4L}{3}$,
$v = \lambda f = \frac{4L}{3} \times f = \frac{4\times750}{3} \times 0.3 = 300$ms^{-1}.

119. A tuning fork of frequency 340Hz is vibrated just above a cylindrical tube of height 1.2m.If water is slowly poured into the tube,at what maximum height will resonance occur. [Speed of sound in air=340ms^{-1}] **A.0.95m** B.0.60m C.0.50m.
First position of resonance is equal to the fundamental note,$L = \frac{\lambda}{4}$,L=length of air column,v=340ms^{-1}, f=340Hz.
$v = \lambda f = 4L \times f$, $L = \frac{v}{4f} = \frac{340}{4\times340} = 0.25$m,



The maximum height of the tube at which resonance will will occur = height or length of water column L_w.
Height of the tube h=1.2m,
Length of air column L=0.25m, $L_w = h - L$, L_w =1.2−0.25 =0.95m.

120. In a resonance tube closed at one end, the first two lengths of the air column that vibrate in resonance with a tuning fork of frequency 316Hz are 20.6cm and 73.1cm respectively.Calculate the velocity of sound in air to 3 significant figures A.296ms^{-1} B.325ms^{-1} C.330ms^{-1} **D.332ms^{-1}** E.338ms^{-1}.
f=316Hz, L_1=20.6cm, L_2=73.1cm, v=?,
$v = 2f(L_2 - L_1) = 2\times316(73.1-20.6)$,
$v = 632(0.525) = 331.8 = 332$ms^{-1}.

121. The shortest length of the air column in a resonance tube at resonance is 0.11m and the next resonant length is 0.36m.Calculate the frequency of vibration, given that the speed of sound in air is 340ms^{-1} A.340Hz B.347Hz **C.680Hz** D.694Hz E.1360Hz
v= 340ms^{-1}, L_1=0.11m, L_2=0.36m, f=?
$v = 2f(L_2 - L_1)$, $f = \frac{v}{2(L_2-L_1)} = \frac{340}{2(0.36-0.11)}$
$f = \frac{340}{2\times0.25}$ =680Hz.

122. Which of the following is a percussion instrument? A.Flute B.Guitar C.Piano D.Trumpet **E.Xylophone.**
Flute and trumpet are wind instruments while guitar and piano are string instruments.

123. Which of the following pairs of musical instruments works on the principle of vibration in air pipes? A.Guitar and piano B.Guitar and flute **C.Trumpet and flute** D.Organ and piano E.Violin and trumpet
Air column is vibrated in trumpet and flute, string is vibrated in guitar,piano,violin and organ.

124. Which of the following instruments has a pure tone A.guitar B.vibrating strings **C.tuning fork** D.siren.
A tuning fork has a pure tone or sound as it

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produces sound with single or fundamental frequency without being excited i.e free vibration, vibrating at its own natural or fundamental frequency.

Jamb past questions on sound waves :
[1982/6,47,1983/1,1984/14,15,1985/27,28, 1986/26,35,37,1987/29,30,31,1988/21,24,25, 26,2728,29,1989/20,21,1990/21,1991/30,31, 32,36,1992/22,1993/20,21,24,1994/29,30,31, 1995/30,31,1997/29,1998/24,25,26,27,29, 1999/12,19,21,2000/18,24,2001/12,18,2002/ 13,15,29,2003/18,19,2004/4,2008/29,2009/28 ,29,2011/26,31,2013/27,2014/27]

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Chapter 17 – Light wave and Reflection of light from plane and curved/spherical surface

● **Light wave** – Light wave is an electromagnetic wave i.e wave which do not require a material medium for its propagation or that can travel through a vacuum.

- Light wave is a transverse wave. Light is a visible for for energy that enables vision.

- Sources of light – It can be natural e.g sun, stars,fire flies e..tc or artificial or e.g electric lamp, candle light e.t.c.

- Luminous source – source that generate and emit light by itself. Natural or self luminous sources are sun,stars,glowing worms,fire flies and incandescent lamps. Artificial or man-made luminous sources are candle light,torch light, car head lamps,matches and electric lamps.

- Non-luminous source – source that does not generate light by itself but due to light falling on it from another object that it reflects i,e they are seen only when they reflect light from a luminous body. e.g moon,road signs, earth and pages of a book.

- Transmission of light – Parts of the light incident on a body is transmitted through the body while other parts are either reflected or absorbed by the body. The amount of light transmitted through a body depends on its nature: transparent,transluscent and opaque.

- Transparent body – body that allows light to pass through it and that can be seen through e.g plane glass and clean water.

- Translucent body – body that allows light to pass through but cannot be seen through e.g window panes,obscured/tinted glass sheets and tissue paper.

- Opaque body – body that do not allow light to pass through and cannot be seen through e,g cardboard sheets,brick walls,metal sheets, wood or planks.

- Ray of light is the direction or path taken by light energy.

- Beam of light is a collection of rays or stream of light energy.

- Parallel beam is produced by a search light, convergent beam is produced by a hand lens while divergent beam is produced by lamps.

- Rectilinear propagation of light is the phenomenon of light rays travelling in straight lines.

- Diffraction of light is also called non-rectilinear propagation of light.

- Application of principle of rectilinear propagation of light: Formation of shadow, eclipse and operation of a pin-hole camera.

1.Shadow – images formed when source of light is blocked by an opaque material. **Full shadow or Umbra or black region** is formed from **a point source of light i.e small lightsource**. **Partial shadow or Penumbra or grey region** is formed from an **extended light source i.e large source of light**.

- The sharpness of the boundary of a shadow isdetermined by the intensity of light stricking the object.

2.Eclipse – shadow formed when an opaque celestial body (earth or moon) blocks a defined light source i.e the sun or the effect produced when one heavenly body cast its shadow on another.

- Solar eclipse or eclipse of the sun – sun is blocked by moon and the shadow of the moon is formed on earth.

- Lunar eclipse or eclipse of the moon – sun is blocked by earth and the shadow of the earth is cast on the moon

- Total eclipse(Umbra or total darkness): formed where the extremes lines from the sun touch the earth.

- Partial eclipse(Penumbra):formed where the intersecting lines from the sun touch the earth.

- Annular eclipse – formed outside the earth when the extremes rays from the sun required for total eclipse intersect before reaching the earth. It is seen beyond the vertex of the cone of the moon produced (Umbra cone) and a bright ring of light or annulus is formed around the shadow of the moon.

3.Pin-hole camera – It is made from an opaque box, one side of which is made of a translucent sheet of paper and a tiny hole opposite it.

- Images formed by a pinhole camera :

I.The image is real (as it can be caught on a screen) and inverted

II. With a pinhole or small tiny hole, the image will be sharp,clear and dim or not very bright due to limited amount of light entering it.

III. With a large hole or many pinholes, the image will be blurred, overlapping and bright.

IV. For large magnification(enlarged image), the object distance must be small i.e the object should be close to the pin hole and the image distance should be large i.e the distance between the pinhole and the screen is increaded.

- Hence, reducing the object distance will increase the height of the image. The image formed in this case is enlarged but less bright due to ditribution of the light over a greater are of the enlarged image. $\boldsymbol{Image\ height} \propto \frac{\boldsymbol{image\ distance}}{\boldsymbol{object\ distance}}$, $\boldsymbol{H_I \propto \frac{v}{u}}$ **thus, bigger and brighter image.**

- When a pinhole camera is used in taking still

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pictures the screen is replaced by a photographic film.

- External light is excluded from the pinhole camera by covering it with a dark cloth,inoder for a clearer image to be seen on the screen.

- Magnification produced by a pinhole camera – Magnification m is the ratio of the size(height) of the image to the size(height) of the object.

$\boldsymbol{m = \frac{image\ height\ H_I}{object\ height\ H_O} = \frac{image\ distance\ v}{object\ distance\ u}}$,

$\boldsymbol{m = \frac{length\ of\ camera}{object\ distance} = \frac{L}{u}}$.

● Reflection of light waves

- Reflection – Reflection of light occurs when light falls on an opaque body. It can either be regular or diffused.

- Polished or smooth surfaces e.g mirrors, produces regular reflection.

- Rough surface e.g a page of a book, produce diffused or scattered reflection.

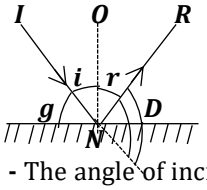
- Laws of reflection :

1.The incident ray, the reflected ray and the normal at the point of incidence, all lie in the same plane.

2.The angle of incidence $\boldsymbol{i}$ with the normal is equal to the angle of reflection $\boldsymbol{r}$ from the surface, $\boldsymbol{i = r}$. **NB :** Normal is not a ray.

- The principle of reversibility of light states that the path of light ray is reversible. It is a general principle in optics

- Reflection in plane mirrors : A plane mirror is made when one side of a plane surface is coated.



IN is the incidient ray,
RN is the reflected ray,
ON isthe normal.

- The angle of incidence i is equal to the angle of reflection r , $\boldsymbol{i = r}$.

-Angle between the incident ray and reflected ray= $\boldsymbol{i + r = 2i = 2r}$.

- Glanging angle or angle of glance $\boldsymbol{g}$ is the angle between the incident ray and the mirror or the angle between the reflected ray and the mirror. $\boldsymbol{g + i = 90°}$, $\boldsymbol{g + r = 90°}$.

- Angle of deviation $\boldsymbol{D}$ is the angle through which the incident ray has deviated from its original path after being reflected.

$\boldsymbol{D = 180° - (i + r) = 180° - 2i = 180° - 2r}$.

- Angle of deviation is twice the glancing angle or angle of glance. $\boldsymbol{D = 2g}$.

- If a ray is incidented perpendicularly or normally on the mirror, the angle of incidence and angle of reflection are 0°, as the ray follows through the same path.

- Images formed by a plane mirror:

1.The image is virtual – behind the mirror and cannot be caught on a screen.

2.Erect or upright.

3.Image is at the same distance as the object from the mirror – image distance v is equal to the object distance u, $\boldsymbol{v = u}$.

4.Image is the same size as object – image height equals object height and magnification is unity, $\boldsymbol{m = 1}$, $\boldsymbol{H_I = H_O}$.

5.Laterally inverted – inversion on the side only,the right hand of the object appears as the left hand on the mirror image. No change in the vertical placement of the image, top is top and bottom is bottom.

- Images in mirrors are formed due to reflection only. Multiple images are formed in plane mirrors due to the thickness of the mirror.

- Inclined plane mirrors – The number of images formed when two formed when two mirrors are inclined at an angle θ to each other is given by; $\boldsymbol{n = \frac{360}{\theta} - 1}$.

- When two mirros are parallel to each other e.g mirrors in a barber's shop, the angle between them is 0° and they form infinite (multiple) images. $n = \frac{360}{\theta} - 1 = \alpha - 1 = \alpha$ i.e Infinite.

- Perpendicularly inclined plane mirrors – When two mirrors M_1 and M_2 are inclined perpendicularly i.e at 90° to each other, the angle of incidence or reflection in M_1 is equal to the glancing angle of M_2 while the angle of incidence or reflection in M_2 is equal to the glancing angle in M_1.

- The angle of deviation of the emergent ray by two mirrors inclined at 90° to each other is always 180°, regardless of the angle of incidence.

- Plane mirror rotation – When a ray of light is incident on a plane mirror and the mirror is rotated through an angle θ, with the incident ray being kept constant, **the reflected ray is rotated through an angle 2θ i.e twice the angle of rotation of the mirror.**

- The rotation of the mirror can be either in an anticlockwise direction or clockwise direction depending on the one described in a particular question.

- For anticlockwise rotation of mirrors :

1.The new angle of incidence or reflection after rotation = $\boldsymbol{i + \theta}$.

2.The new angle between the incident and reflected ray = $\boldsymbol{2(i + \theta)}$.

- For clockwise rotation of mirrors :

1.The new angle of incidence or reflection after rotation = $\boldsymbol{\theta - i}$.

2.The new angle between the incident and reflected ray =$\boldsymbol{2(\theta - i)}$.

- The angle between the new reflected ray after rotation and the normal before rotation in both clockwise and anticlockwise rotation of the mirror = $\boldsymbol{i + 2\theta}$. i=initial incident angle, θ=angle

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of rotation of the mirror.

- The displacement of the beam of light due to rotation of the mirror from its initial position on a screen is given by – $\tan 2\theta = \frac{D_B}{D_{MS}}$, D_B=Displacement of beam or spot on the screen, D_{MS}=distance between the mirror and screen.

- Applications of plane mirrors –

1.Simple periscope – Two plane mirrors are placed **parallel to each other i.e at 0°** and each of the mirrors is inclined at 45° to the wall. It is used for observing above or below obstacles or normal view of vision. The major disadvantage of a simple periscope is that it produces infinite or multiple images.

2.Sextant – Used in navigation and in measuring the angle of elevation of the sun.

3.Kaleidoscope – Two plane mirrors are inclined at 60° to each other. It is used in textile industries for producing colour patterns and in producing symmetrical patterns when pieces of coloured glasses acting as object are arranged by shaking the tube.

4.Mirror galvanometer – A galvanometer with no physical pointer but with the principle of rotation of reflected ray

5.Theatrical stage – Optical illusion e.g a man walking with no head or a candle burning inside a bottle filled water.

● Reflection from Spherical/curved surface

- Concave or Converging mirror – The outer surface is coated with silver while the inner surface is the reflecting part.

- The nature of image formed depends on the object position. As the object moves closer to the mirror or approaches the principal focus, the image becomes bigger and moves away from the mirror ($u \propto \frac{1}{v}$, $u \propto \frac{1}{H_I}$, $v \propto H_I$).

- Real images are formed on a screen and are always inverted, are seen in the same side of the mirror while Virtual image cannot be caught on a screen and are always erect or upright, are seen behind the mirror.

- F = principal focus, C= centre of curvature, f= focal length, r= radius of curvature, P= pole, m= magnification.

- Images formed by a concave mirror:

1.Object at infinity ($u = \infty$) – image at $F(v = f)$, real inverted and diminished image.

2.Object at $C(u > r$ or$2f)$ – image is between F and $C(f < v < r)$, real,inverted and diminished image.

3.Object at $C(u = r$ or $2f)$ – image at $C(v = r$ *or* $2f)$, real,inverted and same size.

4.Object between F and $C(r > u > f)$ – image is beyond C $(v > r$ or$2f)$, real,inverted and magnified image.

5.Object at $F(u = f)$ – image at infinity ($v = \infty$), the nature of the image cannot be captured e.g car-head lamps, torchlights and searchlight.

6.Object between F and P or mirror ($u < f$) – image is formed behind the mirror e.g dentist mirror, make-up or shaving mirror.

- If the **object is beyond C**,the image is always **diminished ($m < 1$).**

- If the **object is at C**,the image is **same size as theobject ($m = 1$).**

- If the **object is behind C**,the image is always **magnified ($m > 1$).**

- Concave mirrors are used shaving mirrors when the object is placed close to the face or at a distance less than the focal length.

- Parabolic mirrors are concave mirrors which produces parallel beam of constant intensity when a small light source is placed at its focus. It is used in searchlights, car headlamps and torch lights.

- A concave mirror used for shaving should be narrow while a concave mirror used for head lamps should be wide.

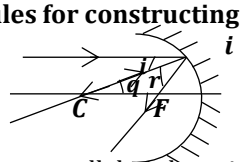
- Convex or Diverging mirror – The inner surface is coated with silver while the outer surface is the reflecting part.

- The closer the object to the mirror, the closer the image to the mirror ($u \propto v$).

- The image formed by a convex mirror is always virtual,erect and diminished **(VED)** and between the principal focus and the pole of the mirror, **at all object positions**.

- Convex mirrors are used in driving mirros, monitoring mirros and security mirrors in super-markets, as it gives a wide field of view of objects within a large angle and forms an erect image of the object.

- Rules for constructing ray diagram :



$i = r$, $i = q$, $r = q$

1.Rays parallel to the principal axis are reflected through are reflected through the principal focus.

2.Rays passing through the principal focus are reflected parallel to the principal axis.

3.Rays passing through the centre of curvature are reflected back through the centre of curvature i.e at angle of incidence or reflection of 0°.

- Mirror formula : 1. $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$. **2.** $v = \frac{uf}{u-f}$.

3. $u = \frac{vf}{v-f}$. **4.** $f = \frac{r}{2}$.

- Magnification formula:

1. $m = \frac{v}{u} = \frac{H_I}{H_O}$. **2.** $m = \frac{f}{u-f} = \frac{r}{2u-r}$.

3. $m = \frac{v}{f} - 1 = \frac{2v}{r} - 1$.

- Points to note when calculating any question

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on mirrors:

1.Object distance u is positive for both concave and convex mirror.

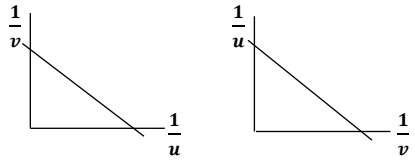
2.Image distance v is positive for real and inverted images and negative for virtual and erect images.

3.Focal length f of concave mirror is positive while that of convex mirror is negative.

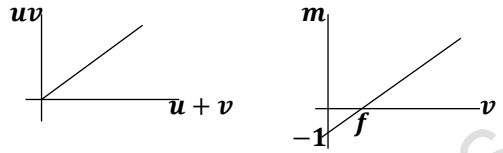
4.Magnification m is positive for real and inverted images and negative for virtual and erect images.

5.The distance between an object and its image is given by : $\boldsymbol{D = u \pm v}$**. It is positive for virtual images and negative for real images.**

- Graphical illustration of mirror formulas (this also apply to lenses) – This shows the graphical relationship between the variables in mirror formula.



- The intercept on either axis, of both graphs above is equal to the reciprocal of the focal length $\frac{1}{f}$. Hence, **the focal length is equal to the reciprocal of the intercept.**



- The slope of the graph of uv against $u + v$, is equal to the **focal length** $\boldsymbol{f}$**,** $\boldsymbol{uv = f(u + v)}$.

- The slope of the graph of M against v is equal to the reciprocal of the **focal length** $\frac{1}{f}$, the intercept on $\boldsymbol{m}$**-axis** is equal to $-\mathbf{1}$, while the intercept of $\boldsymbol{v}$**-axis**,is equal to **focal length.**

- Comparison between plane and convex mirror in driving mirrors :

Plane mirror	Convex mirror
1.Image of same size as the object (Advantage)	Diminished image i.e image are smaller than object (Disadvantage).
2.Lateral inversion (Disadvantage)	No lateral inversion (Advantage).
3.Correct distance of image can be estimated i.e image distance equals objects distance (Advantage)	Correct distance of image can't be estimated,as the image seems farther way (Disadvantage).
4.Erect image	Erect image.

- Spherical aberration in mirrors is the inability of a curved mirror to reflect wide beam or rays of light that are far from the principal axis through a common focus but through a point nearer than the focus.It is prevented by using a parabolic mirror.

Examples :

1. A luminous object is one that A. gives off dim blue-green light only in the dark **B. gives out light of its own** C. shines by reflected light only D. glows only in the presence of light.

2. Which of the following is not self luminous? A.incandescent electric bulb B.the moon C.the sun **D.lighted candle** E.incandescent fluorescent tube

The moon is a non-luminous object, hence it depends on light falling on it from a luminous object. Sun,incandescent electric bulb and fluorescent bulb are luminouse object.

2.Which of the following is not a luminous object? A.the sun **B.the moon** C.a star D.a lighted candle E.a lighted bulb.

3. I.Moon II.Sun III.Street light IV.Stars. Which of the above is a natural source of light? A.II, III and IV B.I,II and III C.III and IV **D.II and IV**

Moon is a non-luminous object. Street light is an artificial or man-made luminous source. Sun and Stars are natural luminous sources.

4. Non-luminous objects can be seen because they A.are polished B.are near **C.reflect light** D.emit light

Non-luminous objects are seen by the light they reflect. Luminous objects emit light.

5. What happens to parallel beam of light? A.diverge as they travel **B.they meet at infinity** C.they intersect D.they converge as they travel

The light rays of a parallel beam are parallel to one another and they do not meet or intersect. The light rays of a convergent beam converge(meets) at a point as they travel while that of a divergent beam diverge(spread out from the source) as they travel.

6. Which of the following statements supports the assumption that light travels in a straight lines? A.light can be diffracted **B.a source of light produces distinct shadows of opaque objects** C.a source of light produces interference patterns on suitably placed screen D.light can be refracted

Rectilinear propagation of light is a phenomenon which states that light travels in a straight line. It is applied in the formation of shadows and eclipse and in the operation of a pinhole camera.

7. Shadow and eclipse result from the A.Refraction of light B.Reflection of light **C.Rectilinear propagation of light** D.Diffraction of light.

8. Light travels in straight lines. In which of the following is this principle manifested? I.pin hole camera II.formation of shadows III.diffraction iv.occurence of eclipse A.I and III B.II and III C.I,II and III **D.I,II and IV** E.I,II,III and IV .

9. The sharpness of the boundary of the shadow

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of an object is determined by the A.rays of light passing through the object **B.intensity of light striking the object** C.opacity of object D.nature of the object

A bright extended source of light will gives less sharp and brighter shadow while a small or point source of light gives a sharper shadow.

10. I.Total internal reflection of light II.Conservation of light energy III.Relative motion of the earth,sun amd moon IV.Rectilinear propagation of light. Which of the above is a phenomenon of total solar eclipse? A.I and IV B.II and IV C.I and III **D.III and IV**

Eclipse is as a result of rectilinear propagation of light. Eclipse of the sun or moon is based on the position of the sun,moon and earth relative to one another.

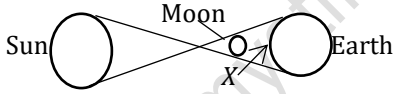
11. An eclipse of the sun may occur when A.the sun passes between the moon snd the earth **B.the moon passes between the earth and the sun** C.the earth passes between the moon and the sun D.the moon and the earth rotate together E.a part of the earth rotates away from the sun

Solar eclipse or eclipse of the sun occurs when the moon passes between the earth and the sun, such that the shadow of the moon is cast on the earth.

12. The eclipse of the sun occurs when the A.moon is note completely hidden in the earth's shadow B.moon's umbra falls on some part of the earth **C.moon is between the sun and the earth** D.earth is between the sun and the moon.

13. Total eclipse of the sun occurs when the A.earth is between the moon and the sun B.sun is between the moon and the earth **C.moon is between the sun and the earth** D.ozone layer threathened.

14.



When the sun,the moon and the earth are as showm in the diagram above, an observer standing at *X* is in A.penumbra region and sees a partial eclipse B.penumbra region and sees a total eclipse C.umbra and sees a partial eclipse **D.umbra and sees a total eclipse.**

Partial eclipse is seen at the edges or penumbra region while total eclipse is seen at the centre or umbra region.

15. Annular eclipse occurs when the A.earth is between the sun and the moon **B.moon is between the sun and the earth** C.extreme rays of light from the sun intersect after reaching the earth D.sun is between the moon and the earth.

In annular eclipse the extreme rays of the sun required for total eclipse intersect before reaching the earth not after reaching the earth.

16. An annular eclipse of the sun is seen by an observer at a point A.in the umbra produced by the moon B.in the umbra produced by the earth C.in the penumbra produced by the moon **D.beyond the vertex of the umbra cone of the moon.**

Annular eclipse is seen on earth beyond the intersectiong of the extreme lines required for total eclipse or umbra i.e beyond the the vertex of the umbra cone.

17. The image in a pinhole camera is A.erect and formed by refraction through a lens B.virtual and formed by dispersion C.erect and gets sharper as the hole becomes larger **D.inverted and formed by the light from each point travelling in a straight line**

The image in a pinhole camera is real and inverted. It is formed due to rectilinear propagation of light.

18. If the size of the table of a pinhole camera is increased, the image formed becomes **A.brighter and blurred** B.brighter and larger C.brighter and sharper D.blurred and duller E.blurred and larger

Increasing the holes of a pinhole camera increases the amount of light entering it hence the brightness increases leading to overlapping images. Thr image then becomes brighter,less sharp and blurred.

19. The pin-hole camera is less sharply defined image when the **A.pin-hole is larger** B.illumination is less C.screen is further from the pin-hole D.object is further from the pin-hole.

Less-sharply defined or blurred image is due to increased brightness when pin-hole is larger .

20. What is the effect of the increase in the size of the hole of a pin-hole camera on the image? It **A.gives a blurred image** B.corrects chromatic aberration C.magnifies the image D.brings the image into a sharper.

21. As an object moves towards a pin-hole camera,the image becomes A.brighter and smaller B.less brighter and bigger C.brighter but remains the same size D.less bright and diminishing in size **E.brighter and bigger**

Object moving towards or closer to the pinhole camera decreases the object distance hence the image height increases i.e becomes larger as $image\ height \propto \frac{image\ distance}{object\ distance}$, $H_I \propto \frac{v}{u}$. Object moved towards the pin hole reduces the sharpness of the image and increases the brightness.

22. If the distance between the object of a pinhole camera is reduced by half, the size of the image of the object **A.is doubled** B.is quadrupled C.remains the same D.is halved

$image\ height \propto \frac{image\ distance}{object\ distance}$, $H_I \propto \frac{v}{u}$, $\therefore H_I \propto \frac{1}{u}$, when the object distance is halved, then image size or height $H_I = \frac{1}{1/2} = 2$ i.e doubled.

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22. A man 1.5m tall is standing 3m in front of a pinhole camera whose distance between the hole and the screen is 0.1m. What is the height of the man on the screen? A.0.05m B.0.15m C.0.30m D.1.00m

object height O_H=1.5m, object distance O_D=3m, image distance I_D=0.1m, image height I_H=?,

$m = \frac{image\ height\ H_I}{object\ height\ H_O} = \frac{image\ distance\ v}{object\ distance\ u}$,

$\frac{H_I}{H_O} = \frac{v}{u}$, $\frac{H_I}{1.5} = \frac{0.1}{3}$, $H_I = \frac{0.1\times1.5}{3} = 0.05$m.

23. An object of height 4cm is placed in front of a cuboid pinhole camera of size 6cm. If the image formed is 2cm high, how far is the object from the pinhole? A.3.0cm B.8.0cm **C.12.0cm** D.16.0cm

H_O=4cm, L=6cm, H_I=2cm, u=?,

$m = \frac{image\ height\ H_I}{object\ height\ H_O} = \frac{Length\ of\ camera\ L}{object\ distance\ u}$,

$\frac{H_I}{H_O} = \frac{L}{u}$, $\frac{2}{4} = \frac{6}{O_D}$, $O_D = \frac{6\times4}{2} = 12$cm.

24. An object 2.00m high is placed at a distance of 5.00m from a pin-hole camera.If the image height is 4.00cm,what is the distance of the film from the pin hole? A.2.50cm B.5.00cm **C.10.00cm** D.20.00cm E.100.00cm .

H_I=4cm, H_O=2m=200cm, u=5m=500cm, v=?,

$m = \frac{H_I}{H_O} = \frac{v}{u}$, $\frac{4}{200} = \frac{v}{500}$, $v = \frac{4\times500}{200} = 10.00$cm.

25. Which of the following cannot be explained by using the laws of reflection? A.Formation of images by plane mirrors B.Formation of images by curved mirrors C.Action of the periscope **D.Formation of shadows** E.None of the above

Formation of shadows is explained by rectilinear propagation of light and not by reflection.

26. A ray of light makes an angle of 35° with a plane mirror. What is the angle of reflection? A.55° B.35° C.70° D.65°.

The angle a ray of light makes with the mirror is the glancing angle or angle of glance g.

g=35°, angle of reflection r=?, $i + g = 90°$ or $r + g = 90°$, $r = 90° - g = 90° - 35° = 55°$.

27. A reflected ray of light makes an angle of 27° with the surface of a plane mirror, what is the angle of incidence of the ray? A.27° B.54°**C.63°** D.117° E.153°

g=27°, angle of incidence i=?, $i = 90° - g$,

$i = 90° - 27° = 63°$.

28. A ray of light strikes a plane mirror at a glancing angle of 50°. Calculate the angle between the incident and reflected rays A.20° B.40° C.50° D.80° E.100°.

g=50°, angle between the incident and reflected rays θ=?, $i = 90° - g = 90° - 50°$,

$i = 40°$, $i = r = 40°$, angle between the incident and reflected rays $\theta = i + r$,

$\theta = 40° + 40° = 80°$.

29. A ray of light is incident on a plane mirror such that the angle of reflection is 25°. What is the angle of deviation of the ray after reflection from the mirror? A.25° B.50° C.65° D.115°**E.130°**

r=25°, angle of deviation D=?, $D = 180° - 2r$,

$D = 180° - 2\times25° = 180° - 50° = 130°$.

30. A ray of light incident on a plane mirror is inclined to the mirror at an angle of 50°. What is the angle of deviation of the ray? A.50° B.80° C.90° **D.100° .**

g=50°, $D = 2g = 2\times50° = 100°$. Alternatively, $i = 90° - g = 90° - 50° = 40°$, $D = 180° - 2i$,

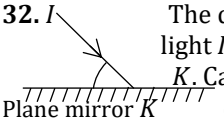
$D = 180° - 2\times40° = 180° - 80° = 100°$.

31. An engineer intends to deviate a light ray from its path 120° through reflection from plane mirror. Calculate the angle of incidence A.20° **B.30°** C.40° D.60°.

D=120°, i=?, $D = 180° - 2i$, $2i = 180° - D$,

$2i = 180° - 120° = 60°$, $i = \frac{60°}{2} = 30°$.

32.



The diagram above shows a ray of light IK incident on plane mirror at K. Calculate the angle of deviation on of the ray after reflection A.35° B.55° **C.70°** D.105° E.145°.

g=35°, $D = 2g = 2\times35° = 70°$.

33. Which of the following statements is not true of the image in a plane mirror? The image is A.the same size as the object B.laterally inverted C.virtual **D.magnified** E.the same distance behind the mirror as the object is in front

The image formed by a plane mirror is virtual, erect or upright,laterally inverted,same size as the object and the same distance behind the mirror as the object in front. The image of a plane mirror is same size as the object, hence can not magnified.

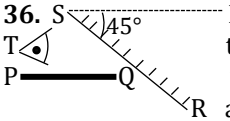
34. Which of the following does not describe the image formed by a plane mirror? A.Erect B.Laterally inverted C.Same distance from the mirror as object **D.Magnified** E.Virtual.

35. Which of the following diagrams shows the correct image of the letters *LF* as seen through a plane mirror.

A.LF FL B.LF ⅎL C.LF ⅃ⅎ **D.LF** ⅁⅃

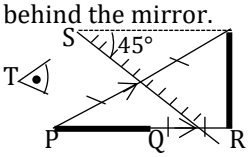
Images formed by a plane mirror is laterally inverted i.e truned through 180° or the left side of an object turns to the right and the right turns to the left.

36.



PQ is a thin rod on a horizontal table,RS is plane mirror inclined at 45° to the horizontal as shown in the figure. The image of PQ as seen in the mirror by the eye at T is A.Horizontal B.Parallel to the mirror C.At infinity **D.Vertical** E.Highly magnified.

On drawing the ray diagram the distance of each point of the object in front of the mirror is same as the the distance from its respective image behind the mirror.



Hence, the virtual image of the rod same size as the object, appears vertical.

Demystified Series by Dr. Timothy

37. The magnification of an object 2cm tall when placed 10cm in front of a plane mirror is A.6.0 **B.1.0** C.0.7 D.0.6.

Object height H_O=2cm, object distance u=10cm, In a plane mirror, object height equals image height and object distance equals image distance. Thus, image height H_I=2cm, image distance v=10cm. Magnification $m = \frac{H_I}{H_O} = \frac{v}{u} = 1$.

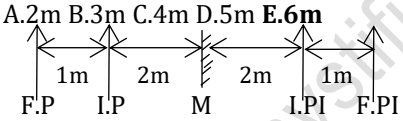
38. A boy looks at the image of an object in a plane mirror. He observes two images, a main bright one and the other faint. The observed image result from **A.Reflection only** B.Refraction only D.Reflection and refraction D.Diffraction and interference.

Images formed in mirror are due to reflection. Multiple images are formed in mirrors due to the thickness of the mirror. Refraction takes place in lenses and glasses. Reflection and refraction can take place in thick glasses, hence the reason for multiple images.

39. Which of the following descriptions of the image of a real object formed by a plane mirror is not correct? It is A.same size as the object B.laterally inverted C.virtual D.at the same distance behind the mirror as the object in front **E.formed on a screen**

Plane mirrors forms virtual images. A virtual image is one through which light rays do not actually pass through but is still visible to the eye and cannot be formed on a screen. A real image is one through which light rays actually pass through and can be caught on a screen.

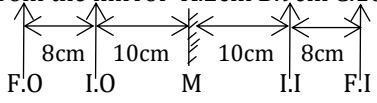
40. A barber, standing at a distance of 2m from a large plane mirror, walks 1m farther way from the mirror. How far is his image now from him A.2m B.3m C.4m D.5m **E.6m**



F.P=final position, I.P=initial position, M=mirror, I.PI=Initial position's image, F.PI=final position's image. Object distance in a plane mirror is equal to the image distance.

The distance of the barber from his image = F.PI to F.P = 1+2+2+1 = 6m.

41. An object is placed 10cm in fornt of a plane mirror. If it is moved 8cm further away from the mirror, determine the distance of the final image from the mirror A.2cm B.9cm C.16cm **D.18cm**

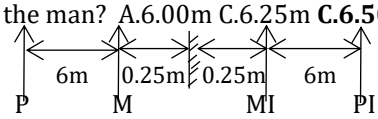


Distance of the final image from the mirror = F.I to M = 8+10 = 18cm.

42. A man stands 4m in front of a plane mirror If the mirror is moved 1m towards the man, the distance between him and his new image is A.3m B.5m **C.6m** D.10m.

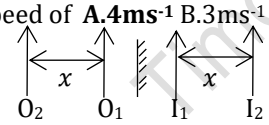
Initial object distance from the mirror=4m, when the mirror moves 1m towards the man, the new object distance u = 4−1=3m. New image distance v=3m. The distance between the man and his image = $u + v$ = 3+3 = 6m.

43. A man at the back of a crowd watches a parade by holding a plane mirror just above his head. The parade passes 6m behind his head and the mirror is 0.25m in front of the man. How far does the image in the mirror appear to be from the man? A.6.00m C.6.25m **C.6.50m** D.6.75m



P=parade, M=man, MI=man's image, PI=parade's image. The distance of the image of the parade in the crowd from the man = PI to M = 6+0.25+0.25 = 6.50m.

44. If a boy walks away from a plane mirror at a speed of 2ms^{-1}, his image moves from him at a speed of **A.4ms^{-1}** B.3ms^{-1} C.2ms^{-1} D.1ms^{-1}

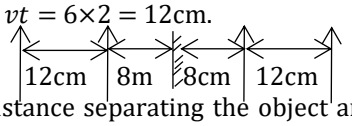


If the body walks away from the mirror from O_1 to O_2 at ts his distance from the mirror , $x = 2t$m, distance moved by image from I_1 to I_2, $x = 2t$m. Distance moved by the the image from the man = $2t + 2t = 4t$m.

Speed at which the image moves way from him = $\frac{4t}{t}$ = 4ms^{-1}.

45. An object 8cm in front of a plane mirror moves at a speed of 6cms^{-1} away from the mirror. What will be the distance separating the object and the image after 2s? A.8cm B.12cm C.20cm D.28cm **E.40cm**

With a speed of 6cms^{-1}, distance moved after 2s = $vt = 6\times2 = 12$cm.



Distance separating the object and the image = 12+8+8+12 = 40cm.

46. An object is placed in front of two plane mirrors inclined at 60° to each other. Determine the number of images formed A.2 B.3 C.5 D.7

θ=60°, n=?, $n = \frac{360^\circ}{\theta} - 1 = \frac{360^\circ}{60^\circ} - 1 = 6-1$,

n = 5 images.

47. An object is positioned between two plane mirros inclined at right angles to each other.The object is 1 unit distance from each mirror.The number of images formed is A.1 **B.3** C.2 D.5 E.4

θ=90° i.e at right angles, n=?, $n = \frac{360^\circ}{\theta} - 1$,

$n = \frac{360^\circ}{90^\circ} - 1 = 4-1 = 3$ images.

48. A man standing between two parallel mirrors in a barber's shop will see the following number of his own image A.Eight B.Two C.Four D.One **E.Infinite**

θ=0° i.e parallel, n=?, $n = \frac{360^\circ}{\theta} - 1$,

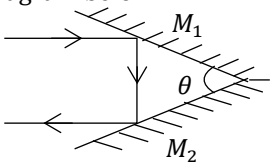
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$n = \frac{360^\circ}{0^\circ} - 1 = \alpha - 1 = \alpha$ i.e infinite.

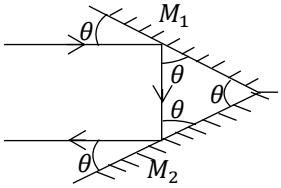
49. Two plane mirrors are inclined to each other such that an object placed between them has 11 images. Determine the angle of inclination **A.30°** B.45° C.60° D.90°

n=11 images, θ=?, $n = \frac{360^\circ}{\theta} - 1$, $11 = \frac{360^\circ}{\theta} - 1$, $\frac{360^\circ}{\theta} = 11+1 = 12$, $\theta = \frac{360^\circ}{12}$, $\theta = 30^\circ$.

50. An incidenet ray is made to travel in the opposite direction parallel to itself by means of two mirrors M_1 and M_2 set at an angle θ as shown in the diagram below

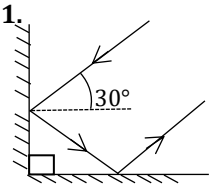


What is the magnitude of θ? A.30° B.45° **C.60°** D.90° E.120° .



$\theta + \theta + \theta = 180^\circ$,
$3\theta = 180^\circ$,
$\theta = 180/3 = 60^\circ$.

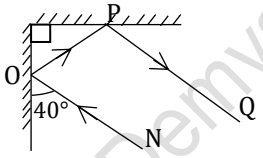
51.



Two mirrors m_1, m_2 are inclined at right angles as shown above. Calculate the the angle of reflection of the ray at m_2
A.30° B.45° **C.60°** D.90° .

Angle of incidence of m_1, i_1=30°. For two mirrors inclined at right angles, the angle of incidence of m_1 is equal to the glancing angle of m_2 while the angle of incidence of m_2 is equal to glancing angle of m_1. $i_1 = g_2 = 30^\circ$, $r = 90^\circ - g$, $r_2 = 90^\circ - g_2$ $r_2 = 90^\circ - 30^\circ = 60^\circ$.

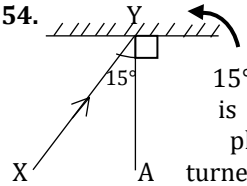
52.



Two mirrors of the same length are arranged as shown in the diagram below. A ray of light NO strikes the system at O and emerges along PQ.
The emergent ray has been deviated through.
A. 230° B. 220° C. 210° D. 200° **E. 180°.**
The angle of deviation of the emergent ray by two mirrors inclined at 90° to each other is always 180°, regardless of the angle of incidence.

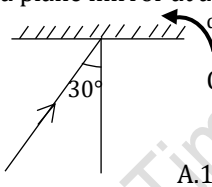
53. A ray of light strikes a plane mirror at an angle of 35°. If the mirror is rotated thrpugh 10°, through what angle is the reflected ray rotated?
A.70° B.45° C.25° **D.20°**
If a mirror is rotated through angle θ, the reflected ray is rotated through an angle 2θ.
Angle of rotation of the mirror θ=10°,
The reflected ray is rotated through angle = 2θ = $2\times10 = 20^\circ$.

54.



In the fugure an incident ray XY makes an angle of 15° with a fixed lines AY which is normal to the surface of a plane mirror. If the mirror is turned as indicated by the arrow through 40°, what angle will the reflected make with AY? A.25° B.40° C.15° D.55° **E.95°**
The arrow indicates anticlockwise rotation.
Angle of incidence before rotation i=15°, angle of rotation of the mirror θ=40°, For anticlockwise rotation: Angle the reflected ray after rotation makes with the normal AY before rotation = $i + 2\theta = 15^\circ + 2\times40^\circ = 15^\circ + 80^\circ = 95^\circ$.

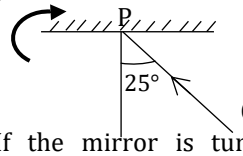
55. In the diagram below, a ray of light is incident on a plane mirror at an angle of 30°,



Calculate the new angle of incidence if the mirror is rotated through 10°
A.10° B.20° C.30° **D.40°** E.60°.
For anticlockwise rotation of mirror :The new angle of incidence or reflection = $i + \theta$, i=30°, θ=10°, New angle of incidence = $30^\circ + 10^\circ = 40^\circ$.

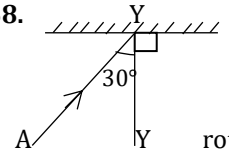
56. A ray of light is incident normally on a plane mirror. If the incident ray is kept fixed while the mirror is rotated through an angle of 30°, determine the initial and final angle of reflection respectively. **A.0° and 60°** B.90° and 30° C.0° and 30° D.90° and 60°.
When a ray is incidented normally or perpendiculaly on a plane mirror, the angle of incidence or the angle of reflection is 0°, $i = r = 0^\circ$. When the mirror is rotated through 30°, the new angle of reflection is equal to $2\times30^\circ = 60^\circ$. Thus, the initial and final angle of reflection are 0° and 60° respectively.

57. A ray OP is incident at an angle of 25° on a plane mirror as shown in the diagram below.



If the mirror is turned through 40° in the direction indicated, the angle between the incident ray and the new relected ray is **A.30°** B.50° C.80° D.130°.
For clockwise rotation of a mirror: The new angle of incidence or reflection = $\theta - i$ and The angle between the incident ray and the new reflected ray = $2(\theta - i)$, θ=40°, i=25°.
The angle between the incident ray and the new reflected ray = $2(40^\circ - 25^\circ) = 2(15^\circ) = 30^\circ$.

58.

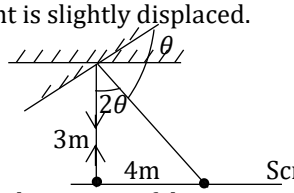


In the diagram above, an incident ray AY makes an angle of 30° with the normal XY. If the mirror is rotated anticlockwise about Y

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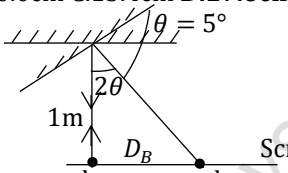
through an angle of 20°, while AY is fixed, what angle will the reflected ray now make with the incident ray? A.70° B.80° **C.100°** D.120°.
For anticlockwise rotation: The angle between the incident ray and new reflected ray $=2(i+\theta)$, i=30°, θ=20° , The angle between the incident ray and the new reflected ray $= 2(30° + 20°) = 2(50) = 100°$.

59. A plane mirror gives a spot of light on a screen which is 3m from a mirror. The screen is perpendicular to the initial ray of light. When the mirror is rotated,the spot of light moves a distance of 4m across the screen. Calculate the angle of rotation of the mirror. **A.26.6°**B.53.1° C.36.9° D.18.5°
The reflected gives a spot of light on the screen and on rotation of the mirror the spot of light is slightly displaced.



Displacement of beam or spot on the screen D_B=4m, distance between the mirror and screen D_{MS}=3m, $\tan 2\theta = \frac{D_B}{D_{MS}} = \frac{4}{3} = 1.333$,
$2\theta = tan^{-1}1.333 = 53.12°$, $\theta = \frac{53.12}{2} = 26.6°$

60. An incident ray is reflected normally by a plane mirror onto a screen where it forms a bright spot.The mirror and screen are parallel and 1m apart.If the mirror is rotated through 5°,calculate the displacement of the spot A.8.7cm B.10.0cm C.15.4cm **D.17.6cm.**



Distance between the mirror and screen D_{MS}=1m=100cm, angle of rotation of the mirror θ=5°, displacement of beam or spot on the screen D_B=?, $\tan 2\theta = \frac{D_B}{D_{MS}}$, $D_B = D_{MS}\tan 2\theta$,
$D_B = 100\tan 2(5°) = 100\tan 10° = 100\times 0.176$,
$D_B = 17.6$cm.

61. The simple periscope is an optical instrument which I.comprises two parallel plane mirrors, suitably arranged II.magnifies images of objects III.is used for viewing behind obstacles. Which of the statements above is/are correct? A.I,II and III **B.I and III** C.I and II D.II E.I.
A simple periscope is made of plane mirrors, whose magnification is unity i.e not a magnifier. The two plane mirros are parallel i.e at 0° to each other. It is used for viewing objects over or behind barriers/obstacles. Simple periscopes are used by soldiers in armoured tanks.

62. Two plane mirrors in a kaleidoscope are usually placed **A.At angle of 60°** B.Parallel to one another C.Perpendicular to one another D.At an angle of 45°
A kaleidoscope consists of two plane mirrors inclined at 60° to each other. It is used in textile industries for obtaining coloured patterns.

63. The instrument used by designers to obtain different colour patterns is called A.episcope B.periscope **C.kaleidoscope** D.sexant.

64. The instrument used for measuring the angle of elevation of the sun is called A.Altimeter B.Kaleidoscope C.Periscope **D.Sexant** E.Telescope.
Sexant is used in navigation for measuring the angle of elevation of the sun. Its operation is based on the principle of rotation of plane mirrors.

65. Which one of the following is common to both the sextant and the periscope? They both A.depend on the laws of refraction B.employ converging lenses of short focal lengths C.use totally reflecting prisms **D.depend on law of reflection of light** E.are used to measure the angle of elevation.
Sexant and periscope are both made of plane mirrors, hence they depend on law of reflection of light. Sexant is used to measure the angle of elevation of the sun while periscope is used for viewing objects above or below an obstacle.

66. The inside portion of part of a hollow metal sphere of diameter 20cm is polished.The portion will therefore form A.concave mirror of focal length 5cm B.concave mirror of focal length 10cm **C.convex mirror of focal length 5cm** D.convex mirror of focal length 20cm
Diameter of the sphere d=20cm, radius of curvature of the sphere $r = \frac{d}{2} = \frac{20}{2} = 10$cm.
The focal length of the mirror obtained from the sphere is half the radius of curvature. $f = \frac{r}{2} = \frac{10}{2} = 5$cm. The mirror is a convex mirror as the inner potion is polished.

67. An image formed on a screen is always **A.inverted** B.magnified C.upright D.virtual
Real images can be caught on a screen and they are always inverted. Virtual images are cannot be caught ona screen and are always erect or upright. Realand virtual images can both either be magnified or diminished.

68. Which of the following is applicable to a real image formed by a concave mirror? I.It can be observed on a screen II.It is always inverted and in front of the mirror III.It only seems to exist IV.It is formed by the actual converging of rays of light A.I and II B.I and III C.I,II and III **D.I,II and IV** E.All four statements
Real images can be observed or caught on a screen, are always inverted and in front of the mirror i.e same side of the mirror with the object.

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They are formed by the actual intersection of light rays.

69. Which of the following statements is/are correct for light reflected from a spherical mirror? I.incident rays parallel to the principal axis pass through the centre of curvature after reflection II.rays through the centre of curvature are returned along the same path III.rays through the principal focus are sent parallel to the principal axis A.I B.I and III **C.II and III** D.I and II E.I,II and III

Rays of light incident parallel to the principal axis are reflect through the principal focus. Rays of light incident through the principal focus are reflected parallel to the principal to the principal axis. Rays of light incident through the centre of curvature are reflected along the same path.

70. A ray of light which passes through the centre of curvature of a concave mirror is reflected by

the mirror at an angle of **A.0**°B.45° C.90° D.180°

A ray of light passing through the centre of curvature is reflected along the same path i.e at angle 0°.

6. After reflection from the concave mirror, rays of light from the sun converges A.at the radius of curvature **B.at the focus** C.beyond the radius of curvature D.between the focus and the radius of curvature.

Rays of light from the sun are at infinity, hence they converge at the principal focus of the concave mirror after reflection.

71. Heat from the sun is to be used to burn a piece of paper with the aid of a concave mirror of radius of curvature 32cm.To enable the paper to start burning in the building in the shortest possible time,the paper is coloured X and positioned at a distance Y from the mirror.What are X and Y respectively? A.black and infinity B.white and 32cm **C.black and 16cm** D.black and 32cm E.white and 16cm Colouring the paper black will allow the sun to burn the paper as black body are good absorber and radiator of heat. Heat from the sun are parallel rays of light from infinity and pass through the principal focus , focal length of the mirror, $f = \frac{r}{2} = \frac{32}{2} = 16\text{cm}$. Thus, the rays of light from the sun converges at the focus and the heat from it is absorbed by the paper painted black and is burnt.

- Colouring the paper white will prevent the heat from penetrating the paper,as white bodies are good reflectors.

72. In order to produce a parallel beam of light, a lamp is placed at a distance from the concave mirror equal to A.the radius of curvature **B.half the radius of curvature** C.half the focal length D.twice the focal length E.twice the radius of curvature.

Placing a lamp at the principal focus i.e at a distance equal to the focal length or half the radius of curvature will produce parallel beam of light, so that the image is at infinity.

73. At what position will an object be placed in front of a concave mirror in order to obtain an image at infinity? A.At centre of curvature B.Between the principal focus and the centre of curvature C.At the pole of the mirror **D.At the principal focus.**

When an object is placed at the principal focus of a concave mirror the image is formed at infinity.

74. A parallel beam of light can be produced by placing a small bright source of light **A.at the principal focus of a converging mirror** B.at the principal focus of a diverging mirror C.at the centre of curvature of a converging mirror D.at the centre of curvature of a diverging mirror E.one diameter away from a circular plane mirror.

75. Where on the principal axis of a concave mirror must an object be placed to obtain its image at the centre of curvature **A.At the centre of curvature** B.Between the centre of curvature and the principal focus C.At the principal focus D.Between the focus and the pole E.At infinity.

When an object is placed at the centre of curvature of a comcave mirror a real, inverted image same size of the object is formed at the centre of curvature. Magnification is unity m=1 in this condition.

76. If an object stands at the centre of curvature of a converging mirror, the image produced will be A.inverted and virtual **C.inverted and real** C.erect and virtual D.magnified and real

The image formed if an object is at the centre of curvature is real, inverted, same size as the object and at the centre of curvature.

77. When an object is placed between the centre of curvature and the focus of a converging mirror, the image formed is A.real, dimisnihsed and erect **B.real, magnified and inverted** C.real, magnified and erect D.virtual, diminished.

An object placed between the centre of curvature and the principal focus of a concave mirror will form an image beyond the centre of curvature which is real, inverted and magnified.

78. The image formed by a concave mirror is real, inverted and magnified, when the object is placed A.at the focus B.at the centre of curvature C.beyond the centre of curvature **D.between the centre of curvature and the focus.**

79. When an object is placed very close to the pole of a concave mirror, the virtual image obtained is A.Diminished and upright B.Diminished and inverted C.Enlarged and inverted **D.Enlarged and upright.**

A virtual,erect or upright and magnified or enlarged image is formed by a concave mirror when the object is placed between the focus and

Demystified Series by Dr Timothy

the pole of the mirror.

80. An object O lies at a distance m in fornt of a concave mirror of focal length f. If m$<f$, then the final image obtained will be A.Virtual and diminished **B.Magnified and erect** C.Real and inverted D.Diminished and erect.

When object is placed at a distance m from a concave mirror which is less then its focal length f i.e between the principal focus and the mirror, a virtual,erect and magnified image is formed behind the mirror.

81. A pencil is placed vertically between a concave mirror and its focal point. The image in the mirror will be A.Real, inverted and diminished B.Virtual, inverted and enlarged C.Real, erect and enlarged **D.Virtual, erect and enlarged.**

82. A real object object is placed at a distance $u$cm in front of a curved mirror of focal length $f$cm. If the image is upright and magnified, then the mirror is A.Convex and $u=2f$ B.Concave and $u=2f$ C.Convex and $u=\frac{f}{2}$ **D.Concave and $u=\frac{f}{2}$.**

An upright or erect and magnified image is formed by a concave mirror when the objects distance is less than the focal length of the mirror i.e between the focus and the pole. An object distance $u=\frac{f}{2}$ is less than the focal length f.

83. A boy on looking into a mirror observed that his face appeared to have grown bigger. The boy must have been looking at a A.convex mirror with his face at the focus **B.concave mirror with his face between the focus and the mirror** C.concave mirror with his face at the focus D.convex mirror with his face between the focus and the mirror

Object placed between the focus and the concave mirror will form a virtual, erect and magnified image.

84. A concave mirror used as a shaving mirror should A.have a virtual focal length **B.be narrow** C.have a small curvature D.have a long focal length

Concave mirror used as shaving or dressing mirror is narrow while concave mirror used for search light and car head lamps os wide.

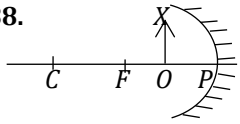
85. Where can a man place his face to get an enlarged image when using a concave mirror to shave? A.Between the centre of curvature and the principal focus B.At the principal focus **C.Between the principal focus and the pole** D.At the centre of curvature.

Shaving,dressing or make-up mirror is made of concave mirror. The person's face position during shaving should be between the focus and the pole i.e within the focal length in order to obtain a virtual, erect and magnified image.

86. The most suitable type of mirror used for the construction of a searchlight is the A.concave mirror B.convex mirror C.spherical mirror **D.parabolic mirror.**

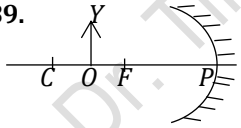
Parabolic mirror produced wide parallel beam of light.

87. What type of mirrors are capable of producing parallel beams of light such as those arising from the headlamps of a car? A.cylindrical mirrors **B.parabolic mirrors** C.spherical mirrors D.plane mirrors.

88. 

The image of the object OX formed by the mirror in the figure shown is A.real, upright and diminished **B.virtual, upright and magnified** C.virtual, upright and diminished D.real, inverted and magnified.

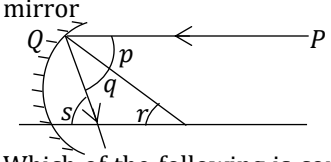
Object placed between principal focus F and the mirror, will form a virtual, erect or upright, magnified image which is behind the mirror or the right of the pole P.

89. 

The image of the object OY formed by the mirror in the figure shown is located A.between F and P B.to the right of P **C.to the left of C** D.between C and F .

Object placed between the centre of curvature C and the principal focus F,will form real, inverted and magnified image which is beyond C i.e to the left of C.

90. In the diagram below, an incident ray, parallel and close to the prin cipal axis of a converging mirror, passes thrpugh the focus F after reflection. R is the centre of curvature of the mirror



Which of the following is correct? A.$q=s$ B.$p=s$ C.$r=s$ D.$p=q=s$ **E.$p=q=r$.**

Angle of incidence $i=p$, angle of reflection $r=q$. Law of reflection : angle of incidence equal angle of reflection, $i=r$. $\therefore p=q$. $p=r$ (alternate angles). $\therefore p=q=r$. $p+q=s$.

91. Which of the following is true for the image formed by a convex mirror? I.The image is always virtual II.The image is always erect III.The image lies between the pole and focus IV.The image is never magnified V.The focal length is negative A.I B.I and II C.II and III D.I,II,III and IV **E.All five.**

The image formed by a convex mirror is always virtual, erect and diminished i.e cannot be magnified. It is formed between the pole abd the focus. The focal length of a convex mirroir is negative.

92. Which of the following best describes the images formed by a convex mirror? **A.Virtual,**

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erect and diminished B.Real, erect and diminished C.Virtual, inverted and magnified D.Real, inverted and magnified E.Virtual, inverted and diminished.

93. Convex mirrors are used as driving mirriros because images formed are **A.erect, virtual and diminished** B.erect, real and dimisnised C.erect, virtual and magnified D.inverted, virtual and diminished.

94. A convex mirror is used as a driving mirror because I.Its image is erect II.It has a large field of view III.It has a long focal length. Identify the correct statement(s) A.I **B.I and II** C.I and III D.II and III E.I,II and III.

Convex mirror is used as a car driving mirror because its forms an upright or erect image of objects and has a wide or large field of view.

95. Which of the following properties make the convex mirrors useful as driving mirror? I.the image is real II.the image is erect III.It has a wide field of view IV.the image is magnified A.I and III **B.II and III** C.I,II and III D.I,II and IV.

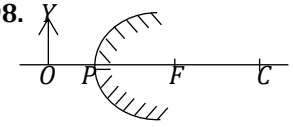
96. Which of the following mirrors can be used as a driving mirror? I.A plane mirror II.A converging mirror III.A diverging mirror A.I B.II C.I and II **D.I and III** E.II and III

A plane mirror and a convex or diverging mirror can be used as driving mirror but convex or diverging mirrors are mostly used due to its wider field of view..

97. Which of the following is a advantage of the diverging mirror over the plane mirror when used as a driving mirror? **A.the field of view is wider** B.the images are magnified C.the correct distance of the car behind can be estimated D.the image is virtual E.all the images formed are erect

A convex or diverging mirror has a wider field of view than a plane mirror. A convex or diverging mirror forms a diminished image, hence the correct position or size of the object cannot be estimated i.e a disadvantage of convex mirror. Plane mirrors form images of same size or distance as the object, hence the correct position or size of an object can be estimated i.e an advantage of plane mirror. Images in plane mirror are laterally inverted i.e a disadvantage of plane mirror. Both convex and plane mirror forms erect images.

98.



The image of the object OX formed by the convex mirror in the figure shown is A.at C B.between F and C **C.between P and F** D.between O and P

The image formed by a convex mirror is always virtual,erect or upright and diminished and is always formed between the principal focus *F* and the pole *P*, at all positions of the object.

99. An object is placed 12cm in front of a concave mirror of radius 32cm.The image formed is **A.virtual and 48cm behind the mirror** B.real and 48cm in front of the mirror C.virtual and 19.2cm behind the mirror D.real and 19.2cm in front if the mirror

Object distance u=12cm, radius of curvature r=32cm, $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$, $v=\frac{uf}{u-f}$, $f=\frac{r}{2}$

$f=\frac{r}{2}=\frac{32}{2}=16\text{cm}$, $v=\frac{uf}{u-f}=\frac{12\times16}{12-16}=\frac{192}{-48}$,

$v=-48\text{cm}$. The image is virtual, as image distance is negative i.e 48cm behind the mirror.

100. An object is placed 10cm in front of a concave mirror of focal length 15cm. What is the position and nature of the image formed? A.30cm and virtual B.6cm and real C.6cm and virtual D.30cm and real.

u=10cm, f=15cm, $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$, $v=\frac{uf}{u-f}$,

$v=\frac{10\times15}{10-15}=\frac{150}{-5}=-30\text{cm}$, the image is 30cm from the lens as virtual as the image distance is negative.

101. An object is placed 10cm in front of a concave mirror of focal length 15cm. The image formed is **A.virtual, erect and magnified** B.real, inverted and magnified C.virtual, erect and diminished D.virtual, inverted and magnified E.real, inverted and diminished.

u=10cm, f=15cm, $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$, $v=\frac{uf}{u-f}$,

$v=\frac{10\times15}{10-15}=\frac{150}{-5}=-30\text{cm}$, the image is virtual as the image distance is negative. Virtual images are always erect. Virtual image in a concave mirror is magnified or magnification $m=\frac{v}{u}=-\frac{30}{10}=-3$ i.e magnified. Magnification is negative as the the image is virtual

102. An object which is 5cm high is placed vertically 10cm in front of a concave mirror. If this object produces an image 40cm from the mirror, the height of the image is A.0.75cm B.4.00cm C.8.00cm **D.12.00cm.**

Object height H_O=3cm, u=10cm, v=40cm, image height H_I=?, $m=\frac{H_I}{H_O}=\frac{v}{u}$,

$\frac{H_I}{3}=\frac{40}{10}$, $H_I=\frac{40\times3}{10}=12\text{cm}$.

103. A concave mirror of radius of curvature 40cm forms a real image twice as large as the object. The object distance is A.60cm B.40cm **C.30cm** D.10cm.

r=40cm, $f=\frac{r}{2}=\frac{40}{2}=20\text{cm}$, m=2, u=?,

$m=\frac{f}{u-f}$, $2=\frac{20}{u-20}$, $u-20=\frac{20}{2}=10$,

$u=10+20=30\text{cm}$. Alternatively,

$m=\frac{r}{2u-r}$, $2=\frac{40}{2u-40}$, $2u-40=\frac{40}{2}=20$,

$2u=20+40=60$, $u=\frac{60}{2}=30\text{cm}$.

104. Which of the following expressions gives the linear magnification produced by a concave mirror of radius of curvature r, if u and v are the

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object and image distance respectively? A.$\frac{v}{r}-1$ **B.$\frac{2v}{r}-1$** C.$\frac{u}{r}-r$ D.$\frac{2u}{r}-1$

$m=\frac{v}{f}-1$, $f=\frac{r}{2}$, $m=\frac{v}{r/2}-1=\frac{2v}{r}-1$.

105. A dentist obtains a linear magnification of 4 of a hole in a tooth by placing a concave mirror at a distance of 2.0cm from the tooth. The radius of curvature of the mirror is A.5.3cm **B.3.2cm** C.2.7cm D.1.6cm.

m=4, u=2.0cm, r=?, $m=\frac{r}{2u-r}$, $4=\frac{r}{2\times2-r}$,

$4=\frac{r}{4-r}$, $4(4-r)=r$, $16-4r=r$,

$r+4r=16$, $5r=16$, $r=\frac{16}{5}=3.2$cm.

106. A real image three times the size of an object is formed 24cm from a converging mirror. What is the focal length of the mirror? A.8cm B.16cm **C.6cm** D.12cm

m=3, v=24cm, f=?, $m=\frac{v}{f}-1$, $3=\frac{24}{f}-1$,

$\frac{24}{f}=3+1=4$, $f=\frac{24}{4}=6$cm.

107. An object placed 10cm from a concave mirror of focal length 5cm would have its image A.at infinity B.10cm from the focus **C.at the centre of curvature** D.at the radius of curvature.

u=10cm, f=5cm, v=?, $v=\frac{uf}{u-f}=\frac{10\times5}{10-5}=\frac{50}{5}$,

$v=10$cm. Image distance equals object distance, $v=u=10$cm. Hence the object is placed at the centre of curvature and the image will also be formed at the centre of curvature.

108. An object of height 5cm is placed at 20cm from a concave mirror of focal length 10cm. The image height is A.20cm B.15cm C.10cm **D.5cm**

Object height H_O=5cm, u=20cm, f=10cm, image height H_I=?, $m=\frac{H_I}{H_O}=\frac{f}{u-f}$,

$\frac{H_I}{5}=\frac{10}{20-10}$, $\frac{H_I}{5}=\frac{10}{10}=1$, $H_I=5$cm.

109. Calculate the position of an object placed in front of a convex mirror having focal length of 12cm such that an image is formed 6cm away from the mirror A.3.0cm B.4.0cm C.9.0cm **D.12.0cm.**

f=−12cm (negative focal length), v=−6cm (convex mirror forms virtual image), u=?,

$u=\frac{vf}{v-f}=\frac{-6\times-12}{-6-(-12)}=\frac{72}{-6+12}=\frac{72}{6}=12$cm.

110. The driving mirror of a car has a radius of curvature of 1m. A vehicle behind the car is 4m from the mirror. Find the image distance behind the mirror A.$\frac{8}{7}$m **B.$\frac{4}{9}$m** C.$\frac{9}{2}$m D.$\frac{4}{7}$m

Car driving mirror is a convex mirror. r=1m, $f=\frac{r}{2}=\frac{1}{2}=-0.5$m (negative focal length), u=4m, $v=\frac{uf}{u-f}=\frac{4\times-0.5}{4-(-0.5)}=\frac{-2}{4+0.5}=\frac{-2}{4.5}$,

$v=\frac{-20}{45}=-\frac{4}{9}$,image distance is negative as the image is virtual. Image distance behind the mirror $v=\frac{4}{9}$m.

111. The image of an object is located 6cm behind a convex mirror. If its magnification is 0.6, calculate the focal length of the mirror? A.3.75cm B.6.60cm C.10.00cm **D.15.00cm.**

v=−6cm(virtual image), m=−0.6 (virtual image), f=?, $m=\frac{v}{f}-1$, $-0.6=-\frac{6}{f}-1$,

$-\frac{6}{f}=-0.6+1$, $-\frac{6}{f}=0.4$, $f=-\frac{6}{0.4}$,

$f=-15$cm, focal length of a convex mirror is negative.

112. A convex mirror produces an image which is one-quarter the size of an object placed in front of it. If the radius of curvature of the mirror is 32cm,the distance between the object and its image is **A.60cm** B.48cm C.32cm D.12cm

$m=-\frac{1}{4}$ (virtual image), r=32cm, $f=\frac{r}{2}=\frac{32}{2}$,

f=−16cm (convex mirror), u=?, v=?,

$m=\frac{f}{u-f}$, $-\frac{1}{4}=-\frac{16}{u-(-16)}$, $-\frac{1}{4}=-\frac{16}{u+16}$,

$-1(u+16)=4\times-16=-64$,

$-u-16=-64$, $-u=-64+16=-48$,

$u=48$cm. $m=\frac{v}{u}$, $v=Mu=-\frac{1}{4}\times48$, $v=-12$cm (virtual image), $v=12$cm.

Distance between an object an its image, $D=u\pm v$, it is positive for virtual image and negative for real image. $D=u+v=48+12=60$cm.

113. A candle is placed 20cm from a concave mirror of focal length 10cm. The distance between the candle and its image os A>40cm B.20cm C.10cm **D.0cm.**

u=20cm, f=10cm, v=?, $v=\frac{uf}{u-f}=\frac{20\times10}{20-10}$,

$v=\frac{200}{10}=20$cm. Distance between the candle and its image $D=u-v$ (real image),

$D=20-20=0$cm.

114. An object is placed at a distance of 15cm from a concave mirror of focal length 10cm. The distance between the object and its image is **A.15cm** B.30cm C.45cm D.infinite.

u=15cm, f=10cm, v=?, $v=\frac{uf}{u-f}=\frac{15\times10}{15-10}$,

$v=\frac{150}{5}=30$cm. Distance between the object and its image $D=v-u$ (real image and $v>u$), $D=30-15=15$cm.

Jamb past questions on Light wave and reflection of light :

[1978/45,1979/16,18,32,1981/12,1982/7,18, 23,32,36,1984/6,1985/30,32,1986/33,1988/31 ,32,1989/26,27,1990/23,24,25,1991/25,26, 1992/24,25,26,1993/25,26,1995/35,1997/34, 1997/34,1998/29,30,31,2001/11,14,2002/19, 2003/21,23,2004/2,2005/34,37,2006/35,37, 2007/5,2008/30,32,33,2009/31,32,33,2100/29 ,2011/30,2012/30,31,32,33,2013/29,2014/33, 2015/49]

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Chapter 18 – Refraction of light

● Refraction of light rays from plane surface

- Refraction is the bending of light (change in direction) when it passes through a **transparent media** due to difference in speed and wavelength as it passes from one medium to another. **Frequency is constant during refraction.**

- Laws of refraction:

1.The incident ray,the refracted ray and the normal at the point of incidence, all lie in the same plane.

2.Snell's law – The ratio of sine of angle of incidence to sine of angle of refraction is constant for a given pair of media.This constant is the refractive index $\boldsymbol{n}$ of the medium.

Snell's law : $\boldsymbol{n = \frac{\sin i}{\sin r}}$

- Refractive index $\boldsymbol{n}$ of a medium is the extent of refraction of light in that medium with respect to another medium.

$\boldsymbol{n = \frac{\sin i}{\sin r} = \frac{speed\ of\ light\ in\ medium\ 1}{speed\ light\ in\ medium\ 2} = \frac{v_1}{v_2}}$.

$\boldsymbol{v \propto \lambda}$**, as frequency $\boldsymbol{f}$ is constant thus,**

$\boldsymbol{n = \frac{wavelength\ of\ light\ \ in\ medium\ 1}{wavelength\ of\ light\ in\ medium\ 2} = \frac{\lambda_1}{\lambda_2}}$.

- For a light moving from air to glass : The refractive index of glass n_g is given by –

$\boldsymbol{n_g = n(\text{air-glass}) = n_{(a-g)} = \frac{speed\ in\ air}{speed\ in\ glass}}$,

$\boldsymbol{n_g = n_{(a-g)} = \frac{v_a}{v_g}}$.

- For a ray of light travelling from medium of refractive index n_1 to another medium of refractive index n_2, the refractive index of medium 1 in respect to medium 2 $n_{(1-2)}$ is given by – $\boldsymbol{n_{(1-2)} = \frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}}$.

- Based on the principle of reversibility of light – $\boldsymbol{n_{(1-2)} = \frac{1}{n_{(2-1)}}}$.

- When a ray of light is refracted from one medium to another with parallel boundaries or interface e.g (air-glass-water), then the product the refractive index and the sine of angle of incidence of the light in each media are equal or constant. $\boldsymbol{n \sin i = \text{constant.}}$

$\boldsymbol{\therefore n_a \sin i_a = n_g \sin i_g = n_w \sin i_w}$.

- The higher the optical density or refractive index of a medium, the lower the speed of light (and wavelength) in that medium. Hence when light travels between two media of different densities, it is either refracted away or towards the normal depending on the densities or refractive index of the media. Thus :

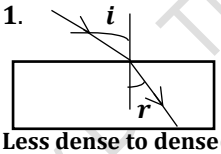
1.When a ray of light moves from a less dense medium to a denser medium or lower refractive index to that with higher refractive index, it bends towards the normal and the angle of incidence is greater than the angle of refraction e.g a ray of light moving from water ($n = 1.33$) to glass ($n = 1.5$).

2.When a ray of light moves from a denser medium to a less dense medium or higher refractive index to that with lower refractive index, it bends towards the normal and the angle of incidence is less than the angle of refraction e.g a ray of light moving from glass ($n = 1.5$) to air ($n = 1$).

3.If the incident ray is incident normally i.e at angle 90° to the side of the block the angle of incidence is zero ($i = 0°$), it passes straight through the glass, emerging at the other face without a change in direction or displacement.

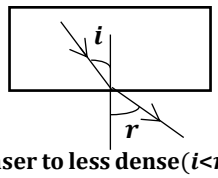
- The diagrams below illustrates the change in direction of light after refraction at the interface or boundary between glass and air –

1.



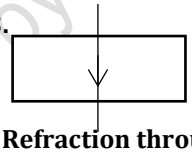
Less dense to dense(i>r).

2.



Denser to less dense(i<r).

3.



Incident ray along the normal and emerges without being displaced.

- Refraction through a glass block –

1.The angle of refraction r increases proportionately with the angle of incidence i.

2.The angle of incidence i at the first face is equal to the angle of emergence e at the opposite face.

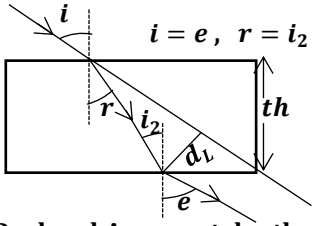
3.The angle of refraction r at the first face is equal to the angle of incidence at the second or opposite face i_2.

4.The incident ray is parallel to the emergent ray, hence the emergent is not deviated from the incident ray and the total angle of deviation of is 0°.

- The **angle of deviation/displacement $\boldsymbol{D}$** after refraction in the first face, $\boldsymbol{D = i - r}$.

5.The incident ray changes from the original path at any of the faces and it is called **Lateral deviation or displacement d_L ,** $\boldsymbol{d_L = \frac{th \sin(i-r)}{\cos r}}$.

th=thickness of the block, i=angle of incidence, r=angle of refraction.



- Real and Apparent depth – It is observed that a mark on a piece of paper under a glass block

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appears nearer to top of the glass block when viewed. The actual depth of the glass block is called real depth while the depth as it appears to the eye is called the appararent depth. $\boldsymbol{n = \frac{Real\ depth}{Apparent\ depth} = \frac{R}{A}}$. Real depth of a glass block is equal to its thickness, $\boldsymbol{R = th.}$

- The apparent depth is displaced from the real depth by a certain value which is equal to the **Lateral or appararent displacement** $\boldsymbol{d}$.

$\boldsymbol{R = A + d,\ A = R - d.}$ **Thus,** $\boldsymbol{n = \frac{R}{A} = \frac{R}{R-d}}$,

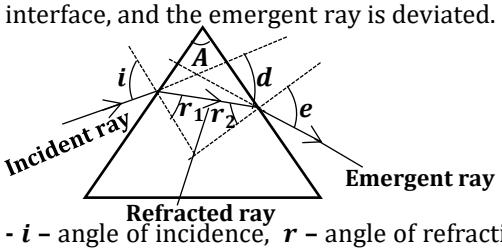
$\boldsymbol{d = \frac{R(n-1)}{n} = R\left(1 - \frac{1}{n}\right)}$.

- **Percentage of the lateral or apparent displacement** $= \boldsymbol{\frac{lateral\ displacement}{Real\ depth} \times 100}$,

$\boldsymbol{\%d = \frac{d}{R} \times 100 = \frac{R-A}{R} \times 100.}$

- Refraction through triangular block/prism –

A ray of light passing through is refracted twice i.e air-prism interface and opposite prism-air interface, and the emergent ray is deviated.



- $\boldsymbol{i}$ – angle of incidence, $\boldsymbol{r}$ – angle of refraction,$\boldsymbol{A}$ – refracting angle. $\boldsymbol{A=2r.}$ **For an equilateral traingular prism,** $\boldsymbol{A = 60°}$.

- There is always an angle between the incident and emergent rays, which is the angle of deviation $\boldsymbol{d}$. For equilateral prism, total angle of deviation. $\boldsymbol{D = d_1 + d_2, d_1 = (i_1 - r_1)}$,

$\boldsymbol{d_2 = (i_2 - r_2)}$, $\boldsymbol{D = (i_1 - r_1) + (i_2 - r_2)}$,

$\boldsymbol{r_1 + r_2 = A,\ \ D = (i + e) - A.}$

- The deviation of a ray of light passing from air into triangular prism is determined by – **i.**angle of incidence**ii.**refracting angle(A) of the prism **iii.**refractive index of the material of the prism.

- At minimum deviation $\boldsymbol{D_{min}}$ **:**

1.The light rays passes symmetrically through the prism i.e the rays within the prism are parallel to the base of the prism.

2.Angle of incidence equals the angle of emergence, $\boldsymbol{i = e}$.

3.Angle of refraction in the first face r_1 is equal to the angle of incidence in the second face r_2.

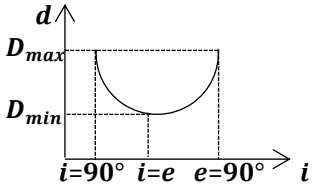
4.Refracting angle $\boldsymbol{A = r_1 + r_2 = 2r.}$

- Minimum deviation is same as the angle through which a ray is least deviated.

- The refractive index of the glass prism n – $\boldsymbol{i = \frac{1}{2}(A + D_{min}),\ \ D_{min} = 2i - 2r = 2i - A}$,

$\boldsymbol{r = \frac{1}{2}A}$, $\boldsymbol{n = \frac{\sin i}{\sin r} = \frac{sin\frac{1}{2}(A+D_{min})}{sin\frac{A}{2}}}$.

- As the angle of incidence increases, the angle of deviation first decreases to a minimum value called **minimum deviation** and then increases giving rise to a curved graph. **At minimum deviation** $\boldsymbol{D_{min}}$ **the angle of incidence equals the angle of emergence** $(\boldsymbol{i = e})$**. At maximum deviation** $\boldsymbol{D_{max}}$ **the angle of incidence and angle of emergence are equal to 90°.**



- Critical angle and Total internal reflection –

- When a ray travels from a less dense to a dense medium,the ray is always reflected and refracted.

- When a ray travels from a dense to a less dense medium – **I.**with a small angle of incidence,the ray is strongly refracted and weakly reflected i.e decrease in brightness or intensity or amplitude. **II.**as the angle of incidence, the ray becomes weakly refracted and strongly reflected. **III.**when the angle of incidence is increased such that the angle of refraction is 90°, the angle of incidence is called **critical angle** and the ray is strongly refracted and weakly reflected. **IV.**with angle of incidence greater than the critical angle,the ray is not refracted but strongly reflected inside the dense medium.

- **Critical angle** $\boldsymbol{C}$ is the angle of incidence at which the angle of refraction is 90° when light passes from denser to less dense medium.

- The refractive index of the medium is given by : $\boldsymbol{n = \sin C}$ (When light travel from dense to less dense medium) OR $\boldsymbol{n = \frac{1}{\sin C}}$ (When light travel from dense to less dense medium).

- When the emergent ray grazes the surface of the prism it means the ray makes an angle of 90° with the normal (angle of refraction is 90°). Thus, the the angle of incidence equals the critical angle.

- Total internal reflection is the reflection of an incident ray at the interface between the medium of incidence and another medium of lower refractive index when the angle of incidence exceeds the critical angle.

- For glass of refractive index 1.5, total internal reflection occurs when the angle of of incidence is greater than 42°, its critical angle.

- For total internal reflection to occur –

1.Light must pass from a dense medium to a less dense medium.

2.The angle of incidence in the denser medium must be greater than the critical angle.

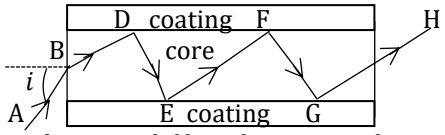
- Application of Total internal reflection;

1.Submarine periscope – Right-angled isosceles prisms (reflecting prism) are used to replace plane mirrors in simple periscopes, to prevent occurence of multiple images, tarnishing and deterioration associated with plane mirrors.

2.Optical fibres – Thin flexible glass tube e.g silica glass,fluoride glass,phosphate glass or chalcogenide glass,of high refractive index

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through which light signals can travel over a long distance by means of total internal reflection.It used in telecommunication to transmit signals over longer distances at higher band widths(data base) than metal wires which were previously used.



In the optical fibre digragram above, ray AB is refracted as BD, to meet the interface between the core and coating at an angle freater than the critical angle. Total internal reflection occurs repeatedly and emerge as ray GH.

3.Appearance of shinning object e.g diamond

4.Prism binoculars oe short telescope – Device for looking at small distant object by magnifying the objects uising lenses for eye.

5.LASER(Light Amplification by Stimulated Emission of Radiation) – Device which emits light through a process of optical modulation/ amplification based on a stimulation of eledtromagnetic signal. **Uses of Laser : i.**Communication:hologram production,optical fibres,space communication,data transfer,VCD and DVD playing. **ii.**Medicine:Servical cancer surgery,boring holes in skull,monitoring glucose level for diabetic patient **iii.**Militray/security: generation of isotopes fro nuclear weapons, aircrafts and satellites,guidiance system missles.

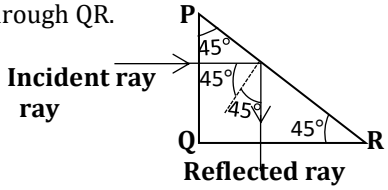
- Total Internal reflection in nature –

1.Fish's eye view – Fishes or a swimmer moving on his back under water,has its field of view widened and can see everything above water surface at a cone range twice the critical angle, while objects in water and outside this range are seen by total internal reflection.

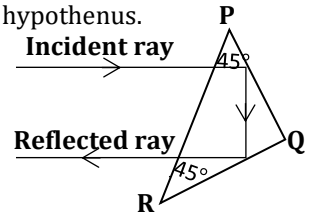
2.Mirrage – An optical illusion caused by a combined effect of refraction(majorly) and total internal reflection,whereby shining pool of water is seen on a tarmac road on a sunny day and disappears on getting close to the spot.

- Totally reflecting prism – They are right angled isosceles triangular prism.

1.Turning of incident ray through 90° : When a ray is incident normally (at right angles) on one side of the prism PQ **(but not the hypotenuse)**, the ray is totally internally replected as it strikes the hypothenuse PR at angle 45° greater than the critical angle of glass,42° and emerges normally through QR.



2.Turning of incident ray through 180° or inverting a beam of light: When a ray is incident normally on the hypothenuse PR of a prism, the ray is totally reflected twice, at the interface PQ and QR and emerge normally through the hypothenus.



● Refraction from spherical surfaces/ lenses

- Convex or Converging lens is thickest in middle and thinnest in edges.

- Concave or Diverging lens is thinnest in middle and thickest in edges.

- Real images in lens is formed on opposite side of the lens i.e behind the lens while Virtual image is formed in same side of the lens with the object i.e infront of the lens.

- Images formed by a convex lens :

1.Object at infinity – image at F, real,inverted and diminished **e.g burning glass.**

2.Object beyond $2F$ – image between F and $2F$, real,inverted and diminished e.g lens camera.

3.Object at $2F$ – image at $2F$,real,inverted and same size **e.g copying or duplicating camera**.

4.Object between F and $2F$ – image beyond $2F$, real,inverted and magnified **e.g slide projectors and objective lens of microscope**.

4.Object at F – image at infinity, real, inverted and magnified **e.g theatre and stadium spot lights.**

5.Object between F and opical centre C of the lens – image is beyond object i.e same side of the lens with object, virtual,erect or upright,magnified e.g **simple microscope or magnifying glass.**

- Images formed by concave lens :

1.Image is always virtual,erect or upright, diminished **(VED)**, at all objects positions.

2.Image is always between the principal focus and the lens.

- Graphical representation, lens formula and magnification formula is same as that of mirrors (**Page 207)**.

- The focal length of convex lens is positive while that of concave lens is negative.

- Image distance is positive for real images (inverted) and negative for virtual images (erect or upright).

- Distance between an object and its image is given by – $\boldsymbol{D = v \pm u}$**. It is positive for real, inverted images and negative for virtual, erect images.** $\boldsymbol{D = u - v}$**(for concave lens).**

- Spherical aberration in lenses – The inability of a lens to refract wide beam or rays of light that

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are far from the principal axis through a common focus but through a point nearer than the focus. It can be minimized by allowing light to pass through the optical centre ***C*** (centre of the lens) only.

- Rules for constructing ray diagrams in lens:

1.A ray from the object, parallel to the principal axis emerges through the principal focus after refraction by the lens.

2.A ray from the object through the principal focus emerges parallel to the principal axis after refraction by the lens.

3.A ray from the object passing through the optical centre of the lens passes through undeviated.

- The existence conjugate points one on either side of a lens is based on principle of reversibility of light.

- The magnifying power of a lens depends on its focal length.

- The shorter the focal length the greater the power of the lens and the longer the focal length lesser the power of the lens.

- Power of lens *P* is equal to the reciprocal of its focal length in meters**. Power of a lens is in Dioptres *D*.** $P = \frac{1}{f} = \frac{u+v}{uv}$ **.**

- Power of a convex lens is positive while that of a concave lens is negative.

Examples :

1.When a ray of light passes from water to glass,it is A.bent away from the normal **B.bent towards the normal** C.totatally internally reflected D.absorbed by the glass E.broken into pure spectrum.

When light passes from less dense medium or medium with low refractive index(water) to a dense medium or medium with high refractive index(glass),it bends towards the normal and the angle of incidence in water is greater than the angle of refraction in glass.

2. A ray of light which strikes a glass slab normal incidence passes through the slab **A.undeviated and undisplaced through the slab** B.deviated and undisplaced at a lower speed C.deviated and displaced at a lower speed D.undeviated and displaced at a faster speed

A ray incidente normally on a glass slab and will pass through undeviated and undisplaced emerging normally at the other face.

3. When a ray of light is incident normally on air-glass interface, its angle of refraction is **A.0°** B.42° C.45° D.60° E.90°

When a ray of light is incident normally, the angle of incidence and the angle of refraction is 0°.

4. When light passes through two media x and y of refractive index 1.51 and 1.33 respectively, the speed of light in A.x is same as in y B.x and y is same as in vacuum C.is higher than in y**D.y is higher than in x .**

The speed of light is high is lessss dense medium (low refractive index) than in dense medium (high refractive index), optical density or refractive index varies inversely with the speed of light. Hence, speed of light in y ($n = 1.33$) is higher than in x ($n = 1.51$).

5. Light of velocity 3.0×10^8ms^{-1} is incident on a material of refractive index n. If the velocity is reduced to 2.4×10^8ms^{-1} in the material, what is n? A.2.33 B.2.25 C.1.33 **D.1.25**

Velocity in air v_a=3.0×10^8ms^{-1}, velocity in medium v_m=2.4×10^8ms^{-1}, refractive index of the medium n_m=?, $n_m = n_{(a-m)} = \frac{v_a}{v_m}$,

$n_m = n_{(a-m)} = \frac{3.0\times10^8}{2.4\times10^8} = 1.25.$

6. If the speed of light in water is 2.26×10^8ms^{-1} and the velocity of light in air is 3.0×10^8ms^{-1}. Calculate the refractive index of water A.1.33 B.1.23 C.1.45 D.1.56

v_w=2.26×10^8ms^{-1}, v_a=3.0×10^8ms^{-1}, n_w=?,

$n_w = n_{(a-w)} = \frac{v_a}{v_w} = \frac{3.0\times10^8}{2.26\times10^8} = 1.327 = 1.33.$

7. A ray of light passes through glasss of refractive index 1.50. If the speed of light in vacuum is 3.00×10^8ms^{-1}, determine the spoeed of light in glass A.1.50×10^8ms^{-1} **B.2.00×10^8ms^{-1}** C.2.25×10^8ms^{-1} D.4.50×10^8ms^{-1}

v_a=3.00×10^8ms^{-1}, n_g=1.50, v_g=?,

$n_g = n_{(a-g)} = \frac{v_a}{v_g}$, $1.50 = \frac{3.00\times10^8}{v_g}$,

$v_g = \frac{3.00\times10^8}{1.50}$ =2.00×10^8ms^{-1}.

8. The speed of light in vacuum is 3.0×10^8ms^{-1} If the refractive index of transpartent liquid is 4/3, then the speed of light in the liquid is A.0.44×10^8ms^{-1} B.2.25×10^8ms^{-1} C.3.0×10^8ms^{-1} D.4.0×10^8ms^{-1} E.4.33×10^8ms^{-1}

v_v=3.0×10^8ms^{-1}, n_L=4/3, v_L=?,

$n_L = \frac{v_v}{v_L}$, $v_L = \frac{v_v}{n_L} = \frac{3.0\times10^8}{4/3} = \frac{3.0\times10^8\times3}{4}$,

v_L =2.25×10^8ms^{-1}.

9. Light of frequency 6.0×10^{14}Hz travelling in air is transmitted through glass of refractive index 1.5m, calculate the frequency of light in the glass A.4.0×10^{14}Hz **B.6.0×10^{14}Hz** C.7.5×10^{14}Hz D.9.0×10^{14}Hz.

Refraction occurs when light travels through different media of different densities or refractive indices. Frequency of light in both media is the same, as frequency is constant during refraction. Thus, $f_a = f_g$ =6.0×10^{14}Hz.

10. A light wave of frequency 5×10^{14}Hz moves through water which has a refractive index of 4/3. Calculate the wavelength in water if the vcelocity of light in air is 3.0×10^8ms^{-1} A.6.0×10^{-7}m **B.4.5×10^{-7}m** C.2.2×10^{-6}m D.1.7×10^{-6}m

Frequecy of light in air is same as that in water.

f_a =f_w = 5×10^{14}Hz, n_w=4/3, v_a=3.0×10^8ms^{-1},

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λ_w=?, $n_w = \frac{v_a}{v_w} = \frac{v_a}{\lambda_w f_w}$,
$\frac{4}{3} = \frac{3.0\times10^8}{5\times10^{14}\lambda_w}$, $\lambda_w = \frac{3.0\times10^8\times3}{5\times10^{14}\times4}$ =4.5×10^{-7}m.

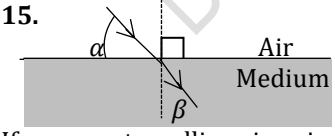
11. Light of wavelength 5000×10^{-8}cm travels in free space with a velocity of 3×10^{8}ms^{-1}. What is its wavelength in glass of refractive index 1.5? **A.3333×10^{-8}cm** B.5000×10^{-8}cm C.6666×10^{-8}cm D.7500×10^{-8}cm
λ_a=5000×10^{-8}cm, v_a=3.0×10^{8}ms^{-1}, n_g=1.5, λ_g=?,
$n_g = \frac{v_a}{v_g} = \frac{\lambda_a}{\lambda_g}$, $1.5 = \frac{5000\times10^{-8}}{\lambda_g}$,
$\lambda_g = \frac{5000\times10^{-8}}{1.5}$ =3333×10^{-8}cm.

12. A glass plate 0.9cm thick has a refractive index of 1.5. How long does it take for a pulse of light to pass through the plate?
A.3.0×10^{-11}s B.4.5×10^{-11}s C.3.0×10^{-10}s **D.4.5×10^{-10}s** (c=3.0×10^{8}ms^{-1})
th=0.9cm=0.009m, n_g=1.5, v_a=3.0×10^{8}ms^{-1},
$v_g = \frac{th}{t}$, t=time taken for the pulse to pass through the glass plate. $n_g = \frac{v_a}{v_g}$, $v_g = \frac{v_a}{n_g}$,
$v_g = \frac{d}{t} = \frac{th}{t} = \frac{v_a}{n_g}$, $\frac{0.009}{t} = \frac{3.0\times10^8}{1.5}$, $t = \frac{0.09\times1.5}{3.0\times10^8}$,
$t = 4.5\times10^{-10}$s.

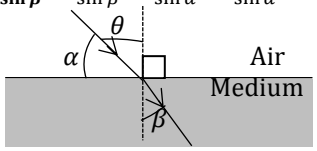
13. A ray of light passes from air to water to glass to air. Given that the refractive index for light passing from air to water is $\frac{4}{3}$ and air to glass is $\frac{3}{2}$, calculate the refractive index of glass relative to water A.0.50 B.0.67 C.0.75 **D.1.13.**
$n_{(a-w)}$=n_w= $\frac{4}{3}$, $n_{(a-g)}$=n_g= $\frac{3}{2}$, refractive index of glass relative to water $n_{(g-w)}$=?,
$n_{(w-g)} = \frac{n_g}{n_w} = \frac{3/2}{4/3} = \frac{3}{2}\times\frac{3}{4} = \frac{9}{8} = 1.13$.

14. The absolute refractive indices of glass and water are 3/2 and 4/3 repsectively. The refractive index at the interface when a ray travels from water to glass is A.1/2 B.8/9 **C.9/8** D.17/12 E.2 .
n_g= $\frac{3}{2}$, n_w= $\frac{4}{3}$, $n_{(w-g)}$=?, $n_{(w-g)} = \frac{n_g}{n_w}$,
$n_{(w-g)} = \frac{n_g}{n_w} = \frac{3/2}{4/3} = \frac{3}{2}\times\frac{3}{4} = \frac{9}{8}$.

15.

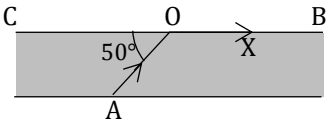


If a ray travelling in air is incident on a transparent medium as shown in the diagram the refractive index of the medium is given as **A.$\frac{\cos\alpha}{\sin\beta}$** B.$\frac{\sin\alpha}{\sin\beta}$ C.$\frac{\cos\beta}{\sin\alpha}$ D.$\frac{\sin\beta}{\sin\alpha}$.

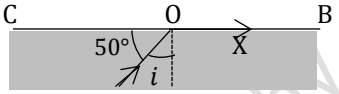


Angle of incidence$i = \theta =$ 90−α, angle of refraction $r = \beta$, refractive index of the medium n=?, $n = \frac{\sin i}{\sin r} = \frac{\sin(90-\alpha)}{\sin\beta}$, $\sin(90-\alpha) = \cos\alpha$, $n = \frac{\cos\alpha}{\sin\beta}$.

16.



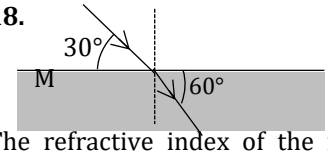
The diagram above shows an incident ray AO inclined at angle of 50° to the interface CB. The refracted ray OB is found to lie along the surface. What is the refractive index of the medoium X repect to air? A.$\frac{\sin 50°}{\sin 40°}$ B.$\frac{\sin 40°}{\sin 50°}$ C.$\frac{\sin 90°}{\sin 50°}$ **D.$\frac{\sin 40°}{\sin 90°}$** E.$\frac{\sin 90°}{\sin 40°}$.



50° + i =90°, i = 90° − 50° = 40°, r = 90°,
$n = \frac{\sin i}{\sin r}$, $n = \frac{\sin 40°}{\sin 90°}$.

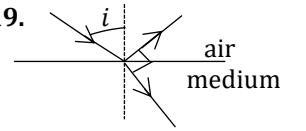
17. If the refractive index of glass is $\sqrt{2}$, calculate the angle of incidence of a ray of light travelling from air to glass when the angle of refraction is 30° ($\sin 30° = \frac{1}{2}$, $\sin 45° = \frac{\sqrt{2}}{2}$, $\sin 60° = \frac{\sqrt{3}}{2}$, $\sin 90° = 1$) A.0° B.30° C.45° D.60° E.90°
n_g=$\sqrt{2}$, r=30°, i=?, $n_g = \frac{\sin i}{\sin r}$,
$\sqrt{2} = \frac{\sin i}{\sin 30°}$, $\sin i = \sqrt{2}\sin 30°$, $\sin i = \sqrt{2}\times\frac{1}{2}$,
$\sin i = \frac{\sqrt{2}}{2}$, $i = \sin^{-1}\frac{\sqrt{2}}{2} = 45°$.
A.$\frac{\cos\alpha}{\sin\beta}$ B.$\frac{\sin\alpha}{\sin\beta}$ C.$\frac{\cos\beta}{\sin\alpha}$ D.$\frac{\sin\beta}{\sin\alpha}$.

18.

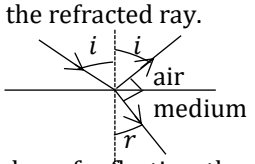


The refractive index of the medium M in the diagram above is A.2/$\sqrt{3}$ B.1/$\sqrt{3}$ C.2$\sqrt{3}$ **D.$\sqrt{3}$.**
i = 90−30 = 60°, r = 90−60 = 30°, n_M=?,
$n_M = \frac{\sin i}{\sin r}$, $n_M = \frac{\sin 60°}{\sin 30°} = \frac{\sqrt{3}/2}{1/2} = \frac{\sqrt{3}}{2}\times 2 = \sqrt{3}$.

19.



From the digarm above, the correct equation for the refractive index n of the medium is
A.$n = \tan i$ B.$n = \frac{\sin i}{\sin 90°}$ C.$n = \frac{\sin i}{\cos(90°-i)}$
D.$n = \sin 90°$.
The light ray in thediagram is **polarized,** as the reflected ray is perpendicular or at right angles to the refracted ray.



By law of reflection, the angle of incidence equals the angle of reflection. r=angle of refraction. i + 90° + r = 180°, r = 90° − i ,

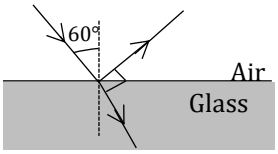
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$n = \frac{\sin i}{\sin r} = \frac{\sin i}{\sin(90° - i)}$, $\sin(90° - i) = \cos i$,
$n = \frac{\sin i}{\cos i} = \tan i$ (Brewster's law).

20. A ray of light is incident on air-glass boundary at an angle θ. If the angle between the partially reflected ray and the refracted ray is 90°, calculate θ, given that the refractive index of glass is 1.50 A.28.2 B.46.5 **C.56.3** D.64.7

The partially reflected ray and refracted ray are at right angles i.e 90° to each other, thus the light ray is polarized. n=1.5, i=θ=?,
$n = \tan i$, $1.5 = \tan\theta$, $\theta = \tan^{-1} 1.5 = 56.3°$.

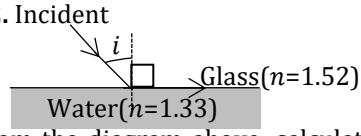
21.



In the figure above,a ray of light in air strikes a glass plate at an angle of incidence of 60°.The reflected ray is observedbe perpendicular to the refracted ray.What is the refractive index of glass? **A.1.73** B.1.50 C.0.87 D.0.57.

The light ray is polarized, as the reflected ray is perpendicular or at right angles to the refracted ray. i=60°, $n = \tan i = \tan 60°$, $n = 1.73$.

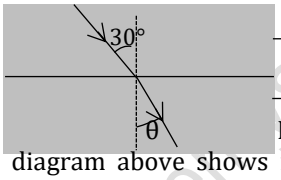
22. Incident



From the diagram above, calculate the incident angle A.41° B.49° C.55°**D.61**°

n_g=1.52, n_w=1.33, r=90°, i=?,
$n_{(g-w)} = \frac{n_w}{n_g} = \frac{\sin i}{\sin r}$, $\frac{1.33}{1.52} = \frac{\sin i}{\sin 90°} = \frac{\sin i}{1}$,
$\sin i = \frac{1.33}{1.52} = 0.875$, $i = \sin^{-1} 0.875 = 61°$.

23.



The diagram above shows ray of light passing from medium X to medium Y. Calculate angle θ to the nearest degree A.18° B.22°**C.26**° D.30°

i=30°, n_X=4/3, n_Y=3/2, r=θ=?,
$n_{(X-Y)} = \frac{n_Y}{n_X} = \frac{\sin i}{\sin r}$, $\frac{3/2}{4/3} = \frac{\sin 30°}{\sin\theta}$, $\frac{3}{2} \times \frac{3}{4} = \frac{0.5}{\sin\theta}$,
$\frac{9}{8} = \frac{0.5}{\sin\theta}$, $\sin\theta = \frac{8 \times 0.5}{9}$, $\sin\theta = 0.444$,
$\theta = \sin^{-1} 0.444 = 26.38° = 26°$.

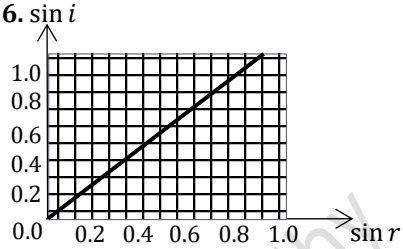
24. A ray of light travelling from glass into ethyl alcohol is incident at the boundary at angle of incidence 30°. Calculate the angle of refraction. [Refractive index of glass=1.5 ; refractive index of ethyl alcohol=1.36] A.27.0° B.51.7° **C.33.4**° D.72.8°.

n_E=1.36, n_g=1.5, i=30°, r=? $n_{(g-E)} = \frac{n_E}{n_g} = \frac{\sin i}{\sin r}$,
$\frac{1.36}{1.5} = \frac{\sin 30°}{\sin r} = \frac{0.5}{\sin r}$, $\sin r = \frac{1.5 \times 0.5}{1.36} = 0.551$,
$r = \sin^{-1} 0.551 = 33.4°$.

25. The velocity of light in air and glass are 3.0×10^8ms^{-1} and 1.8×10^8ms^{-1} respectively. Calculate the sine of angle of incidence that will produce an angle of refraction of 30° of ray of light incident on glass A.1.2 B.1.0 **C.0.8** D.0.6 E.0.3.

v_a=3.0×10^8ms^{-1}, v_g=1.8×10^8ms^{-1}, r=30°, $\sin i$=?,
$n_g = \frac{v_a}{v_g} = \frac{\sin i}{\sin r}$, $\frac{3.0\times10^8}{1.8\times10^8} = \frac{\sin i}{\sin 30°}$, $\frac{5}{3} = \frac{\sin i}{0.5}$,
$\sin i = \frac{5 \times 0.5}{3} = 0.833 = 0.8$.

26. $\sin i$



The graph above shows the relationship between $\sin i$ and $\sin r$ for a ray of light in air incident on a liquid surface. If i and r are respectively angles of incidence and refraction deduce the refractive index from the graph. A.1.63 B.1.50 C.1.33 **D.1.25** E.0.80.

Refrcative index n is give by the slope of the graph of $\sin i$ against $\sin r$. n = slope of the graph.
$n = \frac{\Delta \sin i}{\Delta \sin r} = \frac{1-0}{0.8-0} = \frac{1}{0.8} = 1.25$.

27. A measuring cylinder has a depth of 4.2cm when filled with a liquid. A coin dropped inside it appears to be at a depth of 3.5cm when viewed from above. Calculate the refractive index of the liquid A.2.20 B.1.83 **C.1.20** D.0.83 E.0.70

Real depth R=4.2cm, Apparent depth A=3.5cm, n=?, $n = \frac{R}{A} = \frac{4.2}{3.5} = 1.2$.

28. A nail is placed at the bottom of a bucket which is filled with water to a depth of 20cm. At what depth will the nail appear to be when viewed from the top? (Refractive index of water=1.33) A.5.0cm B.7.5cm C.10.0cm D.12.5cm **E.15.0cm**.

Real depth R=20cm, Apparent depth A=?, n=1.33=4/3, $n = \frac{R}{A}$, $A = \frac{R}{n} = \frac{20}{4/3} = 20 \times \frac{3}{4}$,
$A = 15.0$cm.

29. A coin is placed at the bottom of a cube of glass $t$cm thick. If the refraction index of the glass is μ, how high does the coin appear to be raised to an observer looking perpendicularly into the glass? A.$\frac{1}{t-\mu}$ **B.**$\frac{t(\mu-1)}{\mu}$ C.$t(1+\frac{1}{\mu})$ D.$\frac{t}{\mu}$ E.μt.

Real depth R=Thickness t, n= μ, height appear to be raised =apparent displacement d,
$n = \frac{R}{A} = \frac{R}{R-d}$, $n(R-d) = R$, $nR - nd = R$,
$nR - R = nd$, $R(n-1) = nd$, $d = \frac{R(n-1)}{n}$,
$d = \frac{t(\mu-1)}{\mu}$.

30. A travelling microscope is focused on a mark on a table. If a glass block of thickness 18.0cm is placed on the mark, by how many centimetres

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will the microscope be moved upwards so as to focus the mark once again? (Refractive iondex of glass=1.5) A.27cm B.24cm C.9cm **D.6cm**

R=th=18.0cm, n=1.5, d=?, $d = \frac{R(n-1)}{n}$,

$d = \frac{18(1.5-1)}{1.5} = \frac{18(0.5)}{1.5} = 6$cm.

31. An object is placed directly below a glass block of thickness 3.0cm. Calculate the lateral displacement of the object if the refractive index of the glass is 1.5 **A.1.0cm** B.1.5cm C.2.0cm D.2.5cm.

R=th=3.0cm, n=1.5, d=?, $d = \frac{R(n-1)}{n}$,

$d = \frac{3(1.5-1)}{1.5} = \frac{3(0.5)}{1.5} = 1$cm.

32. The difference between the apparent and real depths of a swimming pool filled with water is 1.5m.Calculate the depth of the swimming pool if the refractive index of water is 1.33? A.4cm B.3cm C.5cm **D.6cm**

$d = R - A = 1.5$m, $A = R - d$, n=1.33,

$n = \frac{R}{A} = \frac{R}{R-d}$, $R = n(R-d)$, $R = nR - nd$,

$nR - R = nd$, $R(n-1) = nd$, $R = \frac{nd}{n-1}$,

$R = \frac{1.33\times1.5}{1.33-1} = \frac{1.995}{0.33} = 6.05\text{cm} = 6\text{cm}$.

33. The apparent depth of stone viewed from the top of a rectangular glass block was found to be 10.5cm. Calculate the percentage displacement of the stone, when displaced 3.5cm from the base of the block **A.25%** B.33% C.66% D.75%

A=10.5cm, d=3.5cm, $R = A + d$, percentage displacement of the stone $= \frac{d}{R} \times 100$,

$\%d = \frac{d}{A+d} \times 100 = \frac{3.5}{10.5+3.5} \times 100 = \frac{3.5}{14} \times 100$,

$\%d = 25\%$.

34. A submarine is observed to rise from a real depth of 80m to 60m in water. Calculate the change in apparent depth. (Refractive index of water=4/3) A.60m **C.15m** C.80m D.45m

R_1=80m, R_2=60m, n=4/3, change in apparent depth ΔA=?, $n = \frac{R}{A}$, $A = \frac{R}{n}$, $\Delta A = A_1 - A_2$,

$\Delta A = \frac{R_1}{n} - \frac{R_2}{n} = \frac{1}{n}(R_1 - R_2)$,

$\Delta A = \frac{1}{4/3}(80-60) = \frac{3}{4} \times 20 = 15$cm.

35. Which of following statement about a ray of light which passes through a rectangular glass block and emerges from the opposite faces are correct? I.The angle of incidence at the first face = the angle of emergence from the opposite face II.The angle of refraction at the first face = the angle of incidence at the opposite face III.The incident is parallel to emergent ray IV.The angle of refraction at the first face = the angle of emergence from the opposite face **A.I,II and III** B.I,II and IV C.I,III and IV D.II,III and IV E.I,II,III and IV .

36. A ray of light from air is incident at angle of 30.0° on one face of a glass prism.What is the angle of deviation of theray as it passes from air to glass. (refractive index of glass = 1.5) **A.10.5°** B.19.5° C.20.0° D.21° E.49.5° .

i=30°, n_g=1.5, angle of deviation d=?,

$n_g = \frac{\sin i}{\sin r}$, $\sin r = \frac{\sin i}{n_g} = \frac{\sin 30^\circ}{1.5} = \frac{0.5}{1.5}$,

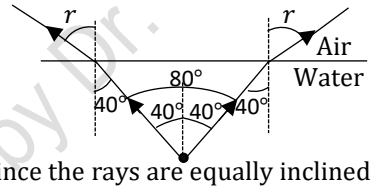
$\sin r = 0.333$, $r = \sin^{-1} 0.333 = 19.5^\circ$.

$d = i - r = 30 - 19.5 = 10.5^\circ$.

37. A ray of light is incident on one side of a parallel-sided glass block of refractive index 1.50 and emerges from the other side after refraction through the block. If the angle of incidence is 40°,the angular deviation of the emergent ray is A.60° B.39° **C.0°** D.28°.

The incident ray passing through a rectangular block is parallel to the emergent ray,hence the emergent ray is not deviated and angular deviation is 0° .

38. Two rays of light which are equally inclined to the vertical emanate from a point below the surface of water(refractive index=4/3).If the two rays are inclined to each other at angle of 80° in the water,the angle between the rays when they emerge into air is A.59° B.105°**C.118**°D.160°



Since the rays are equally inclined to the vertical and are inclined at 80° to each other, each ray will be inclined at 40° to the vertical.

The ray travels from water to air :

$n_w = n_{(a-w)}$, $n_{(w-a)} = \frac{1}{n_{(a-w)}} = \frac{1}{n_w}$, $n_{(w-a)} = \frac{\sin i}{\sin r}$

$\frac{1}{4/3} = \frac{\sin 40^\circ}{\sin r}$, $\sin r = \frac{4}{3} \times 0.643 = 0.857$,

$r = \sin^{-1} 0.857 = 59^\circ$. The angle between the rays when they emerge into the air will be the sum of their angle of refraction. Their angle of refraction are equal. $\theta = 2\times59^\circ = 118^\circ$.

39. If the refractive index of water relative to air is 1.33, what is the critical angle for ray travelling from water to air? A.39° B.41° C.42° **D.49°.**

$n_{(a-w)}$=n_w=1.33, C=?, Air to water is less dense to dense, $n_w = \frac{1}{\sin C}$, $1.33 = \frac{1}{\sin C}$,

$\sin C = \frac{1}{1.33} = 0.752$, $C = \sin^{-1} 0.752$,

$C = 48.75^\circ = 49^\circ$.

40. Calculate the critical angle for medium whose refractive index is 1.25 A.41° 21' B.42° 18' C.49° 0' D.52° 6' **E.53° 8' .**

n=1.25, C=?, $n = \frac{1}{\sin C}$, $1.25 = \frac{1}{\sin C}$,

$\sin C = \frac{1}{1.25} = 0.8$, $C = \sin^{-1} 0.8 = 53.13^\circ$

$C = 53.13^\circ = 53^\circ + 0.13(60') = 53^\circ\ 8'$.

41. Calculate the critical angle in glass for light travelling from glass to water [Refractive index of water=1.33, refractive index of glass=1.50] A.41.8° B.48.6° C.51.4° D.60.0°**E.62.5°**

n_w=1.33, n_g=1.50, C=?, A ray of light moving glass

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to water is dense to less dense medium thus, $n_{(g-w)} = \sin C$, angle of refraction at critical angle=90°, sin 90°=1. $n_{(g-w)} = \frac{n_w}{n_g} = \sin C$, $\frac{1.33}{1.5} = \sin C$, $\sin C = 0.8867$, $C = \sin^{-1} 0.8867$ $C = 62.46° = 62.5°$.

42. A ray of light incident at an angle θ in a rectangular prism **grazes** the surface of the prism on emerging from the prism.Determine the value of θ. [Absolute refractive index of the material of the prism is 1.5] **A.41.8°** B.48.2° C.60.8° D.90.0°.

When emergent ray grazes the surface of the prism, it means it makes angle 90° with the normal (angle of refraction is 90°). Angle of refraction is 90° at critical angle.

Angle of incidence θ = critical angle C,

$n = \frac{1}{\sin C} = \frac{1}{\sin \theta}$, $\sin \theta = \frac{1}{n} = \frac{1}{1.5} = 0.667$,

$\theta = \sin^{-1} 0.667 = 41.8°$.

43. If the refractive index of glass is 1.5,what is the critical angle at the air-glass interface? A.$\sin^{-1}$ (1/2) **B.$\sin^{-1}$ (2/3)** C.$\sin^{-1}$ (3/4) D.$\sin^{-1}$ (8/9) .

$n_g = n_{(a-g)} = \frac{1}{\sin C}$, n_g= 1.5 ,

$\sin C = \frac{1}{n} = \frac{1}{1.5} = \frac{10}{15} = \frac{2}{3}$, $C = \sin^{-1}(\frac{2}{3})$.

44. The velocity of light in air is 3×10^8ms^{-1}. Calculate the velocity of light in glass if the critical angle for light travelling from glass to air to 42°. A.3.00×10^8ms^{-1} B.2.70×10^8ms^{-1} C.2.45×10^8ms^{-1} **D.2.01×10^8ms^{-1}** E.1.40×10^8ms^{-1}

v_a=3×10^8ms^{-1}, C=42°, v_g=?, glass to air is dense to less dense. $n_{(g-a)} = \frac{v_g}{v_a} = \sin C$,

$\frac{v_g}{3\times10^8} = \sin 42°$, $v_g = \sin 42° \times 3\times10^8$,

$v_g = 0.6691\times3\times10^8 = 2.01\times10^8$ms^{-1}.

45. Which of the following is/are necessary for the phenomenon of total internal relfection to occur? I.Light must travel from an optically dense to an optically less dense medium II.The angle of incidence in the optically dense medium must be greater than the critical angle between the two media III.The refracted ray must travel in the dense medium at 90° to the normal. A.I B.II **C.I and II** D.I and III E.II and III.

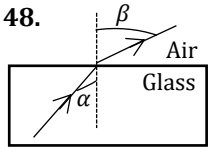
Total internal reflection occurs when a ray of light moves from a denser medium to a less denser or rarer medium and the anlge of incidence must be greater than the critical angle.

46. Which of the following conditions are necessary and sufficient for total internal reflection to take place at the boundary between two media? I.light is passing from optically dense medium to optically less dense medium II.light is passing from optically less dense medium to optically dense medium III.angle of incidence is greater than critical angle IV.angle of incidence is less than critical angle A.I and II B.II and III C.I and III **D.I and III** E.II and IV.

Total internal reflection occurs when light moves from an optically dense to optically less dense medium and the angle of incidence os greater than the critical angle(or critical angle is less than the angle of incidence).

47. Total internal reflection will not occur when light travels from **A.water into glass** B.glass into air C.glass into water D.water to air

Total internal reflection occurs when light travels from optically dense to optically less dense medium i.e glass to water, glass to air, water to air e.t.c, and not the reverse like air to glass.

48. 

In the above diagram, a ray of light passes from glass into air. If the angle of incidence is α and the angle of refraction in air is β, which of the following is true? A.$\alpha = \beta$ **B.the refractive index of glass is $\frac{\sin\beta}{\sin\alpha}$** C.the ray suffers total internal reflection if β is greater than the critical angle D.the ray will suffer total internal reflection, if α is less than the critical angle E.the refractive index of glass is $\frac{\sin\alpha}{\sin\beta}$.

When light travels from glass to air i.e dense to less dense medium, it is refracted away from the normal and the angle of refraction β is greater than the angle of incidence α , $\beta > \alpha$. The refractive index of glass is the refractive index from air to glass not glass to air thus, $n = \frac{\sin\beta}{\sin\alpha}$ or $\frac{1}{n} = \frac{\sin\alpha}{\sin\beta}$. Total internal internal reflection occurs when the angle of incidence in the denser medium is greater than the critical angle, $\alpha > C$.

49. In which of the following optical instruments is used made of total internal reflection? I.sextant II.kaleidoscope III.prism binoculars IV.submarine periscope A.I and II B.II and III **C.III and IV** D.I and IV E.I,II and III.

Prism binoculars are shorts telescopes with two converging lens for refraction and totally reflecting prism for total internal reflection. It produces an upright image and achieve a high magnification of an object. Submarine periscope are made of reflecting prism which reduces the distortion and formation of multiple images associated with plane mirrors in a simple periscope.

50. The operation of optical fibre is based on the principle of A.dispersion of light B.interference of light **C.refraction of light** D.polarization of light .

Optical fibre is operated based on total internal relfection majorly but refraction also occurs at the face of incidence.

51. Which of the following is not caused by refraction alone? A.a meter rule appears bent when immersed in water **B.there appear to be**

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pool of water on tarred road on hot days C.a pool of water appears to be shallower than it really is D.light passing through a triangular prism is bent towards the base E.a parallel beam of light is made to converge to a point by a convex lens.

The appearance of pool of water on a tarred broads on hot or sunny days is called mirage which can be explained by refraction and total internal reflection.A meter rule which appears bent when immersed in water,pool of water appearing shallower and images formed by a triangular prism and lens are explained by refraction only.

52. Which of the following may be used to explain mirage? I.layers of air near the road surface have varying refractive indices in hot weather II.road surfaces sometimes become good reflectors in hot water III.light from the sky can be reflected upwards after coming close to the road surface A.II B.II and III **C.I and III** D.I,II and III.

Mirage doesn't occur due to the nature of the road but due to decreasing refractive index or increasing temperature of the media through which light from the sky passes.

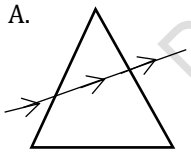
Refraction concept – mirage is the progressive bending of light as it travels through warm layers of decreasing refractive index unitl it becomes parallel to the ground.

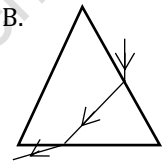
Reflection concept – mirage is the progressive bending of light from the sky as it passes through successive layers of optically less dense air of increasing temperature near the hot ground until it is fnally totally internally reflected.

53. Which of the following phenomena explains the formation of mirage? I.reflection II.refraction III.diffraction **A.I and II** B.II and III C.I and III D.I,II and III.

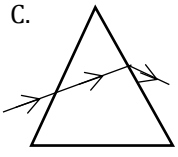
54. Which of the following diagrams below correctly illustrates the path of a ray of light through a glass prism?

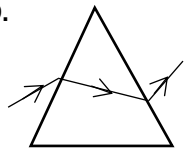
A. B.





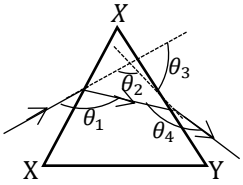
C. **D.**





The ray of light refracts towards the normal on the incident face as it moves from air to glass i.e less dense to dense. It refracts away from the normal on emerging at the other face from glasss to air i.e dense to less dense (option D).

55. In the diagram below, which of the angles θ_1, θ_2, θ_3 and θ_4, is the angle of deviation of a ray of light passing through glass prism XYZ?



A.θ_1 B.θ_2 **C.$\boldsymbol{\theta_3}$** D.θ_4.

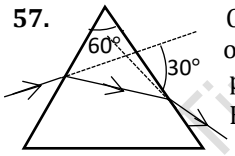
56. A ray of light undergoes minimum deviation at 40° when its is incident on an equilateral triangular glass prism. Calculate the refractive index of the prism A.148 B.1.50 **C.1.53** D1.67

Minimum deviation D_{min}=40°, refracting angle of an equilateral prism A=60°, n=?,

$$n=\frac{sin\frac{1}{2}(A+D_{min})}{sin\frac{A}{2}}=\frac{sin\frac{1}{2}(60°+40°)}{sin\frac{60°}{2}}=\frac{\sin\frac{1}{2}(100°)}{\sin 30°},$$

$$n=\frac{\sin 50°}{\sin 30°}=\frac{0.766}{0.5}=1.53.$$

57.



Calculate the refractive index of the material for the glass prism in the diagram A.3/2 B.4/3 C.$\sqrt{2}$ /2 **D.$\boldsymbol{\sqrt{2}}$.**

A=60°, D_{min}=30°, n=?, $n=\frac{sin\frac{1}{2}(A+D_{min})}{sin\frac{A}{2}}$,

$$n=\frac{sin\frac{1}{2}(60°+30°)}{sin\frac{60°}{2}}=\frac{\sin\frac{1}{2}(90°)}{\sin 30°}=\frac{\sin 45°}{\sin 30°}=\frac{1/\sqrt{2}}{1/2},$$

$$n=\frac{1}{\sqrt{2}}\times\frac{2}{1}=\frac{2}{\sqrt{2}}=\frac{2}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}}=\frac{2\sqrt{2}}{2}=\sqrt{2}\,.$$

58. Calculate the angle of minimum deviation for a ray which is refracted through an equiangular prism of refractive index 1.4 **A.29°** B.60° C.90° D.99°.

A=60°(equiangular prism), n=1.4, D_{min}=?,

$$n=\frac{sin\frac{1}{2}(A+D_{min})}{sin\frac{A}{2}},\ 1.4=\frac{sin\frac{1}{2}(60°+D_{min})}{sin\frac{60°}{2}},$$

$$1.4=\frac{sin\frac{1}{2}(60°+D_{min})}{sin\,30°}=\frac{sin\frac{1}{2}(60°+D_{min})}{0.5},$$

$$sin\frac{1}{2}(60°+D_{min})=1.4\times0.5=0.7,$$

$$\frac{1}{2}(60°+D_{min})=\sin^{-1}0.7=44.43°,$$

$$(60°+D_{min})=44.43°\times2=88.86°,$$

$$D_{min}=88.86°-60°=28.86°=29°.$$

Alternatively, $A=2r$, $r=\frac{A}{2}=\frac{60°}{2}=30°$, n=1.4, i=?, $n=\frac{\sin i}{\sin r}$, $\sin i=n\sin r$, $\sin i=1.4\times\sin 30°$

$$\sin i=1.4\times0.5=0.7,\ i=\sin^{-1}0.7=44.43°,$$

$$D_{min}=2i-2r=2\times44.43°-2\times30°,$$

$$D_{min}=88.86°-60°=28.86°=29°.\ \text{OR}$$

$$D_{min}=2i-A=2\times44.43°-60°=88.86°-60°$$

$$D_{min}=28.86°=29°.$$

59. A ray of light is incident on an equilateral glass prism of refractive index 3/2.Calculate the angle through which the ray is minimally deviated in the prism. A.30.0° **B.37.2°**C.42.0° D.48.6°.

A=60°(equilateral prism), n=3/2, D_{min}=?,

$$n=\frac{sin\frac{1}{2}(A+D_{min})}{sin\frac{A}{2}},\quad\frac{3}{2}=\frac{sin\frac{1}{2}(60°+D_{min})}{sin\frac{60°}{2}},$$

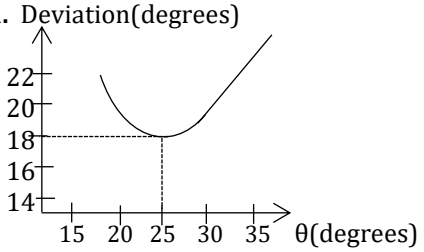
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$\frac{3}{2} = \frac{sin\frac{1}{2}(60°+D_{min})}{sin\ 30°} = \frac{sin\frac{1}{2}(60°+D_{min})}{0.5}$,
$sin\frac{1}{2}(60° + D_{min}) = \frac{3}{2} \times 0.5 = 0.75$,
$\frac{1}{2}(60° + D_{min}) = \sin^{-1} 0.75 = 48.59°$,
$(60° + D_{min}) = 48.59° \times 2 = 97.18°$,
$D_{min} = 97.18° - 60° = 37.18° = 37.2°$.

60. A ray of light is incident at an angle of 30° on a glass prism of refractive index 1.5. Calculate the angle through which the ray is minimally deviated in the prism A.10.5° B.19.5° **C.21.1°** D.38.9° E.40.5°

i=30°, n=1.5, D_{min}=?, $n = \frac{\sin i}{\sin r}$, $\sin r = \frac{\sin i}{n}$,
$\sin r = \frac{\sin 30°}{1.5} = \frac{0.5}{1.5} = 0.333$, $r = \sin^{-1} 0.333$
$r = 19.45°$, $D_{min} = 2i - 2r$,
$D_{min} = 2\times30° - 2\times19.45° = 60° - 38.9°$,
$D_{min} = 21.1°$.

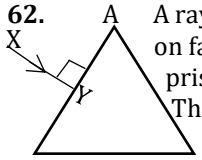
61. Deviation(degrees)



The figure above shows a plot of angles through a glass prism when light is incident at θ degrees on the prism. The incident angle that produces the minimum deviation is **A.25°** B.18° C.35° D.20° .

From the graphs above, at minimum deviation of 18°, angle of incidence is 25°

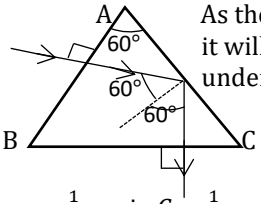
62. A ray of light XY is incident normally on face AB of an equilateral glass prism ABC of refractive index 1.50. The ray will



BC

A.emerge on the face AC,deflected away from the normal **B.be totally internally reflected at the face AC and emerge normal to the face BC** C.emerge on face AC,deflected towards the normal D.be totally internally reflected at the face AC and emerge deflected away from the normal on face BC.

The prism is an equilateral triangular prism with a refracting angle of 60°.



As the ray is incident normally, it will pass through the face AB undeflected and undeviated.

$n = \frac{1}{\sin C}$, $\sin C = \frac{1}{n}$, n=1.5, $\sin C = \frac{1}{1.5}$,
$\sin C = 0.667$, $C = \sin^{-1} 0.667 = 41.83°$,
$C = 42°$. The light ray strikes the face AC within the prism at angle of incidence of 60°. Since the angle of incidence 60°, is greater than the critical angle 42°, the ray is totally internally reflected with angle of reflection of 60° towards the face BC where it is incident again normally and emerges normally i.e at 90° to the face BC.

63. A 45° triangular glas prism can be used as reflector of light because A.refraction never takes place in such a prism B.the angle of reflection equals the angle of incidence C.the refractive index of glass is less than 1 D,it is transparent **E.the critical angle for glass is less than 45°.**

A ray of light that strikes one face of the triangular prism normally or perpendicularly meets the other face at angle of incidence 45° which is greater than the critical angle of glass 42°, thus the ray is totally internally reflected.

64. In the day time, it is possibles to see under shady areas such as under a tree, because light has undergone A.internal reflection B.refraction **C.diffraction** D.diffuse reflection

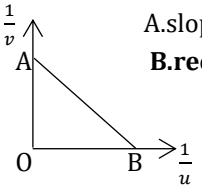
Diffraction of light is the property of light to bend round obstacles or spread out through tiny gaps. Thus, light rays from the sun spreads out through the tiny gaps in trees by diffraction such that one can see under the shady area in the tree. Refraction and total internal reflection is incorrect as only one media is involved in this case.

65. A ray incident on a glass prism undergoes minimum deviation when the A.incident angle is equal to angle of refraction B.refraction angle equals 90° C.incident angle equals 90°**D.incident angle equals angle of emergence**.

66. In a ray diagram for a thin converging lens, a ray that is not parallel to the optic axis but passes through the optic centre will **A.pass through undeviated** B.pass through the centre of curvature after refraction C.emerge parallel to the principal axis D.pass through the principal focus after refraction.

Ray of light pssing through the optical centre of a lens pass through undeviated i.e angular difference of 0°. Ray of light parallel to principal axis of a lens will be refracted through the principal focus. Ray passing through the principal focus will be refracted parallel to the principal axis.

67. In an experiment with a convex lens, a graph of the reciprocal of image distance $(1/v)$ is plotted against the reciprocal of the object distance $(1/u)$ as shown in the figure. The focal length is given by the



A.slope of the line AB
B.reciprocal of either OA and OB
C.length of OA or OB
D.ratio of OA/OB.

The focal length of a graph of $\frac{1}{v}$ vs $\frac{1}{u}$, is equal to

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the reciprocal to the intercept i.e OA or OB, on either axis ($\frac{1}{v}$ – axis or $\frac{1}{u}$ – axis).

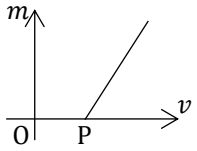
68. The correct shape of the graph of uv against $(u+v)$ for an object distance u and image distance v in an experiment to find the focal length of a convex lens is given as

A. uv $(u+v)$ B. uv $(u+v)$

C. uv $(u+v)$ D. uv $(u+v)$.

The graph of uv against $(u+v)$ is a straight line graph with a positive slode equal to the focal length f, $uv = f(u+v)$, as shown in option A.

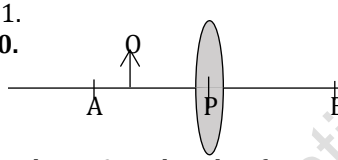
69. In an experiment ysing a converging lens to produce real imagees on a screen, the linear magnification m is plotted against the image distance v as illustrated in the diagram below. The distance OP represents the



A.focal length of the lens
B.radius of curvature
C.thickness of the lens
D.diameter of the lens.

$m = \frac{v}{f} - 1 = \frac{1}{f}(v) - 1$, the slope of the graph is equal to the reciprocal of the fical length $\frac{1}{f}$, the intercept on the v-axis OP is equal to the focal length f, the intercept on the m-axis is equal to -1.

70.



An object O is placed in front of a convex lens of focal length AP=PB in the figure above. The image is formed A.between A and P B.between P and B **C.beyond A** D.beyond B .

Object between principal focus A and optical centre P,forms a virtual,erect and magnified image(VEM) and is formed beyong the object i.e beyond A.

71. As an object approaches the focus of a converging lens from infinity, its image I.move closer to the lens II.moves away from the lens III.remains the same IV.becomes smaller. Which of the above is/are true? **A.II** B.III C.IV D.I and III E.III and IV .

In a conex or converging lens, $u \propto \frac{1}{v} \propto \frac{1}{H_I}$, as the object approaches the focus from infinity i.e decreasing object distance, the images moves away from the lens i.e increasing image distance, and the images becomes larger in size i.e increasing image size/height.

72. Which of the following is correct? As an object is moves from the centre of curvature towards the focus of a converging lens, its image A.moves closer to the lens B.remains stationary **C.becomes larger** D.becomes smaller E.remains the same size.

73. If a document is to be produced using a duplicating camera, where should the camera be placed in order to obtaib an exact copy? **A.at the point which is twice the focal length of the lens of the camera** C.between the principal focus and a point twice the focal length of the lens of the camera D.Beyond a poibt twice the focal length of the lens of the camera.

At point $2F$ i.e the point which is twice the focal length of the lens, an image same size as the object is formed on $2F$ at the opposite side of the lens. Thus, it can be used as a duplicating or copying camera when an object is placed at the centre of curvature $2F$ or at a point twice the focal length.

74. A converging lens of focal length f will act as a simple microscope if the object is placed A.beyond $2f$ fromn the lens B.at $2f$ from the lens C.between $2f$ and f from the lens D.at f from the lens **E.between f and the lens.**

A converging or convex lens functions as a simple microscope when the object is placed between the focal length and the lens, to produce a virtual, erect and magnified image.

75. When an object is placed between the principal focus and the optical centre of a convex lens, it could be used as a A.reflecting lens B.compound microscope C.projector **D.simple microscope.**

When an object is placed between the principal focus and the optical centre of a convex lens, the lens functions as a simple microscope.

76. A converging lens of focal length 5cm forms an image which is 10cm from the lens and on the same side of the lens as the object.How far from the lens is the object? A.0.3cm **B.3.3cm** C.5.0cm D.10.0cm E.15.0cm

f=5cm, v=−10cm(virtual image are on the same side of the lens as the object,hence image distance is negative), u=?,

$u = \frac{vf}{v-f} = \frac{-10\times5}{-10-5} = \frac{-50}{-15} = \frac{10}{3} = 3.3$cm.

77. An object is placed 14cm in front of a converging lens of focal length 10cm. The image is formed A.$5\frac{5}{6}$cm on the same side as the object B.$5\frac{5}{6}$cm on the opposite side from the object C.24cm on the opposite side from the object D.35cm on the ssame side as the onbject E.35cm on the opposite side from the object

u=14cm, f=10cm, v=?, $v = \frac{uf}{u-f}$,

$v = \frac{14\times10}{14-10} = \frac{140}{4} = 35$cm. The image formed is a real image and is formed on the opposite side of the lens from the object.

Demystified Series by Dr Timothy

78. An object is placed 5.0cm from a thin converging lens of focal length 10.0cm. The image formed will be A.3.3cm from the lens, erect, virtual and on the same side of the lens B.3.3cm from the lens, inverted, real and on the other side of the lens C.5.0cm from the lens, erect, virtual and on the same side of the lens **D.10.0cm from the lens, erect, virtual and on the same side of the lens** E.10.0cm from the lens, erect, real and the other side of the lens.

u=5.0cm, f=10.0cm, v=?, $v=\frac{uf}{u-f}$,

$v=\frac{5\times10}{5-10}=\frac{50}{-5}=-10$cm, the image distance is negative, thus the image is virtual, erect and on the same side of the lens with the object.

79. An object placed 50cm away from the focus of an converging lens of focal length 15cm, produces a focused image on a screen. Calculate the distance between the object and the screen A.42.3cm B.65.6cm C.76.5cm **D.84.5cm.**

Distance of the object from the principal focus of the lens=50cm, focal length f=15cm, object distance u=50+15=65cm, v=?, $v=\frac{uf}{u-f}$,

$v=\frac{65\times15}{65-15}=\frac{975}{50}=19.5$cm. Distance between the object and the screen $=u+v=65+19.5$

Object distance from the screen $=84.5$cm.

80. An object is placed 30.0cm from a converging lens. If the real image is formed 90.0cm from the object, calculate the focal length of the lens A.20.0cm **B.22.5cm** C.40.0cm D.60.0cm.

u=30.0cm, v=90.0cm, f=?, $f=\frac{uv}{u+v}$,

$f=\frac{30\times90}{30+90}=\frac{2700}{120}=22.5$cm.

81. A converging lens of focal length 5cm forms a virtual image which is 20cm from the lens. How far fron the lens is the object? A.2.0cm **B.4.0cm** C.6.7cm D.20.0cm E.25.0cm.

f=5cm, v=−20cm(virtual image), u=?,

$u=\frac{vf}{v-f}=\frac{-20\times5}{-20-5}=\frac{-100}{-25}=4.0$cm.

17. An object 1cm high placed on the axis 15cm from a converging lens forms an image 30cm from the lens. The size of the image is A.0.5cm B.1.5cm C.2cm D.2.5cm E.3cm

H_O=1cm, H_I=?, u=15cm, v=30cm,

$m=\frac{H_I}{H_O}=\frac{v}{u}$, $\frac{H_I}{1}=\frac{30}{15}$, $H_I=2$cm.

82. An object placed on the principle axis of a convex lens of focal length 10cm, produces a real image of double magnification. The image distance from the lens is **A.30cm** B.25cm C.20cm D.15cm E.10cm.

f=10cm, m=2, v=?, $m=\frac{v}{f}-1$, $2=\frac{v}{10}-1$,

$\frac{v}{10}=2+1=3$, $v=3\times10=30$cm.

83. An object is placed in front of a converging lens of focal length 20cm.The image is virtual and has a magnification of 2. What is the distance of the object from the lens A.5cm **B.10cm** C.30cm D.40cm.

f=20cm, m=−2(virtual image), u=?,

$m=\frac{f}{u-f}$, $-2=\frac{20}{u-20}$, $u-20=\frac{20}{-2}$,

$u-20=-10$, $u=-10+20=10$cm.

84. An object 15cm from a converging lens forms a real image of magnification 3. Calculate the focal length of the lens A.1.33cm B.1.76cm C.3.75cm **D.11.25cm** E.22.50cm.

u=15cm, m=3, f=?, $m=\frac{f}{u-f}$, $3=\frac{f}{15-f}$,

$3(15-f)=f$, $45-3f=f$, $45=f+3f$,

$45=4f$, $f=\frac{45}{4}=11.25$cm.

85. An object is placed 10cm from a converg-ing lens of focal length 15cm.Calculate the magnification of the image formed **A.3.0** B.1.5 C.0..6 D.0.3

u=10cm, f=15cm, m=?, $m=\frac{f}{u-f}=\frac{15}{10-15}$,

$m=\frac{15}{-5}=-3$, the negative sign indicates that the image is virtual.

86. A converging lens of focal length 10cm forms an erect image three times the size of a real object. Calculate the distance between the object and the image A.6.6cm B.11.1cm **C.13.3cm** D.20.0cm.

f=10cm, m=−3(erect image are virtual), D=?,

$m=\frac{v}{f}-1$, $-3=\frac{v}{10}-1$, $\frac{v}{10}=-3+1=-2$,

$v=-2\times10=-20$cm (virtual image),

$m=\frac{v}{u}$, $-3=\frac{-20}{u}$, $u=\frac{-20}{-3}=6.67$cm,

Disatnce between the object and its virtual image $D=v-u=20-6.67=13.3$cm.

87. A lens of focal length 12.0cm forms an upright image three times the size of a real object.The distance between the object and the image is **A.16.0cm** B.24.0cm C.8.0cm D.32.0cm.

Only a convex lens can produce an upright (erect) and magnified image.Concave lens form erect and diminished image. Erect or upright images are virtual.

m=−3, f=12cm, D=?, $m=\frac{f}{u-f}$,

$-3=\frac{12}{u-12}$, $u-12=\frac{12}{-3}=-4$, $u=-4+12$,

$u=8$cm, $m=\frac{v}{u}$, $v=mu=-3\times8=24$cm

Distance between the object and its virtual image $D=v-u$ =24−8=16cm.

88. An object is placed 5.6×10^{-2}m in front of a converging lens of focal length 1.0×10^{-1}m.The image formed is A.real,erect and magnified **B.virtual,erect and magnified** C.real,inverted and magnified D.virtual,erect and diminished.

$u=5.6\times10^{-2}$m=0.056m, $f=1\times10^{-1}$m=0.1mk, $v=\frac{uf}{u-f}=\frac{0.056\times0.1}{0.056-0.1}=\frac{0.056}{0.044}=-1.273$cm,

Image distance is negative, thus it is a virtual. Virtual images are always erect. Convex (converging lens) forms virtual images that is magnified. Alternatively,

Demystified Series by Dr Timothy

$m = \frac{f}{u-f} = \frac{0.1}{0.056-0.1} = \frac{0.1}{0.044} = -2.3$

Image is virtual as magnification is negative, virtual images are always erect. The image is also magnified as magnification M=2.3.

89. A converging lens and a screen are placed 20cm and 80cm respectively from an object in a straight line so that a sharp image of the object is formed on the screen. If the object is 3cm high, calculate the height of the image formed A.1cm **B.9cm** C.12cm D.15cm.

Object distance from the the lens u=20cm, distance of object from the screen=80cm, image distance from the lens v=80−20=60cm,

H_O=3cm, H_I=?, $m = \frac{H_I}{H_O} = \frac{v}{u}$, $\frac{H_I}{3} = \frac{60}{20}$,

$H_I = 3\times3 = 9$cm.

90. The magnification of an object produced with a converging lens of focal length 15cm is 5. If the image is real, what is the separation between the object and the image? A.13cm B.18cm C.72cm D.90cm **E.108cm.**

f=15cm, m=5, D=?, $m = \frac{v}{f} - 1$, $5 = \frac{v}{15} - 1$,

$5+1 = \frac{v}{15}$, $6 = \frac{v}{15}$, $v = 6\times15 = 90$cm,

$m = \frac{v}{u}$, $u = \frac{v}{m} = \frac{90}{5} = 18$cm,

Distance between the object and its real image $D = v + u = 90 + 18 = 108$cm.

91. The image formed by a diverging lens is always A.magnified, virtual and erect B.diminished, real and inverted C.diminished, virtual and erect **D.diminished, virtual and erect.**

The image formed by a concave or diverging lens is always virtual,erect or upright and diminished and is formed between the object and the optical centre of the lens.

92. The image of a real object formed by a diverging lens is always A.inverted B.real **C.situated between the object and the lens** D.larger than the object.

93. An object is placed at a point X between the focal point F and the optical centre C of a diverging lens. If F^1 is the focal point on the other side of the lens, the image of the object is formed between A.F and X **B.X and C** C.C and F^1 D.F^1 and $2F^1$.

The image formed by a concave or diverging lens is always between the object X and optical centre C of the lens, regardless the position of the object.

94. Which of the following instruments can be used to form a virtual image of an object? I.Converging lens II.Diverging lens III.Concave mirror A.I B.II C.III D.II and III **E.I,II and III.**

A converging lens forms a virtual image when the object is placed between the focus and the lens.A diverging lens forms virtual image always. A concave mirror forms a virtual image when the object is placed between the focus and the pole of the mirror.

95. Which of the following correctly describes the image of an object 4cm from a diverging lens of focal length 12cm? A.The image is virtual, 3cm in front of the lens B.The image is real, 6cm behind the lens **C.The image is virtual, 6cm in front of the lens** D.The image is real, 3cm in front of the lens.

u=4cm, f=−12cm(concave lens), v=?, $v = \frac{uf}{u-f} = \frac{4\times-12}{4-(-12)} = \frac{-48}{4+12} = \frac{-48}{16} = -3$cm. The image is virtual image. Virtual image is formed in fornt of the lens. Alternatively, without solving, image formed by a concave or diverging lens is always virtual and closer to the lens than the object.

96. An object is placed 15cm from a diverging lens of focal length 12cm. The image of the object formed by the lens is A.Real and 6.67cm from the lens **B.Virtual and 6.67cm from the lens** C.Real and 60.00cm from the lens D.Virtual and 60.00cm from the lens

u=15cm, f=−12cm, v=?, $v = \frac{uf}{u-f} = \frac{15\times-12}{15-(-12)}$

$v = \frac{-180}{15+12} = \frac{-180}{27} = -6.67$cm (virtual image). Alternatively, image formed by a concave or diverging lens is always virtual and closer to the lens than the object.

97. An object is placed at a distance $2f$ from a diverging lens of focal length f.The image distance is A.$2f$ **B.$2f/3$** C.f D.$f/4$.

u=$2f$, $f = -f$, v=?, $v = \frac{uf}{u-f} = \frac{2f\times-f}{2f-(-f)}$,

$v = \frac{-2f^2}{2f+f} = \frac{-2f^2}{3f} = \frac{-2f}{3}$ (virtual image).

[Option A would have been correct,if was a convex or converging lens].

98. A diverging lens of focal length 30cm produces an image 20cm from the lens. Determine the object distance A.10cm B.12cm C.50cm **D.60cm.**

f=−30cm, v=−20cm, u=?, $u = \frac{vf}{v-f}$,

$u = \frac{-20\times-30}{-20-(-30)} = \frac{600}{-20+30} = \frac{600}{10} = 60$cm.

99. A concave lens of focal length 15cm forms an image 1/3 of the size of the object. The object distance is A.10cm B.20cm **C.30cm** D.60cm.

m=$-\frac{1}{3}$ (magnification is negative for concave lens), f=−15cm, u=?, $m = \frac{f}{u-f}$, $-\frac{1}{3} = \frac{-15}{u-(-15)}$,

$-\frac{1}{3} = \frac{-15}{u+15}$,

$-1(u + 15) = 3\times-15$, $-u - 15 = -45$,

$-u = -45+15 = -30$, $u = 30$cm.

100. A thin converging lens has a power of 4 dioptres. Determine its focal length **A.0.25m** B.0.03m C.2.50m D.5.00.

P=$4D$(positive for converging or convex lens),

$P = \frac{1}{f}$, $f = \frac{1}{P} = \frac{1}{4}$ =0.25m =25cm.

101. A thin lens is placed 50cm from an illuminated object. The image produced has

Demystified Series by Dr Timothy

linear magnification of $\frac{1}{4}$. Calculate the power of the lens in diopters A.2.5D B.5.0D **C.10.0D** D.25.0D

u=50cm, m=$\frac{1}{4}$, P=?, $m = \frac{f}{u-f}$, $\frac{1}{4} = \frac{f}{50-f}$,

$50-f = 4f$, $50 = 4f + f = 5f$, $f = \frac{50}{5}$,

f = 10cm=0.1m, $P = \frac{1}{f} = \frac{1}{0.1} = 10.0D$.

102. A convex lens of power 5 dioptres forms an erect image of magnification 3 of an object. Find the object and image distances respectively. A.13.33cm and 20cm **B.13.33cm and 40cm** C.26.66cm and 20cm D.26.66cm and 40cm.

P=5D, m=-3(erect images are virtual), $P = \frac{1}{f}$,

$f = \frac{1}{P} = \frac{1}{5}$ =0.2m =20cm.

$m = \frac{v}{f} - 1$, $-3 = \frac{v}{20} - 1$, $-3+1 = \frac{v}{20}$,

$-2 = \frac{v}{20}$, $v = -2\times20 = -40$cm.

$m = \frac{v}{u}$, $u = \frac{v}{m} = \frac{-40}{-3} = 13.3$cm. $m = \frac{f}{u-f}$,

$-3 = \frac{20}{u-20}$, $u - 20 = \frac{20}{-3} = -6.67$,

u =20$-$6.67=13.33cm

103. A converging lens has a focal length of 5cm. Dtermine its power A.+20.0D B.+0.2D C.$-$20.0D D.$-$0.2D

f=5cm=0.05m, P=?, $P = \frac{1}{f} = \frac{1}{0.05} = +20.0D$.

104. A diverging lens has a focal length of 10cm. Calculate its power A.$-$0.1D B.+0.1D **C.$-$10.0D** D.+10.0D.

f=$-$10cm=$-$0.1m, P=?, $P = \frac{1}{f} = \frac{1}{-0.1}$,

$P = -10D$ or -10 dioptres.

105. An object 20cm from a convex lens has an image 25cm away from the lens.Calculate the power of the lens A.45D **B.9D** C.20D D.5D.

u=20cm=0.2m, v=25cm=0.25m, f=?, P=?

$P = \frac{1}{f} = \frac{u+v}{uv} = \frac{0.2+0.25}{0.2\times0.25} = \frac{0.45}{0.05} = 9D$.

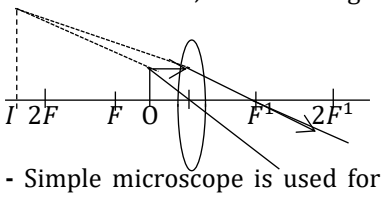
Jamb past questions on refraction of light :

[1978/42,1979/8,42,1980/14,25,35,1981/26, 32,40,44,1982/8,24,1983/7,46,1984/32,1985/ 31,33,1986/28,1987/34,35,1988/30,33,1990/ 26,27,1991/27,1991/29,1992/29,1993/29, 1994/34,36,1995/33,1997/35,38,1998/33, 1999/34,2000/16,19,2001/15,2002/14,16, 2003/16,29,31,38,2004/1,5,7,2005/39,2006/ 38,2008/34,35,2009/25,34,2010/30,2011/32, 2012/34,35,2013/31,2015/7,16]

Demystified Series by Dr. Timothy

Chapter 19 – Optical instruments, application of light waves

● **Simple microscope or magnifying glass or hand lens –** It consists of a single convex lens used to **magnify near objects i.e make it appear bigger.**The object $O$is placed between the principal focus F and the lens, the image I formed is virtual,erect and magnified **(VEM).**



- Simple microscope is used for studying or to view biological specimens and for reading small prints.

- The magnification of a simple microscope is given by – $\boldsymbol{m = \frac{D}{f} - 1}$, D=least distance of vision and it is negative.

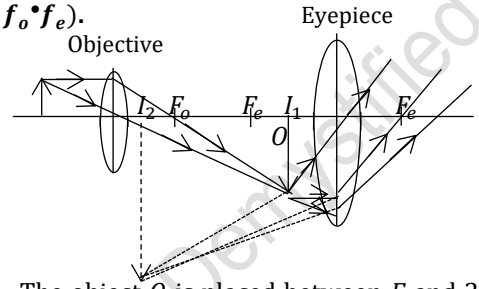
For a normal eye the least distance is 25cm.

$\boldsymbol{m = \frac{v}{f} - 1 = \frac{f}{u-f}}$ **. m and v are negative,as a virtual image is formed, while u and f are positive.**

- Simple microscope gives both linear and angular magnification.

● **Compound microscope –** It consists of two converging(convex lens) lenses; an objective lens which is closer to the object and an eyepiece lens(or **ocular)** which is close to the observer's eye.

Both lens have **short focal lengths** with the **focal length of the eye piece $f_e$being greater than that of the objective $f_o(f_e > f_o)$ or $(f_o < f_e$, $f_o \bullet f_e)$.**



- The object O is placed between $F_o$and $2F_o(u > f_o)$, of the objective lens. The image formed I_1-(object of the eye piece) is real,inverted and magnified **(RIM)** and is beyond $2F_o$, which is less than the focal length of the eye piece,hence the final image I_2 is virtual,inverted and magnified **(VIM)** and is formed beyond the object**.**

-The eye piece acts as a magnifier or magnifying glass.

- Compound microscope can produce both **linear m_L and angular magnification m_A.**

- Magnifying power of a compound microscope, $\boldsymbol{M = m_L = m_L = m_o \times m_e}$.

magnification by the objective lens $\boldsymbol{m_o = \frac{v_o}{u_o}}$.

magnification by the eye piece lens $\boldsymbol{m_e = \frac{v_e}{u_e}}$.

v_o=image distance of objective lens, u_o=object distance of objective lens, v_e=image distance of eye piece lens, u_e=object distance of eye piece lens.

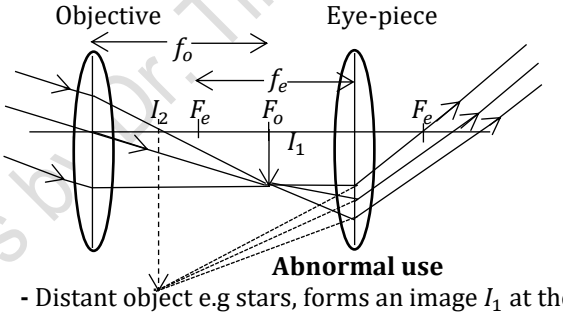
- Image distance of the eye-peiece v_e is equal to least distance of vision D, $\boldsymbol{v_e = D}$**.**

- m_o is positive as it forms real image while m_e is negative as it forms virtual images.

Alternatievly, $\boldsymbol{m_L = m_A = \left(\frac{v_o}{f_o} - 1\right)\left(\frac{v_e}{f_e} + 1\right)}$

- Distance of separation between the lenses: $\boldsymbol{D = v_o + u_e}$.

● **Astronomical telescope –** It consists of two convex lenses ; the objective lens of long focal length and eyepiece lens(or **ocular)** of short focal length $(\boldsymbol{f_o > f_e})$**, for viewing distant object, celestial or heavenly objects e.g moon,stars e.t.c, making them appear larger and closer.**



Abnormal use

- Distant object e.g stars, forms an image I_1 at the focus f_o ,of the object lens which is real, inverted and diminished**(RID)**. The objective image I_2 which lies within the focus f_e of the eye piece, acts as the eyepiece object.Hence,a final virtual,inverted and magnified image **(VIM)** is formed at the near point (25cm from the eyepiece) of the observer.

- Distance of separation between lenses or length of an astronomical telescope abnormal adjustment: $\boldsymbol{D = f_o + u_e}$.

- The magnifying power of astronomical telescope is due to the fact that the angle substended at the eye by the final image at infinity is very much greater than that susbended by the distant object.

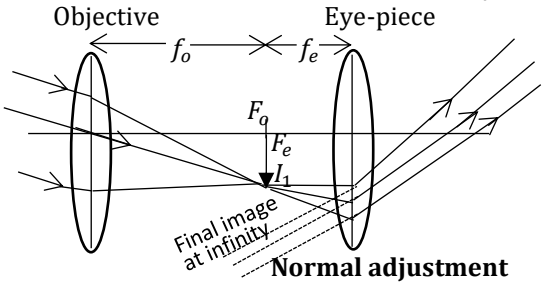
- The inverted image formed by the astronomical telescope is a disadvantaged when viewing the earth but can be tolerated when viewing heavenly bodies eg stars.

- In normal adjustment or normal use–

1.The focus of the objective$f_o$coincides with the focus of eyepiecef_e. **2**.The final image is virtual, inverted and magnified but formed **at infinity. 3**.The distance of separation between the lenses or length of the telescope: $\boldsymbol{D = f_o + f_e}$. **4**.The magnifying power of the astronomical telescope is equal ratio of the focal length of the objective

Demystified Series by Dr Timothy

to the focal length of the eyepiece: $\boldsymbol{M = \frac{f_o}{f_e}}$.



Normal adjustment

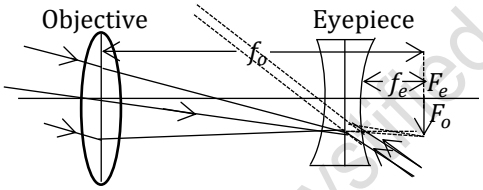
● **Galilean or Galileo's telescope or Opera glass –** It consists of a converging lens of long focal length as the objective and a diverging lens of short focal lengthas the eye piece.

- Normal adjustment – The objective lens collects parallel rays from a distant object to form a real,inverted and diminished image at objective focus.The parallel rays are however intercepted by the eyepiece diverging lens before they converge at the objective focus, which causes the rays to diverge,producing a final virtual,erect and magnified image **(VEM),** at infinity.

- Abnormal use or adjustment – The Galilean telescope is adjusted such that the final image is formed at near point.

- Distance of separation between the lenses or length of a Galilean telescope is equal to the digfference in the focal length of the lenses. $\boldsymbol{D = f_o - f_e}$.

- Magnifying power of a Galilean telescope is given by the ratio of the focal length of the objective to the that of the eyepiece. $M = \frac{f_o}{f_e}$

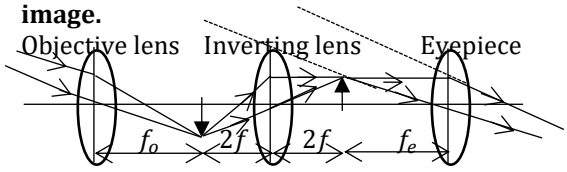


- Galilean telescope can be used on earth as it forms erect image i.e its advantage over astronomical telescope. It has small field of view and poor magnification.

- Galilean telescope is smallest in size and shortest in length i.e its advantage over other telescope.

- Galilean telescope in abnormal use can be used to view actors on the stage in a theatre.

● **Terrestrial telescope –** It consists on extra convex lens(inverting lens) placedbetween the objective and eyepiece lens.**The extra lens helps to re-invert the image to form a final erect image.**



- The object lens forms a real,inverted and diminished**(RID)** image at the objective focus which coincides with the centre of curvature $\boldsymbol{2F}$of the inverting lens,to produce another image(real,re-inverted to become erect and same size) at the same centre of curvature **2F**. The $\boldsymbol{2F}$of the inverting lens coincides with the focus of the eyepiece, hence the final image is virtual,erect and maginified**(VEM)** image at infinity.

- The distance between the objective and eyepiece lens: $\boldsymbol{D = f_o + 4f + f_e}$.

- The advangtage of a terrestrial telescope is the production of erect(upright) image, hence can be used on earth. Its disadvantage is that the inverting lens extends the length of the telescope by $\boldsymbol{4f}$, making it too long.

● **Slide Projector or Projection lantern –** This is an instrument for producing a magnified or enlarged image of a transparent object (slide) on a screen.

- It consists of – **i.**The **power light source** situated at the centre of curvature the concave mirror, which illuminates the object(slide). **ii.The concave mirror** directs and reflects the light onto the condenser. **iii.The condenser** of two plano convex lenses, converge the light through the slide to produce a bright image. It usually has an **infra red filter** between the lenses to prevent the passage of heat to the slide. **iv.**The slide(object) is **between $\boldsymbol{f}$ and $\boldsymbol{2f}$** $(\boldsymbol{f < u < 2f})$of the adjustable projection lens thus, a **real, inverted and magnified image** is formed on the screen which is placed beyond $\boldsymbol{2f}$ of the lens.

- Magnification in a projector m is given by ;

$\boldsymbol{m = \frac{image\ height\ H_I}{object\ height\ H_O} = \frac{image\ breadth\ B_I}{object\ breadth\ B_O} = \frac{image\ length\ L_I}{object\ length\ L_O}}$. $\boldsymbol{m = \frac{v}{f} - 1}$.

- Length of a hall where a slide projector is used=distance between the slide(object) and the screen= $\boldsymbol{u + v}$.

● **Human eye – Parts of the human eye ; i.Cornea** -: Transparent membrane covering the eye in front. **ii.Pupil** -: Adjustable opening in the iris through which light passes into the eye. **iii.Iris** -: Muscular coloured part of the eye which controls the size of the pupil **iv.Converging lens** -: An elastic lens of variable thickness or focal length for focusing object on the retina. **v.Retina** -:Photosensitive part of the eye where image is formed.

- Near and Far points – Near point is the nearest distance of clear vision while the Far point is the farthest distance of clear vision. **In a normal eye,the near point is 25cm whilefar point is at infinity(∞).**

- Accomodation – It is the ability of the eye lens to focus on near and far objects by alteration of its focal length.

- To focus on near objects,the ciliary muscle

Demystified Series by Dr Timothy

relax,buldges the lens and shorten its focal length. To focus on far objects, the ciliary muscles contracts,stretches the lens and increases the focal length. **The ciliary muscles is primarily responsible for accommodation.**

- Persistence of vision or persistent vision – This is the retention of an impression of vision of image on the retina for a period of time after object has been removed i.e making light seen appear to have costant intensity.**This phenomenon is used in the showing and production of motion pictures in cinema films.**

- Binocular or Stereoscopic vision -: Blending or overlapping of two images formed by both eyes to form one distinct image in the brain.**It is imposible with one eye or when the field of view don't overlap.**

- Eye defects –

1.Astigmatism – This occurs due to loss of power of accommodation.It is the ability to see objects clearly(blurring) in one direction than in one direction than the other or inability of the eye to focus on both far and near objects.**It can be corrected using a cylindrical lens.**

2.Presbyopia – The inability of the eye to focus near objects or the loss of power of accommodation due to loss of elasticity in the ciliary muscle due to old age or hardenening/ weakening of the eye lens.**It can be corrected using a bifocal lens(combination of concave and convex lens).**

3.Myopia (Short-sightedness) – Ability of the eye to see near objects clearly and inability to see distant object clearly,due to the eye ball being too long or strong eye lens(short focal length).Image of distant object is focused in front of the retina.**It can be corrected using a concave (diverging) lens.**

4.Hypermetropia or Hyperopia (Long-sightedness) – Ability of the eye to see distant objects clearly and inability to see near objects clearly,due to the eye ball being too short or weak eye lens(long focal length).Image of near object is focused in behind the retina.**It can be corrected using a convex (converging) lens.**

For a short-sighted person – Image distance= distant object or distance that cannot be seen clearly. **Image distance is negative and a concave lens of focal length equal to the image distance(far point of the sufferer) is needed.**

- For a long-sighted person – Image distance= closer object=distance cannot be seen clearly. **Image distance is also negative.**

- Lens camera – It consists of : i.Lens: convexlens whose distance from the film is adjustable **ii.Aperture:** opening through which light enters the camera **iii.Diaphragm:** It regulate the size of the aperture, therby controlling the amount of light entering the camera. **iv.Shutter:** It opens to expose the film to light for a short period of time **v.Photographic film:** Light-sensitive sheet where the image is formed.

Comparison of the human eye and lens camera

Human eye	Lens camera
1.Flexible biological lens	Rigid non-biological lens
2.Iris	Diaphragm
3.Retina	Photographic film
4.Pupil	Aperture
5.Distance between lens and retina is fixed	Distance between lens and film is adjustable/variable
6.Eye lid	Shutter
7.Focal length of lens is variable	Focal length of lens is fixed

- Both lens camera and the human eye form a of real,inverted and dimished image.

Examples :

1. The optical instrument for viewing near objects is the A.telescope B.sextant **C.microsope** D.binoculars.

Simple microscope is used magnifying near objects in order to view them.

2. The magnified and virtual image of a near object is produced by A.Prism binoculars B.Astronomical telescope C.Periscope **D.Simple microscope.**

The image formed by a simple microscope is virtual, erect and magnified.

3. A man uses a converging lens of focal length 10cm as a magnifying glass. How far away from the lens must an object be placed form him to see its image at 25cm from the lens? A.6.7cm **B.7.1cm** C.12.9cm D.15.0cm.

f=10cm, v=−25cm(virtual image), u=?,

$u = \frac{vf}{v-f} = \frac{-25\times10}{-25-10} = \frac{-250}{-35} = 7.1$cm.

4. A converging lens of focal length 3.0cm is used as a simple microscope. If the final image is formed at a distance 30cm from the lens, calculate the linear magnification A.0.1 **B.11.0** C.1.0 D.9.0 E.10.0.

f=3.0cm, v=−30cm, m=?,

$M = \frac{v}{f} - 1 = -\frac{30}{3} - 1 = -10 - 1 = -11$

m=11(the image is virtual, as magnification is negative).

5. A simple magnifying glass is used to view an object. At what distance from the lens must the object be placed so that an image 5 times the size of the object is produced 20cm from the lens? A.2.0cm B.3.3cm **C.4.0cm** D.5.0cm E.10.0cm.

m=−5, v=−20cm, u=?, $m = \frac{v}{u}$, $u = \frac{v}{m}$,

$u = \frac{-20}{-5} = 4.0$cm.

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6. A magnifying glass is used to view a biological specimen.The distance between the lens and the specimen is 10cm. If the image is formed at a distance of 25cm from the lens, calculate the magnification A.2/5 B.0.4 **C.2.5** D.1.

u=10cm, v=−25cm, m=?, $m=\frac{v}{u}=\frac{-25}{10}=-2.5$.

8. A simple microscope forms an image 10cm from an eye close to the lens.If the object is 6cm from the eye, calculate the focal length of the lens **A.15.00cm** B.3.75cm C.4.00cm D.16.00cm

v=−10cm, u=6cm, f=?, $f=\frac{uv}{u+v}=\frac{6\times-10}{6+(-10)}$,

$f=\frac{-60}{6-10}=\frac{-60}{-4}=15$cm.

9. An observer with normal eyes views an object with a magnifying glass of focal length 5cm.The angular magnification is **A.−6** B.−5 C.5 D.6

f=5cm, D=least distance of vision=−25cm(for a normal eye) , $m=\frac{D}{f}-1=\frac{-25}{5}-1=-5-1=-6$.

10.. A compound microscope makes use of A.two diverging lenses to produce an enlarged image of a very distant object B.two converging lens to produce a diminished image C.two diverging lenses to produce a diminished image of a near object **D.two converging lenses to produce an enlarged,inverted image of a near object** E.a combination of converging and diverging lenses to produce the image of both distant and near objects.

Compound microscope uses two converging lenses of short focal lengths to produce a real, inverted and magnified image for the objective and virtual,inverted and magnified image in the eye piece,of anear object placed close to the objective.

11. A compound microscope makes use of A.two diverging lenses to produce an enlarged image of a very distant object B.two converging lenses to produce a diminished and erect image of a distant object C.two diverging lenses to produce a diminished image of a near object **D.two converging lenses to produce an enlarged inverted image of a near object** E.a combination of converging and diverging lenses to produce the images of both distant and near objects.

12. If F represents the principal focus of the objective lens of a compound microscope, the object to be viewed is placed A.at F**B.between F and $2F$** C.between F and the objective lens D.at $2F$ E.beyond $2F$.

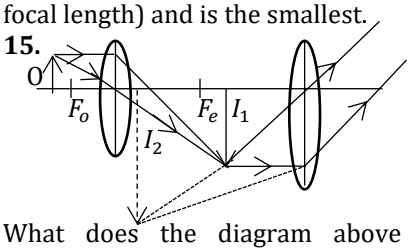
The object is placed between F and $2F$ of the objective lens, and the image is formed beyond $2F$ of the objective.

13. In a compound microscope, the objective and the eyepiece focal lengths are A.long **B.short** C.the same D.at infinity.

14. Of the four lenses whose focal lengths are provided below, the most suitable for use as a microscope objective? A.−4mm B.−4cm **C.+4mm** D.+4cm.

The converging lenses in compound microscope have short focal lengths,with that of the objective lens being shorter than the eye piece. The focal length will be +4mm(convex lens have positive focal length) and is the smallest.

15.



What does the diagram above represents? A.microscope in normal use B.telescope in abnormal use **C.microscope in abnormal use** D.telescope in normal use.

The object is closer to the lens i.e near object and a microscope is used for near objects. In abnormal use of a compound microscope,the image is formed within the lens i.e 25cm from the eye. **In normal use of a compound microscope,the image distance of the objective lens coincides with the focal length of the eye piece and the final image is formed at infinity.**

16. Which of the following relations about the focal length f_o of the objective and focal length f_e of the eyepiece of a compound microscope is correct? A.$f_o=f_e$ **B.$f_o<f_e$** C.$f_o>f_e$ D.$f_o=2f_e$.

In a compound microscope the focal length of the eyepiece is greater than that of the objective or the focal length of the objective is less than that of the eyepiece. $f_e>f_o$ or $f_o<f_e$.

17. A compound microscope has lenses of focal length 1.0cm and 5.0cm. An object is placed 1.2cm from the objective lens,if a virtual image is formed 25cm from the eye piece,calculate the distance between the two lenses? **A.10.2cm** B.12.3cm C.11.5cm D.10.5cm.

Object distance of objective lens u_o=1.2cm, focal length of objective lens f_o=1cm, image distance of objective lens v_o=?

$v_o=\frac{u_of_o}{u_o-f_o}=\frac{1.2\times1}{1.2-1}=\frac{1.2}{0.2}=6$cm.

Object distance of eye piece lens u_e=?, focal length of eye piece f_e=5cm, image distance of eye piece lens v_e=−25cm (the image of eye piece is virtual), $u_e=\frac{v_ef_e}{v_e-f_e}=\frac{-25\times5}{-25-5}=\frac{-125}{-30}$,

u_e = 4.16cm = 4.2cm. Distance of separation between lenses in a compound microscope is equal to the sum of the image distance of the objective lens and object distance of the eye piece. $D=v_o+u_e$ =6+4.2 =10.2cm.

18. If the linear magnification of the objective and eyepiece convex lenses of a compound microscope are 4 and 7 repectively, calculkate the angular magnification of the microscope A.2 B.3 C.11 **D.28.**

Demystified Series by Dr Timothy

Magnification of objective lens m_o=4, magnification of eyepiece m_e=7, anular magnification or magnifying power of compound microscope M=?, $M = m_o \times m_e = 4\times7 = 28$.

19. In a compound microscope, the initial image is twice the object while the final image is four times the object. Calculate the magnifying power of the microscope A.0.5 B.2.0 **C.4.0** D.6.0 E.8.0.

The final image is four times the object (not the initial image, the magnifiying power, $M = 4.0$. Alternatively, m_o=2, $m_e = \frac{4}{2} = 2$, as the final image is four times the object and the initial image is twice the object. $M = m_o \times m_e = 2\times2 = 4.0$.

20. The focal length of the objective and eye piece in a compound microscope are +0.50cm and +2.00 cm respectively.The instrument is focused on an object 0.52cm from the objective lens and a virtual image is viewed by the eye at a distance of 25cm.Calculate the magnifying power of the microscope. A.313 B.334 C.364 **D.338.**

f_o=0.5cm, f_e=2cm, u_o=0.52cm, v_e=D=25cm

$v_o = \frac{u_o f_o}{u_o - f_o} = \frac{0.52\times0.5}{0.52-0.5} = \frac{0.26}{0.02} = 13$cm.

$M = m_o \times m_e = \left(\frac{v_o}{f_o} - 1\right)\left(\frac{v_e}{f_e} + 1\right)$,

$M = \left(\frac{13}{0.5} - 1\right)\left(\frac{25}{2} + 1\right) = (26-1)(12.5+1)$,

$M = (25)(13.5) = 337.5 = 338$.

21. Which of the following is not true about an astronomical telescope in normal adjustment A.the focal length of the objective lens is greater than that of the eye piece B.the distance between the objective lens and the eyepiece is the sum of their focal lengths C.its final image of a star is virtual,erect and at infinite distance **D.its final image of a star is real,erect and at an infinite distance** E.its angular magnification is the ratio of the focal length of the objective to that of the eyepiece.

In normal adjustment, the final image of a heavenly body e.g start, is virtual, inverted magnified but at infinity or infinite distance.

22. A telescope is said to be in normal adjustment when the A.focal length of objective is greater than that of the eyepiece **B.objective focal point coincides with that of the eyepiece** C.focal length of the eye piece is greater than that of the objective D.focal length of the eyepiece is equal to that of the objective.

In normal adjustment the focus of the objective coincides with that of the eyepiece.

23. Which of the following statements about the compound microscope and astronomical telescope is.are correct? I.The final image of a compound microscope is located at infinity II.The final image of the astronomical telescope is always erect III.The objective of a compound microscope has a very short focal length A.I B.II **C.III** D.I and III E.II and III.

The final image of a compound microscope is inverted and within the lenses or at near point from the eye. The final image of an astronomical telescope is always inverted. The objective of a compound microscope has a shorter focal length than the eyepiece.

24. An astronomical telescope is said to be in normal adjustment when the **A.Final image is at infinity** B.Final image is at the near point of eye C.Focal length of the objective is longer that that of eyepiece D.Eye is accommodated.

25. What optical instrument can best be constructed with converging lens of focal lengths 50cm and 5cm? A.Compound microscope B.Terrestrial telescope **C.Astronomical telescope** D.Galileo's telescope.

Astronomical telescope uses two converging lenses: the objective with long focal length (50cm) and eyepiece with short focal length (5cm).

26. An astronomical telescope has objective and eyepiece lenses of focal lengths 3.5m and 5cm respectively. Determine the magnifying power of the telescope when in normal adjustement **A.70.0** B.17.5 C.7.0 D.0.7.

f_o=3.5m=350cm, f_e=5cm, M=?, $M = \frac{f_o}{f_e}$,

$M = \frac{350}{5} = 70.0$.

27. An astronomical telescope has an eyepiece of focal length 5cm. If the angular magnificat- ion in normal adjustment is 10, what is the distance between objective and eyepiece in cm? A.110cm **B.55cm** C.50cm D.45cm.

f_e=5cm, M=m_A=10, f_o=?

$M = \frac{f_o}{f_e}$, $f_o = Mf_e = 10\times5 = 50$cm.

Distance between lenses in normal adjustment $D = f_o + f_e = 50+5 = 55$cm.

28. An astronomical telescope has an angular magnification of 7. If the distance between the lenses in normal adjustment is 45cm, calculate the focal length of the lenses.

M=m_A=7, D=45cm, f_o=?, f_e=?,

$M = \frac{f_o}{f_e}$, $f_o = Mf_e = 7\times f_e = 7f_e$,

$D = f_o + f_e$, $45 = 7f_e + f_e = 8f_e$,

$f_e = \frac{45}{8} = 5.625$cm, $f_o = Mf_e = 7\times f_e = 7f_e$,

$f_o = 7\times5.625 = 39.375$cm.

- Question Drill – 1. The focal length of the lenses of an atronomical telescope are 5.0cm and 100cm respectively.Calculate the length of the telescope, if the object is at infinity and the final image is formed 25.0cm from the eye. **Ans=104cm.**

Hint: It is a case of abnormal adjustment,as the final image is formed at near point i.e 25cm for a normal eye. For normal adjustment the final image is formed at infinity. **2.**An astronomical telescope in normal use has an angular magnification of 4 and the lenses are at a distance

Demystified Series by Dr Timothy

of 25cm, calculate the focal length of the eyepiece. **Ans=5cm.**

29. In a Galilean telescope,the objective and eyepiece have focal lengths of 60cm and 10cm respectively.What is the separation of these lenses when the telescope is used in normal adjustment? A.6cm B.10cm **C.50cm** D.60cm E.70cm.

f_o=60cm, f_e=10cm, Distance of separation of lenses in a Galilean telescope $D = f_o - f_e$,

$D = 60-10=50$cm.

30. An optical instrument was made with a convex lens of focal length 50cm and a concave lens of focal length 5cm.This instrument must be **A.a Galilean telescope** B.an astronomical telescope C.a slide projector D.a compound microscope.

31. The Galilean telescope has an advantage over other telescopes in that it A.produces an erect image **B.is shortest in length** C.has the best magnification D.produces the sharpest image

32. Which of the following statements about the Galilean telescope is not correct? **A.The final image is inverted** B.It is shorter than terrestrial telescope C.The final image is erect D.It has a small field of view.

33. The terrestrial telescope has one extra convex lens more than astronomical telescope. The extra lens is for A.magnification of the image **B.erection of the image** C.improving the sharpness D.creating an inverted image.

34. Which of the following makes use of a concave mirror? A.camera B.kaleidoscope C.periscope D.simple microscope **E.slide projector**

35. Which of the following is not a component of a slide projector? A.concave mirror B.screen C.condenser **D.convex mirror** E.light source.

36. The purpose of the condenser in a film projector is to A.cast an image of the film on the screen B.prevent the image from being blurred **C.make the image brighter** D.invert the image laterally E.turn the image upright on the screen.

37. At which of the following distances from the lens should a slide be placed in a slide projector if f is the focal length of the projection lens A.less than $2f$ **B.greater than f but less than $2f$** C.greater than $2f$ D.equal to f E.equal to $2f$.

38. In a projection lantern of focal length f, the object distance u is such that A.$u > 2f > f$ B.$u < f < 2f$ C.$u = f < 2f$ **D.$f < u < 2f$**.

39. A lantern gives an image 3m square of slide 7.26cm square on a screen. If the screen is 10m from the projection lens of the lattern,calculate the focal length of the lens A.40.3cm B.26.1cm **C.24.7cm** D.0.7cm E.0.3cm.

Image 3m square: image length L_I =3m= 300cm, image breadth B_I=3m= 300cm, Object (slide) 7.26cm square: object length L_O=7.26cm, object breadth B_O=7.26cm,

v= distance of the screen from lens=10m =1000cm, u=?, f=? $m = \frac{L_I}{L_O} = \frac{B_I}{B_O} = \frac{v}{u}$, $\frac{300}{7.62} = \frac{1000}{u}$, $u = \frac{7.62\times1000}{300} = 25.4$cm.

$f = \frac{uv}{u+v} = \frac{25.4\times1000}{25.4+1000} = \frac{25400}{1025.4} = 24.7$cm.

40. A projection lantern is used to give the image of a slide on a screen. If the image is 24 times as large as the slide and the screen is 72.0m from the projecting lens, what is the position of the slide from the lens? A.4.0m B.3.5m **C.3.0m** D.0.3m.

m=24, v=72.0m, u=?, $m = \frac{v}{u}$, $u = \frac{v}{m} = \frac{72.0}{24} = 3.0$.

41. The use of lenses is not applicable in the A.projector B,human eye **C.periscope** D.telescope.

42. If the focal length of a camera lens is 20cm,the distance from the film at which the lens must be set to produce a sharp image of an object 100cm away is A.17cm B.20cm **C.25cm** D.100cm

f=20cm(lens of a camera is a convex lens), u=100cm, distance of the film from the lens= image distance=v=?, $v = \frac{uf}{u-f} = \frac{100\times20}{100-20}$,

$v = \frac{2000}{80} = 25$cm.

43. The following are parts of the eye I.retian II.pupil III.iris. The correct equivalent in the camera in the same order are **A.film, aperture, diaphragm** B.film, diaphragm, aperture C.diaphragm, aperture, film D.aperture, diaphragm, film.

Retina is similar to the film,pupil is similar to the aperture and iris is similar to the diaphragm.

44. The eye controls the amount of light reaching the retina by adjusting **A.Iris** B.Optic nerve C.Cornea D.Retina.

45. Certain similarities between the eye and the camera is/are listed below.Which of the following statements is/are not true? I.both the camera and the eye have a light sensitive screen II.inverted images are formed on the screen in both III.the distance between the lens and the screen is fixed in both IV.the amount of light entering both can be adjusted A.I B.II **C.III** D.II and III E.I and IV.

The distance between the lens and the screen is fixed in the human eye but can be adjusted in a camera.

46. The image of an object on the retina of the human eye is A.virtual and diminished **B.inverted and real** C.erect and diminished D.erect and real.

47. The ability of the human eye to focus objects on the retina is referred to as the power of A.interference B.diffraction **C.accommodation** D.superposition.

48. The property of the eye knon as its power of accommodation is controlled by the A.pupil B.vitreous humour C.iris **D.ciliary muscles.**

Demystified Series by Dr. Timothy

49. The range of accommodation of a normal human eye is A.0m to 0.1m B.0m to 0.25m C.0.10m to 0.25m D.0.10 t0 infinity **E.0.25m to infinity.**
Range of vision for the human eye is from near point i.e 25cm(0.25m) to far point i.e infinity(∞).
50. Prebyopia is a defect of the eye resulting from A.weak ciliary muscle B.short eye ball **C.loss of sphericity of the lens** D.long eye ball Presbyopia is due to the weakening or loss sphericity of the eye lens or loss of elasticity in the ciliary muscle at old age.
51. The lack of power of accommodation which is mainly due to hardening of th eye lens and is usual with the onset of old age is called A.myopia B.hypertropia **C.presbyopia** D.astigmatism.
52. When the human eye loses its power of accommodation, the effect is known as A.astigmatism B.long-sightedness C.short-sightedness **D.presbyopia**.
53. Which of the following eye defects can be corrected using a cylindrical lens? **A.Astigmatism** B.Presbyopia C.Chromatic aberration D.Myopia.
54. For a short sighted person, light rays from a point on a very distant object is focused **A.In front of the retina** B.On the retinal by a converging lens C.Behind the retina by a diverging lens D.In front of the retina at distance $2F$ from the lens.
A short-sighted person can focus near objects clearly on the retina but cannot focus distant object. Distant object are focused in front of the retina and can be made to focus on the retina by a diverging or concave lens.
55. Which of the following is used for the correction of short-sightedness? **A.Concave lens** B.Concave mirror C.Convex mirror D.Convex lens.
56. For the correction of myopic defects in the human eye we reguire A.a convex lens **B.a concave lens** C.a combination of concave and convex lenses D.a prism E.a concave mirror.
57. For correcting long sight defects in the hu7man eye we require a **A.Converging lens** B.Diverging lens C.Microscope D.Periscope E.Plane glass sheet.
58. Which of the following is not correct? A.The average range of distinct vision for normal eye is from the far distance(infinity) up to about 25cm in front of the eye B.Longsighted people have difficulty in making the eye lens sufficiently powerful to focus nearby objects C.Shortsighted people cannot accommodate distant objects **D.Long sighted people need diverging spectacles lenses** E.A person with long sight has his eye-ball too short and the image of an object is therefore formed behind the retina.
The range of distinct vision if from infinity i.e far point, to 25c in front of the eye i.e near point. Longsighted people have short eye-ball and near objects are focused behind retina. Longsightedness is corrected by converging or convex lens. Shortsighted people can see near objects clearly but cannot see far or distant objects clearly.
59. Which of the following statements about defects of vision is/are correct? I.for a long-sighted person,close object appear blurred II.for a short-sighted person,distant object appear blurred III.short-sight is corrected using a pair of converging lens A.I B.II **C.I and II** D.II and III E.I,II and III.
Short-sight is corrected using a diverging lens (concave) while Long-sight is corrected using a converging lens(convex).
60. Which of the following statements is correct about a long sighted boy who does not put on glasses A.he cannot see distant object clearly B.rays of light from close object are focused in front of the retina C.his eye ball is too long D.parallel rays of light are focused behind his retina **(None of the options are correct).**
A long-sighted person can see distant object clearly and cannot see near objects clearly i.e they appear blurred,eye ball is too short,rays of light form a close object are focused behind the retina while parallel rays of light from distant object falls on the retina. The opposite holds for a short-sighted person.
61. A far sighted person cannot see objects that are less than 100m away. If this person wants to read book 25cm away,what type and focal length of lens does he need? **A.convex, 33cm** B.convex, 20cm C.concave, 33cm D.concave, 20cm.
u=25cm, v=−100cm(image distance is negative)
f=? , $f = \frac{uv}{u+v} = \frac{25\times-100}{25+(-100)}$,
$f = \frac{-2500}{25-100} = \frac{-2500}{-75} = 33.3$cm
A converging or convex lens of focal length 33cm is required.
62. A man wears convex lens glasses of focal length 30cm in order to correct his eye defect. Instead of optimum 25cm, his least distance of vision is A.14cm B.28cm C.75cm **D.150cm.**
f=30cm, u=25cm, v=?, $v = \frac{uf}{u-f} = \frac{25\times30}{25-30}$,
$v = \frac{750}{-5} = -150$cm. The image will appear to be at 150cm which will be clearly seen by the man.
63. A certain far-sighted person cannot see object that are closer to the eye than 50cm clearly. Determine the power of the converging lens which will enable him to see at 25cm A.0.03D B.0.04D C.0.06D **D.0.02D.**
u=25cm, v=−50cm, f=?, P=?
$P = \frac{1}{f} = \frac{u+v}{uv} = \frac{25+(-50)}{25\times-50} = \frac{-25}{-1250} = 0.02D$.
64. A patient with a sight defect has a least

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distance of distinct vision of 150cm.For him to be able to read a material placed at a distance 25cm,what is the focal length of the glasses he should wear? A.15.0cm **B.30.0cm** C.17.6cm D.21.4cm.

u=25cm, v=−150cm, f=?, $f = \frac{uv}{u+v}$,

$f = \frac{25\times-150}{25+(-150)} = \frac{-3750}{-125} = 30$cm.

65. A person can focus an object only when it lies within 200cm from him. Which spectacles should be used to increase his maximum distance of distinc vision to infinity? **A.Concave lens** B.Plane glasses C.Binoculars D.Convex lens.

The person is shortsighted, as he cannot focus image of objects greater than 200cm, thus his maximum distance of distinc vision can be increased from 200cm to infinity by concave lens.

66. A near sighted person cannot see objects clearly when they are beyond 70cm from his eyes.What is the focal length of the correcting lens he should use? A.−25cm B.25cm **C.−70cm** D.70cm.

A concave lens is used to correct short-sightedness,as it makes object placed at infinity, u=∞, appear to be at the persons far point, v=−70cm. $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$, $\frac{1}{\infty}+\frac{1}{-70}=\frac{1}{f}$,

$\frac{1}{\infty}=0$, $\frac{1}{-70}=\frac{1}{f}$, $f=-70$cm (concave lens)

In myopic(short-sighted) persons generally,a concave mirror of focal length equal to the persons far-point or image distance is needed.

$f=v=-70$cm(from the question above).

67. A myopic man with a far point of 2m.What will be the prescription for the eye glasses he will use? A.−0.1 diopter **B.−0.5 diopter** C.0.5 diopter D.1.0 diopter.

v=−2m (far point), u=∞, $f=v=-2$m,

$P=\frac{1}{f}=\frac{1}{-2}=-0.5D$.

Jamb past questions on optical instruments and application of light waves :

[1979/11,1984/31,1986/30,1987/37,1999/31, 2006/33,2014/31,32]

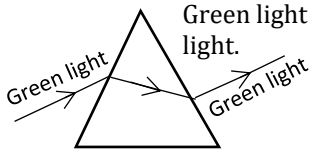
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Chapter 20 – Dispersion of light, Electromagnetic Spectrum

● **Dispersion –** Dispersion is the separation or splitting of a beam of white light into its components colours or spectra.

- Angular dispersion of light by a prism is due to difference in colour (wavelength), speed and refractive indices of the light.

- Wavelength is the characteristic of light that determines its colour. A monochromatic light is a light of one colour or wavelength. e.g green light. When a monochromatic light is passed through a prism, refraction occurs without dispersion.



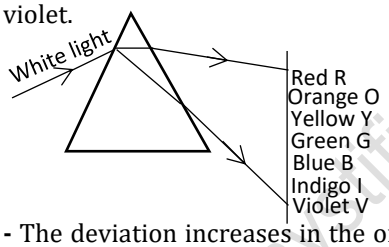
Green light is a monochromatic light.

- All colours of light travels with same speed in a vacuum.

-The component colours or spectrum of white light are red,orange,yellow,green,blue,indigo and violet **(ROYGBIV).**

- When white light is passed into a glass prism,it is separated into a spectrum on a screen consisting of red,orange,yellow,green, blue,indigo and violet (ROYGBIV).

- Red is least deviated,has the fastest speed in glass,highest wavelength,least refractive index and least change of speed. The opposite holds for violet.



- The deviation increases in the order ROYGBIV, with red being least deviated while violet is most deviated. The variation can be determined using the expression below:

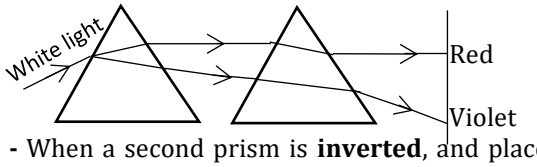
$\boldsymbol{D \propto \frac{n\Delta v}{v\lambda}}$ **,** D=deviation, n=refractive index, Δv=change in speed, v=speed in glass, λ=wavelength of light.

Thus, as deviation increases from red to violet, their refractive index and change in speed of the light colours increases while their speed in glass and wavelength decreases.

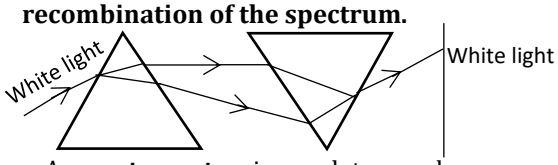
- Rainbow is formed by the dispersion of sunlight by spherical raindrops floating in the air which acts like tiny prisms.

- The spectrum of white light produced by one prism is an **impure spectrum as the adjacent colours overlap.**

- To obtain a **wider spectrum of colours**, another prism identical to the other is placed between it and the screen.



- When a second prism is **inverted**, and placed between the first prism and the screen, a spot of white light is formed on the screen due to **recombination of the spectrum.**



- A **spectrometer** is used to produce a pure spectrum. **Pure spectrum is that in which the colours are clearly separated or distinct from each other i.e they do not overlap.**

-A spectrometer consists of:

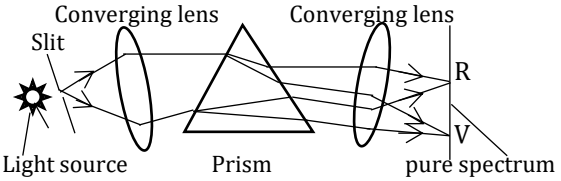
1.A divergent beam of light behind a narrow slit placed at the principal focus of the first converging lens in order to produce parallel rays. **The slit produces a series of narrow coloured images which minimizes overlapping of colours.**

2.The converging lens which produces parallel rays that are incident on the prism.

3.A 60° prism that disperses the white light into coloured spectrum of light.

4.The second converging lens which collects the parallel spectrum of light and brings them to focus on the screen,which is at the focal plane of the lens.

- Inorder to produce a pure spectrum, a slit, two converging or convex lens and triangular prism is required.



- Recombination of colours – The spectrum of white light (ROYGBIV) can be recombined to give white light by : **1.**Inverting prism **2.**Plane mirror **3.**Newton's colour disc **-**

Colour of objects –The colour of an **opaque object** is the colour of light it **reflects** while the colour of a **transparent object** is the colour of light it **transmits.**

- An object will appear white if all colours of white light are reflected and will appear black if no colour of white light is reflected i.e its absorbs all the clours.

- Classification of colours of lights(pure colours) and pigments(impure colours):

1.Primary colours – Colours that can be combined to form another colour e.g red,green and blue in light and red,yellow and blue in

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pigments.

2.Secondar colours – Colours obtained by mixing any two primary colours e.g

In lights:	**In pigments or paints:**
Red+Green=Yellow	Red+Yellow=Orange
Red+Blue=Magenta	Red+Blue=purple or violet
Blue+Green=Cyan	Blue+Yellow=Green

3.Complementary colours – Two colours which give a white colour when mixed e.g
Red+Cyan=White, Green+Magenta=White, Blue+Yellow=White,**Red+Blue+Green=White.**

- Colour mixing in pigments or paints – Pigments or paints absorbs certain colours from white light and reflect the rest.They are impure colours. E.g yellow paint(a mixture of yellow,red and green colours) absorbs blue light and reflects red and green light while a blue paint(a mixture of blue and green light) absorbs red and yellow lights but reflects green light.A mixture of yellow and blue paint thus reflects green light and absorbs blue,red and yellow light. i.e yellow paint +blue paint =green paint.

- Colour filters – are transparent objects used in analysis of colours in a transmitted light. They allow light of one colour to pass through them while absorbing light of other colours e.g If a white light is incident on a red filter, red light is transmitted while other colours are absorbed.

● **Electromagnetic Spectrum –** It consists of electromagnetic waves which are also transverse waves.

- The table below shows the types,source and properties of electromagnetic spectrum:

Spectrum	Source	Properties
1.Gamma rays	Radioactive materials.	Highly penetrating.
2.X-rays	X-ray tube, fast moving electrons stopped by a target.	Penetrating screen and stopped by bones.
3.Ultra violet rays	Sunlight,quartz lamps,gas discharge tubes.	Skin pigmentation, Fluorescence.
4.Light rays (visible ray)	Sunlight,lamps.	Vision.
5.Infra red rays	Heat radiated by hot bodies	Heat sensation, for taking photographs at night or in haze condition.
6.Radio waves	Transmitter	Radio signals

- They travel in vacuum with the same speed i.e speed of light($3\times10^{8}ms^{-1}$).

- The spectrum consists of gamma rays,x-rays, ultraviolet rays,light ray(visible light),infra red ray,radio waves and microwaves**(GXULIRM).**

- The frequency, penetration power and energy decreases while wavelength decreases in the order GXULIRM. Thus, gamma rays has the shortest wavelength & highest frequency while mircowaves has the longest wavelength & lowest frequency.

- Visible rays, infra red ray and ultra violet rays can be detected in the sun's spectrum.

Examples :

1. Dispersion of whoite light is the ability of white light to A.penetrate air,water and glass B,move in a straight line C.move around corners **D.separate to its component colours.**

2. Which of the following statements is not correct? **A.dispersion is a phenomenon by which light bends round corners** B.dispersion results from the difference in speed of light components within a material medium C.the violet component of white light has the lowest speed in any refracting medium D.all components of white light travel with the same speed in vacuum

Diffraction is the bending of light round obstacles not dispersion. Dispersion is as a result of the difference in speeds of the various components of light in glass. All colours of white light (ROYGBIV) travel with same speed in vacuum.

Deviation $\propto \frac{n\Delta v}{v\lambda}$,violet which is the most deviated colour in the spectrum, has the greatest refractive index and change in speed, and lowest speed and wavelength.

3.Dispersion of light by a glass prism is due to the A.Different hidden colours of the glass **B.Different speeds of the various colours in glass** C.Defects in the glass D.High density of glass.

4. A narrow beam of white light can be split into diffrerent colours by a glass prism. The correct explanation is that A.White light is an electromagnetic wave B.The prism has all the colours of white light C.White light has undergone total internal reflection in the prism **D.Different colours of white light travel with different speeds in glass** E.White light consists of yellow, green and red colours.

5. The angular dispersion of a prism depends on A.the angle of incidence as well as the index of refraction **B.the index of refraction only** C.the angle of incidence only D.the thickness of prism E.the prism angle

Dispersion of a prism depends on refractive index(or index of refraction) of the prism.

6. Rainbow formation is a natural phenomenon illustrating that white light A.splits into three components after rainfall B.is dispersed by glass

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prisms C.splits into its constituent colours before rainfall **D.is dispersed into its constituent colours by raindrops.**

7. A rainbow is formed when sunlight is incident on water droplets suspended in the air due to A.diffraction B.refraction **C.dispersion** D.interference

8. The spectrum of white light consists of coloured lights arranged in the following order A.Blue,red,green,yellow,indigo,violet,orange **B.Red,orange,yellow,green,blue, indigo,violet** C.Red,yellow,indigo,green,blue,violet D.Indigo,green,blue,violet,yellow,red,orange E.yellow,blue,green,violet,orange, indigo,red.

The spectrum of white is given by the acronym ROYGBIV, ranging from red to violet.

9. The visible part of light consists consists of the following colours: A.red,indigo,infra red, violet,yellow,green and blue **B.red,green, blue,violet,indigo,orange and yellow** C.blue, ultraviolet, infra red,red,yellow,indigo,violet, green and orange D.infra red and ultra violet E.red,orange, yellow,green,blue,indigo and infra red .

The visible part of light ranges from red to violet (ROYGBIV). The invisible part are infra red and ultra violet.

10. When white light is dispersed by a spectrometer, the component having the shortest wavelength is A.Orange B.Green **C.Violet** D.Red.

Deviation $\propto \frac{n\Delta v}{v\lambda}$, deviation increases in the order : ROYGBIV. Violet is most deviated hence will have the shortest wavelength.

11. Which of the following paurs of colours gives the widest separation in the spectrum of white light? **A.Red and violet** B.Green and yellow C.Red and indigo D.Yellow and violet

Red and violet are most widely separated in the spectrum(ROYGBIV) than the other pairs given.

12. Which of the following pairs of light rays show the widest separation in the spectrum of white light? A.Green and yellow B.Indigo and violet C.Orange and red **D.Blue and red** Red and blue are most widely separated in the spectrum(ROYGBIV) than the other pairs given.

13. A beam of light consisting of orange and blue colours is incident on a glass prism,which of the following is(are) correct? I.Orange is more deviated than blue light II.Blue is more deviated than orange light III.The refractive index of glass is higher for orange than for blue light IV.The refractive index is higher for blue than for orange light A.I B.II C.I and III **D.II and IV**

Deviation $\propto \frac{n\Delta v}{v\lambda}$, deviation increases in the order : ROYGBIV. Blue light is more deviated than orange light, hence will have a higher refractive index of glass than orange light.

14. The angle of deviation of light of various colours passing through a triangular prism increases in the order **A.red→green→blue** B.green→violet→blue C.blue→red→green→ D.blue→green→red

Deviation of the spectrum increases in the order – ROYGBIV: red,orange,yellow,green, blue,indigo and violet.

15. The angle of deviation of light of varous colours passing through a glass prism decreases in the order of **A.blue,orange and red** B.red, blue and orange C.blue,red and orange D.red, orange and blue E.orange,blue and red

Deviation of the spectrum increases in the order – ROYGBIV: red,orange,yellow,green, blue,indigo and violet. Thus, deviation decreases in the order: VIBGYOR.

16. A spectrum of white light is not a pure spectrum when **A.the different colours in it overlap** B.it is no red colour in it C.it is made up of only three colours D.it does not contain all the colours

The spectrum is said to be impure when some of the colours in the middle overlap one another.

17. A spectrum of sunlight is said to be impure when A.It is made of only three colours B.It has no red colour in it **C.The different colours overlap** D.It does not contain white light.

18. Which of the following arrangement in the sequence shown can be used to obtain a pure spectrum of white light is A.Source,slit,prism, diverging lens,screen **B.Source,slit,converging lens,prism,converging lens,screen** C.Source, converging lens,prism,diverging lens,screen D.Source,slit,diverging lens,screen

The slit provides a narrow beam,which is collimated, i.e made parallel by converging lens before it is incident on the prism.After refraction by the prism,the spectrum is brought to a focus on a screenplaced at the focal plane of the second converging lens.

19. The production of pure s[ectrum could easily be achieved using a A.triangular prism only B.triangular prism with two concave lenses **C.triangular prism with two convex lenses** D.glass prism with a pin.

20. In the production of pure spectrum of white light using a spectrometer, the function of the converging lens in a collimator is to A.produce a convergent beam which is incident on the prism **B.produce a parallel beam of light which is incident on the prism** C.diverge the incident beam a bit before dispersion D.focus the dispersed beam on the screen.

21. A source of light,a glass prism, and two converging lenses are suitably arranged to produce a pure spectrum of white light on a screen. The source of white light is then replaced with a sodium vapour lamp. The screen will now

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show A.the colours of the rainbow with dark lines crossing them **B.a large patch of yellow light** C.a narrow line of yellow light D.a red line and yellow line E.a white patch of light

The set up will produce a pure spectrum of light from a source. Sodium vapour lamp emits light of yellow colour(a monochromatic light), thus when passed through the set up a pure spectrum of yellow light widely separated is obtained.

22. Which of the following ecplains why a green leaf appears green in bright daylight? I.It absorbs only the green componene t of sunlight II.It absorbs all colours in sunlight except green III.I It reflects only green component of sunlight A.I B.II C.I and III **D.II and III.**

The colour of the green leaf i.e opaque body, appears green because it absorbs all other colours from sunlight but reflects thje green component.

23. A red flower appears red in sunlight because it A.transmits red light B.refracts red light C.absorbs red light **D.reflects red light**

The colour of opaque objects e.g flower, is seen by the colour it reflects. The colour of a transparent objects e.g glass,light filter, is seen by the colour it transmits.

24. In sunlight, a blue flower looks blue because we see the flower by the light it A.receives from our eye B.disperses C.arbsorbs D.refracts **E.reflects.**

25. When a yellow card is viewed in blue light, it will appear **A.black** B.orange C.purple D.red E.white

A yellow card appears yellow in white light because it reflects only yellow light(a mixture of red and green light) and absorbs light of other colours. Pure blue light is absorbed when incident on a yellow card and the card appears black. **If a green light was used the card will appear green,as it can reflect green from the card. If a red light was used the card will appear red,as it can reflect red from the card.**

26. When a yellow card is observed through a blue glass, the colour would appear **A.Black** B.Green C.Red D.White.

27. What is the apparent colour of a red shirt when viewed in pure green light? A.Red B.Green C.Yellow **D.Black** E.Blue

The green light is absorbed by the red short as they have no colour in common, thus the shirt appears black as no light is transmitted.

28. A white screen is illuminated simultaneously by red and green light beams. The colour seen is **A.yellow** B.blue C.white D.black

The red and green light both combine to form yellow light on the screen.

29. When a red rose flower is observed in blue light, what colour does the observer see? A.Magenta B.Yellow C.Red D.Blue **None of the options are correct**

The red rose flower will absorb the blue light, thus it appears black.

30. When light is incident on an object which is magenta in colour, which of the following colours will be absorbed? A.Red and blue **B.Green only** C.Red and green D.Red only

Magenta is a secondary light colour of red and blue light. Thus, when light is incident on the magenta coloured object, red and blue lights are reflected while green light is absorbed.

31. A red pen is placed on a yellow table mat.In green light, the pen and the mat will appear repectively A.red and yellow **B.black and green** C.black and white D.blue and green

In green light, the red pen absorbs green and appear black while the yellow table mat,which can reflect red and green light, appears green.

32.



White object

E,*F* and *G* are yellow, red and cyan transparent light filters repectively. The colour of a white opaque object when viwed through the arrangement of the filters abvove is A.Yellow B.Red **C.Black** D.Blue

Yellow light filter *E* transmits yellow(a mixture of red and green) from the white object (ROYGBIV). The red light filter *F* transmits only red and absorbs green. The cyan light filter *G* (transmits only blue and green), absorbs the red light and appears black.

33.

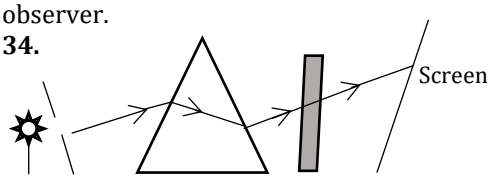


White object

In the diagram above, a white object is observed through green(G) and yellow(Y) light filters.The colour seen is A.black B.yellow C.blue **D.green**

The green light filter G transmit green from the white object(ROYGBIV). The yellow filter Y transmit the green colour which is seen by the observer.

34.



White light source prism Green light filter

In the diagram above light from a white light source is incident on a glass prism. The colour that will appear on the screen on passing through the green light filter is A.black B.white C.yellow **D.green.**

The white on passing through the prism is separated into the spectrum of white light ROYGBIV, which is then incident on the green filter. The green light filter absorbs all other colours except green light which it transmits to

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the screen.

35. If the Nigerian flag(green, white, green) is viewed in pure yellow light, which of the following set of colours would be observed on the flag? **A.green,yellow,green** B.red,yellow, red C.black,yellow,black D.green,white,green

Green part reflect green component from the yellow light and absorbs the red component while the white part reflects the yellow light.

36. Secondary colours of light A.red,green and blue B.green,white and red **C.cyan,magenta and yellow** D.yellow,blue and magenta

The secondary colours of pigemnts or paints are orange, green and purple or violet.

37. Which of the following is a secondary colour? A.Blue **B.Orange** C.Red D.Green

Green is a secondary colour in pigments or paints but a primary colour in light. Orange is a secondary colour in paints/pigments.

38. Which of the following is not part of the electromagnetic spectrum? I.Radio waves II.Beta rays III.Gamma rays IV.Alpha rays A.I and II B.I and III **C.II and IV** E.III and IV.

Alpha rays and beta rays are not electromagnetic waves, they are radioactive radiations.

39. Of the following which is different from the others? A.x-rays B.gamma rays **C.cathode rays** D.ultraviolet rays E.infrared rays

All other options apart from cathode rays are electromagnetic waves. Cathode rays are streams of fast moving electrons.

40. Which of the following electromagnetoic waves has the shortest wavelength? A.Radio waves B.X-rays C.Infra red D.Blue light E.Ultraviolet

Wavelength increases in the order – GXULIRM Thus, X-rays has the shortest wavelength from the options given.

41. In which of the following arrangements is the wavelength in an increasing order? A.Gamma rays,infra red rays,x-rays,radio waves **B.Gamma rays,x-rays,infra-red rays, radio waves** C.Radio waves,x-rays,gamma rays,infra-red rays D.Infra-red rays,radio waves,x-rays,gamma rays

Wavelength increases in the order – GXULIRM i.e gamma rays,x-rays,ultraviolet rays,light rays,infra-red rays,radio waves and micro waves.

42. The range of wavelengths of the visible spectrum is 400nm–700nm. The wavelength of gamma rays is A.Longer than 700nm B.Shorter than 700nm but longer than 400nm C.550nm **D.Shorter than 400nm** E.Infinite

Wavelength increases in the order – GXULIRM The wavelength of gamma rays is shorter than that of light waves or visible spectrum, thus it should be shorter than its lower limit (400nm)

43. Which of the following electromagnetic waves have the lowest frequency? **A.radio waves** B.infra-red C.ultra-violet D.x-rays

Frequency decreases in the order – GXULIRM. Radio waves have the lowest frequencby from the options given.

44. Which of the following electromagnetic waves has the highest frequency? A.Infra-red rays **B.X-rays** C.Ultraviolet rays D.Radio waves

Frequency decreases in the order – GXULIRM. X-rays has the highest frequency from the options given.

45. Which of the following electromagnetic waves is least energetic? **A.Infra-red rays** B.X-rays C.Ultra-violet rays D.Gamma rays

The energy in the spectrum decreases in the order – GXULIRM, hence gamma rays is most energetic while Infra-red is least energetic in the options.

46. Which of the following statements is true of ultraviolet, visible and infra-red rays? A.They are all electromagnetic waves with the same wavelength **B.Ultraviolet rays have shorter wavelengths than infra red rays and produce fluorescence** C.Infra-red rays have shorter wavelength than visible light and produce sensation of heat D.The wavelengths increase in sequence: visible,ultraviolet,infra- red

They are all electromagnetic waves with different wavelengths. Their wavelength increases in the order:ultraviolet rays,visible light and infra red. Thus, the ultraviolet has a shorter wavelength than visible light and infra-red while visible light has a shorter wavelength than infa-red. Ultraviolet rays causes fluorescence in bodies while onfra-red causes heat sensation.

47. Which of the following are true about infra-red radiation? I.It is invisible II.It is called heat ray III.Its frequency is higher than that of blue light IV.It travels as a transverse A.I,II,III and IV **B.I,II and IV** C.I,III and IV D.II,III and IV

Only light rays(visible light) are visible in the electromagnetic spectrum,others invisible i.e infra-red ray(also called heat ray) is invisible. Frequency decreases in the order: GXULIRM, i.e Blue light(light ray) has higher frequency than infra-red ray. All electromagnetic rays are transverse.

48. Radio carrier waves are generated by causing electrons to oscillate rapidly in A.a microphone B.a telephone camera C.an antennaC.coaxial cables **E.a transmitter.**

Jamb past questions on dispersion of light and electromagnetic spectrum :

[1979/40,1980/37,1983/13,1984/40,1985/34, 1986/29,1988/35,1989/29,1990/30,1992/30, 1993/31,1994/39,2010/32,2011/29,2012/37, 2013/28].

Demystified Series by Dr Timothy

Chapter 21 – Electrostatics, Electric field, Capacitors

● **Electrostatics –** This involves the aspect of electricity that deals with charges at rest.

- Electric Charges – It represents the amount or quantity of electricity in a body. There are two types of charges : positive and negatice charges. **Positive charge means deficit of electrons while negative charges means excess of electrons.**

- The quantity or magnitude of any charge Q is a multiple n of the charge of an electron e. $Q = ne$, $e = 1.6\times10^{-19}$**C.**

- Charge carriers in solids are electrons.

- Charge carriers in liquids are ions.

- Charge carriers in gases are electrons and ions.

- Fundamental law of electrostatics states that like charges repel while unlike charges attracts.

- Charging or Electrification of bodies – In charging or electrification, charges are neither created nor destroyed but can only be transferred or induced.

- **Methods of charging bodies –** By friction, contact or electrostatic induction.

1. Charging by friction – For charging insulators and conductors. Rubbing two uncharged bodies together results the to exchange of **equal magnitude of electrons to produce opposite charges. A glass rod or cellulose acetate strip rubbed with silk becomes positively charged while an ebonite or polythene rod rubbed with fur becomes negatively charged.** The glass rod losses electrons to silk to become positive and the silk negative.The ebonite gains electrons from fur to become negative and fur positive.

- No charge is produced when glass rod is rubbed with fur or ebonite rod rubbed with silk.

2. Charging by contact – For charging conductors only. Transferring of chargesof the **same sign but of less quantity** from an insulated charged body to an uncharged body.

3. Charging by electrostatic induction – For charging conductors only. A charged body is brought close to an uncharged body placed on an insulating base**(both bodies do not touch)**, an **induced charge of equal magnitude but opposite sign** is formed on the uncharged body.

- The charge in the charged rod(inducer) is preserved. No charges are lost nor transferred between both bodies.

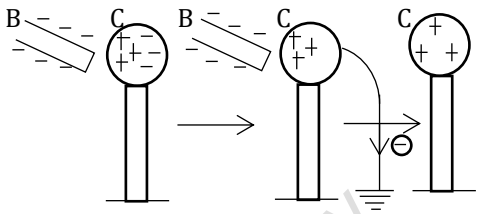
- Charging a single body by induction –

1.A charged rod(inducing charge) B is brought close to a conductor C which sits on an insulating base.

2.Electrons are repelled to the other end of the conductor i.e distribution of opposite charges.

3.Earthing e.g by touching with the finger,to conduct some charges to the earth.Then removal of the finger.

4.The charged rod(inducing charge) is removed, while the induced charges of opposite sign distributes itself in the conductor.



In summary – Placement of charged body → Earthing → Separation(unearthing) → Removal of charged body.

- If a positively charged rod is used, a negative charge is induced in the conductor and if a negatively charged rod is used, a positive charge is induced in the conductor.

- Charging two bodies in contact by induction

1.A charged rod is brought close to the one end of both conductors.

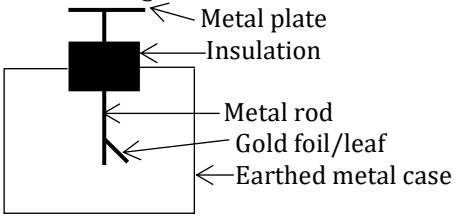
2.Charges of opposite kinds is induced in both conductors with the conductor close to the charged rod acquiring opposite sign as it, while that farther from the charged rod acquires same sign as the charged rod.

3.Separation of the conductors through short distance with the charged rod still still in place.

4.The charged rod is removed.

- The end close to the charged rod acquires opposite charges by inductionwhile the end far from the charged rod acquires same charge as the charged rod.

- Gold leaf electroscope – It is used to detect and test for small charges.



- A gold leaf electroscope can be charged by contact or induction.

Charging by induction – **i.**A charged object is placed near the metal cap **ii.**The charged rod still in placed,the cap of the electroscope is earthed temporarily. **iii.**After breaking the earthing, the charged rod is withdrawn **iv.**The gold leaf and metal rod acquire similar charge and they diverge from each other.

- Gold leaf electroscope is used for – 1.Comparing the insulating property of

Demystified Series by Dr. Timothy

different materials/distinguishing between conductors and insulators : When the metal cap of a charged electroscope is touched with a body ,a rapid collapse of the leaves indicates the body is a conductor, gradual or slow collapse indicates the body is a poor conductor while no change in divergence/collapse indicates the body is an insulator.

2.Comparing the sign and magnitude of charges : Increase in divergence of a charged electroscope is seen when a charged rod of the same charge is placed near it e.g the electroscope is positive and the charged rod is also positive. Decrease in divergence of a charged electroscope is seen when a charged rod of opposite charge or a neutral(uncharged) rod is placed near it. **Increase in divergence is the true test of sign of a charge.**

- Bringing opposite charges to a charged electroscope – divergence increases at a lesser height (close to it) while divergence decreases at a greater height (far from it).

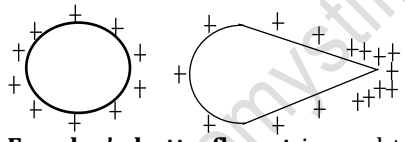
- The degree of divergence of the leaves of the electroscope is a measure of the relative magnitudes of the charges on the bodies.

- The leaf in an electroscope can be made of thin sheet of gold or aluminium,as they can diverge easily.

- Charge density – It is charge per unit area of a surface.

- Charge density is highestat the sharpestpoint of a conductor, as the sharpest point has the smallest area and greatest concentration of charges.

- Charges are mostly concentrated at the outermost part(not innermost part) of hollow conductor.



- Faraday's butterfly net is used to determine the distribution of charges between the innermost and outermost part of a hollow conductor.

- Proof plane – For transferring charges by contact (conduction).

- Electrophorus – It is referred to as the **simplest electrostatic generator.** It is used to produce charges(by induction), store or secure charges and transfer charges.

- Discharging bodies – Charges in a body can be removed by earthing,neutraliziation by opposite charges from surrounding air(atmospheric ionization), by point action and ionization of flames.

- Lightning – Lightning is caused by a storm cloud which is higly charged as a result of the the cloud rubbing against the air i.e charging by friction. Lightning flash results from a sudden flow of charge between the cloud and the earth.

- Lightning conductor – It consists of an iron rod with spikes or copper strip with a sharply pointed end.It is used for protecting a building from being destroyed by lightning.

- If a positively charged cloud passes over the building,negative electric charges are induced at the copper strip i.e the tip, of the lightning,leading to a dipole with the cloud. The atoms of the air undergo ionization due to high electric field between the cloud and tip of the conductor, with negative ions drifting towards the cloud and neutralizes the positive charge in the cloud.

- Coulumb's law – It states that the force of attraction or repulsion F between two point charges q_1 and q_2 is proportional to the product of their charges and inversely proportional to the square of their distance r apart.

$\boldsymbol{F \propto \frac{q_1 q_2}{r^2}}$, $\boldsymbol{F = \frac{Kq_1 q_2}{r^2}}$, $\boldsymbol{F = \frac{1}{4\pi\varepsilon} \times \frac{q_1 q_2}{r^2}}$.

$K=\frac{1}{4\pi\varepsilon}=9.0 \times 10^9 \text{Nm}^2\text{C}^{-2}$,$\varepsilon$=permittivity of medium=$8.854\times10^{-12}\text{C}^2\text{N}^{-1}\text{m}^{-2}$ or 8.854×10^{-12} Farad per meter.

- Coulomb's law is valid only for point charges i.e charges whose value is much smaller compared to the distance between them.

- The greater the force the smaller the deflection or distance between the charges.

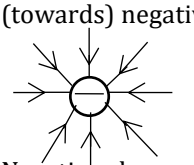
-The force acting on the charges increases as the distance between the charges decreases.

$\boldsymbol{F \propto \frac{1}{r^2}}$, $\boldsymbol{F_1 {r_1}^2 = F_2 {r_2}^2}$.

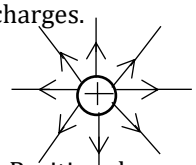
- Electric field – It is a region or space round a charged body where electric force or a region where electrostatic induction takes place when a conductor is placed within. **- Electric flux $\boldsymbol{\phi}$ –** These are imaginary lines that represents the direction of the electric field.

- Electric force and electric field are both vector quantities.

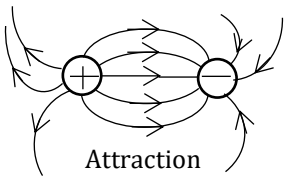
- Electric flux spreads radially outwards for positive charges and spreads radially inwards (towards) negative charges.



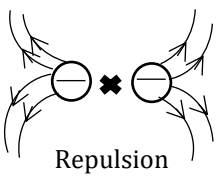
Negative charge radially inwards



Positive charge radially outwards



Attraction



Repulsion

- The total electric flux through a closed surface due to any charge not enclosed by the surface is zero.

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- Electric flux density E – The electric flux ϕ per unit area A perdendicular to the direction of the electric field. $\boldsymbol{E = \frac{\phi}{A}}$.

- Electric flux density is same as electric field intensity.

- Electric field intensity or electric field strength$E$ – This is the force per unit positive charge at any point or potential difference across a unit distance. **It represents the magnitude of the electric field.** $\boldsymbol{E = \frac{F}{Q} = \frac{KQ}{r^2} = \frac{Q}{4\pi\varepsilon r^2} = \frac{V}{r}}$, E=electric field intensity=$N\text{C}^{-1}$ or $V\text{m}^{-1}$, Q=point charge, V=potential difference.

- When two charges of different magnitude are placed at the same distance each other in the same medium but different electric field intensities, the electric field intensity and the charges are related by – $E = \frac{Q}{4\pi\varepsilon r^2}$, ε is constant in the same medium, r is constant at the same distance, thus : $\boldsymbol{\frac{E_1}{Q_1} = \frac{E_2}{Q_2}}$.

- Electric potential V – It is the workdone in bringing a unit charge from infinity to a point in an electric field. **A positively-charged body has a positive potential, a negatively-charged body has a negative potential while a neutral body (e.g the earth) has a zero potential.**

- Electric potential difference between two points is the workdone W in taking a unit charge q,from other point to the other.

$\boldsymbol{W = QV,\ V = \frac{W}{Q} = \frac{KQ}{r} = \frac{Q}{4\pi\varepsilon r}}$.

- For an isolated positive or negative charge the potential energy is zero, as it not placed in an external electric field.

- When two like charges are brought together, work is done against the field,as like charges repel each other.So work has to be done against the field to bring them together.

- When two opposite charges are borught together, work is done by the field, as unlike or opposite charges attract each other. So work is done by the field to bring them together.

- If work is done against the electric field,the electric potential is positive and the potential energy increases.

-If work is done by the electric field, the electric potential is negative and the potential energy decreases.

- Electric current is due to the flow of charges i.e positive charges or negative charges (electrons). Positive charges flows from higher to lower potential (i.e direction of electric field) while negative charges (electrons) flow from lower to higher potential(i.e opposite to the electric field).

- Properties of electric field –

1.In a uniform field, the lines of force are straight, parallel and uniformly spaced.

2.Lines of force do not cross each other i.e as electric field do not point in different direction at the same point.

3.Lines of force starts only on positive charges and end only on negative charges,with the number of field lines starting or ending being proportional to the magnitude of the charge.

4.The closer the field lines are together, the stronger the electric field at that point. i.e the greater the force,the less the distance or deflection or the lesser the force,the greater the deflection or distance.

5.The force on charges can be repulsive or attractive.

6.The force is exerted in same direction with the field and it deflects charges, change the speed or kinetic energy of the charges.

Examples :

1. Which of the following statements is correct? A.charges generated on glass rubbed with silk are called negative B.charges in various media can be carried by protons detached from their atoms **C.the magnitude of any charge is a multiple of the charge of an electron** D.the intensity of an electric field is a scalar quantity E.if 500 Joules work are required to move 2 coulombs charge between two points, the potential difference between the two points is 100V

Glass rod rubbed with silk acquires positive charge. Charges in various media are not carried by protons i.e electrons in solids, ions in liquids, electrons and ions in gases. The intensity or strength of all fields (electric, magnetic and gravitational) are vector quantity. Potential difference= $\frac{Workdone}{Charge}$, $V = \frac{W}{Q} = \frac{500}{2} = 250$ Joule per coulomb or 250 volts. The magnitude of any charge is a multiple of the charge of electrons i.e $Q = ne$.

2. Which of the following is true of an electrical charge? A.positive charge means deficit of electrons B.negative charge means excess of electrons C.electric current means movement of electrons **D.all of the above** E.A and B only

Positive charge means excess of protons and deficit of electrons. Negative charges means deficit of protons and excess of electrons.

3. The production and type of charge of a body can only be accounted for by the A.equilibrium in protons and neutrons **B.deficiency or surplus of electrons** C.deficiency or surplus of protons D.deficiency or surplus in neutrons

Surplus of electrons indicates negative charge while deficiency of electrons indicates positive charge.

4. An electric charge could be transmitted through A.wood,rubber and stone B.paper, clay and plastic C.glass,acid and cloth **D.the human**

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body,water and metals E.all of the above.

5. When a biro pen rubbed on a dry silk cloth is moved very closed to a piece of paper on a dry table, the pen is found to pick up the paper. This is because A.Both the pen and the cloth are magnetized B.The pen is magnetized but the cloth is not C.The pen is charged whule the cloth is magnetized **D.Both the pen and the cloth are charged.**

The biro rubbed on the dry silk cloth is charged by friction and on moving it closed to a piece of paper it induces opposite charges on the paper, thus it attracts it and picks it up.

6. When an ebonite rod is rubbed with fur, it has A.No charge at all **B.A negative charge** C.A positive charge D.Negative and positive charges

Ebonite rod rubbed with fur acquires negative charge.

7. An insulated rod when rubbed with a material acquires A.a negative charge if it is made of glass rubbed with silk **B.no charge if it is made of glass and rubbed with fur** C.no charge if it is made of copper and rubbed with silk D.a positive charge if it is made of copper rubbed with fur.

Glass(an insulator) rubbed with silk acquires positive charge. Ebonite(insulator) rubbed with fur acquires negative charge. No charge is acquired when glass is rubbed with fur or ebonite rubbed with silk. Copper is a conductor not an insulator,as stated in the question.

8. An ebonite rod rubbed with fur attracts a glass rod rubbed wth silk because **A.Ebonite has a negative charge while glass has a positive charge** B.Ebonite has a positive charge while glass has a negative charge C.Both have negative charges D.Both have positive charges.

Ebonite rod acquires negative charges when rubbed with fur while glass rod acquires positive charges when rubbed with silk. Unlike charges attract, thus negatively charged ebonite rod attracts the positively charged glass rod.

9. A short chain is usually attached to the rear side of a petrol tanker trailing behind it to ensure that A.the heat generated in the engine is conducted to the earth **B.the charges generated by friction in the tanker are conducted to the earth** C.the tanker does not exceed a certain speed limit D.the chain vibrates in resonance with the tanker engine. The friction due to the rubbing action between the petrol tanker and air results in the generation of electric charges that collects on the petrol tankers body, which are conducted to the earth through the chain.

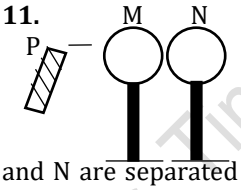
10. In the process of electrification by induction, the amount of induced charges in relation to the inducing charge is A.same sign and quantity B.same sign but greater C.opposite sign but less D.opposite sign and greater **E.opposite sign and equal.**

Charging an uncharged conductor by induction produces charges of opposite sign and equal magnitude as the inducing charge. Charging by friction also produces equal magnitude of charges but of opposite signs.

Charging by contact produces charges of same sign but of less magnitude.

10. What is the correct sequence of steps for charging an insulated conductor by induction? I.bringing a charged body near to the conductor II.removing the charged body III.touching the conductor with a finger IV.removing the finger A.I,II,III,IV **B.I,III,IV,II** C.II,I,III,IV D.III,IV,I,II E.I,III,II,IV .

Placement of charged body → Earthing (touching with finger) → Separation (removing the finger i.e unearthing) → Removal of the charged body.

11. 

In the figure, M and N are insulated metal spheres in contact. A negatively charged ebonite rod P is is brought near M, when M and N are separated and the rod is taken away, which of the following statement is correct? A.Both M and N have acquired negative charges B.Both M and N have acquired positive charges C.M has acquired negative charges and N positive charges **D.M has acquired positive charge and N negative charge.**

The end close to the charged rod acquires opposite charges by inductionwhile the end far from the charged rod acquires same charge as the charged rod.Thus, *M* is positively charged while *N* is negatively charged.

12. The magnitude of charges on two bodies are to be compared. Which of the following instruments would be suitable? **A.Gold leaf electroscope** B.Capacitor C.The electrophorus D.Ebonite rod.

13. The sign of the charge on a charged glass rod may be determined with A.an electrometer **B.a charged electroscope** C.a galvanometer D.an uncharged electroscope.

Increase in divergence is a true test of sign.

14. A positively charged rod *X* is brought near an uncharged metal *Y* and is then touched by a finger with *X* still in place. When the finger is removed, the result is that *Y* hasd A.A negative charge and a positive potential B.No charge and zero potential **C.A negative charge and a negative potential** D.A positive charge and a zero potential.

The positively charged rod induces negative charge in the uncharged metal. Negative charge is at a negative potential.

15. A negatively-charged rod is brought near the cap of a gold leaf electroscope.The cap is earthed momentarily with the rod near to it. The rod is then removed. Which of the following is correct?

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A.the cap will a positively charged and the negatively charged leaves will diverge B.the cap will be negatively charged and the positively charged leaves will diverge **C.the cap and the leaves will be positively charged** D.the cap and the leaves will be negatively charged E.the leaves will not diverge because the cap has been earthed.

Negative charged rod induces positive charges on the cap and leaf of the electroscope. Earthing conducts excess electrons away from the leaf, thus divergence of the leaf increases.

16. What happens when a positively charged insulator is placed near the disc of a positively charged gold leaf electroscope, and is then removed? A.the leaf falls, and then rises when the body is removed **B.the leaf rises further, and remains in that position when the body is removed** C.the leaf falls, and remains in that position when the body is removed D.the leaf rises further and falls to its original position when the body is removed E.nothing happens

Leaf diverges(rises) further when a positively charged rod is brought close to a positively charged electroscope and remains in this position when the body is removed.The leaf can be made to return to its original position by earthing.

17. As a negatively charged rod is brought nearer to the cap of a positively charged gold leaf electroscope,the divergence of the leaf will **A.decreases to zero** B.steadily increase C.decrease to zero and then increase D.remain constant E.increase to a maximum and then decrease.

Divergence of the leaf decreases when a negatively charged rod is brought close to a positively charged electroscope. Option B would have been correct if both the electroscope and charged rod bears same the charge.

18. As a positively charged rod is brought nearer to the top cap of a positively charged gold leaf electroscope, the divergence of the levaes will A.decrease to zero **B.steadily increase** C.decrease to zero and then increase D.remain constant E.increase to maximum and then decrease.

Divergence increases as the charged rod and the gold leaf electroscope are positively charged.

19. A source emitting alpha particlesis brought near the cap of a negatively charged gold-leaf electroscope. The gold leaf will A.diverge and remain there **B.fall and remain in its new position** C.fall and then rises up again to its initial position D.not be affected.

Alpha particles are positively charged. Thus, when brought close to a negatively charged electroscope, the divergence decreases and the gold leaf fall.

20. A charged ball is placed in contact with the inner wall of a hollow conducting sphere. The resulting electric field is A.inside the sphere B.on the surface of the ball C.inside and outside the sphere **D.outside the sphere**.

Excess charges are concentrated on the outside(surface) of the sphere and the electric field inside the hollow sphere is zero.

21. The field that results from the contact of a positively charged ball with the inner wall of an uncharged hollow conducting sphere is A.Inside the sphere B.On the surface of the ball C.Inside and outside the sphere **D.Outside the sphere.**

Charges are concentrated in the outer part of a hollow conductor.

22. Which of the following statements are true of an insulated charged body carrying bpositive charges? I.It contains excess positive charges II.It has an electric field III.It possesses potential energy IV.It carries an electric current **A.I and II** B.I,II and III C.II and III D.I,II,III and IV.

Positively charged bodies contains more protons than electrons. The excess positive charge(protons) creates an electric field. For an isolated positive or negative charge the potential energy is zero, as it not placed in an external electric field.

23. Two unequal insulated charged spheres carrying unlike charges are connected together with a short wire.Charges will flow from one sphere to the other until both spheres A.carry equal charges of opposite signs B.carry equal charges of the same sign **C.are at the same potential** D.become uncharged

Charges/current always flow between two points of different electric potentials until their potential equalizes i.e equipotential.

24. The instrument used for securing a large number of similar charges by induction is called A.electroscope B.proof plane **C.electrophorus** D.capacitor

Electrophorus stores (secures), separate, produce and transfers charges by induction.

Proof plane transfers charges by conduction (contact). Capacitor stores charges when a cell(battery) is connected across its terminal.

Electrophorus is the simplest electrostatic generator.

25. When a conductor is mounted on an insulating stand is charged and left for sometime the conductor eventually losses all of its charges. This is because the A.electric charges evaporate from the surface of the conductor **B.charges on the conductor are neutralized by opposite charges from the surrounding air** C.chrges ionizes the surrounding air D.charges are conducted to the earth.

26. Which of the following statements about lightning conductor is not correct? A.It initiates

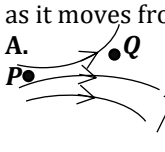
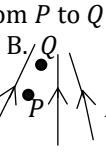
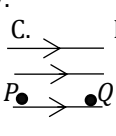
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point action B.The spikes gain an induced charge opposite to that of the cloud C.It consists of a thick metal strip connecting sharp points to a metal plate **D.The metal strip receives charges and discharges them to the air through the sharp points**

27. Which of the following statements are correct about lightning conductor? I.they prevent tall buildings from being damaged by lightning II.they each consist of a sharp pointed rod at one end connected to earth through a thick cable III.charges opposite to those in the thunder cloud are induced in the lightning conductors IV.the induced charge flows down the cable and neutralizes the charge in the cloud A.I and II B.II and III **C.I,II and III** D.I,II and IV E.I,III and IV.

Ligthning conductors consist of a sharp pointed rod or strip made of copper, at the top and is connected to the earth by a thick iron metal cable.They prevents tall buildings from collapsing by lightning. Charges opposite to the cloud are induced in the lightning conductors. The induced charge i.e the charge at the sharp point, is conducted to the earth.It is the charge produced from the ionization of air that neutralizes the charge in the cloud not the induced charge.

28. The diagram below show lines of force in various electric fields. In which of the diagrams would a positive test charge experience less force as it moves from P to Q.

A.  B.  C.  D. 

Curved lines represents non-uniform fields (fields that change directions from one place to the other). The greater deflection of the charge or distance moved by the charge indicates a lesser force applied to the test charge ($F \propto 1/r^2$) . The test charge deflects through a larger angle in option A.

29. If the distance between two stationary charged particles is doubled, the magnitude if the electrostatic force between them will be A.doubled B.halved **C.a quarter of its former value** D.unchanged E.four times the former value

$F_1=F, r_1=r, F_2=?, r_2=2r,\ F \propto \frac{1}{r^2}$,

$F_1{r_1}^2 = F_2{r_2}^2,\ F \times r^2 = F_2 \times (2r)^2$,

$Fr^2 = F_2 \times 4r^2$, $F_2 = \frac{Fr^2}{4r^2} = \frac{F}{4}$ i.e quarter of its former or original value. Alternatively,

$F \propto \frac{1}{r^2},\ r$ is doubled thus, $F = \frac{1}{2^2} = \frac{1}{4}$ i.e quarter of its orginal or former value.

30. If the distance between two point charges is increased by a factor of four, the magnitude of the electrostatic force between them will be A.1/2 of its former value **B.1/16 of its former value** C.1/4 of its former value D.4 times of its former Value.

$F_1=F, r_1=r, F_2=?, r_2=4r,\ F \propto \frac{1}{r^2}$,

$F_1{r_1}^2 = F_2{r_2}^2,\ F \times r^2 = F_2 \times (4r)^2$,

$Fr^2 = F_2 \times 16r^2$, $F_2 = \frac{Fr^2}{16r^2} = \frac{F}{16}$ i.e 1/16 of its former or original value. Alternatively,

$F \propto \frac{1}{r^2},\ r$ is doubled thus, $F = \frac{1}{4^2} = \frac{1}{16}$ i.e 1/16 of its orginal or former value.

31. The electric force between two unlike charges, each of magnitude q, separated by a distance $d$in air of permittivity ε_o is A.$\frac{q^2}{4\pi\varepsilon_o d}$ B.$-\frac{q^2}{4\pi\varepsilon_o d}$ C.$\frac{q^2}{4\pi\varepsilon_o d^2}$D.$-\frac{\boldsymbol{q^2}}{\boldsymbol{4\pi\varepsilon_o d^2}}$

$q_1=q$, $q_2=-q$, $r=d$, $F = \frac{1}{4\pi\varepsilon} \times \frac{q_1 q_2}{r^2}$,

$F = \frac{1}{4\pi\varepsilon_o} \times \frac{q \times -q}{d^2} = -\frac{q^2}{4\pi\varepsilon_o d^2}$.

32. The ratio of electrostatic force F_E to gravitational forcen F_G between two protons each of charge e and mass m, at a distance d is A.$\frac{e}{4\pi\varepsilon_o Gm}$ B.$\frac{e^2}{Gm^2}$ C.$\frac{Gm^2}{4\pi\varepsilon_o e^2}$ **D.**$\frac{\boldsymbol{e^2}}{\boldsymbol{4\pi\varepsilon_o Gm^2}}$**.**

Electrostatic force of repulsion $F_E = \frac{e^2}{4\pi\varepsilon_o d^2}$,

Gravitational force of attraction $F_G = \frac{Gm^2}{d^2}$,

$F_E{:}F_G = \frac{F_E}{F_G} = \frac{\frac{e^2}{4\pi\varepsilon_o d^2}}{\frac{Gm^2}{d^2}} = \frac{e^2}{4\pi\varepsilon_o d^2} \times \frac{d^2}{Gm^2} = \frac{e^2}{4\pi\varepsilon_o Gm^2}$

33. The force of repulsion between two point positive charges 5μC and 8μC separated at a distance of 0.02m apart is A.1.8×10^{-10}N B.9.0×10^{-8}N **C.$\boldsymbol{9.0\times10^{2}}$N** D.$4.5\times10^{3}$N.

($\frac{1}{4\pi\varepsilon} = 9.0 \times 10^9 \text{Nm}^2\text{C}^{-2}$)

$q_1=5\mu C=5\times10^{-6}$C, $q_2=8\mu C=8\times10^{-6}$C, $r=0.02$m

$F = \frac{Kq_1q_2}{r^2} = \frac{1}{4\pi\varepsilon} \times \frac{q_1q_2}{r^2}$,

$F = \frac{9.0\times10^9\times5\times10^{-6}\times8\times10^{-6}}{0.02^2} = \frac{360\times10^{-3}}{4\times10^{-4}}$,

$F = 900N = 9.0\times10^2$N.

34. Two bodies each carrying a charge of 2.00×10^{-10}C are 5cm apart. Calculate the magnitude of the force between the charges. ($\frac{1}{4\pi\varepsilon}=9.0 \times 10^9 \text{Nm}^2\text{C}^{-2}$) **A.$\boldsymbol{1.44\times10^{-7}}$N**

B.7.20×10^{-11}N C.1.44×10^{-11}N D.7.20×10^{-9}N.

$q_1=q_2=q= 2\times10^{-10}$C, r = 5cm = 0.05m,

$F = \frac{Kq_1q_2}{r^2} = \frac{1}{4\pi\varepsilon} \times \frac{q_1q_2}{r^2}$,

$F = \frac{9\times10^9\times(2\times10^{-10})^2}{0.05^2} = \frac{9\times10^9\times4\times10^{-20}}{2.5\times10^{-3}}$,

$F = 1.44\times10^{-7}$N.

35. Two identical charges situated 20cm apart in vacuum repel each other with a force of 1.0N. Calculate the value of each charge. [$(4\pi\varepsilon_0)^{-1} = 9\times10^9 \text{Nm}^2C^{-2}$] A.$3.6\mu$C B.$4.0\mu$C **C.$\boldsymbol{2.1\mu}$C** D.$4.4\mu$C.

$q_1=q_2=q=?$, r=20cm=0.2m, F=1.0N,

$(4\pi\varepsilon_0)^{-1} = 9\times10^9 \text{Nm}^2\text{C}^{-2}$, $F = \frac{1}{4\pi\varepsilon_o} \times \frac{q_1q_2}{r^2}$,

$1.0 = \frac{9\times10^9\times q^2}{0.2^2}$, $1.0 = \frac{9\times10^9\times q^2}{0.04}$,

$0.04 = 9\times10^9\times q^2$, $q^2 = \frac{0.04}{9\times10^9} = 4.44\times10^{-12}$,

$q = \sqrt{4.44\times10^{-12}} = 2.1\times10^{-6}\text{C} = 2.1\mu\text{C}$.

Demystified Series by Dr. Timothy

36. Calculate the electric field strength at a point 0.05m from a point charge of magnitude 3.14×10^{-6}C in a medium of permittivity 2.5×10^{-11}Fm^{-1}. (π=3.14) A.$2.0\times10^{5}NC^{-1}$ **B.4.0×10^{6}NC^{-1}** C.1.1×10^{7}NC^{-1} D.2.2×10^{8}NC^{-1}.

Q=3.14×10^{-6}C, r=0.05m, ε=ε_0=2.5×10^{-11}Fm^{-1}, π=3.14, $E=\frac{Q}{4\pi\varepsilon_o r^2}=\frac{3.14\times10^{-6}}{4\times3.14\times2.5\times10^{-11}\times(0.05)^2}$

$E=\frac{3.14\times10^{-6}}{4\times3.14\times2.5\times10^{-11}\times2.5\times10^{-3}}=4\times10^{6}$NC^{-1}.

37. A charge of 4μC is placed 3m away from appoint a point P, calculate the magnitude of the electric field intensity, due to the charge at the point P A.$8.1\times10^{4}V$m^{-1} B.$2.7\times10^{-4}V$m^{-1} C.$1.2\times10^{4}V$m^{-1} **D.$4.0\times10^{4}V$m^{-1}**

($\frac{1}{4\pi\varepsilon}$=$9.0\times10^{9}$Nm^{2}C^{-2}).

Q=4μC=4×10^{-6}C, r=3m, $\frac{1}{4\pi\varepsilon}$=$9.0\times10^{9}$Nm^{2}C^{-2},

$E=\frac{KQ}{r^2}=\frac{Q}{4\pi\varepsilon r^2}=\frac{9\times10^{9}\times4\times10^{-6}}{3^2}$,

$E=\frac{9\times10^{9}\times4\times10^{-6}}{9}=4\times10^{4}V$m^{-1}.

38. A charge of 2.0×10^{-5}C experiences a force of 80N in a uniform electric field. Calculate the magnitude of the electric field intensity A.8.0×10^{6}NC^{-1} **B.4.0×10^{6}NC^{-1}** C.4.0×10^{-4}NC^{-1} D.2.0×10^{4}NC^{-1}.

E=?, F=80N, Q=2.0×10^{-5}C, $E=\frac{F}{Q}$,

$E=\frac{80}{2.0\times10^{-5}}$=$4.0\times10^{6}$NC^{-1}.

38. Calculate the force acting on an electron of charge 1.6×10^{-19}C placed in an electric field of intensity $10^{8}V$m^{-1} **A.1.6×10^{-11}N** B.1.6×10^{-14}N C.1.6×10^{-16}N D.1.0×10^{-16}N.

Q=1.6×10^{-19}C, E=$10^{8}V$m^{-1}, F=?, $E=\frac{F}{Q}$,

$F=EQ=10^{8}\times1.6\times10^{-19}=1.6\times10^{-11}$N.

39. Two parallel plates at a distance of 9×10^{-3}m 700V. The electric field strength between them is A.6.30×10^{0}Vm^{-1} B.1.26×10^{-4}Vm^{-1} C.7.78×10^{1}Vm^{-1} D.6.30×10^{3}Vm^{-1} **E.7.78×10^{4}Vm^{-1}**

V=$700V$, r=9×10^{-3}m, E=?, $E=\frac{V}{r}=\frac{700}{9\times10^{-3}}$,

E=$7.78\times10^{4}V$m^{-1}.

40. A charge of 50μC has an electric field strength of 360NC^{-1} at a certain point. The electric field strength due to another charge 120μC kept at the same distance apart and in the same medium is A.18NC^{-1} B.144NC^{-1} **C.864NC^{-1}** D.150NC^{-1}

$E=\frac{Q}{4\pi\varepsilon r^2}$, $4\pi\varepsilon r^2$=constant i.e same medium ε and same distance r apart. $K=\frac{1}{4\pi\varepsilon r}=\frac{E}{Q}$,

$\frac{E_1}{Q_1}=\frac{E_2}{Q_2}$, E_1=360NC^{-1}, Q_1=50μC, E_2=?, Q_2=120μC,

$\frac{360}{50}=\frac{E_2}{120}$, $E_2=\frac{360\times120}{50}=864$NC^{-1}.

41. The expression for the potential at a point, distance r from a charge q, in an electric field is A.$\frac{q^2}{4\pi\varepsilon_o r^2}$ B.$\frac{q^2}{4\pi\varepsilon_o r}$ C.$\frac{q}{4\pi\varepsilon_o r^2}$ **D.$\frac{q}{4\pi\varepsilon_o r}$.**

$V=\frac{Q}{4\pi\varepsilon r}=\frac{q}{4\pi\varepsilon_o r}$.

42. Calculate the electric potential at a distance of 20.0cm from a point charge of 0.015C placed in air of permittivity ε_0 A.3.40×10^{9}V **B.6.75×10^{8}V** C.6.75×10^{6}V D.3.40×10^{5}V E.8.30×10^{-12}V

(take; $\frac{1}{4\pi\varepsilon_o}$=$9.0\times10^{9}$Nm^{2}C^{-2}).

Q=0.015C, r=20cm=0.2m, $V=\frac{KQ}{r}=\frac{Q}{4\pi\varepsilon r}$,

$V=\frac{9\times10^{9}\times0.015}{0.2}$=$6.75\times10^{8}V$.

43. Find the workdone in moving a 2C charge between two points X and Y in an electric field if the potential difference is 100 volts A.50J B.100J **C.200J** D.400J.

Q=2C, V=$100V$, $V=\frac{W}{Q}$, $W=QV$,

$W=2\times100=200$J.

44. A point charge of magnitude 2.5μC is moved through a distance of 0.04m against a uniform electric field of intensity $15V$m^{-1}. Calculate the workdone on the charge **A.1.5×10^{-6}J** B.1.0×10^{-6}J C.3.5×10^{-6}J D.4.0×10^{-6}J E.6.0×10^{-6}J.

Q=2.5μC=2.5×10^{-6}C, r=0.04m, E=15Vm^{-1}

Workdone on charge=Force×distance

$W=F\times r$, $F=QE$, $W=QEr$

$W=2.5\times10^{-6}\times15\times0.04=1.5\times10^{-6}$J.

45. Why is it impossible for the lines of force of an electric field to cross one another? A.like charges repel B.a charged particle is deflected by an electric field C.like charges are arranged on the same lines of force **D.a positively charged particle can travel in only one direction at anytime** E.none of these is correct.

45. When two like charges are brought together, the nature potential energy of the system formed will **A.increase because work is done against the field** B.increase because work is done by the field C.decrease because work is done by the field D.decrease because work is done against the field

When two like charges are brought together, work is done against the field,as like charges repel each other.So work has to be done against the field to bring them together.

If work is done against the electric field,the electric potential is positive and the potential energy increases. If work is done by the electric field,the electric potential is negative and the potential energy decreases.

46. When a charge moves through an electric circuit in the direction of an electric force,it A.gains both potential energy and kinetic energy B.gains potential energy and lose kinetic energy **C.loses potential energy and gains kinetic energy** D.loses both potential and kinetic energy

Charges moving in the direction or perpendicular to an electric field or force, gains kinetic energy(speed) and loses potential energy.

● Capacitors

- Capacitor – A capacitor is a device used to store electric charges and energy. It consists two

Demystified Series by Dr Timothy

conductors with equal and oppositecharges which are separated by an insulating material called the **dielectric**.

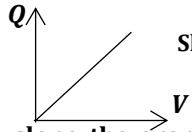
- The equal and opposite charges are called **dipoles.** The ability of a capacitor to store charges is called **Capacitance.**

-The dielectric material must be an insulator e.g air, glass, paraffin wax, paper foil e.t.c.

- Parallel plate capacitor – Common type of capacitor with two parallel plates separated by an insulating medium(dielectrics). Charges are stored due to the decrease in the potential difference and increase in the capacitance of one of the conductors (charged) when another conductor(uncharged but earthed) is brought close to it.

- Capacitance C – Capacitance of a capacitor is the ratio of the charge on either conducor to the potrential difference between the conductors. $\boldsymbol{C = \frac{Q}{V}}$ **,** $\boldsymbol{Q = CV}$**.**

- Graphical relationship with the charge Q stored by a capacitor and the potential difference V across it : Charges stored by a capacitor is directly proportional to the potential difference, $\boldsymbol{Q \propto V}$.



Slope=Capacitance C.

- The slope the graph above is equal to the capacitance across the capacitor.

- The area of the graph above is equal to the workdone in stroring charges or energy stored by a capacitor.

- Factors affecting capacitance of a capacitor:

1. Areaof the plates – Capacitance increases with increase in area of the plates ($\boldsymbol{C \propto A}$)**.**

2. Distance of separation between the plates – Capacitance decreases as the distance of separation between the plates increases ($\boldsymbol{C \propto \frac{1}{d}}$)**.**

3.Shape of plates/conductors.

4.Dielectrics (insulating medium) between the plates – Thickness or density of the medium. Capacitance increases as the thickness or density of the dielectrics increases. ($\boldsymbol{C \propto \varepsilon}$)**, e.g capacitance with air is less than that with paper.** $\boldsymbol{C \propto \frac{\varepsilon A}{d}}$, $\boldsymbol{C = \frac{\varepsilon A}{d}}$, C=capacitance, d=distance, A=area, ε=dielectric constant or permittivity of the medium. $\varepsilon = \varepsilon_0$ i.e in vacuum(free space) or air.

- Relative permittivity $\boldsymbol{\varepsilon_r}$is the ratio of the permittivity in a medium ε_m to the permitivity in vacuum/air ε_o. It is also expressed as the ratio of the capacitance in a medium C_m to the capacitance in vacuum/air C_o. $\boldsymbol{\varepsilon_r = \frac{\varepsilon_m}{\varepsilon_o} = \frac{C_m}{C_o}}$**.**

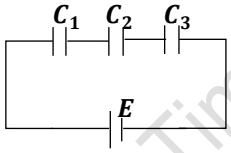
- The capacitance of an isolated sphere of radius r which carries a charge Q is – $\boldsymbol{C = 4\pi\varepsilon_o r}$.

- Charging a capacitor – The plates of the capacitor are connected to a cell (battery) for some time,so that electrons are attracted by the positive terminal of the cell and drifted towards one of the plates leaving opposite charges($\pm Q$) on both plates. The energy stored in the capacitor is half of the total energy supplied by the cell, while the remaining half is converted to heat energy.

- Discharging a capacitor – The plates are connected or in contact by wires, which ensures the continuous back flow of electrons round the circuit and neutralise the positive charged plate and charges on the plates.

- Arrangement of capacitors :

1.Series arrangement – When capacitors C_1, C_2 and C_3 are connected in series, as shown below :



I.Same charge flows through each of them i.e $\boldsymbol{Q_1 = Q_2 = Q_3 = Q}$.

II.Voltage across each capacitor is different and their arithmetic sum is equal to the total voltage or E.M.F of the cell. $\boldsymbol{E = V_T = V_1 + V_2 + V_3}$ **.**

III. The effective or total capacitance is equa; to the reciprocal sum of the capacitance of each capacitor – $\boldsymbol{\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$**.**

- For two capacitors in series – $\boldsymbol{C_T = \frac{C_1 C_2}{C_1 + C_2}}$**.**

- If the capacitance of n capacitors are of equal magnitude, then – $\boldsymbol{C_T = \frac{C}{n}}$ **.**

IV. Their individual voltage is given by – $\boldsymbol{V_1 = \frac{Q}{C_1}}$**,** $\boldsymbol{V_2 = \frac{Q}{C_2}}$ **,** $\boldsymbol{V_3 = \frac{Q}{C_3}}$**.** Alternatievly, for two capacitors in series, Q is the same or constant, hence; $\boldsymbol{V \propto \frac{1}{C}}$ **i.e the smaller capacitance will have the highest voltage.**

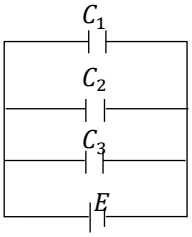
$\boldsymbol{V(\text{across } C) = \frac{C \text{ of other capacitor}}{\text{sum of capacitance}} \times V_T}$

$\boldsymbol{V_1(\text{voltage across } C_1) = \frac{C_2}{C_1 + C_2} \times V_T}$.

$\boldsymbol{V_2(\text{voltage across } C_2) = \frac{C_1}{C_1 + C_2} \times V_T}$ **.**

- When capacitors are stored in series, the equivalent capacitance is less than the individual capacitance and even the small individual capacitance.

- Parallel arrangements – When capacitors C_1, C_2 and C_3 are in parallel as shown below :



Demystified Series by Dr Timothy

I.Voltage or potential difference across each of the capacitors is the same and equal to the E.M.F. $\boldsymbol{V_1 = V_2 = V_3 = V_T = E}$ **.**

II. The effective or total capacitance is the arithmethic sum of each of the capacitance – $\boldsymbol{C_T = C_1 + C_2 + C_3}$ **.**

III. The total or effective charges is given by the arithmetic sum of the individual charges – $\boldsymbol{Q_T = Q_1 + Q_2 + Q_3 = C_T V}$.

IV. Their individual charges is given by – $\boldsymbol{Q_1 = C_1 V,\ Q_2 = C_2 V,\ \ Q_3 = C_3 V.}$ Alternatively, for capacitors in parallel, V is same or constant,hence; $\boldsymbol{Q \propto C.}$ **i.e the higher capacitance will store highest charge.**

$\boldsymbol{Q}$**(in** $\boldsymbol{C}$**)**$= \frac{\boldsymbol{C\ of\ the\ capacitor}}{\boldsymbol{sum\ of\ capacitance}} \times \boldsymbol{Q_T}$

$\boldsymbol{Q_1}$**(charge in** $\boldsymbol{C_1}$**)** $= \frac{C_1}{C_1+C_2} \times \boldsymbol{Q_T}$ **.**

$\boldsymbol{Q_2}$**(charge in** $\boldsymbol{C_2}$**)** $= \frac{C_2}{C_1+C_2} \times \boldsymbol{Q_T}$

- Workdone in charging a capacitor or energy stored by a capacitor **–The energy stored in a capacitor is electrostatic potential energy.**

$\boldsymbol{W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}}$ **.**

- The ratio of energy stored in two parallel capacitors is equal to the ratio of their capacitance.

- Workdone on charges – $\boldsymbol{W = QV = CV^2 = \frac{Q^2}{C}}$ **.**

- Othertypes of capacitors includes – paper capacitor, variable air capacitor and eletctrolytic capacitor.

- Uses/application of capacitors –1. Tuning in radio circuits. **2.**Ignition system of a motor vehicle. **3.**Cancels sparking in switches or circuits with inductance e.g induction coil. **4.**Blocking noise in a.c amplifiers. **5.**Storing electrical energy and large quantities of charge. **6.**Filters out alternate current(A.C) from direct current(D.C) i.e having high resistance for D.C and low resistance for A.C.

Examples :

1. If the charges on an object is measured as 4.0×10^{-18}C, how many excess electrons does the object posses, given that the charge of an electron is 1.6×10^{-19}C? A.18 B.19 **C.25** D.37

Q=charge on object=4.0×10^{-18}C, e=electronic charge or charge of an electron=1.6×10^{-19}C, n=number of electrons, $Q = ne,\ n = \frac{Q}{e}$,

$n = \frac{4.0\times10^{-18}}{1.6\times10^{-19}} = 25.$

2. A capacitor of capacitance 2μF is charged from a 12V battery and discharged through a milliammeter in 0.02s. Determine the current recorded by the milliammeter A.$4.80\times10^{-7}A$ B.$8.33\times10^{-6}A$C.$1.20\times10^{-6}A$**D.**$\boldsymbol{1.20\times10^{-3}A}$

$Q = It$, Q=charge, I=current, t=time=0.02s

$Q = CV$, C=capacitance=2μF=2×10^{-6}F,

V=voltage=$12V$, $Q = CV = It$,

$2\times10^{-6}\times12 = I\times0.02,\quad I = \frac{2\times10^{-6}\times12}{0.02}$,

$I = 1.20\times10^{-3}A$.

3. The purpose of dielectric material in a parallel plate capacitor is to **A.increase the capacitance** B.dcrease its capacitance C.insulate the plates from each other D.increase the magnetic field between the plates.

$C \propto \frac{\varepsilon A}{d}$, Capacitance increases as permittivity of the mediumε increases, area of of the plates A increases and distance between the plates d decreases. Permitivity of the medium increases as the thickness or density of the dielectric material increases.

4. When a piece of rectangular glasss block is inserted between two parallel plate capacitor, at constant plate area and distance of separation, the capaciatance of the capacitor will **A.Increase** B.Decrease C.Decrease then increase D.Remain constant

$C = \frac{\varepsilon A}{d}$, at constant area A and distance of separation, $C \propto \varepsilon$ i.e capacitance increases as the permittivity of the medium increases. Inserting a rectangular block with high density and thickness between the parallel plate capacitor, increases the permittivity and thus, increasing the capaciatance.

5. How would the capacitance of a parallel plate capacitor be affected if the distance of separation of its plates is decreased. It will **A.increase in value** B.decrease slightly C.remain unchanged D.drop to zero

$C \propto \frac{1}{d}$, capaciatance increases as the distance of separation between the plates decreases.

6. If the capacitance of a capacitor is $C = \frac{\varepsilon A}{d}$, which of the following parameters is varied when a variable capacitor is used for tuning radio sets? **A.Area and distance** B. constant (ε) only C.Constant (ε) and araea D.Distance .

In a variable capacitor, the permittivity of the medium is constant while the area and distance is varied or adjusted.

7. Which of the following materials is not used as a dielectric in capacitors? A.Glass **B.Brass** C.Paper foil D.paraffin wax

Dielectric substance in a capacitor are insulators Brass is a conductor hence, cannot be used as a dielectric. **Water is not a good dielectric substance, because it can function as a good insulator only when it is pure.**

8. A parallel capacitor has a common plate area of 5×10^{-8}m^2 and plate separartion of 2×10^{-3}m. Assuming free space, what is the capacitance? A.2.25×10^{-17}F B.4.55010^{-17}F **C.**$\boldsymbol{2.25\times10^{-16}}$**F** D.$4.50\times10^{-16}$F ($\varepsilon_o$=$9.0\times10^{-12}$C^{2}N^{-1}m^{-2})

A=5×10^{-8}m^2, d=2×10^{-3}m, ε_o=9.0×10^{-12}C$^2N^{-1}$m^{-2}, C=?, $C = \frac{\varepsilon_o A}{d} = \frac{9\times10^{-12}\times5\times10^{-8}}{2\times10^{-3}} = 2.25\times10^{-16}$F .

Demystified Series by Dr Timothy

9. A parallel plate capacitor of plate area $10cm^2$ in vacuum has a capapcitance of $10^{-2}\mu F$. What is the distance between the plates? (ε_o=$9.0\times10^{-12}Fm^{-1}$) A.9×10^{-13}m **B.9×10^{-7}m** C.9×10^{-3}m D.9×10^{7}m.

A=$10cm^2$=$10\times10^{-4}m^2$, C=$10^{-2}\mu F$=10^{-8}F,

ε_o=$9.0\times10^{-12}Fm^{-1}$, d=?, $C=\frac{\varepsilon_o A}{d}$, $d=\frac{\varepsilon_o A}{C}$,

$d=\frac{9\times10^{-12}\times10\times10^{-4}}{10^{-8}}=9\times10^{-7}$m .

10. A parallel plate capacitor consists of two plates each of area $9.6\times10^{-3}m^2$, separated by a dielectric of thickness 2.25×10^{-4}m and dielectric constant 900. Calculate the capacitance of the capacitor. (ε_0=permittivity of free space= $8.85\times10^{-12}Fm^{-1}$) A.$3.8\times10^{-10}$F B.$3.8\times10^{-7}$F **C.$3.4\times10^{-7}$F** D.$3.4\times10^{-10}$F .

A=$9.6\times10^{-3}m^2$, d=thickness of the dielectric= 2.25×10^{-4}m, relative permittivity or dielectric constant ε_r=900, permittivity of free space/ air ε_o=$8.85\times10^{-12}Fm^{-1}$, $\varepsilon_r=\frac{\varepsilon_m}{\varepsilon_o}$, $\varepsilon_m=\varepsilon_r\varepsilon_o$.

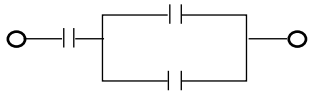
permitivity of the medium ε_m=?, $C=\frac{\varepsilon_m A}{d}=\frac{\varepsilon_r\varepsilon_o A}{d}$

$C=\frac{900\times8.85\times10^{-12}\times9.6\times10^{-3}}{2.25\times10^{-4}}=3.4\times10^{-7}$F.

11. The capacitance of a parallel plate capacitor is $20\mu F$ in air and $60\mu F$ in the presence of a dielectric. What is the dielectric constant? A.6.0 B.2.0 C.0.3 **D.3.0.**

Capacitance with a medium C_m=60μF, Capacitance in air/free space or vacuum C_o=20μF, ε_r=relative permittivity or dielectric constant=? $\varepsilon_r=\frac{C_m}{C_o}=\frac{60}{20}=3$.

12.

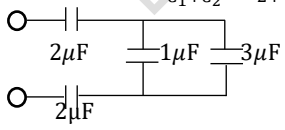


In the circuit shown above, each capacitor has a capacitance of 2μF. The equvalent capacitance is **A.$4/3\mu F$** B.$3/2\mu$F C.3μF D.6μF.

Parallel capacitor : Equivalent capacitance= arithmetic sum of individual capacitance,

Total capacitance=$2+2=4\mu F$. This is now in series with the first capacitor, the equivalent capacitance, $C_T=\frac{C_1C_2}{C_1+C_2}=\frac{2\times4}{2+4}=\frac{8}{6}=\frac{4}{3}\mu F$.

13.

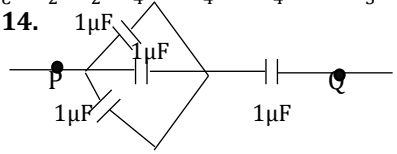


The equivalent capacitance for the circuit shown above is A.8.0μF B.4.5μF **C.$0.8\mu F$** D.0.5μF.

The 1μF and 3μF capacitors are in parallel arrangement, Total capacitance=$1+3=4\mu$F. The 4μF is now in series with the two 2μF, equivalent capacitance:

$\frac{1}{C}=\frac{1}{2}+\frac{1}{2}+\frac{1}{4}=\frac{2+2+1}{4}=\frac{5}{4}$, $C=\frac{4}{5}=0.8\mu$F.

14.

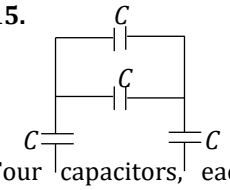


Calculate the effective capacitance between the points P and Q in the diagram above A.$0.25\mu F$ **B.$0.75\mu F$** C.1.33μF D.1.66μF E.4.00μF.

The first three 1μF capacitors are in parallel, Total capacitance = $1+1+1=3\mu$F.

The $3\mu F$ is now in series with the last 1μF, Effective capacitance, $C=\frac{3\times1}{3+1}=\frac{3}{4}$ =0.75μF.

15.



Four capacitors, each of capacitance C, are connedted as shown in the diagram above. The equivalent capacitance is A.$C/4$ B.$5C/2$ **C.$2C/5$** D.$4C$

Total capacitance of the parallel connection, $C_P=C+C=2C$. This $2C$ is now in series with the other two C capacitor, the equivalent capacitance; $\frac{1}{C_T}=\frac{1}{2C}+\frac{1}{C}+\frac{1}{C}=\frac{1+2+2}{2C}=\frac{5}{2C}$,

$C_T=\frac{2C}{5}$.

16. Three $2\mu F$ capacitors are to be connected in such a ways as to give an effective capacitance of $3\mu F$. This can be done by connecting A.two parallel capacitors in series with the third **B.two series capacitors in parallel with the third capacitor** C.all three capacitors in series D.all three capacitors in parallel.

In option B, two $2\mu F$ are in series,

Total capacitance= $\frac{2\times2}{2+2}=\frac{4}{4}=1\mu$F.

The third 2μF in parallel with 1μF, equivalent capacitance, $C=2+1=3\mu$F.

In option A, two 2μF are in parallel, Total capacitance=$2+2$ = 4μF. The third 2μF in series with $4\mu F$, equivalent capacitance, $C=\frac{2\times4}{2+4}=\frac{8}{6}=\frac{4}{3}\mu$F. In option C, the three 2μF in series, equivalent capacitance $C=\frac{2}{3}\mu$F. In option D, the three 2μF are in parallel, equivalent capacitance, C=$2+2+2=6\mu$F.

17. The plates of a 0.5μF uncharged capacitor are connected to a 10V battery. The charge on the capacitor after a long time is **A.$5.0\mu C$** B.10.0μC C.20.0μC D.50.0μC.

C=0.5μF=0.5×10^{-6}F, V=10V, Q=?, $Q=CV$,

$Q=0.5\times10^{-6}\times10=5\times10^{-6}=5\mu$C.

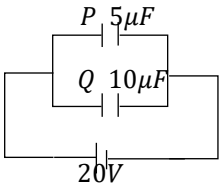
18. Two 50μF parallel plate capacitors are connected in series. The combined capacitor is then connected across a 100V battery. The charge on each plate of the capacitor is A.5.00×10^{-5}C **B.2.5×10^{-3}C** C.1.25×10^{-3}C D.1.00×10^{-2}C.

The charges on each of the two capacitors are equal and is equal to the total charge,as they are in series. If the capacitors are of equal

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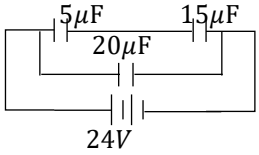
magnitude, then their effective capacitance, $C_T = \frac{C}{n}$, number of capacitors n=2,
$C_T = \frac{50}{2} = 25\mu\text{F} = 25\times10^{-6}\text{F}$, V=E.M.F of battery =100V, $Q_T = CV = 25\times10^{-6}\times100 = 2.5\times10^{-3}\text{C}$.

19.



The diagram above shows two capacitors P and Q of capacitances 5μF and 10μF. Find the charges stored in P and Q respectively. A.200μC and 100μC **B.100μC and 200μC** C.4μC and 2μC D.2μC and 4μC.
V_P=V_Q=V=20V(parallel capacitors), C_P=5μF= $5\times10^{-6}F$, C_Q=10μF=$10\times10^{-6}F$,
$Q_P = C_PV$ =$5\times10^{-6}\times20$=100×10^{-6}C=100μC.
$Q_Q = C_QV$ =$10\times10^{-6}\times20$=200×10^{-6}C=200μC
Alternatively, $Q \propto C$ (parallel capacitors). Thus, the higher capacitance will store the highest charge. C_T =10+5 =15μF
$Q_T = C_TV$ =15μ ×20=300μC.
$Q_P = \frac{C_P}{C_P+C_Q}\times Q_T = \frac{5}{5+10}\times300\mu$ =100μC
$Q_Q = \frac{C_Q}{C_Q+C_P}\times Q_T = \frac{5}{5+10}\times300\mu$ =200μC.

20.



In the network shown,determine the potential difference across the 5μF capacitor A.3V B.6V C.12V **D.18V.**
Voltage V(across 20μF)=Voltage V(across 5μF and 15μF)=V_T=24V, as they are in parallel.
5μF and 15μF are connected in series –
$C = \frac{5\times15}{5+15} = \frac{75}{2} = 3.75\mu F$.
The charge in the series capacitors –
$Q = CV = 3.75\mu\times24 = 90\mu$C.
Q(5μF)=Q(15μF), series capacitors. $V = \frac{Q}{C}$,
V(across 5μF)= $\frac{90\mu}{3.75\mu} = 18V$. Alternatively,
$V \propto \frac{1}{C}$ (series capacitors) thus, the smaller capacitance will have the highest voltage.V_T=24V, V(across 5μF) = $\frac{C_{15}}{C_5+C_{15}}\times V_T$,
V(across 5μF)= $\frac{15}{5+15}\times24 = 18V$.

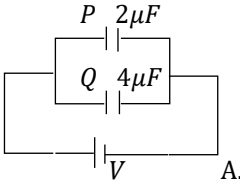
21. The plates of a parallel-plate capacitor with air separating the plates are connected to a battery so that the potential difference between the plates is constant. If the separation between the plates is decreased I.the capacitance of the capacitor decreases II.the capacitance of the capacitor increases III.the charge on each plate decreases IV.the charge on each plate increases Which of the above statements are correct? A.I and IV B.II and III C.I and III **D.II and IV.**
$C \propto \frac{\varepsilon A}{d}$, Capacitance varies inversely with the distance of separation of the plates. $Q = CV$, V=constant, $\therefore Q \propto C$. i.e Charge varies directly with the capacitance. Capacitance increases as the distance of separation of the plates decreases hence, charge on each plates also increases.

23. The energy stored in a capacitor of capacitance 10μF carrying a charge of 100μC is A.4×10^{-3}J **B.5×10^{-4}J** C.5 $\times10^{4}$J D.4×10^{2}J.
C=10μF=$10\times10^{-6}F$, Q=100μC=100×10^{-6}C
$W = \frac{Q^2}{2C} = \frac{(100\times10^{-6})^2}{2\times10\times10^{-6}} = \frac{(10^{-4})^2}{2\times10^{-5}} = \frac{10^{-8}}{2\times10^{-5}}$,
W =5×10^{-4}J.

24. 2×10^{-4}J of energy is stored in a 0.01μF capacitor. The charge on each plates is A.$2\times10^{-6}\mu$C **B.2μC** C.$2\times10^{3}\mu$C D.$2\times10^{8}\mu$C.
W=2×10^{-4}J, C=0.01μF=0.01×10^{-6}F, Q=?,
$W = \frac{Q^2}{2C}$, Q^2 =$2WC$ = $2\times2\times10^{-4}\times0.01\times10^{-6}$,
Q^2 =4×10^{-12} , $Q = \sqrt{4\times10^{-12}} = 2\times10^{-6}$C ,
Q = 2 μC.

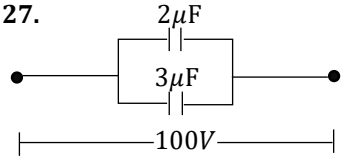
25. A capacitor 8μF, is charged to a potential difference of 100V. The energy stored by the capacitor is A.1.0×10^{4}J B.8.0×10J C.1.25 ×10J **D.4×10^{-2}J.**
C=8μF=$8\times10^{-6}F$, V=100V, W=?
$W = \frac{1}{2}CV^2 = \frac{1}{2}\times8\times10^{-6}\times(100)^2$,
$W = 4\times10^{-6}\times10^{4} = 4\times10^{-2}$J.

26. The diagram below shows two capacitors P and Q of capacitances 2μF and 4μF respectively connected to a d.c source.The ratio of the energy stored in P to Q is



A.4:1 **B.1:2** C.1:4 D.2:1.
C_P=2μF, C_Q=4μF, $W = \frac{1}{2}CV^2$,
Energy stored in P, $W_P = \frac{1}{2}C_PV^2$,
Enegry stored in Q, $W_Q = \frac{1}{2}C_QV^2$,
Ratio of energy stored in P to $Q = \frac{W_P}{W_Q}$,
$\frac{W_P}{W_Q} = \frac{\frac{1}{2}C_PV^2}{\frac{1}{2}C_QV^2} = \frac{C_P}{C_Q} = \frac{2}{4} = \frac{1}{2}$, W_P:W_Q =1:2.
Alternatively, the ratio of energy stored in two parallel capacitors is equal to the ratio of their capacitance.

27.



In the circuit above, the potential across each capacitor is 100V. The total energy stored in the

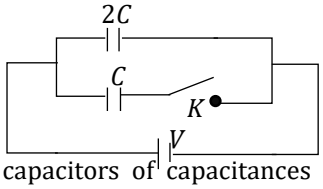
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two capacitors is A.3.0×10^{4}J B.3.0×10^{2}J **C.2.5×10^{-2}J** D.6.0×10^{-3}J.

Effective capacitance C= $2+3=5\mu$F=5×10^{-6}F

$W=\frac{1}{2}CV^2=\frac{1}{2}\times5\times10^{-6}\times100^2=2.5\times10^{-2}$J.

28.



Two capacitors of capacitances C and $2C$, are connected across a voltage as shown in the diagram. With the key K open, the energy stored by the $2C$ capacitor is E. If the key is now closed and the system attains equilibrium, the total energy stored in the capacitors is A.$2E/3$ B.E **C.$3E/2$** D.$3E$.

With the key open, the capacitor C is cut off from the connection, Energy stored in $2C=E=\frac{1}{2}\times2C\times V^2=CV^2$.

With the key closed, both capacitors $2C$ and C, are involved in the connection. Total capacitance=$2C+C=3C$.

Energy stored at equilibrium in capacitors combined $=\frac{1}{2}\times3C\times V^2=\frac{3}{2}CV^2=\frac{3E}{2}$.

Therefore, the energy stored when the key is closed=$3E/2$ or $1.5E$.

Jamb past questions on electrostatics, electric field and capacitors :

[1978/29,1980/9,21,1982/26,32,1983/47,1984/7,35,1987/40,1988/42,1989/32,1990/36, 1991/39,1992/39,41,1995/37,43,1997/39, 2001/25,26,30,2000/38,2004/50,2005/17,19, 2006/32,2008/38,2009/39,2010/33,34,2013/ 36,37,2014/33,2014/34,37,2015/11,15]

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Chapter 22 – Electric cells, Conduction of electricity through liquids and gases

● **Electric cells –** These are cells which produce current that flows in one direction i.e direct current, by the conversion of chemical energy to electric energy. It consists of two dissimilar electrodes and an electrolyte.

- The electrolyte dissociates into positive and negative ions. On passage of current, positive ions are attracted to the negative electrode, where they are neutralized by accepting electrons while the negative ions are attracted to the positive electrode where they are neutralized by losing electrons.

- Current is generated due to the flow of ions.

- Current flows from the positive to the negative electrode in the external conductor and from negative to the positive electrode in the electrolyte.

- Cells are classified based on the E.M.F i.e strength of the cell,amount of current produced and internal resistance of the cell.

- Electromotive force E.M.F or *E* of a cell – It is the total energy reuired to send a unit positive charge round a complete circuit.It is the driving force for the current in a circuit or the potential difference between two electrodes when no current is flowing.

- Internal resistance r of a cell – It is the opposition to the flow of current in the cell producing the current.

- Types of electric cells –

1.Primary cells: Cells that the process through which electric current is generated is irreversible e.g simple voltaic cell, Daniel cell, Lechlanche cells(wet and dry).

2.Secondary cells (or Accumulators): Cells in which the chemical process through which current is generated is reversible i.e it can be recharged by passing current into the cell in a reverse direction e.g acid cell(Lead-acid cell), alkaline cell(nickel-iron cell).

-The internal resistance of Daniel cell (D) , simple voltaic cell (S), Leclanche cell (L) and accumulator (A), increases in the order – $r_A < r_D < r_L < r_S$. **Thus, accumulators have the lowest internal resistance and will produce the largest amount of current.**

- Simple Voltaic cell – Copper(Cu) is the positive electrode(anode), zinc(Zn) is the negative electrode(cathode), while sulphuric acid(H_2SO_4) is the electrolyte.

- All primary cells are affected by **polarisation and local action.**

1. Polarisation – Accumulation of hydrogen bubbles on the copper plate electrode or anode as current is supplied.It causes a reduction in the E.M.F of the cell and an opposing or back or E.M.F to the original E.M.F.The internal resistance of the cell also increases and reduces the amount of current produced.

- Polarisation can be prevented by adding a depolariser e.g manganese dioxide, which oxidises hydrogen to water or by brushing the copper electrode frequently.

2. Local action – Dissolution of zinc electrode in the electrolyte to liberate hydrogen, due to impurities such as **iron and carbon**,which set up tiny local cells at zinc surface.

- Local action can be prevented by cleaning in sulpuric acid and rubbing mercury on the zinc (i.e amalagamating the zinc electrode) or by using a pure zinc electrode.

- Daniel cell – Copper is the positive electrode in copper sulphate solution(copper plating), zinc is the negative electrode in sulphuric acid (electrolyte). The copper sulphate solution is the depolariser.It has a low E.M.F of 1.1V and **provides only a small/low current for a short time.It keeps a steady current and can be used as a standard cell.**

- Leclanche cell (wet and dry cell) – Carbon is the positive electrode, zinc is the negative electrode, ammonium chloride(solution for wet type and paste for dry type) is the electrolyte. The depolariser is a mixture of powdered carbon and manganese dioxide.

- It keeps an unsteady current and is used for supplying intermittent current in battery(cell) used in torch-lights and transistor radios.

- Lead acid accumulator – Lead dioxide(PbO_2) is the positive electrode,lead is the negative electrode and sulphuric acid is the electrolyte.

- The relative density of the acid is 1.25 and the E.M.F is about 2.2V when fully charged and drops as it discharges. It is used in car batteries, phone batteries e.t.c.

- Discharging (supplying current) – The strength of the electrolyte diminishes i.e relative density drops, and the E.M.F also drops, while the ammeter reading rises. Lead sulphate is formed on the surface of electrodes.

- Recharging – Current is passed in opposite and reverse direction by a direct current(D.C) source or alternate current(A.C) source with a rectifier (it converts the A.C to D.C).The ammeter reading falls as the E.M.F and relative density of the electrolyte rises to their original value.

- Maintenance/Care for acid cells(batteries) –

1.Topping up charge by regular recharging of the cell or removal of electrolyte and drying the cell

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when not in used.
2.Avoiding shorting of terminals i.e connecting a wire directly across the terminals of a cell or improper terminal connections during recharging(positive pole of a cell to the negative terminal of the D.C source),as it produces a relatively low resistance and large current which might be harmful to handle.
3.Avoiding over discharging of the cell for a long period of time.
4.Cleaning battery terminals with water and smeared with grease to prevent current leakage
5.Regular topping of electrolyte level i.e maintaining the electrolyte level with water or compensating the loss of water through evaporation.**Acid should not be added to the cell except when spillage occurs.**
- Nickel-Iron (NIFE) accumulator – Nickel(Ni) or nickel hydroxide is the positive electrode, iron (Fe) is the negative electrode and caustic potash (or potassium hydroxide solution) is the electrolyte.
- The NIFE accumulator supply larger current (used for heavy duty vehicle) and last longerthanlead-acid accumulator. They don't need special maintenance as they can be left discharged for a long time without spoiling. **- Capacity of an accumulator or battery capacity is measured in Ampere-hours AH.**
Capacity of an accumulator or battery capcaicty = Current I × recharge time t.
Current is in amperes A while time is in hours.
- Electrolytes –These are substance that dissociates ito positive ions(cations) and negative ions(anions) in aqueous solution.
- Strong electrolytes are either strong acids, alkali or salts e.g sulphuric acid(H_2SO_4), caustic soda(NaOH), brine(NaCl) e.t.c.
- Weak ekectrolytes are weak acids,alkalis and their salts e.g ethanoic acid, grape juice e.t.c
- Non-electrolytes are covalent substance e.g glucose(sugar), alcohol, hydrocarbons e.t.c.

● Conduction through liquids

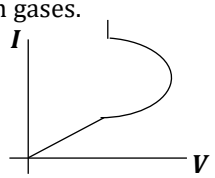
- Electrolysis – Electrolysis is the chemical decomposition of electrolytes and migration of its ions due to the flow of direct current (D.C) through it. It is demonstrated in a **volatmeter,** which contains an electrolyte, anode(positive electrode) and cathode (negative electrode).
- These ions moves randomly when no current is passing. When current flows through the voltameter, the positive ions are attracted to the cathode while the negative ions are attracted to the anode.
- If the electrodes are similar metals and the electrolyte is a salt of the metal,atoms of the metals from the anode are deposited on the cathode and the colour and density of the electrolyte is unchanged. e.g copper electrodes in copper sulphate solution.
- Faraday's laws of electrolysis :
1. Faraday's first law of electrolysis – The mass m of a substance liberated during electrolysis is directly proportional to the quantity of electricity Q passing through the circuit. $\boldsymbol{m \propto Q}$, $\boldsymbol{m = ZQ}$, $\boldsymbol{Q = It}$, $\boldsymbol{\therefore m = ZIt}$. Z=electrochemical equivalent of the substance (e.c.e)= **mass of the substance liberated by 1coulumb of electricity.**
$\boldsymbol{m = \rho V}$, $\boldsymbol{V = A \times th}$, ρ=density of the substance, V=volume of the substance, A=area of the substance, th=thickness of the substance.
2. Faraday'ssecond law of electrolysis – If the same quantity of electricity is passed through different electrolytes, the masses m of elements liberated are proportional to
their chemical equivalent $C.e$.
$\boldsymbol{m \propto}$**C.e, C.e** $= \frac{RAM}{v}$ **or mass deposited by 1Faraday.** $\boldsymbol{RAM}$=relative atomic mass,
v=valency of the substance.
- Application of electrolysis –
1.Electroplating – The metal to be electroplated is the cathode, the coating or electroplating metal is the anode and the electrolyte is a salt of the anode metal.
2.Extraction of metals e.g sodium.
3.Purification of metals.
4.Calibration of ammeter – The ammeter connected in series with a voltameter measures the current $\boldsymbol{I_A}$, flowing through the circuit for a known time $\boldsymbol{t}$and mass deposited $\boldsymbol{m}$is measured. The exact current value $\boldsymbol{I}$is calculated from faraday's first law.
Error in ammeter reading= $\boldsymbol{I - I_A}$ **.**
Percentage error in ammeter reading = $\frac{I - I_A}{I} \times 100$**.**

● Conduction in gases

- Discharge through gases – Under normal conditions gases are insulators.
- Gases conduct electricity better at high voltage(or potential difference),low pressure and with a small quantity of air. Pumping of air out of the tube reduces the pressure.
- Rate of emission or Conductivity of gases is affected by applied voltage V,pressure of the gas P, and the quantity (mass) of gas present m. Conductivity $\propto \frac{V}{Pm}$ **.**
-At ordinary or high pressure and high voltage gases do not conduct but produce a spark or discharge.
- At low pressure and high voltage gases produce a glow due to steady conduction of current rather than sparks or discharge.Further reduction of the pressure gives a dark space which splits into bands extending through the length of the tube

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and the glass glows green.
- Conduction in gases at low pressure and high voltage is referred to as **cold cathode or field**
- The graph below is the current(I)-voltage(V) relationship in the conduction of electricity through gases.



- The colour of light emitted depends on the nature of the gas used in th discharge tube e.g Sodium vapour is yellow, Air is pink, Merucry vapour is light blue e.t.c.

Examples :

1. A discharging electrochemical cell converts A.electrical to chemical energy B.electrical to light energy **C.chemical to electrical energy** D.chemical to light energy
During discharge,it supplies electricity from chemical reactions in the cell and the process reverses during re-charge as a direct current source supplied electricity to the cell.

2. When a storage cell is re-charged A.chemical energy is converted into heat energy B.heat energy is converted into chemical energy C.chemical energy is converted into electrical energy **D.electrical is converted into chemical energy.**

3.Which of the following chemicals is used in simple voltaic cells? A.ammonium chloride B.manganese dioxide C.powdered carbon D.copper sulphate **E.sulphuric acid**
Sulphuric acid is the electrolyte in a simple voltaic cell.

4. In a simple cell, the zinc electrode is amalgamated with mercury inorder to prevent **A.local action** B.polarization C.overheating D.corrosion
Amalgamation prevents local action.

5. Which of the following is not consequence of hydrogen bubbles covering the copper plate of a primary cell? A.polarization **B.local action** C.formation of hydrogen electrode D.increase in the resistance of the cell E.generation of less current by the cell
Polarization results when hydrogen bubbles accumulate on the copper electrode of a simple voltaic cell.This forms an hydrogen electrode i.e an opposing or back E.M.F, thereby increasing the internal resistance of the cell and reducing the amount of current produced.

6. A light bulb is connected to a simple voltaic cell, when the circuit is closed the bulb glows brightly for a short while and then becomes dim. This is because A.hydrogen gas is produced at the cathode **B.hydrogen bubbles accumulate on the anode** C.copper sulphate is formed in the solution D.the filament of the bulb gradually burns out.

7. The most suitable cell used for short interval switches in electric bells is a A.lead acid accumulator B.Daniel cell **C.Leclanche cell** D.Nickel-iron accumulator
Leclanche cells keeps unsteady current and for supplying intermittent current.

8. Which of the following can be used to reduce local action in a Leclanche cell? A.a carbon rod as the positive pole **B.pure zinc as the negative pole** C.potassium permanganate solution in contact with the positive pole D.common salt solution
Local action is caused by the presence of impurities, thus using a pure zinc prevent its occurenece.
- Potassium permanganate can act as a depolarizer to prevent polarization.
- **Primary and secondary cells store chemical energy which is converted into electrical energy when supplying current.**

9. In a Leclanche cell, the role of granulated carbon mixed with manganese(IV) oxide is to I.increase the emf of the cell II.prevent local action in the cell III.prevent polarization in the cell IV.reduce the internal resistance of the cell. Which of the above is/are correct? A.II B.III C.I and II **D.III and IV.**

10. Which of the following is not a component of a dry Leclanche cell? A.carbon rod **B.paste of dilute sulphuric acid** C.zinc container D.manganese dioxide mixed with powdered carbon E.carboard disc at the bottom, separating the positive pole from the negative pole.
The electrodes of Leclanche cell(wet or dry type) are carbon rod(positive electrode) and zinc case(negative electrode).The electrolyte of a dry Leclanche cell is paste of ammonium chloride.Cardboard disc at the bottom of the cell prevents the positive carbon pole from touching the negative zin container to prevent short circuiting of the cell.

11. A dry form of Leclanche cell slowly deteriorates even when not in use due to the A.presence of zinc case which is lined with absorbent paper **B.internal chemical action caused by the presence of small impurities in the zinc plate** C.presence of hydrogen bubbles formed on the cathode when current flows through the cell D.presence of depolarizing agent which oxidizes hydrogen to water.
Local action is caused by the presence of iron and carbon impurities in the zinc electrode, and they set up tiny cells which leads to the wearing away of the zinc case.**It occurs either when the cell is supplying current or not in use. Polarization only occurs when the cell is supplying**

Demystified Series by Dr Timothy

current.

12. Which of the underlisted factors determine the emf of a voltaic cell? I.Quantity of the electrolyte II.Type of electrodes III.Dimensions of the cell A.I **B.II** C.II and III D.I,II and III.

The emf of a cell is independent of the quatity of the electrolyte and the dimension or size of the cell. E.M.F of a primary cell depends on the nature of the electrolyte and electrode (plates) and the more the anode is prone to polarization the lesser the emf produced.

13. The internal resistance of a cell depends on I.cell's arrangement II.the arramgement of the electrodes III.the quantity of the electrolyte IV.the conductivity of the electrolyte. Which of the above are true? A.I and III B.II and IV C.I,II and IV **D.I,II,III and IV.**

14. Which of the following has the lowest internal resistance when new? A.Leclanche cell B.Daniel cell **C.Accumulator** D.Torce cell

- Torch cell is a dry Leclanche cell.

15. An accumulator is rated in the unit of ampere-hour. This is equivalent to the unit of A.power B.energy **C.charge** D.potential difference.

A battery with capacity of one ampere-hour will deliver a current of one ampere in one hour. This is equal to a charge flow of 3600 coulombs i.e 1A×3600s.

16. A car battery rated $45AH$ is charged with a charger whose rating is $2.5A$. How long will it rake to charge the battery fully? A.20hrs B.10hrs C.25hrs D.18hrs

Capacity of the car battery=$45AH$, current I=$2.5A$, reachrge time t=?,

Capacity of a cell $= I(A) \times t(H)$, $45 = 2.5 \times t$, recharge time $t = \frac{45}{2.5} = 18$hrs.

17. A $60AH$ uncharged car battery is to be re-charged using a current of 12A. The time required for the battery to acquire a full charge is **A.5h** B.12h C.60h D.720h

Capacity of the accumulator=60Ah, I=$12A$, reacharge time t=?,

Capacity of a cell $= I(A) \times t(H)$, $60 = 12 \times t$, recharge time $t = \frac{60}{12} = 5$hrs.

18. Which of the following statements is not correct about a lead-acid accumulator? A.It can be recharged **B.it has an emf of 2.0V, which remains steady on discharge** C.it can deliver more current than a battery of single cells with an equal emf D.it can be discharged by leaving the two terminals connected for a long time E.formation of excessive lead sulphate may occur under certain conditions.

The emf and relative density of a lead-acid accumulator falls slowly when discharging.

19. Which of the following statements is not true? A.a chemical action in a primary cell is irreversible B.a secondary cell can be recharged C.as a lead acid accumulator is being charged the density of the acid in it increases **D.a lead acid accumulator can deliver a large current because of its very large internal resistance** E.the conventional current in a circuit flows from the positive to the negative terminal of the battery.

Lead-acid accumulator can deliver a large current due to its low internal resistance.

20. A safety precautiondesigned toprolong the life of a lead-acid accumulator is A.by topping up the cells with concentrated acid B.the use of naked flame when charging the battery **C.keeping the specific gravity of the electrolyte within specific range** D.by connecting wires to the terminals directly when testing the level of charging.

21. In order to check whether an accumulator needs recharging, one may first **A.determine the relative density of the liquid** B.check the level of the liquid C.short circuiting the terminals wit a wire and observe the sparks D.measure the temperature of the liquid E.none of the above

22. Eight dry cells when connected in series produce the same emf as a 12volt car battery, but they will not be suitable for the starting of a are connected in series **C.high internal resistance of the cells** D.of the high internal resistance of the starter motor E.of the high emf of the starter motor

The dry cells cannot supply the high current needed to start the car due to its high internal resistance. A car battery(accumulator) can supply a high current needed to start the car due to its low internal resistance.

23. Which of the following leads to a deacreas in the internal resistance of a lead-acid accumulator? I.increasing the areas of the electrodes II.decreasing the separation of the electrodes III.increasing the separation of the electrodes IV.decreasing the area of the electrodes **A.I and II** B.II and III C.II and IV D.III and IV

Internal resistance of a cell is affected by the separation(distance) of the electrodes and area of the electrodes. Internal resisatnce decreases when the area of the electrodes increases ($r \propto 1/A$) and reducing the distance of separation between the plates ($r \propto d$).

24. Which of the following properties is an advantage of lead-acid accumulator over an alkaline accumulator? A.Possesses low internal resistance B.Can be recharged C.Has shorter life span **D.Possess higher emf.**

The emf of a lead-acid cell is about 2V while that of an alkaline cell is about 1.25V. Alkaline cells have longer life span than lead-acid cells. Both cells possess low internal resistance and can be

Demystified Series by Dr Timothy

recharged.

25. In the conductance of aqueous potassium tetraoxosulphate(VI) solution, current carriers are the **A.ions** B.electrodes C.hydrated ions D.hydrated electrons.

Aqueous potassium tetraoxosulphate(VI) solution is an electrolyte. Charge or current carriers in liquids are ions.

26. Which of the following is an electrolyte? A.sugar solution B.alcohol C.paraffin **D.grape juice.**

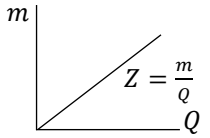
27. Which of the following statements about electrolytes is not correct? A.the substances in solution in the electrolyte become ionized B.ions are electrically charged and are attracted towards electrodes when a potential difference is applied C.ions may be discharged at the electrodes, forming bubbles or deposits D.the mass deposited depends upon the length of time for which current flows **E.the mass of gas set free or metal deposited is proportional to the square of the current.**

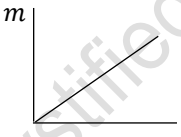
The mass of substance deposited is directly proportional to the current(I) and the time(t).

$m \propto Q$, $Q = It$, $m \propto It$ or $m \propto I$ and $m \propto t$.

28. The unit of quantity of electricity is called A.the ampere B.the volt **C.the coulomb** D.the ammeter E.electromotive force.

29. In Faraday's law of electrolysis, a graph of mass deposited against the quantity of electricity is plotted. The slope of the graph gives A.the current flowing B.the energy released C.the charge released **E.the electrochemical equivalent.**





The graph of the mass m deposited during electrolysis and the quantity of electricity Q is a straight line raph with a slope equal to the electrochemical equivalent.

30. The electrochemical equivalent of a metal is 0.126×10^{-6}Kg/C. The mass of the metal that a current of 5A will deposit from a suitable bath in 1 hour is A.0.0378×10^{-3}Kg B.0.227×10^{-3}Kg C.0.378×10^{-3}Kg D.0.595×10^{-3}Kg **E.2.268×10^{-3}Kg.**

Z=0.126×10^{-6}Kg/C, I=5A, t=1 hour=3600s, m=?

$m = ZIt$, m =$0.126\times10^{-6}\times5\times3600$,

$m = 2268\times10^{-6}$=2.268×10^{-3}kg .

31. In copper plating, a current of 0.5A is allowed for a cathode area 100cm^2. If the current is maintained constant for 100minutes the thickness of the copper deposited will be approximately A.10^{-2}mm B.10^{-3}mm C.10^{-4}mm **D.10^{-5}mm** E.10^{-6}mm (the electrochemical equivalent of copper is 0.00033g/C, density of copper is 10^4g/cm^3) I=0.5A, A=100cm^2, t=100min=100×60=6000s, Z=0.00033g/C, $\rho=10^4$g/cm^3, thickness th=?, $m = ZIt$,

$m = \rho V = \rho(A \times th)$, $\rho \times A \times th = ZIt$,

$10^4\times100\times th = 0.00033\times0.5\times6000$,

$th = \frac{0.00033\times0.5\times6000}{10^4\times100} = 0.99\times10^{-6}$cm

th $\cong 1.0\times10^{-6}$cm=10mm×10^{-6}=10^{-5}mm.

32. Copper of density 10^4g/cm^3 and electrochemical equivalent 3.3×10^{-4}g/C is to be plated on the cathode of a voltameter. If the total surface area of the cathode is 66cm^2 and a current of 50A is maintained for 1 hour, the thickness of copper plated will be approximately A.1.80mm **B.0.90mm** C.0.18mm D.0.09mm

$\rho=10^4$g/cm^3=$10^4\times1000$kg/m^3=10^7kg/m^3,

Z=3.3×10^{-4}g/C, A=66cm^2=66×10^{-4}m^2, I=50A, t=1hr=3600s, th=?, $m = ZIt$,

$m = \rho V = \rho(A \times th)$, $\rho \times A \times th = ZIt$,

$10^7\times66\times10^{-4}\times th = 3.3\times10^{-4}\times50\times3600$,

$th = \frac{3.3\times10^{-4}\times50\times3600}{10^7\times66\times10^{-4}} = 0.0009\text{m} = 0.90\text{mm}.$

33. A charge of one coulomb liberates 0.0033g of copper in an electrolytic process. How long will it take a current of 2A to liberate 1.98g of copper in such a process? **A.5 minutes** B.30 minutes C.50 minutes D.60 minutes E.120 minutes

Z=electrochemical equivalent=mass liberated by 1 coulomb of electricity=0.0033g, I=2A, m=1.98g, t=? $m = ZIt$, $1.98 = 0.0033\times2\times t$,

$t = \frac{1.98}{0.0033\times2} = 300\text{s} = \frac{300}{60} = 5$ minutes.

34. 1g of copper is liberated by a charge flow of 10^3C in a voltameter. In order to liberate 7.2g of copper in the same voltameter, a current of 2A should be maintained for A.6min B.30min **C.60min** D.120min

m_1=1g, $Q_1=I_1t_1=10^3$C, m_2=7.2g, I_2=2A, t_2=?

$Z_1=Z_2=Z$ i.e constant, as both are copper.

$m = ZIt$, $Z = \frac{m}{It}$, $\frac{m_1}{I_1t_1} = \frac{m_2}{I_2t_2}$,

$\frac{1}{10^3} = \frac{7.2}{2\times t_2}$,$t_2 = \frac{10^3\times7.2}{2\times1} = 3600\text{s} = 60\text{min}.$

35. During the electrolysis of copper(II) tetraoxosulphate(VI) solution, an ammeter shows a steady current reading of 1.0A for 30 minutes while 6.6×10^{-4}kg of copper is liberated. Calculate the error in the ammeter reading [The electrochemical equivalent of copper is 3.30×10^{-7}kgC^{-1}] A.0.001A B.0.011A **C.0.111A** D.1.000A E.1.100A

Faulty ammeter reading I_A=1.0A,m=6.6×10^{-4}kg, t=30min=30×60=1800s, Z= 3.30×10^{-7}kgC^{-1}, Actual current=I=? $m = ZIt$, $I = \frac{m}{Zt}$,

$I = \frac{6.6\times10^{-4}}{3.3\times10^{-7}\times1800} = 1.111A$,

Error in ammeter reading= $I - I_A = 1.111-1$,

Error in ammeter reading = $0.111A$.

36. In the process of silver-plating a metal M, the metal is at the A.anode and a direct current is used B.cathode and an alternate current is used

Demystified Series by Dr Timothy

C.cathode and an alternate current is used **D.cathode and a direct current is used** The metal to be plated is placed at the anode, the plating metal i.e silver in this case, is the anode while the electrolyte is a soluble salt of the plating metal(silver). A direct current and not an alternate current, is used in electrolytic process.

37. Gases conducts electricity under A.high pressure and high voltage B.low pressure and low voltage C.normal pressure and low voltage **D.low pressure and high voltage.**

Gases conduct electric current under low pressure and high volatage. Charge carriers in gases are ions and free electrons.

38. Which of the following is not true of a discharge tube? A.the pressure of the gas must be very low B.the glass tube must have two electrodes at the ends **C.the gas in the tube must be carbon dioxide** D.air is gradually pumped out of the tube E.the coating on the walls of the tube at a certain stage glows.

Any gas is used and the colour of light emitted is dependent on the nature of the gas used.

39. Which of the following affects the conductivity of gases in a tube I.the pressure of the gas II.mass of the air III.applied voltage A.I B.III C.I and III **D.I,II and III.**

Gases conducts electricity under high voltage, low pressure and small quantity(mass) of gas.

Jamb past questions on electric cells, conduction of electricity thrpugh liquids and gases :

[1979/30,1984/11,1986/42,1989/44,1991/3644,1992/35,1993/36,46,1998/47,2000/48,2001/29,37,41,2002/40,2003/39,2005/3,5,31,2006/22,2007/21,2010/45,2011/40,46,2012/38,2013/45,2014/46,2015/39]

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Chapter 23 – Current Electricity, Electric Power and Energy

● **Current electricity –** Current electricity involves the flow of electrons or charges round a conductor.

- Electric current I – It is the rate of flow of electric charge round a circuit or the quantity of electricity Q, flowing per unit time t. $I = \frac{Q}{t}$, I = **[Cs^{-1} or A].** The unit for current is ampere(A).

- Terminal potential difference or voltage V – It is the workdone or energy in volt in moving a unit charge from one point in a circuit to the other. It is the P.d or voltage in a ciruit delivering current i.e closed circuit.

- Electromotive force(emf) E – It is the total workdone or energy in volt in moving a unit charge round an electrical circuit. It is also the voltage or p.d in a circuit not delivering current (open circuit). $E = \frac{W}{Q}$, E = **[JC^{-1} or V].** The unit of potential difference is volt (V).

- Factors affecting the emf of a cell –

1.Nature(quality) of chemicals or materials used in a cell. It does not depend on the quantity (amount) of chemicals used or the size of the cell. i.e the stronger the electrolyte,the lower the internal resistance and the greater the emf.

2.It depends on the number of cells, time and temperature. E.M.F decreases with time and temperature e.g primary cells runs down with time.

- Internal resistance of a cell r – It is the opposition to the flow of current by the cell producing the current.

- Factors affecting the internal resistance of a cell – It depends on E.M.F of the cell and :

1.Nature of chemicals or electrodes.

2.Amount of chemicals or electrodes and the size of the cell.

3.Area of the electrodes(plates), $r \propto \frac{1}{A}$.

4.Separation/distance between the electrodes $r \propto d$.

- Internal resistance determines the maximum current that can be supplied by a cell i.e the more the internal resistance, the lesser current the cell can supply.

$I_{max} = \frac{E}{r}$, I_{max}=maximum current, E=E.M.F, r=internal resistance of the cell.

- Lost volt v – It is the potential difference across the internal resistance of a cell.

- External Resistance or Electrical resistance R – It is the opposition to the flow of current in a material.

- Good conductors offers low resistance to current, while poor conductors (insulators) offers high resistance to current.

- Factors affecting the resistance of a conductor :

1.Area $R \propto \frac{1}{A}$ **:** Resistance decreases with area of the wire i.e thick(large area) or thin wire(small area).

2.Length, $R \propto L$ **:** Resistance increases as length of the wire increases.

3.Temperature, $R \propto T$ **:** Resistance of a metallic wire increases with temperature.

4.Nature of the material i.e resistivity , ρ.

$R \propto \frac{L}{A}$, $R = \frac{\rho L}{A}$ **.** Thus, **a short thick wire has a lesser resistance than a long thick wire of the same material.**

- For non-metallic conductors like graphite, their resistance decreases with temeperature. Manganin and Constantine are used for making standard resistors as they little affected by temperature changes.

- Conductivity σ of a material is the reciprocal of its resistivity, $\sigma = \frac{1}{\rho}$.

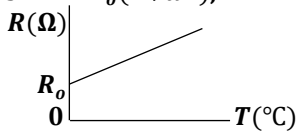
- The resistance and resistivity of a metallic conductor increases with temperature. If the resistance R_o and resistivity ρ_o at **0**℃ and R and ρ at T then, $R = R_o(1+\alpha T)$, $\rho = \rho_o(1+\alpha T)$. **α is the temperature coefficient of resistance.**

- Positive temperature coefficient of resistance – The resistance increases with temperature. e.g metals.

- Negative temperature coefficient of resistance – The resistance decreases with temperature e.g semi-conductors (silicon) and insulators.

- Zero temperature coefficient of resistance – The resistance if not affected by temperature e.g manganin and constatine.

- A graph of the resistance R of a metallic wire with temperature T is a straight line graph with a positive slope ($R_o\alpha$) and intercept R_o at the R-axis. $R = R_o(1+\alpha T)$, $R = R_o + R_o\alpha(T)$.



- Ohm's law – It states that the potential difference V across a metallic conductor is directly proportional to the current I flowing through it, provided temperature and other physical conditions like length and area of the conductor, remains constant. $V \propto I$, $V = IR$, **Resistance is the constant of proportionality,** $R = \frac{V}{I}$ **.**

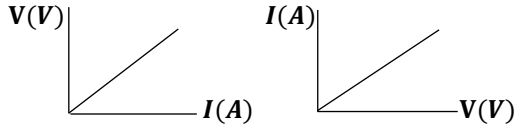
- Ohmic conductors are the ones that obeys ohm's law i.e they have a linear graph between voltage and current e.g **all metals and resistors.**

- Non-ohmic conductors do not obey ohm's law

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i.e they have a non-linear graph between voltage and current e.g electrolytes, diodes, transistors.

-Ohm's law graph is a straight line graph showing the relationship of current I and voltage V.



- Both graphs above represents ohm's law but their slope is different. **The slope of the first graph($V-I$ graph) is equal to resistance R. The slope of the second graph($I-V$ graph) is equal to conductance G in $(\Omega)^{-1}$ or siemens..**

- A resistors is a wire that offers resistance to current. **When current flows through a resistor, it converts electrical energy to heat energy.**

- A rheostat is a variable resistor.Its resistance can be varied. **Rheostat can also be used as a potential divider to produce varying p.d.**

- Current is measured by an **ammeter of low resistance**, which is connected in **series** with the circuit.

- Voltage (terminal P.d and emf), is measure by a **voltmeter of high resistance**, which is connected in **parallel** with the circuit.

- Resistance is measured by an **ohmmeter.**

- **Multimeter** is an instrument that measures current,voltage and resistance e.g **Avometer.**

- **Series and Parallel arrangements of cells and resistors in a circuit –**

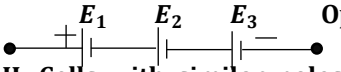
1. Cells in parallel – The emf the cells are equal and is equal to the net emf. $\boldsymbol{E_1 = E_2 = E_3 = E_T \ldots\ldots\ldots E_n}$ **2.**The net internal resistance r_T is given by – $\boldsymbol{r_T = \frac{r}{n}}$, n=number of cells.

- The advantage of connecting cells in parallel is that the effective internal resistance is decreased, the current supplied increases (produces large amount of current), supplies steady current for a long period and the efficiency is high.

2. Cells in series – Cells arranged in series can be connected with similar poles or opposite poles.

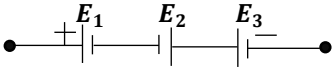
- The net internal resistance for both arrangement is the algebraic sum of their individual resistance. $\boldsymbol{r_T = r_1 + r_2 + r_3}$. Alternatively, $\boldsymbol{r_T = nr}$(if the cells are the same).

I. Cells with opposite poles connected: The net emf is the algebraic sum of their individual e.mf. $\boldsymbol{E_T = E_1 + E_2 + E_3}$ OR $\boldsymbol{E_T = nE}$(if the cells are the same).



Opposite poles.

II. Cells with similar poles connected: Cells which are wrongly connected (like poles) are given a negative emf. $\boldsymbol{E_T = E_1 - E_2 + E_3}$.



Cell E_2 will have a negative emf as it is wrongly connected with its negative pole connected to the negative pole of cell E_1 and its positive pole connected to the positive pole of cell E_3.

3. Resistors in parallel –

I.The p.d across each resistor is the same and is equal to the total voltage/p.d, $\boldsymbol{V_1 = V_2 = V_3 = V}$.

II. The current flowing through each resistor is different and the total current is equal to the sum of individual currents, $\boldsymbol{I = I_1 + I_2 + I_3}$.

III. Their effective resistance is given by the reciprocal sum of their individual resistance, $\boldsymbol{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$. For two resistors, $\boldsymbol{R = \frac{R_1 R_2}{R_1 + R_2}}$.

- For cells in parallel, $\boldsymbol{I_1 = \frac{V}{R_1}}$, $\boldsymbol{I_2 = \frac{V}{R_2}}$, $\boldsymbol{I_3 = \frac{V}{R_3}}$.

Alternatively $\boldsymbol{I \propto \frac{1}{R}}$ or $\boldsymbol{I \propto G}$, as V=constant**.** Thus, **the higher resistance (lower conductance) will have lesser current flowing through it.** Conductance $\boldsymbol{G = \frac{1}{R}}$.

$$\boldsymbol{I(\text{in } R) = \frac{R \text{ in other resistors}}{\text{sum of resistance}} \times I_T}$$

- For resistors of resistance R_1 and R_2 with conductance G_1 and G_2, the currects are :

$$\boldsymbol{I_1(\text{in } R_1) = \frac{R_2}{R_1 + R_2} \times I_T} \text{ OR } \boldsymbol{\frac{G_1}{G_1 + G_2} \times I_T}$$

$$\boldsymbol{I_2(\text{in } R_2) = \frac{R_1}{R_1 + R_2} \times I_T} \text{ OR } \boldsymbol{\frac{G_2}{G_1 + G_2} \times I_T}$$

4. Resistors in series –

I.The p.d across each resistor is different and the net p.d is equal to the sum of their individual p.d, $\boldsymbol{V = V_1 + V_2 + V_3}$.

II. The current flowing through each resistor is the same and equal to the total current. $\boldsymbol{I_1 = I_2 = I_3 = I}$.

III. Their effective resistance is equal to the algebraic sum of their individual resistance. $\boldsymbol{R = R_1 + R_2 + R_3}$.

- For cells in series $\boldsymbol{V_1 = IR_1, V_2 = IR_2, V_3 = IR_3}$. Alternatively $\boldsymbol{V \propto R}$, as I=constant. Thus, **the higher resistance will have a higher voltage.**

$$\boldsymbol{V(\text{across } R) = \frac{R \text{ in the resistor}}{\text{sum of resistance}} \times V_T}.$$

- For resistor with resistance R_1, R_2 and R_3 ,

$$\boldsymbol{V_1(\text{across } R_1) = \frac{R_1}{R_1 + R_2 + R_3} \times V_T}.$$

$$\boldsymbol{V_2(\text{across } R_2) = \frac{R_2}{R_1 + R_2 + R_3} \times V_T}.$$

$$\boldsymbol{V_3(\text{across } R_3) = \frac{R_3}{R_1 + R_2 + R_3} \times V_T}.$$

- The resistance of an open circuit (a cell no delivering current) is infinite. $V = IR$, I=0, $R = \frac{V}{I} = \frac{V}{0} = \infty$(infinite).

- The emf(also called **norminal voltage)** of a cell is the algebraic sum the p.d acting in a circuit. Emf is equal to the sum of all the terminal potential difference and the lost volt.

$$\boldsymbol{E = V + v}, \quad \boldsymbol{V = IR}, \quad \boldsymbol{v = Ir}, \quad \boldsymbol{E = IR + Ir}$$

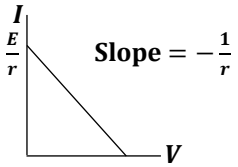
$$\boldsymbol{E = I(R + r)}, \quad \boldsymbol{\frac{V}{E} = \frac{IR}{I(R+r)}}, \quad \boldsymbol{\frac{V}{E} = \frac{R}{R+r}}.$$

- The current-voltage ($I-V$) graph of a cell with

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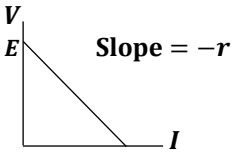
internal resistance is descending line graph with a negative slope. $E = V + v$, $v = E - V$, $v = -V + E$, $v = Ir$, $Ir = -V + E$, $\boldsymbol{I = \left(\frac{-1}{r}\right)V + \frac{E}{r}}$, Comparing with equation of a straight line ; $y = mx + c$. m=slope, c=intercept

The graph of I against V has a negative slope of $-\frac{1}{r}$ (negative reciprocal of internal resistance), and the intercept on the I-axis is $\frac{E}{r}$ (ratio of emf to the internal resistance or **the maximum current I_{max}) .**



- The voltage-current ($V - I$) for a cell with internal resistance is also a descending line graph. $V = E - Ir$, $\boldsymbol{V = -Ir + E}$,

The graph of V against I has a negative slope of $-r$ and the intercept on the V-axis is E.



- The efficiency of cell ϵ is the ratio of electric power output to power input.

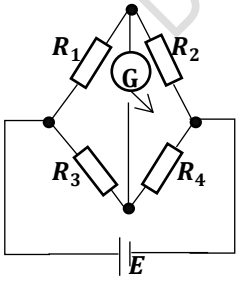
Efficiency $= \frac{Power\ output}{Power\ input} \times 100 = \frac{P_o}{P_i} \times 100$,

$\epsilon = \frac{I^2R}{I^2(R+r)} \times 100 = \frac{R}{R+r} \times 100.$

Efficiency $\epsilon = \frac{Terminal\ p.d}{E.M.F} \times 100 = \frac{V}{E} \times 100.$

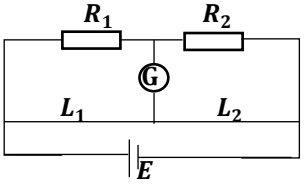
- Wheatstone bridge and Meter bridge – Both instruments are used for measuring the resistance of a material. A galvanometer is attached to the instrument for a null deflection or balance point(where the potential in the circuit is the same or p.d between two points is zero and the galvanometer shows no deflection i.e no current is flowing).

Wheatstone bridge :



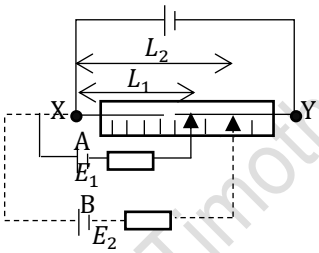
$\boldsymbol{I_1 = I_2}$(as $R_1$and $R_2$are in series),$\boldsymbol{I_3 = I_4}$(as R_3 and R_3 are in series). $\boldsymbol{V_1 = V_3}$(as $R_1$and R_3 are parallel), $\boldsymbol{V_2 = V_4}$(as R_2 and R_4 are parallel). $\frac{R_1}{R_2} = \frac{R_3}{R_4}$. The effective or total resistance R_T in a wheatsone bridge is : $\boldsymbol{R_T = \frac{(R_2+R_2)(R_3+R_4)}{R_1+R_2+R_3+R_4}}$.

Meter bridge :



$\frac{R_1}{L_1} = \frac{R_2}{L_2}$ OR $\frac{R_1}{R_2} = \frac{L_1}{L_2}$, $\boldsymbol{L_2 = 100 - L_1}$.

- Potentiometer – It used to determine actual or accurate value of the emf of a cell by comparing with the emf of a standard cell or battery. It acts as a voltmeter of infinite resistance or ideal voltmeter. It can also measure terminal p.d and very small internal resistance.



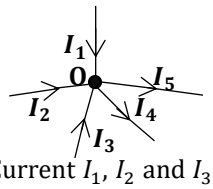
$\frac{E_1}{E_2} = \frac{L_1}{L_2}$. E_1=emf of cell A, E_2=emf of cell B, L_1=balance point of cell A, L_2=balance point of cell B.

- Advantages of a potentiometer over a voltmeter for measuring emf – 1. Potentiomter does not use a pointer unlike a voltmeter **2.**Potentiometer measures emf to a high degree of precision than the voltmeter which may have a zero error. **3.**Potentiometer measures emf when the cell is not delivering current i.e in open circuit.It gives the actual measurement of emf since, no correction is made for lost volt (v= Ir).

- Kirchhoff's Laws – It is used to find the current or voltage of complex circuits.There are two laws; **I.Kirchhoff's current law II.Kirchhoff's voltage law**

1.Kirchhoff's first(current) law KCL – It is also called the **junction law or nodal analysis.** It states that the algebraic sum of currents meeting at a junction or a point is zero i.e $\sum \boldsymbol{I = 0}$ or the total current entering a junction equals the total current leaving the region , $\sum \boldsymbol{I_{in}} = \sum \boldsymbol{I_{out}}$.

-Kirchhiff's current law is based on the law of conservation of charge.



Current I_1, I_2, I_3, I_4 and I_5 meet at a junction O.

Current I_1, I_2 and I_3 flow towards the junction O while current I_4 and I_5 flow away from the junction O. $\boldsymbol{I_1 + I_2 + I_3 - I_4 - I_5 = 0}or\boldsymbol{I_1 + I_2 + I_3 = I_4 + I_5}$.

- Currents flowing towards the junction are positive while currents flowing away from the junction are negtative.

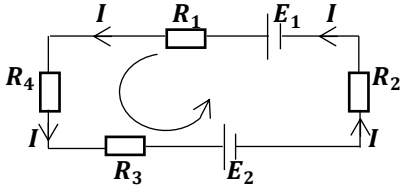
2.Kirchhoff's second(voltage) law KVL – It is

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also called loop analysis.It states that the algebraic sum of all voltages or potential difference in a closed circuit or loopis zero. i.e $\sum \boldsymbol{V} = \boldsymbol{0}$or the algebraic sum of voltage across drops in any closed loop is equal to the sum of emf's in that loop , $\sum \boldsymbol{IR} = \sum \boldsymbol{E}$.

- This law is based on the law of conservation of energy or conservation of electrostatic force.

- In KVL,the sign assigned to the voltage (E and IR), is dependent on the direction (clockwise or anticlockwise) as indicated in the circuit.
- Generally, when travelling a loop, if higher potential point(positive pole) is entered the voltage takes +ve sign and if a lower potential point(negative pole) is entered the voltage takes a −ve sign.
- Emf is positive when the direction moves from −ve terminal to the +ve terminal of the cell and negative when the direction moves from +ve to −ve.
- Voltage across drop(IR) is −ve if the direction travels through the resistor in the same direction as current and +ve if the direction travels in opposite direction with the current.



In the diagram above, the arrow indicates anticlockwise loop and the current also moving in anticlockwise direction :

$\therefore \boldsymbol{E_1 - E_2 - IR_1 - IR_2 - IR_3 - IR_4 = 0}$,

$\boldsymbol{E_1 - E_2 = IR_1 + IR_2 + IR_3 + IR_4}$.

● Electrical energy, Power and Fuse wire

- Electrical energy $\boldsymbol{E}$ – It is the workdone in moving a given quantity of electricity Q through a conductors across a p.d V.

$\boldsymbol{E = QV = IVt = I^2Rt = \frac{V^2t}{R} = Pt}$.

- The trade unit of electrical energy is kilowatt-hour (KWh). $1KW$h$=3.6\times10^6$ joules.

- Electrical power $\boldsymbol{P}$ – It the rate at which electrical energy is dissipated.

$\boldsymbol{P = IV = I^2R = \frac{V^2}{R}}$.

- Joule's law of electrical heating – It states that the quantity of heat energy produced in a metallic conductor as a result of current flowing through it, is proportional to the resistance of the conductor $\boldsymbol{R}$, the square of the current $\boldsymbol{I^2}$ and the time for which it flows $\boldsymbol{t}$.$\boldsymbol{H \propto I^2Rt}$, $\boldsymbol{H = I^2Rt}$.

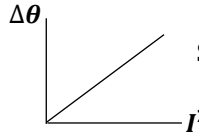
1 calorie=4.3joules.

- Electrical calorimetry – Electrical energy is utilized to heat energy, which is transferred to a body either to change it temperature ($\boldsymbol{mc\Delta\theta}$)or to change its state ($\boldsymbol{mL}$).

$\boldsymbol{H = I^2Rt = mc\Delta\theta}$ **or** $\boldsymbol{mL}$.

- In an experiment to show the heating effect of electric current, the relationship between the temperature rise $\Delta\boldsymbol{\theta}$and the current drawn is a straight line graph with a slope equal to $\frac{Rt}{mc}$.

$\boldsymbol{I^2Rt = mc\Delta\theta}$, $\boldsymbol{\Delta\theta = I^2\frac{Rt}{mc}}$, Thus, $\Delta\boldsymbol{\theta} \propto \boldsymbol{I^2}$



Slope $= \frac{Rt}{mc}$.

- The relationship between the current flowing through a heating element and the time of flow is – $\boldsymbol{H = I^2Rt}$, $\boldsymbol{\frac{H}{R} = I^2t}$, $\boldsymbol{\frac{H}{R}}$ **= constant,** $\boldsymbol{\therefore I^2 \propto \frac{1}{t}}$, $\boldsymbol{I^2{}_1t_1 = I^2{}_2t_2}$.

- House wiring – The house and appliances are protected from sparks,shocks and possible fires that might arise from a faulty equipment by colour coding, use of wires of different thickness,contact breakers, fuses and arrangement of lamps.

- Colour coding – The live wire is either red or brown, neutral wire is either blue or black while the earth wire is either green or yellow.

- The earth wire is connected to the casing of the appliance for safety precautions. When short circuiting occurs i.e the casing of the appliance is in contact with the life wire, the earth wire conducts the sufficient current to the earth,hence blowing the fuse and breaking the circuit.
- Electrical current is delivered to a house using – Live wire through current flows in and neutral wire through current returns to the power station.

- Circuit breakers – They are made with bimetallic strip thermostat. They are used in high-current circuits and they switch off with **over** current and manually switched on when fault is corrected.

- Fuse – A fuse is a thin strip of short lengh wire made of low melting point(fusion point) material such as an alloy of tin and lead. It is connected to the **live wire** of an appliance.

- Fuse rating is the maximum safe current permitted to flow in a fuse before it breaks. e.g if the fuse rating of a fuse is 13A and the appliance draws a current higher than 13A,the fuse will melts as a result of overheating(i.e blowing of the fuse) and breaks the circuit.

- Fuse rating should be a bit higher or equal to the normal current.

- Electrical lamps or bulbs in use –

1.When lamps or bulbs are connected in series the same flows through each lamp but the p.d across each lamp is different and depends on resistance. A single switch controls all the lamps

Demystified Series by Dr Timothy

and if one of the lamps dies, it breaks the circuit and the other lamps will not lit e.g at parties and tiny bulbs in christmass tree.

2.When lamps or bulbs are connected in parallel different current which depends on their resistance flows through them but the p.d across each of them is the same.They are controlled by different switch and malfunctio- ning of one of the bulbs does not affect the others. e.g in homes or house wiring.

- Series combination of bulbs – The total power is given by the sum of the reciprocal of their individual powers. $\frac{1}{P_T}=\frac{1}{P_1}+\frac{1}{P_2}+\frac{1}{P_3}$.

-The bulb of lesser power glows brighter than that of higher power i.e $P=I^2R$, $P\propto R$ (as same current flows for series), hence the lamp of higher power will offer a high resistance to current and its brightness reduces.

- Also their power is related to their and resistance by– $\frac{P_1}{R_1}=\frac{P_2}{R_2}=\frac{P_3}{R_3}$.

- Parallel combination of bulbs – The total power is given by the algebraic sum of their individual powers.$P_T=P_1+P_2+P_3$.

- The lamp of higher power glows brighter than that of lower power i.e $P=\frac{V^2}{R}$, $P\propto\frac{1}{R}$ (as the voltage across each is the same), hence the lamp of higher power will offer a low resistance to current and its brightness increases.

- Also their power is related to their resistance by – $P_1R_2=P_2R_2=P_3R_3$.

- Two advantages of parallel connection over series are – 1. Less voltage is used i.e the voltage across each lamps are equal. **2.**Damage to any circuit component does not affect other components.

Examples :

1. A current of 180mA passes through a conductor for 5 minutes. The quantity of electricity transported is **A.54C** B.0.9C C.3.6×10^{-2} C D.6.0×10^{-4}C .

I=180mA=180×10^{-3}A, t=5min=5×60=300s,

$Q=It$ =180×10^{-3}×300=54C.

2. 60mC of electricity passes through a conductor for 2 minutes. The current through the conductor is A.7.2A B.120.0mA C.30.0mA **D.0.5mA**

Q=60mC=60×10^{-3}C, t=2min=2×60=120s, I=?,

$I=\frac{Q}{t}=\frac{60\times10^{-3}}{120}=0.5\times10^{-3}A=0.5\text{m}A$.

3. The internal resistance of an electric cell of 2.6V is able to deliver a maximum current of 4A is A.2.00Ω B.0.41Ω **C.0.65Ω** D.0.26Ω

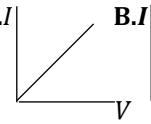
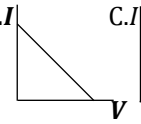
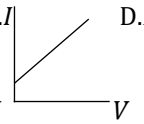
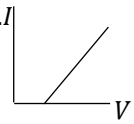
Maximum current I_{max}=4A, emf E=2.6V, internal resistance r=?,

$I_{max}=\frac{E}{r}$, $r=\frac{E}{I_{max}}=\frac{2.6}{4}=0.65\Omega$.

4. Ohm's law states that the ratio V/I is constant for I.steel II.an electrolyte III.silver IV.a diode. Which of the above are correct? A.I and II **B.I and III** C.III and IV D.I,II,III and IV.

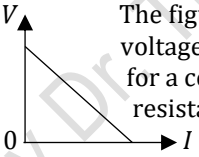
All metals e.g steel, silver, and resistors obey ohm's law.

5. Which of the following graphs correctly illustrates the relationship between the current (I) and the voltage (V) for cell **with appreciable internal resistance**?

A.  B.  C.  D. 

The $(I-V)$ of a cell with internal resistance is a graph with a negative slope= $-\frac{1}{r}$, the intercept on the I-axis is equal to the maximum current I_{max} the cell can supply= $\frac{E}{r}$ and the intercept on the V-axis is equal to the emf E. $I=-\frac{1}{r}(V)+\frac{E}{r}$.

Option A is Ohm's law graph of metals and resistors. Option D is $(I-V)$graph of conduction in liquids (electrolytes).

6. 

The figure avove represents the voltage-current$(V-I)$ relationship for a cell with appreciable internal resistance.

The intercept on the V-axis represents the **A.e.m.f of the cell** B.external resostance of the circuit C.lost volts of the cell D.maximum current E.potntial difference of the external resistance.

From the relation : $V=E-Ir=-Ir+E$, a graph of V against I is a straight line graph with a negative slope which represents the internal resistance$(-r)$ and the intervept on the V-axis represents the emf E of the cell, when the current is zero I,e no current is flowing in the circuit.

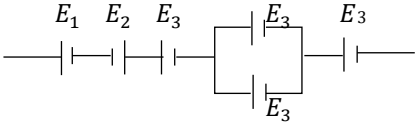
7. The total energy required to send a unit positive charge round a complete electrical circuit is the A.Kinetic energy B.Potential difference **C.Electromotive force** D.Electrical energy.

8. A radio is operated by eight dry cells each of emf 2.0V connected in series. If two of the cells are wrongly connected, the net emf of the radio is A.8V B.10V C.12V D.16V

The cells which are connected wrongly are given a negative emf. Six cells are connected properly while two cells are connected wrongly. Emf of cells properly connected, $E_P=6\times2=12V$. Emf of cells wrongly connected $E_w=2\times-2=-4V$.

Net emf $=E_P+E_w=12-4=8V$.

9. The diagram below shows the supply source of an electrical circuit.



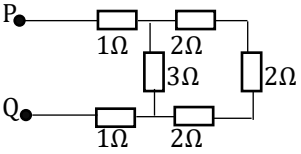
Calculate the total effective emf of the source A.$E_1+E_2+3E_3$ B.$E_1+E_2+4E_3$ **C.$E_1-E_2+3E_3$**

Demystified Series by Dr Timothy

D.$E_1 - E_2 + 4E_3$ E.$E_1 + E_2 - E_3$.
The cell E_2 is given a negative emf as it is wrongly connected i.e its negative pole is connected to the negative pole of cell E_1 and its positive pole is connected to the positive pole of the first cell E_3. The net emf of cells in parallel is equal the emf of one of the cell i.e E_3. Total effective emf
$E = E_1 - E_2 + E_3 + E_3 + E_3 = E_1 - E_2 + 3E_3$.

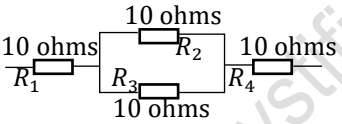
10. Two resistors R_a and R_b are connected in parallel. If $R_a < R_b$, the combined resitance is A.greater than R_a **B.less than R_b** C.$R_a + R_b$ D.greater than R_a but less than R_b
R_a and R_b are connected in parallel, their combined or net resistance R is given by :
$R = \frac{R_a R_b}{R_a + R_b} = R_a\left(\frac{R_b}{R_a + R_b}\right)$. The quantity in the bracket is less than 1. Thus, R is less than R_a and R_b. Generally, the effective resistance of a number of parallel resistors is always less than the least individual resistance.

11.. The total resistance measured at PQin the digram below is?

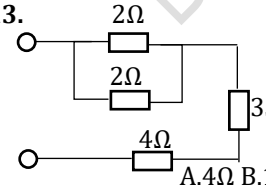


A.18.0Ω B.11.0Ω **C.4.0Ω** D.2.0Ω
The three 2Ω resistors are in series,
R =2+2+2 =6Ω. This 6Ω is now in parallel with the 3Ω resistor, $R = \frac{6\times3}{6+3} = \frac{18}{9} = 2\Omega$. This 2Ω is now in series with the two 1Ω resistor
R = 2+1+1=4Ω.

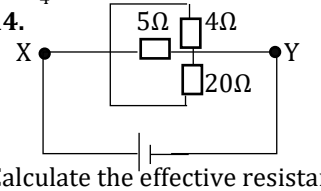
12. What is the total resistance in the below diagram? A.5Ω B.15Ω **C.25.0Ω** D.35Ω E.45Ω



The total resistance $R = R_1 + \frac{R_2 R_3}{R_2 + R_3} + R_4$,
$R = 10 + \frac{10\times10}{10+10} + 10 = 10 + \frac{100}{20} + 10$,
$R = 10 + 5 + 10 = 25\Omega$.

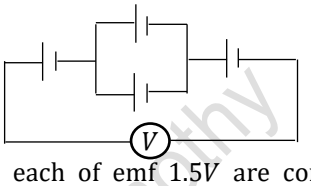
13. 

What is the resistance of the circuit shown in the figure
A.4Ω B.11Ω C.$\frac{19}{5}$ Ω D.$\frac{17}{4}$Ω E.8Ω
Effective resistance $R = \frac{2\times2}{2+2} + 3 + 4$,
$R = \frac{4}{4} + 7 = 1 + 7 = 8\Omega$.

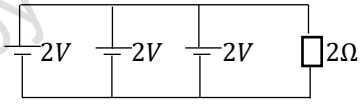
14. 

Calculate the effective resistance between the points X and Y in the circuit diagram shown
A.29.0Ω B.19.0Ω C.11.0Ω **D.2.0Ω** E.0.5Ω
The current entering through junction X branches into three, one through 5Ω, one through 4Ω and one through 20Ω i.e they are all parallel.
$\frac{1}{R} = \frac{1}{5} + \frac{1}{4} + \frac{1}{20} = \frac{4+5+1}{20} = \frac{10}{20}\frac{1}{R} = \frac{1}{2}$, R =2Ω.

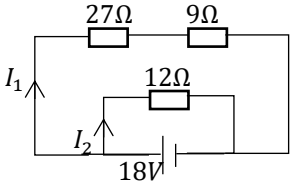
15. Three 6Ω resistors are connected in parallel. If a potential difference of $24V$ is appliued across the combination,the current in each resistor is A.12.0A **B.4.0*A*** C.1.5A D.0.8A
p.d across the combination V =p.d across each resistor=24V(as they're parallel), R_1=R_2=R_3=6Ω, $V = IR$, $I = V/R = 24/6 = 4A$.

16. 

Four cells each of emf 1.5V are connected as shows in the digram. The reading on the voltmeter, V is A.1.5V B.3.0V **C.4.5*V*** D.6.0V
The voltmeter reading=Net emf of the combined cells. The two cells in parallel have an equivalent emf of 1.5V. The net emf of is given by the algebraic sum of all the emf as the cells.
E =1.5+1.5+1.5 =4.5V.

17. 

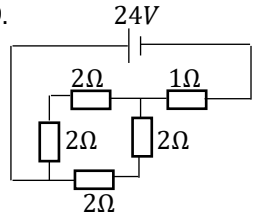
An electric circuit is connected as illustrated above. Determine the equivalent emf and current flowing through the circuit respectively,neglecting the internal resistance of the cells **A.2*V*, 1.0*A*** B.2V, 4.0A C.6V, 0.3A D.6V, 3.0A .
The equivalent emf is equal to the emf of one cell, as the cells are parallel. $E = 2V$, Neglecting the internal resistance, $E = IR$, $I = \frac{E}{R}$, R=2Ω,
$I = \frac{2}{2} = 1.0A$.

18. 

In the circuit shown,the current delivered, by the battery isA.0.4A B.0.5A C.1.5A **D.2.0*A***
Total current delivered $I = I_1 + I_2$,
Effective resistance resistors in series,
R = 27+9 = 36Ω. V(across 36Ω) = V(across 12Ω), as they now in parallel. Current in the series resistors $I_1 = \frac{V}{R} = \frac{18}{36} = 0.5A$.
Current in the parallel resistors $I_2 = \frac{V}{R} = \frac{18}{12} =$ 1.5A. Alternatively, total resistance of the circuit, $R_T = \frac{36\times12}{36+12} = \frac{432}{48} = 9\Omega$. $I = \frac{V}{R_T} = \frac{18}{9} = 2A$.

Demystified Series by Dr Timothy

19.

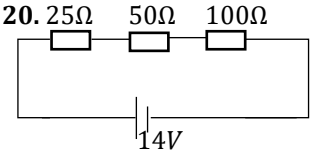


In the circuit shown, the current through the 1Ω resistor is A.0.4A B.0.5A **C.8.0*A*** D.9.6A

The two 2Ω in both outside the cell are in series, R=2+2=4Ω. Also, the two 2Ω resistos inside the cell are in series, $R = 2+2=4\Omega$.

These two 4Ω resistors are now in parallel to each other, $R = \frac{R}{n} = \frac{4}{2} = 2\Omega$ or $R = \frac{4\times4}{4+4} = 2\Omega$ This 2Ω resistor is now in series with the 1Ω resistor, and the current flowing through each conductor is the same ($I_{2\Omega} = I_{1\Omega}$) and is equal to the total current flowing through the circuit. $V = IR,\ I = \frac{V}{R} = \frac{24}{3} = 8.0A$.

20.



In the circuit diagram shown above, the p.d across the 25Ω resistor is A.1V **B.2*V*** C.4V D.14V

For cells in series, the potential difference across each resistor is different but the same current flows through each resistor.

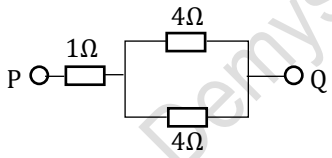
R_T =25+50+100 = 175Ω.$V = IR$,

$I_T = \frac{V_T}{R_T} = \frac{14}{175} = 0.08A.V_{25\Omega} = I_T R$,

$V_{25\Omega}$ = 0.08×25= 2A. Alternatively,For cells in series, $V \propto R$, $I = K, V_{25\Omega} = \frac{R_{25\Omega}}{R_T} \times V_T$,

$V_{25\Omega} = \frac{25}{25+50+100} \times 14 = \frac{25}{175} \times 14 = 2V$.

21. In the figure below, a voltage V is applied across the terminals P and Q.



The voltage across the 1Ω is A.$V/8$ B.$V/4$ **C.*V*/3** D.$V/2$.

Net resistance of the parallel resistors, $R = \frac{4\times4}{4+4}$, $R = \frac{16}{8} = 2\Omega$. The 2Ω resistor is now in series with the 1Ω resistor. Total resistance of the circuit $R_T = 2+1 = 3\Omega$. Total voltage=V, total current flowing through the circuit $I_T = \frac{V}{R_T} = \frac{V}{3}$. Current in the 1Ω = current in 2Ω = total current. $I_{1\Omega} = I_{2\Omega} = I_T = \frac{V}{3}$.

22. Three resistors with resistance 250Ω,500Ω and 1kΩ are connected in series. A 6V battery is connected to either end of the combination Calculate the potential difference between the ends of the 250Ω resistor A.0.20V**B.0.86*V*** C.1.71V D.3.543V.

The three resistiors, 250Ω,500Ω and 1kΩ (1000Ω) are in series.$V \propto R$, I = constant or the same for series resistors. V_T=6V,

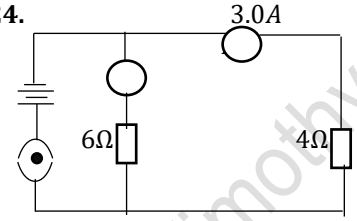
$V_{250\Omega} = \frac{R_{250\Omega}}{R_T} \times V_T = \frac{250}{250+500+1000} \times 6$,

$V_{250\Omega} = \frac{250}{1750} \times 6 = 0.857V = 0.86V$.

23. The resistance of an open circuit is A.zero B.half the normal resistanceof the circuit C.double the normal resistance of the circuit D.one **E.Infinite.**

An open circuit is a circuit not delivery current i.e I=0. $R = \frac{V}{I} = \frac{V}{0} = \infty$(infinite).

24.

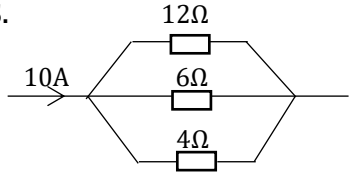


The ammeter A_2 shown in the circuit diagram read 3.0A. Determine the reading of the ammeter A_1 A.12A B.9A C.7A D.2A

The ammeter A_1 records the current through 6Ω while the ammteter A_2 records the current through 4Ω. I_1=?, R_1=6Ω, I_2=3.0A, R_2=4Ω. The p.d across the 6Ω and the 4Ω resistors is the same as they are parallel. $V_1 = V_2$, $V = IR$,

$\therefore I_1R_1 = I_2R_2$, $I_1 \times 6 = 3\times4$, $I_1 = \frac{3\times4}{6} = 2A$.

25.



The current in the 6Ω resistor shown in the diagram above is A.1.7A B.2.7A **C.3.3*A*** D.60.0A.

I_T=10A, $\frac{1}{R_T} = \frac{1}{12} + \frac{1}{6} + \frac{1}{4} = \frac{1+2+3}{12} = \frac{6}{12} = \frac{1}{2}$,

$R_T = 2\Omega$. $V = I_T R_T = 10\times2 = 20V$.

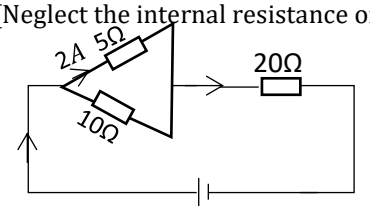
P.d across 6Ω resistor=Total voltage (parallel resistors)=20V, $I_{6\Omega} = \frac{V}{R_{6\Omega}} = \frac{20}{6} = 3.3A$.

Alternatively, for cells in parallel, $I \propto \frac{1}{R}$ or $I \propto G$, V =constant. Coductance $G = \frac{1}{R}$.

$I_{6\Omega} = \frac{G_{6\Omega}}{G_{12\Omega}+G_{6\Omega}+G_{4\Omega}} \times I_T = \frac{\frac{1}{6}}{\frac{1}{12}+\frac{1}{6}+\frac{1}{4}} \times 10$,

$I_{6\Omega} = \frac{\frac{1}{6}}{\frac{1+2+3}{12}} \times 10 = \frac{\frac{1}{6}}{\frac{6}{12}} = \frac{1}{6} \times \frac{12}{6} \times 10 = 3.3A$.

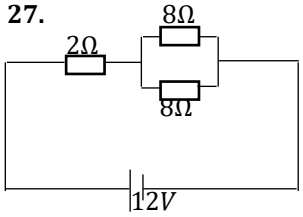
26. Using the diagram below, calculate the potential difference across the 20Ω resistor. {Neglect the internal resistance of the cell]



Demystified Series by Dr Timothy

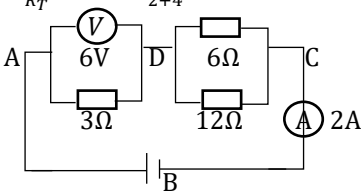
A.5V B.10V C.20V **D.60V.**
p.d across 5Ω $V_{5\Omega}$ = p.d across 10Ω $V_{10\Omega}$, as they are parallel. Current through 5Ω I_1=2A, $V_{5\Omega} = V_{10\Omega}$, $I_1R_1 = I_2R_2$, $2\times5 = I_2\times10$, $I_2 = \frac{2\times5}{10} = 1A$. Current flowing through the 20Ω resistor is the total current I_T. $I_T = I_1 + I_2 = 2+1 = 3A$. P.d across 20Ω resistor $V_{20\Omega} = I_TR_{20\Omega} = 3\times20 = 60V$.

27.



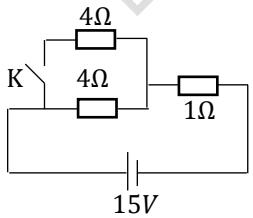
In the circuit diagram shown above, the p.d across the 2Ω resistance is A.12V B.8V **C.4V** D.2V
The two 8Ω parallel resistance gives a combined resistance of, $R = \frac{R}{n} = \frac{8}{2} = 4\Omega$.
The 4Ω resistor is now in series with the 2Ω resistor, net resistance, $R_T = 4+2 = 6\Omega$.
$V_T = 12V$. $V_T = IR_T$, $I = \frac{V_T}{R_T} = \frac{12}{6} = 2A$.
$I_{2\Omega} = I_{4\Omega} = I$ (series resistors) $= 2A$.
$V_{2\Omega} = IR_{2\Omega} = 2\times2 = 4A$. Alternatively, for cells in series, $V \propto R$, I =constant.
$V_{2\Omega} = \frac{R_{2\Omega}}{R_T}\times V_T = \frac{2}{2+4}\times12 = 4V$.

28.



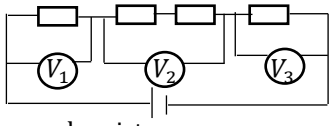
In the circuit diagram shown above, the battery B have negligible internal resistance. The p.d between $D$and $C$is A.2V B.4V C.6V**D.8V**
Net resistance of 6Ω and 12Ω parallel resistor $R = \frac{6\times12}{6+12} = \frac{72}{18} = 4\Omega$. $I_{3\Omega} = I_{4\Omega} = I_T$ =ammeter reading =2A, p.d between D and $C = V$(across 4Ω) resistor, $V_{4\Omega} = 2\times4 = 8V$.
- The emf E of the battery B equals the total voltage in the cell, $E = 6+8 = 14V$.

29.



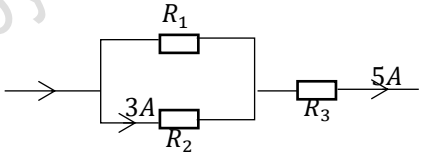
If the key in the circuit shown above is closed, the current delivered by the battery will **A.increase by 2A** B.decrease by 2A C.increase by 3A D.decrease by 3A .
When the key is open, current flows only through the 4Ω(the one below) and 1Ω resistors which are in series, $R = 4+1=5\Omega$
The current when the key is open, $I_1 = \frac{15}{5} = 3A$.
When the key is closed, the two 4Ω resistance are now in parallel and gives an effective resistance of $R = \frac{4}{2} = 2\Omega$. This 2Ω resistor is now in series with the 1Ω resistor to give a total resistance of $R = 2+1 = 3\Omega$. The current which flows when the key is closed, $I_2 = \frac{15}{3} = 5A$.
Closing the key will increase the delivered by = $I_2 - I_1 = 5-3 = 2A$.

30.



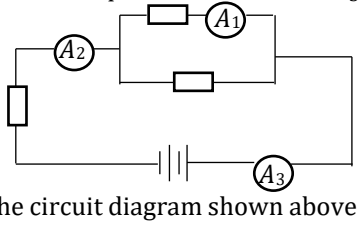
Four equal resistances are connected in series as shown in the circuits diagram above V_1, V_2, V_3 are voltmeters. Which of the following equations is correct? A.$V_1 = 2V_2 = V_3$ **B.$V_1 = V_3 = V_2/2$** C.$V_1 = V_2 = V_3/2$ D.$V_1 = V_2 = V_3$.
The same current I flows through each resistor, as they are in series. The resistance of the resistors are equal R. $V_1 = IR$, $V_2 = I(R + R) = 2IR$, $V_3 = IR$
$\therefore V_1 = V_3 = V_2/2$ or $2V_1 = 2V_3 = V_2$.
Alternatively, for cells in series, voltage drop across any resistor is directly proportional to the resistance $V \propto R$, $R_1 = R_2 = R$ hence, $V_1 = V_3$. $R_2 = 2R$ hence, $V_2 = 2V_1 = 2V_3$ or $V_2/2 = V_1 = V_3$.

31.



Which of the following statements are true of the diagram shown above? I.The current in R_1 is 8A II.The current in R_2 is 2A III.Resistance R_1 is greater than R_2 IV.P.d across R_2=P.d across R_3 A.I and III **B.II and III** C.II and IV D.III,III and IV
The current through the R_3 =Total current flowing through the circuit=Sum of current flowing through R_1 and R_2. Hence, current in $R_1 = 5-3 = 2A$. Recall $I \propto R$, since the current in R_1 is less than that in R_2, resistance R_1 is greater than R_2. P.d across R_1 =P.d across R_2, as both resistors are parallel. P.d across R_3,$V_3 = 5R_3$.

32.

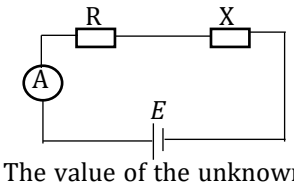


In the circuit diagram shown above A_1, A_2 and A_3 are ammeters. Which of the following statements about the reading is/are correct? I.The reading of A_1 is less than that of A_3 II.The reading of A_2 is twice of A_1 III.The readings of A_2 and A_3 are equal IV.The reading of A_3 is the sum of A_1 and A_2 A.I **B.I and III** C.I,II and III D.I,II,III and IV
In a circuit the ammeter is connected in series in the circuit while the voltmeter is connected in

Demystified Series by Dr Timothy

parallel. The same current flows through ammeter A_2 and A_3hence,will have the same reading.The current branches into two at the junction of the parallel resistors hence, the ammeter A_1 reading decreases, due to decrease in the current. The reading of A_1 is less than that of A_1 and A_3.

33. In the circuit diagram below, the ammeter reads a current of 3 Amps when R is 5 Ohms and reads 6 Amps when R is 2 Ohms.



The value of the unknown resistance X is **A.1Ω** B.2Ω C.3Ω D.4Ω

$I=3A$, when $R=5\Omega$, $I=6A$, when $R=2\Omega$

Net resistance $R_T = R + X$, $V= E=$ constant as the same cell(battery) is used.

$V = E = IR$, $E = 3(5+X)$, $E = 6(2+X)$,

$3(5+X) = 6(2+X)$, $5+X = 2(2+X)$,

$5+X = 4+2X$, $2X - X = 5-4$, $X = 1\Omega$.

34. The resistance of a wire depends on A.the length of the wire B.the diameter of the wire C.the temperature of the wire D.the resistivity of the wire **E.all of the above**

Resistance of a conductor depends on it length, area(a measure of the diameter), temperature and nature of the material (resistivity), $R = \frac{\rho L}{A}$.

35. If the temperature of a resistance wire is gradually increased, its resistance will **A.increase gradually** B.decrease gradually C.first increase and then decrease D.first decrease and then increase E.remain the same.

The resistance of a metallic conductor increases with temperature i.e $R \propto T$.

36. A nichrome wire is to be used in making a resistor. Which of these will give the highest resistance A.long and thick B.short and thick **C.long and thin** D.short and thin

$R = \frac{\rho L}{A}$, $R \propto \frac{L}{A}$. The resistance of a material increases (i.e the current flowing decreases) when the length of the wire increases and the area of the wire decreases. Hence, a long (increased length) and thin(decreased area) has a high resistance and current flowing through it is low/drops.

37. Calculate the resistivity of a wire 4.00m long, which has a cross-section of 0.500mm^2 and a resistance of 3.50Ω A.1.75×10^{-7}Ωm B.2.80×10^{-7}Ωm **C.4.38×10^{-7}Ωm** D.4.38×10^{-6}Ωm D.4.38×10^{-5}Ωm

R=3.5Ω, L=4m, A=0.5mm^2=0.5×(10^{-3})^2 = 0.5×10^{-6}m^2, ρ=? $R = \frac{\rho L}{A}$, $\rho = \frac{RA}{L}$,

$\rho = \frac{3.5\times0.5\times10^{-6}}{4} = 4.38\times10^{-7}\Omega m$.

38. Wire X has twice the diameter and twice the length of wire Y,made of the same material. The ratio of the resistance of X to Y is A.1:4 B.4:1 **C.1:2** D.1:1

$D_X=2D_Y$, $L_X=2L_Y$, resistivity ρ=constant, as both wires are made of same material. Area varies with the square of diameter, $A \propto D^2$

$\rho = \frac{RA}{L}$, $\rho_X=\rho_Y$, $\frac{R_XA_X}{L_X} = \frac{R_YA_Y}{L_Y}$, $\frac{R_XD_X^2}{L_X} = \frac{R_YD_Y^2}{L_Y}$

$\frac{R_X}{R_Y} = \frac{L_X}{L_Y} \times \frac{D_Y^2}{D_X^2} = \frac{2L_Y}{L_Y} \times \frac{D_Y^2}{(2D_Y)^2} = \frac{2}{1} \times \frac{1}{4} = \frac{1}{2}$,

$R_X:R_Y = 1:2$. Alternatively,$R \propto \frac{L}{A}$, $R \propto \frac{L}{D^2}$,

Length and diameter are doubled, $R = \frac{2}{2^2}$,

$R = \frac{1}{2}$ i.e resistance is reduced bby half or $R_X:R_Y = 1:2$.

39. A metal wire of length 1cm and of uniform diameter has a resistance of 0.8Ω. What is the resistance of a 50cm wire coil made of the same material but having twice its diameter? A.250Ω B.125Ω C.40Ω **D.10Ω**

L_1=1cm, $D_1=D$, R_1=0.8Ω, L_2=50cm, $D_2=2D$, R_2=?, resistivity is constant, as they are of same material. $\rho_1=\rho_2$=constant

$\rho = \frac{RA}{L}$, $\rho_1=\rho_2$, $\frac{R_1A_1}{L_1} = \frac{R_2A_2}{L_2}$, $\frac{R_1D_1^2}{L_1} = \frac{R_2D_2^2}{L_2}$

$\frac{R_1}{R_2} = \frac{L_1}{L_2} \times \frac{D_2^2}{D_1^2}$, $\frac{0.8}{R_2} = \frac{1}{50} \times \frac{(2D)^2}{D^2} = \frac{4}{50} = \frac{2}{25}$,

$R_2 = \frac{25\times0.8}{2} = 10\Omega$.

40. A 4m long wire of cross-sectional area 2×10^{-8}m^2 has a resistance of 5Ω. The conductivity of the wire is A.2.5×10^{-8}Ω^{-1}m^{-1} B.4.0×10^{-8}Ω^{-1}m^{-1} C.2.5×10^{7}Ω^{-1}m^{-1} **D.4.0×10^{7}Ω^{-1}m^{-1}**

R=5Ω, A=2×10^{-8}m^2, L=4m, conductivity σ=?

$\rho = \frac{RA}{L}$, $\sigma = \frac{1}{\rho} = \frac{L}{RA}$, $\sigma = \frac{4}{5\times2\times10^{-8}}$,

$\sigma = 4\times10^7\Omega^{-1}m^{-1}$.

41. 0.5m of wire A is connected in series in a circuit to 1.0m of wire B. Wire A and wire B are of the same thicknesss and have resistivities of 7.0×10^{-5}Ωm and 3.5×10^{-5}Ωm respectively. Calculate the ratio of p.d across wire A to that across wire B when current flows **A.1:1** B.1:2 C.2:1 D.1:4 E.4:1

L_A=0.5m, L_B=1.0m, ρ_A=7.0×10^{-5}Ωm, ρ_B=3.5×10^{-5}Ωm, thickness of wire A and B is equal i.e constant area, $A_A=A_B$. For cells in series, the same current flows, $V \propto R$,

$\frac{V_A}{R_A} = \frac{V_B}{R_B}$, $\frac{V_A}{V_B} = \frac{R_A}{R_B}$, $R = \frac{\rho L}{A}$, $R = \rho L$ i.e area$A$ is constant,as both wires are of the same thickness,

$\frac{V_A}{V_B} = \frac{\rho_A L_A}{\rho_B V_B} = \frac{7.0\times10^{-5}\times0.5}{3.5\times10^{-5}\times1}$, $\frac{V_A}{V_B} = \frac{1}{2}$, $V_A:V_B = 1:2$.

42. An electric cell of internal resistance 0.5Ω delivers a current of 2.0A when a resistance of 3Ω is connected across it. The electromotive force of the cell is A.1.0V B.1.5V C.2.5V **D.7.0V**

$I=2A$, R=3Ω, r=0.5Ω, E=?, $E = I(R + r)$,

$E = 2(3+0.5) = 2(3.5) = 7.0V$.

43. The potential difference between the terminals of a battery is 2.8V. With a 5Ω resistor connected across the terminals of the cell, the p.d

Demystified Series by Dr. Timothy

is 2.5V.The internal resistance of the cell A.0.30Ω B.0.50Ω **C.0.60Ω** D.5.00Ω

P.d when the cell is open or not delivery current or not connected to a resistor = emf E = 2.8V, P.d when the cell is closed or delivering current or connected to a resistor = terminal p.d V = 2.5V, external resistance R=5Ω, internal resistance r=?, current delivered by the cell, $I = \frac{V}{R} = \frac{2.5}{5}$, $I = 0.5A$. $E = I(R + r)$, $2.8=0.5(5+r)$, $\frac{2.8}{0.5} = 5+r$, $5.6 = 5+r$, $r = 5.6-5 = 0.6\Omega$.

44. A cell of internal resistance 2Ω and emf 12V is connected to a resistor of 4Ω. The terminal p.d of the cells is A.12V B.10V **C.8V** D.6V

E=12V, r=2Ω, R=4Ω, V=?, $\frac{V}{E} = \frac{R}{R+r}$, $\frac{V}{12} = \frac{4}{4+2}$, $\frac{V}{12} = \frac{4}{6}$, $V = \frac{4\times12}{6} = 8V$.

45. A cell delivers a current of 0.2A through a resistance of 10Ω. When the resistance is changed to 6Ω the current delivered is 0.3A. The internal resistance of the cell is **A.2.0Ω** B.1.5Ω C.1.0Ω D.0.7Ω

$E = I(R + r)$,

When R=10Ω, I=0.2A, $E = 0.2(10+r)$

When R=6Ω, I=0.3A, $E = 0.3(6+r)$

The cells are the same hence, E = constant.

$0.2(10+r) = 0.3(6+r)$, $2+0.2r = 1.8+0.3r$, $0.3r - 0.2r = 2-1.8$, $0.1r = 0.2$, $r = 2\Omega$.

46. A battery delivering a current of 1.0A has a terminal potential difference of 2V. When delivering a current of 1.5A, the terminal p.d is 1V. The internal resistance of the battery is A.0.5Ω B.1.0Ω **C.2.0Ω** D.4.0Ω

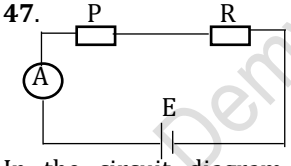
$E = V + $v, lost volt v= Ir, $E = V + Ir$,

When I=1.0A, V=2V, $E = 2+r$,

When I=1.5A, V=1V, $E = 1+1.5r$,

$2+r = 1+1.5r$, $1.5r - r = 2-1$, $0.5r = 1$, $r = \frac{1}{0.5} = 2\Omega$.

47.



In the circuit diagram shown, the ammeter Aregisters a current of 2A, when$P$is 6Ω and 4A when P is 2Ω. The value of resistance R is **A.2Ω** B.3Ω C.4Ω D.6Ω

When P=6Ω, I=2A, $E = 2(6+R)$,

When P=2Ω, I=4A, $E = 4(2+R)$,

$2(6+R) = 4(2+R)$, $6+R = 2(2+R)$, $6+R = 4+2R$, $2R - R = 6-4$, $R = 2\Omega$.

48. Three cells each of emf 1.1V and internal resistance 2Ω are connected in parallel across a 3Ω resistor. The current in the resistor is A.0.12A **B.0.30A** C.0.39A D.0.90A

The three cells in parallel have equal emf,

$E_T = E_1 = E_2 = E_3 =$ 1.1V, net internal resistance $r_T = \frac{r}{n}$, $r_t = \frac{2}{3} = 0.67\Omega$, R=3Ω, I=?

$I = \frac{E}{R+r} = \frac{1.1}{3+0.67} = \frac{1.1}{3.67} = 0.3A$.

49. A cell of emf 1.2V and internal resistance 1.5Ω is connected in series with an ammeter of resistance 0.5Ω and a load of resistance 4Ω. The current in the circuit is **A.0.20A** B.0.24A C.0.30A D.0.40A

E=1.2V, r=1.5Ω, total resistance is the sume of the resistamce of the load and that of the ammteter. $R_T = R_{load} + R_{ammeter}$,

$R_T = 4+0.5 = 4.5\Omega$, I=?, $I = \frac{E}{R+r}$,

$I = \frac{1.2}{4.5+1.5} = \frac{1.2}{6} = 0.2A$.

50. The voltage across the terminals of a cell drops to three-quarters of its **norminal value** when a resistance R is connected across the cell. The cells internal resistance is A.$R/4$ B.$R/3$ **C.$R/2$** D.R

Norminal voltage of the cell is same as emf of the cell.

$V = \frac{3}{4}E$, $E = \frac{4}{3}E$, $E = \frac{4}{3}IR$ $[V = IR]$,

$E = V +$v$= IR + Ir$, $Ir = E - IR$,

$r = \frac{E-IR}{I} = \frac{\frac{4}{3}IR-IR}{I} = \frac{\frac{4IR-3IR}{3}}{I} = \frac{\frac{IR}{3}}{I} = \frac{IR}{3}\times\frac{1}{I}$,

$r = \frac{R}{3}$.

51. A resistance R is connected across the terminal of an electric cell of internal resistance 2Ω and the voltage was reduced to 3/5 of its nominal value. The value of R is **A.6Ω** B.3Ω C.2Ω D.1Ω

r=2Ω, $V = \frac{3}{5}E$, $E = \frac{5}{3}V = \frac{5}{3}IR$ $[V = IR]$,

$E = V +$v$= IR + Ir$, $Ir = E - IR$,

$Ir = \frac{5}{3}IR - IR = \frac{5IR-3IR}{3} = \frac{2}{3}IR$,

$R = \frac{3Ir}{2I} = 3r = 3\times2 = 6\Omega$.

52. An electric bell with nominal voltage E has a resistance 3Ω connected across it. If the voltage falls to 0.6E, the internal resistance of the cell is A.4Ω B.3Ω **C.2Ω** D.1Ω

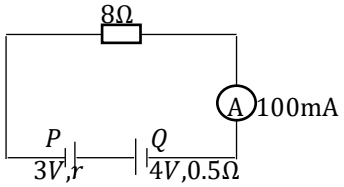
Nominal voltage or emf E, R=3Ω, V=0.6E, $E = \frac{V}{0.6} = \frac{5V}{3} = \frac{5IR}{3}$, r=?, $E = V +$v,

$E = IR + Ir$, $Ir = E - IR = \frac{5IR}{3} - IR$,

$Ir = \frac{5IR-3IR}{3} = \frac{2}{3}IR$, $r = \frac{2IR}{3I} = \frac{2R}{3} = \frac{2\times3}{3}$,

$r = 2\Omega$.

53.



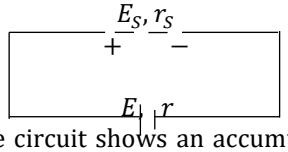
In the circuit shown above, cell P has an emf of 3V and an unknown internal resistance r while cell Q has an emf of 4V and an internal resistance of 0.5Ω. If the ammeter registers a current 100mA, then r is A.0.5Ω B.1.0Ω **C.1.5Ω** D.2.0Ω

E_P=3V, r_P=r, E_Q=4V, r_Q=0.5Ω,I=100mA=0.1AR=8Ω. The cells are

Demystified Series by Dr. Timothy

connected wrongly i.e the positive pole of cell P is connected to the positive pole of cell Q.
Net emf , $E_T = E_Q - E_P$ =4−3 =1V,
$r_T = r_P + r_Q = r$ +0.5, $E = I(R + r)$,
$1 = 0.1[8+(0.5+r)]$, $\frac{1}{0.1}$ =8+0.5+r,
10 =8.5+r, $r = 10-8.5$ =1.5Ω.

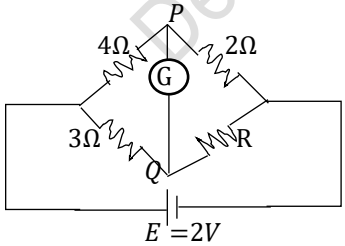
54.



The circuit shows an accumulator of emf E and internal resistance r being charged by a d.c supply of emf E_S and internal resistance r_S. The charging current measured by the ammeter A is given by A.$\frac{E_S+E}{r_S+r}$ B.$\frac{E_S-E}{r_S-r}$ **C.$\frac{E_S-E}{r_S+r}$** D.$\frac{E-E_S}{r_S+r}$
In charging an accumulator, the d.c supply source and the accumulator are connected in series,with similar poles connected as shown above and the emf of the d.c supply must be greater that of the accumulator inorder to charge it. The net emf $E_T = E_S - E$, net internal resistance $r_T = r_S + r$, $E_T = Ir_T$, $I = \frac{E_T}{r_T} = \frac{E_S-E}{r_S+r}$.

55. The internal resistance of a cell supplying current to a 4Ω resistor is 1.5Ω. Calculate the efficiency of the cell A.70.5% B.80.2% C.75.5% **D.72.7% .**
R=4Ω, r=1.5Ω, Efficiency ϵ=?,,
$\epsilon = \frac{Power\ output}{Power\ input} = \frac{IV}{IE} = \frac{I^2R}{I^2(R+r)} = \frac{R}{R+r}$,
$\epsilon = \frac{4}{4+1.5} \times 100 = \frac{4}{5.5} \times 100 = 72.7\%$.

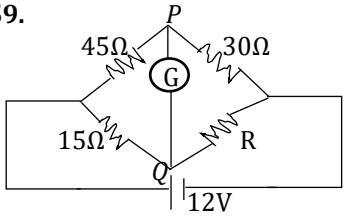
56. A cell of internal resistance r supplies current to a 6.00Ω resistor and its efficiency is 75%. Find the value of r A.8.0Ω B.4.5Ω **C.2.0Ω** D.1.0Ω
R=6.00Ω, ϵ=75%, r=?, $\epsilon = \frac{R}{R+r} \times 100$,
$75 = \frac{6}{6+r} \times 100$, $6+r = \frac{6\times100}{75} = 8$,
$r = 8-6 = 2.0\Omega$.
In the diagram above, the galvanometer indicates a null-deflection. Use the information to answer question 57 and 58.



57. The value of R is **A.1.50Ω** B.2.66Ω C.3.99Ω D.4.50Ω E.6.00Ω
R_1=4Ω, R_2=2Ω, R_3=3Ω, R_4=R, Wheatstone bridge : $\frac{R_1}{R_2} = \frac{R_3}{R_4}$, $\frac{4}{3} = \frac{2}{R}$, $R = \frac{3\times2}{4} = 1.50\Omega$.

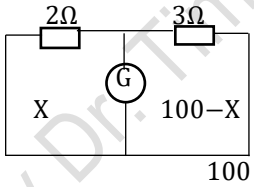
58. The potential difference btween P and Q is **A.0.0V** B.1.5V C.2.0V D.3.0V E.4.0V
The galvanometer indicates a null deflection, as no current is flowing along PQ, because point P and Q are at the same potential. Thus, the potential difference between P and Q = 0.0V .

59.



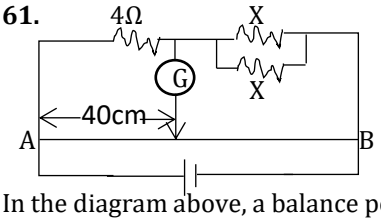
Calculate the resistance of R required to balance the bridge in the circuit above A.5Ω **B.10Ω** C.15Ω D.22.5Ω
R_1=45Ω, R_2=30Ω, R_3=15Ω, R_4=R,
Wheatstone bridge : $\frac{R_1}{R_2} = \frac{R_3}{R_4}$, $\frac{45}{15} = \frac{30}{R}$,
$3 = \frac{30}{R}$, $R = \frac{30}{3} = 10\Omega$.

60. In a meter bridge experiment, two resistors 2Ω and 3Ω occupy the left and right gaps respectively. Find the balance point from the left side of the bridge A.20cm **B.40cm** C.60cm D.80cm



X is the balance point from the left side or zero end of the bridge, R_1=2Ω, L_1=X, R_2=3Ω
L_2=100−L_1=100−X ,
Meter bridge : $R \propto L$, $\frac{R_1}{L_1} = \frac{R_2}{L_2}$, $\frac{R_1}{R_2} = \frac{L_1}{L_2}$,
$\frac{2}{3} = \frac{X}{100-X}$, $2(100-X) = 3X$, $200-2X = 3X$,
$200 = 3X + 2X = 5X$, $X = \frac{200}{5} = 40$cm.

61.



In the diagram above, a balance point is obtained at 40cm from left end. The value of X is A.2Ω B.3Ω C.6Ω **D.12Ω**
Balance point from the left end L_1=40cm, Balance point from the left hand L_2=100−L_1 ,
L_2=100−40=60cm, R_1=4Ω, R_2= $\frac{X}{2}$ ($R_T = \frac{R}{n}$, i.e parallel resistors). $\frac{R_1}{L_1} = \frac{R_2}{L_2}$, $\frac{4}{40} = \frac{X/2}{60}$,
$\frac{X}{2} = \frac{4\times60}{40} = 6$, $X = 6\times2 = 12\Omega$.

62. A calibrated potentiometer can be used to measure an unknown emf E of a cell only when I.the source of the emf is greater than E II.the cell's internal resistance is negligible III.no current flows through the cell IV.the potentiometer wire has a uniform diameter. Which of the above statements are correct? A.I and II B.III and IV C.II,III and IV **D.I,III and IV**
Potentiometer is used in measuring the emf of a cell at null point i.e when no current is flowing

Demystified Series by Dr. Timothy

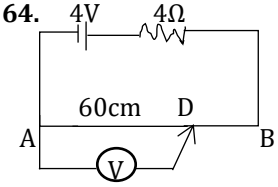
through the cell. The internal resistance is not considered, as no current is flowing through the cell. The potentiometer must have a uniform diameter, for p.d to be proportional to the length of te wire ($V \propto L$).

63. A standard cell of emf $1.6V$ gives a balance point of 40cm on a potentiometer wire. If the standard cell is now replaced by a cell of emf $2.4V$, the new balance point is at A.26.7cm B.50.0cm **C.60.0cm** D.64.0cm

E_1=$1.6V$, L_1=40cm, E_2=$2.4V$, L_2=?,

$\frac{E_1}{L_1} = \frac{E_2}{L_2}$, $\frac{1.6}{40} = \frac{2.4}{L_2}$, $L_2 = \frac{40\times2.4}{1.6} = 60$cm.

64.



In the circuit diagram shown above, AB is a uniform wire of length 1m and resistance 6Ω. Neglecting the internal resistance of the cell, the reading on the voltmeter V is **A.1.44V** B.1.89V C.4.00V D.6.00V

The net resistance in the circuit= Resistance of the resistor +Resistance of the wire AB,R_T = 4+6=10Ω. Emf of the cell, E= V_T= $4V$, Emf of the wire AB or voltage across wire AB=?, R=10Ω, R_{AB}=6Ω, $V \propto R$,

$V_{AB} = E_{AB} = \frac{R_{AB}}{R_T} \times V_T = \frac{6\times4}{10} = 2.4V$.

Emf of the wire AB can also be gotten by:

Total current in the circuit $I = \frac{V_T}{R_T} = \frac{4}{10} = 0.4A$Emf or voltage along the wire AB $E_{AB} = IR_{AB}$, E_{AB} =0.4×6 =$2.4V$. Voltmeter reading=Emf or voltage across wire AD E_{AD}, L_{AD}=60cm, E_{AB}=$2.4V$, L_{AB}=1m=100cm, $E \propto L$, $\frac{E_{AB}}{L_{AB}} = \frac{E_{AD}}{L_{AD}}$ $\frac{2.4}{100} = \frac{E_{AD}}{60}$, $E_{AD} = \frac{2.4\times60}{100}$ =$1.44V$.

Alternatively, Resistance of a wire is directly proportional to its length, $R \propto L$. L_{AD}=60cm, R_{AD}=?, L_{AB}=1m=100cm, R_{AB}=6Ω, $\frac{R_{AB}}{L_{AB}} = \frac{R_{AD}}{L_{AD}}$, $\frac{6}{100} = \frac{R_{AD}}{60}$, $R_{AD} = \frac{6\times60}{100}$ =3.6Ω.

R_{AB}=6Ω, R(of the resistor)=4Ω, R_T=6+4=10Ω (all are in series). The cells are in series hence, voltage varies directly with the resistance. $V \propto R$.

$V_{AD} = \frac{R_{AD}}{R_T} \times V_T = \frac{3.6}{10} \times 4 \;\; 1.44V$.

65. A current of $0.5A$ flows through a resistor when connected to a $40V$ battery. How much energy is dissipated in 2minutes? A.1200J B.1500J **C.2400J** D.96000J .

I=$0.5A$, V=$40V$, t=2 minutes=2×60=120s

Electrical energy E=$QV = IVt = V^2t/R = I^2Rt$.

$E = IVt$ = $0.5\times40\times120$ =$2400J$.

66. A potential difference of $6V$ is used to produce a current of $5A$ for 200s through a heating coil. The heat produced is A.4800cal B.6000cal C.2400J D.240kcal **E.6000J**

I=$5A$, V=$6V$, t=200s, H=electrical heat energy

$H = IVt = Pt = I^2Rt = V^2t/R$

$H = IVt$ = $5\times6\times200$=$6000J$ or $\frac{6000}{4.2}$ =1428.6 calorie or 1428.6cal or 1.428kcal.

67. A standy generator is connected to fifteen $40W$ lamps and a muscician's $600W$ amplifying system. How much energy is used if the generator runs for 6hours? A.3.84kWh **B.7.20kWh** C.8.40kWh D.12.00kWh E.14.56kWh

Power of fifteen $40W$ lamps=15×40=$600W$,

Power of musician's amplifiying system=$600W$,

Total power=600+600=$1200W$=1.2kW. t=5hrs

Energy, $E = P \times t$ = 1.2×5=7.2kWh.

68. The heat produced in a conductor carrying an electric current is A.iversely proportional to the current and the resistance B.directly proportional to the current,resistance and the time C.inversely proportional to the square of the resistance,current and time **D.directly proportional to the square of the current, the resistance and the time**

Joule's law of electrical heating, $H \propto I^2Rt$ or $H \propto I^2$, $H \propto R$, $H \propto t$.

69. It takes 4 minutes to boil a quantity of water electrically. How long will it take to boil same quantity of water using the same heating coil but with the current doubled? (Neglect any external heat lossed) A.64minutes B.32minutes C.8minutes D.2 minutes **E.1minute**

I_1=I, t_1=4 minutes, I_2=$2I$, t_2=?, The relationship between the current flowing through a heating element and the time of flow is – $H = I^2Rt$,

$\frac{H}{R} = I^2t$, $\frac{H}{R}$ = constant, $\therefore I^2 \propto \frac{1}{t}$, ${I_1}^2t_1 = {I_2}^2t_2$.

$I^2 \times4 = (2I)^2 \times t_2$ $4I^2 = 4I^2 \times t_2$,

$t_2 = \frac{4I^2}{4I^2} = 1$minute.

70. An electric iron is rated at 1000watts, $250V$.The corresponding maximum resistance and accompanying current is A.62.5A, 4.0Ω B.16.0A, 62.5Ω C.62.5A, 16.0Ω **D.4.0A, 62.5Ω** E.4.0A, 250Ω

P=$1000W$, V=$250V$, R=?, I=?,

$P = IV = I^2R = \frac{V^2}{R}$, $P = IV$, 1000= $I \times250$, $I = \frac{1000}{250}$ =$4A$. $P = \frac{V^2}{R}$, $1000 = \frac{250^2}{R}$,

$R = \frac{62500}{1000} = 62.5\Omega$.

Alternatively, $P = I^2R$, $1000 = 4^2 \times R$,

$R = \frac{1000}{16} = 62.5\Omega$.

71. A $6V$ battery of internal resistance 0.75Ω delivers current through an external load of 1.25Ω. What is the power supplied? **A.18.0W** B.72.0W C.12.0W D.28.8W

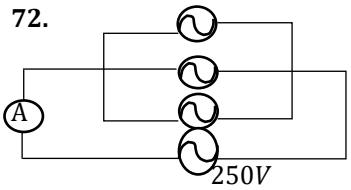
E=$6V$, R=1.25Ω, r=0.75Ω, Power supplied by battery=Power input P_i.

$P_i = IE = I^2(R + r) = \frac{E^2}{R+r}$,

$P_i = \frac{E^2}{R+r} = \frac{6^2}{1.25+0.75} = \frac{36}{2} = 18W$

Demystified Series by Dr Timothy

72.



In the diagram above, three identical lamps each of $100W$ are connected in parallel across a p.d of $250V$. Calculate the reading of the ammeter A A.7.5A B.2.5A **C.1.2*A*** D.0.8A

Total power of lamps is parallel is the algebraic sum of their individual powers.

The lamps are identical thus, P_1=P_2=P_3=P.

Total power P_T =$3P$ =3×100 =$300W$,

p.d across each lamp is the same as they are parallel, V=$250V$, Ammeter reading=Total current flowing in the circuit I, $P = IV$,

$I = \frac{P}{V} = \frac{300}{250}$ =$1.2A$.

73. In homes, electrical appliances and lamps are connected in parallel because **A.less current will be used** B.less voltage will be used C.parallel connection does not heat up the wires D.series connection uses high voltage.

The current flowing through lamps connected in series are the same and is equal to the total current thus, a high current will be used. The current flowing through lamps connected in parallel branches and flows thrpugh each lamp, thus the current decreases and less current will be used.

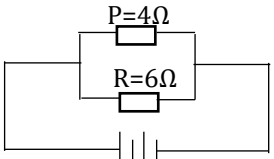
74. A lamp Y is connected across a battery. If a second(identical) lamp Z is now connected in series with Y, A.Y will become brighter **B.*Y* will become dimmer** C.the brightness of Y will remain unchanged D.the lamps will not glow due to short circuit

The brightness of a lamp is determined by the current flowing through it. Connecting lamp Z in series with lamp Y increases the total resistance and the current flowing through lamp Y decreases,hence the brightness decreases and the lamp become dimmer. If lamp Z was connected parallel to lamp Y, the bright of Y will remain unchanged as the current i.e voltage/resistance of lamp Y is constant.

75. A lamp is connected across a battery. A piece of low conductivity wire is connected in parallel with it. The lamp A.glow brighter B.glow dimmer **C.remain unchanged** D.quicly burn out.

A material of low conductivity will have high resistivity, hence high resistance.The brightness of lamp Y remains unchanged as the current flowing through it is constant. Option B would have been correct, if the wire was connected in series with the lamp, due to reduction in the current.

76. In the diagram below, P and R are 4Ω and 6Ω respectively. If the power dissipated in P is $10W$, then what is the power dissipated in R?

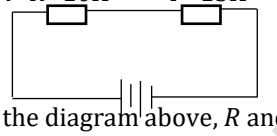


A.2.4W B.4.0W C.6.0W **D.6.7W** E.15.0W

Resistors in parallel are of equal voltage,thus voltage is constant. $P = \frac{V^2}{R}$,$P \propto \frac{1}{R}$, V^2=constant, $P_1R_1 = P_2R_2$, R_1=4Ω, R_2=6Ω, P_2=$10W$, P_2=?, 10×4= P_2 ×6, $P_2 = \frac{10\times4}{6}$,

P_2 =$6.7W$.

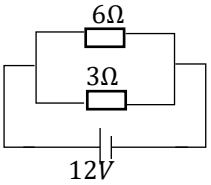
77. R=10Ω T=15Ω



In the diagram above, R and T are resistance 10Ω and 15Ω respectively. If the power dissipation in R is $40W$, then what is the power dissipation in T? **A.60*W*** B.40W C.30W D.50W

Resistors in series are of equal current thus, current is constant. $P = I^2R$, $P \propto R$, I^2=constant, $\frac{P_1}{R_1} = \frac{P_2}{R_2}$, P_1=$40W$, R_1=10Ω, P_2=?, R_2=15Ω, $\frac{4}{10} = \frac{P_2}{15}$, $P_2 = \frac{4\times15}{10} = 60W$.

78. In the diagram below, the ratio of the electric power dissipated in the $6W$ and $3W$ resistors respectively is



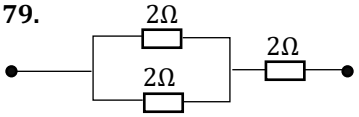
A.2:3 **B.1:2** C.1:3 D.2:1.

R_1=6Ω, R_2=3Ω,P_1:P_2=?, Resistors in parallel are of equal voltage thus, voltage is constant.

$P = \frac{V^2}{R}$, $P \propto \frac{1}{R}$, V^2=constant, $P_1R_1 = P_2R_2$,

$\frac{P_1}{P_2} = \frac{R_2}{R_1}$,$\frac{P_1}{P_2} = \frac{3}{6} = \frac{1}{2}$, P_1:P_2 = 1:2.

79.



In the diagram above, if each of the resistors can dissipate maximum of $18W$ without becoming excessively heated, what is the maximum power the circuit can dissipate? A.27W B.18W C.9W D.5W

Power in each resistor P=$18W$. The total maximum power dissipated in the circuit P_T = $\frac{18\times18}{18+18} + 18 = \frac{324}{36} + 18 = 9+18 = 27W$.

80. Two lamps rated $40W$ and $220V$ are connected in series. The totak power dissipated in both lamps is A.10W **B.20*W*** C.40W D.80W E.none of the above

P_1=P_2=$40W$. For two lamps connected in series, The total power $P_T = \frac{P_1P_2}{P_1+P_2} = \frac{40\times40}{40+40}$,

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$P_T = \frac{1600}{80} = 20W$. Alternatively, $P_T = \frac{P}{n}$, as their power rating is equal. $P_T = \frac{40}{2} = 20W$.

81. Two resistors 5Ω and 10Ω are arranged first in series and later in parallel to a $24V$ volt source. The ratio of total power dissipated in the series and parallel arrangements respectively **A.2:9** B.3:5 C.5:3 D.1:50

R_1=5Ω, R_2=10Ω, V=$24V$.

When the resistors are connected in series :

$R_S = R_1 + R_2 = 5+10 = 15Ω$,

Power dissipated in series $P_S = \frac{V^2}{R_S} = \frac{24^2}{15}$,

$P_S = \frac{576}{15} = 38.4W$.

When the resistors are connected in parallel :

$R_P = \frac{R_1R_2}{R_1+R_2} = \frac{5\times10}{5+10} = \frac{50}{15} = \frac{10}{3}Ω$.

Power dissipated in parallel $P_P = \frac{V^2}{R_P} = \frac{24^2}{10/3}$,

$P_P = 576\times\frac{3}{10} = 172.8W$.

P_S:$P_P = \frac{P_S}{P_P} = \frac{38.4}{172.8} = \frac{1}{4.5} = \frac{10}{45} = \frac{2}{9} = 2{:}9$.

82. A generator with power output of 12kW operating on 2kV distributes power to a workshop through cables of resistance 10Ω. Calculate the power lost in the cables. A.6W B.18W C.60W **D.360W** E.600W .

Power loss $= I^2R$, R=10Ω, Power output P_o =12kW=12000W, V=2kV= 2000V, $P_o = IV$,

$I = \frac{P_o}{V} = \frac{12000}{2000} = 6A$. Power loss $= I^2R$, Power loss =$6^2\times10$ =36×10 =$360W$.

83. A 40kW electric cable is used to transmit electricity through a resistor of resistance 2Ω at 800V. The power loss as internal energy is A.5.0×10^2W B.4.0×10^3W C.5.0×10^3W D.4.0×10^2W

Power output P_o=40kW=40000W, V=800V,$P_o = IV$, $I = \frac{P_o}{V} = \frac{40000}{800} = 50A$,

R=2Ω,Power loss $= I^2R = 50^2\times2$,

Power loss $= 2500\times2 = 5000W = 5.0\times10^3W$.

84. An electrical generator has an e.m.f of $240V$ and an internal resistance 1Ω. If the current supplied by a generator us $20A$ when the terminal voltage is $220V$, find the ratio of the power supplied to the power dissipated A.11:1 B.1:11 C.12:11 D.11:12

E=$240V$, r=1Ω, I=$20A$, V=$220V$,

Power supplied=power input $P_i = IE$,

$P_i = 20\times240 = 4800W$. Power dissipated = power output $P_o = IV = 20\times220 = 4400W$.

Ratio of power supplied to power dissipated, $\frac{P_i}{P_o} = \frac{4800}{4400} = \frac{12}{11}$, P_i:$P_o = 12{:}11$.

85. The electricity meters in houses measure energy consumed in A.volt **B.kilowatt-hour** C.ampere D.coulomb.

86. PHCN measures its electric energy in **A.KWh** B.Wh C.J D.W .

87. If NEPA charges 5k per kWh, what is the cost of operating for 24hours a lamp requiring $1A$ on a $200V$ line? **A.24k** B.55k C.40k D.26k E.32k

Cost per unit=5k per kWh, t=24hrs, I=$1A$, V=$200V$, Power $P = IV = 1\times200 = 200W$,

$P = 0.2\text{k}W$. Cost of electricity = Power(kW) ×time(hrs)×cost per unit = $0.2\times24\times5 = 24$k.

88. A land lord has eight $40W$ electric light bulbs, four $60W$ bulb and two $100W$ bulb in his house. If he has all the points on five hours daily and if NEPA charges 5k per unit, his bill for 30days is **A.₦5.70** B.₦7.25 C.₦3.65 D.₦8.05 E.₦4.50

Total power consumed, P =(8×40+4×60+2×100)=320+240+200 =$760W = 0.76\text{k}W$, t = 5hr/day×30days= 150hrs, cost per unit=5k per unit,

Cost of electricity=Power(kW)×time(hrs) ×cost per unit ,

Cost of electricity=0.76 ×150×5=570k=₦5.70

89. The maximum power transfer occurs in a cell when the external resistance is **A.the same as the internal resistance of the cell** B.greater than the internal resistance of the cell C.less than the internal resistance of the cell D.twice the internal resistance of the cell.

Power transferred or power dissipated or power output P_o is maximum when the external resistance equals the internal resistance of the cell, $R = r$. Power output is maximum when power input is minimum.

90. A cell of internal resistance r is connected to an external resistor R. The condition for maximum power transfer is A.$R < r$ B.R=$2r$ C.$\boldsymbol{R = r}$ D.$R > r$.

91. The function of a $6A$ fuse included in a circuit supplying a house hold refrigerator with power is to keep A.temperature of the refrigerator low and constant **B.current supplied to the refrigerator below $6A$** C.voltage supply is constant D.current supplied to the refrigerator constant and above $6A$.

A fuse wire is a material of low melting point, which melts and breaks off a circuit when flowing through an appliance is above its fuse rating. A fuse limits the total current drawn from the mains supply and keeps the circuit below fuse rating.

92. Every house supplied with electricity is provided with a box of fuses so that A.the consumption of electricity can be recorded B.people residing in the house will not have an electric shock if they touch a live wire **C.the total current drawn from the mains can be limited** Eit serves as a standby supply when there is a power failure.

93. Which of the following is an essential physical property of the wires used for making fuses? A.Low density B.High thermal conductivity C.Low electrical resistivity **D.Low melting point**

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Fuse wires are made of materials with low melting point or fusion point.

94. The fuse in an electric device is always connected to the A.neutral side of an electric supply B.earth side of an electric supply **C.live side of an electric supply** D.terminal side of an electric supply

A fuse is connected to the live lineofan electric supply.

In the wiring of houses, the fuse is connected to the wire coloured A.blue **B.brown** C.yellow D.yellow and green

A fuse coloured red or brown. The earth wire is coloured yellow or green while the neutral wire is coloured blue or black.

95. Electrical aplliances in homes are normally earthed so that **A.a person touching the appliances is safe from electric shock** B.both a.c and d.c sources can be used C.the appliances are maintained at a higher p.d than the earth D.the appliances are maintained at a lower p.d than the earth.

96. Which of the following is most suitable for protecting the circuit of a $2000W$ electric iron connected to a $250V$ mains? A.$13A$**B.$8A$** C.$5A$ D.$3A$

P=$2000W$, V=$250V$, I=?, $P = IV$, $I = \frac{P}{V}$,

$I = \frac{2000}{250} = 8A$. A fuse with a fuse rating equal or a bit above the current drawn from the main circuit i.e $13A$ fuse is most suitable for protecting the electric iron.

97. A $3000W$ electric cooker is to be used on a $200V$ mains circuit. Which of the fuses below can be used safely with the cooker? A.$2A$ B.$5A$ C.$10A$**D.$20A$.**

P=$3000W$, V=$200V$, I=?, $P = IV$, $I = \frac{P}{V}$,

$I = \frac{3000}{200} = 15A$. A fuse with a fuse rating equal or a bit above the current drawn from the main circuit should be used i.e $20A$ fuse.

98. Which of the following apparatus will require the smallest fuse rating for its protection? A.$60W$,$240V$ B.$60W$,$40V$ C.$40W$, $12V$ D.$40W$, $5V$

For $60W$, $240V$ – $I = \frac{P}{V} = \frac{60}{240} = 0.25A$

For $60W$, $40V$ – $I = \frac{P}{V} = \frac{60}{40} = 1.5A$

For $40W$, $12V$ – $I = \frac{P}{V} = \frac{40}{12} = 3.33A$

For $40W$, $5V$ – $I = \frac{P}{V} = \frac{40}{5} = 8A$.

The apparatus with the smallest current will have the smallest fuse rating. ,

99. Which of the following sets of appliance will blow a $13A$ fuse in the mains when plugged into $220V$ sockets and switched on I.two 2kW electric cookers II.three 1.2kW air conditioners III.two 1.2kW electric irons A.III B.I and II C.I and III D.II and III **E.I,II and III**

$P = IV$, V=$220V$, $I = \frac{P}{V}$,

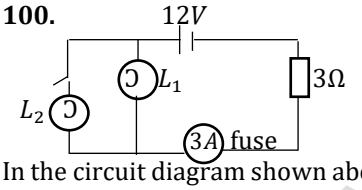
Power of two 2kW electric cookers=2×2 = 4kW = $4000W$, $I = \frac{4000}{220}$ =$18.18A$,

Power of three 1.2kW air condition= 3×1.2 = 3.6kW = $3600W$, $I = \frac{3600}{220}$ =$16.36A$

Power two 1.6kW electric irons=2×1.6 = 3.2kW = $3200W$, $I = \frac{3200}{220}$ =$14.55A$

The currents in each of the appliances are greater than the fuse rating of the fuse i.e $13A$, hence they will all blow the fuse.

100.

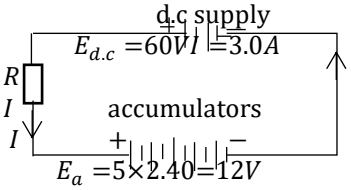


In the circuit diagram shown above, L_1 and L_2 are lamps each, of resistance 1.5Ω. The $4A$ fuse will cut when **A.K is closed** B.K is opened C.the 3Ω resistor is doybled with K closed D.the 3Ω is doubled with K opened.

When the key K is opened, no current flows through L_2, Total circuit resistance= $L_1 + R$ = 1.5+3 =4.5Ω, $I = \frac{V}{R} = \frac{12}{4.5}$ =$2.6A$. The fuse will not melt and cut as the current ($2.6A$) is less than its fuse rating ($4A$).

When the key is closed,current flows round the circuit, L_1 and L_2 are parallel and of equal resistance, their effective resistance= $\frac{1.5}{2}$ =0.75Ω, Total circuit resistance=3+0.75 =3.75Ω , $I = \frac{V}{R} = \frac{12}{3.75}$ =$3.2A$. The fuse will melt and cut the circuit, as the current($3.2A$) is greater than its fuse rating($3A$).

101. Five accumulators of negligible resistance each of emf $2.40V$, are to be charged in series from $60V$ d.c supply. If the charging current is $3.0A$, find the value of the series resistance required **A.16Ω** B.24Ω C.14Ω D.12Ω .



NB: In charging an accumulator by a d.c source similar poles are connected. i.e −ve to −ve or +ve to +ve.

Emf of accumulators E_a =5×2.4=$12V$

Emf of d.c supply $E_{d.c}$ =$60V$, Current in the circuit$I$ =$3A$,resistor of resistance R in series.

The current is flowing anticlokwisely and assuming an anticlockwise direction, aplly Kirchhoff/s voltage law : $E_{d.c} - E_a - IR = 0$. $E_{d.c}$ is +ve, as the direction moves from −ve to +ve terminal, E_a is −ve,as the direction moves from +ve to −ve terminal, voltage drop (IR) is negative as current flows in in the anticlockwise direction assumed.

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$E_{d.c} - E_a = IR$, $60-12 = 3R$, $48 = 3R$,
$R = \frac{48}{3} = 16\Omega$.

Jamb past questions on current electricity and electrical power and energy :
[1978/25,27,28,1979/12,15,27,29,1980/21,46, 47,1981/13,14,18,21,45,1982/38,1983/19,23, 24,39,43,1984/10,46,47,1985/38,40,41,4244, 1986/39,40,1987/37,43,1988/39,40,43,44, 1989/35,38,40,41,42,1990/37,38,40,1991/35, 38,39,40,41,1992/36,37,38,1993/37,38,39,42, 1994/42,44,45,1995/41,44,45,1997/40,45, 1998/44,1999/29,30,42,2000/27,33,40,49, 2001/22,24,28,31,32,36,2002/22,31,32,34,36, 2003/28,30,32,2004/20,45,2005/18,21,2006/ 25,2007/17,26,2009/37,40,41,2010/35,36,37, 38,2011/34,35,37,39,2012/39,2013/38,39,40, 41,2014/37,39,2015/10,29, 43]

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Chapter 24 – Magnetic field,Magnetism and Electromagnetism

● **Magnet –** They attract and repel magnetic materials. Natural magnet is Loadstone or magnetite. Artificial magnets are made from magnetic materials e.g iron,steel,nickel and cobalt.

- When a magnet is suspended freely, it always comes to rest in the **north-south direction.**

- Magnetic poles are regions where magnetic force is strongest in a magnet. North-seeking end is the north pole while south-seeking pole is the south pole.

- Magnetic length is the distance between the poles of a bar magnet.

- Types of magnets –

1.Permanent or hard magnets – Magnets that are not easily magnetized or demagnetized i.e retain its magnetism for a long period.They are made from steel and its allow which contains nickel,cobalt and aluminium.

2.Temporary or soft magnets – Magnets that are easily magnetized or demagnetized i.e does not retain its magnetism for a long period. They are made from soft iron(pure iron). It is used in electromagnets.

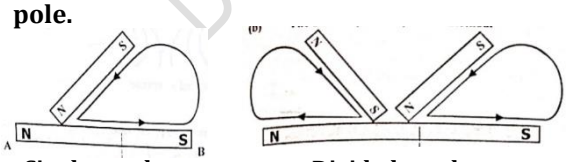
- Induced magnetism in iron is greater than that in steel i.e irons fillings clings more to iron end that steel.

- The law of magnetism – Like poles repel while unlike poles attract. i.e South pole repels south pole but attracts north pole.

- Magnetization – Magnetic materials can be manetized by:

1.Solenoid or electrical method – a magnetic material is placed inside a solenoid conncted to a battery or d.c source. **Clockwise direction of current gives S-pole while anticlockwise direction of current gives N-pole.**

2.Stroking method – i.Single touch ii.Divided touch. **The end of the material last touched acquiresa pole opposite to that of the stroking pole.**



Single touch **Divided touch**

3.Hammering the magnetic material in the north-south(N-S) direction to the earth.

- Demagnetization – Magnetic materials and magnets can be demagnetized by:

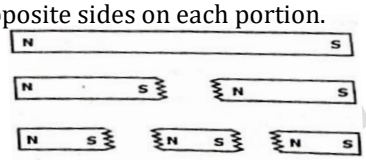
1.Solenoid or electrical method – a magnet or magnetic material is placed inside a solenoid connected to an alternate current(a.c) source and gradually withdrawn in the east-west(E-W) direction through a long distance or reducing the current gradually to zero and placing in the E-W direction.

2.Heating the magnet and allowing it to cool, lying in the E-W direction.

3.Hammering or rough handling in the E−W direction.

- Demagnitized materials are always kept in the E−W and not N−S direction, inorder to prevent induced magnetism from the earth's magnetic field which is oriented N−S direction.

- Breaking a magnet into portions does not destroy the magnet, but unlike poles appear on opposite sides on each portion.



- Induced magnetism or Magnetization induction – An unmagnetized iron piece brought close to the pole of a permanent magnet is magnetized and is attracted to the magnet. The pole of the magnet induces an opposite pole on the iron end near the magnet. When the magnet is removed, the iron piece loses its magnetism.

- Domain Theory of magnetism – It describes a magnetic material as having several parallel group of identical small magnets with like poles adjacent to each other. This forms open chains between the magnets and has a strong resultant magnetism. It describes a non-magnetic material as having several parallel group of identical small magnets arranged end to end or parallel with unlike poles adjacent to each other. This forms closed chains between the magnets and has no resultant magnetism outside it.

- Magnetic field – The region round a magnet where a magnetic force is felt.

- Magnetic flux or magnetic line of force – Imaginary curved lines along which magnetic force acts. Direction of a magnetic flux at any point is the path an imaginary North pole will take if placed at that point and free to move. **It spreads radially outwards for north pole and radialy inwards/towards a south pole i.e starts from N-pole and ends at S-pole and are closest where magnetic field is strongest.**

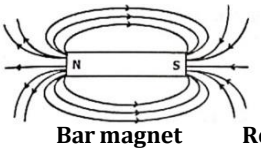
- Magnetic flux density or magnetic induction B – The magnetic flux that passes through a perpendicular area in the field. $\boldsymbol{B = \frac{\emptyset}{A}}$. It is measured in **tesla(T) or webers per meter square(Wb/m^2).**

- Neutral point – It is the point where the resultant magnetic flux density is zero or point where the horizontal component of the earth's flux density is equal and opposite to the magnetic flux density of a magnet.It is also the point where

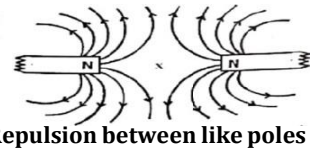
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the earth's magnetic field neutralizes the field due to the magnet.

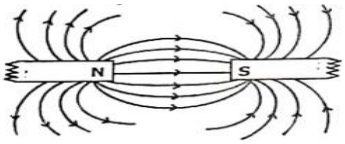
- Maximum density point – It is the point wherethere is a greater concentration of fileds in the same direction or the pont where the field due to the magnet is equal and in the same direction to the earth's magnetic field. It is always opposite the neutral point.



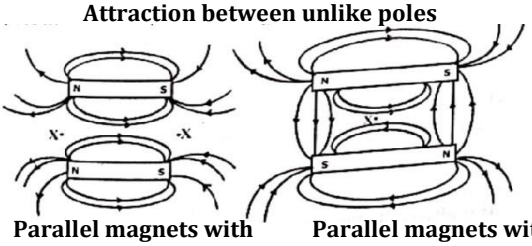
Bar magnet



Repulsion between like poles



Attraction between unlike poles



Parallel magnets with similar poles,with two neutral points X at the mid points(open chains, strong fields).

Parallel magnets with opposite poles,with one neutral point X at the mid points (closed chains, weak fields).

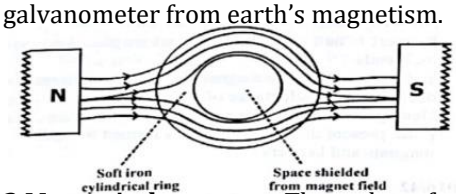
- Direction of magnetic fields – Magnetic compass needle and iron fillings can to determine the direction of magnetic field flux.

1.Iron fillings is used for less sensitive and strong fields e.g poles of a magnet and current carrying wires.

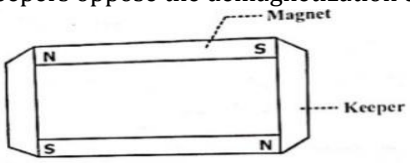
2.Magnetic compass needle is used for sensitive and weak fields e.g earth's field, fields between two magnets and points round a magnet but not the poles.

●Application of magnetic field –

1.Magnetic shielding/screening – It involves the use of soft iron rings(or any other magnetic material) to divert and protect an object placed within it from an unwanted external magnetic field. It is used to protect a moving coil galvanometer from earth's magnetism.



2.Magnetic keepers – The poles of a magnet tend to induce internal demagnetization field when kept alone or due to self demagnetization Demagnetization effects is greater,when the magnetic length(distance between poles) is shorter. Magnetic keepers are soft iron bars placed across the ends of bar magnets placed in pairs with adjacent unlike poles.Induced poles in the keepers oppose the demagnetization effect.



- Chatacteristic of magnetic field lines – 1. They are imaginary lines that do not cross one another. **2.**They begin from the north pole and end at the south pole of a bar magnet. **3.**They are lines which an imaginary N-pole will move if placed at that point.

- Classification of magnetic materials –

1.Paramagnetic materials – These are materials that sets its shortest axis perpendicular or its longest axis in line or parallel to the direction of a stroong uniform magnetic field. It tends to move from weaker to stronger parts of a non-uniform magnetic field.

2.Diamagnetic materials –These are materials that sets its shortest axis parallel or in line or its longer axis at right angles or perendicular to the direction of a non-uniform magnetioc field. It tends to move from stronger to weaker parts of a non-uniform field.

3.Ferromagnmetic materials – This the property of a material to aling with an external magnetic field or capable of being magnetized by weak magnetic fields.

- Current carrying conductors – A current carrying wire has a magnetic field around it. Compass needle placed close to a current-carrying wire is deflected and deflects in opposite direction if the current is reversed.

- The direction of magnetic field in a straight current carrying conductor is given by:

1.Right hand grip rule or Clenched fist rule – It states that if a wire is griped with the right hand with the tumb pointing in the current direction, the curled fingers points in the direction of magnetic field.

2.Maxwell'scockscrew rule – It states that if a right hand cockscrew turns such that its tip points in the current direction, the direction of rotation of the screw is the direction of the magnetic field.

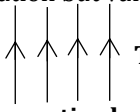
-The direction of the field in a current carrying wire can also be represented by clockwise direction for into,cross,downwards or southwards current and anticlockwise direction for onto,dot,upwards or northwards current.

NB – Clin C down south : Cl=clockwise, in=into, C=cross, down=downwards, south=southward. The opposite hold for Northwards or upwards current. **Anon D up north:** An=anticlockwise, on=onto, D=dot, up=upwards, north=northward.

- Earth's magnetic field – The earth behaves like it has a giant bar mangnet with its south pole at the geographic north and south pole at

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geophraphic north.Its magnetic field is a uniform field of parallel lines pointing northwards i.e direction,field strength or density is constant at a location but varies from place to place.



The Earth's magnetic field

- Magnetic element of the earth –

1. Magnetic meridian – It is a vertical plane about which a freely suspended magnet comes to rest or a vertical plane passing through the magnetic axis of a freely suspended magnet under the influence of the earth's magnetic field.

2. Geographic meridian – It is a vertical plane containing the geographic north and south poles of the earth.

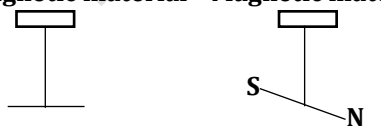
3. Angle of declination or variation – It is the angle between the geographic and magnetic meridian or axis or angle between the direction of the magnetic north and the direction of geographic north at a place. **Compass needle points to the direction of magnetic north.**

- If a compass gives a bearing of **Nθ°E**and the angle of declination at that place is **Nα°E**, the true bearing of the compass needle=**N($\theta+\alpha$)°E, same direction.** If a compass gives a bearing of $N\theta°E$ and the angle of declination at that place is $N\alpha°W$, the true bearing of the compass needle=**N($\alpha-\theta$)°E, opposite direction.**

4. Angle of dip or inclination – It is the angle between the direction of the earth's resultant magnetic field and the horizontal. It is measured by a **dip circle**. **Angle of dip is 0° at the magnetic equator and 90° at the magnetic pole.**

- Angle of dip is used to distinguish between magnetic and non-magnetic materials by suspending each at its mid-point. A freely suspended magnetic material comes to rest in the $N-S$ direction with its N-pole pointing downwards at angle of dip to the horizontal while a non-magnetic material will balance horizontally.

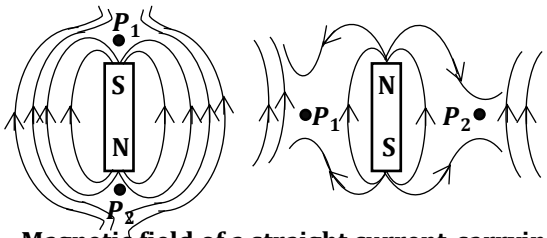
Non-magnetic material **Magnetic material**



5. Component of earth magnetic field – vertical component, $\boldsymbol{V = B\,sin\phi}$**,** horizontal component, $\boldsymbol{H = Bcos\phi}$**.** $\boldsymbol{\phi}$=angle of dip, B=magnetic flux density or magnetic induction. $\boldsymbol{tan\phi = \frac{V}{H}}$.

- Bar magnet in earth's magnetic field –Two neutral points are on the magnetic equator with N-pole of the magnet pointing northwards while two neutral points are on the magnetic axis with S-pole of the magnet pointing northwards.

P_1 and P_2 are the neutral points



- Magnetic field of a straight current-carrying conductor and horizontal component of earth's magnetic field – Both fields interact to produce one neutral points and one maximum density points depending on the current direction, which can be determined by clenched fist rule :

1.Southwards or downwards current gives clockwise,into,cross field with maximum density point due west and neutral point due east **i.e Clin C down south-west.**

2.Northwards or upwards current gives anticlockwise,onto, dot field with maximum density point due east and neutral point due west. **i.e Anon D up north-east.**

● Magnetic effect of electric current

- Magnetic effect of electric current – This is the production of temporary magnets or electromagnets from current. It explains how current behaves like a magnet by having poles and obeying the fundamental law of attraction or repulsion. The fields of a short coil or long coil(solenoid) is similar to that of a bar magnet and its polarity is determined by point through which current enters or leaves the coil i.e **The end through which current enters acquires south pole while the end through which current leaves acquires north pole.**

- Electromagnets – It is made when current is flowing through a coil wire wound round a U-shaped soft iron bar in opposite direction. **Soft iron is used in making electromagnets as the a strong magnetic field is produced and it is easily magnetized and demagnetized.** The magnetism remains as long as current is flowing.

-The strength of an electromagnet B increases with the number of turns in the iron bar N and current flowing in the coil I and increases as the poles are made closer.

- Electromagnets are used in electric bells, telephone receiver i.e earpiece,magnetic lifting devices and magnetic relays.

- Telephone earpiece has a permanet magnet in addition to the electromagnet.

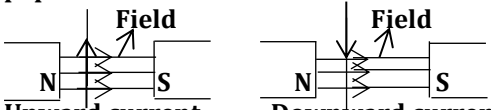
- Force on a current-carrying conductor – The force experience by a current carrying conductor is directly proportional to the strength of the magnetic field B, current in the conductor I, length of the conductor L and position of the conductor in respect to the field θ.

- The force is maximum when the current

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direction or conductor is perpendicular to the field and minimum or zero when the current direction or conductor is parallel or in the direction of the field. $F = BILsin\theta$, $F = BIL$**(maximum at 90°).**

- The direction of the force is determined by **Fleming's left hand rule or motor rule –** It states that if the thumb,forefinger and the middle finger of the left hand are mutually perpendicular **f**orefinger point in the direction of the **f**ield and the se**c**ond or middle finger pointing in the **c**urrect direction,then the thu**m**b points in the direction of the force or **m**otion of the conductor.
- Based on Fleming's left hand rule, the direction of the force or motion of the conductor is **into the paper for upward current and is out of the paper for downward current.**

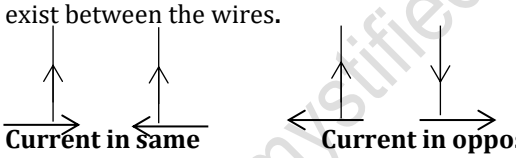


Upward current **Downward current**

- Force between two parallel current carrying wires – Each wire behaves like a magnet and experiences a force i.e concentric lines of force arounf each. They tend to move if not prevented from moving. **It is used in Ampere balance.**

1. Current flowing in the **same direction**, the **force is directed towards each other i.e attraction** and the field between the wires is weaker than the field outside. A neutral point exist between the wires.

2. Current flowing in **opposite direction**, the **force is directed away from each other i.e repulsion** and the field between the wires is stronger than the field outside. No neutral point exist between the wires**.**



Current in same direction **Current in opposite direction**

- The force acting between two parallel current-carrying wires is proportional to the product of their current and inversely proportional to the square of their distance apart.The force in one wire is caused by the current in the other wire.

$F \propto \frac{I_1 I_2}{r^2}$ or $F \propto \frac{I^2}{r^2}$(when equal current flows).

- Force per unit length in both wires is equal and opposite, $\frac{F}{L}$ **= constant.** $\frac{F_1}{L_1} = \frac{F_2}{L_2}$ **.**

Force per unit length, $\frac{F}{L} = \frac{\mu_o I_1 I_2}{2\pi r}$**.**

μ_o=magnetic permeability or permittivity in a vacuum=$4\pi \times 10^{-7}$henry per meter, Hm^{-1}.

- Current I_1 is due to magnetic field or flux density due to I_2 while current I_2 is due to magnetic field or flux density due to I_1 .
- Force due to current I_1, $F_1 = B_1 I_1 L_1$.
- Flux density or magnetic field due to current I_1, $B_1 = \frac{\mu_o I_2}{2\pi r}$.
- Force due to current I_2, $F_2 = B_2 I_2 L_2$.
- Flux density or magnetic field due to current I_2, $B_2 = \frac{\mu_o I_1}{2\pi r}$.

- Force on a current-carrying loop or a rectangular flat coil – Two equal and opposite forces acting on the arms of the coil constitute as couple, hence a torque and a rotational movement is produced. **The resultant force is zero.** The movement of the coil is towards or away from the N or S-poles of a magnet and in accordance with Fleming's left hand rule.

I. Maximum torque is produced with perpendicular arms and parallel planes.

II. No torque is produced with parallel arms and perpendicular planes.

- At the vertical position or when the arms of the coil is parallel to the magnetic field, the net force or toque is zero.
- The torque, τ on acting on the coil is given by – $\tau = BANIcos\theta$,B=magnetic flux density A=area of the coil=Lb, L= length of the coil, b=width of the coil/separation between forces, N=number of turns in the coil, I=current flowing through the coil, θ=angle between the magnetic field and plane of the coil.

- If θ= 0°, when plane of the coil is parallel with field and τ **is maximum,** $\tau = BANI$ **.**

- If θ=90° when plane of coil is perpendicular to field and τ **is zero.**

- Force on a moving charge within a magnetic field – Charges in motion experience a force when they enert a magnetic field.This force causes the charges to move in a spiral or circular path. $F = qvBsin\theta$**.** q=charge, v=velocity, θ=angle between v and B.

1.F is maximum when charge enters at 90° to the field i.e θ=90°. The charge moves in a circle in this case.

2.F is zero when charge enters in the direction of the field or parallel to the field i.e θ=0° or 180°. The charge moves in a straight line.

3.F is zero when the charge is stationary i.e v=0.

4.F deflects charges with no change in speed or kinetic energy.

- The radius of the circular path through the charge moves is given by – $r = \frac{mv}{qB}$

r=radius of cicular path,m=mass of charge.

- The period or time taken for the charge q of mass m to complete one orbit T in the circular path is given by – $T = \frac{2\pi r}{v} = \frac{2\pi m}{Bq}$ **.**

- Electric motor – It converts electrical energy (electric current) to mechanical energy (rotational energy). It consists of a permanent magnet,carbon brushes,armature and a split-ring commutator.

- As the plane of the coil passes through the

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vertical position, the split-ring commutators changes contact from the carbon brushes, thereby reversing the direction of the current and force inorder to maintain a continual clowkise rotation of the coil.

- The split-ring commutator – **i.**reverses the direction of current **ii.**reverses the direction of the forces acting on the side of the coil **iii.**ensures that the coil keeps rotating in the same direction or maintains the direction of rotation of the coil.

- Energy loses in electric motors – 1. Heat losses by current and resistance($H = I^2R$) in the windings of the coil. **2.** Eddy current in the soft iron armature.It is reduced by laminating the iron cylinder. **3.**Workdone against friction in the commutator or bearings.

- Moving coil galvanometer – It is used to detect and measure small direct current(d.c) only and cannot measure alternate current (a.c). It has a regular or uniform scale provided by a radial/uniform magnetic field, not affected by stray magnetic fields due to its stron field and can be adapted to read ranges of current and p.d, by fitting a suitable resistance.

- It consists of – **I.Control hair spring;** provides control couple for the coil and compensates for thermal expansion. **II.Soft iron amature and magnet;** provides a radial magnetic field. **III.Pointer;** attached to the armature to measure the angle of rotation of the armature which is proportional to the current, $\boldsymbol{\theta \propto I}$**.**

- Factors increasing the sensitivity of a moving coil galvanometer –

1.The permanent magnet is made of a special alloy to give a strong magnetic field or high magnetic flux.

2.Large number of turns and area in the coil

3.Control hair spring is made weak and thinner to give a small control couple and light pointer.

-In summary sensitivity , $\boldsymbol{S \propto BAN}$**.**

- Factors increasing the sensitivity of a suspended or reflecting/optical lever galvanometer –

1.Thinner suspension with small torsional control, such that a small deflection of the coil provides a large movement of a pointer(light beam).

2.Large number of turns and area in the coil.

3.Stronger magnetic field or magnetic flux.

- Current sensitivity of a galvanometer S_I :

$\boldsymbol{S_I = \frac{\theta}{I} = \frac{BAN}{K}}$, θ=angle of rotation or deflection of pointer, I=current, B=magnetic flux density, A=area of the coil, N=number of turns in the coil, K=force or stiffness constant of the spring.

- Voltage sensitivity of a galvanometer S_V :

$\boldsymbol{S_V = \frac{\theta}{V} = \frac{\theta}{IR} = \frac{BAN}{KR}}$, V=voltage, R=resistance.

-Conversion of a galvanometer/milliammeter to an ammeter – A **low resistance shunt** is connected in **parallel** to the galvanometer.

$\boldsymbol{I = \frac{i(S+G)}{S}}$**.** I=larger current to be measured, G=resistance of galvanometer, i=current in the galvanometer S=resistance of the shunt.

- Conversion of a galvanometer/ milliammeter to a voltmeter – A **high resistance multiplier** is connected in **series** to the galvanometer. $\boldsymbol{V = i(M + G)}$**.** V=voltage to be measure, M=resistance of the multiplier.

- Moving iron ammeter – It is used to measure both d.c and a.c, it gives the root mean square values of a.c. Its deflection is proportional to the square of the current $\boldsymbol{\theta \propto I^2}$**,** hence it has a non-linear(curved) or non-uniform scale. It is affected by stray magnetic fields.

● Electromagnetic induction –

- Electromagnetic induction – It is the production of electric current or emf(voltage) from a magnet(without a cell or battery) due to relative motion between a conductor and a magnet or when a conductor is placed in magnetic field of varying flux density.

- Induced current or emf is produced is a coil whenever there is a change in the magnetic lines of force i.e flux linkage linking the coil.

-No current is induced when the conductor and the magnet are both stationary or moving in the same direction with the same speed.

- The magnitude induced emf or current depends on number of turns of the coil, strength of the magnet,relative speed between the magnet and the coil,area of cross-section of the coil and presence of a soft iron core inside the coil.

$\boldsymbol{E \propto \omega BAN}$**.**

- Total flux linkage, $\boldsymbol{\phi = BAN cos\theta}$**.** Magnetic flux linking a coil depends on magnetic field strength B, area of each turn A and the number of turns in the coil N.

- Laws of electromagnetic induction –

1.Faraday's law of eletcromagnetc induction or Neuman's law – It states that a changing or varying magnetic flux linking a coil or circuit, induces an emf the magnitude of which is proportional to the rate of change of magnetic flux linking. i.e

$\boldsymbol{E \propto \frac{d\phi}{dt}}$, $\boldsymbol{E = N\frac{d\phi}{dt} = \frac{N(\phi_2 - \phi_1)}{t} = \frac{NA(B_2 - B_1)}{t}}$**.**

E=induced emf, $\frac{d\phi}{dt}$=rate of change of magnetic flux(weber per second, wbs^{-1}), N=number of turns, A=area of the coil, B=magnetic field density.

2.Lenz's law of electromagnetic induction – It states that the induced emf or current opposes the motion/change producing it. **Lenz's law is based on conservation of energy (mechanical to electrical energy).**

- The direction of the induced emf in a straight

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conductor is determnined by **Fleming's right hand rule or dynamo rule –** It states that if the thumb,forefinger and middle finger of the right hand are mutually perpendicular, with the **f**orefinger pointing in the direction of the **f**ield, and thu**m**b pointing in the direction of motion, then the **m**iddle finger will point in the direction of induced **e**mf or current.

- The magnitude of the induced emf is given by, $\boldsymbol{E = BLv sin\theta}$ or $\boldsymbol{E = BLv}$ **(maxiumum at 90°).** E=induced emf, L=length of the conductor, v=velocity of conductor.

I. E is maximum when L and v are both perpendicular to the field B and **E is zero when L or v is parallel to the field B.**

- Electromagnetic generator – They operate on electromagnetic induction. **A generator converts mechanical energy to electrical energy. It requires an external supply of energy to rotate the coil. D.C generators** are called **dynamos** while **A.C generators** are called **alternators.**

- An a.c generator can be converted to a d.c generator or electric motor by replacing the slip rings in a.c with split ring or commutator.

-An a.c generator and d.c generator both have two carbon brushes,armature,field magnet in common, but **an a.c generator hasslip rings while d.c generator has split rings or commutators.**

- Comparism between electric motors and generators – 1. Current is supplied by a cell to rotate the coil in electric motor while current is produced or induced in a generator due to rotation of the coil **2.**Electric motor convert electrical to mechanical energy while generator convert mechanical to electrical energy **3.**Electric motors operate on the force on a current carrying conductor while generator operates on electromagnetic induction **4.**Electric motors is based on Fleming's left hand rule(motor rule) while generators is based on Fleming's right hand rule(dynamo rule) **5.**Electric motors and generators are of the same structure except that no cell in the external circuit of a generator as current is induced or produced not supplied.

- When the coil of an electric motor rotates, it functions as a d.c generator(dynamo) and an emf known as **back emf E_B** is induced in the coil which opposes the emf supplied E by the cell. $\boldsymbol{E_B = \frac{2}{\pi}\omega BAN}$.

- The energy stored in the loop of the coil, W is given by – $\boldsymbol{W = I(E - E_B)t}$. I=current flowing through the coil, t=duration of current flow.

- The instantaneous or effective value of induced emf in a coil of N number of turns is given by – $\boldsymbol{E = E_o sin\omega t}$, E_o=amplitude or peak or maximum induced emf, $\boldsymbol{E_0 = \omega BAN}$, E=instantaneous emf.

- Transformers – It is used to increase(step-up) or decrease(step-down) an alternating emf or voltage from an alternating current mains. **It does not work with a dry cell i.e steady or direct voltage as direct current will not produces changes in magnetic flux.**

- Transformers operates on the principle of mutual inductance and obeys the principle of conservation of energy. When a.c current flows through the primary coil P, a varying magnetic flux is set up which induces an a.c emf in the secondary coil. The emf on each winding is proportional to the number of turn of the coil and inversely proportional to the current, $\boldsymbol{E \propto \frac{N}{I}}$, $\boldsymbol{\frac{E_S}{E_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}}$

- In a **step-up transformer**, number of turns in the secondary coil is greater than that in primary coil and **steps-up voltage or emf but steps-down current, $\boldsymbol{N_S > N_P, E_S > E_P, I_S < I_P}$.**

- In a **step-down transformer**,number of turns in the secondary coil is less than that in primary coil and **steps-down voltage or emf but steps-up current**, $\boldsymbol{N_S < N_P, E_S < E_P, I_S > I_P}$.

- Turn ratio N_R is the ratio of the number of turns in secondary coils to number of turns in primary coil. $\boldsymbol{N_R = \frac{N_S}{N_P}}$ **. In a step-up transformer, $N_R > 1$, while in a step-down transformer, $N_R < 1$.**

- Efficiency $\epsilon$of a transformer is the ratio of the power in the secondary coil (power output) to the power in the primary coil (power input). **Efficiency** $= \boldsymbol{\frac{Power\ output}{Power\ input} \times 100}$**,**

$\boldsymbol{\epsilon = \frac{P_o}{P_i} \times 100 = \frac{I_S E_S}{I_P E_S} \times 100}$**.**

- The efficicency of a transformer is not 100% due to energy loses by :

1.Eddy current loses – Enegry loses in form of heat due to eddy current set up in the core by varying magnetic flux. **It is reduced or minimized by laminating the iron core i.e iron core made of insulated sheets rather than a solid block.**

2.Joule effects or Copper loses(I^2Rloses) – Energy loses due to heat produced in the primary and secondary coils as a result of their resistance. **It is minimized by using a low resistance or thick copper coils in the primary and secondary coils.**

3.Magnetic flux leakage or Flux design loses – Energy loses due to leakage of the magnetic flux between the primary and the secondary coils or due to least flux linkage between the primary and secondary coil. **It is minimized by using an effective core design or special form of coil winding.**

4.Hysteresis or lag response loses – Energy loses due to magnetic reversals i.e reversing

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magnetization of core. **It is minimized by the use of soft iron cores or the use of special alloys in the core of the primary coils.**

- Power transmission – Power transmission through cables can either be high tension or low tension transmission.

1.High tension transmission(HTT) – It is the transmission of **alternate current over long distance** at **high voltage** and **low current,** with a **minimum energy loss.** At the point the point of use or consumption, the voltage is stepped down by a transformer.

2.Low tension transmission(LTT) – It is the transmission of **direct current over short distance** at **low voltage and high current,** with a **maximum or high energy loss.**

- Power transmission is done with a.c rather than d.c because, a.c voltage can be stepped up or down by transformer, heat(I^2R) loses is reduced,a.c is cheaper and easily produced.

- Eddy current or Foucault current – Induced current in a conductor resulting from a varying magnetic flux linked to it. It is also produced from back emf induced in a coil mounted a metal frame. e.g electric motor acting as a dynamo when rotating and current-carrying coil mounted on metal frame

- Eddy currents cause energy losses in form of heat and it can be minimized by laminating the soft iron core.

- Eddy current is applied or used in – **i.**Pointer of moving coil galvanometer **ii.**Brakes of large electric motors **iii**.Sensitive mass balances **iv.**Speedometers of automobiles **v.**metal detectors **vi.**Induction furnance .

- Carbon microphone – It is a device which converts sound energy to electrical energy. It is used in a telephone mouthpiece system. It consists of: **i.**carbon blocks whose resistance varies with change in pressure,hence a change in current. **ii.**a diaphragm which vibrates incoming sound waves.

- Induction coil or Ruhmkorff coil – It is a transformer for stepping up direct current (d.c) i.e producing a very high intermittent emf from a low emf d.c source by electromagnetic induction.

- It consists of **– 1.**Primary coil is of few turns and thick wire while secondary coil is of many turns and thin wire,both are wound on the same core. This ensures that a large output emf is formed, $\boldsymbol{N \propto \frac{E}{th}}$. N=number of turns, E=emf, th=thickness.

2.Make-and-break device in the primary coil which gives a rapid decaying or varying magnetic flux which induces a large voltage pulse in the secondary **3.**Capacitor in the primary which prevents sparks in the make-and-break device so as reduced back emf effect generated in the primary.It also promotes a more rapid decaying magnetic flux to induce a larger secondary emf.

- The strength of secondary current is directly proportional to the rate of change of the primary current ($\boldsymbol{I_S \propto \frac{dI_P}{dt}}$), for circuits with primary and secondary coils e.g transformers and induction coils.

- Induction coil is used in motor car ignition system and operation of X-ray tubes.

- Inductance – Inductance is the property by which a varying current in a coil induces an emf in the coil. It is the property of material to oppose changes in current through it.

It is of two types: Self inductance and Mutual inductance.

1.Self inductance L– It is the production of induced emf in a coil due to change in the magnetic flux in the coil by a changing or alternate current in the coil itself.

$\boldsymbol{E = L\frac{dI}{dt}}$, L=self inductance, $\frac{dI}{dt}$=rate of change of current, E=induced emf.

- Flux linkage per current $\boldsymbol{L = \frac{N\emptyset}{I}}$.

2.Mutual inductance M – It is the production of induced emf in a coil due to change in the magnetic flux by a an alternating or changing current in a neighbouring or another coil.

$\boldsymbol{E = M\frac{dI_A}{dt}}$, E=induced emf in secondaery coils, $\frac{dI_A}{dt}$=rate of change of flux in the primary varying coil.

- Inductors : Inductors are the coils in which emf is induced. They are used in electric circuits to prevent changes or variation in current through the circuit. **The unit of inductance is Henry (H).**

● Arrangement of inductors –

I.Series combination – $\boldsymbol{L = L_1 + L_2 + L_3}$

II.Parallel combination – $\boldsymbol{\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}$.

$\boldsymbol{L = \frac{L_1 L_2}{L_1 + L_2}}$ (for two inductors only).

- Energy is stored in an inductor as work is done against back emf and this energy is drawn from the battery. $\boldsymbol{E = \frac{1}{2}LI^2}$.

Examples :

1.Which of the following about magnets is correct? A freely suspended magnet points in an east west direction B.Brass is a magnetic material C.Like poles attract each other **D.Permanent magnet are made of steel** D.Unlike poles repel each other.;

A freely suspended magnets points in a N-S direction. Brass is not a magnetic material. Like poles repel while unlike poles attracts.

2. Which of the following is true of magnetism? **A.Iron filings cling mainly round the ends of a bar magnet** B.The freely suspended bar magnet comes to rest in the geographic north-south direction C.like poles attract D.Loadstone is a non-magnetic oxide.

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The poles of a magnet are the points where the magnetic effect is strongest. Freely suspended magnets balances in the magnetic N-S direction but at an angle to the geographic N-S. Loadstone is a natural magnet and a magnetic oxide of iron.

3. Which of the following statements is not correct? A.soft iron rather than steel is always used in making electromagnets B.steel is more difficult to magnetize than soft iron **C.steel loses its induced magnetism more quickly and more easily than soft iron** D.the distance between the poles of a bar magnet is called the magnetic length E.nickel, iron and steel are magnetic materials.

Soft iron is easily magnetized and loses its induced magnetism more quickly than steel.

4. A magnet is similar to a bi-polar charged system in the following respects – I.It has two opposite poles- the N and S poles II.The earth's magnetic N-pole is close to the geographic S-pole and the earth's magnetic S-pole is close to the geopraphic N-pole III.The force of attraction or repulsion between opposite or like poles obeys a law identical to Coulomb's law for electric charges. Which of the above statements is/are true? A.I B.II **C.I and III** D.I,II and III.

Magnet has two opposite poles – N-pole and S-pole i.e similar to a bipolar charged system of positive and negative charges. Like poles repel and unlike poles attract i.e similar to Coulombs law for electrostatic charges.

5. The place in a magnet where the resultant attractive force appears to be concentrated are called the A.curvatures **B.poles** C.bars D.lodestone.

6. Which of the following methods cannot be used to convert a bar into magnet? I.Passing of an electric current through a solenoid II.Repeated stroking of the steel bar with a magnet III.Heating of the steel bar IV.Hammering the steel bar in the earth's magnetic field A.I and II B.III and IV C.I,II and III **D.I,II and IV.**

Heating of magnet or magnetic material demagnetizes it. Hammering in a magnetic material in the earth's magnetic field i.e in the N-S direction, magnetizes it.

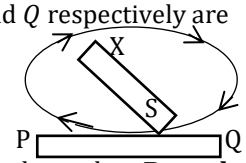
7. Which of the following methods will effectively demagnetize a bar magnet? I.Passing an electric current through the magnet II.Bringing its S-pole in contact with the S-pole of a strong bar magnet III.Heating the magnet A.I B.III **C.I and III** D.I,II and III.

Statement II will increase the resultant magnetism and not demagnetize it.

8. When a number of identical small magnets are arranged in a line, the strength of the resultant magnet A.is largest when they are arranged end to end **B.is greatest when they are arranged parallel with like poles adjacent to each other** C.depends only on the number of magnets provided they are parallel D.is greater when the magnets are arranged on a wooden surface than on a metal surface.

This statement is in accordance with domain theory of magnetism. Magnets placed in parallel with like poles adjacent to each other results in a strong resultant magnetism.

9.The diagram shows a magnet X with its south pole moved along a soft iron bar PQ in the direction as shown. After some time, the poles at P and Q respectively are



A.north-north **B.north-south** C.south-north D.south-south.

The end of the iron bar PQ last touched acquires opposite pole while the end first touched acquires a similar pole as the stroking pole. Hence, P has N-pole as it is the end last touched while Q has S-pole as it is the end first touched.

10. The direction of the magnetic field at a point in the vicinity of a bar magnet is A.always towards the north pole of the magnet B.always away from the south pole of the magnet C.along the line joining the point to the neutral point **D.in the direction the north pole of a compass needle would point.**

The direction of magnetic field is towards the S-pole and away form the N-pole.

11. Which of the following statements about magnetic lines of force is not correct? A.a magnetic line of force is an imaginary line which the north pole of a magnet would describe if it is free to move B.magnetic lines of force do not cross one another **C.magnetic lines of force are closely-packed together at neutral points** D.the presence of magnetic lines of force in a region indicates the presence of magnetic field.

Magnetic lines of force are closest together where the magnetic field is strongest and farther apart at neutral point i.e where the magnetic field is completely neutralized.

12. A magnetic field is said to exist at a point if a force is A.deflected at the point B.straightened at the point **C.exerted on a moving charge at a point** D.exerted on a stationary charge at a point Charges in motion has a magnetic field around it.

13. At different locations on earth's surface, the earth's magnetic field is **A.different in magnitude but not in direction** B.the same in magnitude and direction C.different in both magnitude and direction D.the same in magnitude but different in direction

The earth's magnetic field is a uniform field which always comprises of parallel lines pointing northwards at different locations. The strength

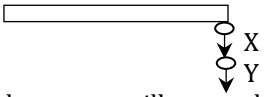
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and magnitude of the field is constant in a particular location but varies from one place to the other.

14. The name given to the characteristic behavior of a substance which set wit their length perpendicular to a strong magnetic field instead of in line with it is A.paramagnetism **B.diamagnetism** C.ferromagnetism D.demagnetization E.magnetization

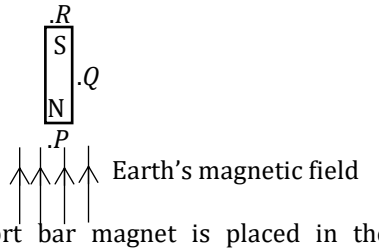
Diamagnetic materials set its length or longer axis perpendicular or its shorter axis parallel to a strong uniform magnetic field. Paramagnetic materials sets its length or longer axis in line or parallel or its shorter axis perpendicular to a strong uniform magnetic field.

15.



The bar magnet illustrated in the diagram above picks up an unmagnetized ssteel nail X. In turn, X picks up a similar nail Y and so on. The nails are said to be magnetized by A.conduction B.single touch **C.induction** D.divided touch.

16.



A short bar magnet is placed in the earth's magnetic field as shown in the above diagram. Where is/are the possible neutral positions of the neutral point? A.P B.R C.Q **D.P and R.** Neutral points exists at a point where the field of the bar magnet is equal and opposite to the earth's field(acting upwards). The field lines of the bar magnet are directly downwards i.e opposite to the earth's field lines, at point P close to N-pole and at point R close to the S-pole.

17. The magnetic meridian at a place is the A.plane at right angles to the direction of the earth's magnetic flux B.plane containing the longitude of the place and the earth's axis C.horizontal plane containing the latitude of the place D.direction in which a dip needle set when placed at that place **E.vertical plane containing the magnetic axis of a freely suspended magnet under the action of the earth's field**

Magnetic meridian of a place is also a plane containing the magnetic N-S line.

18. The angle between the geographic and the mahentic meridians at a point on the magnetic equator is the A.angle of inclination B.angle of dip **C.angle of declination** D.longitude of the point.

Angle of declination or variation or magnetic delcination is zero at the geographic north-pole.

19. The reading of a magnetic compass at the N-pole would not be useful because A.the magnetic field at the N-pole is very weak **B.the magnetic field at the N-pole is very strong** C.the angle of declination at the north pole is zero D.the extremely low temperature affects the magnetism of the compass needle E.the magnetic field at the north pole is vertical.

Compass needle always points in the direction of magnetic north. Magnetic compass is used for plotting the direction of weak and sensitive fields rather than strong fields at the poles of a magnet.

20. A sailor observes that his mariners' compass reads N10°W at a place where the angle of variation is N15°E. Calculate the true bearing of the place. A.N25°W B.N5°W C.N25°E **D.N5°E.**

θ=angle of declination or variation=N15°E,
α=compass bearing=N10°W,
True bearing = N(α−θ)°E = N(15−10)°E
True bearing = N5°E.

21. At a location on earth where the declination is 15°E, a compass needle indicates the direction N50°E. The true geographic bearing is **A.N65°E** B.N50°E C.N35°E D.N15°E**.**

θ=angle of declination or variation=15°E,
α=compass bearing=$N60°E$,
True bearing =N(θ+α)°E=N(60+15)°E=N65°E.

22. The angle between the direction of the earth's magnetic field and the horizontal is called the A.angle of deviation **B.angle of dip** C.magnetic declination D.magnetic meridian.

23. The angle of dip varies from **0° at the equator to 90° at the poles**. This can be explained by the fact that A.lines of force are horizontal at the equator and vertical at the poles B.lines of force are vertical at the equator and horizontal at the poles **C.the north pole of a magnet dips downwards at the equator** D.a magnet freely balanced at its centre of gravity will set horizontally at the poles E.the south pole of a magnet dips downwards at the equator.

24. Which of the following statements is not correct? A.A magnetic field is a region in which a magnetic force can be detected B.A line of force is a line along which a magnetic north pole would tend to move if it wre free **C.Magnetic fields are scalar quantities** D.Neutral points are obtained where the earth's magnetic field is exactly equal and opposite to that due to a magnet E.The dip angle is the angle between the horizontal and the earth's total magnetic field.

Magnetic field and all other fields are vector quantities.

25. The anle of dip at Ibadan, Nigeria is 6°S. This means that if a bar magnet is freely suspended at its centre of gravity, its **A.south pole will point towards the earth at an angle of 6° to the horizontal** B.magnetic axis would make an angle of 6° with the earth's magnetic axis C.south pole would point towards the earth at an angle of 6° to the vertical D.north pole would point towards the

Demystified Series by Dr Timothy

earth at an angle of 6° to the vertical.

26. If the total earth's field is Q at a place where the angle of dip is θ, then the horizontal component of the field is A.$Qsin\theta$ **B.$Qcos\theta$** C.$Qtan\theta$ D.Q E.$Q \div cos\theta$.

Option A represents the vertical component of the earth's field.

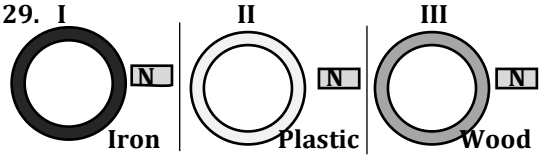
27. The horizontal component of the earth's magnetic fields is I.zero at the magnetic equator II.zero at the magnetic poles III.maxiumum at the magnetic equator IV.maximum at the magnetic poles. Which of the following is/are correct? A.I B.II C.II and III D.I and IV.

Angle of dip, φ is 0° at the equator and 90° at the poles. Horizontal component, $H=Bcos\varphi$.

At the equator,$H=Bcos0°=B \times 1=B$. i.e maximum value. At the poles, $H=Bcos90° = B \times 0=0$ i.e minimum value. The earth's magnetic field is entirely vertical at the magnetic pole and entirely horizontal at the magnetic equator.

28. In storing magnets, keepers are used to **A.reduce self demagnetization** B.cancel the effect of the earth's magnetic field C.protect the magnet from stray electric fields D.increase the strength of the magnets.

29.



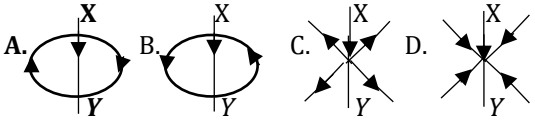
A magnet is placed successively near the rings made of the materials indicated in the diagram above. The polarity of the magnet is also indicated. Inside which of therings will a magnetic field be observed? A.I B.III C.I and III **D.II and III** D.I,II and III.

Magnetic fields are screened or shielded by diverting the field lines away from the rings by a magnetic material e.g soft iron rings.Magnetic fields lines will pass through non-magnetic field materials e.g wood and plastic.

30. A plotting compass is placed near a I.stationary positive charge II.bar magnet III.wire carrying an electric current, In which of the above would a deflection of the compass occur? A.III B.I and II C.I and III **D.II and III** E.I,II and III.

Magnetic compass is plotted to detect magnetic field in a region and will deflect in presence of magnetic field. Bar magnets, current-carrying wires and charges in motion(not stationary) experience magnetic field.

31. Which of the following diagrams shows the correct direction of the magnetic lines of force of a vertical wire XY carrying a current I?



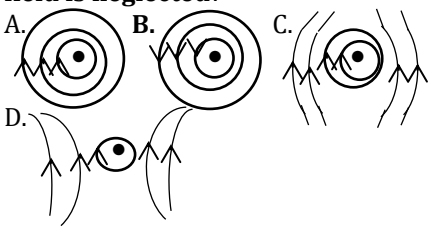
Applying the clenched fist rule or Maxwell's cockscrew rule, downward or southward current gives a clockwise field i.e option A.ALternatievly, applying the mnemonic: Clin C down south :Cl=clockwise, in=into, C=cross, down=downwards, south=southward i.e clockwise field is produced from southward or downward or into the plane, cross current.

32. Which of the following statements is not true of the magnetic field around an infinite straight current carrying conductor A.the lines of magnetic flux are in concentric circles B.the lines lie in a plane perpendicular to the conductor C.the strength of the magnetic field depends on the current flowing in the conductor **D.the lines of magnetic flux are parallel to each other in the plane of the conductor** E.the magnetikc flux decreases as the perpendicular distance from the conductor increases

The magnetic lines of force or lines of magnetic flux in a straight current-carrying conductor are perpendicular(not parallel) to the plane of the conductor.

33. A current-carrying conductor experiences a force when placed in a magnetic field because the A.conductor is magnetized **B.magnetic field of current interacts with external magnetic field** C.force due to motor principle D.electric field of current interacts with external magnetic field.

34. Which of the following diagrams correctly illustrates the magnetic field pattern around a straight conductor which carries current **out of the planeof the paper,if the earth's magnetic field is neglected**?



Applying clenched fist rule or Maxwell's cockscrew rule, current out of the plane of the paper is same as upward current and will give an anticlockwise field. i.e option B, the magnetic field of the current carrying wire **alone.** Alternatively, Applying the mnemonic: Anon D up north:An=anticlockwise, on=onto, D=dot, up=upwards, north=northward i.e anticlockwise field is produced from northward or upward or onto(out of) plane, dot current.

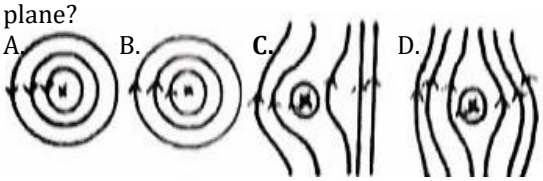
- The earth's magnetic field is neglected, hence no neutral or maximum density point is formed i.e option C and D is not correct.

35. The magnetic field around a straight wire X carrying current **into the plane of the paper interacts with the horizontal component of the earth's magnetic field**. Which of the following diagrams correctly illustrates the resultant magnetic field pattern in the horizontal

Demystified Series by Dr Timothy

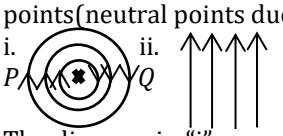
plane?

A. B. **C.** D.



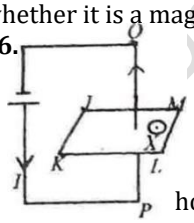
A straight wire carrying current into the plane of the paper, cross, is same as downward or southward current, and will give a clockwise field(According to clenched fist rule or Maxwell's cockscrew rule). On super-imposing or interacting with the earth's magnetic field, at some point both fields points in the same direction(maximum density point due west) and points in opposite direction at some other points(neutral points due east) i.e Option C.

i. ii.



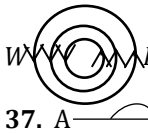
The diagram in "i" represents the magnetic field due to the straight current-carrying wire only, carrying current into the plane of the paper. On interacting with the earth's magnetic field "ii", at point P, the magnetic field direction due to the wire(moving upward) is in the same direction as the earth's magnetic field(moving upward also) hence, point P which is due west is the maximum density point.At point Q, the magnetic field direction due to the wire(moving downward) is in opposite direction to the earth's field(moving upward) hence, point Q which is due eastis the neutral point. Alternatively, applying the mnemonic: Clin C down south-westi.e Southwards or downwards current gives clockwise,into,cross field with maximum density point due west and neutral point due east. Anon D up north-east i.e Northwards or upwards current gives anticlockwise,onto, dot field with maximum density point due east and neutral point due west.

- A current carrying wire behaves like a magnet whether it is a magnetic material or not.

36. 

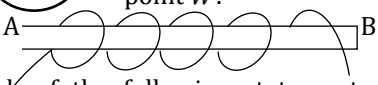
The figure shows a straight vertical conductor PQ which carries a current I in the direction indicated X is a compass mounted on a horizontal cardboard east of the conductor. Neglecting the earth's magnetic field, the compass needle will point **A.northward** B.southward C.eastward D.westward.

The diagram shows a notthward or upward current, hence it will give an anticlockwise field. At point X which is due east E ,of the conductor, the magnetic field direction is pointing due North or upwards while at point due west W, of the conductor, the magnetic field direction is due South or upward.



At point E, field lines points due North and points due South at point W.

37. A B



Which of the following statements about the electromagnet shown above is correct? A.when the current flows from A to B, the end A will be in north pole **B.when the current flows from *B* to *A*, the end *B* will be south pole** C.when the current flows from B to A, the end B will be north pole D.when the alternating current is passed through the coil, the end A will remain a north pole E.the polarity of the electromagnet cannot be determined by the direction of current.

The end through which current enters (starting end) acquires south pole while the end through which current leaves(terminating end) acquires north pole. Hence, if current flows from A to B, end A acquires S-pole while end B acquires N-pole. But if current flows from B to A, end A acquires N-pole while end B acquires S-pole.

38. An equipment that functions as a result of the magnetic effect of electric current is A.rheostat B.thermostat C.carbon microphone **D.electric bell.**

39. The magnetic relay is a device for A.timing in sports B.storing magnetic field C.reading the magnetic flux **D.controlling another circuit carrying larger current.**

Magnetic relay devices are used as control switches.

40. A motor effect occurs when A.an electric current flowing in a conductor stes up a magnetic flux round the conductor B.the flux from a permanent magnet produces a magnetic force on a conductor **C.a current carrying conductpor in a magnetic field experiences a force** D.a current carrying conductor is moved parallel to the magnetic flux from a permanent magnet E.the flux from a conductor carrying a current nullifies the flux from a permanent magnet.

Motor effect is same as the force on a current carrying conductor. It is experienced when a current carrying wire is placed in a strong magnetic field.

41. A free current carrying conductor will move when it is placed in a magnetic field. The force causing motion depends on I.the mass of the conductor II.the strength of the magnetic field III.the length of the conductor IV.the strength of the current A.I,II and III B.I,II and IV **C.I,III and IV** D.II,III and IV E.I,II,III and IV.

The force on a current carrying conductor depends on the strength of the magnetic field B, the strength of the current I and the length of the conductor L. i.e $F = BILsin\theta$. NB: The force is greatest or maximum when θ=90° and minimum or zero when θ=0° or 180°.

Demystified Series by Dr Timothy

42. The direction of motion of a current-carrying conductor placed within the opposite poles of a magnet is determined using **A.Fleming's left-hand rule** B.Fleming's right-hand rule C.Maxwell's cockscrew rule D.Ampere's rule.

43. A wire carrying current of $10A$ and 2.5m in length is placed in a field of flux density $0.14T$, what is the force in the wire if it is placed 60° to the field? **A.3.03N** B.20.5N C.15.3N D.10.5N .

B=0.14T, I=10A, L=2.5m, F=? θ=60°,

$F = BILsin\theta = 0.14\times10\times2.5\ \sin60° = 3.03$N.

44. A conductor of length 2m carries a current of $0.8A$ while kept in a magnetic field of magnetic flux density $0.5T$. The maximum force acting on it is A.0.2N **B.0.8N** C.3.2N D.8.0N.

B=0.5T, L=2m, I=0.8A, F=?, Maximum force, $F = BIL = 0.5\times0.8\times2 = 0.8$N.

45. A current of $5A$ passes through a straight wire in a uniform magnetic field density$2.0\times10^{-3}T$. Calculate theforce per unit exerted on the wire when it is inclined at 30° to the field. A.0.0025Nm^{-1} B.0.0005Nm^{-1} **C.0.005Nm^{-1}** D.0.00025Nm^{-1}.

I=5A, B=$2.0\times10^{-3}T$, θ=30°, $F = BILsin\theta$,

Force per unit length, $\frac{F}{L} = BIsin\theta$

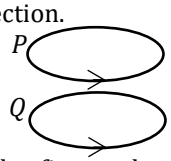
$\frac{F}{L} = 2.0\times10^{-3}\times5\ \sin30° = 0.005$Nm^{-1}.

46. The force experienced by a current-carrying conductor moving in an electric field is employed in the working of the I.moving coil ammeter II.electric bell III.electric motor A.I and II B.II and III **C.I and III** D.I,II and III.

47. Which of the following statements is not correct about two parallel conductors carrying equal currents in the same direction A.each of the conductors will move if not prevented from doing so B.each of the conductors will experience a force C.the forces on the conductor will be equal D.there are concentric lines of force around each conductor **E.the two conductors will repel each other** .

For two parallel current-carrying conductors placed in a magnetic field, repulsion occurs when current flows in opposite direction while attraction occurs when current flows in the same direction.

48.



In the figure above, P and Q are two parallel circular coils, each carrying current as shown. Then A.P and Q repel each other **B.P and Q attracts each other** C.P moves, Q stays still D.Q moves, P stays still E.P and Q both turn around .

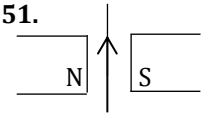
The diagrams shows that current flows in the same direction in coil P and Q which are parallel to each other hence, they will attract each other.

49. Two long parallel wires X and Y carry currents $3A$ and $5A$ respectively. If the force experienced per unit length by wire X is $5\times10^{-5}N$, the force per unit length experienced by wire Y is **A.5×10^{-5}N** B.5×10^{-4}N C.3×10^{-6}N D.3×10^{-5}N .

The force per unit length, $\frac{F}{L}$ of two parallel current-carrying wires are equal and opposite.

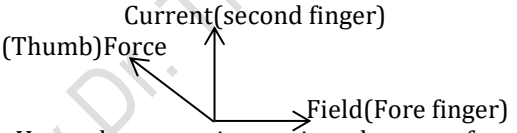
50. The flux density, B at a distance r from a long straight wire carrying a current, I is given by the A.$\mu_0 I/2r$ B.$\mu_0 I/\pi r$ C.$\mu_0 I/\pi r^2$ **D.$\mu_0 I/2\pi$.**

51.



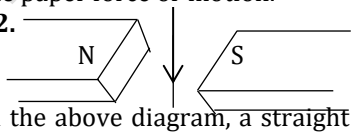
The diagram shows a current -carrying copper placed between the horse-shoe magnet The wire will tend to move A.out of the paper **B.into the paper** C.towards the N-pole D.towards the S-pole.

Applyging Fleming's left hand rule – The fore-finger pointing in the direction of the field i.e due east from N-S , the second or middle finger points in the direction of the current i.e upward, then the thumb points in the direction of the force or motion of the conductor i.e into the paper.



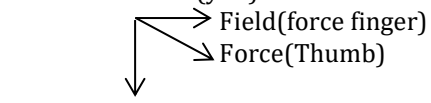
- Upward current gives an into the paper force or motion while downward current produce out of the paper force or motion.

52.

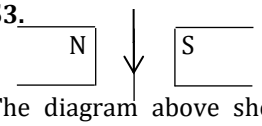


In the above diagram, a straight vertical wire is placed between two opposite poles which are separated horizontally by a short distance. If a current is passed down the wire, in which direction will move? **A.toward the observer** B.away from the observer C.towards the N-pole D.towards the south pole E.upwards.

Applying Fleming's left hand rule – The fore finger points in the direction of the field i.e due east from N—S, the second or middle finger points in the direction of the current i.e southward, then the thumb points in the direction of the force or motion of the conductor i.e towards the observer(you).



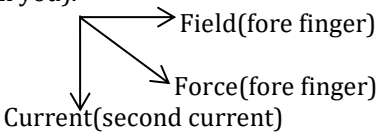
53.



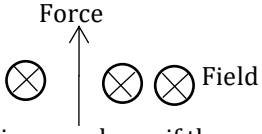
The diagram above shows a current carrying wire between the poles of a magnet. In which direction would the wire tend to move? A.into the paper **B.out of the paper** C.towards the south pole of the magnet D.towards the top of the page .

Demystified Series by Dr. Timothy

Applying Flemings left hand rule – The fore finger points in the direction of the field i.e due east from $N-S$, the second or middle finger points in the direction of the current i.e southward, then the thumb points in the direction of the force or motion i.e out of the paper(your finger points away from you).

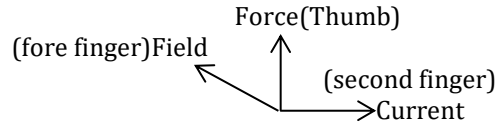


54.



In the diagram above, if the magnetic field points into the paper and the force on a current carrying conductor points upwards, what is the direction of the current? **A.right** B.left C.downwards D.upwards .

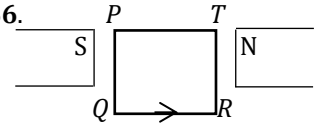
Applying Fleming's left hand rule – The current points to the right as the force or motion points upwards and the field is into the paper(your finger points towards you).



55. Which of the following is/are correct about a plane rectangular current-carrying coil placed in a uniform magnetic field? I.The forces in the two parallel sides produces a resultant translational motion II.The torque is maximum when the plane of the coil is parallel to the direction of the field III.The forces on the two parallel sides are equal and opposite A.I B.II C.III D.I and III **E.II and III .**

The forces in the two parallel sides of a plane rectangular coil are equal and in opposite direction but the currents are in opposite direction. The forces constitute a couple, which produces a rotational motion of the coil. The torque decreases as the angle between the plane of the coil and the field increases i.e torque is maximum when the plane of the coil is parallel or the arms is perpendicular to the field and minimum or zero when the plane of the coil is perpendicular or the arms is parallel to the field.

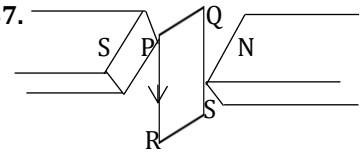
56.



In the figure above $PQRT$ is a rectangular coil situated in a magnetic field. If the current flows in the direction indicated, the sides PQ will tend to move A.towards the S-pole B.toward the N-pole C.out of the paper **D.into the paper .**

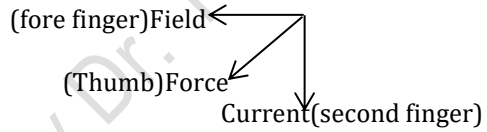
The field is due west or leftwards from N-S, and the current is downwards from P to Q, hence the force or motion of the side PQ is into the paper i.e towards you(observer). In side TR, the current is flowing upward from R to T, hence the force or motion of side TR is out of the paper i.e away from you (observer).

57.



PQRS is a rectangular coil with PQ perpendicular to the magnetic field as shown in the diagram above. If a current flows in the direction PQRS, in what direction will the coil move? A.upward **B.downwards** C.towards the north D.towards the south pole.

According to Fleming's left hand rule – Considering the side QR, the fore finger points in the direction of the field which is due west from N-S, the second finger points in the direction of the currect i.e downwards, hence the force is directed into the paper or towards the observer i.e downwards.



58. A vertical reactangular coil of sides 3cm and 2cm has 20 turns and carries a current of $2A$. Calculate the torque on the coil when it is placed in a uniform horizontal magnetic field of $0.1T$ with its **plane parallel to the field.**

A.1.0×10^{-3}Nm **B.2.4×10^{-3}Nm** C.4.8×10^{-2}Nm D.6.0×10^{-2}Nm.

Length L=3cm=0.03m, breadth b=2cm=0.02m $A=Lb$, I=2A, B=0.1T, N=20turns, θ=0° (planes parallel to the field), Torque τ=?

$\tau = BANIcos\theta$, Maximum torque, $\tau = BANI$

$\tau = BLbNI = 0.1\times0.03\times0.02\times20\times2$

$\tau = 0.00024\text{Nm} = 2.4\times10^{-4}\text{Nm}$.

- The magnitude of the troque increases with the strength of the magnetic field B, area of the coil A i.e length and breadth, number of turns in the coil N and current in the coil, $\tau \propto BANI$.

59. An electron is projected at an angle of 50° to a magnetic field of flux density $15T$ with a velocity of $7.0\times10^{7}\text{ms}^{-1}$, calculate the magnitude of the force exreted on it [e=1.6×10^{-19}C] A.1.1×10^{9}N B.4.9×10^{7}N **C.1.3×10^{-10}N** D.2.6×10^{-11}N.

q=e=1.6×10^{-19}C, v=$7.0\times10^{7}\text{ms}^{-1}$, θ=50°, B=15T, F=? $F = qvBsin\theta$,

$F = 1.6\times10^{-19}\times7\times10^{7}\times15\ \sin50° = 1.3\times10^{-10}$N.

60. An electric charge of 0.05C moves with a speed of 2ms^{-1} at right angle to a uniform magnetic field of flux density $0.04T$. Calculate the magnitude of the magnetic force acting on the charge A.0.000N **B.0.004N** C.0.001N D.1.600N.

q=0.05C, v=2ms^{-1}, θ=90°(charge at right angles to the field), B=0.04T, F=?,

Demystified Series by Dr. Timothy

$F = qvBsin\theta$, Maximum force, $F = qvB$.,
$F = 0.05\times2\times0.04 = 0.004$N.

61. In order to make a moving electron follow a circular path **A.a magnetic field is applied perpendicular to its path** B.a magnetic field is applied parallel to its path C.an electric field is applied parallel to its path D.an electric field is applied perpendicular to its path.

62. An electron moves with a speed of 2.00×10^{7}ms^{-1} in an orbit in a uniform magnetic field of $1.20\times10^{-3}T$. Calculate the radius of the orbit [mass of an electron = 9.11×10^{-31}kg, charge of an electron= 1.61×10^{-19}C] A.4.72×10^{-2}m **B.9.43×10^{-2}m** C.1.89×10^{-2}m D.3.78×10^{-2}m.

v=2.00×10^{7}ms^{-1}, B=$1.20\times10^{-3}T$, m=9.11×10^{-31}kg, q=e=1.61×10^{-19}C, r=?, $r = \frac{mv}{qB}$,

$r = \frac{9.11\times10^{-31}\times2.00\times10^{7}}{1.61\times10^{-31}\times1.2\times10^{-3}} = 9.43\times10^{-2}$m.

63. If a charge ion goes through combined electric and magnetic fields, the resultant emergent velocity of the ion is **A.$\frac{E}{B}$** B.EB C.$\frac{B}{E}$ D.$E - B$.

Electric force on charge, $F_E = qE$, q=charge, E=electric field intensity. Magnetic force on charge, $F_M = qvB$. When both fields interact,

$F_E = F_M$, $qE = qvB$, $v = \frac{qE}{qB} = \frac{E}{B}$.

64. A straight horizontal length of copper wire has a current of $28A$ through it. Find the magnitude of the minimum field needed to suspend the wire. The linear density of the wire is 46.6g/m **A.$1.6\times10^{-2}T$** B.$1.5\times10^{-2}T$ C.$1.4\times10^{-2}T$ D.$1.3\times10^{-2}T$.

I=$28A$, g=10ms^{-2}, Linear density=mass per unit length= $\frac{m}{L}$ =46.6g/m=0.0466kg/m

On suspending the wire in the magnetic field, the force on the current carrying wire is completely balanced by the weight of the wire $F = W$, $F = BIL$, $W = mg$, $BIL = mg$, $B = \frac{g(m/L)}{I}$,

$B = \frac{10(0.0466)}{28} = 1.6\times10^{-2}T$.

65. On which of the following is the operation of a moving coil galvanometer based? A.magnetic effect of electric current B.electromagnetic induction C.chemical reaction porudced by electricity **D.force on a current carrying conductor in a magnetic field** E.electrical energy and power .

Moving coil and moving irons instruments works on the motor effect i.e force on a current carrying conductor but also depends on magnetic effect of electric current.

66. Wich of the following devices does bot make use of magnetic effect of an electric current? A.moving coil ammeter **B.hot wire ammeter** C.moving iron ammeter D.moving coil loudspeaker E.telephone receiver .

Hot wire ammeter works on heating effect of magnetic current. It has a non-linear scale and is used for measuring high frequency a.c and d.c.

67. Which of the following actios will not increase the sensitivity of a moving coil galvanometer? **A.Using strong temporary magnet** B.Increasing the area and number of turns of the coil C.Using weak spiral spring D.Using light pointer .

The sensitivity of a moving coil galvanometer is increased with a special alloy permanent magnet, coil of large area and large number of turns,use of weak hair springs(strength of the spring) and light pointer.

I.The soft iron cylinder determine the magnitude of the radial magnetic field. **II.**Rotation of the coil which is indicated by the pointer depends on the strength of current i.e $\theta \propto$ pointer $\propto I$.

68. Which of the following statements is incorrect? The sensitivity of a galvanometer which incorporates an optical lever arrangement may be increased by providing A.a larger number of turns in the coil CB.a thinner suspension with a small torsional control C.a special ally permanent magnet which gives a high magnetic flux D.all of the above **E.a light source with greater intensity.**

Sensitivity of any galvanometer is increased by increasing the strength of magnetic field (using a special alloy permanent magnet) and number of turns in the coil.

Sensitivity of a reflecting or suspended coil galvanometer i.e galvanometer with optical lever arrangementor light beam as a pointer, is increased by using a thin suspension with small torsional control. Sensitivity of a moving coil galvanometer is increased by using a weak hair spring with small torsional couple.

69. The instrument used for measuring both a.c and d.c is A.an inverter **B.a moving iron ammeter** C.a current balance D.moving coil ammeter .

Moving iron ammeter measures low frequency a.c and d.c by the magnetic effect of electric current.

Moving coil galvanometer measures d.c only by force on a current carrying conductor.

Current balance or ampere balance measures d.c by force between parallel current carrying wires.

Inverter changes the magnitude of voltage or current betweeb input and output.

70. A rectangular coil of wire can rotate in a magnetic field. The ends of the coil are soldered to the two halves of a split-ring. Two carbon brushes are made to press lightly against the split ring and when these are connected in circuit with a battery and rheostat, the coil rotates. This is a description of A.a suspended-coil galvanometer B.a moving coil ammeter C.d.c generator **D.an electric motor** E.an induction coil.

Demystified Series by Dr Timothy

In an electric motor, an external supply of current from battery is needed to rotate the coil while in a d.c generator is rotates when supplied energy(not current) and current is induced in the generator.

71. Which of the following is not a part of a d.c electric motor? A.field magnet B.ammature C.commutator **D.transformer.**

72. What is the effect of using split rings in a simple d.c motor? A.The direction of rotation of the coil is reversed B.The current in the coil flows in the same direction C.The current in the coil becomes alternating **D.The direction of the force is reversed** E.The contact with the voltage source is cut off .

A split ring commutator half changes contact between carbon brushes, thereby reversing the current and force direction and maintains the direction of rotation of the coil.

73. Which of the following statements are not true of a moving coil milliammeter? I.It can be used to measure alternating current II.It has a linear scale III.It can be adapted to read higher values of current IV.A resistor connected in parallel with the milliammeter would convert it to a voltmeter **A.I and IV** B.II and III C.III and IV D.I,II and III E.I,III and IV .

Moving coil ammeter is used to measure direct current only and it has a linear scale. It can be adapted to measure higher values of current by connecting a low resistance shunt in parallel to it. It is converted to voltmeter, by connecting a high resistance resistor (multiplier) in series(not parallel) to it.

74. A galvanometer can be converted into an ammeter by connecting **A.a low resistance shunt in parallel with the galvanometer** B.a low resistance shunt in series with the galvanometer C.a high resistance multiplier in parallel with the galvanometer D.a high resistance multiplier in series with the galvanometer

Option D is used to convert a galvanometer into a voltmeter.

75. A moving coil galvanometer of 300Ω resistance gives a full scale deflection fro 1.0mA. The resistance R of the shunt that is required to convert the galvanometer into a 3.0A ammeter is A.899.7Ω B.10.00Ω **C.0.10Ω** D.0.01Ω .

i=1.0mA=0.001A, G=300Ω, I=3A, S=R=?,

$I = \frac{i(S+G)}{R}$, $3 = \frac{0.001(R+300)}{R}$,

$3R = 0.001(R+300)$, $\frac{3R}{0.001} = R + 300$,

$3000R = R + 300$, $3000R - R = 300$,

$2999R = 300$, $R = \frac{300}{2999} = 0.10\Omega$.

76. A galvanometer of resistance 38Ω is to be connected with a shunt that 1/20 of the current in a circuit passes through the galvanometer. The shunt resistance required is A.1.9Ω **B.2.0Ω** C.36.2Ω D.38.0Ω .

i=$\frac{1}{20}I$, $\frac{I}{i}$=20, G=20Ω, S=?, $I = \frac{i(S+G)}{S}$, $\frac{I}{i} = \frac{S+G}{S}$,

$20 = \frac{S+38}{S}$, $20S = S + 38$, $20S - S = 38$,

$19S = 38$, $S = \frac{38}{19} = 2\Omega$.

77. A milliammeter of resistance 100Ω and full scale deflection of 50mA is to be converted into voltmeter with a full scale of 100V. The resistance required is A.0.05Ω in parallel with the milliammeter B.0.05Ω in series with the milliammeter C.1,900Ω in parallel with the milliammeter **D.1,900Ω in series with the milliammeter**

V=100V, i=50mA=0.05A, G=100Ω, M=?,

$V = i(M + G)$, $100 = 0.05(M + 100)$,

$\frac{100}{0.05} = M + 100$, $2000 = M + 100$,

$M = 2000 - 100 = 1900\Omega$.

78. A moving coil meter of internal resistance 20Ω has a full scale deflection of 10mA. A series resistance of 980Ω is connected to the galvanometer in order to convert it into a voltmeter. The full scale deflection of the voltmeter is A.1V **B.10V** C.20V D.100V.

i=10mA=0.01A, G=20Ω, M=980Ω, V=?,

$V = i(M + G)$, $V = 0.01(980+20)$,

$V = 0.01(1000) = 10V$.

79. A moving coil galvanometer has a full scale deflection of 3A equivalent to 30° deflection. The sensitivity of the instrument is A.0.1° **B.10.0°** C.33.0° D.90.0°.

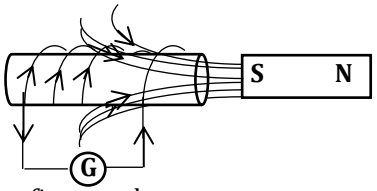
Current, I=3A, angle of rotation or deflection, θ=30°, current sensitivity S_I=? , $S_I = \frac{\theta}{I}$,

$S_I = \frac{30}{3} = 10°A^{-1}$.

80. A bar magnet is placed close to and lying along the axis of a closed coil. There will be no current in the coil when **A.both the coil and the magnet are held stationary** B.the coil is moved towards the stationary magnet C.the coil is moved away from the stationary magnet D.the magnet is moved away from the stationary coil .

Emf(and current) is induced in a coil whenever there is relative motion between the magnet and the coil. No emf or current is induced when both the magent and the coil are stationary or moving in the same direction with the same speed.

81.



The figure above represents an insulated cylindrical coil copper wire wound round a paper tube. The ends of the coil are connected to a sensitive galvanometer G. When a magnet was plunged into te coil it was observed that the

Demystified Series by Dr Timothy

galvanometer needle gave a deflection. Which of the following statements correctly describes this experiment? A.the strength of the current produced us inversely proportional to the speed with which the magnet is plunged into the coil **B.the effect illustrated is induced electromotive force** C.the direction of the current is always such that it enhances the change producing it D.the galvamometer gave a permanent deflection E.on removing the magnet from the coil the galvanometer deflects in the same direction .

The strength of the induced emf (hence current) is directly proportional to relative speed of the magnet,number of turns in the coil, area of the coil and the strength of the magnetic field i.e $E \propto \omega BAN$. The direction of the induced current(or emf) is walways such that it opposes the change producing it i.e Lenz's law. The galvanometer deflects with the induced current and returns to its original position i.e opposite direction, on removal of the current due to removal of the magnet.

82. A circuit has an area of 0.4m^2 and contains 50 loops of wire. If the loops are twisted and allowed to rotate at a constant angular velocity of 10rads^{-1} in a uniform magnetic field of $0.4T$, the amplitude of the induced voltage is **A.80*V*** B.20V C.8V D.16V.

ω=10rads^{-1}, B=0.4T, A=0.4m^2, N=50 loops, E_o=?.

Effecctive value of induced voltage, $E = E_o \sin\omega t$,

amplitude of induced voltage $E_o = \omega BAN$

E_o =10×0.4×0.4×50 =80V.

83. Induced emfs are best explained using **A.Faraday's law** B.Coulumb's law C.Lenz's law D.Ohm's law.

Induced emfs are explained by Faraday's law or Neuman's law of electromagnetic induction while induced currents is explained by Lenz's law.

84. Which of the following will not lead to an increase in the induced emf in a coil of wire rotating between the poles of a magnet? Increasing the A.strength of magnet B.number of turns of coil C.speed of rotation of coil **D.gap between poles of the magnet.**

85. Lenz's law is a law of the conservation of A.electric current B.momentum **C.energy** C.electric charge.

86. The magnetic flux in a coil having 200 turns changes at the time rate of 0.08Wbs^{-1}. The induced emf in the coil is A.250.0V B.25.0V C.16.0V D.1.6V

N=200 turns, $\frac{d\phi}{dt}$=0.08Wbs^{-1}, E=?,

Farday's law of electromagnetic induction, induced emf is proportional to the rate of change of magnetic flux, $E \propto \frac{d\phi}{dt}$, $E = N\frac{d\phi}{dt}$,

$E = 200\times0.08 = 16V$.

87. A band of 500 rectangular loops of wire of area 20cm by 20cm, encloses a region of magnetic field which changes from 0.4T to 1.0T within 5 seconds, calculate the induced emf A.5.60V B.24.00V C.0.24V **D.2.40*V*.**

N=500 loops, A=Lb, L=20cm=0.2m, b=20cm=0.2m, B_2=1T, B_1=0.4T, t=5s, E=?

$E = N\frac{d\phi}{dt} = N\frac{A\Delta B}{dt} = \frac{NA(B_2-B_1)}{t}$,

$E = \frac{500\times(0.2\times0.2)\times(1-0.4)}{5} = 100\times0.04\times0.6 = 2.4V$.

88. The coil of an electric generatir has 500 turns and 8.0cm diameter. If it rotates in a magnetic field of density 0.25T, calculate the angular speed when its peak voltage is 480V. (π=3.142)

A. 191rads^{-1} B. 382rads^{-1} **C. 764rads^{-1}**

D. 1528rads^{-1}.

N=500turns, d=8cm, r=4cm=4×10^{-2}m, B=0.25T, E_o=480V, ω=?, $E_o = \omega BAN$, A=πr^2,

$E_o = \omega B(\pi r^2)N$,

$480 = \omega \times0.25\times \pi \times16\times10^{-4}\times500$,

$480 = 0.2\pi\omega$, $\omega = \frac{480}{0.2\pi} = 763.94 = 764$rads^{-1}.

89. A conductor of length 1m moves with a velocity of 50ms^{-1} at an angle of 30° to the direction of a uniform magnetic field of flux density 1.5Wbm^{-2}. What is the emf induced in the conductor? A.50.5V **B.37.5*V*** C.75.0V D.80.5V.

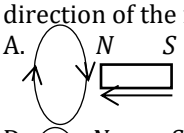
L =1m, v=50ms^{-1}, θ=30°, B=1.5Wbm^{-2} or 1.5T, E=?, $E = BLv\ sin\theta = 1.5\times1\times50\ sin30° = 37.5V$.

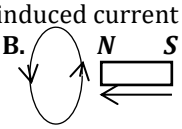
90. A wire of length 30cm is moved with a speed of 2ms^{-1} at right angles to a magnetic field of flux density 0.4T. Calculate the emf in the wire A.0.15V **B.0.24*V*** C.2.70V D.24.00V

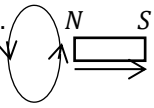
L=30cm=0.3m, v=2ms^{-1}, B=0.4T, E=?,

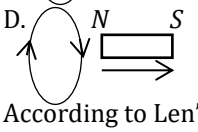
$E = BLv = 0.4\times0.3\times2 = 0.24V$.

91. A horizontal bar magnet is being moved relative to a vertical coil olaced close to it. Which of the folllwing diagrams correctly illustrates the direction of the induced current in the coil?.

A. 

B. 

C. 

D. 

According to Len'z law – the direction of induced current is in such a way that it opposes the changes producing it. If the N-pole of a magnet is moving closer to the coil, the magnetic flux linking the coil induces current in the coil such that it opposes the motion of the magnet towards the coil, hence anticlockwise current direction(Option B),which in turns forms a N-pole in the coil and the magnet is repelled from the coil.

92. A dynamo primarily converts **A.mechanical energy to electrical energy** B.electrical energy into kinetic energy C.potential energy into kinetic energy D.kinetic energy into potential energy

Demystified Series by Dr. Timothy

An electric motor converts electrical energy into mechanical energy.

93. Which of the following methods can be used to increase the emf produced by a simple dynamo? I.Increase the speed of rotation of the coil II.Using a longer field magnet III.Increasing the cross sectional area of the coil IV.Increasing the number of coil windings A.I and III **B.I and IV** C.I,III and IV C.I,II,III and IV .

The induced emf produced by a dynamo(d.c generator) increases with the strength of the magnetic field(not the magnetic length of the magnet), the number of coil windings,area of the coil(not the cross sectional area of the coil wire) and the speed of rotation of the coil , $E \propto \omega BAN$. Winding of the coil on a soft iron armature/core increases the magnetic flux and strength through the coil.The resistance of the coil e.g copper coil, does not affect the value of the induced emf.

94. On an a.c generator which of the following does not apply? A.field magnet B.slip rings **C.commutators** D.armature .

Commutators or split rings are found in d.c generators only.

95. A generator manufacturing company was contracted to produce an a.c dynamo but inadvertently produced a d.c dynamo. To correct this error, A.armature coil should be made of silver B.commutator should be replaced with split rings C.armature coil should be made of aluminium **D.commutator should be replaced with slip rings** .

Slip rings is in a.c generator while split rings is in d.c generator. A.c generator can be converted to a d.c generator when the slip ring of the a.c is replaced woth a split ring.

96. In a simple generator, the main function of a commutator is to A.change the direction of rtation of the armature B.provide a uniform magnetic field around the armature **C.change the direction of current in the coil** D.increase the magnetic flux linking the coil.

97. Which of the following statements about a generator is not correct? A.It can produce direct current B.It can produced alternate current C.It requires an external supply of energy to rotate the coil **D.It requires an external supply of current to the coil** E.It may require the use of a commutator.

98. A steady current of $2A$ flows in a coil of emf $12V$ for 0.4s. A back emf of $3V$ was induced during this period. The energy stored in the loop that can be utilized is **A.7.2J** B.12.0J C.4.7J D.9.6J.

Induced emf in coil E=$12V$, back emf E_B=$3V$, I=$2A$, t=0.4s, The energy stored in the loop of the coil , $W = I(E - E_B)t$.

$W = 2\times(12-3)\times0.4 = 0.8\times9 = 7.2$J.

99. Carbon particles are used in carbon microphine because A.they mught amplify sound B.they are good conductors C.their resistance changes wth temperature **D.their resistance canges with pressure .**

100. Which of the following devices converts sound into electrical energy? **A.a microphone** B.telephone earpiece C.an electric bell D.a horn .

A telephone mouthpiece and a microphone converts sound energy into electrical energy while telephone earpiece converts electrical to sound energy.

101. The back emf induced in a current-carrying coil causes A.resistance B.inductance **C.eddy currents** D.resonance .

Eddy current is produced form back emf set up.

102. Which of the following devices does not make use of eddy current for its action? A/an A.galvanometer B.speedometer C.induction furnance **D.induction coil.**

103. Which of the following statements about a transformer is correct? A.The input is a.c and the output is d.c **B.The input is a.c and the output is a.c** C.The input is d.c and the output is a.c D.Energy loss is increased by laminating the core E.The core is usually made of steel .

A transformer works with a.c and not d.c, hence the input at the primary coil is a.c and the output at the secondary coil is also a.c.

104. When a transformer has more secondary windings than primary windings, it **A.has smaller secondary current** B.has a greater output power C.is a step down transformer D.increases the total enrgy output.

Step-up transformer has more secondary windings than the primary winding ($N_S > N_P$) or turn ratio, $\frac{N_S}{N_P} > 1$. A step-up transformer steps up emf and steps down current , $E_S > E_P$ and $I_S < I_P$. The opposite holds for step-down. A transformer steps-up or step-down a.c voltage(emf) without altering the frequency ,power or energy.

105. An ideal step down transformer steps up A.power B.energy **C.current** D.voltage.

106. The main purpose of the transformer in an a.c radio set is to A.increase power to the radio B.convert energy from a.c to d.c C.step down voltage D.step up voltage **.**

Step-down transformer is used in various electronic devices to step-down voltage for utilization in homes.

107. The efficiency of a transformer can be increased if A.high resistance copper coils are used B.the core is designed to increase eddy currents C.the core is designed to make sure that little primary flux is linked with the secondary **D.the core is made of soft magnetic materials thereby reducing the energy for magnetic reversals .**

The heat energy loss in a transformer is reduced inorder to increase its efficiency.

Demystified Series by Dr Timothy

Use of low resistance or thick copper coils reduce copper losses or joules effect.
Use of laminated iron core(iron sheets) reduced eddy current loses.
Use of efficienct core design to reduce flux leakage and to ensure maximum flux linkage or reduce design losses.
Use of soft iron or special allow magnetic materials as core to reduce hysteresis loses due to magnetic reversals.

108. What precaution should a manufacturer take to ensure that energy loss in a transformer is minimized? A.the winding of the transformer should be made of high resistance wires **B.the core should be made of thin sheets of metal** C.no magnetic material should be used to make the core D.the flux linking the primary with the secondary coil should be minimum.

109. A transformer has primary coil with 500 turns and a secondary coil with 2500 turns. When the voltage input to the primary coil is $120V$, the output is A.$6000V$ **B.$600V$** C.$240V$ D.$60V$ E.$24V$.

N_P=5000 turns, N_S=2500 turns, Input voltage= primary voltage E_P=$120V$, Output voltage= secondary voltage E_S=?, $\frac{N_S}{N_P}=\frac{E_S}{E_P}$,

$\frac{2500}{500}=\frac{E_S}{120}$, $E_S=5\times120=600V$.

110. A dry battery of $6V$ is joined to the primary coil of a commercial transformer. The secondary has twice the number of turns of the primary. The primary resistance is 2Ω. The voltage obtained in the secondary is A.$1.5V$ B.$3V$ C.$12V$ **D.$0V$** E.$6V$

N_S=$3N_P$, A transformer is operates with a.c hence, connecting a dry battery(d.c source) to the primary will not induce any voltage or current in the secondary.

111. A transformer steps down from $240V$ to $12.0V$ and supplies a $12.0V$ $24.0W$ lamp. What is the efficiency of the transformer if the current drawn from te $240V$ supply is $0.120A$. A.57.2% B.66.6% **C.83.3%** D.100%.

E_P=$240V$, I_P=$0.12A$, E_S=$12V$, Power input= power in primary=I_PE_P , power output=Power in secondar=I_SE_S=$24W$, Efficiency ϵ=?,

Efficiency $=\frac{Power\ output}{Power\ input}\times100=\frac{P_o}{P_i}\times100$,

$\epsilon=\frac{I_SE_S}{I_PE_P}\times100=\frac{24}{0.12\times240}\times100=83.3\%$.

112. A transformer has an efficiency of 92.5%. The ratio of the number of turns in the primary coil to that in the secondary coil is 128:45. If the current passing through the secondary coil is $9.0A$, calculate the current passing through the primary coil. A.$2.93A$ B.$3.16A$ **C.$3.42A$** D.$27.68A$.

ϵ=92.5%, $\frac{N_P}{N_S}=\frac{128}{45}$, $\frac{N_P}{N_S}=\frac{E_P}{E_S}=\frac{128}{45}$, $\frac{E_S}{E_P}=\frac{45}{128}$,

I_S=$9A$, I_P=?, $\epsilon=\frac{P_o}{P_i}\times100=\frac{I_SE_S}{I_PE_P}\times100$,

$\epsilon=\frac{I_S}{I_P}\times\frac{E_S}{E_P}\times100$, $92.5=\frac{9}{I_P}\times\frac{45}{128}\times100$,

$I_P=\frac{9\times45\times100}{92.5\times128}=3.42A$.

113. A transformer has 400 turns and 200 truns in the primary and secondary windings respectively. If the current in the primary and secondary windings are $3A$ and $5A$ respectively, calculate the efficiency of the transformer A.85.0% **B.83.3%** C.37.5% D.30.0% .

N_P=400 turns, N_S=200 turns, I_P=$3A$, I_S=$5A$,

$\frac{N_S}{N_P}=\frac{200}{400}=\frac{1}{2}$, $\frac{N_S}{N_P}=\frac{E_S}{E_P}=\frac{1}{2}$,

$\epsilon=\frac{P_o}{P_i}\times100=\frac{I_SE_S}{I_PE_P}\times100=\frac{I_S}{I_P}\times\frac{E_S}{E_P}\times100$,

$\epsilon=\frac{5}{3}\times\frac{1}{2}\times100=83.3\%$.

114. Which of the following can be described as high tension transmission? A.high resistance and low voltage **B.low current and high voltage** C.high current and low voltage D.high voltage and zero current E.high current and low resistance .

High tension transmission is transmission of power at high voltage and low current while low tension transmission is transmission of power at low voltage and high current.

115. It is usual to transmit electric power at high voltage and low current. Which of rhe following are possible advantageous of the method? I.heat losses are reduced because the currents are small II.thin wires can be used because small currents are flowing III.the power can flow faster because the voltage is high A.I B.II and III **C.I and II** D.I and III E.I,II and III.

High tension transmission involes high voltage and low current, as heat losses are reduced due to low current i.e $H\propto I^2$. A thin wire will offer high resistance to current [$R\propto\frac{1}{A}$] hence, only small current flows.

116. Power supply is transmitted in high voltage and low current in order to A.increase the power supply **B.prevent overheating of coil** C.make it travel fast D.make it pass through transformers .

117. Which of the following modes is the most economical mode of transmitting electrical power over long distances? **A.Alternating current at high voltage and low current** B.Alternating current at low voltage and high current C.Alternating current at high voltage and high current D.Direct current at low voltage and high current.

118. The induction coil works on the same principle as **A.transformer** B.a d.c generator C.an a.c generator D.an inductor.

Induction coil and transformer both works on mutual inductance of emf in a secondary coil due to varying magnetic flux or alteration in current in a primary coil. A.c and d.c generators works on self induction. But they all are electromagnetic induction.

119. An induction coil is generally used to A.rectify an alternating current B.produce a large intput voltage C.smoothen a pushing direct

Demystified Series by Dr Timothy

D.modulate an incoming radio signal **E.produce a large output voltage .**
Induction coil produces a large out voltage in the secondary when a low d.c voltage source is connected to the primary i.e steps up d.c voltage.
120. The best material for the core of the primary and secondaery coils of an induction coil is A.copper, because it is a good conductor of electricity B.steel, because it becomes magnetic and retains its magnetism for a long time C.rubber, because it is a good insulator and the user is prevented from shock **D.soft iron, because it becomes magnetic but easily loses its magnetism** E.glass, because it is transparent .
The iron core of an induction coil is made from bundle of wire or laminated into sheets to minimize eddy currents.
121. The principle of operation of an induction coil is based on A.Ohm's law **B.Faraday's law** C.Ampere's law D.Coulomb's law .
Operation of an induction coil is based on Faraday's law of electromagnetic induction.
122. Which of the following reduces the effect of the back emf generated in the primary coil of an induction coil? I.The capacitor in the circuit II.The make-and-break contact in the circuit III.The ratio of turns in the secondary **A.I** B.II C.I and II D.II and III E.I,II and III .
The capacitor in the primary circuit prevents sparks in the make-and-break device by reducing the back emf generated in the primary. The make-and-break device gives a rapid decaying or varying magnetic flux which induces a large ouput d.c voltage in the secondary. The ratio of turns in the secondary determined the strength of the the induced emf , $N \propto \frac{E}{th}$.
123. The device used to prevent wearing away of the make-and-break contacts of an induction coil is called a/an A.fuse B.electrscope C.resistor **D.capacitor .**
124. Given three inductors of inductance 5mH, 10mH and 20mH connected in series, the effective inductance is A.0.35mH **B.35.00mH** C.3.50mH D.2.90mH**.**
For inductors in series, effective inductance is the arithmetic sum of their individual inductance, $L = L_1 + L_2 + L_3 = 5+10+20 = 35\text{mH}$.
125. Two inductors of inductance $4H$ and $8H$ are arranged in series and a current of $10A$ is passed through them. What is the energy stored in them? A.133J B.250J **C.600J** D.50J.
Effective inductance, $L = 4+8=12\text{H}$, $I=10A$,
Energy stored in an inductor, $E = \frac{1}{2}LI^2$
$E = \frac{1}{2}\times12\times10^2 = 6\times100 = 600\text{J}$.
126. The energy stored in aninductor of inductance $5\text{m}H$ when a current of $6A$ flows through it is A.1.4×10^{-2}J B.1.8×10^{-2}J C.1.4×10^{-3}J **D.9.0×10^{-2}J.**
$L=5\text{m}H=5\times10^{-3}H$, $I=6A$, $E=?$, $E = \frac{1}{2}LI^2$,
$E = \frac{1}{2}\times5\times10^{-3}\times6^2 = 9\times10^{-2}J$.

Jamb past questions on magnetic field, magnetism and electromagnetism :
[1978/30,36,1979/26,33,1980/6,1981/9,45,47, 48,1982/39,1983/33,40,1984/33,34,1985/37, 43,45,1986/45,47,48,1987/38,45,1988/37,45, 1989/34,45,1990/34,1991/33,42,43,1992/31, 32,1993/33,34,43,44,1998/46,1999/36,2000/ 47,2001/43,46,48,50,2002/42,44,47,2003/43, 2004/46,2005/1,6,15,20,2006/30,2008/42,43, 2010/42,2011/47,2012/41,2014/40,41,42,43, 2015/18,44]

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Chapter 25 – Simple A.C circuits

- Direct current(d.c) produced by a battery (cell) or d.c generator(dynamo), has a constant value and flows in one direction unless the battery terminals are reversed.

- Alternate current(a.c) produced by an a.c generator(alternator), has a varying value and direction with time.

- Instantaneous value of emf or current is its value at a particular instant of time.

$\boldsymbol{I = I_o \sin(\theta + \emptyset) = I_o \sin(\omega t + \emptyset)}$.

$\boldsymbol{V = V_o sin(\theta + \emptyset) = V_o sin(\omega t + \emptyset)}$.

I or V=instantaneous value of current and voltage, I_o or V_o=maximum or amplitude or peak value of current and voltage, ω=angular frequency or angular velocity=$2\pi f = \frac{2\pi}{T}$, f=frequency, T=period, $\emptyset$=phase angle of phase difference or phase shift between I and V.

- At $\boldsymbol{t = T}$ or **1cycle**, $\boldsymbol{\theta = 360°}$. At $\boldsymbol{t = \frac{T}{2}}$ or $\boldsymbol{\frac{1}{2}}$ **cycle**, $\boldsymbol{\theta = 180°}$. At $\boldsymbol{t = \frac{T}{4}}$ or $\boldsymbol{\frac{1}{4}}$ **cycle**, $\boldsymbol{\theta = 90°}$. At $\boldsymbol{t = \frac{T}{8}}$ or $\boldsymbol{\frac{1}{8}}$ **cycle,** $\boldsymbol{\theta = 45°}$.

- Effective or root mean square(rms) value of voltage and current is the value in which a.c is quoted. **The effective or root mean square (rms) value of an a.c is its d.c equivalent i.e has the same heating effect with a direct current.** $\boldsymbol{V_{rms} = \frac{V_o}{\sqrt{2}}}$, $\boldsymbol{I_{rms} = \frac{I_o}{\sqrt{2}}}$,

- Phasor diagrams – Vector diagrams that shows the relationship between I and V.

1.Resistance in a.c circuit i.e purely resistive circuit – Current and voltage are in phase i.e phase angle=0°.

2.Inductor in a.c circuit i.e purely inductive circuit – Current lags voltage or voltage leads current by 90° or $\frac{\pi}{2}$ radians or $\frac{1}{4}$ cycle.

3.Capacitor in a.c circuit i.e purely capacitive circuit – Current leads voltage or voltage lags by 90° or $\frac{\pi}{2}$ radians or $\frac{1}{4}$ cycle.

- Series *R-L-C* circuit – In a *R-L-C* circuit, current and voltage are out-of-phase by a phase angle between 0 to 90°.

- Reactance $\boldsymbol{X}$ – This is the opposition to the flow of a.c and is numerically equal to the ratio of the amplitude of voltage to the amplitude of current in inductive or capacitive circuit. $\boldsymbol{X = \frac{V_o}{I_o}}$.

1.Resistance and Inductance – Current lags voltage(or voltage leads current) at $\emptyset$. The resistance offered by the inductor is **inductive inductance ,** $\boldsymbol{X_L = \omega L = 2\pi f L}$.

- The energy stored is due to the inductance (stored in the inductor only). $\boldsymbol{E = \frac{1}{2} L I^2}$.

2.Resistance and capacitance – Current lead voltage(or voltage lags current) at $\emptyset$. The resistance offered by the capacitor is **capacitive inductance,** $\boldsymbol{X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}}$.

- The energy stored is due to the capacitance (stored in the capacitor). $\boldsymbol{E = \frac{1}{2} C V^2}$.

3.Inductance and capacitace – The circuit is predominantly inductive if $X_L > X_C$ and current lags voltage(or voltage leads current). The cicuit is predominantly capacitive if $X_L < X_C$ and current leads voltage(or voltage lags current).

- The energy stored is due to both the inductance and capacitance. $\boldsymbol{E = \frac{1}{2} L I^2 + \frac{1}{2} C V^2}$.

- In a series *R-L-C*, circuit **current and voltage are in phase.**

- Impendance *Z* – It is the total opposition to the flow of current in an a.c. It is the ratio of the rms voltage to rms current in an a.c circuit.

$\boldsymbol{Z = \frac{V_{rms}}{I_{rms}} = \frac{V_o}{I_o}}$. $\boldsymbol{Z^2 = R^2 + X^2}$, $\boldsymbol{X = X_L}$ or $\boldsymbol{X_C}$ or $\boldsymbol{X_L - X_C}$ **or** $\boldsymbol{X_C - X_L}$.

- $\boldsymbol{Z = X = X_L - X_C}$ or $\boldsymbol{X_C - X_L}$(for an a.c circuit containing inductor and capacitor only).

- The total voltage V_T – $\boldsymbol{V_T{}^2 = V_R{}^2 + V_L{}^2}$ or $\boldsymbol{V_T{}^2 = V_R{}^2 + V_C{}^2}$ or $\boldsymbol{V_T{}^2 = V_R{}^2 + (V_L - V_C)^2}$.

- $\boldsymbol{V_T = V_L - V_C}$ or $\boldsymbol{V_C - V_L}$(for an a.c circuit containing inductor and capacitor only).

- Voltage across resistor $\boldsymbol{V_R = IR}$.

-Voltage across inductor $\boldsymbol{V_L = IX_L}$.

-Voltage across capacitor $\boldsymbol{V_C = IX_C}$.

- Phase angle $\emptyset$ between current and voltatge is given by the quations below –

1. $\boldsymbol{\sin\emptyset = \frac{X_L - X_C}{Z} = \frac{V_L - V_C}{V_T}}$.

2. $\boldsymbol{\cos\emptyset = \frac{R}{Z} = \frac{V_R}{V_T}}$

3. $\boldsymbol{\tan\emptyset = \frac{X_L - X_C}{R} = \frac{V_L - V_C}{V_R}}$

- Power in an a.c circuit – Power is dissipated only in a resistor in an a.c circuit. Thus, power is not dissipated in an inductor or capacitor.

- Average power $P_{av}$of a *R-L-C* a.c circuit, $\boldsymbol{P_{av} = IV\cos\emptyset}$.*I*=rms current,*V*=rms voltage.

- Power factor $\boldsymbol{= \cos\emptyset = \frac{R}{Z} = \frac{V_R}{V_T}}$.

- Power factor factor of a purely reactive or inductive or capacitive circuit is zero or phase angle $\emptyset$=90° and average power is zero $\boldsymbol{P_{av} = 0}$.

- The power factor of a DC circuit is always 1.

- Instantaneous power $\boldsymbol{P = IV}$, *I*=instantaneous current, *V*=instantaneous voltage.

- In a purely reistive circuit or circuit with current and voltage in phase, $\boldsymbol{P_{av} = I^2 R = IV = \frac{V^2}{R}}$. *I* and *V* are rms value.

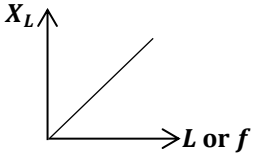
- Power factor of a purely resistive circuit is unity or phase angle $\emptyset$=0°.

- Graphical representation of the relationship

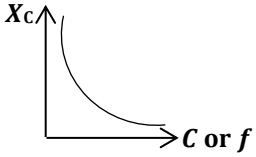
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betweeen reactance,frequency, capacitance and inductance –

1. $X_L = 2\pi fL$ then, $\boldsymbol{X_L} \propto \boldsymbol{L}$ and $\boldsymbol{X_L} \propto \boldsymbol{f}$. Inductive reactance is directly proportional to inductance and frequency i.e Inductive reactance increases with inductance and frequency.



2. $X_C = \frac{1}{2\pi fC}$ then, $\boldsymbol{X_C} \propto \frac{\boldsymbol{1}}{\boldsymbol{C}}$ and $\boldsymbol{X_C} \propto \frac{\boldsymbol{1}}{\boldsymbol{f}}$. Capacitive reactance is inversely proportional to capacitance and frequency i.e capacitive reactance decreases with capacitance and frequency.



- Resonance in *R-L-C* a.c circuit – Resonance is said to occur in an a.c circuit when ;

1.Inductive reactance equals capacitive reactance ($X_L = X_C$) or Net reactance is zero ($X = X_L - X_C = 0$).

2.Current and power are maximum.

3.Impendance is minimum and is equal to resistance($Z = R$)**.**

4.Phase angle is zero, $\emptyset = 0°$ or current and voltage are in phase.

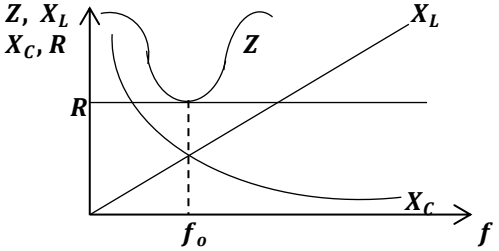
5.Power factor is unity, $cos\emptyset = \frac{R}{Z} = 1$.

6.Volage across the inductor equals the voltage across the capacitor or inductive voltage equal capcitive voltage ($V_L = V_C$) and the net voltage is equal to the voltage across a resistor ($V_T = V_R$).

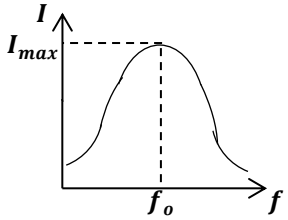
- At resosnance, the frequency called **resonance frequency $\boldsymbol{f_o}$** is given by –

$\boldsymbol{f_o = \frac{1}{2\pi\sqrt{LC}}}$, $\boldsymbol{\omega_o = \frac{1}{\sqrt{LC}}}$.

- Graphical representation of the relationship between impendance Z, inductive reactance X_L, capacitive reactance X_C and resistance R with frequency at resonance : $\boldsymbol{Z}$, $\boldsymbol{X_L}$, $\boldsymbol{X_C}$and $\boldsymbol{R}$ **all coincides at resonance frequency $\boldsymbol{f_o}$. Resistance $\boldsymbol{R}$ is independent of frequency i.e not affected by frequency.**



- Graphical representation of the relationship between current and frequency at resonance – **At frequencies above or below resonant frequency $\boldsymbol{f_o}$, the current is minimum or less than the maximum current. Current is maximum at $\boldsymbol{f = f_o}$.**



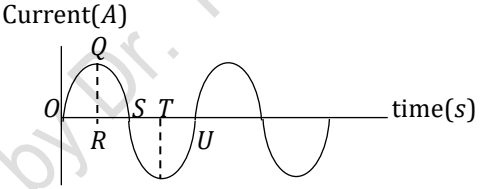
- In summary **– 1.** At frequency lower than resonance frequency, $f < f_o$: $X_L < X_C$or $X_C > X_L$ or $V_L < V_C$ and I leads V or V lags I.

2. At frequency higher than resonance frequency , $f > f_o$: $X_L > X_C$ or $X_C < X_L$ or $V_L > V_C$ and I lags V or V leads I.

3. At frequency equal to resonance frequency , $f = f_o$: $X_L = X_C$ or $V_L = V_C$ and I and V are in phase.

Examples :

Use the diagram below to answer question 1 and 2 ;



1. An alternating current $I = I_0 sin\omega t$has the waveform shown in the figure above, the quantity I_0 is equal to A.OT B.OS **C.$\boldsymbol{QR}$** D.SU.

I_0 is the amplitude or peak current and its represented by the maxium displacement of the waveform from the equilibrium position i.eQR.

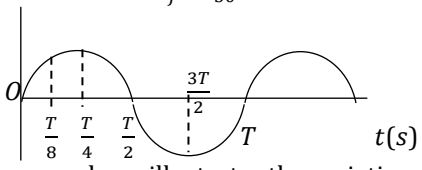
2. The length OU in the diagram is equal to A.ω B.$\frac{\omega}{2\pi}$ C.$\frac{\omega}{\pi}$ **D.$\frac{2\pi}{\omega}$** .

The length OU is the period T i.e time taken to complete one cycle. $T = \frac{2\pi}{\omega}$.

3. An alternating voltage with a frequency of 50Hz has a period of **A.0.02s** B.0.05s C.0.20s D.0.50s .

f=50Hz, period, $T = \frac{1}{f} = \frac{1}{50}$ =0.02s.

4.



The diagram above illustrates the variation of the voltage V in an a.c circuit with time t. If the peak value of voltage is 180V and T is the period, what is the instantaneous voltage at time, $t = \frac{T}{8}$?

A.45V B.90V **C.$\boldsymbol{90\sqrt{2}V}$** D.180V E.180$\sqrt{2}V$

V_o=180V, $t = \frac{T}{8}$, $\omega = \frac{2\pi}{T}$, V=instantaneous voltage=? $V = V_o sin\omega t = V_0 sin\frac{2\pi}{T}t$,

$V = 180 sin\frac{2\pi}{T} \times \frac{T}{8} = 180 \sin\frac{\pi}{4}$ (π rad=180°)

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$V = 180sin\frac{180^\circ}{2} = 180sin45^\circ$ $(sin45^\circ=\frac{1}{\sqrt{2}})$

$V = 180\times\frac{1}{\sqrt{2}} = \frac{180}{\sqrt{2}} = \frac{180}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}} = \frac{180\sqrt{2}}{2} = 90\sqrt{2}V.$

5. The instantaneous value of induced emf as a function of time $\varepsilon = \varepsilon_0 sin\omega t$ where ε_0 is the peak value of emf. The instantaneous value of the emf, one quarter of the period is A.$\frac{\varepsilon_o}{4}$ **B.$\boldsymbol{\varepsilon_o}$** C.0 D.$\frac{\varepsilon_o}{2}$.

$t = \frac{T}{4}$, $\varepsilon = \varepsilon_o sin\omega t = \varepsilon_o sin\frac{2\pi}{T}t,$

$\varepsilon = \varepsilon_o sin\frac{2\pi}{T}\times\frac{T}{4} = \varepsilon_o \sin\frac{\pi}{2}$ (π rad=180°)

$\varepsilon = \varepsilon_o sin\frac{180^\circ}{2} = \varepsilon_o sin90^\circ = \varepsilon_0.$

6. The peak value of the p.d in an AC circuit is $240V$. The instantaneous p.d at a phase angle of 90° is **A.$\boldsymbol{240V}$** B.$120V$ C.$60V$ D.$0V$

Phase angle ωt=90° or $\frac{\pi}{2}$ or $\frac{1}{4}$ cycle, V_0=$240V$,

$V = V_o sin\omega t = 240sin90^\circ = 240V.$

7. In an a.c circuit, the ratio of r.m.s value to peak value of current is A.2 B.$\sqrt{2}$ **C.$\boldsymbol{\frac{1}{\sqrt{2}}}$** D.$\frac{1}{2}$.

$I_{rms} = \frac{I_o}{\sqrt{2}}$, $\frac{I_{rms}}{I_o} = \frac{1}{\sqrt{2}}$. $V_{rms} = \frac{V_o}{\sqrt{2}}$, $\frac{V_{rms}}{V_O} = \frac{1}{\sqrt{2}}$.

8. The peak voltage of a $110V$ rms mains supply is A.$220V$ **B.$\boldsymbol{156V}$** C.$120V$ D.$110V$

V_{rms}=$110V$, V_o=?, $V_{rms} = \frac{V_o}{\sqrt{2}}$, $V_o = V_{rms}\sqrt{2}$,

$V_o = 110\sqrt{2} = 155.56V = 156V.$

9. The rms value of an alternating current is $2A$. What is the peak value of the current? A.4.0A B.3.4A **C.2.8$\boldsymbol{A}$** D.1.4A.

$I_{rms} = \frac{I_o}{\sqrt{2}}$, $I_o = I_{rms}\sqrt{2} = 2\sqrt{2}$=2.8$A$.

10. In a radio circuit, the a.c voltage across a resistor is given by $V = 4sin60\pi ft$. Calculate the root mean square(rms) value of the a.c. **A.$\boldsymbol{2\sqrt{2}V}$** B.$3\sqrt{2}V$ C.$4V$ D.$6V$.

Comparing $V = 4\sin60\pi ft$ with $V = V_o sin\omega t$, $V = V_o sin2\pi ft$, V_o=$4V$, $V_{rms} = \frac{V_o}{\sqrt{2}} = \frac{4}{\sqrt{2}}$,

$V = \frac{4}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2}}{2} = 2\sqrt{2}V.$

11. The voltage of the domestic electric supply is represented by the equation, $V = 311sin314.2t$. Determine the frequency of the a.c supply **A.50.00Hz** B.100.0Hz C.311.0Hz D.314.2Hz (π=3.142) .

Comparing $V = 311sin314.2t$ with $V = V_o sin\omega t$ or $V = V_o sin2\pi ft$, angular frequency or angular velocity ω=314.2rads^{-1}. $\omega = 2\pi f$, $314.2 = 2\pi f$,

$f = \frac{314.2}{2\pi} = \frac{314.2}{2\times3.142} = 50$Hz.

12. The current through a resistor in an a.c circuit is given as $2\sin\omega t$. Determine the d.c equivalent of the current A.$\frac{1}{\sqrt{2}}A$ B.$2\sqrt{2}A$ C.$2A$ **D.$\boldsymbol{\sqrt{2}A}$.**

D.c equivalent of the a.c current is same as effective or rms square value of current.

$I = I_o sin\omega t$, $I = 2\sin\omega t$, I_o=$2A$,

$I_{rms} = \frac{I_o}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \frac{2}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2}A.$

13. An a.c supply lights a lamp with the same brightness as a $12V$. What is the peak voltage of the a.c supply? A.0.12V B.8.49V **C.16.97$\boldsymbol{V}$** D.20.10V

The a.c supply with rms value will have the same brightness(or heating effect) as the $12V$ battery(d.c source).V_{rms}=$12V$, V_o=?

$V_{rms} = \frac{V_o}{\sqrt{2}}$, $V_o = V_{rms}\sqrt{2} = 12\sqrt{2} = 16.97V.$

14. In an a.c circuit, a voltage supply of $20V$ rms is connected to a 5Ω resistor. Find the peak value of the current A.2.50A B.4.00A C.6.00A**D.5.66$\boldsymbol{A}$**

V_{rms}=$20V$, R=5Ω, I_o=?, $V_{rms} = I_{rms}R$,

$I_{rms} = \frac{V_{rms}}{R} = \frac{20}{5} = 4A$, $I_{rms} = \frac{I_o}{\sqrt{2}}$, $I_o = I_{rms}\sqrt{2} = 4\sqrt{2}$=5.66$A$.

15. Calculate the amount of heat generated in an external load of resistance 8Ω if an a.c of peak value 5A is passed through it for 100s. **A.$\boldsymbol{1.0\times10^4}$J** B.$2.0\times10^4$J C.$5.0\times10^4$J D.$4\times10^4$J.

I_o=5A, R=8Ω, t=100s, I_{rms}=?

$I_{rms} = \frac{I_o}{\sqrt{2}} = \frac{5}{\sqrt{2}} = 3.54A.$

The heat energy, $H = I^2Rt = 3.54^2\times8\times100$,

$H = 10025.28 = 1.0\times10^4$J.

16. When an a.c voltage is applied to a capacitor, no power is consumed because the current A.and voltage are 180° out of phase B.and voltage are in phase C.lags behind the voltage by 90° **D.leads the voltage by 90°** D.leads the voltage by 90°.

In a purely capacitive circuit, current leads voltage or voltage lags behind current by 90°. Thus, power dissipated is zero.

17. In a purely inductive circuit, the current **A.lags behind the voltage in phase by 90°** B.leads the voltage in phase by 90° C.is in the same phase with tge voltage D.leads the voltage by 180°.

In a purely inductive circuit, current lags behind voltage or voltage leads current in phase by 90°. Thus, power is dissipated in is zero.

18. In a d.c circuit, a 10microfarad(μF) capacitor is placed in series with a 10Ω resistor. The total resistance of the combination is A.10Ω B.1Ω C.0Ω D.20Ω **E.infinite .**

In a d.c circuit, current flows in a constant direction,hence it does not oscillate and has no frequency.f=0Hz, R=10Ω, X_C=? ,

$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi\times0\times10\times10^{-6}} = \frac{1}{0} = \infty$ or infinite.

In a d.c circuit, the total resistance in series,

$R_T = R + X_C = 10+\infty = \infty$ or infinite.

19. An a.c 2.5A passes through a capacitor when the potential difference across its terminals is 450V. If the angular frequency is 200rads^{-1}, determine its capacitance A.2.78μF B.4.42μF C.14.30μF **D.2.80μF.**

V_C=450V, I=2.5A, ω=200rads^{-1}, C=?

$V_C = IX_C$, $X_C = \frac{V_C}{I} = \frac{450}{2.5} = 180V.$

$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$, $180 = \frac{1}{200\times C}$, $C = \frac{1}{180\times200}$

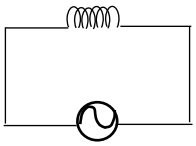
$C = 27.8\times10^{-6}$F = 27.8μF.

20. Determine the inductive reactance when 30.0mH inductor when a negligible resistance is

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connectd to a 1.30×10^3Hz oscillator A.39.0Ω B.122.5Ω **C.245.0Ω** D.3900.0Ω
L=30mH=$3\times10^{-3}H$, f=1.30×10^3Hz, X_L=?
$X_L = \omega L = 2\pi fL = 2\times3.142\times1.30\times10^3\times3\times10^{-3}$
$X_L = 245\Omega$.

21. L=0.2H The reactance of the inductor shown in the figure is 100Ω. The frequency of the A.C source is A.$\frac{500}{\pi}$Hz **B.$\frac{250}{\pi}$Hz** C.50Hz D.20Hz .

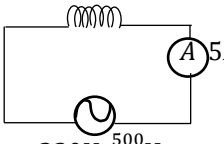


L=0.2H, X_L=100Ω. f=?, $X_L = 2\pi fL$,
$f = \frac{X_L}{2\pi L} = \frac{100}{2\pi\times0.2} = \frac{250}{\pi}$Hz.

22. A 120V, $\frac{60}{\pi}$ Hz a.c source is connected across a 2μF capacitor. Calculate the current passing through the circuit A.$4.9\times10^{-2}A$ **B.$2.9\times10^{-2}A$** C.$2.2\times10^{-2}A$ D.$2.0\times10^{-2}A$.

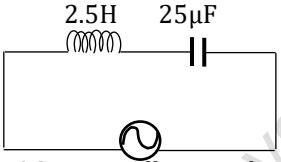
V_{rms}=120V, f=$\frac{60}{\pi}$Hz, C=2μF=$2\times10^{-6}F$, I_{rms}=?,
$X_C = \frac{1}{2\pi fC}$, $X_C = \frac{1}{2\pi\times\frac{60}{\pi}\times2\times10^{-6}} = \frac{10^6}{240} = 4166.7\Omega$.
$V_{rms} = IX_C$, $I = \frac{V_{rms}}{X_C} = \frac{240}{4166.67} = 2.9\times10^{-2}A$.

23. L The inductance L of the coil in the A.C circuit shown above is A.30.00H B.17.60H C.4.40H **D.0.04H.**



V_{rms}=220V, f=$\frac{500}{\pi}$Hz, I_{rms}=5A, L=?,
X_L =2πfL, $V_{rms} = I_{rms}X_L = I_{rms}(2\pi fL)$,
$L = \frac{V_{rms}}{2\pi f I_{rms}} = \frac{220}{2\pi\times\frac{50}{\pi}\times5} = 0.04H$.

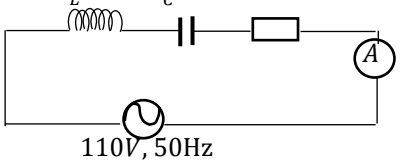
24. 2.5H 25μF



The AC circuit illustrated in the diagram above has a frequency of $\frac{100}{\pi}$Hz. The effective reactance of the circuit is A.700Ω B.500Ω **C.300Ω** D.200Ω
L=2.5H, C=25μF=$25\times10^{-6}F$, f=$\frac{100}{\pi}$Hz ,
X_L =$2\pi fL = 2\pi\times\frac{100}{\pi}\times2.5$= 500Ω.
$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi\times\frac{100}{\pi}\times25\times10^{-6}} = \frac{10^6}{5000}$ =200Ω.
Effective or net reactance $X = X_L - X_C$,
$X = 500-200 = 300\Omega$.

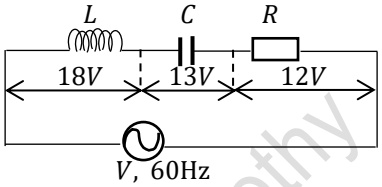
25. In an electrical circuit, an inductor of inductance 0.5H and resistance 50Ω is connected to an alternating current source of frequency 60Hz. Calculate the impendance of the circuit A.50.0Ω B.150.5Ω **C.195.0Ω** D.1950.1Ω .
L=0.5H, R=50Ω, f=50Hz, X_L =2πfL ,
X_L = 2×3.142×60×0.5 = 188.52Ω
Impendance, $Z^2 = {X_L}^2 + R^2$ =188.52^2+50^2 ,
Z^2 =35539.8+2500 =38039.8,
$Z = \sqrt{38039.8}$, Z = 195.0Ω.

26. X_L=5Ω X_C=8Ω R=4Ω



In the R-L-C circuit above, the current indicated by the ammeter A is **A.22.0A** B.15.7A C.10.7A D.6.5A .
Z^2= R^2+$(X_L-X_C)^2$ or R^2+$(X_C-X_L)^2$,
Z^2= 4^2+$(8-5)^2$ = 16+9 = 25, $Z = \sqrt{25}$ =5Ω,
$V = IZ$, $I = \frac{V}{Z} = \frac{110}{5} = 22A$.

27.



The applied voltage V in the R-L-C circuit digram shown above is A.43V B.19V C.17V **D.13V**
V_L=18V, V_C=13V, V_R=12V, V_T=?
${V_T}^2 = {V_R}^2 + (V_L - V_C)^2 = 12^2 + (18-13)^2$
${V_T}^2$ = 144+25 = 169, $V_T = \sqrt{169} = 13A$.

28. A series R-L-C circuit has inductance 12mH, capacitance 1.6μF and resistance 1.5Ω. Find the time when the amplitude of charge oscillating will be 50% of its initial value **A.11ms** C.10ms C.8ms D.9ms.
The charge on the capacitor in a series R-L-C circuit decreases exponentially with time and is given by the equation; $Q = Q_o e^{-\frac{Rt}{2L}}$
Q=charge after time t, Q_o=maximum or initial charge, R=resistance, L=inductance.
Q =50%$Q_o = \frac{50}{100}\times Q_o = \frac{Q_o}{2}$, L=12mH= 0.012H,
R=1.5Ω, C=1.6μF=1.6×10^{-6}F,
$\frac{Q_o}{2} = Q_o e^{-\frac{1.5\times t}{2\times0.012}}$, $\frac{1}{2} = e^{-\frac{1.5t}{0.024}}$ (take natural logarithm i.e In, of both sides).
$\ln\frac{1}{2} = -\frac{1.5t}{0.024}$, $-0.693 = -\frac{1.5t}{0.024}$,
$t = \frac{-0.693\times0.024}{-1.5} = 0.011\text{s} = 11\text{ms}$.

28. An alternating current with a peak value of 5A passes through a resistor of resistance 10Ω. Calculate the rate at which energy is dissipated in the resistor A.50W **B.125W** C.200W D.250W E.500W.
Rate at which energy is dissipated = Power.
For a purely resistive circuit, the average power is given by – $P = {I_{rms}}^2 R = \frac{1}{2}{I_o}^2 R$,
I_o=5A, R=10Ω, P=?, $P = \frac{1}{2}\times5^2\times10$,
P =25×5=125W.

29. In a pure resistive a.c circuit, the current $I = I_o in\omega t$ and the voltage $V = V_o sin\omega t$. Caclulate the instantaneous power dissipated in the circuit in time t. A.I_oV_o B.$2I_oV_o sin\omega t$ C.$\frac{I_oV_o}{2}$ **D.$I_oV_o sin^2\omega t$** E.${I_o}^2{V_o}^2$

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Instantaneous power, $P = IV$, where I and V are instantaneous value,
$P = I_o sin\omega t \times V_o sin\omega t = I_o V_o (sin\omega t)^2$,
$P = I_o V_o sin^2 \omega t$.

30. The power dissipated in an a.c circuit with an rms current of $5A$, rms voltage of $10V$ and a phase angle of 60° is **A.25W** B.70W C.120W D.125W.
Average power dissipated in an a.c circuit, is given by – $P = IVcos\emptyset$, $I$and V are the rms values. I=5A, V=10V, ∅=60°,
$P = 5\times10cos60° = 25$W.

31. When an alternating current given by $I = 10sin(120\pi)t$ passes through a 12Ω resistor, the power dissipated in the resistor is A.1200W **B.600W** C.120W D.30W.
Compare $I = 10sin(120\pi)t$ with the equation
$I = I_o sin2\pi ft$, I_o=10A, R=12Ω,
$P = P = {I_{rms}}^2 R = \frac{1}{2}{I_o}^2 R = \frac{V^2}{Z}R$,
$P = \frac{1}{2}\times10^2\times12 = 100\times6 = 600$W.

32. A coil of resistance 3Ω has an inductive reactance of 4Ω. The power dissipated in the coil when connected to an A.C supply of $50V_{rms}$ is **A.300W** B.400W C.500W D.600W.
R=3Ω, X_L=4Ω, V=50V, The circuit is non-purely resistive i.e I and V are not phase, the average power is given by – $P = IVcos\emptyset$,
$P = IV\frac{R}{Z}$ ($cos\emptyset = \frac{R}{Z}$), R=3Ω, X_L=4Ω, Z=?,
${Z^2} = {X_L}^2 + R^2 = 3^2+4^2$=9+16 =25 ,
$Z = \sqrt{25} = 5\Omega$, $V_{rms} = I_{rms}Z$, V_{rms}=50V,
$I_{rms} = \frac{V_{rms}}{Z} = \frac{50}{10}$ =10A. $P = IV\frac{R}{Z}$,
$P = 10\times50\times\frac{3}{5} = 300W$.

33. The current is a series R-L-C circuit attains its maximum value when the A.impendance is greater than the capacitive reactance **B.inductive reactance is equal to capacitive reactance** C.inductive reactance is greater than resistance D.capacitive reactance is less than resistance E.inductive reactance is greater than the capacitive reactance .
At resonance, current is maximum, impendance is minimum and equal to the resistance, inductive reactance equals capacitive reactance and net reactance is zero.

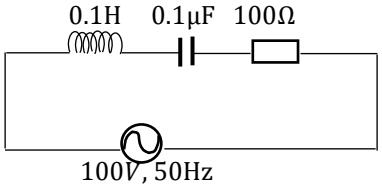
34. In alternating current circuit at resonance, the angle of lead or lag is A.$\frac{\pi}{2}$ **B.0°** C.$\frac{\pi}{3}$ D.π .
At resonance, phase angle or angle of lead or lag is 0° i.e current and voltage are in phase. Power factor $cos\emptyset$ is unity at resonance.

35. In a series L-C circuit, the inductance and the capacitance are 0.5H and 20μF respectively. Calculate the resonant frequency of the circuit A.24.2Hz B.36.7Hz **C.50.3Hz** D.60.5Hz E.80.0Hz .
L=0.5H, C=20μF=20×10^{-6}F, f_o=?
$f_o = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.5\times20\times10^{-6}}} = \frac{1}{2\pi\sqrt{10^{-5}}}$,
$f_o = 50.3$Hz.

36. A circuit contains 100μF capacitor, 10Ω resistor and an inductor L, all connected in series if the resonant frequency of the circuit is 100Hz, find the value of L A.4π^2H B.2π^2H C.$\frac{1}{2\pi}$H **D.$\frac{1}{4\pi^2}$H.**
C=100μF=100×10^{-6}F, R=10Ω, f_o=100Hz, L=?
At resonance, $X_L = X_C$, $2\pi fL = \frac{1}{2\pi fC}$,
$2\pi\times100\times L = \frac{1}{2\pi\times100\times100\times10^{-6}}$,
$L = \frac{1}{2\pi\times100\times2\pi\times100\times100\times10^{-6}} = \frac{1}{4\pi^2}H$.
Alternatively, $f_o = \frac{1}{2\pi\sqrt{LC}}$,
$100 = \frac{1}{2\pi\sqrt{L\times100\times10^{-6}}}$, $200\pi = \frac{1}{\sqrt{L\times100\times10^{-6}}}$,
$(200\pi)^2 = \frac{1}{L\times100\times10^{-6}}$, $4\times10^4\pi^2 = \frac{10^{-4}}{L}$,
$L = \frac{10^{-4}}{4\times10^4\pi^2} = \frac{1}{4\pi^2}$ H.

37. The resonant frequency of an a.c circuit is 1000kHz. If each of the capacitance and inductance is reduced by 50%, and no other changes are made, the resonant frequency will become A.250kHz B.750kHz C.1000kHz **D.2000kHz.**
Initial resonant frequency f_o =1000kHz,
L is reduced by 50% = 50% = $\frac{50}{100} = 0.5$.
C is reduced by 50% = 50%= $\frac{50}{100} = 0.5$.
$f_o = \frac{1}{2\pi\sqrt{LC}}$, $f_o \propto \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.5\times0.5}} = \frac{1}{\sqrt{0.25}} = \frac{1}{0.5} = 2$,
Thus, the new resonance frequency will be twice the initial = 2×1000 = 2000kHz. Alterntaively :
$f_o = \frac{1}{2\pi\sqrt{LC}}$, $f_o\sqrt{LC} = \frac{1}{2\pi}$ = constant,
Initial values : f_o=1000kHz, L=L, C=C ,
Final values : f_o=?, L=50%L=0.5L, C=50%C=0.5C,
$1000\times\sqrt{LC} = f_o\times\sqrt{0.5L\times0.5C}$,
$f_o = \frac{1000\times\sqrt{LC}}{\sqrt{0.25LC}} = 1000\times\sqrt{\frac{LC}{0.25LC}} = 1000\times\sqrt{\frac{1}{0.25}}$,
$f_o = 1000\times\frac{1}{0.5} = 2000$kHz.

37. In the circuit shown below, calculate the energy stored in the inductor at resonance.



A.0.100J **B.0.050J** C.0.010J D.0.005J.
L=0.1H, C=0.1μF, R=100Ω, V=110V, f=50Hz,
At resonance, $Z = R$, $I = \frac{V}{Z} = \frac{V}{R} = \frac{100}{100}$ =1A
Energy stored in an inductor, $E = \frac{1}{2}LI^2$
$E = \frac{1}{2}\times0.1\times1^2$= 0.05$J$.

Jamb past questions on simple a.c circuits : [1992/45,46,1994/49,1995/48,1998/45,1999/35,39,482000/36,2001/38,40,2002/46,2004/22,29,2006/27,2007/39,46,2009/44,2012/43].

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Chapter 26 – Elementary Atomic Physics

● Atomic theories and discovery of atoms

- Dalton's atomic theory – **1.**Matter is composed of discrete units called atoms, which are the smallest, indivisible part of an element. **2.**Atoms can neither be created nor destroyed. Its limitation is seen in radioactivity. **3.**Atoms of the same element are identical in every aspect but differ from atoms of all other elements.

- Limitations of Dalton's theory – 1. It does not recognize the constituents of atoms i.e proton, electron and neutron. **2.**It is limited to matter,elements,molecules and atoms. **3.**The discovery of isotopes shows that atoms of the same element are not exactly the same.

- Thompson's atomic model or plum-pudding or diffuse model – 1. Atoms consists of a sphere of positive charge surrounded by electrons. **2.**Electrons were all alike i.e their specific charge or charge per mass ratio is constant. **3.**Masss of electron is approximately two thousandth the mass of hydrogen.

- Limitations of Thompson's model – 1. It cannot explain the nucleus and its constituents i.e atomic phenomena. **2.**It is limited to atoms and electrons only.

- Rutherford's atomic model or planetary model – 1. Atoms consists of a central positively-charged core called nucleus with electrons revolving round it. **2.**Electrons are negatively charged and are located around the periphery of the nucleus. **3.**Mass of an atoms is concentrated in the nucleus. **4.**A wide empty gap exists between the nucleus of an atom and the orbiting electron.

- Limitations to Rutherford's model – 1. It cannot explain atomic spectra e.g line spectra **2.**It suggest that electrons in atoms revolves round the nucleus and radiates energy continuously and spiral into the nucleus,which is not so. **3.**It is limited to unstable atoms i.e where energy is radiated continuously.

- Bohr's atomic model – 1. Electrons occupy restricted orbit or revolve round the nucleus in certain obits or energy level of definite energy. Electrons are stable and do not emit radiations at these orbits. **2.**Discrete amount of energy is absorbed or radiated when electrons move from one energy level or orbit to the other i.e electron gain discrete amount of energy to move to a higher orbit or energy level and lose same amount of energy in form of radiation to return to the lower orbut or energy level. **3.**Electron in an atom with a single electron e.g hydrogen atom, can occupy any energy level surrounding it,without radiating energy as it moves despite orbital acceleration. **4.**The angular momentum and energy of an electron in any given orbit are quantized. **5.**It explains line spectra of hydrogen and hydrogen-like atoms with one electron. **6.**It can explain all atomic phenomena, stability of atoms

- Limitations of Bohr's model – 1. It cannot explain the spectra lines of atoms with more than one electron. **2.**It provides no insight into what happens during transition form one orbit to another. **3.**An electron moving in one of Bohr's circular orbits forms a current loop and should produce a magnetic moment, but hydrogen atom in the ground state has no magnetic moment due to orbital motion.

- Electron cloud model – It describes an atom as consisting of a small massively positive-charged nucleus,with electron moving rapidly round the nucleus and spending most of its time in high probability region. The density of electron cloud is greater in the region nearer the nucleus than the region farther from the nucleus.

- Atomic structure and properties – An atoms is made up of a tiny missive nucleus and electrons revolving round it.

- The nucleus consists of protons and neutrons. Particles in the nucleus are called nucleons.

- In a neutral atom, the number of protons is same as the number of electrons.

-The mass of protons is same as the mass of neutrons.

-Neutrons forms a shield around the positively charged protons. Hence, the net charge on an atom is electrically zero or neutral.

-An ion is formed when an atom loses or gains electrons.

- An atom of element X is represented by $^{A}_{Z}X$ where A is the mass or nucleon number and Z is the atomic or proton number.$^{A}_{Z}X$ is called the nuclide of X. The number of neutrons in the atom is given by, $n = A - Z$.

● Energy Quqnatization and line spectra

- Energy quantization – Electrons occupy discrete(quantized) energy levels and an energy level can be occupied by only two electrons of opposite spins.

-The gaps between energy levels are called forbidden energy gap i.e they cannot be occupied by electrons.

- Electron in the lowest energy level is said to be in the **ground state** E_0. Excitation of the electron i.e when it absorbs a certain amount of energy, it moves to a higher energy level called the **excited**

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state.

- Line spectra : Absorption and emission

- Atomic spectra – When light of certain wavelength(colours) is passed through the vapour of an atom in the ground state, light in form of radiation is absorbed and they are excited to a higher energy level.This appears as **dark lines** in the radiation and the absorbed radiation is called **absorption spectrum**. An atom in the excited state is untable and would readily emit radiation form of light of certain wavelength(colour) called **emission spectrum,** as they return to their stable ground state. **Emission spectrum is perculiar to a particular atoms and is used for the identification of the vapour of an atom.**

- Line spectra (absorption and emission spectra) are of one wavelength(colour) and are exhibited by atoms of gase(vapours). Continuous spectrum are of range of wavelengths(colours) and are exhibited by atoms of liquids and solids.

- The energy of the radiation is given by planck's equation

– $\boldsymbol{E = hf = \frac{hc}{\lambda}}$ **.** h=planck's constant=$6.63\times10^{-34}Js$, f=frequency of radiation, c=speed of light=$3\times10^{8}ms^{-1}$, λ=wavelength of radiation.

- Energy level – Energy levels occupied by electrons are arranged in an increasing order from bottom to top. The ground state is the first energy level - n=1, $E = E_o$ i.e the first line, second energy level - n=2, E_1 i.e the second line.

- The energy at any energy level is given by –$\boldsymbol{E_n = -\frac{E_o}{n^2}}$**,** $\boldsymbol{n}$=principal quantum number or number of energy level. **-** If an electron is excited from a lower energy level E_1 to a higher energy level E_2, it falls back to E_1 emitting energy which is equal to the energy change or difference given by –

$\boldsymbol{\Delta E = E_2 - E_1 = hf = \frac{hc}{\lambda}}$**.**

- The energy difference or gap between any two energy levels decreases, as the energy level increases from ground state to excited state i.e bottom to top.

$\boldsymbol{E_\infty}$ - - - - - - - - - - - - - **$n=\infty$ (ionization)**
$\boldsymbol{E_3}$ ———————— **$n=4$**
$\boldsymbol{E_2}$ ———————— **$n=3$**
$\boldsymbol{E_1}$ ———————— **$n=2$**

$\boldsymbol{E_o}$ ———————— **$n=1$ (ground state)**

- The ground state energy of hydrogen $\boldsymbol{E_o = -13.6eV}$.

- Transition from higher to lower energy level produce emission spectra while transition from lower to higher energy level produce absorption spectra.

- Emission of greatest energy is with shortest wavelength and highest frequency. $\boldsymbol{E \propto \frac{f}{\lambda}}$**.**

- Ionization energyis the energy change required to remove an electron completely from an atom. Ionization is the transition from the ground state E_0 to the highest energy level E_∞, hence Ionization energy is given by,

$\boldsymbol{E_i = \Delta E = E_\infty - E_o}$**. Ionization energy is the positive value of the ground state energy.**

- The energy of an atom is constant unless the atom is excited and no intermediate energy change occurs.

Examples :

1. Which of the porperties of atomic models does not recognize the consituuents of atom **A.Dalton** B.J.J Thompson C.Rutherford D.Bohr .

2. Which of the following agree with Thompson's model of atom? I.an atom consisted of a sphere of positive charge of electricity with the electrons surrounding the sphere II.electrons were like cathode rays III.electrons were all alike and have the same properties IV.number of electrons per atom is constant and is of the order of the atomic weight V.mass of electron was approximately $\frac{1}{1830}$ of mass of hydrogen i.e nearly two thousandths of hydrogen A.I,II and III B.I,II and V C.I,III and V D.I,II,III and IV **E.I,II,III,IV and V.**

3. Which of the following models produce valid statements throughout except that it cannot explain all atomic phenomena A.Rutherford **B.J.J Thompson** C.Bohr D.Dalton.

Thompson's atomic model explains the properties of electrons but it cannot explain the nucleus and neutrons i.e atomic phenomena.

4. In the Rutherford scattering experiment, a beam of alpha particles was fired at a thin gold film with some of the particles being considered deflected. This shows that A.a gold nucleus contains protons,neutrons and electrons uniformly distributed in a tiny volume **B.the gold nucleus is positively charged and is concentrated in a tiny volume** C.the gold nucleus emitted alpha particles D.the gold nucleus is concentrated in a tiny volume and contains alpha particles

In Rutherford's scattering experiment, alpha particles which are positively charged are fired at the nucleus of a thin gold film, some of the alpha particles which comes in contact with the nucleus are deflected as the nucleus is positively charged and concentrated in a tiny volume.

5. Which of the following is not correct about Rutherford's model of the atom? A.the mass of an atom is concentrated B.the nucleus of an atom is positively charged C.the nucleus is surrounded by electron cloud **D.the model is applicable to atoms with only one electron in the other shell** E.the diameter of the nucleus is of order of 10^{-15}m.

It is Bohr's model that is applicable to atoms with

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only one electron in the outer shell e.g hydrogen atom.

6. Bohr's model of the atom could explain the A.composition of the nucleusB.distribution of charges in an atom C.penetration of an alpha particle thorugh a metal foil D.repulsion of an alpha particle through a metal foil **E.stability of an atom.**

Bohr's model explains that atoms are stable and radiate energy in form of line spectra which are of dicrete values rather than continuous spectra, which denotes instability of an atom.

7. Bohr's atomic model proves most successful for the explanation of the I.structur of the hydrogen atom II.line spectra of the hydrogen atom III.multi-electron atoms. Which of the following statements above is/are correct A.I B.II **C.I and II** D.I and III.

8. Which of the following statements are true in the structure of atoms? A.the atoms of the different elements differ only in the number of protons and electrons **B.the number of protons equals the number of electrons** C.atoms are constructed according to the plan of our solar system D.all of the above E.A and C only.

Atoms of different elements also differ in their number of neutrons. Number of protons equals number of electrons and mass of protons equals mass of neutrons.Atoms are constructed based on the nuclear force binding its constituents together. **Option C is the planetary model by Rutherford.**

9. It is known that a neutron exists in a light atomic nucleus. Which of the following also exists in the nucleus? A.an electron B.a β-particle **D.a proton** E.an atom.

10. An atom is usually electrically neutral because it contains equal number of A.protons and neutrons **B.protons and electrons** C.neutrons and electrons D.photons and protons E.photons and electrons.

11. The net charge on an atom is zero because A.the electrons and the protons that are charged are in the nucleus B.the total charge on the electrons is equal to the total charge on the neutron **C.the neutrons form a shield around the charged protons** D.the number of the electrons in an atom is half the number of protons E.A and B.

The net charge on an atom is zero also because the total charge on the electrons is equal to the total charge on the protons.

12. The mass of a nucleus is the **A.total number of its protons and neutrons** B.number of its protons C.mass of its electrons D.total number of its protons and electrons.

13. When an atom loses or gains a charge, it becomes A.an electron B.a neutron C.a proton **D.an ion.**

14. The nucleons number and the proton number of a neutral atom of an element 238 and 92 respectively. What is the number if neutrons of the atom? A.330 B.238 C.146 C.93 **D.193**

Nucleon number=mass number =A=238, Proton number=atomic number=Z=92, Number of neutrons=$n = A - Z$ =238−92, n =193.

15. Which of the following representations is correct fro an atom X with 36 electrons and 40 neutrons? A.$^{40}_{36}X$ B.$^{36}_{40}X$ C.$^{76}_{40}X$ **D.$^{76}_{40}X$** E.$^{40}_{76}X$.

In a neutral atom, proton number Z =electron number=36. Neutron number=40, Nucleon number=mass number A =40+36=76. The neuclide of an atom is represented as $^{A}_{Z}X = {}^{76}_{40}X$.

16. A nuclide represented as $^{70}_{32}X$ has a neutron-proton ratio of A.0.5 B.0.8 **C.1.2** D.1.5

Mass number Z=70, Atomic number A=32, Number of neutrons n=$A - Z$=70−32=38,

Neutron-proton ratio $= \frac{n}{Z} = \frac{38}{32} = 1.875 \approx 1.2$.

17. Which of the following constituents of an atom moves round the nucleus? **A.Electrons** B.Neutrons C.Nucleons D.Photons E.Protons.

18. Which of the following groups of particles determine the mass of an atom? A.Protons and electrons **B.Neutrons and protons** C.Electrons and neutrons D.Neutrons and photons.

19. Which of the following statements about energy levels in atoms is not correct? A.the level can be represented by horizontal lines where each line indicates a particular energy level B.every atom has a characteristic set of energy level **C.an atom can accept 'packets' of energy of all sizes** D.all energy levels have negative values because the energy of an electron at rest outside an atom is taken as zero

An atom accepts discrete packets of energy called quanta i.e they are of definite amount of energy.

20. Calculate the energy of the third level of an atom if the ground state energy is $-24.8eV$ A.$-8.20eV$ **B.$-2.75eV$** C.$-1.75eV$ D.$-9.20e$

E_0=$-24.8eV$, n=3 i.e third level, E_3=?

$E_3 = -\frac{E_o}{n^2} = -\frac{24.8}{3^2} = -\frac{24.8}{9} = -2.75eV$.

21. In which of the following transitions is the largest quantum of energy liberated by an hydrogen atom when the electron changes levels? [n is the quantum number] **A.n=2 to n=1** B.n=1 to n=2 C.n=2 to n=3 D.n=3 to n=2

- Energy is absorbed when electron transitions is from lower energy level to higher energy level e.g from n=1 to n=2 or n=1 to n=3 or n=2 to n=3.

- Energy is liberated when electron transitions is from higher energy level to lower energy level e.g from n=3 to n=1 or n=3 to n=2 or n=2 to n=1.

- The energy difference or gap between any two energy levels decreases, as the energy level increases from ground state to excited state i.e bottom to top i.e the energy change (energy liberated) from, $E_1 - E_0 > E_2 - E_1 > E_3 - E_2$. Hence,

Demystified Series by Dr. Timothy

the largest quantum energy is liberated when is liberated from n=2 to n=1.

22.

E_∞ _____ $0.0eV$
E_3 _____ $-1.6eV$
E_2 _____ $-3.7eV$
E_1 _____ $-5.5eV$

E_0 _____ $-10.4eV$

The energy levels in the mercury atom is illustrated in the diagram above. Calculate the ionization energy of the atom. A.$+2.2eV$ **B.$+10.4eV$** C.$-10.4eV$ D.$-21.2eV$

Ionization energy, $E_i = E_\infty - E_0$, $E_0 = -10.4eV$, $E_\infty = 0.0eV$, $E_i = 0-(-10.4) = +10.4eV$. OR

Ionization energy is the positive value of the ground state energy $-10.4eV$, $E_i = +10.4eV$.

The table below shows the energy distribution for various levels of an atom. Use it to answer 21 and 22.

Energy level(n)	1	2	3	4
Energy(eV)	-13.6	-3.39	-1.51	-0.85

[$h=6.6\times10^{-34}Js$, $e=1.6\times10^{-19}$C, $c=3.0\times10^{8}$ms^{-1}]

23. Calculate the first excitation energy of the atom A.1.60×10^{-19}J B.1.02×10^{-18}J **C.1.60×10^{-18}J** D.1.02×10J.

Excitation energy is the energy absorbed when electrons move from lower to higher energy level. First excitation energy of the atom is the energy it absorbs when it moves from energy level n=1 to energy level n=2.

1st excitation energy=$\Delta E = E_2 - E_1$,

$E_2 = -3.39eV$, $E_1 = -13.6eV$,

$\Delta E = -3.39-(-13.6) = -3.39+13.6 = 10.21eV$.

[$1eV = 1.6\times10^{-19}$J], $\Delta E = 10.21\times1.6\times10^{-19}$,

$\Delta E = 1.63\times10^{-18}$J $\approx 1.60\times10^{-18}$J.

24. If the atom de-excites from n=2 the wavelength of the emitted radiation is A.2.4×10^{-7}m **B.1.2×10^{-7}m** C.2.4×10^{-9}m D.1.2×10^{-9}m.

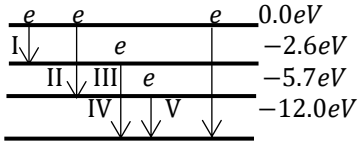
$E_2 - E_1 = 1.63\times10^{-18}$J, $h=6.6\times10^{-34}$Js, $c=3.0\times10^{8}$ ms^{-1}, λ=wavelength. If the atom de-excites from n=2 it comes to n=1, the photon energy ΔE :

$\Delta E = E_2 - E_1 = \frac{hc}{\lambda}$. $1.63\times10^{-18} = \frac{6.6\times10^{-34}\times3\times10^{8}}{\lambda}$,

$\lambda = \frac{6.6\times10^{-34}\times3\times10^{8}}{1.63\times10-18} = 1.2\times10^{-7}$m.

- Photons are packets of light energy.

The diagram below illustrates the energy transition of five electrons of an atom.



Use the diagram to answer questions 23 – 24

25. Which of the transitions will produce the emission of longest wavelength? **A.I** B.II C.III D.IV E.V

Energy change, $\Delta E = E_2 - E_1$

I. $E_2 = 0.0eV$, $E_1 = -2.6eV$, $\Delta E = 0-(-2.6) = 2.6eV$.

II. $E_2 = 0.0eV$, $E_1 = -5.7eV$, $\Delta E = 0-(-5.7) = 5.7eV$.

III. $E_2 = -2.6eV$, $E_1 = -12.0eV$, $\Delta E = -2.6-(-12.0)$, $\Delta E = -2.6+12.0 = 9.4eV$.

IV. $E_2 = -5.7eV$, $E_1 = -12.0eV$, $\Delta E = -5.7-(-12.0)$, $\Delta E = -5.7+12.0 = 6.3eV$.

V. $E_2 = 0.0eV$, $E_1 = -12.0eV$, $\Delta E = 0-(-12.0) = 12.0eV$

I. $2.6eV$ II. $5.7eV$ III. $9.4eV$ IV. $6.3eV$ V. $12.0eV$

The emission of smallest energy will produce the longest wavelength and smallest frequency and vice-versa i.e $E \propto \frac{f}{\lambda}$. Transition in I, with energy change $2.6eV$ will have the longest wavelength and shortest frequency, as it has the smallest energy change.

26. Which will produce the emission if highest frequency? A.I B.II C.III D.IV **E.V.**

Transition in V, with highest energy change $12.0eV$, will produce the highest frequency.

27. The ionization potential of hydrogen is $13.6V$. Calculate the minimum wavelength of the radiation the hydrogen atom can emit. [electronic charge, $e=1.6\times10^{-19}$C, Plancks constant, $h=6.63\times10^{-34}Js$, $c=3.0\times10^{8}$ms^{-1}]

A.1.46×10^{-26}m B.2.71×10^{-24}m **C.9.14×10^{-8}m** D. 1.09×10^{-7}m.

$\Delta E = eV = \frac{hc}{\lambda}$, $\Delta E = E_i$=ionization energy, $V = V_i$=ionization potential, $\lambda = \lambda_i$=minimum wavelength of radiation. The minimum wavelength of the radiation is that which aligns with the ionization potential. $eV_i = \frac{hc}{\lambda_i}$,

$1.6\times10^{-19}\times13.6 = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{\lambda_i}$

$\lambda_I = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{1.6\times10-19\times13.6} = 9.14\times10^{-8}$m.

● **Question Drill –** Calculate the wavelength of the radiation emitted when an electron falls from an energy level of $0.43eV$ to $-13.6eV$. [$1eV = 1.6\times10^{-19}J$, $h=6.63\times10^{-34}Js$, $c=3.0\times10^{8}$ms^{-1}] **Ans=8.86×10^{-8}m.**

28. In a model of the hydrogen atom, the energy levels W_n are given by $W_n = -\frac{R}{n^2}$, n is an integer and R is a constant. Determine the energy released in transition from n=3 to n=2. A.$\frac{R}{5}$ B.$-\frac{R}{4}$ **C.$\frac{5R}{36}$** D.$-\frac{5R}{36}$.

$W_3 = -\frac{R}{3^2} = -\frac{R}{9}$, $W_2 = -\frac{R}{2^2} = -\frac{R}{4}$,

Energy released = $\Delta E = E_3 - E_2 = W_3 - W_2$

$\Delta E = -\frac{R}{9}-(-\frac{R}{4}) = -\frac{R}{9}+\frac{R}{4} = \frac{-4R+9R}{36} = \frac{5R}{36}$.

29. A photon of wavelength λ_o is emitted when an electron in an atom makes a transition form level of energy $2E_K$ to that of E_K. If the electron transits from $\frac{4}{3}E_K$ to E_K level, determine the wavelength of the photon that would be emitted **A.$3\lambda_o$** B.$2\lambda_o$ C.$\frac{3}{4}\lambda_o$ D.$\frac{1}{3}\lambda_o$.

$\Delta E = hf = \frac{hc}{\lambda}$, $\lambda\Delta E = hc$ = constant,

$\lambda_o(2E_K - E_K) = \lambda(\frac{4}{3}E_K - E_K)$, $\lambda_o E_K = \lambda\frac{4E_K - 3E_K}{3}$,

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$\lambda_o E_K = \frac{\lambda E_K}{3}$, $\lambda = \frac{3\times\lambda_o E_K}{E_K} = 3\lambda_o$.

31. When light from a source is sent through a gas I.certain wavelengths are absorbed II.an emission line spectrum results III.a dark line is left. Which of the following is/are correct? A.I B.III C.I and II **D.I and III** E.II and III.

A light sent through a gas produces an absorption spectrum i.e certain wavelengths (colours) are absorbed and a dark line results. It's a result of excitation of an electron in an atom causing transion of the electron from lower to higher energy level.

32. The atoms of a gas are excited to produce a discharge which is examined with a spectrometer. The examination reveals A.a totally dark background B.the colours of the rainbow **C.an emission spectrum** D.an absorption spectrum E.a continuous spectrum.

Emission spectrum is produced when a gas atom is excited by a high voltage to produce a discharge, hence returning from higher energy level to lower energy level. Excitation without any discharge is an absorption spectrum and the background will be totally dark.

33. The minimum energy required to remove an electron from a given atom is known as A.photon energy **B.ionization energy** C.excitation energy D.binding energy E.kinetic energy.

34. The following statements relate to atomic spectrum I.all elements emit and absorb characteristic spectra II.spectra analysis is important method of identifying environmental pollutants III.the chemical composition of stars could be determined using spectra analysis. Which of the statements above are correct? A.I and II **B.I and III** C.II and III D.I,II and III.

● Thermionic emission, X-ray production and Photoelectric emission or phtotoelectricity

- Thermionic emission or Hot cathode emission – It is the emission of electrons from a metallic surface when heated in a vacuum.

- Emission of electron from the anode of a thermionic valve is called secondary emission.

- Heating a valve cathode causes thermal emission from the cathode.

-It is the conversion of heat to electrical energy.

- Anode is made more positive than the cathode i.e anode voltage increases as cathode current decreases.

- The number of electrons emitted per second or rate of emission of electrons R_{Th}depends on the temperature T, mass M, area A and work function W of the cathode, $\boldsymbol{R_{Th} \propto \frac{ATM}{W}}$.

- A **good hot cathode** should have a **high melting point**, **low work function** and **high mechanical strength.**

- Applications of thermionic emission –

1.Thermionic diodes are 'electrical valves' used for **current rectification i.e conversion of a.c to d.c or allowing current to flow in one direction only.**

2.Discharge tube – It produces cathode rays which are stream of fast moving electrons when the cathode is excited.

- Characteristics of cathode rays :

i.They are streams of fast moving electrons which are negatively charged, travelling in a straight line. They are attracted towards the anode,positive plate of an electric field or N-pole of a magnetic field.

ii. They cause greenish glow or flouorescence in glass or zinc sulphide.

iii. They have mass, energy and momentum, hence will turn a light paddle wheel in the tube.

iv. They are highly energetic particles and can produce intense heat on objects which stop them. They can affect photographic plates.

v.They can produce x-rays from high density metals e.g tungsten, when stopped by it.

3. Cathode ray oscilloscope or cathode ray tube – It is used for studying alternate current waveforms, direct measurement of voltage and time and to measure the frequencies and amplitudes of voltages of electronic devices e.g **television picture tubes and computer displays**.

- It contains of – 1.Electron gun – this consists of; heated filament, cathode and anode. The anode is made more positive with respect to the cathode inorder to attract electrons emitted from the cathode and it also acts as a focusing lens to accelerate and focus electron onto a spot. 2.Deflector plates XX' and YY' – they control the path of electron beam, when an alternating p.d is applied across the plates. The XX' pair or X-plates deflects the beam horizontally while the YY' pair oor Y-plates deflects it horizontally. 3.Fluorescent screen.

Similarities between liquid vaporization and thermionic emission :

1.Both processes involves heat.

2.Both occur at the surface of a material.

3.The rate of both phenomenons increases with temperature.

4.Saturation point of particles i.e the particles leaving the surface equals those returning to it, is reached by both processes.

Difference between liquid vaporization and thermionic emission :

1.Vaporization occurs at liquid surface while themionic emission occurs at metallic surface.

2.Vaporization occurs at definte temperation i.e boiling point while thermionic emissionmay not

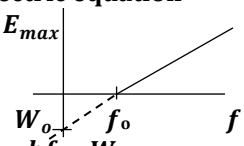
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necessarily occur at a fixed temperature.
3.Water molecules are released in vaporization while electrons are emitted in thermionic emission.
- X-rays or Röentgen rays – They are electromagnetic waves and have no charges. They pass thorugh magnetic or electric fields undeflected. **The production of X-rays is the reverse of photoelectric effect.**
- Characteristics of X-rays ;
1.They are produced when fast moving electrons strike a metal surface or when electrons are accelerated to a heavy target e.g tungsten in a vacuum.
2.They originate from electron change in an atom and X-ray spectra have similar patterns in all metals, as it involves the innermost electrons close to the nucleus. Optical or atomic spectra are different for different metals as it involves the outermost electrons.
3.They are difrracted by crystals as their wavelength is of same order of magnitude as atomic spacings.
4.X-rays are best absorbed by thick lead blocks.
5.They produce ionization of gases and are detected by their effect on a photographic plate or fluorescent screen.
6.Number of electrons emitted per second or rate of emission depends on the intensity while the intensity depends on the hot filaments current.
7.The velocity of the electrons increases with increase in the applied p.d or voltage, $v \propto V$.
8.Hard X-rays have high penetration power, high frequency and short wavelength while Soft X-rays have low penetration power, low frequency and long wavelength.
- X-rays are used – I.for identifying the internal structure of solids. **II.**in medicine for examining bones,teeths and various internal organs. **III.**in industry for revealing flaws in metals and welded joints.
- Photoelectric emission or photoelectric effect – It is the emission of electrons from the surface of a metal by light of appropriate frequency and wavelength or emission of electrons due to elastic collision between incident photon and electrons in an atom of a material. **It is the conversion of light energy to electrical energy.**
- Work function or surface energy of a metal W_o or ϕ is the minimum amount of energy required to liberate photoelectrons from a metallic surface.
- Threshold frequency f_o and **wavelength λ_o** is the minimum amount of frequency wavelength required to liberate photoelectrons.
$W_o = hf_o = \frac{hc}{\lambda_o}$.
- Work function depends on the nature of the metal surface or material of the photocathode e.g alkali metals (group I) such as sodium, caesium e.t.c, emits electrons by visible light while other metals requires ultraviolet lights.
- Threshold frequency depends on work function.
- Photoelectrons are emitted from a metallic surface only when :
i.the frequency of the incident light/photon is greater that the threshold frequency of the metal
ii.energy of the incident light is greater than the workfunction of the metal
iii.the wavelength of the incident light is smaller than the threshold wavelength of the metal.
- Einstein's photoelectric equation –
$E_i = W_o + K.E_{max}$, $hf = hf_o + K.E_{max}$
E_i=energy of incident radiation, W_o=work function or surface energy of the metal surface,$K.E_{max}$=maximum kinetic energy of emitted photoelectrons.
- The number of emitted photoelectrons per second or rate of emission of photoelectrons depends on the intensity of incident light.
- The number of photoelectrons per seconds $\frac{n}{t}$, is equal to the ratio of the photocurrent and electronic charge $\frac{I}{e}$. $\frac{n}{t} = \frac{I}{e}$.
- The maximum kinetic energy of emitted electrons depends on :
I.frequency or wavelength of incident light or photon i.e $K.E_{max} \propto \frac{f}{\lambda}$.
II. Colour of incident photon or light i.e colour of light is dependent on wavelength. e.g the wavelength of the colours of light decreases in the order ; $ROYGBIV$, hence the kinetic energy of emitted photoelectrons increases in that same order. Red light will produce photelectrons of lesser kinetic energy than orange light or green light will photoelectron produce a greater kinetic energy than red light.
III. Work function or nature of the metal surface.
IV. Applied p.d by the cell, $K.E_{max} \propto V$.
- The workdone to accelerate the emitted photoelectron from the cathode towards the anode by an applied p.d V is ; $W = eV$. The rate at which work is done to accelerate the phtotelectron or electrical power of emitted photoelectron is ; $P = IV$. I=photocurrent, e=electronic charge=1.6×10^{-19}C, V=applied p.d.
- The minimum reverse potential required to stop the emission of photoelectrons is **stopping potential or cut off voltage V_o**. It is equivalent to the maximum kinetic energy of photoelectrons. $K.E_{max} = \frac{1}{2}mv_{max}^2 = eV_o$.
- Stopping potential depends on the frequency of the incident radiation , $V_o \propto f$ but independent on the intensity of the incident radiation.

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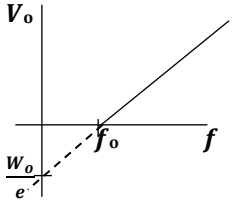
- Graphical representation of Einstein's photoelectric equation –

1. $K.E_{max}$



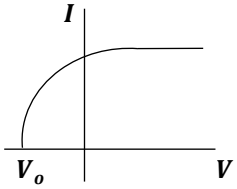
$K.E_{max} = hf - W_o$: $y = mx + c$

Slope of the graph m = **planck's constant** h. Intercept on the $K.E_{max}$-axis = **Work fuction** W_o. Intercept on the f-axis**=Threshold frequency** f_o.



$K.E_{max} = hf - W_o$, $eV_o = hf - W_o$,

$V_o = \frac{h}{e}(f) - \frac{W_o}{e}$. Slope of the graph $m = \frac{h}{e}$, Intercept on the V_o-axis= $\frac{W_o}{e}$. Intercept on the f-axis= f_o.



The graph shows the relationship between photocurrent I and applied p.d V at constant intensity of incident photon or light. Intercept on the negative V-axis = stopping potential V_o

- Application of photoelectric emission or photoelectric effect – e.g photocell

I.Setting off of burglar alarms. **II.**Production of sound from film tracks and in photography to control light intensity. **III.**Control switches in automatic doors or signal devices. **IV.**In photocells. **V.**Photographic light merters.

Examples :

1. The process trough which free electrons leave the surface of a hot metal is known as **A.thermionic emission** B.photo emission C.photon emission D.electron emission.

2. Which of the following is most essential for the production of electrons by thermionic emission? A.electron lens B.deflector plate C.fluorescent screen **D.hot cathode** E.tungsten target .

3. Thermionic emission can occur when A.metal is bombarded by alpha particles **B.certain metals are heated to a very high temperature** C.X-rays fall on a hot surface D.ultra-violet light falls on ametal plate.

4. Which of the following sets will correctly complete the statement below in the order given? In a diode, a heated filament emits …… which may be attracted by the …… causing ….. to flow. A.electrons, cathode, thermions B.electrons, protons, current **C.electrons, anode, current** D.protons, cathode, electrons E.protons, electrode, current.

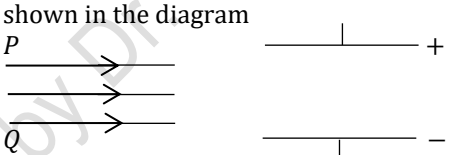
5. Cathode rays I.consists of tiny particles bearing negative electric charges II.consist of fast-moving protons and are deflected in a magnetic field III.are deflected in an electric field but not in a magnetic field IV.can cause the emission of x-rays on hitting a metal target. Which of the above statements are correct? A.I and II B.III and IV **C.I and IV** D.II and III .

Cathode rays are streams of fast-moving electrons and are deflected in both a magnetic and electric field.

6. Which of the following statements about cathode rays is not correct? A.they are bent by a strong field B.they are stream of electrons C.they are bent away by a magnetic field D.they casue fliorescence **E.they are more penetrating than gamma rays**

Gamma rays have the highest penetration power.

7. A stream of cathode rays is projected into an electric field between two plates P and Q as shown in the diagram



Which of the following is correct? The cathode rays A.are accelerated and continued in a straight line B.change direction at the limits of the plate **C.are deflected towards the positive plate** D.are directed towards the negative plate E.are deflected normal to the paper.

Cathode rays are deflected towards the positive plate of an electric field, they are stream of fast-moving electrons which are negatively charged.

8. The cathode ray oscilloscope can be used for the direct measurement of I.resistance II.voltage III.time IV.capacitance A.I and II B.I and III **C.II and III** D.II and IV E.III and IV.

9. In a cathode ray tube, the function of the X-plates is to **A.deflect the electron beam horizontally** B.deflect the electron beam vertically C.reflect the electron beam D.generate suinusoidal waves.

10. Which of the following is not true about X-rays? A.they can destroy living cells B.they are electromagnetic waves C.they are produced when fast moving electrons bombard metal surfaces **D.they are deflected by the magnetic field** E.they travel at the speed of light.

11. Which of the following is not applicable to X-rays? They A.travel in straight lines B.travel with the speed of light **C.are deflected by magnetic field** D.affect photographic plates.

X-rays are electromagnetic waves produced when charged particles are accelerated.They travel in a straight line and move with the speed

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of light, they are not deflected by a magnetic or electric field.

12. The process of detecting a pin mistakenly swallowed by a child is x-ray A.diagnosis B.therapy **C.crystallography** D.mammography

Objects can be detected by diffraction of x-rays through crystals.

13. Which type of radiation does not originate within the nucleus A.Alpha B.Beta C.Gamma D.Neutron **E.X-rays .**

X-rays orginate electron change in the innermost shell of metals.

14. X-rays cannot be used A.to take photographs of bone structure in the body **B.to detect finger prints** C.to detect flaws in metal castings D.to detect alterations to work of art .

Finger prints are transparent to X-rays i.e X-rays can easily pass through finge prints.

15. Which of the following elements is the best absorber of X-rays? A.hydrogen B.oxygen C.calcium D.copper **E.lead .**

X-rays are absorbed by lead, due to its high density.

16. In an operating X-ray tube, a high p.d is applied between the cathode and the anode for the purpose A.heating the cathode B.producing a stream of electrons C.concentrating electrons on the target **D.accelerating the electrons to a very high energy** E.heating the anode.

17. Which of the following phenomena is called phtotelectric effect? A.high energy electons impinge on a metallic anode which then emits photons B.a high energy photon emits proton as it is slowed down **C.a metal absorbs quanta of light and then emits electrons** D.two electrons are created from quantum of light .

Photoelectric emissiom is the emission of electrons from a metal surface, when quantum or packets of light called photons are incident on it.

18. For photoelectric emission, the threshold frequency is the A.the maximum frequency of incident radiation that will eject electron **B.minimum frequency of incident radiation required to eject electrons** C.frequency which results in the emission of the greater number of electrons D.frequency which imparts greatest kinetic energy.

19. In photoelectricity electrons will leave the metal surface when illuminated by light of appropriate frequency, if photon energy is **A.greater than work function** B.less than the work function C.equal to the work function D.equal to the maximum kinetic energy of the electrons .

Inorder for photoelectrons to be emitted from a metal surface ; the energy of incident photon must be greater than the work function or the frequency of incident photon is greater than the threshold frequency.

20. If light with photon energy $3eV$ is incident suitably on the surface of a metal with work function $4eV$, then **A.no electron is emitted** B.the few electrons emitted will have a maximum kinetic energy of $1eV$ C.the few electrons emitted will have maximum kinetic energy $3eV$ D.many eelctrons will be emitted.

Photon energy=energy of incident photon E_i=$3eV$, Workfunction=surface energy W_o=$4eV$. No photoelectron is emitted as the energy of incident photon($3eV$) is less than the work function of the metal($4eV$).

21. Which of the following statements is/are not true about the emission of photoelectrons from metallic surfaces? I.All metallic surfaces emit the same number of photoelectron in a given light II.Most metallic surfaces emit the same number of photoelectrons in visible and infra red light III.A metallic surface emits photoelectrons only if the frequency of the incident light is higher than its own characteristics threshold frequency. A.I B.II C.III **D.I and II** E.II and III .

Statement I and II are not true because the number of photoelectrons emitted from a metallic surface depends on the intentsity of the radiation or incident photon and not on the metal surface i.e work function and threshold frequency of the metal. Only statement III is true.

22. The kinetic energy of a photoelectron ejected from a metal surface illuminated with radiation depends on the I.wavelength of the radiation II.intensity of the radiation III.source of the radiation IV.nature of surface. Which of the aove statements is/are correct? A.I **B.I and IV** C.II and III D.I,II and IV .

The maximum kinetic energy of photoelectron ejected from a metal surface depends on the wavelength of radiation/colour of incident light,frequency of radiation and nature of the surface i.e work function and threshold frequency. It does not depend on the intensity of radiation or source of radiation.

23. The threshold frequency for photoelectric effect depends on the A.frequency of incident light B.intensity of incident light C.p.d between the cathode and the anode **D.material of the photocathode**

Threshold frequency depends on work function which is a characteristic of the metallic surface i.e photocathode.

24. In photoelectric effect, the number of electrons emitted per second from a metallic surface is proportional to the **A.intensity of the incident radiation** B.frequency of incident radiation C.energy of the incident radiation D.work function of the metal .

The rate of emission of photoelectrons or number of emitted photoelectrons per second depends on the intensity of incident photon.

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25. A metallic surface is exposed in turn to green, red and violet lights. The maximum kinetic energy of photoelectrons ejected from the surface increases in the order A.green, red, violet B.violet, green, red **C.red, green, violet** D.violet, red, green.
The wavelength of the colours of light in visible spectrum decreases in the order – red,orange,yellow,green,blue,indigo and violet [$ROYGBIV$], red as the highest wavelength and violet has the lowest wavelength, hence the maximum kinetic energy of electrons increases in that same order as $E \propto \frac{1}{\lambda}$. The maximum kinetic energy of photeelctrons increases in the order red, green and violet.
26. Which of the following statements about photoelectrons is correct? A.A faint green light produces photoelectrons with less kinetic energy than a bright red light B.A red light releases smaller number of electrons than a green light **C.A faint green light produces photoelectrons with greater kinetic energy than a bright red light** D.A red light produces more photoelectrons than a green light.
27. The threshold frequency for a metal surface is 3.0×10^{14}Hz, the work function for the surface is? A.4.54×10^{48}J B.3.00×10^{-16}J C.2.00×10^{-18}J **D.1.98×10^{-19}J.** [Take Planck's constant=6.60×10^{-34}Js] .
f_o=3.0×10^{14}Hz, h=$6.60\times10^{-34}Js$, W_o=?,
Work function or surface energy $W_o = hf_o$,
$W_o = 6.60\times10^{-34}\times3\times10^{15} = 1.98\times10^{-19}$J.
28. The work function of a metal is $2.7eV$. Which of the following pairs correspond to the threshold frequency and wavelength of the metal respectively? [h=$6.60\times10^{-34}Js$, c=3×10^{8}ms^{-1}, $1eV$=$1.6\times10^{-19}J$] A.4.6×10^{-26}Hz, 6.5×10^{34}m B.4.6×10^{-7}Hz, 6.5×10^{14}m **C.6.5×10^{14}Hz, 4.6×10^{-7}m** D.6.5×10^{14}Hz, 4.6×10^{7}m E.6.5×10^{34}Hz, 4.6×10^{-26}m
h=$6.60\times10^{-34}Js$, c=3×10^{8}ms^{-1}, W_o=$2.7eV$,
$1eV$=1.6×10^{-19}J, W_o=$2.7\times1.6\times10^{-19}$=$4.32\times10^{-19}$J,
$W_o = hf_o = \frac{hc}{\lambda_o}$, $W_o = hf_o$, $f_o = \frac{W_o}{h} = \frac{4.32\times10^{-19}}{6.6\times10^{-34}}$,
$f_o = 6.5\times10^{14}$Hz. $W_o = \frac{hc}{\lambda_o}$, $\lambda_o = \frac{hc}{W_o}$,
$\lambda_o = \frac{6.6\times10^{-34}\times3\times10^{8}}{4.32\times10^{-19}} = 4.6\times10^{-7}$m.
29. A metal having a work function of $4.65eV$ is illuminated with a radiation of $6.86eV$. The kinetic energy of the electrons emitted from the metal surface is **A.$2.21eV$** B.$4.24eV$ C.$6.82eV$ D.$11.51eV$
Energy of incident phton, E_i=$6.86eV$, Work function, W_o=$4.65eV$, maximum kinetic energy, $K.E_{max}$=? $E_i = W_o + K.E_{max}$, $K.E_{max} = E_i - W_o$
$K.E_{max} = 6.86-4.65 = 2.21eV$.
31. Caesium has a work function of $3\times10^{-19}J$. The maximum kinetic energy of liberated electrons when it is illuminated by light of frequency of 6.7×10^{14}Hz is **A.$1.42\times10^{-19}J$** B.$3.00\times10^{-19}J$ C.$4.42\times10^{-19}J$ D.$7.42\times10^{-19}J$ [h=$6.60\times10^{-34}Js$]
W_o=$3\times10^{-19}J$, f=6.7×10^{14}Hz, h=$6.60\times10^{-34}Js$,
$E_i = W_o + K.E_{max}$, $K.E_{max} = E_i - W_o$,
$K.E_{max} = hf - W_o$ $[E_i = hf]$
$K.E_{max}$ =$6.60\times10^{-34}\times6.7\times10^{14}-3\times10^{-19}$,
$K.E_{max}$ =$4.422\times10^{-19}-3\times10^{-19}$=$1.42\times10^{-19}$J.
32. Light of energy $5.0eV$ falls on a metal of work function $3eV$and electrons are liberated. The stopping potential is A.15.0V B.8.0V **C.2.0V** D.1.7V.
E_i=$5eV$, W_o=$3eV$, $K.E_{max}$=?, $K.E_{max} = E_i - W_o$,
$K.E_{max} = 5-3 = 2eV$, $K.E_{max} = eV_o$, stopping potential V_o=?, $V_o = \frac{K.E_{max}}{e} = \frac{2eV}{e} = 2V$.
33. Light of wavelength 5.00×10^{-7}m is incident on a material of work function $1.90eV$. Calculate the stopping potential A.0.477V B.0.457V **C.0.575V** D.0.375V [h=6.60×10^{-34}Js, c=3×10^{8}ms^{-1}, $1eV$=1.6×10^{-19}J, e=1.6×10^{-19}C]
λ=5.00×10^{-7}m, W_o=$1.90eV$=$1.90\times1.6\times10^{-19}$,
W_o=$3.04\times10^{-19}J$,c=3×10^{8}ms^{-1}, h=$6.60\times10^{-34}Js$,
electronic charge, e=1.6×10^{-19}C, $E_i = \frac{hc}{\lambda}$,
$K.E_{max} = E_i - W_o$, $eV_o = E_i - W_o = \frac{hc}{\lambda} - W_o$,
$1.6\times10^{-19}\times V_o = \frac{6.6\times10^{-34}\times3\times10^{8}}{5\times10^{-19}} - 3.04\times10^{-19}$,
$1.6\times10^{-19}V_o$ =$3.96\times10^{-19}-3.04\times10^{-19}$,
$1.6\times10^{-19}V_o$ = 0.92×10^{-19},
$V_o = \frac{0.92\times10^{-19}}{1.6\times10^{-19}} = 0.575V$.
33. The energy E of a photon and its wavelength are related by $E\lambda = X$. The numerical value of X is A.6.60×10^{-26} **B.1.99×10^{-25}** C.1.99×10^{-27} D.6.60×10^{-28} [h=$6.63\times10^{-34}Js$, c=3×10^{8}ms^{-1}]
$E = \frac{hc}{\lambda}$, $E\lambda = hc$, $X = E\lambda = hc$,
$X = 6.63\times10^{-34}\times3\times10^{8} = 1.99\times10^{-25}$.
34. An electron is accelerated from rest through a potential difference of 18.2kV in vacuum. If electronic charge=1.6×10^{-19}C and mass of electron=9.1×10^{-31}kg, the maximum speed acquired by the electron is A.4.00×10^{7}ms^{-1} B.5.66×10^{7}ms^{-1} **C.8.00×10^{7}ms^{-1}** D.1.13×10^{8}ms^{-1}.
Kinetic energy of electron=Work done on electron, $K.E_{max} = W$, $\frac{1}{2}mv^2 = eV$,
m=mass of electron=9.1×10^{-31}kg, v=maximum speed of electron=? e=electronic charge or charge on electron=1.6×10^{-19}C, V=accelerating p.d=18.2kV=$18.2\times10^{3}V$,
$\frac{1}{2}\times9.1\times10^{-31}\times v^2$ =$1.6\times10^{-19}\times18.2\times10^{3}$,
$4.55\times10^{-31}v^2$ =29.12×10^{-16},
$v^2 = \frac{29.12\times10^{-16}}{4.55\times10^{-31}} = 6.4\times10^{15}$,
$v = \sqrt{6.4\times10^{15}}$ =8.00×10^{7}ms^{-1}.
35. If the frequency of an emitted X-ray is 1.6×10^{16}Hz, the accelerating potential is A.6.6V **B.66.3V** C.663.0V D.6630.0V
[e=1.6×10^{-19}C, h=6.63×10^{-34}Js]

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Workdone in accelerating electrons, $W = eV$,
Energy of X-rays $E = hf$, $W = E$, $hf = eV$,
$V = \frac{hf}{e} = \frac{6.63\times10^{-34}\times1.6\times10^{16}}{1.6\times10^{-19}} = 6.63V$.

36. A functioning cathode rat tube generates 16.00mA. Determine the number of rays flowing per second [e=1.6×10^{-19}C] A.10^{-15} B.10^{-14} **C.10^{15}** D.10^{14}.
I=16.00mA=0.016A, e=1.6×10^{-19}C,
Number of rays flowing per seconds, $\frac{n}{t} = \frac{I}{e}$,
$\frac{n}{t} = \frac{0.016}{1.6\times10^{-19}} = 10^{15}s^{-1}$.

37. In which of the following is the principle of photoelectric effect put in use? I.Photographic light meters II.Activation of automatic doors III.Setting off of burglary alarms IV.Production of sound from film tracks A.I and II B.III and IV C.I,II and III **D.I,II,III and IV**.

● Radioactivity and Nuclear reactions

- Radioactivity – It is the emission of high energy radiations due to spontaneous decay of the nucleus on an atom. Natural radioactivity occurs in heavy elements i.e elements with unstable nuclei. Artificial radioactivity occurs by bombarding stable nuclide with high energy protons, neutrons e.t.c.

- Radiations emiited during radioactivity –

1.Alpha(α) particle – I. It is a helium nucleus (^4_2He). **II.**It has a large mass, low velocity, high momentum. **III.**It is positively charged. **IV.**It is deflected towards the negative plate in an electric field or S-pole in a magnetic field. It is slightly defelcted due to its mass. **V.**Highest ionizing effect. **VI.**It has the weakest penetrating power hence, stopped by a sheet of paper. **VIII.**Travels with a constant speed different from the speed of light.

2.Beta(β) particle – I. It is a fast moving electrons ($^0_{-1}e$). **II.**It has a small mass, high velocity, low momentum. **III.**It is negatively charged. **IV.**It is deflected towards the positive plate in an electric field or N-pole in a magnetic field. It is more strongly deflected due to its small mass. **V.**Intermediate ionizing effect.
VI. Intermediate penetrating power hence, stopped by an aluminium sheet/foil. **VII.**Travels with a variable speed close to the speed of light.

3.Gamma(γ) rays – I. Electromagnetic radiation of shortest wavelength and highest frequency **II.**It is massless/negligible mass, high velocity and negligible momentum. **III.**It has no charge **IV.**It pass through a magnetic or electric field undeflected in a straight line. **v.**Least ionizing effect. **VI.**Highest penetrating power hence, stopped by a thick lead block. **VII.**Travel with the speed of light.

- Only charged particles e.g cathode rays, electrons, protons, alpha particles and beta particles are deflected or deviated in a magnetic or electric field. Gamma rays, neutrons and X-rays don't have a charge hence are not deflected or deviated in any field.

- Radioactive decay – Spontaneous disintegration of an unsable nuclide to form a stable nuclide with an accompany emission of radiation e.g α,β or γ.

1.Alpha decay – It is the emission of alpha particle represented by helium nucleus (^4_2He). It is accompanied with a decrease in the nucleon or mass number by 4 and decrease in atomic number or nuclear charge by 2. e.g
$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^4_2He$ (α-decay)

2.Beta decay – It is the emission of beta particle represented by electron ($^0_{-1}e$). It is accompanied by an increase in the atomic number or nuclear charge by 1 and no change in the mass number.
$^{234}_{90}Th \rightarrow {}^{234}_{91}Pa + {}^0_{-1}e$ (β-decay)

3.Gamma decay – It is the emission of gamma rays represented as $^0_0\gamma$. It is not accompanied by a change in mass or atomic number. **It can follow any other decay.**

4.Positron decay (positive electron) – It is represented by ^0_1e. It is accompanied by no change is mass number and a decrease in atomic number by 1. $^1_1H \rightarrow {}^1_0n + {}^0_1e$ (positron decay).

- In balancing the equations of nuclear reactions, the total mass numbers in the left hand side must be equal to the total mass numbers on the right hand side and the total atomic numbers on the left hand side must be equal to the total atomic numbers on the right hand side.

- Activity of radioactive elements or count rate or decay rate – It is the numer of emission of radiation or disintegration per unit time.

- Activity is not affected by chemical or physical factors like temperature, pressure, humidity or any external condition.

- Actvity depends on the ionizing ability of the characteristic radiation and the number of atoms originally present. Activity of a radioactive sample also decreases with time.

- Decay constantλ– It is the rate of decay per unit atom of a substance or the ratio of decaying atoms per seconds to the total number of atoms.

- Half life $t_{1/2}$ – It is the time taken for a given quantity of a radioactice material or the activity of a nuclide to decay/reduce to half its original value.

- Longer half life denotes low activity and low decay constant i.e more stable nuclide while shorter half life denotes high activity and high decay constant i.e less stable nuclide.

- The number N of radioactive nuclei present at any time t during radioactive decay is given by –
$\boldsymbol{N = N_o e^{-\lambda t}}$, $\boldsymbol{N = N_o\left(\frac{1}{2}\right)^n}$, $\boldsymbol{\frac{N}{N_o} = \left(\frac{1}{2}\right)^n \frac{N_o}{N} = 2^n}$.

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$\boldsymbol{n = \frac{t}{t_{1/2}}}$. N=Quantity remaining after decay, N_o=Initial quantity, n=number of half lives, t=time taken for decay, $t_{\frac{1}{2}}$=half life.

- Fraction remaining after decay F_R –

$\boldsymbol{F_R = \frac{N}{N_o} = (\frac{1}{2})^n}$.

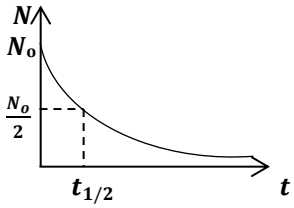
- Fraction lost or decayed after decay F_D –

$\boldsymbol{F_D = 1 - \frac{N}{N_o} = 1 - \left(\frac{1}{2}\right)^n}$.

- Decay constant is related to half life by –

$\boldsymbol{\lambda = \frac{In2}{t_{1/2}} = \frac{0.693}{t_{1/2}}}$.

- Graphical representation of activity or amount of a radioactive sample against time –



- The graph is an exponential curve i.e the amount of radioactive substance decreases with time.
- The time taken for the amount of the sample to be half its initial amount, $\frac{N_o}{2}$ is its half life $t_{1/2}$.
- As the curve does not touch the time axis, the sample can never vanishes completely i.e all the radioactive sample cannot be decayed completely.

- Nuclear reactions – Reactions which involves the neclei of atoms e.g nuclear fission and fusion.

1. Nuclear fission – It is the splitting a heavy nucleus into smaller nuclei(which are also radioactive) by bombarding the **fission element i.e uranium,** with neutrons and is accompanied by the release of huge amount of energy in form of heat.More neutrons are also produced which leads to a **chain reaction** e.g bombarding uranium-235 ($^{235}_{92}U$) by slow moving electrons.

- Nuclear fission is applied in nuclear reactors and atomic bombs.

- Nuclear reactors – Energy realesed nuclear fission due to chain reaction is used generate electricity. It consists of a **nuclear fuel** e.g uranium (fissionable materiel), **moderators** e.g graphite rods and heavy water, which prevents the chain reaction from getting out of control by slowing down fast moving neutrons i.e reduce speed of neutrons, **control rods** e.g boron and cadmium rods, which controls the level of fission by absorbing neutrons intermittently or controls the rate of neutron production, **cooling systems** e.g water or liquid sodium, to reduce excess heat produced in the reaction.

- The major component in a nuclear reactor used to control chain reaction are moderators and control rods.

2. Nuclear fusion – It is the combination of two or more light nuclei to form a heavier nucleus (which are not radioactive) with the release of a very huge amount of energy. **The fusion element is hydrogen.** e.g fusion of deuterium ^2_1H and tritium ^3_1H isotopes of hydrogen to form helium ^4_1He.

- Nuclear fusion occurs in the sun, stars and source of energy in hydrogen bombs or fusion bombs.
- Nuclear fusion occurs at very high temperature hence it is called a **thermonuclear reaction.**

- Nuclear fusion releases much amount of energy than nuclear fission.

- Artificial transmutation – This is the change of the nucleus of an atom to another nucleus by artificial radioactivity when bombarded with neutrons or protons or α-particles , e.g $^{27}_{13}Al + ^4_2He \rightarrow ^{30}_{15}P + ^1_0n$

- Isotopes – They are nuclei of the same element with the same atomic number but difeerent mass number due to difference in the number of neutrons.

- They have the same chemical properties but different physical properties.
- They can be separated by physical process and not chemical processes e.g deflection in a magnetic field or diffusion.
- Isotopes of an element can be formed by irradiation or bombardment artificially by neutron ^1_0n. For example ; if nitrogen (N) has 25 isotopes and the lightest isotope is represented as $^{14}_7N$. We can deduce the other isotopes as follows –

Second isotope $\rightarrow ^{14}_7N + ^1_0n \rightarrow ^{15}_7N$ i.e (1st isotope + 1neutron, 1n).

Third isotope $\rightarrow ^{15}_7N + ^1_0n \rightarrow ^{16}_7N$ or $^{14}_7N + 2^1_0n \rightarrow ^{16}_7N$ i.e (2nd isotope + 1 neutron or 1st isotope + 2 neutrons).

Fourth isotope $\rightarrow ^{16}_7N + ^1_0n \rightarrow ^{17}_7N$ or $^{14}_7N + 3^1_0n \rightarrow ^{17}_7N$ i.e (3rd isotope + 1 neutron or 1st isotope + 3 neutrons)

Hence the twenty-fifth (25th isotope) $\rightarrow ^{14}_7N + 24^1_0n \rightarrow ^{38}_7N$ (1st isotope + 24 neutrons).

- Binding energy – It is the amount of energy that holds or binds nucleons(protons and neutrons) together in the nucleus or the amount energy that will be required to disintegrate the nucleus when formed.

- In nuclear reactions, the mass of the reactants is always greater than the mass of the products. The mass difference is called **mass defect Δm.** Einstein's mass-energy equation shows that the loss in mass is , accounted for the nuclear energy E released in form of heat in nuclear reactions.

$\boldsymbol{E = \Delta mc^2}$. E=energy released, Δm=mass defect or difference between the masses before and after reaction, c=speed of light.

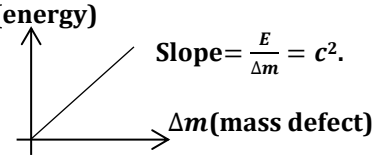
Reactant=mass of protons + mass of neutrons

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Product=mass of nucleus formed. The mass of the products and reactants are in unified atomic mass unit ***u***. $1u$=931MeV, $1MeV$=10^6eV. MeV=mega electron-volt. $1eV$=1.6×$10^{-19}J$.

- Binding energy $= (\boldsymbol{M_R - M_P})\boldsymbol{u}$, M_R=mass of reactants, M_P=mass of products.

- The graphical relationship between energy released and the mass defetct when mass is converted to energy is given as –



- Application of radioisotopes –

1.In medicine – It is used to destroy cancerous cell or treat cancer, iodine-131 is used to threat thyroid disorders, they can also be used as medical tracers for detecting defcts in the body and to sterilize surgical instuments.

2.In agriculture – Mutations induced by radiation is used to yield more desirable and adpted species.

3.In industries – Defects in metals and controlling filling of tins are checked by cobalt-60.

4.In dating – Carbon-14 can be used to predict the age of natural materials e.g rocks,earth, trees e.t.c.

Examples :

1. Which of the following features is not a characteristic of natural radioactivity? A.radioactivity is a nuclear phenomenon **B.The radioactivity of an element is affected by electric and magnetic fields in the surroundings** C.radioactivity is exhibited only by elements of mass number greater than 206 D.radioactive substances emit three radiations; α-rays, β-rays and γ-rays E.the radioactive element is transformed into a new element **.**

The activity of radioactive elements is not affected by physical or chemical factors, electric and magnetic fields. It depends on the ionization ability of the characteristic radiation (with α-particles ionizing most) and the number of atoms originally present.

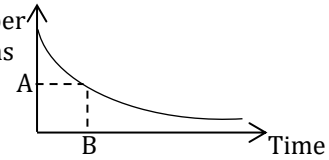
2. The rate of decay of atoms in a radioactive sample is proportional to A.the half-life period B.number of decayed atoms **C.number of undecayed atoms** D.the mass(nucleon) number of the atom E.the atomic(proton) number of the atom

Rate of decay of atoms in a radioactive sample is directly proportional to the number of atoms of the sample present i.e undecayed i.e $-\frac{dN}{dt} \propto N$.

$\frac{dN}{dt} = -\lambda N$.

3. The activity of a radioactive substance depends in A.temperature and purity B.purity and age C.temperature and age D.age, purity and temperature **E.none of the above.**

4. Number of atoms



The points marked A and B respectively represents A.Half number of atoms initially presentand time of decay B.Decay constant and half life C.Half number of decayed atoms and half-life **D.Half number of atoms initially present and half-life .**

A represents half number of atoms originally present $\frac{N_o}{2}$**,** B represents the half- life $t_{1/2}$.

5. Which of the following rays originated from the nucleus of an atom A.X-rays B.cathode rays C.Ultra violet rays **D.Gamma rays** E.Infra red rays.

6. Which of the following is not correct about radioactivity? **A.gamma radiation has an electrical charge** B.gamma rays have negligible charge C.a beta particle has a negative charge D.an alpha particle has a positive charge E.the velocity if gamma rays is the same as that of light.

7. Which of the following is true of the particles emitted in radioactive disintegration? A.the alpha particles is helium nucleus ^3_2He B.the alpha particle is lighter than the beta particle **C.the beta particle is deflected more than alpha particle in a magnetic field** D.the particle is deflected more than beta particle in a magnetic field

Alpha particles is helium nucleus ^4_2He. Alpha particles are heavier than beta particles. Beta particles deflect more than alpha particles in a magnetic and electric field, as they are lighter than alpha particles.

8. Which of the following may be found in light nuclei? I.β-particles II.protons III.neutrons IV.α-particles A.I and II B.I and III C.II and IV **D.II and III** E.I,II and III .

Protons and neutrons make up nucleons which are particles in the nucleus. α and β particles are radiations.

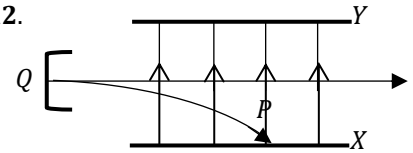
9. Which of the following rays are deviated by magnetic field? I.cathode II.alpha III.beta IV.gamma. **A.I,II and III** B.I,II and IV C.I,III and IV D.II,III and IV E.I,II,III and IV.

10. Of the following types of radiation, one that does not originate from the nucleus is A.alpha B.beta **C.x-rays** D.gamma.

11. Which of the following cannot be deflected by an electric field or a magnetic field? I.α-rays II.β-rays III.γ-rays IV.x-rays A.I and II **B.III and IV** C.I and III D.II and IV.

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12.



A beam of radiation passes through an electric field located between plates X and Y in the figure above. If P is deflected downwards while Q passes undeflected, then P and Q could be, respectively I.β-rays, γ-rays II.α-rays, γ-rays III.β-rays, x-rays IV.α-rays, x-rays. Which of the above is/are correct? A.I B.II **C.I and III** D.II and IV **.**

The electric field lines are directed way from plate X and towards plate Y, hence plate X is positively charged and plate Y is negatively charged.P deflects towards positively charge plate X, hence P is negatively charge i.e β-rays. Q is undeflected i.e x-rays or γ-rays.

13. Which of the following radiations will pass through a sheet of paper? I.Alpha rays II.Beta rays III.Gamma rays. A.I B.II C.III **D.II and III**

Alpha particles have low penerating power, hence is stopped by a sheet of paper and will not pass through it. Beta particles have a fairly high penentration power,hence will pass through a sheet of paper. Gammay rays have the highest penetrating power, hence will pass through a sheet of paper.

14. Which of the following is correct? A.alpha particles have large mass and high penetrating power B.alpha particles have a small mass and a low penetrating power C.beta particles have no mass and a high penetrating power D.beta particles have a large mass and a low penetrating power **E.gamma rays have no mass and a high penetrating power .**

Alpha particles have a large mass and a low penetrating power. Beta particles have low mass and fairly high penetrating power.

15. Which of the following travels in a vacuum with a velocity deifferent from that of light? A.radio waves B.ultraviolet waves **C.alpha rays** D.gamma reays E.x-rays.

16. One of the emissions of radioactive material has the following properties I.momentum ii.deflection by a magnetic field III.fairly high penetrating power. The emission is A.alpha particles **B.beta particles** C.gamma radiation D.X-rays E.neutrons.

Beta particles have low momentum, deflected towards the N-pole in a magnetic field and has a fairly high penetrating power.

17. When an alpha particle, beta particle, a proton and an electron are given the same linear speed in specific direction, the linear momentum of the I.alpha particle is the greatest II.proton is the least III.electron is equal to that of beta particle. Which of the following statements above is/are correct? A.I B.II C.III D.I and II **E.I and III .**

Alpha-particle=$^{4}_{2}He$, Beta-particle=$^{0}_{-1}e$, proton= $^{1}_{1}H$, electron=$^{0}_{-1}e$. Momentum increases with mass. Alpha particle is most massive hence, has the highest momentum. Momentum is least in beta particles, the momentum of proton is intermediate,due to its low mass. Beta particles and electrons have least momentum due to their extremely small mass.

18. The main difference between X-rays and gamma rays lies in their A.ionizing ability B.absorption plate **C.mode of production** D.mode of propagation.

19. Which of the following arrangements of radiations is in order of decreasing penetrating power? A.α, γ, β B.α, β, γ C. β, γ, α **D.γ, β, α.**

19. Which of the following are emited from a radioactive substance without altering the mass number or atomic number of the substance A.proton B.alpha-particles C.neutron **D.gamma rays** E.electrons.

20. Which of the following are true for α-decay I.mass number decreases by four II.atomic number decreases by two III.mass number does not change **A.I and II** B.II and III C.I and III D.I,II and III.

21. In a radioactive series,

$^{235}_{92}L \rightarrow ^{235}_{93}M \rightarrow ^{235}_{94}N \rightarrow ^{231}_{90}O$. The particles emitted are respectively **A.β, β, α particles** B.α, γ, β particles C.γ, α, β particles D.α, α, β particles D. β, α, β particles .

$^{235}_{92}L \rightarrow ^{235}_{93}M + ^{0}_{-1}e$. In this nuclear reaction, the mass number is unchanged but the atomic number increases by 1 i.e β-particle($^{0}_{-1}e$) is emitted.

$^{235}_{93}M \rightarrow ^{235}_{94}N + ^{0}_{-1}e$, the mass number is unchanged but the atomic number increases by 1 i.e β-particle($^{0}_{-1}e$) is emitted.

$^{235}_{94}N \rightarrow ^{231}_{90}O + ^{4}_{2}He$, the mass number decreases by 4 but the atomic number decreases by 2 i.e α-particle($^{4}_{2}He$) is emitted. Therefore, the emission is β, β, α particles.

22. In the radioactive series :

$^{235}_{92}W \rightarrow ^{235}_{93}X \rightarrow ^{231}_{91}Y \rightarrow ^{231}_{92}Z$. The particles emitted are respectively A. α, α, β particles B.α, β, α particles C.β, β, α particles **D.β, α, β particles.**

$^{235}_{92}W \rightarrow ^{235}_{93}X + ^{0}_{-1}e$, the mass number is unchanged but the atomic number increases by 1 i.e β-particle($^{0}_{-1}e$) is emitted.

$^{235}_{93}X \rightarrow ^{231}_{91}Y + ^{4}_{2}He$, the mass number decreases by 4 but the atomic number decreases by 2 i.e α-particle($^{4}_{2}He$) is emitted.

$^{231}_{91}Y \rightarrow ^{231}_{92}Z + ^{0}_{-1}e$, the mass number is unchanged but the atomic number increases by 1 i.e β-particle($^{0}_{-1}e$) is emitted. Therefore, the emission is β, α, β particles.

23. The effect of emission of gamma rays from the atom is to reduce the A.size of the atom

Demystified Series by Dr Timothy

B.energy of the atom C.atomic number of the atom D.mass number of the atom.

24. Which of the folllwoig nstatements is not true of γ-rays? A.γ-rays belong to the electromagnetic spectrum B.γ-rays do not carry an electron charge **C.γ-rays obey the inverse cube law in air** D.the thickness of aluminium which stopes particles has negligible effect on γ-radiation E.the difference between γ-rays and X-rays is that γ-rays originate from energy changes in the nuclei of atoms while X-rays come from energy changes associated with the electron surface structure of atoms.

Gamma rays(γ-rays) is an electromagnetic spectrum. **All electromagnetic waves obey inverse square law.** Gamma rays pass through aluminium sheet due to its high penetrating power while alpha and beta particles are stopped aluminium sheet and will not pass through it.

25. A certain radioisotope ${}^{235}_{92}U$, emits four alpha particles and three beta particles. The mass number and atomic number of the resultant element respectively are **A.219 and 87** B.84 ans 223 C.223 and 87 D.219 and 81 .

${}^{235}_{92}U \rightarrow 4\,{}^{4}_{2}He + 3\,{}^{0}_{-1}e + {}^{A}_{Z}X$

$235=4(4)+3(0)+A$, $92=4(2)+3(-1)+Z$,

$235=16+A$, $92=8-3+Z=5+Z$,

$A=235-16=219$, $Z=92-5=87$.

26. The radioisotope ${}^{235}_{92}U$ decays by emitting two alpha particles, three beta particles and a gamma ray. What is the mass number and atomic number of the resulting daughter element? A.91 and 227 B.92 and 238 **C.227 and 91** D.215 and 88 .

${}^{235}_{92}U \rightarrow 2\,{}^{4}_{2}He + 3\,{}^{0}_{-1}e + {}^{0}_{0}\gamma + {}^{A}_{Z}X$

$235=2(4)+3(0)+A$, $92=2(2)+3(-1)+0+Z$,

$235=8+0+A$, $92=4-3+Z=1+Z$,

$A=235-8=227$, $Z=92-1=91$.

27. Determine the value of x from nuclear reaction below, ${}^{222}_{86}Rn \rightarrow {}^{x}_{88}Ra + 2\beta + \text{energy}$, A.221 **B.222** C.223 D.224 .

$\beta = {}^{0}_{-1}e$, ${}^{222}_{86}Rn \rightarrow {}^{x}_{88}Ra + 2\,{}^{0}_{-1}e + \text{energy}$,

$222 = x + 2(0)$, $x=222$.

28. ${}^{226}_{86}Ra \rightarrow Y + {}^{222}_{86}Rn$. In the nuclear reaction above, Y is A.a beta particle **B.an alpha particle** C.a gamma ray D.an electron.

${}^{226}_{88}Ra \rightarrow {}^{A}_{Z}Y + {}^{222}_{86}Rn$,

$226 = A + 222$, $88 = Z + 86$,

$A=226-222=4$, $Z = 88-86 = 2$.

${}^{A}_{Z}Y = {}^{4}_{2}Y = {}^{4}_{2}He$ i.e alpha particle.

29. Which of the following is the correct definition of half life? A.the average life time of a radioactive material **B.time it takes a radioactive material to decay to half of its original quality** C.half the period a radioactive material decay completely D.half of the decay constant E.the decay constant times half life time.

30. A radioactive substance of initial mass 8g has a half-life of 10days. 7g of the substance would have disintegrated after **A.30 days** B.20 days C.5days D.2½days .

Initial quantity, N_o=8g, quanity decayed N_D=7g, quantity remaining after decay, N=8−7=1g, $n = \frac{t}{t_{1/2}}$, n=number of half-lives, time taken for decay t, half-life $t_{1/2}$=10days.

$\frac{N_o}{N} = 2^n$, $\frac{8}{1} = 2^n$, $8 = 2^n$, $2^3 = 2^n$, n=3,

$n = \frac{t}{t_{\frac{1}{2}}}$, $3 = \frac{t}{10}$, t =3×10=30days.

31. Radium has a half life of 1600years. A lead box contained 24×10⁶atoms of randium in 1976. How many atoms of randium will remain in the box in the year 5176. A.24×10⁶atoms B.18×10⁶atoms C.12×10⁶atoms **D.6×10⁶atoms** E.no atoms $t_{1/2}$=1600yrs, t=5176−1976=3200yrs, N_o=24×10⁶atoms, N=?

$n = \frac{t}{t_{1/2}} = \frac{3200}{1600} = 2$. $\frac{N_o}{N} = 2^n$, $\frac{24\times10^{24}}{N} = 2^2 = 4$,

$N = \frac{24\times10^{24}}{4} = 6\times10^{6}$atoms.

32. A radioactive sample initially contains m atoms. The number of atoms that would have disintegrated after four half-lives is A.$\frac{m}{16}$ B.$\frac{m}{8}$ C.$\frac{7m}{8}$ **D.$\frac{15m}{16}$**.

$N_o=m$, N=?, n=4, $\frac{N_o}{N} = 2^n$, $\frac{m}{N} = 2^4$, $\frac{m}{N} = 16$,

$N = \frac{m}{16}$. Number of atoms decayed or disintegrated = Initial quantity−Number of atoms remaining , $N_D = N_o - N = m - \frac{m}{16}$,

$N_D = \frac{16m-m}{16} = \frac{15m}{16}$.

33. A radioactive material has an initial activity of 7000 counts per second and an activity of 875 counts per second after 6 hours. The half-life od the material is A.6hr B.3hr **C.2hr** D.1hr.

Initial activity A_o=7000 counts per second, Activty after decay or disintegration A=875 counts per second, t=6hrs, $t_{1/2}$=?,

$\frac{A_o}{A} = 2^n$, $\frac{7000}{875} = 2^n$, $8 = 2^n$, $2^3 = 2^n$, n=3,

$n = \frac{t}{t_{1/2}}$, $3 = \frac{6}{t_{1/2}}$, $t_{1/2} = \frac{6}{3} = 2$hrs.

34. A radioactive element has a half-life of 4 days. The fraction that has decayed in 16 days is A.$\frac{3}{4}$ B.$\frac{1}{16}$ **C.$\frac{15}{16}$** D.$\frac{1}{4}$.

t=16 days, $t_{1/2}$=4 days, n=$\frac{t}{t_{1/2}} = \frac{16}{4} = 4$ half-lives,

Fraction lost or decayed after decay

$F_D = 1 - \frac{N}{N_o} = 1 - \left(\frac{1}{2}\right)^n$, $F_D = 1 - \left(\frac{1}{2}\right)^n$,

$F_D = 1 - \left(\frac{1}{2}\right)^4 = 1 - \frac{1}{16} = \frac{16-1}{16} = \frac{15}{16}$.

35. A radioactive element of decay constant λ has a half-life of A.$\frac{In2}{\lambda^2}$ B.$-\frac{In2}{\lambda^2}$ C.$\frac{2In2}{\lambda^2}$ **D.$\frac{In2}{\lambda}$.**

36. A radioactive substance with a half life of 5s has a decay constant of A.0.693s^{-1} B.0.347s^{-1} C.0.218s^{-1} **D.0.139s^{-1}.**

Demystified Series by Dr Timothy

$t_{1/2}$=5s , $\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5}$ =0.139s^{-1}.

37. A radioisotope has a decay constant of $10^{-7}s^{-1}$. The average life of the radioisotope is **A.6.93×10⁶s** B.1.00×10^{-4}s C.1.00×10^{-4}s D.1.00×10^{7}s E.6.93×10^{7}s

Average life of radioisotope is the same as its half life.Decay constant, λ=10^{-7}s^{-1}, $t_{1/2}$=?

$\lambda = \frac{0.693}{t_{1/2}}$, $t_{\frac{1}{2}} = \frac{0.693}{\lambda} = \frac{0.693}{10^{-7}}$ =6.93×10^{6}s.

38. A piece of radioactive material contains 10^{20}atoms. If the half-life of the material is 20 seconds, the number of disintegrations in the first second is **A.3.47×10¹⁸** B.6.93×10^{20} C.3.47×10^{20} D.6.93×10^{18}

$t_{1/2}$=20 seconds, number of atoms N=10^{20} atoms, number of disintegration or decaying atoms per seconds $\frac{dN}{dt}$=? $\frac{dN}{dt} = -\lambda N$, $\lambda = \frac{0.693}{t_{1/2}}$, $\frac{dN}{dt} = \frac{0.693\,N}{t_{1/2}}$, $\frac{dN}{dt} = \frac{0.693\times10^{20}}{20} = 0.03465\times10^{20} \approx 3.47\times10^{18}$

39. Two isotopes of uranium are designated as $^{238}U$and ^{235}U. The numbers 238 and 235 represenrs their A.atomic numbers **B.nucleon numbers** C.proton numbers D.neutron numbers.

41. Which of the following statements are true of isotopes? I.isotopes of an element have the same chemical properties because they have the same number of electrons II.isotopes of elements are normally separated using physical properties III.isotopes of elements have the same number of protons in their nuclei A.I and II B.I and III C.II and III **D.I,II and III.**

Isotopes of an element have the same atomic number, i.e same number of protons and electrons but different mass number, due to difference it the number of neutrons. They have the same chemical properties but different physical properties, hence they can be separated physically.

42. The electrically neutral isotopes of oxygen are different in their I.proton number II.nucleon number III.number of orbital electrons IV.number of neutrons. Which of the statements above are correct? A.I and II B.I and III C.I and IV D.II and III **E.II and IV.**

43. Neutrons are used to induce artificial radioactivity because they **A.are energetic** B.have no charge C.have no mass D.are ionizing.

44. Tin(Sn) has twenty isotopes of which the lightest is represented by the symbol $^{108}_{50}Sn$, knowing that all possible isotopes exist. The symbol of the heaviest isotope is A.$^{109}_{50}Sn$**B.** $^{\mathbf{127}}_{\mathbf{50}}\boldsymbol{Sn}$C.$^{110}_{50}Sn$ D.$^{125}_{50}Sn$.

Isotopes of an element are formed when one of its isotope is bombarded or irradiated artificially neutrons.

First isotope; $^{108}_{50}Sn$

Second isotope; $^{108}_{50}Sn + {}^{1}_{0}n \rightarrow {}^{109}_{50}Sn$, the mass number of the second isotope is increased by 1 i.e 1 neutron.

Therefore,mass number of the 20th isotope (heaviest isotope) is increased by 19 i.e 19 neutrons. $^{108}_{50}Sn + 19{}^{1}_{0}n \rightarrow {}^{127}_{50}Sn$.

45. Which of the following statements is not true of nuclear fission? A.a mother nucleus is broken into two daughter nuclei of roughly equal parts B.neutron is normally used nin bombarding heavy nucleus **C.two ligt nuclei are made to combine to form a heavy nucleus** D.a very enormous energy is released E.there is an apparent loss in mass

In fission process, the mass of the products is less than the mass of the reactants, hence an apparent loss in mass. The number electrons produced by fission process exceeds the number that caused fission, hence a chain reaction is initiated.

46. One of the featured of fission processs is that A.its products are not radioactive **B.it leads to chain reaction** C.neutrons are not released D.the sum of the masses of the reactants equals the sum of the masses of the products .

The products of nuclear fission are also radioactive. It leads to chain reactions due to exceess or additional neutrons produced. The mass of the products is not equal to the mass of the reactants, due to the conversion of mass into energy.

47. When the nucleus of uranium atoms is split into two fragments of nearly equal masses, the sum of te masses of the fragments is less than the mass of the original nucleus. The difference is a neausre of the A.atomic mass B.change of momentum of each fragment C.potential energy D.kinetic energy **E.nuclear energy released**

The difference between the masses of the parent nucleus and the products represents the nuclear energy released in the form of heat.

48. Which of the following are produced after a nuclear fusion process? I.one heavy nucleus II.neutrons III.protons IV.energy A.I and II **B.I and IV** C.II and III D.II and IV .

Nuclear fusion involves the combination of lighter nuclei to for a heavier nucleus with the liberation of tremendous amount of energy. Neutrons are not always produced in nuclear fusion.

50. Which of the following explains the process by which protons are converted into helium atoms with tremendous release of energy? A.photoelectric emission B.thermionic emission **C.thermonuclear fusion** D.field emission E.nuclear fission.

Thermonuclear fusion or nuclear fusion is the combination of lighter nuclei(i.e protons) to form a hevier nucleus(i.e helium nucleus). Protons are represented as hydrogen atoms $^{2}_{1}H$ while helium atoms are $^{4}_{2}He$.

$^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{4}_{2}He$ + energy.

Demystified Series by Dr Timothy

51. In a certain fusion reaction, a deuteron ^2_1H interacts with a tritium ^3_1H and produces an α-particle ^4_2He and a second product. The second product is A.a proton B.an electron **C.a neutron** D.a gamma ray .
$^2_1H + {}^3_1H \rightarrow {}^4_2He + {}^A_ZY$,
$2+3=4+A$, $1+1=2+Z$,
$5=4+A$, $2=2+Z$,
$A=5-4=1$, $Z=2-2=0$.
$^A_ZY = {}^1_0Y = {}^1_0n$. Hence, the second product is a neutron.

52. In a nuclear fusion process four protons each of mass M_P, wre fused to produce a nucleus X of mass M_X. Which of the following equation is correct? **A.$4M_p > M_X$** B.$4M_P = M_X$ C.$4M_P < M_X$ D.$M_P = M_X$.
$4M_p \rightarrow M_X +$ energy. In nuclear reactions, the mass of the reactant is always greater than the mass of the product or the mass of the reactant is always less than that of the products. i.e
$4M_p > M_X$ or $M_X < 4M_P$.

53. Nuclear fission is preferable to nuclear fusion in the generation of energy because A.the raw materials for fusion are not easily obtained **B.very high temperatures are required for fusion** C.the by-products of fusionare very dangerous D.energy obtained from fusion is relatively smaller.

54. Which of the following materials is used to control the rate of neutron production in a nuclear reaction? A.concrete shield B.graphite rods **C.boron rods** D.uranium rods. Concrete shield are biological shield which protects nuclear reactor opeartors from intense exposure to radiation. Graphite rods acts as moderator to slow down the neutrons, without absorbing it, to prevent dying out of the chain reaction. Boron rods are control rods which absorbs excess neutrons produced by the chain reaction, which might speed up the reaction. Uranium rods are fissionable materials used as nuclear fuel.

55. In a modern nuclear reactor, the escaping radiation is prevented from reaching the opeartors by providing them with A.white laboratory jackets B.gas masks **C.lead-line jackets** D.water proof jackets.

56. Which of the following are the essential parts of an atomic bomb? **A.uranium and neutrons** B.radium and polonium C.nitrogen and neutrons D.uranium and α-particles.

57. In a thermonuclear reaction, the total initial mass is 5.02×10^{-27}kg and the total final mass is 5.01×10^{-27}kg. The energy released in the process is A.9.0×10^{-10}J B.9.0×10^{-11}J C.9.0×10^{-12}J **D.9.0×10^{-13}J** [$c=3.0\times10^8$ms^{-1}].
Initial mass=5.02×10^{-27}kg, final mass= 5.01×10^{-27}kg, $c=3.0\times10^8$ms^{-1},Mass defect or loss in mass=Δm, Δm=Initial mass−Final mass
$\Delta m=5.02\times10^{-27}-5.01\times10^{-27}=0.01\times10^{-27}$J,
Energy released, $E = \Delta mc^2$
$E = 0.01\times10^{-27}\times(3.0\times10^8)^2$,
$E = 1.0\times10^{-29}\times9\times10^{16} = 9\times10^{-13}$J.

58. In the fusion of hydrogen isotopes into helium, the decrease in mass is about 0.65%. Calculate the energy obtainable when 1.0g of hydrogen is used. [$c=3.0\times10^8$ms^{-1}] A.5.85×10^9J B.5.85×10^{10}J **C.5.85×10^{11}J** D.5.85×10^{12}J.
Initial mass=1g=1×10^{-3}kg, mass defect or decrease in mass=Δm=0.65% of 1×10^{-3}kg,
$\Delta m=\frac{0.65}{100}\times1\times10^{-3}=6.5\times10^{-6}$kg,
$E = \Delta mc^2=6.5\times10^{-6}\times(3.0\times10^8)^2$,
$E = 6.5\times10^{-6}\times9\times10^{16}=5.85\times10^{11}J$.

57. When a nucleus is formed by bringing protons and neutrons together, the actual mass of the firmed nucleus is less than the sum of masses of the constituent protons and neutrons. The energy equivalent to the mass difference is A.lost energy B.work function C.stability energy **D.binding energy.**

58. In a fission of uranium isotope, a mass defct of 0.25 a.m.u is observed. Calculate the ninding energy of the isotope. A.$1.25\times10^{-19}J$ **B.$3.72\times10^{-11}J$** C.$1.49\times10^{-19}J$ D.$1.86\times10^{-9}J$
[1 a.m.u=$931MeV$, $1eV=1.6\times10^{-19}J$]
Mass defect, Δm= 0.25 a.m.u, $1MeV=10^6eV$, 1a.m.u=$931MeV=931\times10^6eV$,$1eV=1.6\times10^{-19}J$
∴ 0.25 a.m.u=$0.25\times931\times10^6\times1.6\times10^{-19}$
Binding enegy=$3.72\times10^{-11}J$.

59. In the nuclear reaction, $^1_1H + {}^7_3Li \rightarrow 2{}^4_2He$, the atomic masses of hydrogen and lithium are $1.008U$ and $7.106U$ respectively and the atomic mass of the helium atom is $4.004U$. I.energy is released by nuclear fission II.the energy released can be calculated from $E = \Delta mc^2$, where Δm is the mass change produced and c is the speed of light respectively III.the relation $E = \Delta mc^2$, can be verified by measuring the initial energy of the α-particles A.I B.III C.I and II **D.II and III** E.I,II and III
- The nuclear reaction above is an example of nuclear fussion.
- Δm=mass defect=mass of reactants−mass of products. mass of reactants=mass of hydrogen nucleus+mass of lithium nucleus= $1.008+7.016=8.024U$, mass of product= 2(mass of helium nucleus)=$2(4.004)=8.008U$, mass defect, $\Delta m = 8.024-8.008=0.016U$.
- Measuring the initial energy of the alpha particle (^4_2He) can be used to verify $E=\Delta mc^2$.

60. A possible fusion reaction is ;
$^2_1H + {}^2_1H \rightarrow {}^3_1H + {}^1_1H + Q$, where Q is the energy released as a result of the reaction. If $Q=4.03MeV$. Calculate the atomic mass of ^3_1H in atomic mass unit. [$^2_1H=2.01410u$, $^1_1H=1.00783u$, $1u=931MeV$]
A.$3.01604u$ B.$2.01504u$ C.$3.01705u$ D.$4.03702u$
Mass of reactants=$^2_1H + {}^2_1H = 2(^2_1H)$
Mass of reactants=$2(2.01410)=4.02820u$,

Demystified Series by Dr Timothy

Mass of products=$^3_1H + {}^1_1H$, $^3_1H=x$,
Mass of products=$(x + 1.00783)u$,
Mass defect, Δm=mass of reactant−mass of products=$4.02820-(x + 1.00783)$
Mass defect, $\Delta m = 4.02820-1.00783+x$,
Mass defect, $\Delta m = (3.02037+x)u$,
Energy released=Binding energy=$4.03MeV$,
$1u=931MeV$, $\therefore 4.03MeV=\frac{4.03}{931}=0.004329u$.
Energy released=difference between mass of reactants and products=$(M_R-M_P)u$
$0.004329=3.02037+x$,
$x = {}^3_1H = 3.02037-0.004329=3.01604u$.
Alternatively, $^2_1H + {}^2_1H \rightarrow {}^3_1H + {}^1_1H + Q$,
Substitute their given value into the nuclear equation , $2(2.01410)= x + 1.00783+0.004329$,
$4.02820= x +1.01216$, $x = 4.02820-1.01216$
$x = 3.01604u$.

61. The binding energy of helium ^4_2He is
A.2.017u **B.0.333u** C.4.033u D.0.330u.
[atomic mass of proton=1.00783u, atomic mass of neutron=1.00867u]
Atomic mass of proton=1.00783u, Atomic mass of neutron=1.00867u, Number of protons=2, Number of neutrons=4−2=2,
Total mass of reactants M_R=Mass of nucleons =Mass of protons+mass of neutrons,
$M_R=2(1.00783)+2(1.00867)$
$M_R=2.01566+2.01734=4.033u$,
Mass of product M_P =Mass of helium ^4_2He nucleus $=4u$. Binding energy=difference between mass of reactants(nucleons) and product (nucleus)= $M_R-M_P=4.033u-4u$,
Binding energy=$0.033u$.

62. Calculate in joules, the binding energy of ^7_3Li [Atomic mass of ^7_3Li=7.01600u, Masss of neutron=1.00867u, Mass of proton=1.00783u Unified atomic mass unit, u=931MeV, $1eV=1.6\times10^{-19}$J] A.5.17×10^{-11}J **B.6.28×10^{-12}J** C.7.04×10^{-12}J D.8.35×10^{-11}J.
Mass of proton=1.00783u, number of protons =3, Total mass of protons = 3(1.00783)= 3.02349u, Mass of neutron=1.00867u, number of neutrons=7−3=4, Total mass of neutrons = 4(1.00867)= 4.03468u.
Mass of reactants M_R=Mass of nucleons=Mass of protons+Mass of neutrons,
$M_R=3.02349+4.03468=7.05817u$,
Mass of product M_P=Atomic mass of ^7_3Li nucleus=7.01600u.
Binding energy=$(M_R-M_P)u$
Binding energy=$7.05817-7.01600=0.04217u$
$1u=931MeV$, $1MeV=10^6eV$,$1eV=1.6\times10^{-19}$J,
$0.04217u = 0.04217\times931\times10^6\times1.6\times10^{-19}$,
Binding energy in joules = 6.28×10^{-12}J.

● Wave particle duality or wave particle paradox

- Wave particle duality or wave particle paradox of light and matter – It is also called the **principle of complementarity.** It is the fact or idea that light and matter sometimes exists as wave and as particle other times but not simultaneously.
- Wave nature or theory of light – It is explained by the properties of wave i.e **reflection, refraction, diffraction, interference and polarisation.**
- The invention of electron microscope and scanning tunneling microscope(STM) is based on the wave nature of matter.
- Particles used in explaining the wave nature of matter are ; neutrons,electrons/beta particles, alpha particles/helium nucleus and proton/hydrogen atom.
- Louis De Broglie's principle proposed the wave nature of matter just like light, as moving particles with wave properties. Louis De Broglie's wavelength λ, is given by;
$\boldsymbol{\lambda = \frac{h}{P} = \frac{h}{mv} = \frac{h}{mc}}$,
P=momentum of the particle=mv or mc, m=mass, v=velocity, c=velocity of light, h=planck's constant=$6.6\times10^{-34}Js$.
- Electron diffraction – It is used in testing or experimenting the wave nature of matter, by difrracting electrons or waves matter using crystals. Electrons produced by a hot filament are accelerated from rest through a voltage or p.d V. Increasing the voltage V on the anode causes the velocity v, of the electrons to increase. The wavelength λ of the material object is give by ;
$\boldsymbol{\lambda = \frac{h}{p} = \frac{h}{m_e v} = \frac{h}{\sqrt{2m_e eV}} = \frac{hv}{2eV}}$.
$\boldsymbol{v = \sqrt{\frac{2eV}{m_e}}}$. m_e=mass of electron, e=electronic charge or charge on electron, V=accelerating p.d or voltage, v=speed of electron.
- Diffraction of particles heavier than electron e.g neutrons, is done by reducing the velocity. In neutron diffraction is done with slow moving electrons where the velocity is reduced inoder to increase the wavelength and decrease the momentum i.e $\boldsymbol{v \propto \frac{P}{\lambda}}$.
- In the wave nature of electron, increase in speed of a photoelectron will decrease its wavelength and increase it energy. i.e $\boldsymbol{v \propto \frac{1}{\lambda}}$ and **energy** $\boldsymbol{\propto v^2}$.
- Particle nature of light and matter – It is explained by **Brownian motion, diffusion, osmosis, crystal structure, thermionic emission, photoelectric emission, Compton effect, radiation from heated bodies, electron momentum and line spectra (absorption and emission).**
- Heisenberg uncertainty principle or indeterminacy principle – It states that the position $\boldsymbol{x}$ and momentum $\boldsymbol{P}$ of a particle cannot

Demystified Series by Dr Timothy

be accurately determined at the same instant of time **OR** It is impossible to determine acurrately, both the wave properties of matter and its particulate properties at the same time.

$\Delta x \,.\, \Delta P \geq h = \frac{h}{2\pi}$.

Heinsenberg uncertainty principles can also be expressed in terms of energy E and time t or position x and velocity v.

$\Delta E \,.\, \Delta t \geq h = \frac{h}{2\pi}$. $\Delta x \,.\, \Delta v \geq h = \frac{h}{2\pi}$.

- Position and momentum, energy and time, position and velocity are known as **complementary variables.**

- Heisenberg uncertainty principle does not apply to objects above atomic sizes i.e macroscopic matter, but applies to particulate matter e.g atoms, because h is extremely small (6.63×10^{-34}Js).

Examples :

1. Which of the following phenomena best supports the wave nature of matter? **A.Electron diffraction** B.Photoelectric effect C.Electron momentum D.Black body radiation

Electron diffraction is explained in terms of waves. Photoelectric effect, electron momentum and black body radiation are explained in terms of particle nature of matter.

2. Electron passing through crystals are diffracted because they A.are reppeled by the atoms in the crystal B.are attracted by atoms in crystal C.are particles **D.possess wave properties.**

3. Which of the following is not an evidence of the particulate nature of matter? A.Diffusion B.Brownian motion **C.Diffraction** D.Crystal structure E.Photoelectricity.

4. A moving electron possesses the characteristics of A.particles only B.waves only **C.both particles and waves** D.momentum only

A moving electron has momentum and energy(particle nature of matter).Moving electrons also have velocity and undergointerference(wave nature of matter).

5. The particle nature of light is demonstrated by the A.speed of light B.colours of light C.diffraction of light **D.photoelectric effect**

Speed of light, colours(wavelength) of light due to dispersion and diffraction of light are wave properties). Photoelectric effect and spectra(emission and absorption) formation are particle nature of light.

6. The wave theory of light cannot be used to explain **A.phtoelectricity** B.polarization C.diffraction D.interference.

7. Which of the scientists postulated that moving particles exhibit wave properties? A.Niel Bohr B.Ernest Rutherford C.Wemer Heisenber **D.Louis De Broglie** E.Erwin Schroedinger.

8. An electron of mass 9.1×10^{-31}kg moves with a speed of 10^7ms^{-1}. Calculate the wavelength of the associated wave. [$h=6.6\times10^{-34}Js$] A.$9.10\times10^{-24}Js$ B.4.55×10^{-17}m **C.7.25×10^{-11}m** D.6.20×10^{-8}m

$m_e=9.1\times10^{-31}$kg, $v=10^7$ms^{-1}, $h=6.6\times10^{-34}Js$,

$\lambda = \frac{h}{p} = \frac{h}{m_e v} = \frac{6.6\times10^{-34}}{9.1\times10^{-31}\times10^{7}} = 7.25\times10^{-11}$m

9. If electrons are accelerated from rest through a potential difference of 10kV, what is the wavelength of the associated electrons? **A.1.22×10^{-11}m** B.3.87×10^{-10}m C.2.27×10^{11}m D.2.27×10^{14}m [$m_e=9.1\times10^{-31}$kg, $e=1.6\times10^{-19}$C $h=6.6\times10^{-34}$Js]

$V=10\text{k}V=10\times10^3V=1.0\times10^4V$, $m_e=9.1\times10^{-31}$kg

$e=1.6\times10^{-19}$C, $h=6.6\times10^{-34}Js$, λ=?

$\lambda = \frac{h}{\sqrt{2m_e eV}} = \frac{6.6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times1.6\times10^{-19}\times1\times10^4}}$,

$\lambda = 1.22\times10^{-11}$m.

10. Calculate the minimum wavelength of X-rays when a voltage of 60kV is applied to an X-ray tube. A.2.06×10^{-11}m **B.1.03×10^{-11}m** C.3.09×10^{-11}m D.4.12×10^{-11}m [$e=1.6\times10^{-19}$C, $h=6.6\times10^{-34}$Js, $c=3.0\times10^8$ms^{-1}]

$V=60\text{k}V=60\times10^3V=6.0\times10^4V$, $e=1.6\times10^{-19}$C, $h=6.6\times10^{-34}$Js, $v=c=3.0\times10^8$ms^{-1},

$\lambda = \frac{hv}{2eV} = \frac{6.6\times10^{-34}\times3\times10^8}{2\times1.6\times10^{-19}\times6\times10^4} = 1.03\times10^{-11}$m.

11. The mass and wavelength of a moving electron are 9.0×10^{-31}kg and 1.0×10^{-10}m respectively. Calculate the kinetic energy of the electron [$h=6.6\times10^{-34}Js$] A.1.21×10^{-17}J B.2.31×10^{-17}J **C.2.42×10^{-17}J** D.3.24×10^{-17}J.

$m_e=9.0\times10^{-31}$kg, $\lambda=1.0\times10^{-10}$m, $h=6.6\times10^{-34}$Js,

$\lambda = \frac{h}{m_e v}$, $v = \frac{h}{m_e \lambda}$, $K.E = \frac{1}{2}m_e v^2$

$K.E = \frac{1}{2}m_e\left(\frac{h}{m_e\lambda}\right)^2 = \frac{1}{2}\times\frac{m_e}{m_e{}^2}\times\frac{h^2}{\lambda^2} = \frac{h^2}{2m_e\lambda}$,

$K.E = \frac{(6.6\times10^{-34})^2}{2\times9\times10^{-34}\times1\times10^{-10}} = \frac{4.356\times10^{-67}}{1.8\times10^{-50}}$,

$K.E = 2.42\times10^{-17}$J.

12. It is always not possible to determine exactly and simulataneously the position and momentum of a particle. This statement is known as the A.De Broglie's law **B.Heinsenberg's principle** C.Compton's effect D.Frank Hertz experimental law E.wave particle paradox.

13. Which of the following statements is a correct consequence of the uncertainty principle? A.The uncertainty in our knowledge of energy and duration taken to measure it are each less than plank's constant B.The complete knowledge of position of a particle implies incomplete ignorance of its energy C.It is possible to measure exactly both the position and momentum of a particle at the same time D.Both momentum and energy of a particle can be known with absolute certainty **E.A particle's kinetic energy cannot be measured accurately at any time**

Complementary variable in Heinsenberg's uncertainty principles are; position and momentum, energy and time, position and

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velocity. The product of the uncertainty of each complementary variable is greater than or equal to planck's constant.

14. The uncertainty principle can be most usefully apllied to measurement on A.the momentum of a train at a particular position B.the energy of a satellite at a time in space **C.the energy of an electron in a cathode ray tube at a particular time** D.the momentum of a football as it just of an electrons at a particular velocity

Heinsenberg's uncertainty principle only apply to non-massive bodies i.e atomic sizes e.g electrons. Train,satellite and football are massive bodies.

15. If Δx is the uncertainty in the measurement of the position along the x-axis and ΔP_x is the uncertainty in the measurement of linear momentum along the x-axis, then uncertainty principle relation is given as **A.$\Delta P_x \Delta x \geq h$** B.$\Delta P_x \Delta x = h$ C.$\Delta P_x \Delta x = 2\pi$ D.$\Delta P_x \Delta x = 2\pi/h$ E.$\Delta P_x \Delta x = 1$.

16. If the uncertainty in the measurement of the position of a particle is 5×10^{-10}m, the uncertainty in the momentum of the particle is A.1.32×10^{-44}Ns B.3.30×10^{-44}Ns **C.1.32×10^{-24}Ns** D.3.30×10^{-24}Ns [h=6.6×10^{-34}Js]

Uncertainty in position, Δx=5×10^{-10}m, h=$6.6\times10^{-34}Js$,Uncertainty in momentum, ΔP,

$\Delta x \,.\, \Delta P \geq h$, $\Delta P \geq \frac{h}{\Delta x} \geq \frac{6.6\times10^{-34}}{5\times10^{-10}}$,

$\Delta P \geq 1.32\times10^{-24}$Ns.

17. The uncertainty in determining the duration during which an electron remains in a particular energy level before returning to the ground state is 2.0×10^{-9}s. Calculate the uncertainty in detrmining its energy at that level. [Take $\frac{h}{2\pi} = h = 1.054\times10^{-34}$Js] **A.$5.27\times10^{-26}$J** B.$4.16\times10^{-26}$J C.$3.13\times10^{-26}$J D.$2.14\times10^{-26}$J.

Uncertainty in time,Δt = $2.0\times10^{-9}s$,

$\frac{h}{2\pi} = 1.054\times10^{-34}$Js, Uncertainty in energy, ΔE=?,

$\Delta E \,.\, \Delta t \geq \frac{h}{2\pi}$, $\Delta E \geq \frac{h}{2\pi} \times \frac{1}{\Delta t}$,

$\Delta E \geq 1.054\times10^{-34} \times \frac{1}{2.0\times10^{-9}} \geq 5.27\times10^{-26}J$.

18. The uncertainty in the velocity v of a moving electron of mass 10^{-30}kg is 3×10^{6}ms^{-1}. Calculate the uncertainty of the simultaneously measurement of its position x. [h=6.62×10^{-34}Js] A.1.11×10^{-10}m **B.2.21×10^{-10}m** C.3.22×10^{-10}m D.4.12×10^{-10}m

Mass of electron,m=10^{-30}kg, velocity of electron=3×10^{6}ms^{-1}, Uncertainty in momentum, $\Delta P = \Delta(m \times v)$, $\Delta x \,.\, \Delta P \geq h$, $\Delta x \,.\, \Delta(m \times v) \geq h$,

$\Delta x \geq \frac{h}{\Delta(m\times v)} \geq \frac{6.62\times10^{-34}}{10^{-30}\times3\times10^{6}} \geq 2.21\times10^{-10}$m.

Jamb past questions on elementary atomic physics :

[1979/31,1980/4,5,20,41,1981/29,1982/40, 1983/34,1984/25,42,44,1985/49,50,1986/50, 1987/47,48,49,50,1988/49,50,1989/46,47,49, 50,1990/46,48,50,1991/47,1992/47,49,50,199 3/47,49,50,1994/50,1995/49,50,1997/50,1998 /48,49,50,1999/43,44,47,49,2000/28,37,2001/ 38,39,42,44,2002/41,43,45,49,2003/33,40,44, 46,48,2004/23,28,44,2005/5,7,9,12,2006/2,4,8, 2007/10,12,2008/46,2009/45,46,48,2010/46, 47,2011/41,43,2012/47,2013/47,2014/44,45, 48,2015/9,21]

Demystified Series by Dr. Timothy

Chapter 27 – Basic Electronics : Semiconductos, diodes and transistors

● **Semi-conductors –** Materials whose resistivty and conductivity is between that of conductors and insulators e.g silicon and germanium.

- Pure or intrinsic semi-conductors are group IV elements i.e tetravalent or quadrivalent.

- Characteristics of semi-conductors –

1.They exhibit covalent bonds between their atoms.

2. Pure or intrinsic semi-conductors are poor conductors, it will only conduct electricity when a battery is connected across its end.

3. The current carriers are free electrons and holes. **Pure or intrinsic semi-conductors contain equal numbers of holes and free electrons.**

4.Semi-conductors have negative temperature coefficient of resistance or resistivity i.e their resistance decreases with increase in temperature, hence their conductivity increases with temperature increase.

5. They conduct electricity better with impurities(dopants) and heat.

- At absolute zero temperature(0K), no free electrons or holes are found in a pure or intrinsic semi-conductor.

- Energy band theory or Band theory is the theory of electronic conduction in solids, which satisfactorily explains the porperties of conductors,insulators and semi-conductors and also gives distinguishes them.

- Valence band is the low energy region or range of energies of valence electrons.

- Conduction band is the high energy region with free electron accelerated from the valence band or range of energies of free electrons.

- Forbidden energy band or fabrication band is the energy difference or the gap between the valence band and the conduction band.

1. In semi-conductors – I. The valence band contains holes while the conduction band contains free electrons. **II.**The number of holes equals the number of free electrons. **III.**The forbidden energy gap or fabrication band is narrow or small.

2. In conductors – I. The valence band and conduction band contains electrons. **II.**The number of electrons in the conduction band is greater than that in the valence band. **III.**Forbidden energy gap or fabrication band is little or no gap.

3. In insulators – I. The valence band contains electrons while no electrons is present in the conduction band i.e empty. **II.**Forbidden energy gap or fabrication band is large or wide.

- Impure or Extrinsic semi-conductors are formed by doping. Doping is the addition of impurities to a pure or intrinsic semi-conductor inoder to improve its conducting ability.

- Impure or Extrinsic semi-conductors are of two types namely ; Positive type (P-type) and Negative type(N-type).

1.P-type is formed when a pure or intrinsic semi-conductor is doped with a **group III** elements **(acceptor or trivalent impurities)** e.g boron, aluminium,thalium,indium,gallium **[BATIG].**

2.N-type is formed when a pure or intrinsic semi-conductor is dopped with a **group V** element **(donor or pentavalent impurities)** e.g bismuth, phosphorus,arsenic,antimony **[BPA^2].**

- Mnemonic: PTNP – PT : P-type, trivalent, NP : N-type, pentavalent.

- In impure of extrinsic semi-conductors the number of holes is not equal to the number of electrons. The one that is more in number is called the **majority carrier**, while the one that is lesser in number is called **minority carrier.**

- In P-type : majority carrier are holes, while minority carrier are free electrons.

- In N-type : majority carriers are free electrons while minority carriers are holes.

- P-n junction diode or depletion layer or potential barrier – It is the contact surface of a P-type and N-type semi-conductor joined together.

- When voltage is applied i.e connecting a battery, across the junction, **biasing (forward or reverse)** occurs.

1.Forward biased – It occurs when the positive terminal of a cell is connected to the P-type and the negative terminal is connected to the N-type of a junction diode. Forward biasing drives majority carriers (holes for P-type and electrons for N-type), towards the junction and drives minority carrirers (free electrons for P-type and holes for N-type) away from the junction.

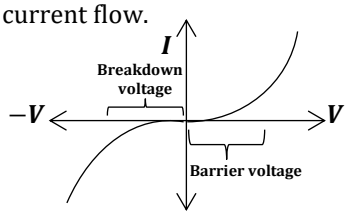
- Features of forward biasing – I. Large current is produced as current flow is due to majority carriers. **II.**Depletion layer reduces in width or narrowed. **III.**Juction offers low resistance to current flow. **IV.**Internal barrier voltage is overcomed. **V.**Potential barrier is reduced.

2.Reverse biased – It occurs when the positive terminal of a cell is connected to the N-type and the negative terminal is connected to the P-type. Reverse biasing drives majority carriers (holes for P-type and electrons for N-type) away from the junction and drives minority carriers (free electrons in P-type and holes in N-type) towards the junction.

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- Features of reverse biasing – I. Extremely small current called reverse saturation current or leakage curret, is produced and current flow is due to minority carriers. **II.**Depletion layer increases in width or widened. **III.**Junction offers a high resistance to current. **IV.**Breakdown voltage is overcomed. **V.**Potential barrier is increased.

- Current - voltage ($I-V$) characteristic curve/graph of a p-n junction diode shows that the junction has a low resistance in forward-bias direction i.e larger current flow and a high resistance in reverse-bias direction i.e small current flow.



- The conductivity of semi-conductors is not affected by its small size.

- Advantages of semi-conductors diodes over thermionic diodes (vacuum diodes) –

1.Small size. **2.**Low power requirements, require no filament power and produce less heat. **3.**Highly durable, less liable to break. **4.**It needs only small p.d or voltage to operate in a radio receiver. **5.**It requires no time to heat up inoder to produce current carriers, as they are already available in semi-conductor.

- Disadvantages of semi-conductors diodes –

1.Easily damaged by high temperature (extremely heat/temperature sensitive). **2.**Destroyed by slight overload for a short time.

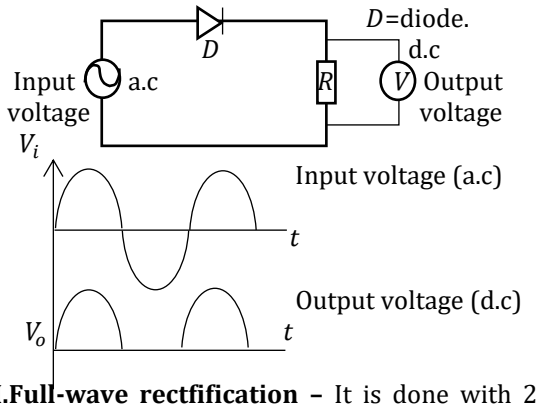
- Zener diodes are diodes which conducts well in both forward and reverse bias direction,but they are operated in reverse bias direction. **Zener diodes are used as voltage regulators and stabilizer for power supply .**

- Uses of p-n junction or semi conductor diodes –

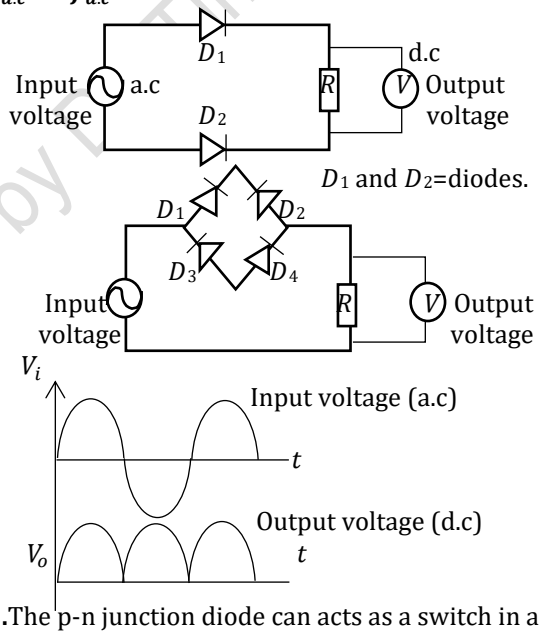
1. Current rectification – It is used as a rectifier in alternate circuit to convert a.c to d.c, because it allows current in one direction only i.e forward biased direction. In a rectifier circuit, the input voltage is a.c while the output voltage is d.c.

- Rectification can be half-wave rectification or full-wave rectification depending on the arrangement of the diodes.

I.Half-wave rectification – It is done with 1 diode connected in series with to am a.c voltage source.It conducts on the positive half of the cycle i.e forward biased direction and cuts off current i.e no current is conducted, on the negative half cycle i.e reverse bias direction.**The frequency of the output d.c $f_{d.c}$ equals the frequency of the input a.c $f_{a.c}$.**



II.Full-wave rectfification – It is done with 2 diodes in series or 4 diodes in bridge arrangement. The negative half of a.c flips to the positive half cycle such that both halves are in same direction. Full wave rectification is of a higher d.c output and efficiency than half wave rectification. **The frequency of the output d.c $f_{d.c}$ is twice the frequency of the input a.c $f_{a.c}$ $f_{d.c}=2f_{a.c}$.**



2.The p-n junction diode can acts as a switch in a circuits. It closes the circuit i.e switch on in the forward biased and opens the circuit i.e switch off in the reverse biased.

3.The p-n junction is used as a regulator or control in circuits, detecting faults and protecting circuits.

● **Transistors –** These are semi-conductor devices with three layers or terminals namely **emitter(E), collector(C)** which forms the outer layersand **base(B)** which forms the middle layer. It is formed when two p-n junctions; emmiter-base junction and collector-base junction are sandwiched together.Hence, it is also called a **bipolar junction device.**

I. Emitter – It is the heavily doped layer or terminal rich in holes.It supplies the majority carriers and is always forward biased in respect to the base with an applied voltage.

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II. Collector – It is the lightly dopped layer or terminal rich in electrons.It collects majority carriers and is always reverse biased in respect to the base with an applied voltage.

III. Base – It is the thinnest slightly doped layer or terminal poor in electrons.

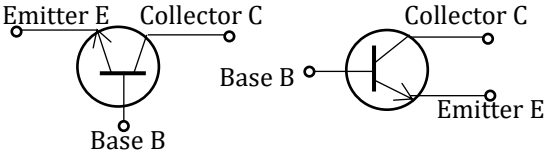
- Transistors are operated by current i.e they are controlled by small current to regulate large current.

- Transistors are of two types; **p-n-p** transistor and **n-p-n** transistor.

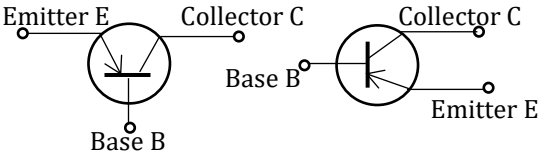
I.p-n-p transistor – It consists of two outer P-type and a N-type placed between them. Acceptor impurities from the P-type diffuses on both sides of the N-type. Current flows into the circuit.

II.n-p-n transistor – It consists of two outer N-type and a P-type placed between them. Donor impurities from the N-type diffuses on both sides of the P-type. Current flows out of the circuit.

- Circuits symbols of transistors : $\boldsymbol{I_E = I_B + I_C}$**.**



n-p-n transistor



p-n-p transistor

- In-P ; Inward arrorw p-n-p, Out-N ; Outward arrow n-p-n.

- The emitter and collector are slant with the emitter bearing an arrow end indicating direction of current flow. **Current flows out** of the circuit in n-p-n transistors while **current flows into** the circuit in p-n-p transistor.

- Uses of transistors –

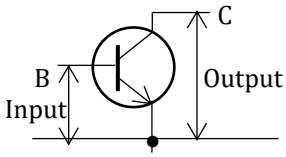
1.Transistors are used as amplifiers for the amplication of a.c current,voltage and power.

2.Switches when amplifying a.c voltage.

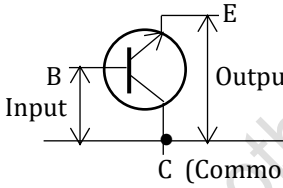
3.Regulates current, voltage or power.

- Transistor amplification configuration – A transistor when used as an amplifier, has 1 of its terminals common to the input and output terminals, hence it has 2 inputs and 2 outputs terminals. There are 3 types of transistors amplification configuration connection:

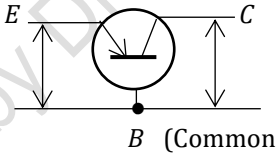
1.Common emitter(C.E) connection – Emitter is common to both the input (emitter and base) and output (emitter and collector). The properties of C.E configuration are **high current, high voltage and high power amplification. Common-emitter provides satisfactory current amplification and is most widely used in audio(low) frequency amplifiers**.



2.Common collector(C.C) or emitter follower connection – Collector is common to both input (base and collector) and output (emitter and collector). The properties of C.C configuration are **high current, medium power amplification and voltage buffering i.e does not amplify voltage.**



3.Common base (C.B) connection – Base is common to both input (emitter and base) and output (base and collector). The properties of C.B configuration are **high voltage and low power amplification.It does not amplify current.**

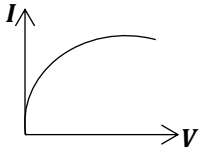


- Tansistor as a switch –

I.Cut off current region – When the switch is opened such that the no current is flowing through the transistor i.e input current is zero,then the output current is zero and the out put voltage is maximum.

II.Saturation or maximum current region – When the switch is closed such that current is flowing through the transistor i.e input current is maximum, then the output current is maximum and the output voltage is zero.

- Current-voltage ($\boldsymbol{I-V}$) characteristic curve of a transistor –



- Transistors and p-n junction diodes are non-ohmic conductors.

Examples :

1. Which of the following is a semi-conductor material? A.Copper **B.silicon** C.Steel D.Iron.

2. Which of the following is a pure semi-conductor? **A.Silicon** B.Phosphorus C.Transistor D.Boron E.Carbon

Pure or intrinsic semi-condutors are group IV elements e.g silicon and germanium. Phosphorus is a group V element. Transistor is an impure or

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extrinsic semi-conductor formed fron the joining two p-n junction together. A pure semi-conductor is an insulator or poor conductor, but graphite(a form of carbon) is a good conductor.

3.Intrinsic semi-conductor atoms are bonded together by **A.covalent bonding** C.ionic bonding C.thermal bonding E.metallic bonding

Intrinsic or pure semi conductor atoms are bonded by covalent bonding.

4. For semi conductors to have negative temperature coefficient of resistance implies that A.their resistance increases with temperature B.their resistance is constantly changing with temperature **C.their resistance decreases with temperature** D.they have electrons and holes at high temperature.

5. Intrinsic semi-conductors A.do not conduct electricity at all **B.conduct better at higher temperature** C.are used as conductors in electronics D.have low resistivity at lower temperatures E.have high resistivity at higher temperatures

The resistance of semi-conductors decreases with increase in temperarure,hence their conductivity increases with temperature.

6. I.small size II.Low power requirement III.Not easily damaged by high temperature IV.Highly durable. Which of the above are the advantages of semi-conductors? A.I,II and III B.II,III and IV **C.I,II and IV** D.I,II,III and IV.

7. In semi-conductors, the carriers of current at room temperature are A.electrons only **B.electrons and holes** C.holes D.electrons and ions.

8. The major difference between a pure semi-conductor and a pure metal is that A.metals are harder than semi-conductors B.while the resistance of metals decreases with temperature, the reverse is the case of semi-conductors C.metals have forbidden gaps while semi-conductors have not **D.the resistance of metals increases with temperature, while for semi-conductors, it is the reverse.**

The resistance of metals increases with temperature as they have positive temperature coefficient of resistance. Metals i.e good conductors don't have energy gap while semi-conductors have narrow or small forbidden energy gap. **9.**

The theory of electronic conduction in solids, which satisfactorily explains the properties of conductors, insulators and semi-conductors is A.particle theory **B.band theory** C.group theory D.string theory.

10. In a pure semi-conductor, the number of electrons in the conduction band is **A.equal to the number of holes in the valence band** B.greater than the number of holes in the valence band C.less than the number of holes in the valence band D.twice the number of bands in the valence band.

11. When a pure semi-conductor is heated, its resistance A.increases **B.decreases** C.remains the same D.increases then decreases.

12. When impurities are added to semi-conductors, the conductivity of the semi-conductor **A.increeases** B.decreases C.remain constant D.increases then decreases.

13. Which of the following statements are true about semi-conductors I.doped semi-conductors are extrinsic semi-conductor II.Germanium can be used as an intrinsic semi-conductor III.Silicon can be used as an extrinsic semi-conductor IV.materials used for doping semi-conductors have four valence electrons **A.I and II** B.II and IV C.I,II and III D.I,II and IV E.II,III and IV

Doped semi-conductors are impure or extrinsic semi-conductors. Pure semi-conductors are intrinsic semi-conductors. Silicon and Germanium are pure or intrinsic semi-conductors. Doping materials have three and five valence electrons i.e group III and V.

14. If a donor impurity is introduced into a silicon lattice, the resulting material is **A.n-type** B.p-type C.richer in electron-hole pairs D.richer in holes only **.**

Donor impurity is a pentavalent or group V element.Doping silicon with a donor impurity forms a n-type semi-conductor.

15. Pure silicon can be converted to p-type material by adding controlled amount of A.pentavalent atoms B.hexavalent atoms C.tetravalent atoms **D.trivalent atoms .**

Doping a pure silicon with a trivalent atom i.e group III element, acceptor impurity, forms a p-type semi-conductor.

16. A certain manufacture wishes to make an n-type semi-conductor. Which materials would he select for doping? **A.Arsenic and antimony** B.Aluminium and indium C.Boron and antinomy D.Helium and boron

Arsenic and antimony are pentavalent or group V elements i.e donor impurities, hence when used to dope a semi-conductor forms a n-type semi-conductor.

17. Silicon doped with aluminium and germanium doped with arsenic become A.n-type semi-conductors B.p-type semi-conductorsC.n- and p-type respectively **D.p- and n-type semi-conductor respectively**

Silicon doped with aluminium(trivalent or group III element or acceptor impurity) forms a p-type semi-conductor while germanium doped with arsenic(pentavalent or group V element or donor impurity) forms a n-type semi-conductor.

18. As the voltage across a junction diode increases in the forward biased direction A.forward resistance increases B.current

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decreases **C.current increases** D.contains remains constant E.potential barrier increases
In the forward biased direction, current increases as voltage across the p-n junction diode increases due to a decrease in the resistance and potential barrier.

19. The current in a reverse-biased junction is due to A.electrons B.majority carriers C.holes **D.minority carriers**
Current in reverse-biased direction is due to minority carriers while current in forward-biased direction is due to majority carriers.

20. The battery connection that permits current to flow across a p-n junction is called A.amplification B.rectification **C.forward biased** D.reverse biased.

21. In a semi-conductor junction diode, as the depletion or barrier layer is forward biased, the layer **A.narrows** B.remains constant C.widens then narrows D.widens
The depletion layer narrows or reduces its width in forward biased, in order to increase the flow of majority carriers, hence increase in current flow.

22. If a reversed-biased voltage is applied across a p-n junction, the depletion layer width is A.halved B.constant C.decreased **D.increased**
The depletion layer widens or increases its width in reverse biased, inroder to decrease the flow of majority carriers and permit the flow of minority carriers, hence decrease in current flow.

23. The p-n junction diode is primarily used as a A.transistor **B.rectifier** C.an amplifier D.a shunt
p-n junction diode is majorly used as a rectifier and minorly as a switch.

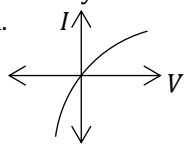
24. A p-n junction diode is used as A.a rectifier in a d.c circuit B.an amplifier in an a.c circuit **C.a rectifier in an a.c circuit** D.an amplifier in an d.c circuit
A p-n junction is used as a rectifier in a.c circuit for the rectification of a.c to d.c.

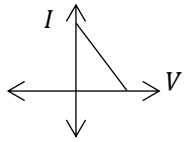
25. A semi-conductor diode is used in rectifying alternating current into direct current mainly because it A.allows current to flow in either direction B.is non-linear C.offers a high input resistance **D.allows current to flow only in one direction**
Semi-conductor diode i.e p-n junction diode is used in rectifying alternate current because it allows current to flow in one direction as it offers low resistance and cuts off current in the other direction as it offers high resistance.

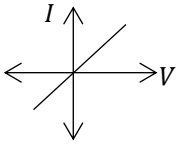
26. The p-n junction can act as rectfiers because they **A.conduct current when forward biased** B.conduct current when reverse biased C.block current when forward biased D.conduct current in both directions
The p-n junction diode can act as rectifiers because they conduct current when forward biased and block current when reverse biased.

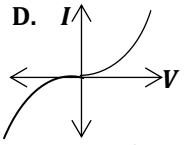
27. A diode is not used for A.regulation **B.amplification** C.detection D.instrument protection E.switching
Transistors are used for amplification not diodes.

28. A typical graph of a p-n junction diode is shown by

A. 

B. 

C. 

D. 

The graph shows that it is forward biased in one direction and reverse biased in the other direction as shown in option D**.**

29. Zener diode is used for A.current amplification B.power amplification **C.voltage regulation** D.energy conversion

30. In bipolar transistors, conduction is by A.charges and current B.electrons and position C.electrons only **D.electrons and holes.**

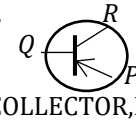
31. A transistor is operated by **A.current** B.voltage C.voltage and current D.resistance
Transistors are controlled by small current to regulate large current.

32. Transistors are used for the A.conversion of a.c to d.c **B.amplification of signals**
C.rectification of signals D.conversion of d.c to a.c.
Transistors are used as amplifiers.

33. The minority carriers in a n-p-n transistor are the **A.holes** B.electrons C.ioms D.electrons and holes
In a n-p-n transistor, the majority carriers are free electrons while the minority carriers are holes. In a p-n=p transistor, the majority carriers are holes while the minority carriers are free electrons.

34. I.For current amplification II.For power amplification III.For voltage stabilization IV.As a switch. Which of the above are the uses of transistor? A.I,II,III and IV B.I,III and IV C.I,II and III **D.I,II and IV.**
Zener diodes are used as voltage regulators or stabilizers.

35. A transistor is used in the amplification of signals because it A.allows doping B.contains electrons and hole carriers C.consumes a lot of power **D.controls flow of current**
Transistors are controlled by small current to regulate large current i.e a sma;; input current produces a large output current.

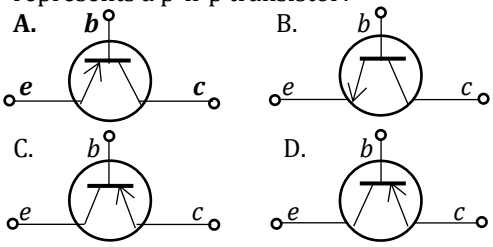
36. 

In the circuit diagram of a transistor above, the terminals P, Q and R represents
A.COLLECTOR,EMITTER and BASE **B.EMITTER,**

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BASE and COLLECTOR C.BASE, EMITTER and COLLECTOR D.BASE, COLLECTOR and EMITTER.

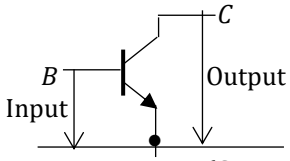
37. Which of the following diagrams correctly represents a p-n-p transistor?



In p-n-p transistor, current flows into the circuit i.e In-P; Inward current p-n-p as shown in option A. Option B represents n-p-n transistor, current flows out of the circuit i.e Out-N; Outward current n-p-n. Current flows in and out of a transistor through the emitter only.

38. In common-emitter transistor amplifier, the input signal is applied between the A.emitter and ground **B.base and emitter** C.base and collector D.emitter and collector E.collector and ground.

In common-emitter connection, the input signal or voltage is through the emitter-base junction or through the base while the output signal or voltage is through the emitter-collector junction or through the collector.

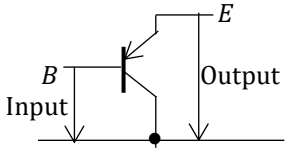


(Common emitter)

39. In common emitter configuration, the output voltage is through the A.base B.emitter **C.collector** D.resistor.

40. In the common collector amplifier circuit, the input signal is applied between the **A.base and the collector** B.emitter and the junction C.collector and the common base D.collector and the emitter junction E.the base and the emitter

In a common collector connection, the input signal or voltage is through the collector-base junction or through the base while the output signal or voltage is through the emitter-collector junction or through the emitter.



(Common collector)

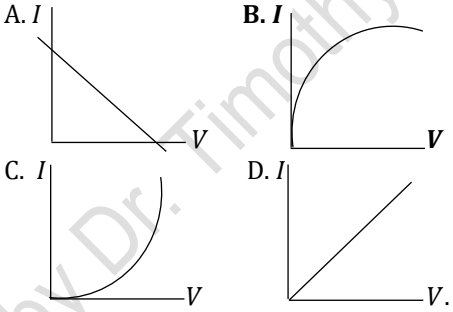
41. In which of the following configurations can a transistor be used as a voltage amplifier? I. common-collector configuration II. common-base configuration III. common-emitter configuration A.I B.II C.III **D.I and II** E.II and III

Common-collector does not amplify voltage.

42. When a base current of the common emitter transistor is kept at zero, it operates in the A.active-on region B.breakdown region **C.cut-off region** D.saturation region

In a common emitter transistor,with switch opened such that the no current is flowing through the transistor i.e. input current(base current) is zero, then the output current (colletor current) is zero and the out put voltage is maximum i.e. cut off region. While with switch closed such that current is flowing through the transistor i.e. input current(base current) is maximum, then the output current (collector current) is maximum and the output voltage is zero i.e. saturation or maximum region.

43. A typical transistor characteristic is represented as



Jamb past questions on Basic electronics :

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